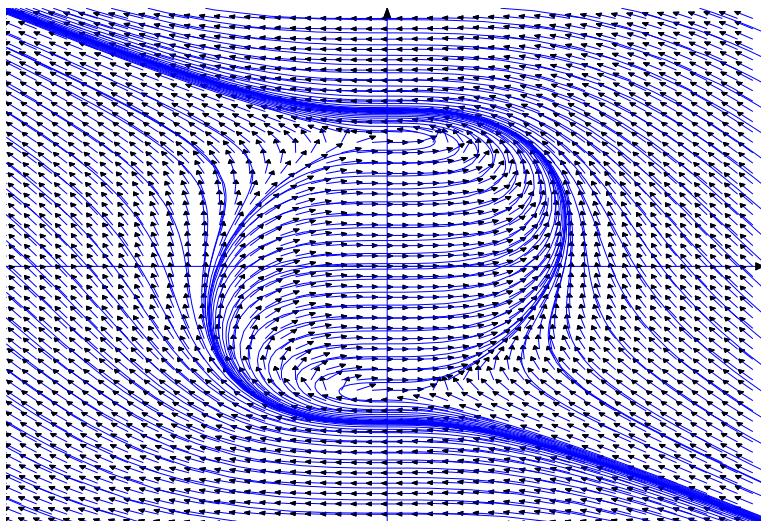


STUDENT SOLUTIONS MANUAL FOR

# ELEMENTARY DIFFERENTIAL EQUATIONS



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**TO BEVERLY**

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# Chapter 1

## Introduction

### 1.2 Basic Concepts

1.2.2. (a) If  $y = ce^{2x}$ , then  $y' = 2ce^{2x} = 2y$ .

(b) If  $y = \frac{x^2}{3} + \frac{c}{x}$ , then  $y' = \frac{2x}{3} - \frac{c}{x^2}$ , so  $xy' + y = \frac{2x^2}{3} - \frac{c}{x} + \frac{x^2}{3} + \frac{c}{x} = x^2$ .

(c) If

$$y = \frac{1}{2} + ce^{-x^2}, \quad \text{then} \quad y' = -2xce^{-x^2}$$

and

$$y' + 2xy = -2xce^{-x^2} + 2x\left(\frac{1}{2} + ce^{-x^2}\right) = -2xce^{-x^2} + x + 2xce^{-x^2} = x.$$

(d) If

$$y = \frac{1 + ce^{-x^2/2}}{1 - ce^{-x^2/2}}$$

then

$$\begin{aligned} y' &= \frac{(1 - ce^{-x^2/2})(-cxe^{-x^2/2}) - (1 + ce^{-x^2/2})cxe^{-x^2/2}}{(1 - ce^{-x^2/2})^2} \\ &= \frac{-2cxe^{-x^2/2}}{(1 - ce^{-x^2/2})^2} \end{aligned}$$

and

$$\begin{aligned} y^2 - 1 &= \left( \frac{1 + ce^{-x^2/2}}{1 - ce^{-x^2/2}} \right)^2 - 1 \\ &= \frac{(1 + ce^{-x^2/2})^2 - (1 - ce^{-x^2/2})^2}{(1 - ce^{-x^2/2})^2} \\ &= \frac{4ce^{-x^2/2}}{(1 - ce^{-x^2/2})^2}, \end{aligned}$$

so

$$2y' + x(y^2 - 1) = \frac{-4cx + 4cx}{(1 - ce^{-x^2/2})^2} = 0.$$

(e) If  $y = \tan\left(\frac{x^3}{3} + c\right)$ , then  $y' = x^2 \sec^2\left(\frac{x^3}{3} + c\right) = x^2 \left(1 + \tan^2\left(\frac{x^3}{3} + c\right)\right) = x^2(1 + y^2)$ .

(f) If

$$\begin{aligned} y &= (c_1 + c_2x)e^x + \sin x + x^2, \quad \text{then} \\ y' &= (c_1 + 2c_2x)e^x + \cos x + 2x, \\ y' &= (c_1 + 3c_2x)e^x - \sin x + 2, \end{aligned}$$

and

$$\begin{aligned} y'' - 2y' + y &= c_1e^x(1 - 2 + 1) + c_2xe^x(3 - 4 + 1) \\ &\quad - \sin x - 2\cos x + \sin x + 2 - 4x + x^2 \\ &= -2\cos x + x^2 - 4x + 2. \end{aligned}$$

(g) If  $y = c_1e^x + c_2x + \frac{2}{x}$ , then  $y' = c_1e^x + c_2 - \frac{2}{x^2}$  and  $y'' = c_1e^x + \frac{4}{x^3}$ , so  $(1 - x)y'' + xy' - y = c_1(1 - x + x - 1) + c_2(x - x) + \frac{4(1 - x)}{x^3} - \frac{2}{x} - \frac{2}{x} = \frac{4(1 - x - x^2)}{x^3}$

(h) If  $y = \frac{c_1 \sin x + c_2 \cos x}{x^{1/2}} + 4x + 8$  then  $y' = \frac{c_1 \cos x - c_2 \sin x}{x^{1/2}} - \frac{c_1 \sin x + c_2 \cos x}{2x^{3/2}} + 4$  and  $y'' = -\frac{c_1 \sin x + c_2 \cos x}{x^{1/2}} - \frac{c_1 \sin x - c_2 \cos x}{x^{3/2}} + \frac{3}{4} \frac{c_1 \sin x + c_2 \cos x}{x^{5/2}}$ , so  $x^2y'' + xy' + \left(x^2 - \frac{1}{4}\right)y = c_1 \left(-x^{-3/2} \sin x - x^{1/2} \cos x + \frac{3}{4}x^{-1/2} \sin x + x^{1/2} \cos x - \frac{1}{2}x^{-1/2} \sin x + x^{3/2} \sin x - \frac{1}{4}x^{-1/2} \sin x\right) + c_2 \left(-x^{-3/2} \cos x + x^{1/2} \sin x + \frac{3}{4}x^{-1/2} \cos x - x^{1/2} \sin x - \frac{1}{2}x^{-1/2} \cos x + x^{3/2} \cos x - \frac{1}{4}x^{-1/2} \cos x\right) + 4x + \left(x^2 - \frac{1}{4}\right)(4x + 8) = 4x^3 + 8x^2 + 3x - 2$ .

**1.2.4.** (a) If  $y' = -xe^x$ , then  $y = -xe^x + \int e^x dx + c = (1 - x)e^x + c$ , and  $y(0) = 1 \Rightarrow 1 = 1 + c$ , so  $c = 0$  and  $y = (1 - x)e^x$ .

(b) If  $y' = x \sin x^2$ , then  $y = -\frac{1}{2} \cos x^2 + c$ ;  $y\left(\sqrt{\frac{\pi}{2}}\right) = 1 \Rightarrow 1 = 0 + c$ , so  $c = 1$  and  $y = 1 - \frac{1}{2} \cos x^2$ .

(c) Write  $y' = \tan x = \frac{\sin x}{\cos x} = -\frac{1}{\cos x} \frac{d}{dx}(\cos x)$ . Integrating this yields  $y = -\ln|\cos x| + c$ ;  $y(\pi/4) = 3 \Rightarrow 3 = -\ln(\cos(\pi/4)) + c$ , or  $3 = \ln\sqrt{2} + c$ , so  $c = 3 - \ln\sqrt{2}$ , so  $y = -\ln(|\cos x|) + 3 - \ln\sqrt{2} = 3 - \ln(\sqrt{2}|\cos x|)$ .

(d) If  $y'' = x^4$ , then  $y' = \frac{x^5}{5} + c_1$ ;  $y'(2) = -1 \Rightarrow \frac{32}{5} + c_1 = -1 \Rightarrow c_1 = -\frac{37}{5}$ , so  $y' = \frac{x^5}{5} - \frac{37}{5}$ . Therefore,  $y = \frac{x^6}{30} - \frac{37}{15}(x - 2) + c_2$ ;  $y(2) = -1 \Rightarrow \frac{64}{30} + c_2 = -1 \Rightarrow c_2 = -\frac{47}{15}$ , so  $y = -\frac{47}{15} - \frac{37}{5}(x - 2) + \frac{x^6}{30}$ .

(e) (A)  $\int xe^{2x} dx = \frac{xe^{2x}}{2} - \frac{1}{2} \int e^{2x} dx = \frac{xe^{2x}}{2} - \frac{e^{2x}}{4}$ . Therefore,  $y' = \frac{xe^{2x}}{2} - \frac{e^{2x}}{4} + c_1$ ;  $y'(0) = 1 \Rightarrow -\frac{1}{4} + c_1 = \frac{5}{4} \Rightarrow c_1 = \frac{5}{4}$ , so  $y' = \frac{xe^{2x}}{2} - \frac{e^{2x}}{4} + \frac{5}{4}$ ; Using (A) again,  $y = \frac{xe^{2x}}{4} - \frac{e^{2x}}{8} - \frac{e^{2x}}{8} + \frac{5}{4}x + c_2 = \frac{xe^{2x}}{4} - \frac{e^{2x}}{4} + \frac{5}{4}x + c_2$ ;  $y(0) = 7 \Rightarrow -\frac{1}{4} + c_2 = 7 \Rightarrow c_2 = \frac{29}{4}$ , so  $y = \frac{xe^{2x}}{4} - \frac{e^{2x}}{4} + \frac{5}{4}x + \frac{29}{4}$ .



(f) (A)  $\int x \sin x dx = -x \cos x + \int \cos x dx = -x \cos x + \sin x$  and (B)  $\int x \cos x dx = x \sin x - \int \sin x dx = x \sin x + \cos x$ . If  $y'' = -x \sin x$ , then (A) implies that  $y' = x \cos x - \sin x + c_1$ ;  $y'(0) = -3 \Rightarrow c = -3$ , so  $y' = x \cos x - \sin x - 3$ . Now (B) implies that  $y = x \sin x + \cos x + \cos x - 3x + c_2 = x \sin x + 2 \cos x - 3x + c_2$ ;  $y(0) = 1 \Rightarrow 2 + c_2 = 1 \Rightarrow c_2 = -1$ , so  $y = x \sin x + 2 \cos x - 3x - 1$ .

(g) If  $y''' = x^2 e^x$ , then  $y'' = \int x^2 e^x dx = x^2 e^x - 2 \int x e^x dx = x^2 e^x - 2x e^x + 2e^x + c_1$ ;  $y''(0) = 3 \Rightarrow 2 + c_1 = 3 \Rightarrow c_1 = 1$ , so (A)  $y'' = (x^2 - 2x + 2)e^x + 1$ . Since  $\int (x^2 - 2x + 2)e^x dx = (x^2 - 2x + 2)e^x - \int (2x - 2)e^x dx = (x^2 - 2x + 2)e^x - (2x - 2)e^x + 2e^x = (x^2 - 4x + 6)e^x$ , (A) implies that  $y' = (x^2 - 4x + 6)e^x + x + c_2$ ;  $y'(0) = -2 \Rightarrow 6 + c_2 = -2 \Rightarrow c_2 = -8$ , so (B)  $y' = (x^2 - 4x + 6)e^x + x - 8$ ; Since  $\int (x^2 - 4x + 6)e^x dx = (x^2 - 4x + 6)e^x - \int (2x - 4)e^x dx = (x^2 - 4x + 6)e^x - (2x - 4)e^x + 2e^x = (x^2 - 6x + 12)e^x$ , (B) implies that  $y = (x^2 - 6x + 12)e^x + \frac{x^2}{2} - 8x + c_3$ ;

$y(0) = 1 \Rightarrow 12 + c_3 = 1 \Rightarrow c_3 = -11$ , so  $y = (x^2 - 6x + 12)e^x + \frac{x^2}{2} - 8x - 11$ .

(h) If  $y''' = 2 + \sin 2x$ , then  $y'' = 2x - \frac{\cos 2x}{2} + c_1$ ;  $y''(0) = 3 \Rightarrow -\frac{1}{2} + c_1 = 3 \Rightarrow c_1 = \frac{7}{2}$ , so  $y'' = 2x - \frac{\cos 2x}{2} + \frac{7}{2}$ . Then  $y' = x^2 - \frac{\sin 2x}{4} + \frac{7}{2}x + c_2$ ;  $y'(0) = -6 \Rightarrow c_2 = -6$ , so  $y' = x^2 - \frac{\sin 2x}{4} + \frac{7}{2}x - 6$ . Then  $y = \frac{x^3}{3} + \frac{\cos 2x}{8} + \frac{7}{4}x^2 - 6x + c_3$ ;  $y(0) = 1 \Rightarrow \frac{1}{8} + c_3 = 1 \Rightarrow c_3 = \frac{7}{8}$ , so  $y = \frac{x^3}{3} + \frac{\cos 2x}{8} + \frac{7}{4}x^2 - 6x + \frac{7}{8}$ .

(i) If  $y''' = 2x + 1$ , then  $y'' = x^2 + x + c_1$ ;  $y''(2) = 7 \Rightarrow 6 + c_1 = 7 \Rightarrow c_1 = 1$ ; so  $y'' = x^2 + x + 1$ . Then  $y' = \frac{x^3}{3} + \frac{x^2}{2} + (x - 2) + c_2$ ;  $y'(2) = -4 \Rightarrow \frac{14}{3} + c_2 = -4 \Rightarrow c_2 = -\frac{26}{3}$ , so  $y' = \frac{x^3}{3} + \frac{x^2}{2} + (x - 2) - \frac{26}{3}$ . Then  $y = \frac{x^4}{12} + \frac{x^3}{6} + \frac{1}{2}(x - 2)^2 - \frac{26}{3}(x - 2) + c_3$ ;  $y(2) = 1 \Rightarrow \frac{8}{3} + c_3 = 1 \Rightarrow c_3 = -\frac{5}{3}$ , so  $y = \frac{x^4}{12} + \frac{x^3}{6} + \frac{1}{2}(x - 2)^2 - \frac{26}{3}(x - 2) - \frac{5}{3}$ .

**1.2.6.** (a) If  $y = x^2(1 + \ln x)$ , then  $y(e) = e^2(1 + \ln e) = 2e^2$ ;  $y' = 2x(1 + \ln x) + x = 3x + 2x \ln x$ , so  $y'(e) = 3e + 2e \ln e = 5e$ ; (A)  $y'' = 3 + 2 + 2 \ln x = 5 + 2 \ln x$ . Now,  $3xy' - 4y = 3x(3x + 2x \ln x) - 4x^2(1 + \ln x) = 5x^2 + 2x^2 \ln x = x^2 y''$ , from (A).

(b) If  $y = \frac{x^2}{3} + x - 1$ , then  $y(1) = \frac{1}{3} + 1 - 1 = \frac{1}{3}$ ;  $y' = \frac{2}{3}x + 1$ , so  $y'(1) = \frac{2}{3} + 1 = \frac{5}{3}$ ; (A)  $y'' = \frac{2}{3}$ . Now  $x^2 - xy' + y + 1 = x^2 - x\left(\frac{2}{3}x + 1\right) + \frac{x^2}{3} + x - 1 + 1 = \frac{2}{3}x^2 = x^2 y''$ , from (A).

(c) If  $y = (1 + x^2)^{-1/2}$ , then  $y(0) = (1 + 0^2)^{-1/2} = 1$ ;  $y' = -x(1 + x^2)^{-3/2}$ , so  $y'(0) = 0$ ; (A)  $y'' = (2x^2 - 1)(1 + x^2)^{-5/2}$ . Now,  $(x^2 - 1)y - x(x^2 + 1)y' = (x^2 - 1)(1 + x^2)^{-1/2} - x(x^2 + 1)(-x)(1 + x^2)^{-3/2} = (2x^2 - 1)(1 + x^2)^{-1/2} = y''(1 + x^2)^2$  from (A), so  $y'' = \frac{(x^2 - 1)y - x(x^2 + 1)y'}{(x^2 + 1)^2}$ .

(d) If  $y = \frac{x^2}{1 - x}$ , then  $y(1/2) = \frac{1/4}{1 - 1/2} = \frac{1}{2}$ ;  $y' = -\frac{x(x - 2)}{(1 - x)^2}$ , so  $y'(1/2) = \frac{(-1/2)(-3/2)}{(1 - 1/2)^2} = 3$ ; (A)  $y'' = \frac{2}{(1 - x)^3}$ . Now, (B)  $x + y = x + \frac{x^2}{1 - x} = \frac{x}{1 - x}$  and (C)  $xy' - y = -\frac{x^2(x - 2)}{(1 - x)^2} - \frac{x^2}{1 - x} = \frac{x^2}{(1 - x)^2}$ . From (B) and (C),  $(x + y)(xy' - y) = \frac{x^3}{(1 - x)^3} = \frac{x^3}{2} y''$ , so  $y'' = \frac{2(x + y)(xy' - y)}{x^3}$ .

**1.2.8.** (a)  $y = (x - c)^a$  is defined and  $x - c = y^{1/a}$  on  $(c, \infty)$ ; moreover,  $y' = a(x - c)^{a-1} =$

$$a(y^{1/a})^{a-1} = ay^{(a-1)/a}.$$

(b) if  $a > 1$  or  $a < 0$ , then  $y \equiv 0$  is a solution of (B) on  $(-\infty, \infty)$ .

**1.2.10.** (a) Since  $y' = c$  we must show that the right side of (B) reduces to  $c$  for all values of  $x$  in some interval. If  $y = c^2 + cx + 2c + 1$ ,

$$\begin{aligned} x^2 + 4x + 4y &= x^2 + 4x + 4c^2 + 4cx + 8c + 4 \\ &= x^2 + 4(1+c)x + 4(c^2 + 2c + 1) \\ &= x^2 + 4(1+c) + 2(c+1)^2 = (x + 2c + 2)^2. \end{aligned}$$

Therefore,  $\sqrt{x^2 + 4x + 4y} = x + 2c + 2$  and the right side of (B) reduces to  $c$  if  $x > -2c - 2$ .

(b) If  $y_1 = -\frac{x(x+4)}{4}$ , then  $y_1' = -\frac{x+2}{2}$  and  $x^2 + 4x + 4y = 0$  for all  $x$ . Therefore,  $y_1$  satisfies

(A) on  $(-\infty, \infty)$ .

## Chapter 2

# First Order Equations

### 2.1 Linear First Order Equations

**2.1.2.**  $\frac{y'}{y} = -3x^2$ ;  $|\ln|y|| = -x^3 + k$ ;  $y = ce^{-x^3}$ .  $y = ce^{-(\ln x)^2/2}$ .

**2.1.4.**  $\frac{y'}{y} = -\frac{3}{x}$ ;  $\ln|y| = -3\ln|x| + k = -\ln|x|^3 + k$ ;  $y = \frac{c}{x^3}$ .

**2.1.6.**  $\frac{y'}{y} = -\frac{1+x}{x} = -\frac{1}{x} - 1$ ;  $|\ln|y|| = -\ln|x| - x + k$ ;  $y = \frac{ce^{-x}}{x}$ ;  $y(1) = 1 \Rightarrow c = e$ ;  
 $y = \frac{e^{-(x-1)}}{x}$ .

**2.1.8.**  $\frac{y'}{y} = -\frac{1}{x} - \cot x$ ;  $|\ln|y|| = -\ln|x| - \ln|\sin x| + k = -\ln|x \sin x| + k$ ;  $y = \frac{c}{x \sin x}$ ;  
 $y(\pi/2) = 2 \Rightarrow c = \pi$ ;  $y = \frac{\pi}{x \sin x}$ .

**2.1.10.**  $\frac{y'}{y} = -\frac{k}{x}$ ;  $|\ln|y|| = -k \ln|x| + k_1 = \ln|x^{-k}| + k_1$ ;  $y = c|x|^{-k}$ ;  $y(1) = 3 \Rightarrow c = 3$ ;  
 $y = 3x^{-k}$ .

**2.1.12.**  $\frac{y_1'}{y_1} = -3$ ;  $\ln|y_1| = -3x$ ;  $y_1 = e^{-3x}$ ;  $y = ue^{-3x}$ ;  $u'e^{-3x} = 1$ ;  $u' = e^{3x}$ ;  $u = \frac{e^{3x}}{3} + c$ ;  
 $y = \frac{1}{3} + ce^{-3x}$ .

**2.1.14.**  $\frac{y_1'}{y_1} = -2x$ ;  $\ln|y_1| = -x^2$ ;  $y_1 = e^{-x^2}$ ;  $y = ue^{-x^2}$ ;  $u'e^{-x^2} = xe^{-x^2}$ ;  $u' = x$ ;  $u = \frac{x^2}{2} + c$ ;  
 $y = e^{-x^2} \left( \frac{x^2}{2} + c \right)$ .

**2.1.16.**  $\frac{y_1'}{y_1} = -\frac{1}{x}$ ;  $\ln|y_1| = -\ln|x|$ ;  $y_1 = \frac{1}{x}$ ;  $y = \frac{u}{x}$ ;  $\frac{u'}{x} = \frac{7}{x^2} + 3$ ;  $u' = \frac{7}{x} + 3x$ ;  $u = 7 \ln|x| + \frac{3x^2}{2} + c$ ;  
 $y = \frac{7 \ln|x|}{x} + \frac{3x}{2} + \frac{c}{x}$ .

$$\mathbf{2.1.18.} \quad \frac{y_1'}{y_1} = -\frac{1}{x} - 2x; \quad \ln|y_1| = -\ln|x| - x^2; \quad y_1 = \frac{e^{-x^2}}{x}; \quad y = \frac{ue^{-x^2}}{x}; \quad \frac{u'e^{-x^2}}{x} = x^2e^{-x^2};$$

$$u' = x^3; \quad u = \frac{x^4}{4} + c; \quad y = e^{-x^2} \left( \frac{x^3}{4} + \frac{c}{x} \right).$$

$$\mathbf{2.1.20.} \quad \frac{y_1'}{y_1} = -\tan x; \quad \ln|y_1| = \ln|\cos x|; \quad y_1 = \cos x; \quad y = u \cos x; \quad u' \cos x = \cos x; \quad u' = 1;$$

$$u = x + c; \quad y = (x + c) \cos x.$$

$$\mathbf{2.1.22.} \quad \frac{y_1'}{y_1} = \frac{4x-3}{(x-2)(x-1)} = \frac{5}{x-2} - \frac{1}{x-1}; \quad \ln|y_1| = 5\ln|x-2| - \ln|x-1| = \ln \left| \frac{(x-2)^5}{x-1} \right|;$$

$$y_1 = \frac{(x-2)^5}{x-1}; \quad y = \frac{u(x-2)^5}{x-1}; \quad \frac{u'(x-2)^5}{x-1} = \frac{(x-2)^2}{x-1}; \quad u' = \frac{1}{(x-2)^3}; \quad u = -\frac{1}{2} \frac{1}{(x-2)^2} + c;$$

$$y = -\frac{1}{2} \frac{(x-2)^3}{(x-1)} + c \frac{(x-2)^5}{(x-1)}.$$

$$\mathbf{2.1.24.} \quad \frac{y_1'}{y_1} = -\frac{3}{x}; \quad \ln|y_1| = -3\ln|x| = \ln|x|^{-3}; \quad y_1 = \frac{1}{x^3}; \quad y = \frac{u}{x^3}; \quad \frac{u'}{x^3} = \frac{e^x}{x^2}; \quad u' = xe^x;$$

$$u = xe^x - e^x + c; \quad y = \frac{e^x}{x^2} - \frac{e^x}{x^3} + \frac{c}{x^3}.$$

$$\mathbf{2.1.26.} \quad \frac{y_1'}{y_1} = -\frac{4x}{1+x^2}; \quad \ln|y_1| = -2\ln(1+x^2) = \ln(1+x^2)^{-2}; \quad y_1 = \frac{1}{(1+x^2)^2}; \quad y = \frac{u}{(1+x^2)^2};$$

$$\frac{u'}{(1+x^2)^2} = \frac{2}{(1+x^2)^2}; \quad u' = 2; \quad u = 2x + c; \quad y = \frac{2x+c}{(1+x^2)^2}; \quad y(0) = 1 \Rightarrow c = 1; \quad y =$$

$$\frac{2x+1}{(1+x^2)^2}.$$

$$\mathbf{2.1.28.} \quad \frac{y_1'}{y_1} = -\cot x; \quad \ln|y_1| = -\ln|\sin x|; \quad y_1 = \frac{1}{\sin x}; \quad y = \frac{u}{\sin x}; \quad \frac{u'}{\sin x} = \cos x; \quad u' =$$

$$\sin x \cos x; \quad u = \frac{\sin^2 x}{2} + c; \quad y = \frac{\sin x}{2} + c \csc x; \quad y(\pi/2) = 1 \Rightarrow c = \frac{1}{2}; \quad y = \frac{1}{2}(\sin x + \csc x).$$

$$\mathbf{2.1.30.} \quad \frac{y_1'}{y_1} = -\frac{3}{x-1}; \quad \ln|y_1| = -3\ln|x-1| = \ln|x-1|^{-3}; \quad y_1 = \frac{1}{(x-1)^3}; \quad y = \frac{u}{(x-1)^3};$$

$$\frac{u'}{(x-1)^3} = \frac{1}{(x-1)^4} + \frac{\sin x}{(x-1)^3}; \quad u' = \frac{1}{x-1} + \sin x; \quad u = \ln|x-1| - \cos x + c; \quad y = \frac{\ln|x-1| - \cos x + c}{(x-1)^3};$$

$$y(0) = 1 \Rightarrow c = 0; \quad y = \frac{\ln|x-1| - \cos x}{(x-1)^3}.$$

$$\mathbf{2.1.32.} \quad \frac{y_1'}{y_1} = -\frac{2}{x}; \quad \ln|y_1| = 2\ln|x| = \ln(x^2); \quad y_1 = x^2; \quad y = ux^2; \quad u'x^2 = -x; \quad u' = -\frac{1}{x};$$

$$u = -\ln|x| + c; \quad y = x^2(c - \ln|x|); \quad y(1) = 1 \Rightarrow c = 1; \quad y = x^2(1 - \ln x).$$

$$\mathbf{2.1.34.} \quad \frac{y_1'}{y_1} = -\frac{3}{x-1}; \quad \ln|y_1| = -3\ln|x-1| = \ln|x-1|^{-3}; \quad y_1 = \frac{1}{(x-1)^3}; \quad y = \frac{u}{(x-1)^3};$$

$$\frac{u'}{(x-1)^3} = \frac{1 + (x-1)\sec^2 x}{(x-1)^4}; \quad u' = \frac{1}{x-1} + \sec^2 x; \quad u = \ln|x-1| + \tan x + c; \quad y = \frac{\ln|x-1| + \tan x + c}{(x-1)^3};$$

$$y(0) = -1 \Rightarrow c = 1; \quad y = \frac{\ln|x-1| + \tan x + 1}{(x-1)^3}.$$

$$\mathbf{2.1.36.} \quad \frac{y_1'}{y_1} = \frac{2x}{x^2-1}; \quad \ln|y_1| = \ln|x^2-1|; \quad y_1 = x^2-1; \quad y = u(x^2-1); \quad u'(x^2-1) = x;$$

$$u' = \frac{x}{x^2 - 1}; \quad u = \frac{1}{2} \ln |x^2 - 1| + c; \quad y = (x^2 - 1) \left( \frac{1}{2} \ln |x^2 - 1| + c \right); \quad y(0) = 4 \Rightarrow c = -4;$$

$$y = (x^2 - 1) \left( \frac{1}{2} \ln |x^2 - 1| - 4 \right).$$

$$\mathbf{2.1.38.} \quad \frac{y_1'}{y_1} = -2x; \quad \ln |y_1| = -x^2; \quad y_1 = e^{-x^2}; \quad y = ue^{-x^2}; \quad u'e^{-x^2} = x^2; \quad u' = x^2 e^{x^2}; \quad u =$$

$$c + \int_0^x t^2 e^{t^2} dt; \quad y = e^{-x^2} \left( c + \int_0^x t^2 e^{t^2} dt \right); \quad y(0) = 3 \Rightarrow c = 3; \quad y = e^{-x^2} \left( 3 + \int_0^x t^2 e^{t^2} dt \right).$$

$$\mathbf{2.1.40.} \quad \frac{y_1'}{y_1} = -1; \quad \ln |y_1| = -x; \quad y_1 = e^{-x}; \quad y = ue^{-x}; \quad u'e^{-x} = \frac{e^{-x} \tan x}{x}; \quad u' = \frac{\tan x}{x};$$

$$u = c + \int_1^x \frac{\tan t}{t} dt; \quad y = e^{-x} \left( c + \int_1^x \frac{\tan t}{t} dt \right); \quad y(1) = 0 \Rightarrow c = 0; \quad y = e^{-x} \int_1^x \frac{\tan t}{t} dt.$$

$$\mathbf{2.1.42.} \quad \frac{y_1'}{y_1} = -1 - \frac{1}{x}; \quad \ln |y_1| = -x - \ln |x|; \quad y_1 = \frac{e^{-x}}{x}; \quad y = \frac{ue^{-x}}{x}; \quad \frac{u'e^{-x}}{x} = \frac{e^{x^2}}{x}; \quad u' = e^x e^{x^2};$$

$$u = c + \int_1^x e^t e^{t^2} dt; \quad y = \frac{e^{-x}}{x} \left( c + \int_1^x e^t e^{t^2} dt \right); \quad y(1) = 2 \Rightarrow c = 2e; \quad y = \frac{1}{x} \left( 2e^{-(x-1)} + e^{-x} \int_1^x e^t e^{t^2} dt \right).$$

**2.1.44. (b)** Eqn. (A) is equivalent to

$$y' - \frac{2}{x} = -\frac{1}{x} \quad (\text{B})$$

on  $(-\infty, 0)$  and  $(0, \infty)$ . Here  $\frac{y_1'}{y_1} = \frac{2}{x}$ ;  $\ln |y_1| = 2 \ln |x|$ ;  $y_1 = x^2$ ;  $y = ux^2$ ;  $u'x^2 = -\frac{1}{x}$ ;  
 $u' = -\frac{1}{x^3}$ ;  $u = \frac{1}{2x^2} + c$ , so  $y = \frac{1}{2} + cx^2$  is the general solution of (A) on  $(-\infty, 0)$  and  $(0, \infty)$ .

(c) From the proof of (b), any solution of (A) must be of the form

$$y = \begin{cases} \frac{1}{2} + c_1 x^2, & x \geq 0, \\ \frac{1}{2} + c_2 x^2, & x < 0, \end{cases} \quad (\text{C})$$

for  $x \neq 0$ , and any function of the form (C) satisfies (A) for  $x \neq 0$ . To complete the proof we must show that any function of the form (C) is differentiable and satisfies (A) at  $x = 0$ . By definition,

$$y'(0) = \lim_{x \rightarrow 0} \frac{y(x) - y(0)}{x - 0} = \lim_{x \rightarrow 0} \frac{y(x) - 1/2}{x}$$

if the limit exists. But

$$\frac{y(x) - 1/2}{x} = \begin{cases} c_1 x, & x > 0 \\ c_2 x, & x < 0, \end{cases}$$

so  $y'(0) = 0$ . Since  $0y'(0) - 2y(0) = 0 \cdot 0 - 2(1/2) = -1$ , any function of the form (C) satisfies (A) at  $x = 0$ .

(d) From (b) any solution  $y$  of (A) on  $(-\infty, \infty)$  is of the form (C), so  $y(0) = 1/2$ .

(e) If  $x_0 > 0$ , then every function of the form (C) with  $c_1 = \frac{y_0 - 1/2}{x_0^2}$  and  $c_2$  arbitrary is a solution of the initial value problem on  $(-\infty, \infty)$ . Since these functions are all identical on  $(0, \infty)$ , this does not contradict Theorem 2.1.1, which implies that (B) (so (A)) has exactly one solution on  $(0, \infty)$  such that  $y(x_0) = y_0$ . A similar argument applies if  $x_0 < 0$ .

**2.1.46. (a)** Let  $y = c_1 y_1 + c_2 y_2$ . Then

$$\begin{aligned} y' + p(x)y &= (c_1 y_1 + c_2 y_2)' + p(x)(c_1 y_1 + c_2 y_2) \\ &= c_1 y_1' + c_2 y_2' + c_1 p(x)y_1 + c_2 p(x)y_2 \\ &= c_1 (y_1' + p(x)y_1) + c_2 (y_2' + p(x)y_2) = c_1 f_1(x) + c_2 f_2(x). \end{aligned}$$

**(b)** Let  $f_1 = f_2 = f$  and  $c_1 = -c_2 = 1$ .

**(c)** Let  $f_1 = f$ ,  $f_2 = 0$ , and  $c_1 = c_2 = 1$ .

**2.1.48. (a)** If  $z = \tan y$ , then  $z' = (\sec^2 y)y'$ , so  $z' - 3z = -1$ ;  $z_1 = e^{3x}$ ;  $z = ue^{3x}$ ;  $u'e^{3x} = -1$ ;  $u' = -e^{-3x}$ ;  $u = \frac{e^{-3x}}{3} + c$ ;  $z = \frac{1}{3} + ce^{3x} = \tan y$ ;  $y = \tan^{-1} \left( \frac{1}{3} + ce^{3x} \right)$ .

**(b)** If  $z = e^{y^2}$ , then  $z' = 2yy'e^{y^2}$ , so  $z' + \frac{2}{x}z = \frac{1}{x^2}$ ;  $z_1 = \frac{1}{x^2}$ ;  $z = \frac{u}{x^2}$ ;  $\frac{u'}{x^2} = \frac{1}{x^2}$ ;  $u' = 1$ ;  $u = x + c$ ;  $z = \frac{1}{x} + \frac{c}{x^2} = e^{y^2}$ ;  $y = \pm \left[ \ln \left( \frac{1}{x} + \frac{c}{x^2} \right) \right]^{1/2}$ .

**(c)** Rewrite the equation as  $\frac{y'}{y} + \frac{2}{x} \ln y = 4x$ . If  $z = \ln y$ , then  $z' = \frac{y'}{y}$ , so  $z' + \frac{2}{x}z = 4x$ ;  $z_1 = \frac{1}{x^2}$ ;  $z = \frac{u}{x^2}$ ;  $\frac{u'}{x^2} = 4x$ ;  $u' = 4x^3$ ;  $u = x^4 + c$ ;  $z = x^2 + \frac{c}{x^2} = \ln y$ ;  $y = \exp \left( x^2 + \frac{c}{x^2} \right)$ .

**(d)** If  $z = -\frac{1}{1+y}$ , then  $z' = \frac{y'}{(1+y)^2}$ , so  $z' + \frac{1}{x}z = -\frac{3}{x^2}$ ;  $z_1 = \frac{1}{x}$ ;  $z = \frac{u}{x}$ ;  $\frac{u'}{x} = -\frac{3}{x^2}$ ;  $u' = -\frac{3}{x}$ ;  $u = -3 \ln |x| - c$ ;  $z = -\frac{3 \ln |x| + c}{x} = -\frac{1}{1+y}$ ;  $y = -1 + \frac{x}{3 \ln |x| + c}$ .

## 2.2 Separable Equations

**2.2.2.** By inspection,  $y \equiv k\pi$  ( $k$  = integer) is a constant solution. Separate variables to find others:  $\left( \frac{\cos y}{\sin y} \right) y' = -\sin x$ ;  $\ln(|\sin y|) = \cos x + c$ .

**2.2.4.**  $y \equiv 0$  is a constant solution. Separate variables to find others:  $\left( \frac{\ln y}{y} \right) y' = -x^2$ ;  $\frac{(\ln y)^2}{2} = -\frac{x^3}{3} + c$ .

**2.2.6.**  $y \equiv 1$  and  $y \equiv -1$  are constant solutions. For others, separate variables:  $(y^2 - 1)^{-3/2} yy' = \frac{1}{x^2}$ ;  $-(y^2 - 1)^{-1/2} = -\frac{1}{x} - c = -\left( \frac{1+cx}{x} \right)$ ;  $(y^2 - 1)^{1/2} = \left( \frac{x}{1+cx} \right)$ ;  $(y^2 - 1) = \left( \frac{x}{1+cx} \right)^2$ ;  $y^2 = 1 + \left( \frac{x}{1+cx} \right)^2$ ;  $y = \pm \left( 1 + \left( \frac{x}{1+cx} \right)^2 \right)^{1/2}$ .

**2.2.8.** By inspection,  $y \equiv 0$  is a constant solution. Separate variables to find others:  $\frac{y'}{y} = -\frac{x}{1+x^2}$ ;  $\ln |y| = -\frac{1}{2} \ln(1+x^2) + k$ ;  $y = \frac{c}{\sqrt{1+x^2}}$ , which includes the constant solution  $y \equiv 0$ .

**2.2.10.**  $(y-1)^2 y' = 2x+3$ ;  $\frac{(y-1)^3}{3} = x^2+3x+c$ ;  $(y-1)^3 = 3x^2+9x+c$ ;  $y = 1 + (3x^2+9x+c)^{1/3}$ .

**2.2.12.**  $\frac{y'}{y(y+1)} = -x$ ;  $\left[ \frac{1}{y} - \frac{1}{y+1} \right] y' = -x$ ;  $\ln \left| \frac{y}{y+1} \right| = -\frac{x^2}{2} + k$ ;  $\frac{y}{y+1} = ce^{-x^2/2}$ ;  $y(2) = 1 \Rightarrow$

$c = \frac{e^2}{2}$ ;  $y = (y+1)ce^{-x^2/2}$ ;  $y(1 - ce^{-x^2/2}) = ce^{-x^2/2}$ ;  $y = \frac{ce^{-x^2/2}}{1 - ce^{-x^2/2}}$ ; setting  $c = \frac{e^2}{2}$  yields

$$y = \frac{e^{-(x^2-4)/2}}{2 - e^{-(x^2-4)/2}}.$$

$$\mathbf{2.2.14.} \quad \frac{y'}{(y+1)(y-1)(y-2)} = -\frac{1}{x+1}; \left[ \frac{1}{6} \frac{1}{y+1} - \frac{1}{2} \frac{1}{y-1} + \frac{1}{3} \frac{1}{y-2} \right] y' = -\frac{1}{x+1}; \left[ \frac{1}{y+1} - \frac{3}{y-1} + \frac{2}{y-2} \right] y' = -\frac{6}{x+1}; \ln|y+1| - 3\ln|y-1| + 2\ln|y-2| = -6\ln|x+1| + k; \frac{(y+1)(y-2)^2}{(y-1)^3} = \frac{c}{(x+1)^6};$$

$$y(1) = 0 \Rightarrow c = -256; \frac{(y+1)(y-2)^2}{(y-1)^3} = -\frac{256}{(x+1)^6}.$$

$$\mathbf{2.2.16.} \quad \frac{y'}{y(1+y^2)} = 2x; \left[ \frac{1}{y} - \frac{y}{y^2+1} \right] y' = 2x; \ln \left( \frac{|y|}{\sqrt{y^2+1}} \right) = x^2 + k; \frac{y}{\sqrt{y^2+1}} = ce^{x^2};$$

$$y(0) = 1 \Rightarrow c = \frac{1}{\sqrt{2}}; \frac{y}{\sqrt{y^2+1}} = \frac{e^{x^2}}{\sqrt{2}}; 2y^2 = (y^2+1)e^{x^2}; y^2(2 - e^{x^2}) = e^{2x^2}; y = \frac{1}{\sqrt{2e^{-2x^2} - 1}}.$$

$$\mathbf{2.2.18.} \quad \frac{y'}{(y-1)(y-2)} = -2x; \left[ \frac{1}{y-2} - \frac{1}{y-1} \right] y' = -2x; \ln \left| \frac{y-2}{y-1} \right| = -x^2 + k; \frac{y-2}{y-1} = ce^{-x^2};$$

$$y(0) = 3 \Rightarrow c = \frac{1}{2}; \frac{y-2}{y-1} = \frac{e^{-x^2}}{2}; y-2 = \frac{e^{-x^2}}{2}(y-1); y \left( 1 - \frac{e^{-x^2}}{2} \right) = 2 - \frac{e^{-x^2}}{2}; y = \frac{4 - e^{-x^2}}{2 - e^{-x^2}}.$$

The interval of validity is  $(-\infty, \infty)$ .

$$\mathbf{2.2.20.} \quad \frac{y'}{y(y-2)} = -1; \frac{1}{2} \left[ \frac{1}{y-2} - \frac{1}{y} \right] y' = -1; \left[ \frac{1}{y-2} - \frac{1}{y} \right] y' = -2; \ln \left| \frac{y-2}{y} \right| = -2x + k;$$

$$\frac{y-2}{y} = ce^{-2x}; y(0) = 1 \Rightarrow c = -1; \frac{y-2}{y} = -e^{-2x}; y-2 = -ye^{-2x}; y(1 + e^{-2x}) = 2; y = \frac{2}{1 + e^{-2x}}.$$

The interval of validity is  $(-\infty, \infty)$ .

**2.2.22.**  $y \equiv 2$  is a constant solution of the differential equation, and it satisfies the initial condition. Therefore,  $y \equiv 2$  is a solution of the initial value problem. The interval of validity is  $(-\infty, \infty)$ .

$$\mathbf{2.2.24.} \quad \frac{y'}{1+y^2} = \frac{1}{1+x^2}; \tan^{-1} y = \tan^{-1} x + k; y = \tan(\tan^{-1} x + k). \text{ Now use the identity}$$

$$\tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B} \text{ with } A = \tan^{-1} x \text{ and } B = \tan^{-1} c \text{ to rewrite } y \text{ as } y = \frac{x+c}{1-cx}, \text{ where } c = \tan k.$$

**2.2.26.**  $(\sin y)y' = \cos x$ ;  $-\cos y = \sin x + c$ ;  $y(\pi) = \frac{\pi}{2} \Rightarrow c = 0$ , so (A)  $\cos y = -\sin x$ . To obtain  $y$  explicitly we note that  $-\sin x = \cos(x + \pi/2)$ , so (A) can be rewritten as  $\cos y = \cos(x + \pi/2)$ . This equation holds if and only if one of the following conditions holds for some integer  $k$ :

$$(B) \ y = x + \frac{\pi}{2} + 2k\pi; \text{ or } (C) \ y = -x - \frac{\pi}{2} + 2k\pi.$$

Among these choices the only way to satisfy the initial condition is to let  $k = 1$  in (C), so  $y = -x + \frac{3\pi}{2}$ .

**2.2.28.** Rewrite the equation as  $P' = -a\alpha P(P - 1/\alpha)$ . By inspection,  $P \equiv 0$  and  $P \equiv 1/\alpha$  are constant solutions. Separate variables to find others:  $\frac{P'}{P(P - 1/\alpha)} = -a\alpha$ ;  $\left[ \frac{1}{P - 1/\alpha} - \frac{1}{P} \right] P' =$

$-a$ ;  $\ln \left| \frac{P - 1/\alpha}{P} \right| = -at + k$ ; (A)  $\frac{P - 1/\alpha}{P} = ce^{-\alpha t}$ ;  $P(1 - ce^{-\alpha t}) = 1/\alpha$ ; (B)  $P = \frac{1}{\alpha(1 - ce^{-\alpha t})}$ .

From (A),  $P(0) = P_0 \Rightarrow c = \frac{P_0 - 1/\alpha}{P_0}$ . Substituting this into (B) yields  $P = \frac{P_0}{\alpha P_0 + (1 - \alpha P_0)e^{-\alpha t}}$ .

From this  $\lim_{t \rightarrow \infty} P(t) = 1/\alpha$ .

**2.2.30.** If  $q = rS$  the equation for  $I$  reduces to  $I' = -rI^2$ , so  $\frac{I'}{I^2} = -r$ ;  $-\frac{1}{I} = -rt - \frac{1}{I_0}$ ; so

$I = \frac{I_0}{1 + rI_0 t}$  and  $\lim_{t \rightarrow \infty} I(t) = 0$ . If  $q \neq rS$ , then rewrite the equation for  $I$  as  $I' = -rI(I - \alpha)$

with  $\alpha = S - \frac{q}{r}$ . Separating variables yields  $\frac{I'}{I(I - \alpha)} = -r$ ;  $\left[ \frac{1}{I - \alpha} - \frac{1}{I} \right] I' = -r\alpha$ ;  $\ln \left| \frac{I - \alpha}{I} \right| =$

$-r\alpha t + k$ ; (A)  $\frac{I - \alpha}{I} = ce^{-r\alpha t}$ ;  $I(1 - ce^{-r\alpha t}) = \alpha$ ; (B)  $I = \frac{\alpha}{1 - ce^{-r\alpha t}}$ . From (A),  $I(0) = I_0 \Rightarrow$

$c = \frac{I_0 - \alpha}{I_0}$ . Substituting this into (B) yields  $I = \frac{\alpha I_0}{I_0 + (\alpha - I_0)e^{-r\alpha t}}$ . If  $q < rS$ , then  $\alpha > 0$  and

$\lim_{t \rightarrow \infty} I(t) = \alpha = S - \frac{q}{r}$ . If  $q > rS$ , then  $\alpha < 0$  and  $\lim_{t \rightarrow \infty} I(t) = 0$ .

**2.2.34.** The given equation is separable if  $f = ap$ , where  $a$  is a constant. In this case the equation is

$$y' + p(x)y = ap(x). \quad (\text{A})$$

Let  $P$  be an antiderivative of  $p$ ; that is,  $P' = p$ .

**SOLUTION BY SEPARATION OF VARIABLES.**  $y' = -p(x)(y - a)$ ;  $\frac{y'}{y - a} = -p(x)$ ;  $\ln |y - a| = -P(x) + k$ ;  $y - a = ce^{-P(x)}$ ;  $y = a + ce^{-P(x)}$ .

**SOLUTION BY VARIATION OF PARAMETERS.**  $y_1 = e^{-P(x)}$  is a solution of the complementary equation, so solutions of (A) are of the form  $y = ue^{-P(x)}$  where  $u'e^{-P(x)} = ap(x)$ . Hence,  $u' = ap(x)e^{P(x)}$ ;  $u = ae^{P(x)} + c$ ;  $y = a + ce^{-P(x)}$ .

**2.2.36.** Rewrite the given equation as (A)  $y' - \frac{2}{x}y = \frac{x^5}{y + x^2}$ .  $y_1 = x^2$  is a solution of  $y' - \frac{2}{x}y = 0$ .

Look for solutions of (A) of the form  $y = ux^2$ . Then  $u'x^2 = \frac{x^5}{(u + 1)x^2} = \frac{x^3}{u + 1}$ ;  $u' = \frac{x}{u + 1}$ ;

$(u + 1)u' = x$ ;  $\frac{(1 + u)^2}{2} = \frac{x^2}{2} + \frac{c}{2}$ ;  $u = -1 \pm \sqrt{x^2 + c}$ ;  $y = x^2(-1 \pm \sqrt{x^2 + c})$ .

**2.2.38.**  $y_1 = e^{2x}$  is a solution of  $y' - 2y = 0$ . Look for solutions of the nonlinear equation of the form  $y = ue^{2x}$ . Then  $u'e^{2x} = \frac{xe^{2x}}{1 - u}$ ;  $u' = \frac{x}{1 - u}$ ;  $(1 - u)u' = x$ ;  $-\frac{(1 - u)^2}{2} = \frac{1}{2}(x^2 - c)$ ;  $u = 1 \pm \sqrt{c - x^2}$ ;  $y = e^{2x}(1 \pm \sqrt{c - x^2})$ .

## 2.3 Existence and Uniqueness of Solutions of Nonlinear Equations

**2.3.2.**  $f(x, y) = \frac{e^x + y}{x^2 + y^2}$  and  $f_y(x, y) = \frac{1}{x^2 + y^2} - \frac{2y(e^x + y)}{(x^2 + y^2)^2}$  are both continuous at all  $(x, y) \neq (0, 0)$ . Hence, Theorem 2.3.1 implies that if  $(x_0, y_0) \neq (0, 0)$ , then the initial value problem has a unique solution on some open interval containing  $x_0$ . Theorem 2.3.1 does not apply if  $(x_0, y_0) =$



$(0, 0)$ .

**2.3.4.**  $f(x, y) = \frac{x^2 + y^2}{\ln xy}$  and  $f_y(x, y) = \frac{2y}{\ln xy} - \frac{x^2 + y^2}{x(\ln xy)^2}$  are both continuous at all  $(x, y)$  such that  $xy > 0$  and  $xy \neq 1$ . Hence, Theorem 2.3.1 implies that if  $x_0 y_0 > 0$  and  $x_0 y_0 \neq 1$ , then the initial value problem has unique solution on an open interval containing  $x_0$ . Theorem 2.3.1 does not apply if  $x_0 y_0 \leq 0$  or  $x_0 y_0 = 1$ .

**2.3.6.**  $f(x, y) = 2xy$  and  $f_y(x, y) = 2x$  are both continuous at all  $(x, y)$ . Hence, Theorem 2.3.1 implies that if  $(x_0, y_0)$  is arbitrary, then the initial value problem has a unique solution on some open interval containing  $x_0$ .

**2.3.8.**  $f(x, y) = \frac{2x + 3y}{x - 4y}$  and  $f_y(x, y) = \frac{3}{x - 4y} + 4\frac{2x + 3y}{(x - 4y)^2}$  are both continuous at all  $(x, y)$  such that  $x \neq 4y$ . Hence, Theorem 2.3.1 implies that if  $x_0 \neq 4y_0$ , then the initial value problem has a unique solution on some open interval containing  $x_0$ . Theorem 2.3.1 does not apply if  $x_0 = 4y_0$ .

**2.3.10.**  $f(x, y) = x(y^2 - 1)^{2/3}$  is continuous at all  $(x, y)$ , but  $f_y(x, y) = \frac{4}{3}xy(y^2 - 1)^{1/3}$  is continuous at  $(x, y)$  if and only if  $y \neq \pm 1$ . Hence, Theorem 2.3.1 implies that if  $y_0 \neq \pm 1$ , then the initial value problem has a unique solution on some open interval containing  $x_0$ , while if  $y_0 = \pm 1$ , then the initial value problem has at least one solution (possibly not unique on any open interval containing  $x_0$ ).

**2.3.12.**  $f(x, y) = (x + y)^{1/2}$  and  $f_y(x, y) = \frac{1}{2(x + y)^{1/2}}$  are both continuous at all  $(x, y)$  such that  $x + y > 0$ . Hence, Theorem 2.3.1 implies that if  $x_0 + y_0 > 0$ , then the initial value problem has a unique solution on some open interval containing  $x_0$ . Theorem 2.3.1 does not apply if  $x_0 + y_0 \leq 0$ .

**2.3.14.** To apply Theorem 2.3.1, rewrite the given initial value problem as (A)  $y' = f(x, y)$ ,  $y(x_0) = y_0$ , where  $f(x, y) = -p(x)y + q(x)$  and  $f_y(x, y) = -p(x)$ . If  $p$  and  $f$  are continuous on some open interval  $(a, b)$  containing  $x_0$ , then  $f$  and  $f_y$  are continuous on some open rectangle containing  $(x_0, y_0)$ , so Theorem 2.3.1 implies that (A) has a unique solution on *some* open interval containing  $x_0$ . The conclusion of Theorem 2.1.2 is more specific: the solution of (A) exists and is unique on  $(a, b)$ . For example, in the extreme case where  $(a, b) = (-\infty, \infty)$ , Theorem 2.3.1 still implies only existence and uniqueness on *some* open interval containing  $x_0$ , while Theorem 2.1.2 implies that the solution exists and is unique on  $(-\infty, \infty)$ .

**2.3.16.** First find solutions of (A)  $y' = y^{2/5}$ . Obviously  $y \equiv 0$  is a solution. If  $y \neq 0$ , then we can separate variables on any open interval where  $y$  has no zeros:  $y^{-2/5}y' = 1$ ;  $\frac{5}{3}y^{3/5} = x + c$ ;  $y = \left(\frac{3}{5}(x + c)^{5/3}\right)$ . (Note that this solution is also defined at  $x = -c$ , even though  $y(-c) = 0$ . To satisfy the initial condition, let  $c = 1$ . Thus,  $y = \left(\frac{3}{5}(x + 1)^{5/3}\right)$  is a solution of the initial value problem on  $(-\infty, \infty)$ ; moreover, since  $f(x, y) = y^{2/5}$  and  $f_y(x, y) = \frac{2}{5}y^{-3/5}$  are both continuous at all  $(x, y)$  such that  $y \neq 0$ , this is the only solution on  $(-5/3, \infty)$ , by an argument similar to that given in Example 2.3.7, the function

$$y = \begin{cases} 0, & -\infty < x \leq -\frac{5}{3} \\ \left(\frac{3}{5}x + 1\right)^{5/3}, & -\frac{5}{3} < x < \infty \end{cases}$$

(To see that  $y$  satisfies  $y' = y^{2/5}$  at  $x = -\frac{5}{3}$  use an argument similar to that of Discussion 2.3.15-2)

For every  $a \geq \frac{5}{3}$ , the following function is also a solution:

$$y = \begin{cases} \left(\frac{3}{5}(x+a)\right)^{5/3}, & -\infty < x < -a, \\ 0, & -a \leq x \leq -\frac{5}{3}, \\ \left(\frac{3}{5}x+1\right)^{5/3}, & -\frac{5}{3} < x < \infty. \end{cases}$$

**2.3.18.** Obviously,  $y_1 \equiv 1$  is a solution. From Discussion 2.3.18 (taking  $c = 0$  in the two families of solutions) yields  $y_2 = 1 + |x|^3$  and  $y_3 = 1 - |x|^3$ . Other solutions are  $y_4 = 1 + x^3$ ,  $y_5 = 1 - x^3$ ,

$$y_6 = \begin{cases} 1 + x^3, & x \geq 0, \\ 1, & x < 0 \end{cases}; \quad y_7 = \begin{cases} 1 - x^3, & x \geq 0, \\ 1, & x < 0 \end{cases};$$

$$y_8 = \begin{cases} 1, & x \geq 0, \\ 1 + x^3, & x < 0 \end{cases}; \quad y_9 = \begin{cases} 1, & x \geq 0, \\ 1 - x^3, & x < 0 \end{cases}$$

It is straightforward to verify that all these functions satisfy  $y' = 3x(y-1)^{1/3}$  for all  $x \neq 0$ . Moreover,  $y'_i(0) = \lim_{x \rightarrow 0} \frac{y_i(x) - 1}{x} = 0$  for  $1 \leq i \leq 9$ , which implies that they also satisfy the equation at  $x = 0$ .

**2.3.20.** Let  $y$  be any solution of (A)  $y' = 3x(y-1)^{1/3}$ ,  $y(3) = -7$ . By continuity, there is some open interval  $I$  containing  $x_0 = 3$  on which  $y(x) < 1$ . From Discussion 2.3.18,  $y = 1 + (x^2 + c)^{3/2}$  on  $I$ ;  $y(3) = -7 \Rightarrow c = -5$ ; (B)  $y = 1 - (x^2 - 5)^{3/2}$ . It now follows that every solution of (A) satisfies  $y(x) < 1$  and is given by (B) on  $(\sqrt{5}, \infty)$ ; that is, (B) is the unique solution of (A) on  $(\sqrt{5}, \infty)$ . This solution can be extended uniquely to  $(0, \infty)$  as

$$y = \begin{cases} 1, & 0 < x \leq \sqrt{5}, \\ 1 - (x^2 - 5)^{3/2}, & \sqrt{5} < x < \infty \end{cases}$$

It can be extended to  $(-\infty, \infty)$  in infinitely many ways. Thus,

$$y = \begin{cases} 1, & -\infty < x \leq \sqrt{5}, \\ 1 - (x^2 - 5)^{3/2}, & \sqrt{5} < x < \infty \end{cases}$$

is a solution of the initial value problem on  $(-\infty, \infty)$ . Moreover, if  $\alpha \geq 0$ , then

$$y = \begin{cases} 1 + (x^2 - \alpha^2)^{3/2}, & -\infty < x < -\alpha, \\ 1, & -\alpha \leq x \leq \sqrt{5}, \\ 1 - (x^2 - 5)^{3/2}, & \sqrt{5} < x < \infty, \end{cases}$$

and

$$y = \begin{cases} 1 - (x^2 - \alpha^2)^{3/2}, & -\infty < x < -\alpha, \\ 1, & -\alpha \leq x \leq \sqrt{5}, \\ 1 - (x^2 - 5)^{3/2}, & \sqrt{5} < x < \infty, \end{cases}$$

are also solutions of the initial value problem on  $(-\infty, \infty)$ .

## 2.4 Transformation of Nonlinear Equations into Separable Equations

**2.4.2.** Rewrite as  $y' - \frac{2}{7x}y = -\frac{x}{7y^6}$ . Then  $\frac{y_1'}{y_1} = \frac{2}{7x}$ ;  $\ln|y_1| = \frac{2}{7}\ln|x| = \ln|x|^{2/7}$ ;  $y_1 = x^{2/7}$ ;  $y = ux^{2/7}$ ;  $u'x^{2/7} = -\frac{1}{7u^6x^{5/7}}$ ;  $u^6u' = -\frac{1}{7x}$ ;  $\frac{u^7}{7} = -\frac{1}{7}\ln|x| + \frac{c}{7}$ ;  $u = (c - \ln|x|)^{1/7}$ ;  $y = x^{2/7}(c - \ln|x|)^{1/7}$ .

**2.4.4.** Rewrite as  $y' + \frac{2x}{1+x^2}y = \frac{1}{(1+x^2)^2y}$ . Then  $\frac{y_1'}{y_1} = -\frac{2x}{1+x^2}$ ;  $\ln|y_1| = -\ln(1+x^2)$ ;  $y_1 = \frac{1}{1+x^2}$ ;  $y = \frac{u}{1+x^2}$ ;  $\frac{u'}{1+x^2} = \frac{1}{u(1+x^2)}$ ;  $u'u = 1$ ;  $\frac{u^2}{2} = x + \frac{c}{2}$ ;  $u = \pm\sqrt{2x+c}$ ;  $y = \pm\frac{\sqrt{2x+c}}{1+x^2}$ .

**2.4.6.**  $\frac{y_1'}{y_1} = \frac{1}{3}\left(\frac{1}{x} + 1\right)$ ;  $\ln|y_1| = \frac{1}{3}(\ln|x| + x)$ ;  $y_1 = x^{1/3}e^{x/3}$ ;  $y = ux^{1/3}e^{x/3}$ ;  $u'x^{1/3}e^{x/3} = x^{4/3}e^{4x/3}u^4$ ;  $\frac{u'}{u^4} = xe^x$ ;  $-\frac{1}{3u^3} = (x-1)e^x - \frac{c}{3}$ ;  $u = \frac{1}{[3(1-x)e^x + c]^{1/3}}$ ;  $y = \left[\frac{x}{3(1-x) + ce^{-x}}\right]^{1/3}$ .

**2.4.8.**  $\frac{y_1'}{y_1} = x$ ;  $\ln|y_1| = \frac{x^2}{2}$ ;  $y_1 = e^{x^2/2}$ ;  $y = ue^{x^2/2}$ ;  $u'e^{x^2/2} = xu^{3/2}e^{3x^2/4}$ ;  $\frac{u'}{u^{3/2}} = xe^{x^2/4}$ ; (A)  $-\frac{2}{u^{1/2}} = 2e^{x^2/4} + 2c$ ;  $u^{1/2} = -\frac{1}{c + e^{x^2/4}}$ ;  $u = \frac{1}{(c + e^{x^2/4})^2}$ ;  $y = \frac{1}{(1 + ce^{-x^2/4})^2}$ . Because of (A) we must choose  $c$  so that  $y(1) = 4$  and  $1 + ce^{-1/4} < 0$ . This implies that  $c = -3e^{1/4}$ ;  $y = \left[1 - \frac{3}{2}e^{-(x^2-1)/4}\right]^{-2}$ .

**2.4.10.**  $\frac{y_1'}{y_1} = 2$ ;  $\ln|y_1| = 2x$ ;  $y_1 = e^{2x}$ ;  $y = ue^{2x}$ ;  $u'e^{2x} = 2u^{1/2}e^x$ ;  $u^{-1/2}u' = 2e^{-x}$ ;  $2u^{1/2} = -2e^{-x} + 2c$ ;  $u^{1/2} = c - e^{-x} > 0$ ;  $y(0) = 1 \Rightarrow u(0) = 1 \Rightarrow c = 2$ ;  $u = (2 - e^{-x})^2$ ;  $y = (2e^x - 1)^2$ .

**2.4.12.** Rewrite as  $y' + \frac{2}{x}y = \frac{y^3}{x^2}$ . Then  $\frac{y_1'}{y_1} = -\frac{2}{x}$ ;  $\ln|y_1| = -2\ln|x| = \ln x^{-2}$ ;  $y_1 = \frac{1}{x^2}$ ;  $y = \frac{u}{x^2}$ ;  $\frac{u'}{x^2} = \frac{u^3}{x^8}$ ;  $\frac{u'}{u^3} = \frac{1}{x^6}$ ;  $-\frac{1}{2u^2} = -\frac{1}{5x^5} + c$ ;  $y(1) = \frac{1}{\sqrt{2}} \Rightarrow u(1) = \frac{1}{\sqrt{2}} \Rightarrow c = -\frac{4}{5}$ ;  $u = \left[\frac{5x^5}{2(1+4x^5)}\right]^{1/2}$ ;  $y = \left[\frac{5x}{2(1+4x^5)}\right]^{1/2}$ .

**2.4.14.**  $P = ue^{at}$ ;  $u'e^{at} = -a\alpha u^2e^{2at}$ ;  $\frac{u'}{u^2} = -a\alpha e^{at}$ ;  $-\frac{1}{u} = -a\int_0^t \alpha(\tau)e^{a\tau}d\tau - \frac{1}{P_0}$ ;  $P = \frac{P_0e^{at}}{1 + aP_0\int_0^t \alpha(\tau)e^{a\tau}d\tau}$ , which can also be written as  $P = \frac{P_0}{e^{-at} + aP_0e^{-at}\int_0^t \alpha(\tau)e^{a\tau}d\tau}$ . Therefore,

$$\lim_{t \rightarrow \infty} P(t) = \begin{cases} \infty & \text{if } L = 0, \\ 0 & \text{if } L = \infty, \\ 1/aL & \text{if } 0 < L < \infty. \end{cases}$$

**2.4.16.**  $y = ux$ ;  $u'x + u = u^2 + 2u$ ; (A)  $u'x = u(u+1)$ . Since  $u \equiv 0$  and  $u \equiv -1$  are constant solutions of (A),  $y \equiv 0$  and  $y = -x$  are solutions of the given equation. The nonconstant solutions

of (A) satisfy  $\frac{u'}{u(u+1)} = \frac{1}{x}$ ;  $\left[\frac{1}{u} - \frac{1}{u+1}\right]u' = \frac{1}{x}$ ;  $\ln\left|\frac{u}{u+1}\right| = \ln|x| + k$ ;  $\frac{u}{u+1} = cx$ ;  
 $u = (u+1)cx$ ;  $u(1-cx) = cx$ ;  $u = \frac{cx}{1-cx}$ ;  $y = \frac{cx^2}{1-cx}$ .

**2.4.18.**  $y = ux$ ;  $u'x + u = u + \sec$ ;  $u'x = \sec u$ ;  $(\cos u)u' = \frac{1}{x}$ ;  $\sin u = \ln|x| + c$ ;  $u = \sin^{-1}(\ln|x| + c)$ ;  $y = x \sin^{-1}(\ln|x| + c)$ .

**2.4.20.** Rewrite the given equation as  $y' = \frac{x^2 + 2y^2}{xy}$ ;  $y = ux$ ;  $u'x + u = \frac{1}{u} + 2u$ ;  $u'x = \frac{1+u^2}{u}$ ;  
 $\frac{uu'}{1+u^2} = \frac{1}{x}$ ;  $\frac{1}{2} \ln(1+u^2) = \ln|x| + k$ ;  $\ln\left(1 + \frac{y^2}{x^2}\right) = \ln x^2 + 2k$ ;  $1 + \frac{y^2}{x^2} = cx^2$ ;  $x^2 + y^2 = cx^4$ ;  
 $y = \pm x\sqrt{cx^2 - 1}$ .

**2.4.22.**  $y = ux$ ;  $u'x + u = u + u^2$ ;  $u'x = u^2$ ;  $\frac{u'}{u^2} = \frac{1}{x}$ ;  $-\frac{1}{u} = \ln|x| + c$ ;  $y(-1) = 2 \Rightarrow u(-1) = -2 \Rightarrow c = \frac{1}{2}$ ;  $u = -\frac{2}{2\ln|x| + 1}$ ;  $y = -\frac{2x}{2\ln|x| + 1}$ .

**2.4.24.** Rewrite the given equation as  $y' = -\frac{x^2 + y^2}{xy}$ ;  $y = ux$ ;  $u'x + u = -\frac{1}{u} - u$ ;  $u'x = -\frac{1+2u^2}{u}$ ;  
 $-\frac{uu'}{1+2u^2} = \frac{1}{x}$ ;  $-\frac{1}{4} \ln(1+2u^2) = \ln|x| + k$ ;  $x^4(1+2u^2) = c$ ;  $y(1) = 2 \Rightarrow u(1) = 2 \Rightarrow c = 9$ ;  
 $x^4(1+2u^2) = 9$ ;  $u^2 = \frac{9-x^4}{2x^4}$ ;  $u = \frac{1}{x^2} \left(\frac{9-x^4}{2}\right)^{1/2}$ ;  $y = \frac{1}{x} \left(\frac{9-x^4}{2}\right)^{1/2}$ .

**2.4.26.** Rewrite the given equation as  $y' = 2 + \frac{y^2}{x^2} + 4\frac{y}{x}$ ;  $y = ux$ ;  $u'x + u = 2 + u^2 + 4u$ ;  
 $u'x = u^2 + 3u + 2 = (u+1)(u+2)$ ;  $\frac{u'}{(u+1)(u+2)} = \frac{1}{x}$ ;  $\left[\frac{1}{u+1} - \frac{1}{u+2}\right]u' = \frac{1}{x}$ ;  $\ln\left|\frac{u+1}{u+2}\right| = \ln|x| + k$ ;  
 $\frac{u+1}{u+2} = cx$ ;  $y(1) = 1 \Rightarrow u(1) = 1 \Rightarrow c = \frac{2}{3}$ ;  $\frac{u+1}{u+2} = \frac{2}{3}x$ ;  $u+1 = \frac{2}{3}x(u+2)$ ;  
 $u\left(1 - \frac{2}{3}x\right) = -1 + \frac{4}{3}x$ ;  $u = -\frac{4x-3}{2x-3}$ ;  $y = -\frac{x(4x-3)}{2x-3}$ .

**2.4.28.**  $y = ux$ ;  $u'x + u = \frac{1+u}{1-u}$ ;  $u'x = \frac{1+u^2}{1-u}$ ;  $\frac{(1-u)u'}{1+u^2} = \frac{1}{x}$ ;  $\tan^{-1}u - \frac{1}{2} \ln(1+u^2) = \ln|x| + c$ ;  
 $\tan^{-1}\frac{y}{x} - \frac{1}{2} \ln\left(1 + \frac{y^2}{x^2}\right) = \ln|x| + c$ ;  $\tan^{-1}\frac{y}{x} - \frac{1}{2} \ln(x^2 + y^2) = c$ .

**2.4.30.**  $y = ux$ ;  $u'x + u = \frac{u^3 + 2u^2 + u + 1}{(u+1)^2} = \frac{u(u+1)^2 + 1}{(u+1)^2} = u + \frac{1}{(u+1)^2}$ ;  $u'x = \frac{1}{(u+1)^2}$ ;  
 $(u+1)^2u' = \frac{1}{x}$ ;  $\frac{(u+1)^3}{3} = \ln|x| + c$ ;  $(u+1)^3 = 3(\ln|x| + c)$ ;  $\left(\frac{y}{x} + 1\right)^3 = 3(\ln|x| + c)$ ;  
 $(y+x)^3 = 3x^3(\ln|x| + c)$ .

**2.4.32.**  $y = ux$ ;  $u'x + u = \frac{u}{u-2}$ ; (A)  $u'x = \frac{u(u-3)}{2-u}$ ; Since  $u \equiv 0$  and  $u \equiv 3$  are constant solutions of (A),  $y \equiv 0$  and  $y = 3x$  are solutions of the given equation. The nonconstant solutions of (A) satisfy  $\frac{(2-u)u'}{u(u-3)} = \frac{1}{x}$ ;  $\left[\frac{1}{u-3} + \frac{2}{u}\right]u' = -\frac{3}{x}$ ;  $\ln|u-3| + 2\ln|u| = -3\ln|x| + k$ ;  $u^2(u-3) = \frac{c}{x^3}$ ;  
 $y^2(y-3x) = c$ .

**2.4.34.**  $y = ux$ ;  $u'x + u = \frac{1+u+3u^3}{1+3u^2} = u + \frac{1}{1+3u^2}$ ;  $(1+3u^2)u' = \frac{1}{x}$ ;  $u + u^3 = \ln|x| + c$ ;

$$\frac{y}{x} + \frac{y^3}{x^3} = \ln|x| + c.$$

**2.4.36.** Rewrite the given equation as  $y' = \frac{x^2 - xy + y^2}{xy}$ ;  $y = ux$ ;  $u'x + u = \frac{1}{u} - 1 + u$ ;  $u'x = \frac{1-u}{u}$ ;

$$\frac{uu'}{u-1} = -\frac{1}{x}; \left[1 + \frac{1}{u-1}\right] u' = -\frac{1}{x}; u + \ln|u-1| = -\ln|x| + k; e^u(u-1) = \frac{c}{x}; e^{y/x}(y-x) = c.$$

**2.4.38.**  $y = ux$ ;  $u'x + u = 1 + \frac{1}{u} + u$ ; (A)  $u'x = \frac{u+1}{u}$ . Since (A) has the constant solution  $u = -1$ ;

$y = -x$  is a solution of the given equation. The nonconstant solutions of (A) satisfy  $\frac{uu'}{u+1} = \frac{1}{x}$ ;

$$\left[1 - \frac{1}{u+1}\right] u' = \frac{1}{x}; u - \ln|u+1| = \ln|x| + c; \frac{y}{x} - \ln\left|\frac{y}{x} - 1\right| = \ln|x| + c; y - x \ln|y-x| = cy.$$

**2.4.40.** If  $x = X - X_0$  and  $y = Y - Y_0$ , then  $\frac{dy}{dx} = \frac{dY}{dX} = \frac{dY}{dX} \frac{dX}{dx} = \frac{dY}{dX}$ , so  $y = y(x)$  satisfies the given equation if and only if  $Y = Y(X)$  satisfies

$$\frac{dY}{dX} = \frac{a(X - X_0) + b(Y - Y_0) + \alpha}{c(X - X_0) + d(Y - Y_0) + \beta},$$

which reduces to the nonlinear homogeneous equation

$$\frac{dY}{dX} = \frac{aX + bY}{cX + dY}$$

if and only if

$$\begin{aligned} aX_0 + bY_0 &= \alpha \\ cX_0 + dY_0 &= \beta. \end{aligned} \tag{B}$$

We will now show that if  $ad - bc \neq 0$ , then it is possible (for any choice of  $\alpha$  and  $\beta$ ) to solve (B). Multiplying the first equation in (B) by  $d$  and the second by  $b$  yields

$$\begin{aligned} daX_0 + dbY_0 &= d\alpha \\ bcX_0 + bdY_0 &= b\beta. \end{aligned}$$

Subtracting the second of these equations from the first yields  $(ad - bc)X_0 = \alpha d - \beta b$ . Since  $ad - bc \neq 0$ , this implies that  $X_0 = \frac{\alpha d - \beta b}{ad - bc}$ . Multiplying the first equation in (B) by  $c$  and the second by  $a$  yields

$$\begin{aligned} caX_0 + cbY_0 &= c\alpha \\ acX_0 + adY_0 &= a\beta. \end{aligned}$$

Subtracting the first of these equation from the second yields  $(ad - bc)Y_0 = \alpha c - \beta a$ . Since  $ad - bc \neq 0$  this implies that  $Y_0 = \frac{\alpha c - \beta a}{ad - bc}$ .

**2.4.42.** For the given equation, (B) of Exercise 2.4.40 is

$$\begin{aligned} 2X_0 + Y_0 &= 1 \\ X_0 + 2Y_0 &= -4. \end{aligned}$$

Solving this pair of equations yields  $X_0 = 2$  and  $Y_0 = -3$ . The transformed differential equation is

$$\frac{dY}{dX} = \frac{2X + Y}{X + 2Y}. \quad (\text{A})$$

Let  $Y = uX$ ;  $u'X + U = \frac{2+u}{1+2u}$ ; (B)  $u'X = -\frac{2(u-1)(u+1)}{2u+1}$ . Since  $u \equiv 1$  and  $u \equiv -1$  satisfy (B),  $Y = X$  and  $Y = -X$  are solutions of (A). Since  $X = x + 2$  and  $Y = y - 3$ , it follows that  $y = x + 5$  and  $y = -x + 1$  are solutions of the given equation. The nonconstant solutions of (B) satisfy  $\frac{(2u+1)u'}{(u-1)(u+1)} = -\frac{2}{X}$ ;  $\left[\frac{1}{u+1} + \frac{3}{u-1}\right]u' = -\frac{4}{X}$ ;  $\ln|u+1| + 3\ln|u-1| = -4\ln|X| + k$ ;  $(u+1)(u-1)^3 = \frac{c}{X^4}$ ;  $(Y+X)(Y-X)^3 = c$ ; Setting  $X = x + 2$  and  $Y = y - 3$  yields  $(y+x-1)(y-x-5)^3 = c$ .

**2.4.44.** Rewrite the given equation as  $y' = \frac{y^3 + x}{3xy^2}$ ;  $y = ux^{1/3}$ ;  $u'x^{1/3} + \frac{1}{3x^{2/3}}u = \frac{u^3 + 1}{3u^2x^{2/3}}$ ;  $u'x^{1/3} = \frac{1}{3x^{2/3}u^2}$ ;  $u^2u' = \frac{1}{3x}$ ;  $\frac{u^3}{3} = \frac{1}{3}(\ln|x| + c)$ ;  $u = (\ln|x| + c)^{1/3}$ ;  $y = x^{1/3}(\ln|x| + c)^{1/3}$ .

**2.4.46.** Rewrite the given equation as  $y' = \frac{2(y^2 + x^2y - x^4)}{x^3}$ ;  $y = ux^2$ ;  $u'x^2 + 2xu = 2x(u^2 + u - 1)$ ; (A)  $u'x^2 = 2x(u^2 - 1)$ . Since  $u \equiv 1$  and  $u \equiv -1$  are constant solutions of (A),  $y = x^2$  and  $y = -x^2$  are solutions of the given equation. The nonconstant solutions of (A) satisfy  $\frac{u'}{u^2 - 1} = \frac{2}{x}$ ;  $\left[\frac{1}{u-1} - \frac{1}{u+1}\right]u' = \frac{4}{x}$ ;  $\ln\left|\frac{u-1}{u+1}\right| = 4\ln|x| + k$ ;  $\frac{u-1}{u+1} = cx^4$ ;  $(u-1) = (u+1)cx^4$ ;  $u(1 - cx^4) = 1 + cx^4$ ;  $u = \frac{1 + cx^4}{1 - cx^4}$ ;  $y = \frac{x^2(1 + cx^4)}{1 - cx^4}$ .

**2.4.48.**  $y = u \tan x$ ;  $u' \tan x + u \sec^2 x = (u^2 + u + 1) \sec^2 x$ ;  $u' \tan x = (u^2 + 1) \sec^2 x$ ;  $\frac{u'}{u^2 + 1} = \sec^2 x \cot x = \cot x + \tan x$ ;  $\tan^{-1} u = \ln|\sin x| - \ln|\cos x| + c = \ln|\tan x| + c$ ;  $u = \tan(\ln|\tan x| + c)$ ;  $y = \tan x \tan(\ln|\tan x| + c)$ .

**2.4.50.** Rewrite the given equation as  $y' = \frac{(y + \sqrt{x})^2}{2x(y + 2\sqrt{x})}$ ;  $y = ux^{1/2}$ ;  $u'x^{1/2} + \frac{1}{2\sqrt{x}}u = \frac{(u+1)^2}{2\sqrt{x}(u+2)}$ ;  $u'x^{1/2} = \frac{1}{2\sqrt{x}(u+2)}$ ;  $(u+2)u' = \frac{1}{2x}$ ;  $\frac{(u+2)^2}{2} = \frac{1}{2}(\ln|x| + c)$ ;  $(u+2)^2 = \ln|x| + c$ ;  $u = -2 \pm \sqrt{\ln|x| + c}$ ;  $y = x^{1/2}(-2 \pm \sqrt{\ln|x| + c})$ .

**2.4.52.**  $y_1 = \frac{1}{x^2}$  is a solution of  $y' + \frac{2}{x}y = 0$ . Let  $y = \frac{u}{x^2}$ ; then

$$\frac{u'}{x^2} = \frac{3x^2(u^2/x^4) + 6x(u/x^2) + 2}{x^2(2x(u/x^2) + 3)} = \frac{3(u/x)^2 + 6(u/x) + 2}{x^2(2(u/x) + 3)},$$

so (A)  $u' = \frac{3(u/x)^2 + 6(u/x) + 2}{2(u/x) + 3}$ . Since (A) is a homogeneous nonlinear equation, we now substitute  $u = vx$  into (A). This yields  $v'x + v = \frac{3v^2 + 6v + 2}{2v + 3}$ ;  $v'x = \frac{(v+1)(v+2)}{2v+3}$ ;  $\frac{(2v+3)v'}{(v+1)(v+2)} = \frac{1}{x}$ ;

$\left[\frac{1}{v+1} + \frac{1}{v+2}\right]v' = \frac{1}{x}$ ;  $\ln|(v+1)(v+2)| = \ln|x| + k$ ; (B)  $(v+1)(v+2) = cx$ . Since  $y(2) = 2 \Rightarrow$

$u(2) = 8 \Rightarrow v(2) = 4$ , (B) implies that  $c = 15$ .  $(v+1)(v+2) = 15x$ ;  $v^2 + 3v + 2 - 15x = 0$ . From the quadratic formula,  $v = \frac{-3 + \sqrt{1+60x}}{2}$ ;  $u = vx = \frac{x(-3 + \sqrt{1+60x})}{2}$ ;  $y = \frac{u}{x^2} = \frac{-3 + \sqrt{1+60x}}{2x}$ .

**2.4.54.** Differentiating (A)  $y_1(x) = \frac{y(ax)}{a}$  yields (B)  $y'_1(x) = \frac{1}{a}y'(ax) \cdot a = y'(ax)$ . Since  $y'(x) = q(y(x)/x)$  on some interval  $I$ , (C)  $y'(ax) = q(y(ax)/ax)$  on some interval  $J$ . Substituting (A) and (B) into (C) yields  $y'_1(x) = q(y_1(x)/x)$  on  $J$ .

**2.4.56.** If  $y = z + 1$ , then  $z' + z = xz^2$ ;  $z = ue^{-x}$ ;  $u'e^{-x} = xu^2e^{-2x}$ ;  $\frac{u'}{u^2} = xe^{-x}$ ;  $-\frac{1}{u} = -e^{-x}(x+1) - c$ ;  $u = \frac{1}{e^{-x}(x+1)+c}$ ;  $z = \frac{1}{x+1+ce^x}$ ;  $y = 1 + \frac{1}{x+1+ce^x}$ .

**2.4.58.** If  $y = z + 1$ , then  $z' + \frac{2}{x}z = z^2$ ;  $z_1 = \frac{1}{x^2}$ ;  $z = \frac{u}{x^2}$ ;  $\frac{u'}{x^2} = \frac{u^2}{x^4}$ ;  $\frac{u'}{u^2} = \frac{1}{x^2}$ ;  $-\frac{1}{u} = -\frac{1}{x} + c = -\frac{1-cx}{x}$ ;  $u = -\frac{x}{1-cx}$ ;  $z = -\frac{1}{x(1-cx)}$ ;  $y = 1 - \frac{1}{x(1-cx)}$ .

## 2.5 Exact Equations

**2.5.2.**  $M(x, y) = 3y \cos x + 4xe^x + 2x^2e^x$ ;  $N(x, y) = 3 \sin x + 3$ ;  $M_y(x, y) = 3 \cos x = N_x(x, y)$ , so the equation is exact. We must find  $F$  such that (A)  $F_x(x, y) = 3y \cos x + 4xe^x + 2x^2e^x$  and (B)  $F_y(x, y) = 3 \sin x + 3$ . Integrating (B) with respect to  $y$  yields (C)  $F(x, y) = 3y \sin x + 3y + \psi(x)$ . Differentiating (C) with respect to  $x$  yields (D)  $F_x(x, y) = 3y \cos x + \psi'(x)$ . Comparing (D) with (A) shows that (E)  $\psi'(x) = 4xe^x + 2x^2e^x$ . Integration by parts yields  $\int xe^x dx = xe^x - e^x$

and  $\int x^2e^x dx = x^2e^x - 2xe^x + 2e^x$ . Substituting from the last two equations into (E) yields  $\psi(x) = 2x^2e^x$ . Substituting this into (C) yields  $F(x, y) = 3y \sin x + 3y + 2x^2e^x$ . Therefore,  $3y \sin x + 3y + 2x^2e^x = c$ .

**2.5.4.**  $M(x, y) = 2x - 2y^2$ ;  $N(x, y) = 12y^2 - 4xy$ ;  $M_y(x, y) = -4y = N_x(x, y)$ , so the equation is exact. We must find  $F$  such that (A)  $F_x(x, y) = 2x - 2y^2$  and (B)  $F_y(x, y) = 12y^2 - 4xy$ . Integrating (A) with respect to  $x$  yields (C)  $F(x, y) = x^2 - 2xy^2 + \phi(y)$ . Differentiating (C) with respect to  $y$  yields (D)  $F_y(x, y) = -4xy + \phi'(y)$ . Comparing (D) with (B) shows that  $\phi'(y) = 12y^2$ , so we take  $\phi(y) = 4y^3$ . Substituting this into (C) yields  $F(x, y) = x^2 - 2xy^2 + 4y^3$ . Therefore,  $x^2 - 2xy^2 + 4y^3 = c$ .

**2.5.6.**  $M(x, y) = 4x + 7y$ ;  $N(x, y) = 3x + 4y$ ;  $M_y(x, y) = 7 \neq 3 = N_x(x, y)$ , so the equation is not exact.

**2.5.8.**  $M(x, y) = 2x + y$ ;  $N(x, y) = 2y + 2x$ ;  $M_y(x, y) = 1 \neq 2 = N_x(x, y)$ , so the equation is not exact.

**2.5.10.**  $M(x, y) = 2x^2 + 8xy + y^2$ ;  $N(x, y) = 2x^2 + \frac{xy^3}{3}$ ;  $M_y(x, y) = 8x + 2y \neq 4x + \frac{y^3}{3} = N_x(x, y)$ , so the equation is not exact.

**2.5.12.**  $M(x, y) = y \sin xy + xy^2 \cos xy$ ;  $N(x, y) = x \sin xy + xy^2 \cos xy$ ;  $M_y(x, y) = 3xy \cos xy + (1 - x^2y^2) \sin xy \neq (xy + y^2) \cos xy + (1 - xy^3) \sin xy = N_x(x, y)$ , so the equation is not exact.

**2.5.14.**  $M(x, y) = e^x(x^2y^2 + 2xy^2) + 6x$ ;  $N(x, y) = 2x^2ye^x + 2$ ;  $M_y(x, y) = 2xye^x(x+2) = N_x(x, y)$ , so the equation is exact. We must find  $F$  such that (A)  $F_x(x, y) = e^x(x^2y^2 + 2xy^2) + 6x$  and (B)  $F_y(x, y) = 2x^2ye^x + 2$ . Integrating (B) with respect to  $y$  yields (C)  $F(x, y) = x^2y^2e^x + 2y + \psi(x)$ .

Differentiating (C) with respect to  $x$  yields (D)  $F_x(x, y) = e^x(x^2y^2 + 2xy^2) + \psi'(x)$ . Comparing (D) with (A) shows that  $\psi'(x) = 6x$ , so we take  $\psi(x) = 3x^2$ . Substituting this into (C) yields  $F(x, y) = x^2y^2e^x + 2y + 3x^2$ . Therefore,  $x^2y^2e^x + 2y + 3x^2 = c$ .

**2.5.16.**  $M(x, y) = e^{xy}(x^4y + 4x^3) + 3y$ ;  $N(x, y) = x^5e^{xy} + 3x$ ;  $M_y(x, y) = x^4e^{xy}(xy + 5) + 3 = N_x(x, y)$ , so the equation is exact. We must find  $F$  such that (A)  $F_x(x, y) = e^{xy}(x^4y + 4x^3) + 3y$  and (B)  $F_y(x, y) = x^5e^{xy} + 3x$ . Integrating (B) with respect to  $y$  yields (C)  $F(x, y) = x^4e^{xy} + 3xy + \psi(x)$ . Differentiating (C) with respect to  $x$  yields (D)  $F_x(x, y) = e^{xy}(x^4y + 4x^3) + 3y + \psi'(x)$ . Comparing (D) with (A) shows that  $\psi'(x) = 0$ , so we take  $\psi(x) = 0$ . Substituting this into (C) yields  $F(x, y) = x^4e^{xy} + 3xy$ . Therefore,  $x^4e^{xy} + 3xy = c$ .

**2.5.18.**  $M(x, y) = 4x^3y^2 - 6x^2y - 2x - 3$ ;  $N(x, y) = 2x^4y - 2x^3$ ;  $M_y(x, y) = 8x^3y - 6x^2 = N_x(x, y)$ , so the equation is exact. We must find  $F$  such that (A)  $F_x(x, y) = 4x^3y^2 - 6x^2y - 2x - 3$  and (B)  $F_y(x, y) = 2x^4y - 2x^3$ . Integrating (A) with respect to  $x$  yields (C)  $F(x, y) = x^4y^2 - 2x^3y - x^2 - 3x + \phi(y)$ . Differentiating (C) with respect to  $y$  yields (D)  $F_y(x, y) = 2x^4y - 2x^3 + \phi'(y)$ . Comparing (D) with (B) shows that  $\phi'(y) = 0$ , so we take  $\phi(y) = 0$ . Substituting this into (C) yields  $F(x, y) = x^4y^2 - 2x^3y - x^2 - 3x$ . Therefore,  $x^4y^2 - 2x^3y - x^2 - 3x = c$ . Since  $y(1) = 3 \Rightarrow c = -1$ ,  $x^4y^2 - 2x^3y - x^2 - 3x + 1 = 0$  is an implicit solution of the initial value problem. Solving this for

$y$  by means of the quadratic formula yields  $y = \frac{x + \sqrt{2x^2 + 3x - 1}}{x^2}$ .

**2.5.20.**  $M(x, y) = (y^3 - 1)e^x$ ;  $N(x, y) = 3y^2(e^x + 1)$ ;  $M_y(x, y) = 3y^2e^x = N_x(x, y)$ , so the equation is exact. We must find  $F$  such that (A)  $F_x(x, y) = (y^3 - 1)e^x$  and (B)  $F_y(x, y) = 3y^2(e^x + 1)$ . Integrating (A) with respect to  $x$  yields (C)  $F(x, y) = (y^3 - 1)e^x + \phi(y)$ . Differentiating (C) with respect to  $y$  yields (D)  $F_y(x, y) = 3y^2e^x + \phi'(y)$ . Comparing (D) with (B) shows that  $\phi'(y) = 3y^2$ , so we take  $\phi(y) = y^3$ . Substituting this into (C) yields  $F(x, y) = (y^3 - 1)e^x + y^3$ . Therefore,  $(y^3 - 1)e^x + y^3 = c$ . Since  $y(0) = 0 \Rightarrow c = -1$ ,  $(y^3 - 1)e^x + y^3 = -1$  is an implicit solution of the initial value problem. Therefore,  $y^3(e^x + 1) = e^x - 1$ , so  $y = \left(\frac{e^x - 1}{e^x + 1}\right)^{1/3}$ .

**2.5.22.**  $M(x, y) = (2x - 1)(y - 1)$ ;  $N(x, y) = (x + 2)(x - 3)$ ;  $M_y(x, y) = 2x - 1 = N_x(x, y)$ , so the equation is exact. We must find  $F$  such that (A)  $F_x(x, y) = (2x - 1)(y - 1)$  and (B)  $F_y(x, y) = (x + 2)(x - 3)$ . Integrating (A) with respect to  $x$  yields (C)  $F(x, y) = (x^2 - x)(y - 1) + \phi(y)$ . Differentiating (C) with respect to  $y$  yields (D)  $F_y(x, y) = x^2 - x + \phi'(y)$ . Comparing (D) with (B) shows that  $\phi'(y) = -6$ , so we take  $\phi(y) = -6y$ . Substituting this into (C) yields  $F(x, y) = (x^2 - x)(y - 1) - 6y$ . Therefore,  $(x^2 - x)(y - 1) - 6y = c$ . Since  $y(1) = -1 \Rightarrow c = 6$ ,  $(x^2 - x)(y - 1) - 6y = 6$  is an implicit solution of the initial value problem. Therefore,  $(x^2 - x - 6)y = x^2 - x + 6$ , so  $y = \frac{x^2 - x + 6}{(x - 3)(x + 2)}$ .

**2.5.24.**  $M(x, y) = e^x(x^4y^2 + 4x^3y^2 + 1)$ ;  $N(x, y) = 2x^4ye^x + 2y$ ;  $M_y(x, y) = 2x^3ye^x(x + 4) = N_x(x, y)$ , so the equation is exact. We must find  $F$  such that (A)  $F_x(x, y) = e^x(x^4y^2 + 4x^3y^2 + 1)$  and (B)  $F_y(x, y) = 2x^4ye^x + 2y$ . Integrating (B) with respect to  $y$  yields (C)  $F(x, y) = x^4y^2e^x + y^2 + \psi(x)$ . Differentiating (C) with respect to  $x$  yields (D)  $F_x(x, y) = e^xy^2(x^4 + 4x^3) + \psi'(x)$ . Comparing (D) with (A) shows that  $\psi'(x) = e^x$ , so we take  $\psi(x) = e^x$ . Substituting this into (C) yields  $F(x, y) = (x^4y^2 + 1)e^x + y^2$ . Therefore,  $(x^4y^2 + 1)e^x + y^2 = c$ .

**2.5.28.**  $M(x, y) = x^2 + y^2$ ;  $N(x, y) = 2xy$ ;  $M_y(x, y) = 2y = N_x(x, y)$ , so the equation is exact. We must find  $F$  such that (A)  $F_x(x, y) = x^2 + y^2$  and (B)  $F_y(x, y) = 2xy$ . Integrating (A) with respect to  $x$  yields (C)  $F(x, y) = \frac{x^3}{3} + xy^2 + \phi(y)$ . Differentiating (C) with respect to  $y$  yields (D)  $F_y(x, y) = 2xy + \phi'(y)$ . Comparing (D) with (B) shows that  $\phi'(y) = 0$ , so we take  $\phi(y) = 0$ .



Substituting this into (C) yields  $F(x, y) = \frac{x^3}{3} + xy^2$ . Therefore,  $\frac{x^3}{3} + xy^2 = c$ .

**2.5.30. (a)** Exactness requires that  $N_x(x, y) = M_y(x, y) = \frac{\partial}{\partial y}(x^3y^2 + 2xy + 3y^2) = 2x^3y + 2x + 6y$ .

Hence,  $N(x, y) = \frac{x^4y}{4} + x^2 + 6xy + g(x)$ , where  $g$  is differentiable.

**(b)** Exactness requires that  $N_x(x, y) = M_y(x, y) = \frac{\partial}{\partial y}(\ln xy + 2y \sin x) = \frac{1}{y} + 2 \sin x$ . Hence,  $N(x, y) = \frac{x}{y} - 2 \cos x + g(x)$ , where  $g$  is differentiable.

**(c)** Exactness requires that  $N_x(x, y) = M_y(x, y) = \frac{\partial}{\partial y}(x \sin x + y \sin y) = y \cos y + \sin y$ . Hence,  $N(x, y) = x(y \cos y + \sin y) + g(x)$ , where  $g$  is differentiable.

**2.5.32.** The assumptions imply that  $\frac{\partial M_1}{\partial y} = \frac{\partial N_1}{\partial x}$  and  $\frac{\partial M_2}{\partial y} = \frac{\partial N_2}{\partial x}$ . Therefore,  $\frac{\partial}{\partial y}(M_1 + M_2) = \frac{\partial M_1}{\partial y} + \frac{\partial M_2}{\partial y} = \frac{\partial N_1}{\partial x} + \frac{\partial N_2}{\partial x} = \frac{\partial}{\partial x}(N_1 + N_2)$ , which implies that  $(M_1 + M_2) dx + (N_1 + N_2) dy = 0$  is exact on  $R$ .

**2.5.34.** Here  $M(x, y) = Ax^2 + Bxy + Cy^2$  and  $N(x, y) = Dx^2 + Exy + Fy^2$ . Since  $M_y = Bx + 2Cy$  and  $N_x = 2Dx + Ey$ , the equation is exact if and only if  $B = 2D$  and  $E = 2C$ .

**2.5.36.** Differentiating (A)  $F(x, y) = \int_{y_0}^y N(x_0, s) ds + \int_{x_0}^x M(t, y) dt$  with respect to  $x$  yields  $F_x(x, y) = M(x, y)$ , since the first integral in (A) is independent of  $x$  and  $M(t, y)$  is a continuous function of  $t$  for each fixed  $y$ . Differentiating (A) with respect to  $y$  and using the assumption that  $M_y = N_x$  yields  $F_y(x, y) = N(x_0, y) + \int_{x_0}^x \frac{\partial M}{\partial y}(t, y) dt = N(x_0, y) + \int_{x_0}^x \frac{\partial N}{\partial x}(t, y) dt = N(x_0, y) + N(x, y) - N(x_0, y) = N(x, y)$ .

**2.5.38.**  $y_1 = \frac{1}{x^2}$  is a solution of  $y' + \frac{2}{x}y = 0$ . Let  $y = \frac{u}{x^2}$ ; then

$$\frac{u'}{x^2} = -\frac{2x(u/x^2)}{(x^2 + 2x^2(u/x^2) + 1)} = -\frac{2xu}{x^2(x^2 + 2u + 1)},$$

so  $u' = -\frac{2xu}{x^2 + 2u + 1}$ , which can be rewritten as (A)  $2xu dx + (x^2 + 2u + 1) du = 0$ . Since  $\frac{\partial}{\partial u}(2xu) = \frac{\partial}{\partial x}(x^2 + 2u + 1) = 2x$ , (A) is exact. To solve (A) we must find  $F$  such that (A)  $F_x(x, u) = 2xu$  and (B)  $F_u(x, u) = x^2 + 2u + 1$ . Integrating (A) with respect to  $x$  yields (C)  $F(x, u) = x^2u + \phi(u)$ . Differentiating (C) with respect to  $u$  yields (D)  $F_u(x, u) = x^2 + \phi'(u)$ . Comparing (D) with (B) shows that  $\phi'(u) = 2u + 1$ , so we take  $\phi(u) = u^2 + u$ . Substituting this into (C) yields  $F(x, u) = x^2u + u^2 + u = u(x^2 + u + 1)$ . Therefore,  $u(x^2 + u + 1) = c$ . Since  $y(1) = -2 \Rightarrow u(1) = -2, c = 0$ . Therefore,  $u(x^2 + u + 1) = 0$ . Since  $u \equiv 0$  does not satisfy  $u(1) = -2$ , it follows that  $u = -x^2 - 1$  and  $y = -1 - \frac{1}{x^2}$ .

**2.5.40.**  $y_1 = e^{-x^2}$  is a solution of  $y' + 2xy = 0$ . Let  $y = ue^{-x^2}$ ; then  $u'e^{-x^2} = -e^{-x^2}\left(\frac{3x + 2u}{2x + 3u}\right)$ , so  $u' = -\frac{3x + 2u}{2x + 3u}$ , which can be rewritten as (A)  $(3x + 2u) dx + (2x + 3u) du = 0$ . Since  $\frac{\partial}{\partial u}(3x + 2u) =$

$\frac{\partial}{\partial x}(2x + 3u) = 2$ , (A) is exact. To solve (A) we must find  $F$  such that (A)  $F_x(x, u) = 3x + 2u$  and (B)  $F_u(x, u) = 2x + 3u$ . Integrating (A) with respect to  $x$  yields (C)  $F(x, u) = \frac{3x^2}{2} + 2xu + \phi(u)$ . Differentiating (C) with respect to  $u$  yields (D)  $F_u(x, u) = 2x + \phi'(u)$ . Comparing (D) with (B) shows that  $\phi'(u) = 3u$ , so we take  $\phi(u) = \frac{3u^2}{2}$ . Substituting this into (C) yields  $F(x, u) = \frac{3x^2}{2} + 2xu + \frac{3u^2}{2}$ . Therefore,  $\frac{3x^2}{2} + 2xu + \frac{3u^2}{2} = c$ . Since  $y(0) = -1 \Rightarrow u(0) = -1, c = \frac{3}{2}$ . Therefore,  $3x^2 + 4xu + 3u^2 = 3$  is an implicit solution of the initial value problem. Rewriting this as  $3u^2 + 4xu + (3x^2 - 3) = 0$  and solving for  $u$  by means of the quadratic formula yields  $u = -\left(\frac{2x + \sqrt{9 - 5x^2}}{3}\right)$ , so  $y = -e^{-x^2} \left(\frac{2x + \sqrt{9 - 5x^2}}{3}\right)$ .

**2.5.42.** Since  $M dx + N dy = 0$  is exact, (A)  $M_y = N_x$ . Since  $-N dx + M dy = 0$  is exact, (B)  $M_x = -N_y$ . Differentiating (A) with respect to  $y$  and (B) with respect to  $x$  yields (C)  $M_{yy} = N_{xy}$  and (D)  $M_{xx} = -N_{yx}$ . Since  $N_{xy} = N_{yx}$ , adding (C) and (D) yields  $M_{xx} + M_{yy} = 0$ . Differentiating (A) with respect to  $x$  and (B) with respect to  $y$  yields (E)  $M_{yx} = N_{xx}$  and (F)  $M_{xy} = -N_{yy}$ . Since  $M_{xy} = M_{yx}$ , subtracting (F) from (E) yields  $N_{xx} + N_{yy} = 0$ .

**2.5.44. (a)** If  $F(x, y) = x^2 - y^2$ , then  $F_x(x, y) = 2x$ ,  $F_y(x, y) = -2y$ ,  $F_{xx}(x, y) = 2$ , and  $F_{yy}(x, y) = -2$ . Therefore,  $F_{xx} + F_{yy} = 0$ , and  $G$  must satisfy (A)  $G_x(x, y) = 2y$  and (B)  $G_y(x, y) = 2x$ . Integrating (A) with respect to  $x$  yields (C)  $G(x, y) = 2xy + \phi(y)$ . Differentiating (C) with respect to  $y$  yields (D)  $G_y(x, y) = 2x + \phi'(y)$ . Comparing (D) with (B) shows that  $\phi'(y) = 0$ , so we take  $\phi(y) = c$ . Substituting this into (C) yields  $G(x, y) = 2xy + c$ .

(b) If  $F(x, y) = e^x \cos y$ , then  $F_x(x, y) = e^x \cos y$ ,  $F_y(x, y) = -e^x \sin y$ ,  $F_{xx}(x, y) = e^x \cos y$ , and  $F_{yy}(x, y) = -e^x \cos y$ . Therefore,  $F_{xx} + F_{yy} = 0$ , and  $G$  must satisfy (A)  $G_x(x, y) = e^x \sin y$  and (B)  $G_y(x, y) = e^x \cos y$ . Integrating (A) with respect to  $x$  yields (C)  $G(x, y) = e^x \sin y + \phi(y)$ . Differentiating (C) with respect to  $y$  yields (D)  $G_y(x, y) = e^x \cos y + \phi'(y)$ . Comparing (D) with (B) shows that  $\phi'(y) = 0$ , so we take  $\phi(y) = c$ . Substituting this into (C) yields  $G(x, y) = e^x \sin y + c$ .

(c) If  $F(x, y) = x^3 - 3xy^2$ , then  $F_x(x, y) = 3x^2 - 3y^2$ ,  $F_y(x, y) = -6xy$ ,  $F_{xx}(x, y) = 6x$ , and  $F_{yy}(x, y) = -6x$ . Therefore,  $F_{xx} + F_{yy} = 0$ , and  $G$  must satisfy (A)  $G_x(x, y) = 6xy$  and (B)  $G_y(x, y) = 3x^2 - 3y^2$ . Integrating (A) with respect to  $x$  yields (C)  $G(x, y) = 3x^2y + \phi(y)$ . Differentiating (C) with respect to  $y$  yields (D)  $G_y(x, y) = 3x^2 + \phi'(y)$ . Comparing (D) with (B) shows that  $\phi'(y) = -3y^2$ , so we take  $\phi(y) = -y^3 + c$ . Substituting this into (C) yields  $G(x, y) = 3x^2y - y^3 + c$ .

(d) If  $F(x, y) = \cos x \cosh y$ , then  $F_x(x, y) = -\sin x \cosh y$ ,  $F_y(x, y) = \cos x \sinh y$ ,  $F_{xx}(x, y) = -\cos x \cosh y$ , and  $F_{yy}(x, y) = \cos x \cosh y$ . Therefore,  $F_{xx} + F_{yy} = 0$ , and  $G$  must satisfy (A)  $G_x(x, y) = -\cos x \sinh y$  and (B)  $G_y(x, y) = -\sin x \cosh y$ . Integrating (A) with respect to  $x$  yields (C)  $G(x, y) = -\sin x \sinh y + \phi(y)$ . Differentiating (C) with respect to  $y$  yields (D)  $G_y(x, y) = -\sin x \cosh y + \phi'(y)$ . Comparing (D) with (B) shows that  $\phi'(y) = 0$ , so we take  $\phi(y) = c$ . Substituting this into (C) yields  $G(x, y) = -\sin x \sinh y + c$ .

(e) If  $F(x, y) = \sin x \cosh y$ , then  $F_x(x, y) = \cos x \cosh y$ ,  $F_y(x, y) = \sin x \sinh y$ ,  $F_{xx}(x, y) = -\sin x \cosh y$ , and  $F_{yy}(x, y) = \sin x \cosh y$ . Therefore,  $F_{xx} + F_{yy} = 0$ , and  $G$  must satisfy (A)  $G_x(x, y) = -\sin x \sinh y$  and (B)  $G_y(x, y) = \cos x \cosh y$ . Integrating (A) with respect to  $x$  yields (C)  $G(x, y) = \cos x \sinh y + \phi(y)$ . Differentiating (C) with respect to  $y$  yields (D)  $G_y(x, y) = \cos x \cosh y + \phi'(y)$ . Comparing (D) with (B) shows that  $\phi'(y) = 0$ , so we take  $\phi(y) = c$ . Substituting this into (C) yields  $G(x, y) = \cos x \sinh y + c$ .

## 2.6 Exact Equations

**2.6.2. (a) and (b).** To show that  $\mu(x, y) = \frac{1}{(x-y)^2}$  is an integrating factor for (A) and that (B) is exact, it suffices to observe that  $\frac{\partial}{\partial x} \left( \frac{xy}{x-y} \right) = -\frac{y^2}{(x-y)^2}$  and  $\frac{\partial}{\partial y} \left( \frac{xy}{x-y} \right) = \frac{x^2}{(x-y)^2}$ . By Theorem 2.5.1 this also shows that (C) is an implicit solution of (B). Since  $\mu(x, y)$  is never zero, any solution of (B) is a solution of (A).

(c) If we interpret (A) as  $-y^2 + x^2 y' = 0$ , then substituting  $y = x$  yields  $-x^2 + x^2 \cdot 1 = 0$ .

(NOTE: In Exercises 2.6.3–2.6.23, the given equation is multiplied by an integrating factor to produce an exact equation, and an implicit solution is found for the latter. For a complete analysis of the relationship between the sets of solutions of the two equations it is necessary to check for additional solutions of the given equation “along which” the integrating factor is undefined, or for solutions of the exact equation “along which” the integrating factor vanishes. In the interests of brevity we omit these tedious details except in cases where there actually is a difference between the sets of solutions of the two equations.)

**2.6.4.**  $M(x, y) = 3x^2 y$ ;  $N(x, y) = 2x^3$ ;  $M_y(x, y) - N_x(x, y) = 3x^2 - 6x^2 = -3x^2$ ;  $p(x) = \frac{M_y(x, y) - N_x(x, y)}{N(x, y)} = -\frac{3x^2}{2x^3} = -\frac{3}{2x}$ ;  $\int p(x) dx = -\frac{3}{2} \ln|x|$ ;  $\mu(x) = P(x) = x^{-3/2}$ ; therefore  $3x^{1/2} y dx + 2x^{3/2} dy = 0$  is exact. We must find  $F$  such that (A)  $F_x(x, y) = 3x^{1/2} y$  and (B)  $F_y(x, y) = 2x^{3/2}$ . Integrating (A) with respect to  $x$  yields (C)  $F(x, y) = 2x^{3/2} y + \phi(y)$ . Differentiating (C) with respect to  $y$  yields (D)  $F_y(x, y) = 2x^{3/2} + \phi'(y)$ . Comparing (D) with (B) shows that  $\phi'(y) = 0$ , so we take  $\phi(y) = 0$ . Substituting this into (C) yields  $F(x, y) = 2x^{3/2} y$ , so  $x^{3/2} y = c$ .

**2.6.6.**  $M(x, y) = 5xy + 2y + 5$ ;  $N(x, y) = 2x$ ;  $M_y(x, y) - N_x(x, y) = (5x + 2) - 2 = 5x$ ;  $p(x) = \frac{M_y(x, y) - N_x(x, y)}{N(x, y)} = \frac{5x}{2x} = \frac{5}{2}$ ;  $\int p(x) dx = \frac{5x}{2}$ ;  $\mu(x) = P(x) = e^{5x/2}$ ; therefore  $e^{5x/2}(5xy + 2y + 5) dx + 2xe^{5x/2} dy = 0$  is exact. We must find  $F$  such that (A)  $F_x(x, y) = e^{5x/2}(5xy + 2y + 5)$  and (B)  $F_y(x, y) = 2xe^{5x/2}$ . Integrating (B) with respect to  $y$  yields (C)  $F(x, y) = 2xye^{5x/2} + \psi(x)$ . Differentiating (C) with respect to  $x$  yields (D)  $F_x(x, y) = 5xye^{5x/2} + 2ye^{5x/2} + \psi'(x)$ . Comparing (D) with (A) shows that  $\psi'(x) = 5e^{5x/2}$ , so we take  $\psi(x) = 2e^{5x/2}$ . Substituting this into (C) yields  $F(x, y) = 2e^{5x/2}(xy + 1)$ , so  $e^{5x/2}(xy + 1) = c$ .

**2.6.8.**  $M(x, y) = 27xy^2 + 8y^3$ ;  $N(x, y) = 18x^2 y + 12xy^2$ ;  $M_y(x, y) - N_x(x, y) = (54xy + 24y^2) - (36xy + 12y^2) = 18xy + 12y^2$ ;  $p(x) = \frac{M_y(x, y) - N_x(x, y)}{N(x, y)} = \frac{18xy + 12y^2}{18x^2 y + 12y^2 x} = \frac{1}{x}$ ;  $\int p(x) dx = \ln|x|$ ;  $\mu(x) = P(x) = x$ ; therefore  $(27x^2 y^2 + 8xy^3) dx + (18x^3 y + 12x^2 y^2) dy = 0$  is exact. We must find  $F$  such that (A)  $F_x(x, y) = 27x^2 y^2 + 8xy^3$  and (B)  $F_y(x, y) = 18x^3 y + 12x^2 y^2$ . Integrating (A) with respect to  $x$  yields (C)  $F(x, y) = 9x^3 y^2 + 4x^2 y^3 + \phi(y)$ . Differentiating (C) with respect to  $y$  yields (D)  $F_y(x, y) = 18x^3 y + 12x^2 y^2 + \phi'(y)$ . Comparing (D) with (B) shows that  $\phi'(y) = 0$ , so we take  $\phi(y) = 0$ . Substituting this into (C) yields  $F(x, y) = 9x^3 y^2 + 4x^2 y^3$ , so  $x^2 y^2(9x + 4y) = c$ .

**2.6.10.**  $M(x, y) = y^2$ ;  $N(x, y) = \left( xy^2 + 3xy + \frac{1}{y} \right)$ ;  $M_y(x, y) - N_x(x, y) = 2y - (y^2 + 3y) = -y(y + 1)$ ;  $q(y) = \frac{N_x(x, y) - M_y(x, y)}{M(x, y)} = \frac{y(y + 1)}{y^2} = 1 + \frac{1}{y}$ ;  $\int q(y) dy = y \ln|y|$ ;  $\mu(y) = Q(y) = ye^y$ ; therefore  $y^3 e^y dx + e^y(xy^3 + 3xy^2 + 1) dy = 0$  is exact. We must find  $F$  such that (A)  $F_x(x, y) = y^3 e^y$  and (B)  $F_y(x, y) = e^y(xy^3 + 3xy^2 + 1)$ . Integrating (A) with respect to  $x$  yields (C)  $F(x, y) =$

$xy^3e^y + \phi(y)$ . Differentiating (C) with respect to  $y$  yields (D)  $F_y(x, y) = xy^3e^y + 3xy^2e^y + \phi'(y)$ . Comparing (D) with (B) shows that  $\phi'(y) = e^y$ , so we take  $\phi(y) = e^y$ . Substituting this into (C) yields  $F(x, y) = xy^3e^y + e^y$ , so  $e^y(xy^3 + 1) = c$ .

**2.6.12.**  $M(x, y) = x^2y + 4xy + 2y$ ;  $N(x, y) = x^2 + x$ ;  $M_y(x, y) - N_x(x, y) = (x^2 + 4x + 2) - (2x + 1) = x^2 + 2x + 1 = (x + 1)^2$ ;  $p(x) = \frac{M_y(x, y) - N_x(x, y)}{N(x, y)} = \frac{(x + 1)^2}{x(x + 1)} = 1 + \frac{1}{x}$ ;  $\int p(x) dx = x + \ln|x|$ ;  $\mu(x) = P(x) = xe^x$ ; therefore  $e^x(x^3y + 4x^2y + 2xy) dx + e^x(x^3 + x^2) dy = 0$  is exact. We must find  $F$  such that (A)  $F_x(x, y) = e^x(x^3y + 4x^2y + 2xy)$  and (B)  $F_y(x, y) = e^x(x^3 + x^2)$ . Integrating (B) with respect to  $y$  yields (C)  $F(x, y) = y(x^3 + x^2)e^x + \psi(x)$ . Differentiating (C) with respect to  $x$  yields (D)  $F_x(x, y) = e^x(x^3y + 4x^2y + 2xy) + \psi'(x)$ . Comparing (D) with (A) shows that  $\psi'(x) = 0$ , so we take  $\psi(x) = 0$ . Substituting this into (C) yields  $F(x, y) = y(x^3 + x^2)e^x = x^2y(x + 1)e^x$ , so  $x^2y(x + 1)e^x = c$ .

**2.6.14.**  $M(x, y) = \cos x \cos y$ ;  $N(x, y) = \sin x \cos y - \sin x \sin y + y$ ;  $M_y(x, y) - N_x(x, y) = -\cos x \sin y - (\cos x \cos y - \cos x \sin y) = -\cos x \cos y$ ;  $q(y) = \frac{N_x(x, y) - M_y(x, y)}{M(x, y)} = \frac{\cos x \cos y}{\cos x \cos y} = 1$ ;  $\int q(y) dy = 1$ ;  $\mu(y) = Q(y) = e^y$ ; therefore  $e^y \cos x \cos y dx + e^y(\sin x \cos y - \sin x \sin y + y) dy = 0$  is exact. We must find  $F$  such that (A)  $F_x(x, y) = e^y \cos x \cos y$  and (B)  $F_y(x, y) = e^y(\sin x \cos y - \sin x \sin y + y)$ . Integrating (A) with respect to  $x$  yields (C)  $F(x, y) = e^y \sin x \cos y + \phi(y)$ . Differentiating (C) with respect to  $y$  yields (D)  $F_y(x, y) = e^y(\sin x \cos y - \sin x \sin y) + \phi'(y)$ . Comparing (D) with (B) shows that  $\phi'(y) = ye^y$ , so we take  $\phi(y) = e^y(y - 1)$ . Substituting this into (C) yields  $F(x, y) = e^y(\sin x \cos y + y - 1)$ , so  $e^y(\sin x \cos y + y - 1) = c$ .

**2.6.16.**  $M(x, y) = y \sin y$ ;  $N(x, y) = x(\sin y - y \cos y)$ ;  $M_y(x, y) - N_x(x, y) = (y \cos y + \sin y) - (\sin y - y \cos y) = 2y \cos y$ ;  $q(y) = \frac{N_x(x, y) - M_y(x, y)}{N(x, y)} = -\frac{2 \cos y}{\sin y}$ ;  $\int q(y) dy = -2 \ln|\sin y|$ ;  $\mu(y) = Q(y) = \frac{1}{\sin^2 y}$ ; therefore  $\left(\frac{y}{\sin y}\right) dx + x\left(\frac{1}{\sin y} - \frac{y \cos y}{\sin^2 y}\right) dy = 0$  is exact. We must find  $F$  such that (A)  $F_x(x, y) = \frac{y}{\sin y}$  and (B)  $F_y(x, y) = x\left(\frac{1}{\sin y} - \frac{y \cos y}{\sin^2 y}\right)$ . Integrating (A) with respect to  $x$  yields (C)  $F(x, y) = \frac{xy}{\sin y} + \phi(y)$ . Differentiating (C) with respect to  $y$  yields (D)  $F_y(x, y) = x\left(\frac{1}{\sin y} - \frac{y \cos y}{\sin^2 y}\right) + \phi'(y)$ . Comparing (D) with (B) shows that  $\phi'(y) = 0$ , so we take  $\phi(y) = 0$ . Substituting this into (C) yields  $F(x, y) = \frac{xy}{\sin y}$ , so  $\frac{xy}{\sin y} = c$ . In addition, the given equation has the constant solutions  $y = k\pi$ , where  $k$  is an integer.

**2.6.18.**  $M(x, y) = \alpha y + \gamma xy$ ;  $N(x, y) = \beta x + \delta xy$ ;  $M_y(x, y) - N_x(x, y) = (\alpha + \gamma x) - (\beta + \delta y)$ ; and  $p(x)N(x, y) - q(y)M(x, y) = p(x)x(\beta + \delta y) - q(y)y(\alpha + \gamma x)$ . so exactness requires that  $(\alpha + \gamma x) - (\beta + \delta y) = p(x)x(\beta + \delta y) - q(y)y(\alpha + \gamma x)$ , which holds if  $p(x)x = -1$  and  $q(y)y = -1$ . Thus  $p(x) = -\frac{1}{x}$ ;  $q(y) = -\frac{1}{y}$ ;  $\int p(x) dx = -\ln|x|$ ;  $\int q(y) dy = -\ln|y|$ ;  $P(x) = \frac{1}{x}$ ;  $Q(y) = \frac{1}{y}$ ;  $\mu(x, y) = \frac{1}{xy}$ . Therefore,  $\left(\frac{\alpha}{x} + \gamma\right) dx + \left(\frac{\beta}{y} + \delta\right) dy = 0$  is exact. We must find  $F$  such that (A)  $F_x(x, y) = \frac{\alpha}{x} + \gamma$  and (B)  $F_y(x, y) = \frac{\beta}{y} + \delta$ . Integrating (A) with respect to  $x$  yields (C)  $F(x, y) = \alpha \ln|x| + \gamma x + \phi(y)$ . Differentiating (C) with respect to  $y$  yields (D)  $F_y(x, y) = \phi'(y)$ .

Comparing (D) with (B) shows that  $\phi'(y) = \frac{\beta}{y} + \delta$ , so we take  $\phi(y) = \beta \ln |y| + \delta y$ . Substituting this into (C) yields  $F(x, y) = \alpha \ln |x| + \gamma x + \beta \ln |y| + \delta y$ , so  $|x|^\alpha |y|^\beta e^{\gamma x} e^{\delta y} = c$ . The given equation also has the solutions  $x \equiv 0$  and  $y \equiv 0$ .

**2.6.20.**  $M(x, y) = 2y$ ;  $N(x, y) = 3(x^2 + x^2 y^3)$ ;  $M_y(x, y) - N_x(x, y) = 2 - (6x + 6xy^3)$ ; and  $p(x)N(x, y) - q(y)M(x, y) = 3p(x)(x^2 + x^2 y^3) - 2q(y)y$ . so exactness requires that (A)  $2 - 6x - 6xy^3 = 3p(x)x(x + xy^3) - 2q(y)y$ . To obtain similar terms on the two sides of (A) we let  $p(x)x = a$  and  $q(y)y = b$  where  $a$  and  $b$  are constants such that  $2 - 6x - 6xy^3 = 3a(x + xy^3) - 2b$ , which holds if  $a = -2$  and  $b = -1$ . Thus,  $p(x) = -\frac{2}{x}$ ;  $q(y) = -\frac{1}{y}$ ;  $\int p(x) dx = -2 \ln |x|$ ;  $\int q(y) dy = -\ln |y|$ ;  $P(x) = \frac{1}{x^2}$ ;  $Q(y) = \frac{1}{y}$ ;  $\mu(x, y) = \frac{1}{x^2 y}$ . Therefore,  $\frac{2}{x^2} dx + 3\left(\frac{1}{y} + y^2\right) dy = 0$  is exact. We must find  $F$  such that (B)  $F_x(x, y) = \frac{2}{x^2}$  and (C)  $F_y(x, y) = 3\left(\frac{1}{y} + y^2\right)$ . Integrating (B) with respect to  $x$  yields (D)  $F(x, y) = -\frac{2}{x} + \phi(y)$ . Differentiating (D) with respect to  $y$  yields (E)  $F_y(x, y) = \phi'(y)$ .

Comparing (E) with (C) shows that  $\phi'(y) = 3\left(\frac{1}{y} + y^2\right)$ , so we take  $\phi(y) = y^3 + 3 \ln |y|$ . Substituting this into (D) yields  $F(x, y) = -\frac{2}{x} + y^3 + 3 \ln |y|$ , so  $-\frac{2}{x} + y^3 + 3 \ln |y| = c$ . The given equation also has the solutions  $x \equiv 0$  and  $y \equiv 0$ .

**2.6.22.**  $M(x, y) = x^4 y^4$ ;  $N(x, y) = x^5 y^3$ ;  $M_y(x, y) - N_x(x, y) = 4x^4 y^3 - 5x^4 y^3 = -x^4 y^3$ ; and  $p(x)N(x, y) - q(y)M(x, y) = p(x)x^5 y^3 - q(y)x^4 y^4$ . so exactness requires that  $-x^4 y^3 = p(x)x^5 y^3 - q(y)x^4 y^4$ , which is equivalent to  $p(x)x - q(y)y = -1$ . This holds if  $p(x)x = a$  and  $q(y)y = a + 1$  where  $a$  is an arbitrary real number. Thus,  $p(x) = \frac{a}{x}$ ;  $q(y) = \frac{a+1}{y}$ ;  $\int p(x) dx = a \ln |x|$ ;  $\int q(y) dy = (a+1) \ln |y|$ ;  $P(x) = |x|^a$ ;  $Q(y) = |y|^{a+1}$ ;  $\mu(x, y) = |x|^a |y|^{a+1}$ . Therefore,  $|x|^a |y|^{a+1} (x^4 y^4 dx + x^5 y^3 dy) = 0$  is exact for any choice of  $a$ . For simplicity we let  $a = -4$ , so (A) is equivalent to  $y dx + x dy = 0$ . We must find  $F$  such that (B)  $F_x(x, y) = y$  and (C)  $F_y(x, y) = x$ . Integrating (B) with respect to  $x$  yields (D)  $F(x, y) = xy + \phi(y)$ . Differentiating (D) with respect to  $y$  yields (E)  $F_y(x, y) = x + \phi'(y)$ . Comparing (E) with (C) shows that  $\phi'(y) = 0$ , so we take  $\phi(y) = 0$ . Substituting this into (D) yields  $F(x, y) = xy$ , so  $xy = c$ .

**2.6.24.**  $M(x, y) = x^4 y^3 + y$ ;  $N(x, y) = x^5 y^2 - x$ ;  $M_y(x, y) - N_x(x, y) = (3x^4 y^2 + 1) - (5x^4 y^2 - 1) = -2x^4 y^2 + 2$ ;  $p(x) = \frac{M_y(x, y) - N_x(x, y)}{N(x, y)} = -\frac{2x^4 y^2 - 2}{x^5 y^2 - x} = -\frac{2}{x}$ ;  $\int p(x) dx = -2 \ln |x|$ ;  $\mu(x) = P(x) = \frac{1}{x^2}$ ; therefore  $\left(x^2 y^3 + \frac{y}{x^2}\right) dx + \left(x^3 y^2 - \frac{1}{x}\right) dy = 0$  is exact. We must find  $F$  such that (A)  $F_x(x, y) = \left(x^2 y^3 + \frac{y}{x^2}\right)$  and (B)  $F_y(x, y) = \left(x^3 y^2 - \frac{1}{x}\right)$ . Integrating (A) with respect to  $x$  yields (C)  $F(x, y) = \frac{x^3 y^3}{3} - \frac{y}{x} + \phi(y)$ . Differentiating (C) with respect to  $y$  yields (D)  $F_y(x, y) = x^3 y^2 - \frac{1}{x} + \phi'(y)$ . Comparing (D) with (B) shows that  $\phi'(y) = 0$ , so we take  $\phi(y) = 0$ .

Substituting this into (C) yields  $F(x, y) = \frac{x^3 y^3}{3} - \frac{y}{x}$ , so  $\frac{x^3 y^3}{3} - \frac{y}{x} = c$ .

**2.6.26.**  $M(x, y) = 12xy + 6y^3$ ;  $N(x, y) = 9x^2 + 10xy^2$ ;  $M_y(x, y) - N_x(x, y) = (12x + 18y^2) - (18x + 10y^2) = -6x + 8y^2$ ; and  $p(x)N(x, y) - q(y)M(x, y) = p(x)x(9x + 10y^2) - q(y)y(12x + 6y^2)$ ,

so exactness requires that (A)  $-6x + 8y^2 = p(x)x(9x + 10y^2) - q(y)y(12x + 6y^2)$ . To obtain similar terms on the two sides of (A) we let  $p(x)x = a$  and  $q(y)y = b$  where  $a$  and  $b$  are constants such that  $-6x + 8y^2 = a(9x + 10y^2) - b(12x + 6y^2)$ , which holds if  $9a - 12b = -6$ ,  $10a - 6b = 8$ ; that is,  $a = b = 2$ . Thus  $p(x) = \frac{2}{x}$ ;  $q(y) = \frac{2}{y}$ ;  $\int p(x) dx = 2 \ln |x|$ ;  $\int q(y) dy = 2 \ln |y|$ ;  $P(x) = x^2$ ;  $Q(y) = y^2$ ;  $\mu(x, y) = x^2 y^2$ . Therefore,  $(12x^3 y^3 + 6x^2 y^5) dx + (9x^4 y^2 + 10x^3 y^4) dy = 0$  is exact. We must find  $F$  such that (B)  $F_x(x, y) = 12x^3 y^3 + 6x^2 y^5$  and (C)  $F_y(x, y) = 9x^4 y^2 + 10x^3 y^4$ . Integrating (B) with respect to  $x$  yields (D)  $F(x, y) = 3x^4 y^3 + 2x^3 y^5 + \phi(y)$ . Differentiating (D) with respect to  $y$  yields (E)  $F_y(x, y) = 9x^4 y^2 + 10x^3 y^4 + \phi'(y)$ . Comparing (E) with (C) shows that  $\phi'(y) = 0$ , so we take  $\phi(y) = 0$ . Substituting this into (D) yields  $F(x, y) = 3x^4 y^3 + 2x^3 y^5$ , so  $x^3 y^3(3x + 2y^2) = c$ .

**2.6.28.**  $M(x, y) = ax^m y + by^{n+1}$ ;  $N(x, y) = cx^{m+1} + dxy^n$ ;  $M_y(x, y) - N_x(x, y) = [ax^{m+1} + (n+1)by^n] - [(m+1)cx^m + dy^n]$ ;  $p(x)N(x, y) - q(y)M(x, y) = xp(x)(cx^{m+1} + dxy^n) - yp(y)(ax^m + by^n)$ . Let (A)  $xp(x) = \alpha$  and (B)  $yp(y) = \beta$ , where  $\alpha$  and  $\beta$  are to be chosen so that  $[ax^{m+1} + (n+1)by^n] - [(m+1)cx^m + dy^n] = \alpha(cx^m + dy^n) - \beta(ax^m + by^n)$ , which will hold if

$$\begin{aligned} c\alpha - a\beta &= a - (m+1)c & \stackrel{\text{df}}{=} & A \\ d\alpha - b\beta &= -d + (n+1)b & \stackrel{\text{df}}{=} & B. \end{aligned} \tag{C}$$

Since  $ad - bc \neq 0$  it can be verified that  $\alpha = \frac{aB - bA}{ad - bc}$  and  $\beta = \frac{cB - dA}{ad - bc}$  satisfy (C). From (A) and (B),  $p(x) = \frac{\alpha}{x}$  and  $q(y) = \frac{\beta}{y}$ , so  $\mu(x, y) = x^\alpha y^\beta$  is an integrating factor for the given equation.

**2.6.30.** (a) Since  $M(x, y) = p(x)y - f(x)$  and  $N(x, y) = 1$ ,  $\frac{M_y(x, y) - N_x(x, y)}{N(x, y)} = p(x)$  and Theorem 2.6.1 implies that  $\mu(x) \pm e^{\int p(x) dx}$  is an integrating factor for (C).

(b) Multiplying (A) through  $\mu = \pm e^{\int p(x) dx}$  yields (D)  $\mu(x)y' + \mu'(x)y = \mu(x)f(x)$ , which is equivalent to  $(\mu(x)y)' = \mu(x)f(x)$ . Integrating this yields  $\mu(x)y = c + \int \mu(x)f(x) dx$ , so  $y = \frac{1}{\mu(x)} \left( c + \int \mu(x)f(x) dx \right)$ , which is equivalent to (B) since  $y_1 = \frac{1}{\mu}$  is a nontrivial solution of  $y' + p(x)y = 0$ .

# Chapter 3

## Numerical Methods

### 3.1 Euler's Method

**3.1.2.**  $y_1 = 1.200000000$ ,  $y_2 = 1.440415946$ ,  $y_3 = 1.729880994$

**3.1.4.**  $y_1 = 2.962500000$ ,  $y_2 = 2.922635828$ ,  $y_3 = 2.880205639$

**3.1.6.**

| $x$ | $h = 0.1$    | $h = 0.05$   | $h = 0.025$  | Exact        |
|-----|--------------|--------------|--------------|--------------|
| 0.0 | 2.000000000  | 2.000000000  | 2.000000000  | 2.000000000  |
| 0.1 | 2.100000000  | 2.169990965  | 2.202114518  | 2.232642918  |
| 0.2 | 2.514277288  | 2.649377900  | 2.713011720  | 2.774352565  |
| 0.3 | 3.317872752  | 3.527672599  | 3.628465025  | 3.726686582  |
| 0.4 | 4.646592772  | 4.955798226  | 5.106379369  | 5.254226636  |
| 0.5 | 6.719737638  | 7.171467977  | 7.393322991  | 7.612186259  |
| 0.6 | 9.876155616  | 10.538384528 | 10.865186799 | 11.188475269 |
| 0.7 | 14.629532397 | 15.605686107 | 16.088630652 | 16.567103199 |
| 0.8 | 21.751925418 | 23.197328550 | 23.913328531 | 24.623248150 |
| 0.9 | 32.399118931 | 34.545932627 | 35.610005377 | 36.665439956 |
| 1.0 | 48.298147362 | 51.492825643 | 53.076673685 | 54.647937102 |

**3.1.8.**

| $x$  | $h = 0.05$  | $h = 0.025$ | $h = 0.0125$ | Exact        |
|------|-------------|-------------|--------------|--------------|
| 1.00 | 2.000000000 | 2.000000000 | 2.000000000  | 2.000000000  |
| 1.05 | 2.250000000 | 2.259280190 | 2.264490570  | 2.270158103  |
| 1.10 | 2.536734694 | 2.559724746 | 2.572794280  | 2.587150838  |
| 1.15 | 2.867950854 | 2.910936426 | 2.935723355  | 2.963263785  |
| 1.20 | 3.253613825 | 3.325627715 | 3.367843117  | 3.415384615  |
| 1.25 | 3.706750613 | 3.820981064 | 3.889251900  | 3.967391304  |
| 1.30 | 4.244700641 | 4.420781829 | 4.528471927  | 4.654198473  |
| 1.35 | 4.891020001 | 5.158883503 | 5.327348558  | 5.528980892  |
| 1.40 | 5.678467290 | 6.085075790 | 6.349785943  | 6.676923077  |
| 1.45 | 6.653845988 | 7.275522641 | 7.698316221  | 8.243593315  |
| 1.50 | 7.886170437 | 8.852463793 | 9.548039907  | 10.500000000 |

3.1.10.

| $x$                   | $h = 0.1$   | $h = 0.05$  | $h = 0.025$ | $h = 0.1$ | $h = 0.05$ | $h = 0.025$ |
|-----------------------|-------------|-------------|-------------|-----------|------------|-------------|
| 1.0                   | 1.000000000 | 1.000000000 | 1.000000000 | 0.0000    | 0.0000     | 0.0000      |
| 1.1                   | 0.920000000 | 0.921898275 | 0.922822717 | -0.0384   | -0.0189    | -0.0094     |
| 1.2                   | 0.847469326 | 0.851018464 | 0.852746371 | -0.0745   | -0.0368    | -0.0183     |
| 1.3                   | 0.781779403 | 0.786770087 | 0.789197876 | -0.1092   | -0.0540    | -0.0268     |
| 1.4                   | 0.722453556 | 0.728682209 | 0.731709712 | -0.1428   | -0.0707    | -0.0351     |
| 1.5                   | 0.669037867 | 0.676299618 | 0.679827306 | -0.1752   | -0.0868    | -0.0432     |
| 1.6                   | 0.621054176 | 0.629148585 | 0.633080163 | -0.2062   | -0.1023    | -0.0509     |
| 1.7                   | 0.578000416 | 0.586740390 | 0.590986601 | -0.2356   | -0.1170    | -0.0583     |
| 1.8                   | 0.539370187 | 0.548588902 | 0.553070392 | -0.2631   | -0.1310    | -0.0653     |
| 1.9                   | 0.504674296 | 0.514228603 | 0.518877246 | -0.2889   | -0.1441    | -0.0719     |
| 2.0                   | 0.473456737 | 0.483227470 | 0.487986391 | -0.3129   | -0.1563    | -0.0781     |
| Approximate Solutions |             |             |             | Residuals |            |             |

3.1.12.

| $x$ | $h = 0.1$    | $h = 0.05$   | $h = 0.025$  | “Exact”      |
|-----|--------------|--------------|--------------|--------------|
| 1.0 | 0.000000000  | 0.000000000  | 0.000000000  | 0.000000000  |
| 1.1 | -0.100000000 | -0.099875000 | -0.099780455 | -0.099664000 |
| 1.2 | -0.199000000 | -0.198243434 | -0.197800853 | -0.197315517 |
| 1.3 | -0.294996246 | -0.293129862 | -0.292110713 | -0.291036003 |
| 1.4 | -0.386095345 | -0.382748403 | -0.380986158 | -0.379168221 |
| 1.5 | -0.470695388 | -0.465664569 | -0.463078857 | -0.460450590 |
| 1.6 | -0.547627491 | -0.540901018 | -0.537503081 | -0.534085626 |
| 1.7 | -0.616227665 | -0.607969574 | -0.603849795 | -0.599737720 |
| 1.8 | -0.676329533 | -0.666833345 | -0.662136956 | -0.657473792 |
| 1.9 | -0.728190908 | -0.717819639 | -0.712718751 | -0.707670533 |
| 2.0 | -0.772381768 | -0.761510960 | -0.756179726 | -0.750912371 |

3.1.14.

| Euler’s method |              |              |              |              |
|----------------|--------------|--------------|--------------|--------------|
| $x$            | $h = 0.1$    | $h = 0.05$   | $h = 0.025$  | “Exact”      |
| 2.0            | 2.000000000  | 2.000000000  | 2.000000000  | 2.000000000  |
| 2.1            | 2.420000000  | 2.440610764  | 2.451962006  | 2.464119569  |
| 2.2            | 2.922484288  | 2.972198224  | 2.999753046  | 3.029403212  |
| 2.3            | 3.524104434  | 3.614025082  | 3.664184099  | 3.718409925  |
| 2.4            | 4.244823572  | 4.389380160  | 4.470531822  | 4.558673929  |
| 2.5            | 5.108581185  | 5.326426396  | 5.449503467  | 5.583808754  |
| 2.6            | 6.144090526  | 6.459226591  | 6.638409411  | 6.834855438  |
| 2.7            | 7.385795229  | 7.828984275  | 8.082588076  | 8.361928926  |
| 2.8            | 8.875017001  | 9.485544888  | 9.837137672  | 10.226228709 |
| 2.9            | 10.661332618 | 11.489211987 | 11.969020902 | 12.502494409 |
| 3.0            | 12.804226135 | 13.912944662 | 14.559623055 | 15.282004826 |



| Euler semilinear method |              |              |              |              |
|-------------------------|--------------|--------------|--------------|--------------|
| $x$                     | $h = 0.1$    | $h = 0.05$   | $h = 0.025$  | "Exact"      |
| 2.0                     | 2.000000000  | 2.000000000  | 2.000000000  | 2.000000000  |
| 2.1                     | 2.467233571  | 2.465641081  | 2.464871435  | 2.464119569  |
| 2.2                     | 3.036062650  | 3.032657307  | 3.031011316  | 3.029403212  |
| 2.3                     | 3.729169725  | 3.723668026  | 3.721008466  | 3.718409925  |
| 2.4                     | 4.574236356  | 4.566279470  | 4.562432696  | 4.558673929  |
| 2.5                     | 5.605052990  | 5.594191643  | 5.588940276  | 5.583808754  |
| 2.6                     | 6.862874116  | 6.848549921  | 6.841623814  | 6.834855438  |
| 2.7                     | 8.398073101  | 8.379595572  | 8.370660695  | 8.361928926  |
| 2.8                     | 10.272163096 | 10.248681420 | 10.237326199 | 10.226228709 |
| 2.9                     | 12.560265110 | 12.530733531 | 12.516452106 | 12.502494409 |
| 3.0                     | 15.354122287 | 15.317257705 | 15.299429421 | 15.282004826 |

| Euler's method |             |             |             |             |
|----------------|-------------|-------------|-------------|-------------|
| $x$            | $h = 0.2$   | $h = 0.1$   | $h = 0.05$  | "Exact"     |
| 1.0            | 2.000000000 | 2.000000000 | 2.000000000 | 2.000000000 |
| 1.2            | 1.768294197 | 1.786514499 | 1.794412375 | 1.801636774 |
| 1.4            | 1.603028371 | 1.628427487 | 1.639678822 | 1.650102616 |
| 1.6            | 1.474580412 | 1.502563111 | 1.515157063 | 1.526935885 |
| 1.8            | 1.368349549 | 1.396853671 | 1.409839229 | 1.422074283 |
| 2.0            | 1.276424761 | 1.304504818 | 1.317421794 | 1.329664953 |
| 2.2            | 1.194247156 | 1.221490111 | 1.234122458 | 1.246155344 |
| 2.4            | 1.119088175 | 1.145348276 | 1.157607418 | 1.169334346 |
| 2.6            | 1.049284410 | 1.074553688 | 1.086419453 | 1.097812069 |
| 2.8            | 0.983821745 | 1.008162993 | 1.019652023 | 1.030719114 |
| 3.0            | 0.922094379 | 0.945604800 | 0.956752868 | 0.967523153 |

| Euler semilinear method |             |             |             |             |
|-------------------------|-------------|-------------|-------------|-------------|
| $x$                     | $h = 0.2$   | $h = 0.1$   | $h = 0.05$  | "Exact"     |
| 1.0                     | 2.000000000 | 2.000000000 | 2.000000000 | 2.000000000 |
| 1.2                     | 1.806911831 | 1.804304958 | 1.802978526 | 1.801636774 |
| 1.4                     | 1.659738603 | 1.654968381 | 1.652547436 | 1.650102616 |
| 1.6                     | 1.540257861 | 1.533652916 | 1.530308405 | 1.526935885 |
| 1.8                     | 1.438532932 | 1.430361800 | 1.426232584 | 1.422074283 |
| 2.0                     | 1.348782285 | 1.339279577 | 1.334486249 | 1.329664953 |
| 2.2                     | 1.267497415 | 1.256876924 | 1.251528766 | 1.246155344 |
| 2.4                     | 1.192497494 | 1.180958765 | 1.175157264 | 1.169334346 |
| 2.6                     | 1.122416379 | 1.110147777 | 1.103988310 | 1.097812069 |
| 2.8                     | 1.056405906 | 1.043585743 | 1.037158237 | 1.030719114 |
| 3.0                     | 0.993954754 | 0.980751307 | 0.974140320 | 0.967523153 |

3.1.18.

| Euler's method |             |             |             |             |
|----------------|-------------|-------------|-------------|-------------|
| $x$            | $h = 0.2$   | $h = 0.1$   | $h = 0.05$  | "Exact"     |
| 0.0            | 1.000000000 | 1.000000000 | 1.000000000 | 1.000000000 |
| 0.2            | 1.200000000 | 1.186557290 | 1.179206574 | 1.171515153 |
| 0.4            | 1.333543409 | 1.298441890 | 1.280865289 | 1.263370891 |
| 0.6            | 1.371340142 | 1.319698328 | 1.295082088 | 1.271251278 |
| 0.8            | 1.326367357 | 1.270160237 | 1.243958980 | 1.218901287 |
| 1.0            | 1.233056306 | 1.181845667 | 1.158064902 | 1.135362070 |
| 1.2            | 1.122359136 | 1.080477477 | 1.060871608 | 1.042062625 |
| 1.4            | 1.013100262 | 0.981124989 | 0.965917496 | 0.951192532 |
| 1.6            | 0.914000211 | 0.890759107 | 0.879460404 | 0.868381328 |
| 1.8            | 0.827848558 | 0.811673612 | 0.803582000 | 0.795518627 |
| 2.0            | 0.754572560 | 0.743869878 | 0.738303914 | 0.732638628 |

| Euler semilinear method |             |             |             |             |
|-------------------------|-------------|-------------|-------------|-------------|
| $x$                     | $h = 0.2$   | $h = 0.1$   | $h = 0.05$  | "Exact"     |
| 0.0                     | 1.000000000 | 1.000000000 | 1.000000000 | 1.000000000 |
| 0.2                     | 1.153846154 | 1.162906599 | 1.167266650 | 1.171515153 |
| 0.4                     | 1.236969953 | 1.250608357 | 1.257097924 | 1.263370891 |
| 0.6                     | 1.244188456 | 1.258241892 | 1.264875987 | 1.271251278 |
| 0.8                     | 1.195155456 | 1.207524076 | 1.213335781 | 1.218901287 |
| 1.0                     | 1.115731189 | 1.125966437 | 1.130768614 | 1.135362070 |
| 1.2                     | 1.025938754 | 1.034336918 | 1.038283392 | 1.042062625 |
| 1.4                     | 0.937645707 | 0.944681597 | 0.948002346 | 0.951192532 |
| 1.6                     | 0.856581823 | 0.862684171 | 0.865583126 | 0.868381328 |
| 1.8                     | 0.784832910 | 0.790331183 | 0.792963532 | 0.795518627 |
| 2.0                     | 0.722610454 | 0.727742966 | 0.730220211 | 0.732638628 |

3.1.20.

| Euler's method |             |             |             |             |
|----------------|-------------|-------------|-------------|-------------|
| $x$            | $h = 0.1$   | $h = 0.05$  | $h = 0.025$ | "Exact"     |
| 0.0            | 1.000000000 | 1.000000000 | 1.000000000 | 1.000000000 |
| 0.1            | 0.700000000 | 0.725841563 | 0.736671690 | 0.746418339 |
| 0.2            | 0.498330000 | 0.532982493 | 0.547988831 | 0.561742917 |
| 0.3            | 0.356272689 | 0.392592562 | 0.408724303 | 0.423724207 |
| 0.4            | 0.254555443 | 0.289040639 | 0.304708942 | 0.319467408 |
| 0.5            | 0.181440541 | 0.212387189 | 0.226758594 | 0.240464879 |
| 0.6            | 0.128953069 | 0.155687255 | 0.168375130 | 0.180626161 |
| 0.7            | 0.091393543 | 0.113851516 | 0.124744976 | 0.135394692 |
| 0.8            | 0.064613612 | 0.083076641 | 0.092230966 | 0.101293057 |
| 0.9            | 0.045585102 | 0.060505907 | 0.068068776 | 0.075650324 |
| 1.0            | 0.032105117 | 0.043997045 | 0.050159310 | 0.056415515 |

| Euler semilinear method |             |             |             |             |
|-------------------------|-------------|-------------|-------------|-------------|
| $x$                     | $h = 0.1$   | $h = 0.05$  | $h = 0.025$ | “Exact”     |
| 0.0                     | 1.000000000 | 1.000000000 | 1.000000000 | 1.000000000 |
| 0.1                     | 0.740818221 | 0.743784320 | 0.745143557 | 0.746418339 |
| 0.2                     | 0.555889275 | 0.558989106 | 0.560410719 | 0.561742917 |
| 0.3                     | 0.418936461 | 0.421482025 | 0.422642541 | 0.423724207 |
| 0.4                     | 0.315890439 | 0.317804400 | 0.318668549 | 0.319467408 |
| 0.5                     | 0.237908421 | 0.239287095 | 0.239902094 | 0.240464879 |
| 0.6                     | 0.178842206 | 0.179812811 | 0.180239888 | 0.180626161 |
| 0.7                     | 0.134165506 | 0.134840668 | 0.135133367 | 0.135394692 |
| 0.8                     | 0.100450939 | 0.100918118 | 0.101117514 | 0.101293057 |
| 0.9                     | 0.075073968 | 0.075396974 | 0.075532643 | 0.075650324 |
| 1.0                     | 0.056020154 | 0.056243980 | 0.056336491 | 0.056415515 |

3.1.22.

| Euler's method |             |             |             |             |
|----------------|-------------|-------------|-------------|-------------|
| $x$            | $h = 0.1$   | $h = 0.05$  | $h = 0.025$ | “Exact”     |
| 2.0            | 1.000000000 | 1.000000000 | 1.000000000 | 1.000000000 |
| 2.1            | 1.000000000 | 1.005062500 | 1.007100815 | 1.008899988 |
| 2.2            | 1.020500000 | 1.026752091 | 1.029367367 | 1.031723469 |
| 2.3            | 1.053489840 | 1.059067423 | 1.061510137 | 1.063764243 |
| 2.4            | 1.093521685 | 1.097780573 | 1.099748225 | 1.101614730 |
| 2.5            | 1.137137554 | 1.140059654 | 1.141496651 | 1.142903776 |
| 2.6            | 1.182269005 | 1.184090031 | 1.185056276 | 1.186038851 |
| 2.7            | 1.227745005 | 1.228755801 | 1.229350441 | 1.229985178 |
| 2.8            | 1.272940309 | 1.273399187 | 1.273721920 | 1.274092525 |
| 2.9            | 1.317545833 | 1.317651554 | 1.317786528 | 1.317967533 |
| 3.0            | 1.361427907 | 1.361320824 | 1.361332589 | 1.361383810 |

| Euler semilinear method |             |             |             |             |
|-------------------------|-------------|-------------|-------------|-------------|
| $x$                     | $h = 0.1$   | $h = 0.05$  | $h = 0.025$ | “Exact”     |
| 2.0                     | 1.000000000 | 1.000000000 | 1.000000000 | 1.000000000 |
| 2.1                     | 0.982476904 | 0.996114142 | 1.002608435 | 1.008899988 |
| 2.2                     | 0.988105346 | 1.010577663 | 1.021306044 | 1.031723469 |
| 2.3                     | 1.009495813 | 1.037358814 | 1.050731634 | 1.063764243 |
| 2.4                     | 1.041012955 | 1.071994816 | 1.086964414 | 1.101614730 |
| 2.5                     | 1.078631301 | 1.111346365 | 1.127262285 | 1.142903776 |
| 2.6                     | 1.119632590 | 1.153300133 | 1.169781376 | 1.186038851 |
| 2.7                     | 1.162270287 | 1.196488725 | 1.213325613 | 1.229985178 |
| 2.8                     | 1.205472927 | 1.240060456 | 1.257146091 | 1.274092525 |
| 2.9                     | 1.248613584 | 1.283506001 | 1.300791772 | 1.317967533 |
| 3.0                     | 1.291345518 | 1.326535737 | 1.344004102 | 1.361383810 |

## 3.2 The Improved Euler Method and Related Methods

3.2.2.  $y_1 = 1.220207973$ ,  $y_2 = 1.489578775$   $y_3 = 1.819337186$

3.2.4.  $y_1 = 2.961317914$ ;  $y_2 = 2.920132727$ ;  $y_3 = 2.876213748$ .

3.2.6.

| $x$ | $h = 0.1$    | $h = 0.05$   | $h = 0.025$  | Exact        |
|-----|--------------|--------------|--------------|--------------|
| 0.0 | 2.000000000  | 2.000000000  | 2.000000000  | 2.000000000  |
| 0.1 | 2.257138644  | 2.238455342  | 2.234055168  | 2.232642918  |
| 0.2 | 2.826004666  | 2.786634110  | 2.777340360  | 2.774352565  |
| 0.3 | 3.812671926  | 3.747167263  | 3.731674025  | 3.726686582  |
| 0.4 | 5.387430580  | 5.285996803  | 5.261969043  | 5.254226636  |
| 0.5 | 7.813298361  | 7.660199197  | 7.623893064  | 7.612186259  |
| 0.6 | 11.489337756 | 11.260349005 | 11.206005869 | 11.188475269 |
| 0.7 | 17.015861211 | 16.674352914 | 16.593267820 | 16.567103199 |
| 0.8 | 25.292140630 | 24.783149862 | 24.662262731 | 24.623248150 |
| 0.9 | 37.662496723 | 36.903828191 | 36.723608928 | 36.665439956 |
| 1.0 | 56.134480009 | 55.003390448 | 54.734674836 | 54.647937102 |

3.2.8.

| $x$  | $h = 0.05$   | $h = 0.025$  | $h = 0.0125$ | Exact        |
|------|--------------|--------------|--------------|--------------|
| 1.00 | 2.000000000  | 2.000000000  | 2.000000000  | 2.000000000  |
| 1.05 | 2.268367347  | 2.269670336  | 2.270030868  | 2.270158103  |
| 1.10 | 2.582607299  | 2.585911295  | 2.586827341  | 2.587150838  |
| 1.15 | 2.954510022  | 2.960870733  | 2.962638822  | 2.963263785  |
| 1.20 | 3.400161788  | 3.411212150  | 3.414293964  | 3.415384615  |
| 1.25 | 3.942097142  | 3.960434900  | 3.965570792  | 3.967391304  |
| 1.30 | 4.612879780  | 4.642784826  | 4.651206769  | 4.654198473  |
| 1.35 | 5.461348619  | 5.510188575  | 5.524044591  | 5.528980892  |
| 1.40 | 6.564150753  | 6.645334756  | 6.668600859  | 6.676923077  |
| 1.45 | 8.048579617  | 8.188335998  | 8.228972215  | 8.243593315  |
| 1.50 | 10.141969585 | 10.396770409 | 10.472502111 | 10.500000000 |

3.2.10.

| $x$                   | $h = 0.1$   | $h = 0.05$  | $h = 0.025$ | $h = 0.1$ | $h = 0.05$ | $h = 0.025$ |
|-----------------------|-------------|-------------|-------------|-----------|------------|-------------|
| 1.0                   | 1.000000000 | 1.000000000 | 1.000000000 | 0.00000   | 0.000000   | 0.000000    |
| 1.1                   | 0.923734663 | 0.923730743 | 0.923730591 | 0.00004   | 0.000001   | -0.000001   |
| 1.2                   | 0.854475600 | 0.854449616 | 0.854444697 | 0.00035   | 0.000068   | 0.000015    |
| 1.3                   | 0.791650344 | 0.791596016 | 0.791584634 | 0.00078   | 0.000167   | 0.000039    |
| 1.4                   | 0.734785779 | 0.734703826 | 0.734686010 | 0.00125   | 0.000277   | 0.000065    |
| 1.5                   | 0.683424095 | 0.683318666 | 0.683295308 | 0.00171   | 0.000384   | 0.000091    |
| 1.6                   | 0.637097057 | 0.636973423 | 0.636945710 | 0.00213   | 0.000483   | 0.000115    |
| 1.7                   | 0.595330359 | 0.595193634 | 0.595162740 | 0.00250   | 0.000572   | 0.000137    |
| 1.8                   | 0.557658422 | 0.557513000 | 0.557479947 | 0.00283   | 0.000650   | 0.000156    |
| 1.9                   | 0.523638939 | 0.523488343 | 0.523453958 | 0.00311   | 0.000718   | 0.000173    |
| 2.0                   | 0.492862999 | 0.492709931 | 0.492674855 | 0.00335   | 0.000777   | 0.000187    |
| Approximate Solutions |             |             |             | Residuals |            |             |

3.2.12.

| $x$ | $h = 0.1$    | $h = 0.05$   | $h = 0.025$  | "Exact"      |
|-----|--------------|--------------|--------------|--------------|
| 1.0 | 0.000000000  | 0.000000000  | 0.000000000  | 0.000000000  |
| 1.1 | -0.099500000 | -0.099623114 | -0.099653809 | -0.099664000 |
| 1.2 | -0.196990313 | -0.197235180 | -0.197295585 | -0.197315517 |
| 1.3 | -0.290552949 | -0.290917718 | -0.291006784 | -0.291036003 |
| 1.4 | -0.378532718 | -0.379013852 | -0.379130237 | -0.379168221 |
| 1.5 | -0.459672297 | -0.460262848 | -0.460404546 | -0.460450590 |
| 1.6 | -0.533180153 | -0.533868468 | -0.534032512 | -0.534085626 |
| 1.7 | -0.598726853 | -0.599496413 | -0.599678824 | -0.599737720 |
| 1.8 | -0.656384109 | -0.657214624 | -0.657410640 | -0.657473792 |
| 1.9 | -0.706530934 | -0.707400266 | -0.707604759 | -0.707670533 |
| 2.0 | -0.749751364 | -0.750637632 | -0.750845571 | -0.750912371 |

3.2.14.

| Improved Euler method |              |              |              |              |
|-----------------------|--------------|--------------|--------------|--------------|
| $x$                   | $h = 0.1$    | $h = 0.05$   | $h = 0.025$  | "Exact"      |
| 2.0                   | 2.000000000  | 2.000000000  | 2.000000000  | 2.000000000  |
| 2.1                   | 2.461242144  | 2.463344439  | 2.463918368  | 2.464119569  |
| 2.2                   | 3.022367633  | 3.027507237  | 3.028911026  | 3.029403212  |
| 2.3                   | 3.705511610  | 3.714932709  | 3.717507170  | 3.718409925  |
| 2.4                   | 4.537659565  | 4.553006531  | 4.557202414  | 4.558673929  |
| 2.5                   | 5.551716960  | 5.575150456  | 5.581560437  | 5.583808754  |
| 2.6                   | 6.787813853  | 6.822158665  | 6.831558101  | 6.834855438  |
| 2.7                   | 8.294896222  | 8.343829180  | 8.357227947  | 8.361928926  |
| 2.8                   | 10.132667135 | 10.200955596 | 10.219663917 | 10.226228709 |
| 2.9                   | 12.373954732 | 12.467758807 | 12.493470722 | 12.502494409 |
| 3.0                   | 15.107600968 | 15.234856000 | 15.269755072 | 15.282004826 |

| Improved Euler semilinear method |              |              |              |              |
|----------------------------------|--------------|--------------|--------------|--------------|
| $x$                              | $h = 0.1$    | $h = 0.05$   | $h = 0.025$  | "Exact"      |
| 2.0                              | 2.000000000  | 2.000000000  | 2.000000000  | 2.000000000  |
| 2.1                              | 2.464261688  | 2.464155139  | 2.464128464  | 2.464119569  |
| 2.2                              | 3.029706047  | 3.029479005  | 3.029422165  | 3.029403212  |
| 2.3                              | 3.718897663  | 3.718531995  | 3.718440451  | 3.718409925  |
| 2.4                              | 4.559377397  | 4.558849990  | 4.558717956  | 4.558673929  |
| 2.5                              | 5.584766724  | 5.584048510  | 5.583868709  | 5.583808754  |
| 2.6                              | 6.836116246  | 6.835170986  | 6.834934347  | 6.834855438  |
| 2.7                              | 8.363552464  | 8.362335253  | 8.362030535  | 8.361928926  |
| 2.8                              | 10.228288880 | 10.226744312 | 10.226357645 | 10.226228709 |
| 2.9                              | 12.505082132 | 12.503142042 | 12.502656361 | 12.502494409 |
| 3.0                              | 15.285231726 | 15.282812424 | 15.282206780 | 15.282004826 |

3.2.16.

| Improved Euler method |             |             |             |             |
|-----------------------|-------------|-------------|-------------|-------------|
| $x$                   | $h = 0.2$   | $h = 0.1$   | $h = 0.05$  | "Exact"     |
| 1.0                   | 2.000000000 | 2.000000000 | 2.000000000 | 2.000000000 |
| 1.2                   | 1.801514185 | 1.801606135 | 1.801629115 | 1.801636774 |
| 1.4                   | 1.649911580 | 1.650054870 | 1.650090680 | 1.650102616 |
| 1.6                   | 1.526711768 | 1.526879870 | 1.526921882 | 1.526935885 |
| 1.8                   | 1.421841570 | 1.422016119 | 1.422059743 | 1.422074283 |
| 2.0                   | 1.329441172 | 1.329609020 | 1.329650971 | 1.329664953 |
| 2.2                   | 1.245953205 | 1.246104819 | 1.246142713 | 1.246155344 |
| 2.4                   | 1.169162994 | 1.169291515 | 1.169323639 | 1.169334346 |
| 2.6                   | 1.097677870 | 1.097778523 | 1.097803683 | 1.097812069 |
| 2.8                   | 1.030626179 | 1.030695880 | 1.030713305 | 1.030719114 |
| 3.0                   | 0.967473721 | 0.967510790 | 0.967520062 | 0.967523153 |

| Improved Euler semilinear method |             |             |             |             |
|----------------------------------|-------------|-------------|-------------|-------------|
| $x$                              | $h = 0.2$   | $h = 0.1$   | $h = 0.05$  | “Exact”     |
| 1.0                              | 2.000000000 | 2.000000000 | 2.000000000 | 2.000000000 |
| 1.2                              | 1.801514185 | 1.801606135 | 1.801629115 | 1.801636774 |
| 1.4                              | 1.649911580 | 1.650054870 | 1.650090680 | 1.650102616 |
| 1.6                              | 1.526711768 | 1.526879870 | 1.526921882 | 1.526935885 |
| 1.8                              | 1.421841570 | 1.422016119 | 1.422059743 | 1.422074283 |
| 2.0                              | 1.329441172 | 1.329609020 | 1.329650971 | 1.329664953 |
| 2.2                              | 1.245953205 | 1.246104819 | 1.246142713 | 1.246155344 |
| 2.4                              | 1.169162994 | 1.169291515 | 1.169323639 | 1.169334346 |
| 2.6                              | 1.097677870 | 1.097778523 | 1.097803683 | 1.097812069 |
| 2.8                              | 1.030626179 | 1.030695880 | 1.030713305 | 1.030719114 |
| 3.0                              | 0.967473721 | 0.967510790 | 0.967520062 | 0.967523153 |

| Improved Euler method |             |             |             |             |
|-----------------------|-------------|-------------|-------------|-------------|
| $x$                   | $h = 0.2$   | $h = 0.1$   | $h = 0.05$  | “Exact”     |
| 0.0                   | 1.000000000 | 1.000000000 | 1.000000000 | 1.000000000 |
| 0.2                   | 1.166771705 | 1.170394902 | 1.171244037 | 1.171515153 |
| 0.4                   | 1.255835116 | 1.261642355 | 1.262958788 | 1.263370891 |
| 0.6                   | 1.263517157 | 1.269528214 | 1.270846761 | 1.271251278 |
| 0.8                   | 1.212551997 | 1.217531648 | 1.218585457 | 1.218901287 |
| 1.0                   | 1.130812573 | 1.134420589 | 1.135150284 | 1.135362070 |
| 1.2                   | 1.039104333 | 1.041487727 | 1.041938536 | 1.042062625 |
| 1.4                   | 0.949440052 | 0.950888923 | 0.951132561 | 0.951192532 |
| 1.6                   | 0.867475787 | 0.868263999 | 0.868364849 | 0.868381328 |
| 1.8                   | 0.795183973 | 0.795523696 | 0.795530315 | 0.795518627 |
| 2.0                   | 0.732679223 | 0.732721613 | 0.732667905 | 0.732638628 |

| Improved Euler semilinear method |             |             |             |             |
|----------------------------------|-------------|-------------|-------------|-------------|
| $x$                              | $h = 0.2$   | $h = 0.1$   | $h = 0.05$  | “Exact”     |
| 0.0                              | 1.000000000 | 1.000000000 | 1.000000000 | 1.000000000 |
| 0.2                              | 1.170617859 | 1.171292452 | 1.171459576 | 1.171515153 |
| 0.4                              | 1.261629934 | 1.262938347 | 1.263262919 | 1.263370891 |
| 0.6                              | 1.269173253 | 1.270734290 | 1.271122186 | 1.271251278 |
| 0.8                              | 1.216926014 | 1.218409355 | 1.218778420 | 1.218901287 |
| 1.0                              | 1.133688235 | 1.134944960 | 1.135257876 | 1.135362070 |
| 1.2                              | 1.040721691 | 1.041728386 | 1.041979126 | 1.042062625 |
| 1.4                              | 0.950145706 | 0.950931597 | 0.951127345 | 0.951192532 |
| 1.6                              | 0.867573431 | 0.868179975 | 0.868331028 | 0.868381328 |
| 1.8                              | 0.794899034 | 0.795364245 | 0.795480063 | 0.795518627 |
| 2.0                              | 0.732166678 | 0.732521078 | 0.732609267 | 0.732638628 |

3.2.20.

| Improved Euler method |             |             |             |             |
|-----------------------|-------------|-------------|-------------|-------------|
| $x$                   | $h = 0.1$   | $h = 0.05$  | $h = 0.025$ | “Exact”     |
| 0.0                   | 1.000000000 | 1.000000000 | 1.000000000 | 1.000000000 |
| 0.1                   | 0.749165000 | 0.747022742 | 0.746561141 | 0.746418339 |
| 0.2                   | 0.565942699 | 0.562667885 | 0.561961242 | 0.561742917 |
| 0.3                   | 0.428618351 | 0.424803657 | 0.423978964 | 0.423724207 |
| 0.4                   | 0.324556426 | 0.320590918 | 0.319732571 | 0.319467408 |
| 0.5                   | 0.245417735 | 0.241558658 | 0.240723019 | 0.240464879 |
| 0.6                   | 0.185235654 | 0.181643813 | 0.180866303 | 0.180626161 |
| 0.7                   | 0.139546094 | 0.136310496 | 0.135610749 | 0.135394692 |
| 0.8                   | 0.104938506 | 0.102096319 | 0.101482503 | 0.101293057 |
| 0.9                   | 0.078787731 | 0.076340645 | 0.075813072 | 0.075650324 |
| 1.0                   | 0.059071894 | 0.056999028 | 0.056553023 | 0.056415515 |

| Improved Euler semilinear method |             |             |             |             |
|----------------------------------|-------------|-------------|-------------|-------------|
| $x$                              | $h = 0.1$   | $h = 0.05$  | $h = 0.025$ | “Exact”     |
| 0.0                              | 1.000000000 | 1.000000000 | 1.000000000 | 1.000000000 |
| 0.1                              | 0.745595127 | 0.746215164 | 0.746368056 | 0.746418339 |
| 0.2                              | 0.560827568 | 0.561515647 | 0.561686492 | 0.561742917 |
| 0.3                              | 0.422922083 | 0.423524585 | 0.423674586 | 0.423724207 |
| 0.4                              | 0.318820339 | 0.319306259 | 0.319427337 | 0.319467408 |
| 0.5                              | 0.239962317 | 0.240339716 | 0.240433757 | 0.240464879 |
| 0.6                              | 0.180243441 | 0.180530866 | 0.180602470 | 0.180626161 |
| 0.7                              | 0.135106416 | 0.135322934 | 0.135376855 | 0.135394692 |
| 0.8                              | 0.101077312 | 0.101239368 | 0.101279714 | 0.101293057 |
| 0.9                              | 0.075489492 | 0.075610310 | 0.075640381 | 0.075650324 |
| 1.0                              | 0.056295914 | 0.056385765 | 0.056408124 | 0.056415515 |

3.2.22.

| Improved Euler method |             |             |             |             |
|-----------------------|-------------|-------------|-------------|-------------|
| $x$                   | $h = 0.1$   | $h = 0.05$  | $h = 0.025$ | “Exact”     |
| 2.0                   | 1.000000000 | 1.000000000 | 1.000000000 | 1.000000000 |
| 2.1                   | 1.010250000 | 1.009185754 | 1.008965733 | 1.008899988 |
| 2.2                   | 1.033547273 | 1.032105322 | 1.031811002 | 1.031723469 |
| 2.3                   | 1.065562151 | 1.064135919 | 1.063849094 | 1.063764243 |
| 2.4                   | 1.103145347 | 1.101926450 | 1.101685553 | 1.101614730 |
| 2.5                   | 1.144085693 | 1.143140125 | 1.142957158 | 1.142903776 |
| 2.6                   | 1.186878796 | 1.186202854 | 1.186075600 | 1.186038851 |
| 2.7                   | 1.230530804 | 1.230088035 | 1.230007943 | 1.229985178 |
| 2.8                   | 1.274404357 | 1.274147657 | 1.274104430 | 1.274092525 |
| 2.9                   | 1.318104153 | 1.317987551 | 1.317971490 | 1.317967533 |
| 3.0                   | 1.361395309 | 1.361379259 | 1.361382239 | 1.361383810 |

| Improved Euler semilinear method |             |             |             |             |
|----------------------------------|-------------|-------------|-------------|-------------|
| $x$                              | $h = 0.1$   | $h = 0.05$  | $h = 0.025$ | “Exact”     |
| 2.0                              | 1.000000000 | 1.000000000 | 1.000000000 | 1.000000000 |
| 2.1                              | 1.012802674 | 1.009822081 | 1.009124116 | 1.008899988 |
| 2.2                              | 1.038431870 | 1.033307426 | 1.032108359 | 1.031723469 |
| 2.3                              | 1.072484834 | 1.065821457 | 1.064263950 | 1.063764243 |
| 2.4                              | 1.111794329 | 1.104013534 | 1.102197168 | 1.101614730 |
| 2.5                              | 1.154168041 | 1.145554968 | 1.143547198 | 1.142903776 |
| 2.6                              | 1.198140189 | 1.188883373 | 1.186728849 | 1.186038851 |
| 2.7                              | 1.242762459 | 1.232984559 | 1.230712361 | 1.229985178 |
| 2.8                              | 1.287441845 | 1.277221941 | 1.274850828 | 1.274092525 |
| 2.9                              | 1.331821976 | 1.321210992 | 1.318753047 | 1.317967533 |
| 3.0                              | 1.375699933 | 1.364730937 | 1.362193997 | 1.361383810 |

| $x$ | $h = 0.1$   | $h = 0.05$  | $h = 0.025$ | Exact       |
|-----|-------------|-------------|-------------|-------------|
| 1.0 | 1.000000000 | 1.000000000 | 1.000000000 | 1.000000000 |
| 1.1 | 1.151019287 | 1.153270661 | 1.153777957 | 1.153937085 |
| 1.2 | 1.238798618 | 1.241884421 | 1.242580821 | 1.242799540 |
| 1.3 | 1.289296258 | 1.292573128 | 1.293313355 | 1.293546032 |
| 1.4 | 1.317686801 | 1.320866599 | 1.321585242 | 1.321811247 |
| 1.5 | 1.333073855 | 1.336036248 | 1.336705820 | 1.336916440 |
| 1.6 | 1.341027170 | 1.343732006 | 1.344343232 | 1.344535503 |
| 1.7 | 1.345001345 | 1.347446389 | 1.347998652 | 1.348172348 |
| 1.8 | 1.347155352 | 1.349355473 | 1.349852082 | 1.350008229 |
| 1.9 | 1.348839325 | 1.350816158 | 1.351261995 | 1.351402121 |
| 2.0 | 1.350890736 | 1.352667599 | 1.353067951 | 1.353193719 |

| $x$  | $h = 0.05$   | $h = 0.025$  | $h = 0.0125$ | Exact        |
|------|--------------|--------------|--------------|--------------|
| 1.00 | 2.000000000  | 2.000000000  | 2.000000000  | 2.000000000  |
| 1.05 | 2.268496358  | 2.269703943  | 2.270043628  | 2.270158103  |
| 1.10 | 2.582897367  | 2.585985695  | 2.586855275  | 2.587150838  |
| 1.15 | 2.954995034  | 2.960992388  | 2.962683751  | 2.963263785  |
| 1.20 | 3.400872342  | 3.411384294  | 3.414355862  | 3.415384615  |
| 1.25 | 3.943047906  | 3.960651794  | 3.965644965  | 3.967391304  |
| 1.30 | 4.614039436  | 4.643018510  | 4.651277424  | 4.654198473  |
| 1.35 | 5.462568051  | 5.510357362  | 5.524069547  | 5.528980892  |
| 1.40 | 6.564985580  | 6.645224236  | 6.668472955  | 6.676923077  |
| 1.45 | 8.047824947  | 8.187384679  | 8.228413044  | 8.243593315  |
| 1.50 | 10.136329642 | 10.393419681 | 10.470731411 | 10.500000000 |

| $x$ | $h = 0.1$   | $h = 0.05$  | $h = 0.025$ | “Exact”     |
|-----|-------------|-------------|-------------|-------------|
| 0.0 | 1.000000000 | 1.000000000 | 1.000000000 | 1.000000000 |
| 0.1 | 0.984142840 | 0.984133302 | 0.984130961 | 0.984130189 |
| 0.2 | 0.965066124 | 0.965044455 | 0.965039117 | 0.965037353 |
| 0.3 | 0.942648578 | 0.942611457 | 0.942602279 | 0.942599241 |
| 0.4 | 0.916705578 | 0.916648569 | 0.916634423 | 0.916629732 |
| 0.5 | 0.886970525 | 0.886887464 | 0.886866778 | 0.886859904 |
| 0.6 | 0.853066054 | 0.852948011 | 0.852918497 | 0.852908668 |
| 0.7 | 0.814458249 | 0.814291679 | 0.814249848 | 0.814235883 |
| 0.8 | 0.770380571 | 0.770143777 | 0.770083998 | 0.770063987 |
| 0.9 | 0.719699643 | 0.719355385 | 0.719267905 | 0.719238519 |
| 1.0 | 0.660658411 | 0.660136630 | 0.660002840 | 0.659957689 |



3.2.30.

| $x$ | $h = 0.1$    | $h = 0.05$   | $h = 0.025$  | "Exact"      |
|-----|--------------|--------------|--------------|--------------|
| 1.0 | 0.000000000  | 0.000000000  | 0.000000000  | 0.000000000  |
| 1.1 | -0.099666667 | -0.099665005 | -0.099664307 | -0.099674132 |
| 1.2 | -0.197322275 | -0.197317894 | -0.197316222 | -0.197355914 |
| 1.3 | -0.291033227 | -0.291036361 | -0.291036258 | -0.291123993 |
| 1.4 | -0.379131069 | -0.379160444 | -0.379166504 | -0.379315647 |
| 1.5 | -0.460350276 | -0.460427667 | -0.460445166 | -0.460662347 |
| 1.6 | -0.533897316 | -0.534041581 | -0.534075026 | -0.534359685 |
| 1.7 | -0.599446325 | -0.599668984 | -0.599721072 | -0.600066382 |
| 1.8 | -0.657076288 | -0.657379719 | -0.657450947 | -0.657845646 |
| 1.9 | -0.707175010 | -0.707553135 | -0.707641993 | -0.708072516 |
| 2.0 | -0.750335016 | -0.750775571 | -0.750879100 | -0.751331499 |

**3.2.32.** (a) Let  $x_i = a + ih$ ,  $i = 0, 1, \dots, n$ . If  $y$  is the solution of the initial value problem  $y' = f(x)$ ,  $y(a) = 0$ , then  $y(b) = \int_a^b f(x) dx$ . The improved Euler method yields  $y_{i+1} = y_i + .5h(f(a + ih) + f(a + (i + 1)h))$ ,  $i = 0, 1, \dots, n - 1$ , where  $y_0 = a$  and  $y_n$  is an approximation to  $\int_a^b f(x) dx$ . But

$$y_n = \sum_{i=0}^{n-1} (y_{i+1} - y_i) = .5h(f(a) + f(b)) + h \sum_{i=1}^{n-1} f(a + ih).$$

(c) The local truncation error is a multiple of  $y'''(\tilde{x}_i) = f''(\tilde{x}_i)$ , where  $x_i < \tilde{x}_i < x_{i+1}$ . Therefore, the quadrature formula is exact if  $f$  is a polynomial of degree  $< 2$ .

(d) Let  $E(f) = \int_a^b f(x) dx - y_n$ . Note that  $E$  is linear. If  $f$  is a polynomial of degree 2, then  $f(x) = f_0(x) + K(x - a)^2$  where  $\deg(f_0) \leq 1$ . Since  $E(f_0) = 0$  from (c) and

$$\begin{aligned} E((x - a)^2) &= \frac{(b - a)^3}{3} - \frac{(b - a)^2 h}{2} - h^3 \sum_{i=1}^{n-1} i^2 \\ &= h^3 \left[ \frac{n^3}{3} - \frac{n^2}{2} - \frac{n(n - 1)(2n - 1)}{6} \right] = -\frac{nh^3}{6} = -\frac{(b - a)h^2}{6}, \end{aligned}$$

$E(f) = -\frac{K(b - a)h^2}{6}$ ; therefore the error is proportional to  $h^2$ .

### 3.3 The Runge-Kutta Method

**3.3.2.**  $y_1 = 1.221551366$ ,  $y_2 = 1.492920208$

**3.3.4.**  $y_1 = 2.961316248$ ;  $y_2 = 2.920128958$ .

3.3.6.

| $x$ | $h = 0.1$    | $h = 0.05$   | $h = 0.025$  | Exact        |
|-----|--------------|--------------|--------------|--------------|
| 0.0 | 2.000000000  | 2.000000000  | 2.000000000  | 2.000000000  |
| 0.1 | 2.232752507  | 2.232649573  | 2.232643327  | 2.232642918  |
| 0.2 | 2.774582759  | 2.774366625  | 2.774353431  | 2.774352565  |
| 0.3 | 3.727068686  | 3.726710028  | 3.726688030  | 3.726686582  |
| 0.4 | 5.254817388  | 5.254263005  | 5.254228886  | 5.254226636  |
| 0.5 | 7.613077020  | 7.612241222  | 7.612189662  | 7.612186259  |
| 0.6 | 11.189806778 | 11.188557546 | 11.188480365 | 11.188475269 |
| 0.7 | 16.569088310 | 16.567225975 | 16.567110808 | 16.567103199 |
| 0.8 | 24.626206255 | 24.623431201 | 24.623259496 | 24.623248150 |
| 0.9 | 36.669848687 | 36.665712858 | 36.665456874 | 36.665439956 |
| 1.0 | 54.654509699 | 54.648344019 | 54.647962328 | 54.647937102 |

3.3.8.

| $x$  | $h = 0.05$   | $h = 0.025$  | $h = 0.0125$ | Exact        |
|------|--------------|--------------|--------------|--------------|
| 1.00 | 2.000000000  | 2.000000000  | 2.000000000  | 2.000000000  |
| 1.05 | 2.270153785  | 2.270157806  | 2.270158083  | 2.270158103  |
| 1.10 | 2.587139846  | 2.587150083  | 2.587150789  | 2.587150838  |
| 1.15 | 2.963242415  | 2.963262317  | 2.963263689  | 2.963263785  |
| 1.20 | 3.415346864  | 3.415382020  | 3.415384445  | 3.415384615  |
| 1.25 | 3.967327077  | 3.967386886  | 3.967391015  | 3.967391304  |
| 1.30 | 4.654089950  | 4.654191000  | 4.654197983  | 4.654198473  |
| 1.35 | 5.528794615  | 5.528968045  | 5.528980049  | 5.528980892  |
| 1.40 | 6.676590929  | 6.676900116  | 6.676921569  | 6.676923077  |
| 1.45 | 8.242960669  | 8.243549415  | 8.243590428  | 8.243593315  |
| 1.50 | 10.498658198 | 10.499906266 | 10.499993820 | 10.500000000 |

3.3.10.

| $x$                   | $h = 0.1$   | $h = 0.05$  | $h = 0.025$ | $h = 0.1$    | $h = 0.05$   | $h = 0.025$   |
|-----------------------|-------------|-------------|-------------|--------------|--------------|---------------|
| 1.0                   | 1.000000000 | 1.000000000 | 1.000000000 | 0.000000000  | 0.000000000  | 0.000000000   |
| 1.1                   | 0.923730622 | 0.923730677 | 0.923730681 | -0.000000608 | -0.000000389 | -0.0000000245 |
| 1.2                   | 0.854443253 | 0.854443324 | 0.854443328 | -0.000000819 | -0.000000529 | -0.0000000335 |
| 1.3                   | 0.791581155 | 0.791581218 | 0.791581222 | -0.000000753 | -0.000000495 | -0.0000000316 |
| 1.4                   | 0.734680497 | 0.734680538 | 0.734680541 | -0.000000523 | -0.000000359 | -0.0000000233 |
| 1.5                   | 0.683288034 | 0.683288051 | 0.683288052 | -0.000000224 | -0.000000178 | -0.0000000122 |
| 1.6                   | 0.636937046 | 0.636937040 | 0.636937040 | 0.000000079  | 0.000000006  | -0.0000000009 |
| 1.7                   | 0.595153053 | 0.595153029 | 0.595153028 | 0.000000351  | 0.000000171  | 0.0000000093  |
| 1.8                   | 0.557469558 | 0.557469522 | 0.557469520 | 0.000000578  | 0.000000309  | 0.0000000179  |
| 1.9                   | 0.523443129 | 0.523443084 | 0.523443081 | 0.000000760  | 0.000000421  | 0.0000000248  |
| 2.0                   | 0.492663789 | 0.492663738 | 0.492663736 | 0.000000902  | 0.000000508  | 0.0000000302  |
| Approximate Solutions |             |             |             | Residuals    |              |               |

3.3.12.

| $x$ | $h = 0.1$    | $h = 0.05$   | $h = 0.025$  | "Exact"      |
|-----|--------------|--------------|--------------|--------------|
| 1.0 | 0.000000000  | 0.000000000  | 0.000000000  | 0.000000000  |
| 1.1 | -0.099663901 | -0.099663994 | -0.099664000 | -0.099664000 |
| 1.2 | -0.197315322 | -0.197315504 | -0.197315516 | -0.197315517 |
| 1.3 | -0.291035700 | -0.291035983 | -0.291036001 | -0.291036003 |
| 1.4 | -0.379167790 | -0.379168194 | -0.379168220 | -0.379168221 |
| 1.5 | -0.460450005 | -0.460450552 | -0.460450587 | -0.460450590 |
| 1.6 | -0.534084875 | -0.534085579 | -0.534085623 | -0.534085626 |
| 1.7 | -0.599736802 | -0.599737663 | -0.599737717 | -0.599737720 |
| 1.8 | -0.657472724 | -0.657473726 | -0.657473788 | -0.657473792 |
| 1.9 | -0.707669346 | -0.707670460 | -0.707670529 | -0.707670533 |
| 2.0 | -0.750911103 | -0.750912294 | -0.750912367 | -0.750912371 |

3.3.14.

| Runge-Kutta method |              |              |              |              |
|--------------------|--------------|--------------|--------------|--------------|
| $x$                | $h = 0.1$    | $h = 0.05$   | $h = 0.025$  | “Exact”      |
| 2.0                | 2.000000000  | 2.000000000  | 2.000000000  | 2.000000000  |
| 2.1                | 2.464113907  | 2.464119185  | 2.464119544  | 2.464119569  |
| 2.2                | 3.029389360  | 3.029402271  | 3.029403150  | 3.029403212  |
| 2.3                | 3.718384519  | 3.718408199  | 3.718409812  | 3.718409925  |
| 2.4                | 4.558632516  | 4.558671116  | 4.558673746  | 4.558673929  |
| 2.5                | 5.583745479  | 5.583804456  | 5.583808474  | 5.583808754  |
| 2.6                | 6.834762639  | 6.834849135  | 6.834855028  | 6.834855438  |
| 2.7                | 8.361979619  | 8.361919939  | 8.361928340  | 8.361928926  |
| 2.8                | 10.226043942 | 10.226216159 | 10.226227891 | 10.226228709 |
| 2.9                | 12.502240429 | 12.502477158 | 12.502493285 | 12.502494409 |
| 3.0                | 15.281660036 | 15.281981407 | 15.282003300 | 15.282004826 |

| Runge-Kutta semilinear method |              |              |              |              |
|-------------------------------|--------------|--------------|--------------|--------------|
| $x$                           | $h = 0.1$    | $h = 0.05$   | $h = 0.025$  | “Exact”      |
| 2.0                           | 2.000000000  | 2.000000000  | 2.000000000  | 2.000000000  |
| 2.1                           | 2.464119623  | 2.464119573  | 2.464119570  | 2.464119569  |
| 2.2                           | 3.029403325  | 3.029403219  | 3.029403212  | 3.029403212  |
| 2.3                           | 3.718410105  | 3.718409936  | 3.718409925  | 3.718409925  |
| 2.4                           | 4.558674188  | 4.558673945  | 4.558673930  | 4.558673929  |
| 2.5                           | 5.583809105  | 5.583808776  | 5.583808755  | 5.583808754  |
| 2.6                           | 6.834855899  | 6.834855467  | 6.834855440  | 6.834855438  |
| 2.7                           | 8.361929516  | 8.361928963  | 8.361928928  | 8.361928926  |
| 2.8                           | 10.226229456 | 10.226228756 | 10.226228712 | 10.226228709 |
| 2.9                           | 12.502495345 | 12.502494468 | 12.502494413 | 12.502494409 |
| 3.0                           | 15.282005990 | 15.282004899 | 15.282004831 | 15.282004826 |

3.3.16.

| Runge-Kutta method |             |             |             |             |
|--------------------|-------------|-------------|-------------|-------------|
| $x$                | $h = 0.2$   | $h = 0.1$   | $h = 0.05$  | “Exact”     |
| 1.0                | 2.000000000 | 2.000000000 | 2.000000000 | 2.000000000 |
| 1.2                | 1.801636785 | 1.801636775 | 1.801636774 | 1.801636774 |
| 1.4                | 1.650102633 | 1.650102617 | 1.650102616 | 1.650102616 |
| 1.6                | 1.526935904 | 1.526935886 | 1.526935885 | 1.526935885 |
| 1.8                | 1.422074302 | 1.422074284 | 1.422074283 | 1.422074283 |
| 2.0                | 1.329664970 | 1.329664954 | 1.329664953 | 1.329664953 |
| 2.2                | 1.246155357 | 1.246155345 | 1.246155344 | 1.246155344 |
| 2.4                | 1.169334355 | 1.169334347 | 1.169334346 | 1.169334346 |
| 2.6                | 1.097812074 | 1.097812070 | 1.097812069 | 1.097812069 |
| 2.8                | 1.030719113 | 1.030719114 | 1.030719114 | 1.030719114 |
| 3.0                | 0.967523147 | 0.967523152 | 0.967523153 | 0.967523153 |

| Runge–Kutta semilinear method |             |             |             |             |
|-------------------------------|-------------|-------------|-------------|-------------|
| $x$                           | $h = 0.2$   | $h = 0.1$   | $h = 0.05$  | “Exact”     |
| 1.0                           | 2.000000000 | 2.000000000 | 2.000000000 | 2.000000000 |
| 1.2                           | 1.801636785 | 1.801636775 | 1.801636774 | 1.801636774 |
| 1.4                           | 1.650102633 | 1.650102617 | 1.650102616 | 1.650102616 |
| 1.6                           | 1.526935904 | 1.526935886 | 1.526935885 | 1.526935885 |
| 1.8                           | 1.422074302 | 1.422074284 | 1.422074283 | 1.422074283 |
| 2.0                           | 1.329664970 | 1.329664954 | 1.329664953 | 1.329664953 |
| 2.2                           | 1.246155357 | 1.246155345 | 1.246155344 | 1.246155344 |
| 2.4                           | 1.169334355 | 1.169334347 | 1.169334346 | 1.169334346 |
| 2.6                           | 1.097812074 | 1.097812070 | 1.097812069 | 1.097812069 |
| 2.8                           | 1.030719113 | 1.030719114 | 1.030719114 | 1.030719114 |
| 3.0                           | 0.967523147 | 0.967523152 | 0.967523153 | 0.967523153 |

3.3.18.

| Runge–Kutta method |             |             |             |             |
|--------------------|-------------|-------------|-------------|-------------|
| $x$                | $h = 0.2$   | $h = 0.1$   | $h = 0.05$  | “Exact”     |
| 0.0                | 1.000000000 | 1.000000000 | 1.000000000 | 1.000000000 |
| 0.2                | 1.171515610 | 1.171515156 | 1.171515152 | 1.171515153 |
| 0.4                | 1.263365845 | 1.263370556 | 1.263370869 | 1.263370891 |
| 0.6                | 1.271238957 | 1.271250529 | 1.271251232 | 1.271251278 |
| 0.8                | 1.218885528 | 1.218900353 | 1.218901230 | 1.218901287 |
| 1.0                | 1.135346772 | 1.135361174 | 1.135362016 | 1.135362070 |
| 1.2                | 1.042049558 | 1.042061864 | 1.042062579 | 1.042062625 |
| 1.4                | 0.951181964 | 0.951191920 | 0.951192495 | 0.951192532 |
| 1.6                | 0.868372923 | 0.868380842 | 0.868381298 | 0.868381328 |
| 1.8                | 0.795511927 | 0.795518241 | 0.795518603 | 0.795518627 |
| 2.0                | 0.732633229 | 0.732638318 | 0.732638609 | 0.732638628 |

| Runge–Kutta semilinear method |             |             |             |             |
|-------------------------------|-------------|-------------|-------------|-------------|
| $x$                           | $h = 0.2$   | $h = 0.1$   | $h = 0.05$  | “Exact”     |
| 0.0                           | 1.000000000 | 1.000000000 | 1.000000000 | 1.000000000 |
| 0.2                           | 1.171517316 | 1.171515284 | 1.171515161 | 1.171515153 |
| 0.4                           | 1.263374485 | 1.263371110 | 1.263370904 | 1.263370891 |
| 0.6                           | 1.271254636 | 1.271251485 | 1.271251291 | 1.271251278 |
| 0.8                           | 1.218903802 | 1.218901442 | 1.218901297 | 1.218901287 |
| 1.0                           | 1.135363869 | 1.135362181 | 1.135362077 | 1.135362070 |
| 1.2                           | 1.042063952 | 1.042062706 | 1.042062630 | 1.042062625 |
| 1.4                           | 0.951193560 | 0.951192595 | 0.951192536 | 0.951192532 |
| 1.6                           | 0.868382157 | 0.868381378 | 0.868381331 | 0.868381328 |
| 1.8                           | 0.795519315 | 0.795518669 | 0.795518629 | 0.795518627 |
| 2.0                           | 0.732639212 | 0.732638663 | 0.732638630 | 0.732638628 |

3.3.20.

| Runge–Kutta method |             |             |             |             |
|--------------------|-------------|-------------|-------------|-------------|
| $x$                | $h = 0.1$   | $h = 0.05$  | $h = 0.025$ | “Exact”     |
| 0.0                | 1.000000000 | 1.000000000 | 1.000000000 | 1.000000000 |
| 0.1                | 0.746430962 | 0.746418992 | 0.746418376 | 0.746418339 |
| 0.2                | 0.561761987 | 0.561743921 | 0.561742975 | 0.561742917 |
| 0.3                | 0.423746057 | 0.423725371 | 0.423724274 | 0.423724207 |
| 0.4                | 0.319489811 | 0.319468612 | 0.319467478 | 0.319467408 |
| 0.5                | 0.240486460 | 0.240466046 | 0.240464947 | 0.240464879 |
| 0.6                | 0.180646105 | 0.180627244 | 0.180626225 | 0.180626161 |
| 0.7                | 0.135412569 | 0.135395665 | 0.135394749 | 0.135394692 |
| 0.8                | 0.101308709 | 0.101293911 | 0.101293107 | 0.101293057 |
| 0.9                | 0.075663769 | 0.075651059 | 0.075650367 | 0.075650324 |
| 1.0                | 0.056426886 | 0.056416137 | 0.056415552 | 0.056415515 |

| Runge–Kutta semilinear method |             |             |             |             |
|-------------------------------|-------------|-------------|-------------|-------------|
| $x$                           | $h = 0.1$   | $h = 0.05$  | $h = 0.025$ | “Exact”     |
| 0.0                           | 1.000000000 | 1.000000000 | 1.000000000 | 1.000000000 |
| 0.1                           | 0.746416306 | 0.746418217 | 0.746418332 | 0.746418339 |
| 0.2                           | 0.561740647 | 0.561742780 | 0.561742908 | 0.561742917 |
| 0.3                           | 0.423722193 | 0.423724084 | 0.423724199 | 0.423724207 |
| 0.4                           | 0.319465760 | 0.319467308 | 0.319467402 | 0.319467408 |
| 0.5                           | 0.240463579 | 0.240464800 | 0.240464874 | 0.240464879 |
| 0.6                           | 0.180625156 | 0.180626100 | 0.180626158 | 0.180626161 |
| 0.7                           | 0.135393924 | 0.135394645 | 0.135394689 | 0.135394692 |
| 0.8                           | 0.101292474 | 0.101293021 | 0.101293055 | 0.101293057 |
| 0.9                           | 0.075649884 | 0.075650297 | 0.075650322 | 0.075650324 |
| 1.0                           | 0.056415185 | 0.056415495 | 0.056415514 | 0.056415515 |

3.3.22.

| Runge–Kutta method |             |             |             |             |
|--------------------|-------------|-------------|-------------|-------------|
| $x$                | $h = 0.1$   | $h = 0.05$  | $h = 0.025$ | “Exact”     |
| 2.0                | 1.000000000 | 1.000000000 | 1.000000000 | 1.000000000 |
| 2.1                | 1.008912398 | 1.008900636 | 1.008900025 | 1.008899988 |
| 2.2                | 1.031740789 | 1.031724368 | 1.031723520 | 1.031723469 |
| 2.3                | 1.063781819 | 1.063765150 | 1.063764295 | 1.063764243 |
| 2.4                | 1.101630085 | 1.101615517 | 1.101614774 | 1.101614730 |
| 2.5                | 1.142915917 | 1.142904393 | 1.142903811 | 1.142903776 |
| 2.6                | 1.186047678 | 1.186039295 | 1.186038876 | 1.186038851 |
| 2.7                | 1.229991054 | 1.229985469 | 1.229985194 | 1.229985178 |
| 2.8                | 1.274095992 | 1.274092692 | 1.274092535 | 1.274092525 |
| 2.9                | 1.317969153 | 1.317967605 | 1.317967537 | 1.317967533 |
| 3.0                | 1.361384082 | 1.361383812 | 1.361383809 | 1.361383810 |

| Runge-Kutta semilinear method |             |             |             |             |
|-------------------------------|-------------|-------------|-------------|-------------|
| $x$                           | $h = 0.1$   | $h = 0.05$  | $h = 0.025$ | "Exact"     |
| 2.0                           | 1.000000000 | 1.000000000 | 1.000000000 | 1.000000000 |
| 2.1                           | 1.008913934 | 1.008900843 | 1.008900041 | 1.008899988 |
| 2.2                           | 1.031748526 | 1.031725001 | 1.031723564 | 1.031723469 |
| 2.3                           | 1.063798300 | 1.063766321 | 1.063764371 | 1.063764243 |
| 2.4                           | 1.101656264 | 1.101617259 | 1.101614886 | 1.101614730 |
| 2.5                           | 1.142951721 | 1.142906691 | 1.142903955 | 1.142903776 |
| 2.6                           | 1.186092475 | 1.186042105 | 1.186039051 | 1.186038851 |
| 2.7                           | 1.230043983 | 1.229988742 | 1.229985397 | 1.229985178 |
| 2.8                           | 1.274156172 | 1.274096377 | 1.274092762 | 1.274092525 |
| 2.9                           | 1.318035787 | 1.317971658 | 1.317967787 | 1.317967533 |
| 3.0                           | 1.361456502 | 1.361388196 | 1.361384079 | 1.361383810 |

|                |      |              |              |              |              |
|----------------|------|--------------|--------------|--------------|--------------|
| <b>3.3.24.</b> | $x$  | $h = .1$     | $h = .05$    | $h = .025$   | Exact        |
|                | 1.00 | 0.142854841  | 0.142857001  | 0.142857134  | 0.142857143  |
|                | 1.10 | 0.053340745  | 0.053341989  | 0.053342066  | 0.053342071  |
|                | 1.20 | -0.046154629 | -0.046153895 | -0.046153849 | -0.046153846 |
|                | 1.30 | -0.153363206 | -0.153362764 | -0.153362736 | -0.153362734 |
|                | 1.40 | -0.266397049 | -0.266396779 | -0.266396762 | -0.266396761 |
|                | 1.50 | -0.383721107 | -0.383720941 | -0.383720931 | -0.383720930 |
|                | 1.60 | -0.504109696 | -0.504109596 | -0.504109589 | -0.504109589 |
|                | 1.70 | -0.626598326 | -0.626598268 | -0.626598264 | -0.626598264 |
|                | 1.80 | -0.750437351 | -0.750437320 | -0.750437318 | -0.750437318 |
|                | 1.90 | -0.875050587 | -0.875050574 | -0.875050573 | -0.875050573 |
|                | 2.00 | -1.000000000 | -1.000000000 | -1.000000000 | -1.000000000 |
| <b>3.3.26.</b> | $x$  | $h = .1$     | $h = .05$    | $h = .025$   | Exact        |
|                | 0.50 | -8.954103230 | -8.954063245 | -8.954060698 | -8.954060528 |
|                | 0.60 | -5.059648314 | -5.059633293 | -5.059632341 | -5.059632277 |
|                | 0.70 | -2.516755942 | -2.516749850 | -2.516749465 | -2.516749439 |
|                | 0.80 | -0.752508672 | -0.752506238 | -0.752506084 | -0.752506074 |
|                | 0.90 | 0.530528482  | 0.530529270  | 0.530529319  | 0.530529323  |
|                | 1.00 | 1.500000000  | 1.500000000  | 1.500000000  | 1.500000000  |
|                | 1.10 | 2.256519743  | 2.256519352  | 2.256519328  | 2.256519326  |
|                | 1.20 | 2.863543039  | 2.863542454  | 2.863542417  | 2.863542415  |
|                | 1.30 | 3.362731379  | 3.362730700  | 3.362730658  | 3.362730655  |
|                | 1.40 | 3.782361948  | 3.782361231  | 3.782361186  | 3.782361183  |
|                | 1.50 | 4.142171279  | 4.142170553  | 4.142170508  | 4.142170505  |

**3.3.28.** (a) Let  $x_i = a + ih$ ,  $i = 0, 1, \dots, n$ . If  $y$  is the solution of the initial value problem  $y' = f(x)$ ,  $y(a) = 0$ , then  $y(b) = \int_a^b f(x) dx$ . The Runge-Kutta method yields  $y_{i+1} = y_i + \frac{h}{6}(f(a + ih) + 4f(a + (2i + 1)h/2) + f(a + (i + 1)h))$ ,  $i = 0, 1, \dots, n - 1$ , where  $y_0 = a$  and  $y_n$  is an approximation to  $\int_a^b f(x) dx$ . But

$$y_n = \sum_{i=0}^{n-1} (y_{i+1} - y_i) = \frac{h}{6}(f(a) + f(b)) + \frac{h}{3} \sum_{i=1}^{n-1} f(a + ih) + \frac{2h}{3} \sum_{i=1}^n f(a + (2i - 1)h/2).$$

(c) The local truncation error is a multiple of  $y^{(5)}(\tilde{x}_i) = f^{(4)}(\tilde{x}_i)$ , where  $x_i < \tilde{x}_i < x_{i+1}$ . Therefore, the quadrature formula is exact if  $f$  is a polynomial of degree  $< 4$ .

(d) Let  $E(f) = \int_a^b f(x) dx - y_n$ . Note that  $E$  is linear. If  $f$  is a polynomial of degree 4, then  $f(x) = f_0(x) + K(x-a)^4$  where  $\deg(f_0) \leq 3$  and  $K$  is constant. Since  $E(f_0) = 0$  from (c) and

$$\begin{aligned} E((x-a)^4) &= \frac{(b-a)^5}{5} - \frac{(b-a)^4 h}{6} - \frac{h^5}{3} \sum_{i=1}^{n-1} i^4 - \frac{2h^5}{3} \sum_{i=1}^n (i-1/2)^4 \\ &= h^5 \left[ \frac{n^5}{5} - \frac{n^4}{6} - \left( \frac{n^5}{15} - \frac{n^4}{6} + \frac{n^3}{9} - \frac{n}{90} \right) - \left( \frac{2n^5}{15} - \frac{n^3}{9} + \frac{7n}{360} \right) \right] \\ &= -\frac{nh^5}{120} = -\frac{(b-a)h^4}{120}, \end{aligned}$$

$E(f) = -\frac{(b-a)h^4}{120}$ ; thus, the error is proportional to  $h^4$ .

## Chapter 4

# Applications of First Order Equations

### 4.1 Growth and Decay

**4.1.2.**  $k\tau = \ln 2$  and  $\tau = 2 \Rightarrow k = \frac{\ln 2}{2}$ ;  $Q(t) = Q_0 e^{-t \ln 2/2}$ ; if  $Q(T) = \frac{Q_0}{10}$ , then  $\frac{Q_0}{10} = Q_0 e^{-T \ln 2/2}$ ;  $\ln 10 = \frac{T \ln 2}{2}$ ;  $T = \frac{2 \ln 10}{\ln 2}$  days.

**4.1.4.** Let  $t_1$  be the elapsed time since the tree died. Since  $p(t) = e^{-(t \ln 2)\tau}$ , it follows that  $p_1 = p_0 e^{-(t_1 \ln 2)/\tau}$ , so  $\ln\left(\frac{p_1}{p_0}\right) = -\frac{t_1}{\tau} \ln 2$  and  $t_1 = \tau \frac{\ln(p_0/p_1)}{\ln 2}$ .

**4.1.6.**  $Q = Q_0 e^{-kt}$ ;  $Q_1 = Q_0 e^{-kt_1}$ ;  $Q_2 = Q_0 e^{-kt_2}$ ;  $\frac{Q_2}{Q_1} = e^{-k(t_2 - t_1)}$ ;  $\ln\left(\frac{Q_1}{Q_2}\right) = k(t_2 - t_1)$ ;  
 $k = \frac{1}{t_2 - t_1} \ln\left(\frac{Q_1}{Q_2}\right)$ .

**4.1.8.**  $Q' = .06Q$ ,  $Q(0) = Q_0$ ;  $Q = Q_0 e^{.06t}$ . We must find  $\tau$  such that  $Q(\tau) = 2Q_0$ ; that is,  $Q_0 e^{.06\tau} = 2Q_0$ , so  $.06\tau = \ln 2$  and  $\tau = \frac{\ln 2}{.06} = \frac{50 \ln 2}{3}$  yr.

**4.1.10.** (a) If  $T$  is the time to triple the value, then  $Q(T) = Q_0 e^{.05T} = 3Q_0$ , so  $e^{.05T} = 3$ . Therefore,  $.05T = \ln 3$  and  $T = 20 \ln 3$ .

(b) If  $Q(10) = 100000$ , then  $Q_0 e^{.5} = 100000$ , so  $Q_0 = 100000 e^{-.5}$

**4.1.12.**  $Q' = -\frac{Q^2}{2}$ ,  $Q(0) = 50$ ;  $\frac{Q'}{Q^2} = -\frac{1}{2}$ ;  $-\frac{1}{Q} = -\frac{t}{2} + c$ ;  $Q(0) = 50 \Rightarrow c = -\frac{1}{50}$ ;  $\frac{1}{Q} = \frac{t}{2} + \frac{1}{50} = \frac{1 + 25t}{50}$ ;  $Q = \frac{50}{1 + 25t}$ . Now  $Q(T) = 25 \Rightarrow 1 + 25T = 2 \Rightarrow 25T = 1 \Rightarrow T = \frac{1}{25}$  years.

**4.1.14.** Since  $\tau = 1500$ ,  $k = \frac{\ln 2}{1500}$ ; hence  $Q = Q_0 e^{-(t \ln 2)/1500}$ . If  $Q(t_1) = \frac{3Q_0}{4}$ , then  $e^{-(t_1 \ln 2)/1500} = \frac{3}{4}$ ;  $-t_1 \frac{\ln 2}{1500} = \ln\left(\frac{3}{4}\right) = -\ln\left(\frac{4}{3}\right)$ ;  $t_1 = 1500 \frac{\ln(4/3)}{\ln 2}$ . Finally,  $Q(2000) = Q_0 e^{-\frac{4}{3} \ln 2} = 2^{-4/3} Q_0$ .



**4.1.16.** (A)  $S' = 1 - \frac{S}{10}$ ,  $S(0) = 20$ . Rewrite the differential equation in (A) as (B)  $S' + \frac{S}{10} = 1$ . Since  $S_1 = e^{-t/10}$  is a solution of the complementary equation, the solutions of (B) are given by  $S = ue^{-t/10}$ , where  $u'e^{-t/10} = 1$ . Therefore,  $u' = e^{t/10}$ ;  $u = 10e^{t/10} + c$ ;  $S = 10 + ce^{-t/10}$ . Now  $S(0) = 20 \Rightarrow c = 10$ , so  $S = 10 + 10e^{-t/10}$  and  $\lim_{t \rightarrow \infty} S(t) = 10$  g.

**4.1.18.** (A)  $V' = -750 + \frac{V}{20}$ ,  $V(0) = 25000$ . Rewrite the differential equation in (A) as (B)  $V' - \frac{V}{20} = -750$ . Since  $V_1 = e^{t/20}$  is a solution of the complementary equation, the solutions of (B) are given by  $V = ue^{t/20}$ , where  $u'e^{t/20} = -750$ . Therefore,  $u' = -750e^{-t/20}$ ;  $u = 15000e^{-t/20} + c$ ;  $V = 15000 + ce^{t/20}$ ;  $V(0) = 25000 \Rightarrow c = 10000$ . Therefore,  $V = 15000 + 10000e^{t/20}$ .

**4.1.20.**  $p' = \frac{p}{2} - \frac{p^2}{8} = -\frac{1}{8}p(p-4)$ ;  $\frac{p'}{p(p-4)} = -\frac{1}{8}$ ;  $\frac{1}{4} \left[ \frac{1}{p-4} - \frac{1}{p} \right] p' = -\frac{1}{8}$ ;  $\left[ \frac{1}{p-4} - \frac{1}{p} \right] p' = -\frac{1}{2}$ ;  $\left| \frac{p-4}{p} \right| = -\frac{t}{2} + k$ ;  $\frac{p-4}{p} = ce^{-t/2}$ ;  $p(0) = 100 \Rightarrow c = \frac{24}{25}$ ;  $\frac{p-4}{p} = \frac{24}{25}e^{-t/2}$ ;  $p-4 = \frac{24}{25}pe^{-t/2}$ ;  $p \left( 1 - \frac{24}{25}e^{-t/2} \right) = 4$ ;  $p = \frac{4}{1 - \frac{24}{25}e^{-t/2}} = \frac{100}{24 - 24e^{-t/2}}$ .

**4.1.22.** (a)  $P' = rP - 12M$ .

(b)  $P = ue^{rt}$ ;  $u'e^{rt} = -12M$ ;  $u' = -12Me^{rt}$ ;  $u = \frac{12M}{r}e^{-rt} + c$ ;  $P = \frac{12M}{r} + ce^{rt}$ ;  $P(0) = P_0 \Rightarrow c = P_0 - \frac{12M}{r}$ ;  $P = \frac{12M}{r}(1 - e^{rt}) + P_0e^{rt}$ .

(c) Since  $P(N) = 0$ , the answer to (b) implies that  $M = \frac{rP_0}{12(1 - e^{-rN})}$ .

**4.1.24.** The researcher's salary is the solution of the initial value problem  $S' = aS$ ,  $S(0) = S_0$ . Therefore,  $S = S_0e^{at}$ . If  $P = P(t)$  is the value of the trust fund, then  $P' = -S_0e^{at} + rP$ , or  $P' - rP = -S_0e^{at}$ . Therefore, (A)  $P = ue^{rt}$ , where  $u'e^{rt} = -S_0e^{at}$ , so (B)  $u' = -S_0e^{(a-r)t}$ . If  $a \neq r$ , then (B) implies that  $u = \frac{S_0}{r-a}e^{(a-r)t} + c$ , so (A) implies that  $P = \frac{S_0}{r-a}e^{at} + ce^{rt}$ . Now  $P(0) = P_0 \Rightarrow c = P_0 - \frac{S_0}{r-a}$ ; therefore  $P = \frac{S_0}{r-a}e^{at} + \left( P_0 - \frac{S_0}{r-a} \right) e^{rt}$ . We must choose  $P_0$  so that  $P(T) = 0$ ; that is,  $P = \frac{S_0}{r-a}e^{aT} + \left( P_0 - \frac{S_0}{r-a} \right) e^{rT} = 0$ . Solving this for  $P_0$  yields  $P_0 = \frac{S_0(1 - e^{(a-r)T})}{r-a}$ . If  $a = r$ , then (B) becomes  $u' = -S_0$ , so  $u = -S_0t + c$  and (A) implies that  $P = (-S_0t + c)e^{rt}$ . Now  $P(0) = P_0 \Rightarrow c = P_0$ ; therefore  $P = (-S_0t + P_0)e^{rt}$ . To make  $P(T) = 0$  we must take  $P_0 = S_0T$ .

**4.1.26.**  $Q' = \frac{at}{1 + btQ^2} - kQ$ ;  $\lim_{t \rightarrow \infty} Q(t) = (a/bk)^{1/3}$ .

## 4.2 Cooling and Mixing

**4.2.2.** Since  $T_0 = 100$  and  $T_M = -10$ ,  $T = -10 + 110e^{-kt}$ . Now  $T(1) = 80 \Rightarrow 80 = -10 + 110e^{-k}$ , so  $e^{-k} = \frac{9}{11}$  and  $k = \ln \frac{11}{9}$ . Therefore,  $T = -10 + 110e^{-t \ln \frac{11}{9}}$ .

**4.2.4.** Let  $T$  be the thermometer reading. Since  $T_0 = 212$  and  $T_M = 70$ ,  $T = 70 + 142e^{-kt}$ . Now  $T(2) = 125 \Rightarrow 125 = 70 + 142e^{-2k}$ , so  $e^{-2k} = \frac{55}{142}$  and  $k = \frac{1}{2} \ln \frac{142}{55}$ . Therefore, (A)

$$T = 70 + 142e^{-\frac{t}{2} \ln \frac{142}{55}}.$$

$$(a) T(2) = 70 + 142e^{-2 \ln \frac{142}{55}} = 70 + 142\left(\frac{55}{142}\right)^2 \approx 91.30^\circ\text{F}.$$

$$(b) \text{ Let } \tau \text{ be the time when } T(\tau) = 72, \text{ so } 72 = 70 + 142e^{-\frac{\tau}{2} \ln \frac{142}{55}}, \text{ or } e^{-\frac{\tau}{2} \ln \frac{142}{55}} = \frac{1}{71}. \text{ Therefore,}$$

$$\tau = 2 \frac{\ln 71}{\ln \frac{142}{55}} \approx 8.99 \text{ min.}$$

(c) Since (A) implies that  $T > 70$  for all  $t > 0$ , the thermometer will never read  $69^\circ\text{F}$ .

**4.2.6.** Since  $T_M = 20$ ,  $T = 20 + (T_0 - 20)e^{-kt}$ . Now  $T_0 - 5 = 20 + (T_0 - 20)e^{-4k}$  and  $T_0 - 7 = 20 + (T_0 - 20)e^{-8k}$ . Therefore,  $\frac{T_0 - 25}{T_0 - 20} = e^{-4k}$  and  $\left(\frac{T_0 - 27}{T_0 - 20}\right) = e^{-8k}$ , so  $\frac{T_0 - 27}{T_0 - 20} = \left(\frac{T_0 - 25}{T_0 - 20}\right)^2$ , which implies that  $(T_0 - 20)(T_0 - 27) = (T_0 - 25)^2$ , or  $T_0^2 - 47T_0 + 540 = T_0^2 - 50T_0 + 625$ ; hence  $3T_0 = 85$  and  $T_0 = (85/3)^\circ\text{C}$ .

**4.2.8.**  $Q' = 3 - \frac{3}{40}Q$ ,  $Q(0) = 0$ . Rewrite the differential equation as (A)  $Q' + \frac{3}{40}Q = 3$ . Since  $Q_1 = e^{-3t/40}$  is a solution of the complementary equation, the solutions of (A) are given by  $Q = ue^{-3t/40}$  where  $u'e^{-3t/40} = 3$ . Therefore,  $u' = 3e^{3t/40}$ ,  $u = 40e^{3t/40} + c$ , and  $Q = 40 + ce^{-3t/40}$ . Now  $Q(0) = 0 \Rightarrow c = -40$ , so  $Q = 40(1 - e^{-3t/40})$ .

**4.2.10.**  $Q' = \frac{3}{2} - \frac{Q}{20}$ ,  $Q(0) = 10$ . Rewrite the differential equation as (A)  $Q' + \frac{Q}{20} = \frac{3}{2}$ . Since  $Q_1 = e^{-t/20}$  is a solution of the complementary equation, the solutions of (A) are given by  $Q = ue^{-t/20}$  where  $u'e^{-t/20} = \frac{3}{2}$ . Therefore,  $u' = \frac{3}{2}e^{t/20}$ ,  $u = 30e^{t/20} + c$ , and  $Q = 30 + ce^{-t/20}$ . Now  $Q(0) = 10 \Rightarrow c = -20$ , so  $Q = 30 - 20e^{-t/20}$  and  $K = \frac{Q}{100} = .3 - .2e^{-t/20}$ .

**4.2.12.**  $Q' = 10 - \frac{Q}{5}$ , or (A)  $Q' + \frac{Q}{5} = 10$ . Since  $Q_1 = e^{-t/5}$  is a solution of the complementary equation, the solutions of (A) are given by  $Q = ue^{-t/5}$  where  $u'e^{-t/5} = 10$ . Therefore,  $u' = 10e^{t/5}$ ,  $u = 50e^{t/5} + c$ , and  $Q = 50 + ce^{-t/5}$ . Since  $\lim_{t \rightarrow \infty} Q(t) = 50$ , the minimum capacity is 50 gallons.

**4.2.14.** Since there are  $2t + 600$  gallons of mixture in the tank at time  $t$  and mixture is being drained at 4 gallons/min,  $Q' = 3 - \frac{2}{t+300}Q$ ,  $Q(0) = 40$ . Rewrite the differential equation as (A)  $Q' + \frac{2}{t+300}Q = 3$ . Since  $Q_1 = \frac{1}{(t+300)^2}$  is a solution of the complementary equation, the solutions of (A) are given by  $Q = \frac{u}{(t+300)^2}$  where  $\frac{u'}{(t+300)^2} = 3$ . Therefore,  $u' = 3(t+300)^2$ ,  $u = (t+300)^3 + c$ , and  $Q = t + 300 + \frac{c}{(t+300)^2}$ . Now  $Q(0) = 40 \Rightarrow c = -234 \times 10^5$ , so

$$Q = t + 300 - \frac{234 \times 10^5}{(t+300)^2}, \quad 0 \leq t \leq 300.$$

**4.2.16. (a)**  $S' = -k_m(S - T_m)$ ,  $S(0) = 0$ , so (A)  $S = T_m + (S_0 - T_m)e^{-k_mt}$ .  $T' = -k(T - S) = -k(T - T_m - (S_0 - T_m)e^{-k_mt})$ , from (A). Therefore,  $T' + kT = kT_m + k(S_0 - T_m)e^{-k_mt}$ ;  $T = ue^{-kt}$ ;

(B)  $u' = kT_me^{kt} + k(S_0 - T_m)e^{(k-k_m)t}$ ;  $u = T_me^{kt} + \frac{k}{k-k_m}(S_0 - T_m)e^{(k-k_m)t} + c$ ;  $T(0) = T_0 \Rightarrow c = T_0 - T_m - \frac{k}{k-k_m}(S_0 - T_m)$ ;  $u = T_me^{kt} + \frac{k}{k-k_m}(S_0 - T_m)e^{(k-k_m)t} + T_0 - T_m - \frac{k}{k-k_m}(S_0 - T_m)$ ;

$$T = T_m + (T_0 - T_m)e^{-kt} + \frac{k(S_0 - T_m)}{(k - k_m)}(e^{-k_m t} - e^{-kt}).$$

(b) If  $k = k_m$  (B) becomes (B)  $u' = kT_me^{kt} + k(S_0 - T_m)$ ;  $u = T_me^{kt} + k(S_0 - T_m)t + c$ ;  $T(0) = T_0 \Rightarrow c = T_0 - T_m$ ;  $u = T_me^{kt} + k(S_0 - T_m)t + (T_0 - T_m)$ ;  $T = T_m + k(S_0 - T_m)te^{-kt} + (T_0 - T_m)e^{-kt}$ .

(c)  $\lim_{t \rightarrow \infty} T(t) = \lim_{t \rightarrow \infty} S(t) = T_m$  in either case.

**4.2.18.**  $V' = aV - bV^2 = -bV(V - b/a)$ ;  $\frac{V'}{V(V - b/a)} = -b$ ;  $\left[ \frac{1}{V - b/a} - \frac{1}{V} \right] V' = -a$ ;  $\ln \left| \frac{V - b/a}{V} \right| = -at + k$ ; (A)  $\frac{V - b/a}{V} = ce^{-at}$ ; (B)  $V = \frac{a}{b} \frac{1}{1 - ce^{-at}}$ . Since  $V(0) = V_0$ , (A)  $\Rightarrow c = \frac{V_0 - a/b}{V_0}$ . Substituting this into (B) yields  $V = \frac{a}{b} \frac{V_0}{V_0 - (V_0 - a/b)e^{-at}}$  so  $\lim_{t \rightarrow \infty} V(t) = a/b$ .

**4.2.20.** If  $Q_n(t)$  is the number of pounds of salt in  $T_n$  at time  $t$ , then  $Q'_{n+1} + \frac{r}{W}Q_{n+1} = rc_n(t)$ ,  $n = 0, 1, \dots$ , where  $c_0(t) \equiv c$ . Therefore,  $Q_{n+1} = u_{n+1}e^{-rt/W}$ ; (A)  $u'_{n+1} = re^{rt/W}c_n(t)$ . In particular, with  $n = 0$ ,  $u_1 = cW(e^{rt/W} - 1)$ , so  $Q_1 = cW(1 - e^{-rt/W})$  and  $c_1 = c(1 - e^{-rt/W})$ . We will show by induction that  $c_n = c \left( 1 - e^{-rt/W} \sum_{j=0}^{n-1} \frac{1}{j!} \left( \frac{rt}{W} \right)^j \right)$ . This is true for  $n = 1$ ; if it is true for a given  $n$ , then, from (A),

$$u'_{n+1} = cre^{rt/W} \left( 1 - e^{-rt/W} \sum_{j=0}^{n-1} \frac{1}{j!} \left( \frac{rt}{W} \right)^j \right) = cre^{rt/W} - cr \sum_{j=0}^{n-1} \frac{1}{j!} \left( \frac{rt}{W} \right)^j,$$

so (since  $Q_{n+1}(0) = 0$ ),

$$u_{n+1} = cW(e^{rt/W} - 1) - c \sum_{j=0}^{n-1} \frac{1}{(j+1)!} \frac{r^{j+1}}{W^j} t^{j+1}.$$

Therefore,

$$c_{n+1} = \frac{1}{W}u_{n+1}e^{-rt/W} = c \left( 1 - e^{-rt/W} \sum_{j=0}^n \frac{1}{j!} \left( \frac{rt}{W} \right)^j \right),$$

which completes the induction. From this,  $\lim_{t \rightarrow \infty} c_n(t) = c$ .

**4.2.22.** Since the incoming solution contains 1/2 lb of salt per gallon and there are always 600 gallons in the tank, we conclude intuitively that  $\lim_{t \rightarrow \infty} Q(t) = 300$ . To verify this rigorously, note that  $Q_1(t) = \exp \left( -\frac{1}{150} \int_0^t a(\tau) d\tau \right)$  is a solution of the complementary equation, (A)  $Q_1(0) = 1$ , and (B)  $\lim_{t \rightarrow \infty} Q_1(t) = 0$  (since  $\lim_{t \rightarrow \infty} a(t) = 1$ ). Therefore,  $Q = Q_1u$ ;  $Q_1u' = 2$ ;  $u' = \frac{2}{Q_1}$ ;  $u = Q_0 + 2 \int_0^t \frac{d\tau}{Q_1(\tau)}$  (see (A)), and  $Q(t) = Q_0Q_1(t) + 2Q_1(t) \int_0^t \frac{d\tau}{Q_1(\tau)}$ .

From (B),  $\lim_{t \rightarrow \infty} Q(t) = 2 \lim_{t \rightarrow \infty} Q_1(t) \int_0^t \frac{d\tau}{Q_1(\tau)}$ , a  $0 \cdot \infty$  indeterminate form. By L'Hospital's rule,

$$\lim_{t \rightarrow \infty} Q(t) = 2 \lim_{t \rightarrow \infty} \frac{1}{Q_1(t)} \bigg/ \left( \frac{-Q'_1(t)}{Q_1^2(t)} \right) = -2 \lim_{t \rightarrow \infty} \frac{Q_1(t)}{Q'_1(t)} = 300.$$

### 4.3 Elementary Mechanics

**4.3.2.** The firefighter's mass is  $m = \frac{192}{32} = 6$  sl, so  $6v' = -192 - kv$ , or (A)  $v' + \frac{k}{6}v = -32$ . Since  $v_1 = e^{-kt/6}$  is a solution of the complementary equation, the solutions of (A) are  $v = ue^{-kt/6}$  where  $u'e^{-kt/6} = -32$ . Therefore,  $u' = -32e^{kt/6}$ ;  $u = -\frac{192}{k}e^{kt/6} + c$ ;  $v = -\frac{192}{k} + ce^{-kt/6}$ . Now  $v(0) = 0 \Rightarrow c = \frac{192}{k}$ . Therefore,  $v = -\frac{192}{k}(1 - e^{-kt/6})$  and  $\lim_{t \rightarrow \infty} v(t) = -\frac{192}{k} = -16$  ft/s, so  $k = 12$  lb-s/ft and  $v = -16(1 - e^{-2t})$ .

**4.3.3.**  $m = \frac{64000}{32} = 2000$ , so  $2000v' = 50000 - 2000v$ , or (A)  $v' + v = 25$ . Since  $v_1 = e^{-t}$  is a solution of the complementary equation, the solutions of (A) are  $v = ue^{-t}$  where  $u'e^{-t} = 25$ . Therefore,  $u' = 25e^t$ ;  $u = 25e^t + c$ ;  $v = 25 + ce^{-t}$ . Now  $v(0) = 0 \Rightarrow c = -25$ . Therefore,  $v = 25(1 - e^{-t})$  and  $\lim_{t \rightarrow \infty} v(t) = 25$  ft/s.

**4.3.4.**  $20v' = 10 - \frac{1}{2}v$ , or (A)  $v' + \frac{1}{20}v = \frac{1}{2}$ . Since  $v_1 = e^{-t/40}$  is a solution of the complementary equation, the solutions of (A) are  $v = ue^{-t/40}$  where  $u'e^{-t/40} = \frac{1}{2}$ . Therefore,  $u' = \frac{e^{t/40}}{2}$ ;  $u = 20e^{t/40} + c$ ;  $v = 20 + ce^{-t/40}$ . Now  $v(0) = -7 \Rightarrow c = -27$ . Therefore,  $v = 20 - 27e^{-t/40}$ .

**4.3.6.**  $m = \frac{3200}{32} = 100$  sl. The component of the gravitational force in the direction of motion is  $-3200 \cos(\pi/3) = -1600$  lb. Therefore,  $100v' = -1600 + v^2$ . Separating variables yields  $\frac{v'}{(v-40)(v+40)} = \frac{1}{100}$ , or  $\left[ \frac{1}{v-40} - \frac{1}{v+40} \right] = \frac{4}{5}$ . Therefore,  $\ln \left| \frac{v-40}{v+40} \right| = \frac{4t}{5} + k$  and  $\frac{v-40}{v+40} = ce^{4t/5}$ . Now  $v(0) = -64 \Rightarrow c = \frac{13}{3}$ ; therefore  $\frac{v-40}{v+40} = \frac{13e^{4t/5}}{3}$ , so  $v = \frac{40(3 + 13e^{4t/5})}{3 - 13e^{4t/5}}$ , or  $v = -\frac{40(13 + 3e^{-4t/5})}{13 - 3e^{-4t/5}}$ .

**4.3.8.** From Example 4.3.1, (A)  $v = -\frac{mg}{k} + \left(v_0 + \frac{mg}{k}\right)e^{-kt/m}$ . Integrating this yields (B)  $y = -\frac{mgt}{k} - \frac{m}{k} \left(v_0 + \frac{mg}{k}\right)e^{-kt/m} + c$ . Now  $y(0) = y_0 \Rightarrow c = y_0 + \frac{m}{k} \left(v_0 + \frac{mg}{k}\right)$ . Substituting this into (B) yields

$$\begin{aligned} y &= -\frac{mgt}{k} - \frac{m}{k} \left(v_0 + \frac{mg}{k}\right)e^{-kt/m} + y_0 + \frac{m}{k} \left(v_0 + \frac{mg}{k}\right) \\ &= y_0 + \frac{m}{k} \left(v_0 - gt + \frac{mg}{k} - \left(v_0 + \frac{mg}{k}\right)e^{-kt/m}\right) \\ &= y_0 + \frac{m}{k}(v_0 - v - gt) \end{aligned}$$

where the last equality follows from (A).

**4.3.10.**  $m = \frac{256}{32} = 8$  sl. Since the resisting force is 1 lb when  $|v| = 4$  ft/s,  $k = \frac{1}{16}$ . Therefore,  $8v' = -256 + \frac{1}{16}v^2 = \frac{1}{16}(v^2 - (64)^2)$ . Separating variables yields  $\frac{v'}{(v-64)(v+64)} = \frac{1}{128}$ , or  $\left[ \frac{1}{v-64} - \frac{1}{v+64} \right]v' = 1$ . Therefore,  $\ln \left| \frac{v-64}{v+64} \right| = t + k$  and  $\frac{v-64}{v+64} = ce^t$ . Now  $v(0) = 0 \Rightarrow c = -1$ ;

therefore  $\frac{v-64}{v+64} = -e^t$ , so  $v = \frac{64(1-e^t)}{1+e^t}$ , or  $v = -\frac{64(1-e^{-t})}{1+e^{-t}}$ . Therefore,  $\lim_{t \rightarrow \infty} v(t) = -64$ .

**4.3.12. (a)**  $mv' = -mg - kv^2 = -mg(1 + \gamma^2 v^2)$ , where  $\gamma = \sqrt{\frac{k}{mg}}$ . Therefore, (A)  $\frac{v'}{1 + \gamma^2 v^2} = -g$ .

With the substitution  $u = \gamma v$ ,  $\int \frac{dv}{1 + \gamma^2 v^2} = \frac{1}{\gamma} \int \frac{du}{1 + u^2} = \frac{1}{\gamma} \tan^{-1} u = \frac{1}{\gamma} \tan^{-1}(\gamma v)$ . Therefore,  $\frac{1}{\gamma} \tan^{-1}(\gamma v) = -gt + c$ . Now  $v(0) = v_0 \Rightarrow c = \frac{1}{\gamma} \tan^{-1}(\gamma v_0)$ , so  $\frac{1}{\gamma} \tan^{-1}(\gamma v) = -gt + \frac{1}{\gamma} \tan^{-1}(\gamma v_0)$ .

Since  $v(T) = 0$ , it follows that  $T = \frac{1}{\gamma g} \tan^{-1} \gamma v_0 = \sqrt{\frac{m}{kg}} \tan^{-1} \left( v_0 \sqrt{\frac{k}{mg}} \right)$ .

(b) Replacing  $t$  by  $t - T$  and setting  $v_0 = 0$  in the answer to the previous exercise yields  $v = -\sqrt{\frac{mg}{k}} \frac{1 - e^{-2\sqrt{\frac{gk}{m}}(t-T)}}{1 + e^{-2\sqrt{\frac{gk}{m}}(t-T)}}$ .

**4.3.14. (a)**  $mv' = -mg + f(|v|)$ ; since  $s = |v| = -v$ , (A)  $ms' = mg - f(s)$ .

(b) Since  $f$  is increasing and  $\lim_{t \rightarrow \infty} f(s) \leq mg$ ,  $mg - f(s) > 0$  for all  $s$ . This and (A) imply that  $s$  is an increasing function of  $t$ , so either (B)  $\lim_{t \rightarrow \infty} s(t) = \infty$  or (C)  $\lim_{t \rightarrow \infty} s(t) = \bar{s} < \infty$ . However, (A) and (C) imply that  $s'(t) > K = g - f(\bar{s})/m$  for all  $t > 0$ . Consequently,  $s(t) > s_0 + Kt$  for all  $t > 0$ , which contradicts (C) because  $K > 0$ .

(c) There is a unique positive number  $s_T$  such that  $f(s_T) = mg$ , and  $s \equiv s_T$  is a constant solution of (A). Now suppose that  $s(0) < s_T$ . Then Theorem 2.3.1 implies that (D)  $s(t) < s_T$  for all  $t > 0$ , so (A) implies that  $s$  is strictly increasing. This and (D) imply that  $\lim_{t \rightarrow \infty} s(t) = \bar{s} \leq s_T$ . If  $\bar{s} < s_T$  then (A) implies that  $s'(t) > K = g - f(\bar{s})/m$ . Consequently,  $s(t) > s(0) + Kt$ , which contradicts (D) because  $K > 0$ . Therefore,  $s(0) < s_T \Rightarrow \lim_{t \rightarrow \infty} s(t) = s_T$ . A similar proof with inequalities reversed shows that  $s(0) > s_T \Rightarrow \lim_{t \rightarrow \infty} s(t) = s_T$ .

**4.3.16. (a)** (A)  $mv' = -mg + k\sqrt{|v|}$ ; since the magnitude of the resistance is 64 lb when  $v = 16$  ft/s,  $4k = 64$ , so  $k = 16 \text{ lb} \cdot \text{s}^{1/2}/\text{ft}^{1/2}$ . Since  $m = 2$  and  $g = 32$ , (A) becomes  $2v' = -64 + 16\sqrt{|v|}$ , or  $v' = -32 + 8\sqrt{|v|}$ .

(b) From Exercise 4.3.14(c),  $v_T$  is the negative number such that  $-32 + 8\sqrt{|v_T|} = 0$ ; thus,  $v_T = -16$  ft/s.

**4.3.18.** With  $h = 0$ ,  $v_e = \sqrt{2gR}$ , where  $R$  is the radius of the moon and  $g$  is the acceleration due to gravity at the moon's surface. With length in miles,  $g = \frac{5.31}{5280} \text{ mi/s}^2$ , so  $v_e = \sqrt{\frac{2 \cdot 5.31 \cdot 1080}{5280}} \approx 1.47$  miles/s.

**4.3.20.** Suppose that there is a number  $y_m$  such that  $y(t) \leq y_m$  for all  $t \geq 0$  and let  $\alpha = \frac{gR^2}{(y_m + R)^2}$ .

Then  $\frac{d^2 y}{dt^2} \leq -\alpha$  for all  $t \geq 0$ . Integrating this inequality from  $t = 0$  to  $t = T > 0$  yields  $v(T) - v_0 \leq -\alpha T$ , or  $v(T) \leq v_0 - \alpha T$ , so  $v(T) < 0$  for  $T > \frac{v_0}{\alpha}$ . This implies that the vehicle must eventually fall back to Earth, which contradicts the assumption that it continues to climb forever.

## 4.4 Autonomous Second Order Equations

**4.4.1.**  $\bar{y} = 0$  is a stable equilibrium. The phase plane equivalent is  $v \frac{dv}{dy} + y^3 = 0$ , so the trajectories are  $v^2 + \frac{y^4}{4} = c$ .

**4.4.2.**  $\bar{y} = 0$  is an unstable equilibrium. The phase plane equivalent is  $v \frac{dv}{dy} + y^2 = 0$ , so the trajectories are  $v^2 + \frac{2y^3}{3} = c$ .

**4.4.4.**  $\bar{y} = 0$  is a stable equilibrium. The phase plane equivalent is  $v \frac{dv}{dy} + ye^{-y} = 0$ , so the trajectories are  $v^2 - e^{-y}(y+1) = c$ .

**4.4.6.**  $p(y) = y^3 - 4y = (y+2)y(y-2)$ , so the equilibria are  $-2, 0, 2$ . Since

$$\begin{aligned} y(y-2)(y+2) &< 0 && \text{if } y < -2 \text{ or } 0 < y < 2, \\ &> 0 && \text{if } -2 < y < 0 \text{ or } y > 2, \end{aligned}$$

0 is unstable and  $-2, 2$  are stable. The phase plane equivalent is  $v \frac{dv}{dy} + y^3 - 4y = 0$ , so the trajectories are  $2v^2 + y^4 - 8y^2 = c$ . Setting  $(y, v) = (0, 0)$  yields  $c = 0$ , so the equation of the separatrix is  $2v^2 - y^4 + 8y^2 = 0$ .

**4.4.8.**  $p(y) = y(y-2)(y-1)(y+2)$ , so the equilibria are  $-2, 0, 1, 2$ . Since

$$\begin{aligned} y(y-2)(y-1)(y+2) &> 0 && \text{if } y < -2 \text{ or } 0 < y < 1 \text{ or } y > 2, \\ &< 0 && \text{if } -2 < y < 0 \text{ or } 1 < y < 2, \end{aligned}$$

0, 2 are stable and  $-2, 1$  are unstable. The phase plane equivalent is  $v \frac{dv}{dy} + y(y-2)(y-1)(y+2) = 0$ , so the trajectories are  $30v^2 + y^2(12y^3 - 15y^2 - 80y + 120) = c$ . Setting  $(y, v) = (-2, 0)$  and  $(y, v) = (1, 0)$  yields  $c = 496$  and  $c = 37$  respectively, so the equations of the separatrices are  $30v^2 + y^2(12y^3 - 15y^2 - 80y + 120) = 496$  and  $30v^2 + y^2(12y^3 - 15y^2 - 80y + 120) = 37$ .

**4.4.10.**  $p(y) = y^3 - ay$ . If  $a \leq 0$ , then  $p(0) = 0$ ,  $p(y) > 0$  if  $y > 0$ , and  $p(y) < 0$  if  $y < 0$ , so 0 is stable. If  $a > 0$ , then

$$\begin{aligned} y^3 - ay = y(y - \sqrt{a})(y + \sqrt{a}) &> 0 && \text{if } -\sqrt{a} < y < 0 \text{ or } y > \sqrt{a}, \\ &< 0 && \text{if } y < -\sqrt{a} \text{ or } 0 < y < \sqrt{a}, \end{aligned}$$

so  $-\sqrt{a}$  and  $\sqrt{a}$  are stable and 0 is unstable. We say that  $a = 0$  is a critical value because it separates the two cases.

**4.4.12.**  $p(y) = y - ay^3$ . If  $a \leq 0$ , then  $p(0) = 0$ ,  $p(y) > 0$  if  $y > 0$ , and  $p(y) < 0$  if  $y < 0$ , so 0 is stable. If  $a > 0$ , then

$$\begin{aligned} y - ay^3 = -ay(y - 1/\sqrt{a})(y + 1/\sqrt{a}) &> 0 && \text{if } y < -1/\sqrt{a} < y < 0 \text{ or } 0 < y < 1/\sqrt{a} \\ &< 0 && \text{if } -1/\sqrt{a} < y < 0 \text{ or } y > 1/\sqrt{a}, \end{aligned}$$

so  $-\sqrt{a}$  and  $\sqrt{a}$  are unstable and 0 is stable. We say that  $a = 0$  is a critical value because it separates the two cases.

**4.4.24.** (a) Since  $v' = -p(y) \geq k$  and  $v(0) = 0$ ,  $v \geq kt$  and therefore  $y \geq y_0 + kt^2/2$  for  $0 \leq t < T$ .

(b) Let  $0 < \epsilon < \rho$ . Suppose that  $y$  is the solution of the initial value problem (A)  $y'' + p(y) = 0$ ,  $y(0) = y_0$ ,  $y'(0) = 0$ , where  $\bar{y} < y_0 < \bar{y} + \epsilon$ . Now let  $Y = y - \bar{y}$  and  $P(Y) = p(Y + \bar{y})$ . Then  $P(0) = 0$  and  $P(Y) < 0$  if  $0 < Y \leq \rho$ . Moreover,  $Y$  is the solution of  $Y'' + p(Y) = 0$ ,  $Y(0) = Y_0$ ,  $Y''(0) = 0$ , where  $Y_0 = y_0 - \bar{y}$ , so  $0 < Y_0 < \epsilon$ . From (a),  $Y(t) \geq \epsilon$  for some  $t > 0$ . Therefore,  $y(t) > \bar{y} + \epsilon$  for some  $t > 0$ , so  $\bar{y}$  is an unstable equilibrium of  $y'' + p(y) = 0$ .

## 4.5 Applications to Curves

**4.5.2.** Differentiating (A)  $e^{xy} = cy$  yields (B)  $(xy' + y)e^{xy} = cy'$ . From (A),  $c = \frac{e^{xy}}{y}$ . Substituting

this into (B) and cancelling  $e^{xy}$  yields  $xy' + y = \frac{y'}{y}$ , so  $y' = -\frac{y^2}{(xy - 1)}$ .

**4.5.4.** Solving  $y = x^{1/2} + cx$  for  $c$  yields  $c = \frac{y}{x} - x^{-1/2}$ , and differentiating yields  $0 = \frac{y'}{x} - \frac{y}{x^2} + \frac{x^{-3/2}}{2}$ , or  $xy' - y = -\frac{x^{1/2}}{2}$ .

**4.5.6.** Rewriting  $y = x^3 + \frac{c}{x}$  as  $xy = x^4 + c$  and differentiating yields  $xy' + y = 4x^3$ .

**4.5.8.** Rewriting  $y = e^x + c(1 + x^2)$  as  $\frac{y}{1 + x^2} = \frac{e^x}{1 + x^2} + c$  and differentiating yields  $\frac{y'}{1 + x^2} - \frac{2xy}{(1 + x^2)^2} = \frac{e^x}{1 + x^2} - \frac{2xe^x}{(1 + x^2)^2}$ , so  $(1 + x^2)y' - 2xy = (1 - x)^2 e^x$ .

**4.5.10.** If (A)  $y = f + cg$ , then (B)  $y' = f' + cg'$ . Multiplying (A) by  $g'$  and (B) by  $g$  yields (C)  $yg' = fg' + cgg'$  and (D)  $y'g = f'g + cg'g$ , and subtracting (C) from (D) yields  $y'g - yg' = f'g - fg'$ .

**4.5.12.** Let  $(x_0, y_0)$  be the center and  $r$  be the radius of a circle in the family. Since  $(-1, 0)$  and  $(1, 0)$  are on the circle,  $(x_0 + 1)^2 + y_0^2 = (x_0 - 1)^2 + y_0^2$ , which implies that  $x_0 = 0$ . Therefore, the equation of the circle is (A)  $x^2 + (y - y_0)^2 = r^2$ . Since  $(1, 0)$  is on the circle,  $r^2 = 1 + y_0^2$ . Substituting this into (A) shows that the equation of the circle is  $x^2 + y^2 - 2yy_0 = 1$ , so  $2y_0 = \frac{x^2 + y^2 - 1}{y}$ .

Differentiating  $y(2x + 2yy') - y'(x^2 + y^2 - 1) = 0$ , so  $y'(y^2 - x^2 + 1) + 2xy = 0$ .

**4.5.14.** From Example 4.5.6 the equation of the line tangent to the parabola at  $(x_0, x_0^2)$  is (A)  $y = -x_0^2 + 2x_0x$ .

(a) From (A),  $(x, y) = (5, 9)$  is on the tangent line through  $(x_0, x_0^2)$  if and only if  $9 = -x_0^2 + 10x_0$ , or  $x_0^2 - 10x_0 + 9 = (x_0 - 1)(x_0 - 9) = 0$ . Letting  $x_0 = 1$  in (A) yields the line  $y = -1 + 2x$ , tangent to the parabola at  $(x_0, x_0^2) = (1, 1)$ . Letting  $x_0 = 9$  in (A) yields the line  $y = -81 + 18x$ , tangent to the parabola at  $(x_0, x_0^2) = (9, 81)$ .

(b) From (A),  $(x, y) = (6, 11)$  is on the tangent line through  $(x_0, x_0^2)$  if and only if  $11 = -x_0^2 + 12x_0$ , or  $x_0^2 - 12x_0 + 11 = (x_0 - 1)(x_0 - 11) = 0$ . Letting  $x_0 = 1$  in (A) yields the line  $y = -1 + 2x$ , tangent to the parabola at  $(x_0, x_0^2) = (1, 1)$ . Letting  $x_0 = 11$  in (A) yields the line  $y = -121 + 22x$ , tangent to the parabola at  $(x_0, x_0^2) = (11, 121)$ .

(c) From (A),  $(x, y) = (-6, 20)$  is on the tangent line through  $(x_0, x_0^2)$  if and only if  $20 = -x_0^2 - 12x_0$ , or  $x_0^2 + 12x_0 + 20 = (x_0 + 2)(x_0 + 10) = 0$ . Letting  $x_0 = -2$  in (A) yields the line  $y = -4 - 4x$ , tangent to the parabola at  $(x_0, x_0^2) = (-2, 4)$ . Letting  $x_0 = -10$  in (A) yields the line  $y = -100 - 20x$ , tangent to the parabola at  $(x_0, x_0^2) = (-10, 100)$ .

(d) From (A),  $(x, y) = (-3, 5)$  is on the tangent line through  $(x_0, x_0^2)$  if and only if  $5 = -x_0^2 - 6x_0$ ,

or  $x_0^2 + 6x_0 + 5 = (x_0 + 1)(x_0 + 5) = 0$ . Letting  $x_0 = -1$  in (A) yields the line  $y = -1 - 2x$ , tangent to the parabola at  $(x_0, x_0^2) = (-1, 1)$ . Letting  $x_0 = -5$  in (A) yields the line  $y = -25 - 10x$ , tangent to the parabola at  $(x_0, x_0^2) = (-5, 25)$ .

**4.5.15. (a)** If  $(x_0, y_0)$  is any point on the circle such that  $x_0 \neq \pm 1$  (and therefore  $y_0 \neq 0$ ), then differentiating (A) yields  $2x_0 + 2y_0y'_0 = 0$ , so  $y'_0 = -\frac{x_0}{y_0}$ . Therefore, the equation of the tangent line is  $y = y_0 - \frac{x_0}{y_0}(x - x_0)$ . Since  $x_0^2 + y_0^2 = 1$ , this is equivalent to (B).

**(b)** Since  $y' = -\frac{x_0}{y_0}$  on the tangent line, we can rewrite (B) as  $y - xy' = \frac{1}{y_0}$ . Hence (F)  $\frac{1}{(y - xy')^2} = y_0^2$  and (G)  $x_0^2 = 1 - y_0^2 = \frac{(y - xy')^2 - 1}{(y - xy')^2}$ . Since  $(y')^2 = \frac{x_0^2}{y_0^2}$ , (F) and (G) imply that  $(y')^2 = (y - xy')^2 - 1$ , which implies (C).

**(c)** Using the quadratic formula to solve (C) for  $y'$  yields

$$y' = \frac{xy \pm \sqrt{x^2 + y^2 - 1}}{x^2 - 1} \quad (\text{H})$$

if  $(x, y)$  is on a tangent line with slope  $y'$ . If  $y = \frac{1 - x_0x}{y_0}$ , then  $x^2 + y^2 - 1 = x^2 + \left(\frac{1 - x_0x}{y_0}\right)^2 - 1 = \left(\frac{x - x_0}{y_0}\right)^2$  (since  $x_0^2 + y_0^2 = 1$ ). Since  $y' = -\frac{x_0}{y_0}$ , this implies that (H) is equivalent to  $-\frac{x_0}{y_0} = \frac{1}{x^2 - 1} \left[ \frac{x(1 - x_0x)}{y_0} \pm \left| \frac{x - x_0}{y_0} \right| \right]$ , which holds if and only if we choose the “ $\pm$ ” so that  $\pm \left| \frac{x - x_0}{y_0} \right| = -\left( \frac{x - x_0}{y_0} \right)$ . Therefore, we must choose  $\pm = -$  if  $\frac{x - x_0}{y_0} > 0$ , so (H) reduces to (D), or  $\pm = +$  if  $\frac{x - x_0}{y_0} < 0$ , so (H) reduces to (E).

**(d)** Differentiating (A) yields  $2x + 2yy' = 0$ , so  $y' = -\frac{x}{y}$  on either semicircle. Since (D) and (E) both reduce to  $y' = \frac{xy}{1 - x^2} = -\frac{x}{y}$  (since  $x^2 + y^2 = 1$ ) on both semicircles, the conclusion follows.

**(e)** From (D) and (E) the slopes of tangent lines from  $(5, 5)$  tangent to the circle are  $y' = \frac{25 \pm \sqrt{49}}{24} = \frac{3}{4}, \frac{4}{3}$ . Therefore, tangent lines are  $y = 5 + \frac{3}{4}(x - 5) = \frac{1 + 3x/5}{4/5}$  and  $y = 5 + \frac{4}{3}(x - 5) = \frac{1 - 4x/5}{-3/5}$ , which intersect the circle at  $(-3/5, 4/5)$   $(4/5, -3/5)$ , respectively. (See (B)).

**4.5.16. (a)** If  $(x_0, y_0)$  is any point on the parabola such that  $x_0 > 0$  (and therefore  $y_0 \neq 0$ ), then differentiating (A) yields  $1 = 2y_0y'_0$ , so  $y'_0 = \frac{1}{2y_0}$ . Therefore, the equation of the tangent line is

$y = y_0 + \frac{1}{2y_0}(x - x_0)$ . Since  $x_0 = y_0^2$ , this is equivalent to (B).

**(b)** Since  $y' = \frac{1}{2y_0}$  on the tangent line, we can rewrite (B) as  $\frac{y_0}{2} = y - xy'$ . Substituting this into (B) yields  $y = (y - xy') + \frac{x}{4(y - xy')}$ , which implies (C).



(c) Using the quadratic formula to solve (C) for  $y'$  yields

$$y' = \frac{y \pm \sqrt{y^2 - x}}{2x} \quad (\text{F})$$

if  $(x, y)$  is on a tangent line with slope  $y'$ . If  $y = \frac{y_0}{2} + \frac{x}{2y_0}$ , then  $y^2 - x = \frac{1}{4} \left( y_0 - \frac{x}{y_0} \right)^2$  so (F)

is equivalent to  $\frac{1}{2y_0} = \frac{y_0 + \frac{x}{y_0} \pm \left| y_0 - \frac{x}{y_0} \right|}{4x}$  which holds if and only if we choose the “ $\pm$ ” so that  $\pm \left| y_0 - \frac{x}{y_0} \right| = - \left( y_0 - \frac{x}{y_0} \right)$ . Therefore, we must choose  $\pm = +$  if  $x > y_0^2 = x_0$ , so (F) reduces to (D), or  $\pm = -$  if  $x < y_0^2 = x_0$ , so (F) reduces to (E).

(d) Differentiating (A) yields  $1 = 2yy'$ , so  $y' = \frac{1}{2y}$  on either half of the parabola. Since (D) and

(E) both reduce to this if  $x = y^2$ , the conclusion follows.

**4.5.18.** The equation of the line tangent to the curve at  $(x_0, y(x_0))$  is  $y = y(x_0) + y'(x_0)(x - x_0)$ .

Since  $y(x_0/2) = 0$ ,  $y(x_0) - \frac{y'(x_0)x_0}{2} = 0$ . Since  $x_0$  is arbitrary, it follows that  $y' = \frac{2y}{x}$ , so  $\frac{y'}{y} = \frac{2}{x}$ ,  $\ln|y| = 2\ln|x| + k$ , and  $y = cx^2$ . Since  $(1, 2)$  is on the curve,  $c = 2$ . Therefore,  $y = 2x^2$ .

**4.5.20.** The equation of the line tangent to the curve at  $(x_0, y(x_0))$  is  $y = y(x_0) + y'(x_0)(x - x_0)$ .

Since  $(x_1, y_1)$  is on the line,  $y(x_0) + y'(x_0)(x_1 - x_0) = y_1$ . Since  $x_0$  is arbitrary, it follows that  $y + y'(x_1 - x) = y_1$ , so  $\frac{y'}{y - y_1} = \frac{1}{x - x_1}$ ,  $\ln|y - y_1| = \ln|x - x_1| + k$ , and  $y - y_1 = c(x - x_1)$ .

**4.5.22.** The equation of the line tangent to the curve at  $(x_0, y(x_0))$  is  $y = y(x_0) + y'(x_0)(x - x_0)$ .

Since  $y(0) = x_0$ ,  $x_0 = y(x_0) - y'(x_0)x_0$ . Since  $x_0$  is arbitrary, it follows that  $x = y - xy'$ , so

(A)  $y' - \frac{y}{x} = -1$ . The solutions of (A) are of the form  $y = ux$ , where  $u'x = -1$ , so  $u' = -\frac{1}{x}$ . Therefore,  $u = -\ln|x| + c$  and  $y = -x\ln|x| + cx$ .

**4.5.24.** The equation of the line normal to the curve at  $(x_0, y_0)$  is  $y = y(x_0) - \frac{x - x_0}{y'(x_0)}$ . Since

$y(0) = 2y(x_0)$ ,  $y(x_0) + \frac{x_0}{y'(x_0)} = 2y(x_0)$ . Since  $x_0$  is arbitrary, it follows that  $y'y = x$ , so (A)

$\frac{y^2}{2} = \frac{x^2}{2} + \frac{c}{2}$  and  $y^2 = x^2 + c$ . Now  $y(2) = 1 \leftrightarrow c = -3$ . Therefore,  $y = \sqrt{x^2 - 3}$ .

**4.5.26.** Differentiating the given equation yields  $2x + 4y + 4xy' + 2yy' = 0$ , so  $y' = -\frac{x + 2y}{2x + y}$  is

a differential equation for the given family, and (A)  $y' = \frac{2x + y}{x + 2y}$  is a differential equation for the

orthogonal trajectories. Substituting  $y = ux$  in (A) yields  $u'x + u = \frac{2 + u}{1 + 2u}$ , so  $u'x = -\frac{2(u^2 - 1)}{1 + 2u}$

and  $\frac{1 + 2u}{(u - 1)(u + 1)}u' = -\frac{2}{x}$ , or  $\left[ \frac{3}{u - 1} + \frac{1}{u + 1} \right]u' = -\frac{4}{x}$ . Therefore,  $3\ln|u - 1| + \ln|u + 1| =$

$-4\ln|x| + K$ , so  $(u - 1)^3(u + 1) = \frac{k}{x^4}$ . Substituting  $u = \frac{y}{x}$  yields the orthogonal trajectories

$(y - x)^3(y + x) = k$ .

**4.5.28.** Differentiating yields  $ye^{x^2}(1 + 2x^2) + xe^{x^2}y' = 0$ , so  $y' = -\frac{y(1 + 2x^2)}{x}$  is a differential

equation for the given family. Therefore, (A)  $y' = -\frac{x}{y(1+2x^2)}$  is a differential equation for the orthogonal trajectories. From (A),  $yy' = -\frac{x}{1+2x^2}$ , so  $\frac{y^2}{2} = -\frac{1}{4}\ln(1+2x^2) + \frac{k}{2}$ , and the orthogonal trajectories are given by  $y^2 = -\frac{1}{2}\ln(1+2x^2) + k$ .

**4.5.30.** Differentiating (A)  $y = 1 + cx^2$  yields (B)  $y' = 2cx$ . From (C),  $c = \frac{y-1}{x^2}$ . Substituting this into (B) yields the differential equation  $y' = \frac{2(y-1)}{x}$  for the given family of parabolas. Therefore,  $y' = -\frac{x}{2(y-1)}$  is a differential equation for the orthogonal trajectories. Separating variables yields  $2(y-1)y' = -x$ , so  $(y-1)^2 = -\frac{x^2}{2} + k$ . Now  $y(-1) = 3 \leftrightarrow k = \frac{9}{2}$ , so  $(y-1)^2 = -\frac{x^2}{2} + \frac{9}{2}$ . Therefore, (D)  $y = 1 + \sqrt{\frac{9-x^2}{2}}$ . This curve intersects the parabola (A)

if and only if the equation (C)  $cx^2 = \sqrt{\frac{9-x^2}{2}}$  has a solution  $x^2$  in  $(0, 9)$ . Therefore,  $c > 0$  is a necessary condition for intersection. We will show that it is also sufficient. Squaring both sides of (C) and simplifying yields  $2c^2x^4 + x^2 - 9 = 0$ . Using the quadratic formula to solve this for  $x^2$  yields  $x^2 = \frac{-1 + \sqrt{1+72c^2}}{4c^2}$ . The condition  $x^2 < 9$  holds if and only if  $-1 + \sqrt{1+72c^2} < 36c^4$ , which is equivalent to  $1 + 72c^2 < (1 + 36c^2)^2 = 1 + 72c^2 + 1296c^4$ , which holds for all  $c > 0$ .

**4.5.32.** The angles  $\theta$  and  $\theta_1$  from the  $x$ -axis to the tangents to  $C$  and  $C_1$  satisfy  $\tan \theta = f(x_0, y_0)$  and  $\tan \theta_1 = \frac{f(x_0, y_0) + \tan \alpha}{1 - f(x_0, y_0) \tan \alpha} = \frac{\tan \theta + \tan \alpha}{1 - \tan \theta \tan \alpha} = \tan(\theta + \alpha)$ . Therefore, assuming  $\theta$  and  $\theta_1$  are both in  $[0, 2\pi)$ ,  $\theta_1 = \theta + \alpha$ .

**4.5.34.** Circles centered at the origin are given by  $x^2 + y^2 = r^2$ . Differentiating yields  $2x + 2yy' = 0$ , so  $y' = -\frac{x}{y}$  is a differential equation for the given family, and  $y' = \frac{-(x/y) + \tan \alpha}{1 + (x/y) \tan \alpha}$  is a differential equation for the desired family. Substituting  $y = ux$  yields  $u'x + u = \frac{-1/u + \tan \alpha}{1 + (1/u) \tan \alpha} = \frac{-1 + u \tan \alpha}{u + \tan \alpha}$ . Therefore,  $u'x = -\frac{1+u^2}{u + \tan \alpha}$ ,  $\frac{u + \tan \alpha}{1+u^2}u' = -\frac{1}{x}$  and  $\frac{1}{2}\ln(1+u^2) + \tan \alpha \tan^{-1} u = -\ln|x| + k$ . Substituting  $u = \frac{y}{x}$  yields  $\frac{1}{2}\ln(x^2 + y^2) + (\tan \alpha) \tan^{-1} \frac{y}{x} = k$ .

# Chapter 5

## Linear Second Order Equations

### 5.1 Homogeneous Linear Equations

**5.1.2. (a)** If  $y_1 = e^x \cos x$ , then  $y_1' = e^x(\cos x - \sin x)$  and  $y_1'' = e^x(\cos x - \sin x - \sin x - \cos x) = -2e^x \sin x$ , so  $y_1'' - 2y_1' + 2y_1 = e^x(-2 \sin x - 2 \cos x + 2 \sin x + 2 \cos x) = 0$ . If  $y_2 = e^x \sin x$ , then  $y_2' = e^x(\sin x + \cos x)$  and  $y_2'' = e^x(\sin x + \cos x + \cos x - \sin x) = 2e^x \cos x$ , so  $y_2'' - 2y_2' + 2y_2 = e^x(2 \cos x - 2 \sin x - 2 \cos x + 2 \sin x) = 0$ .

**(b)** If (B)  $y = e^x(c_1 \cos x + c_2 \sin x)$ , then

$$y' = e^x(c_1(\cos x - \sin x) + c_2(\sin x + \cos x)) \quad (C)$$

and

$$\begin{aligned} y'' &= c_1 e^x(\cos x - \sin x - \sin x - \cos x) \\ &\quad + c_2 e^x(\sin x + \cos x + \cos x - \sin x) \\ &= 2e^x(-c_1 \sin x + c_2 \cos x), \end{aligned}$$

so

$$\begin{aligned} y'' - 2y' + 2y &= c_1 e^x(-2 \sin x - 2 \cos x + 2 \sin x + 2 \cos x) \\ &\quad + c_2 e^x(2 \cos x - 2 \sin x - 2 \cos x + 2 \sin x) = 0. \end{aligned}$$

**(c)** We must choose  $c_1$  and  $c_2$  in (B) so that  $y(0) = 3$  and  $y'(0) = -2$ . Setting  $x = 0$  in (B) and (C) shows that  $c_1 = 3$  and  $c_1 + c_2 = -2$ , so  $c_2 = -5$ . Therefore,  $y = e^x(3 \cos x - 5 \sin x)$ .

**(d)** We must choose  $c_1$  and  $c_2$  in (B) so that  $y(0) = k_0$  and  $y'(0) = k_1$ . Setting  $x = 0$  in (B) and (C) shows that  $c_1 = k_0$  and  $c_1 + c_2 = k_1$ , so  $c_2 = k_1 - k_0$ . Therefore,  $y = e^x(k_0 \cos x + (k_1 - k_0) \sin x)$ .

**5.1.4. (a)** If  $y_1 = \frac{1}{x-1}$ , then  $y_1' = -\frac{1}{(x-1)^2}$  and  $y_1'' = \frac{2}{(x-1)^3}$ , so

$$\begin{aligned} (x^2 - 1)y_1'' + 4xy_1' + 2y_1 &= \frac{2(x^2 - 1)}{(x-1)^3} - \frac{4x}{(x-1)^2} + \frac{2}{x-1} \\ &= \frac{2(x+1) - 4x + 2(x-1)}{(x-1)^2} = 0. \end{aligned}$$

Similar manipulations show that  $(x^2 - 1)y_2'' + 4xy_2' + 2y_2 = 0$ . The general solution on each of the intervals  $(-\infty, -1)$ ,  $(-1, 1)$ , and  $(1, \infty)$  is (B)  $y = \frac{c_1}{x-1} + \frac{c_2}{x+1}$ .

(b) Differentiating (B) yields (C)  $y' = -\frac{c_1}{(x-1)^2} - \frac{c_2}{(x+1)^2}$ . We must choose  $c_1$  and  $c_2$  in (B) so that  $y(0) = -5$  and  $y'(0) = 1$ . Setting  $x = 0$  in (B) and (C) shows that  $-c_1 + c_2 = -5$ ,  $-c_1 - c_2 = 1$ . Therefore,  $c_1 = 2$  and  $c_2 = -3$ , so  $y = \frac{2}{x-1} - \frac{3}{x+1}$  on  $(-1, 1)$ .

(d) The Wronskian of  $\{y_1, y_2\}$  is

$$W(x) = \begin{vmatrix} \frac{1}{x-1} & \frac{1}{x+1} \\ -\frac{1}{(x-1)^2} & -\frac{1}{(x+1)^2} \end{vmatrix} = \frac{2}{(x^2-1)^2}, \quad (\text{D})$$

so  $W(0) = 2$ . Since  $p(x) = \frac{4x}{x^2-1}$ , so  $\int_0^x p(t) dt = \int_0^x \frac{4t}{t^2-1} dt = \ln(x^2-1)^2$ , Abel's formula implies that  $W(x) = W(0)e^{-\ln(x^2-1)^2} = \frac{2}{(x^2-1)^2}$ , consistent with (D).

**5.1.6.** From Abel's formula,  $W(x) = W(\pi)e^{-3\int_\pi^x (t^2+1) dt} = 0 \cdot e^{-3\int_\pi^x (t^2+1) dt} = 0$ .

**5.1.8.**  $p(x) = \frac{1}{x}$ ; therefore  $\int_1^x p(t) dt = \int_1^x \frac{dt}{t} = \ln x$ , so Abel's formula yields  $W(x) = W(1)e^{-\ln x} = \frac{1}{x}$ .

**5.1.10.**  $p(x) = -2$ ;  $P(x) = -2x$ ;  $y_2 = uy_1 = ue^{3x}$ ;  $u' = \frac{Ke^{-P(x)}}{y_1^2(x)} = \frac{Ke^{2x}}{e^{6x}} = Ke^{-4x}$ ;  $u = -\frac{K}{4}e^{-4x}$ . Choose  $K = -4$ ; then  $y_2 = e^{-4x}e^{3x} = e^{-x}$ .

**5.1.12.**  $p(x) = -2a$ ;  $P(x) = -2ax$ ;  $y_2 = uy_1 = ue^{ax}$ ;  $u' = \frac{Ke^{-P(x)}}{y_1^2(x)} = \frac{Ke^{2ax}}{e^{2ax}} = K$ ;  $u = Kx$ . Choose  $K = 1$ ; then  $y_2 = xe^{ax}$ .

**5.1.14.**  $p(x) = -\frac{1}{x}$ ;  $P(x) = -\ln x$ ;  $y_2 = uy_1 = ux$ ;  $u' = \frac{Ke^{-P(x)}}{y_1^2(x)} = \frac{Kx}{x^2} = \frac{K}{x}$ ;  $u = K \ln x$ . Choose  $K = 1$ ; then  $y_2 = x \ln x$ .

**5.1.16.**  $p(x) = -\frac{1}{x}$ ;  $P(x) = -\ln|x|$ ;  $y_2 = uy_1 = ux^{1/2}e^{2x}$ ;  $u' = \frac{Ke^{-P(x)}}{y_1^2(x)} = \frac{Kx}{xe^{4x}} = e^{-4x}$ ;  $u = -\frac{Ke^{-4x}}{4}$ . Choose  $K = -4$ ; then  $y_2 = e^{-4x}(x^{1/2}e^{2x}) = x^{1/2}e^{-2x}$ .

**5.1.18.**  $p(x) = -\frac{2}{x}$ ;  $P(x) = -2 \ln|x|$ ;  $y_2 = uy_1 = ux \cos x$ ;  $u' = \frac{Ke^{-P(x)}}{y_1^2(x)} = \frac{Kx^2}{x^2 \cos^2 x} = K \sec^2 x$ ;  $u = K \tan x$ . Choose  $K = 1$ ; then  $y_2 = \tan x(x \cos x) = x \sin x$ .

**5.1.20.**  $p(x) = -\frac{3x+2}{3x-1} = -1 - \frac{3}{3x-1}$ ;  $P(x) = -x - \ln|3x-1|$ ;  $y_2 = uy_1 = ue^{2x}$ ;  $u' = \frac{Ke^{-P(x)}}{y_1^2(x)} = \frac{K(3x-1)e^x}{e^{4x}} = K(3x-1)e^{-3x}$ ;  $u = -Kxe^{-3x}$ . Choose  $K = -1$ ; then  $y_2 = xe^{-3x}e^{2x} = xe^{-x}$ .

**5.1.22.**  $p(x) = -\frac{2(2x^2-1)}{x(2x+1)} = -2 - \frac{2}{2x+1} + \frac{2}{x}$ ;  $P(x) = -2x - \ln|2x+1| + 2 \ln|x|$ ;  $y_2 = uy_1 = \frac{u}{x}$ ;  $u' = \frac{Ke^{-P(x)}}{y_1^2(x)} = \frac{K(2x+1)e^{2x}}{x^2}x^2 = K(2x+1)e^{2x}$ ;  $u = Kxe^{2x}$ . Choose  $K = 1$ ; then  $y_2 = \frac{xe^{2x}}{x} = e^{2x}$ .

$e^{2x}$ .

**5.1.24.** Suppose that  $y \equiv 0$  on  $(a, b)$ . Then  $y' \equiv 0$  and  $y'' \equiv 0$  on  $(a, b)$ , so  $y$  is a solution of (A)  $y'' + p(x)y' + q(x)y = 0$ ,  $y(x_0) = 0$ ,  $y'(x_0) = 0$  on  $(a, b)$ . Since Theorem 5.1.1 implies that (A) has only one solution on  $(a, b)$ , the conclusion follows.

**5.1.26.** If  $\{z_1, z_2\}$  is a fundamental set of solutions of (A) on  $(a, b)$ , then every solution  $y$  of (A) on  $(a, b)$  is a linear combination of  $\{z_1, z_2\}$ ; that is,  $y = c_1 z_1 + c_2 z_2 = c_1(\alpha y_1 + \beta y_2) + c_2(\gamma y_1 + \delta y_2) = (c_1 \alpha + c_2 \gamma)y_1 + (c_1 \beta + c_2 \delta)y_2$ , which shows that every solution of (A) on  $(a, b)$  can be written as a linear combination of  $\{y_1, y_2\}$ . Therefore,  $\{y_1, y_2\}$  is a fundamental set of solutions of (A) on  $(a, b)$ .

**5.1.28.** The Wronskian of  $\{y_1, y_2\}$  is

$$W = \begin{vmatrix} y_1 & y_2 \\ y'_1 & y'_2 \end{vmatrix} = \begin{vmatrix} y_1 & ky_1 \\ y'_1 & ky'_1 \end{vmatrix} = k(y_1 y'_1 - y'_1 y_1) = 0.$$

nor  $y_2$  can be a solution of  $y'' + p(x)y' + q(x)y = 0$  on  $(a, b)$ .

**5.1.30.**  $W(x_0) = (y_1(x_0)y'_2(x_0) - y'_1(x_0)y_2(x_0)) = 0$  if either  $y_1(x_0) = y_2(x_0) = 0$  or  $y'_1(x_0) = y'_2(x_0) = 0$ , and Theorem 5.1.6 implies that  $\{y_1, y_2\}$  is linearly dependent on  $(a, b)$ .

**5.1.32.** Let  $x_0$  be an arbitrary point in  $(a, b)$ . By the motivating argument preceding Theorem 5.1.4, (B)  $W(x_0) = y_1(x_0)y'_2(x_0) - y'_1(x_0)y_2(x_0) \neq 0$ . Now let  $y$  be the solution of  $y'' + p(x)y' + q(x)y = 0$ ,  $y(x_0) = y_1(x_0)$ ,  $y'(x_0) = y'_1(x_0)$ . By assumption,  $y$  is a linear combination of  $\{y_1, y_2\}$  on  $(a, b)$ ; that is,  $y = c_1 y_1 + c_2 y_2$ , where

$$\begin{aligned} c_1 y_1(x_0) + c_2 y_2(x_0) &= y_1(x_0) \\ c_1 y'_1(x_0) + c_2 y'_2(x_0) &= y'_1(x_0). \end{aligned}$$

Solving this system by Cramers' rule yields

$$c_1 = \frac{1}{W(x_0)} \begin{vmatrix} y_1(x_0) & y_2(x_0) \\ y'_1(x_0) & y'_2(x_0) \end{vmatrix} = 1 \text{ and } c_2 = \frac{1}{W(x_0)} \begin{vmatrix} y_1(x_0) & y_1(x_0) \\ y'_1(x_0) & y_1(x_0) \end{vmatrix} = 0.$$

Therefore,  $y = y_1$ , which shows that  $y_1$  is a solution of (A). A similar argument shows that  $y_2$  is a solution of (A).

**5.1.34.** Expanding the determinant by cofactors of its first column shows that the first equation in the exercise can be written as

$$\frac{y}{W} \begin{vmatrix} y'_1 & y'_2 \\ y''_1 & y''_2 \end{vmatrix} - \frac{y'}{W} \begin{vmatrix} y_1 & y_2 \\ y''_1 & y''_2 \end{vmatrix} + \frac{y''}{W} \begin{vmatrix} y_1 & y_2 \\ y'_1 & y'_2 \end{vmatrix} = 0,$$

which is of the form (A) with

$$p = -\frac{1}{W} \begin{vmatrix} y_1 & y_2 \\ y''_1 & y''_2 \end{vmatrix} \quad \text{and} \quad q = \frac{1}{W} \begin{vmatrix} y'_1 & y'_2 \\ y''_1 & y''_2 \end{vmatrix}.$$

**5.1.36.** Theorem 5.1.6 implies that there are constants  $c_1$  and  $c_2$  such that (B)  $y = c_1 y_1 + c_2 y_2$  on  $(a, b)$ . To see that  $c_1$  and  $c_2$  are unique, assume that (B) holds, and let  $x_0$  be a point in  $(a, b)$ . Then (C)  $y' = c_1 y'_1 + c_2 y'_2$ . Setting  $x = x_0$  in (B) and (C) yields

$$\begin{aligned} c_1 y_1(x_0) + c_2 y_2(x_0) &= y(x_0) \\ c_1 y'_1(x_0) + c_2 y'_2(x_0) &= y'(x_0). \end{aligned}$$

Since Theorem 5.1.6 implies that  $y_1(x_0)y_2'(x_0) - y_1'(x_0)y_2(x_0) \neq 0$ , the argument preceding Theorem 5.1.4 implies that  $c_1$  and  $c_2$  are given uniquely by

$$c_1 = \frac{y_2'(x_0)y(x_0) - y_2(x_0)y'(x_0)}{y_1(x_0)y_2'(x_0) - y_1'(x_0)y_2(x_0)} \quad c_2 = \frac{y_1(x_0)y'(x_0) - y_1'(x_0)y(x_0)}{y_1(x_0)y_2'(x_0) - y_1'(x_0)y_2(x_0)}.$$

**5.1.38.** The general solution of  $y'' = 0$  is  $y = c_1 + c_2x$ , so  $y' = c_2$ . Imposing the stated initial conditions on  $y_1 = c_1 + c_2x$  yields  $c_1 + c_2x_0 = 1$  and  $c_2 = 0$ ; therefore  $c_1 = 1$ , so  $y_1 = 1$ . Imposing the stated initial conditions on  $y_2 = c_1 + c_2x$  yields  $c_1 + c_2x_0 = 0$  and  $c_2 = 1$ ; therefore  $c_1 = -x_0$ , so  $y_2 = x - x_0$ . The solution of the general initial value problem is  $y = k_0 + k_1(x - x_0)$ .

**5.1.40.** Let  $y_1 = a_1 \cos \omega x + a_2 \sin \omega x$  and  $y_2 = b_1 \cos \omega x + b_2 \sin \omega x$ . Then

$$\begin{aligned} a_1 \cos \omega x_0 + a_2 \sin \omega x_0 &= 1 \\ \omega(-a_1 \sin \omega x_0 + a_2 \cos \omega x_0) &= 0 \end{aligned}$$

and

$$\begin{aligned} b_1 \cos \omega x_0 + b_2 \sin \omega x_0 &= 0 \\ \omega(-b_1 \sin \omega x_0 + b_2 \cos \omega x_0) &= 1. \end{aligned}$$

Solving these systems yields  $a_1 = \cos \omega x_0$ ,  $a_2 = \sin \omega x_0$ ,  $b_1 = -\frac{\sin \omega x_0}{\omega}$ , and  $b_2 = \frac{\cos \omega x_0}{\omega}$ .

Therefore,  $y_1 = \cos \omega x_0 \cos \omega x + \sin \omega x_0 \sin \omega x = \cos \omega(x - x_0)$  and  $y_2 = \frac{1}{\omega}(-\sin \omega x_0 \cos \omega x + \cos \omega x_0 \sin \omega x) = \frac{1}{\omega} \sin \omega(x - x_0)$ . The solution of the general initial value problem is  $y = k_0 \cos \omega(x - x_0) + \frac{k_1}{\omega} \sin \omega(x - x_0)$ .

**5.1.42. (a)** If  $y_1 = x^2$ , then  $y_1' = 2x$  and  $y_1'' = 2$ , so  $x^2 y_1'' - 4x y_1' + 6y_1 = x^2(2) - 4x(2x) + 6x^2 = 0$  for  $x$  in  $(-\infty, \infty)$ . If  $y_2 = x^3$ , then  $y_2' = 3x^2$  and  $y_2'' = 6x$ , so  $x^2 y_2'' - 4x y_2' + 6y_2 = x^2(6x) - 4x(3x^2) + 6x^3 = 0$  for  $x$  in  $(-\infty, \infty)$ . If  $x \neq 0$ , then  $y_2(x)/y_1(x) = x$ , which is nonconstant on  $(-\infty, 0)$  and  $(0, \infty)$ , so Theorem 5.1.6 implies that  $\{y_1, y_2\}$  is a fundamental set of solutions of (A) on each of these intervals.

**(b)** Theorem 5.1.6 and (a) imply that  $y$  satisfies (A) on  $(-\infty, 0)$  and on  $(0, \infty)$  if and only if  $y = \begin{cases} a_1 x^2 + a_2 x^3, & x > 0, \\ b_1 x^2 + b_2 x^3, & x < 0. \end{cases}$  Since  $y(0) = 0$  we can complete the proof that  $y$  is a solution of (A) on  $(-\infty, \infty)$  by showing that  $y'(0)$  and  $y''(0)$  both exist if and only if  $a_1 = b_1$ . Since

$$\frac{y(x) - y(0)}{x - 0} = \begin{cases} a_1 x + a_2 x^2, & \text{if } x > 0, \\ b_1 x + b_2 x^2, & \text{if } x < 0, \end{cases}$$

it follows that  $y'(0) = \lim_{x \rightarrow 0} \frac{y(x) - y(0)}{x - 0} = 0$ . Therefore,  $y' = \begin{cases} 2a_1 x + 3a_2 x^2, & x \geq 0, \\ 2b_1 x + 3b_2 x^2, & x < 0. \end{cases}$  Since

$$\frac{y'(x) - y'(0)}{x - 0} = \begin{cases} 2a_1 + 3a_2 x, & \text{if } x > 0, \\ 2b_1 + 3b_2 x, & \text{if } x < 0, \end{cases} \quad \text{it follows that } y''(0) = \lim_{x \rightarrow 0} \frac{y'(x) - y'(0)}{x - 0} \text{ exists if and}$$

only if  $a_1 = b_1$ . By renaming  $a_1 = b_1 = c_1$ ,  $a_2 = c_2$ , and  $b_2 = c_3$  we see that  $y$  is a solution of (A) on  $(-\infty, \infty)$  if and only if  $y = \begin{cases} c_1 x^2 + c_2 x^3, & x \geq 0, \\ c_1 x^2 + c_3 x^3, & x < 0. \end{cases}$

(c) We have shown that  $y(0) = y'(0) = 0$  for any choice of  $c_1$  and  $c_2$  in (C). Therefore, the given initial value problem has a solution if and only if  $k_0 = k_1 = 0$ , in which case every function of the form (C) is a solution.

(d) If  $x_0 > 0$ , then  $c_1$  and  $c_2$  in (C) are uniquely determined by  $k_0$  and  $k_1$ , but  $c_3$  can be chosen arbitrarily. Therefore, (B) has a unique solution on  $(0, \infty)$ , but infinitely many solutions on  $(-\infty, \infty)$ . If  $x_0 < 0$ , then  $c_1$  and  $c_3$  in (C) are uniquely determined by  $k_0$  and  $k_1$ , but  $c_2$  can be chosen arbitrarily. Therefore, (B) has a unique solution on  $(-\infty, 0)$ , but infinitely many solutions on  $(-\infty, \infty)$ .

**5.1.44.** (a) If  $y_1 = x^3$ , then  $y'_1 = 3x^2$  and  $y''_1 = 6x$ , so  $x^2 y''_1 - 6x y'_1 + 12y_1 = x^2(6x) - 6x(3x^2) + 12x^3 = 0$  for  $x$  in  $(-\infty, \infty)$ . If  $y_2 = x^4$ , then  $y'_2 = 4x^3$  and  $y''_2 = 12x^2$ , so  $x^2 y''_2 - 6x y'_2 + 12y_2 = x^2(12x^2) - 6x(4x^3) + 12x^4 = 0$  for  $x$  in  $(-\infty, \infty)$ . If  $x \neq 0$ , then  $y_2(x)/y_1(x) = x$ , which is nonconstant on  $(-\infty, 0)$  and  $(0, \infty)$ , so Theorem 5.1.6 implies that  $\{y_1, y_2\}$  is a fundamental set of solutions of (A) on each of these intervals.

(b) Theorem 5.1.2 and (a) imply that  $y$  satisfies (A) on  $(-\infty, 0)$  and on  $(0, \infty)$  if and only if (C)  $y = \begin{cases} a_1 x^3 + a_2 x^4, & x > 0, \\ b_1 x^3 + b_2 x^4, & x < 0. \end{cases}$  Since  $y(0) = 0$  we can complete the proof that  $y$  is a solution of (A) on  $(-\infty, \infty)$  by showing that  $y'(0)$  and  $y''(0)$  both exist for any choice of  $a_1, a_2, b_1$ , and  $b_2$ . Since  $\frac{y(x) - y(0)}{x - 0} = \begin{cases} a_1 x^2 + a_2 x^3, & \text{if } x > 0, \\ b_1 x^2 + b_2 x^3, & \text{if } x < 0, \end{cases}$  it follows that  $y'(0) = \lim_{x \rightarrow 0} \frac{y(x) - y(0)}{x - 0} = 0$ . Therefore,  $y' = \begin{cases} 3a_1 x^2 + 4a_2 x^3, & x \geq 0, \\ 3b_1 x^2 + 4b_2 x^3, & x < 0. \end{cases}$  Since  $\frac{y'(x) - y'(0)}{x - 0} = \begin{cases} 3a_1 x + 4a_2 x^2, & \text{if } x > 0, \\ 3b_1 x + 4b_2 x^2, & \text{if } x < 0, \end{cases}$  it follows that  $y''(0) = \lim_{x \rightarrow 0} \frac{y'(x) - y'(0)}{x - 0} = 0$ . Therefore, (B) is a solution of (A) on  $(-\infty, \infty)$ .

(c) We have shown that  $y(0) = y'(0) = 0$  for any choice of  $a_1, a_2, b_1$ , and  $b_2$  in (B). Therefore, the given initial value problem has a solution if and only if  $k_0 = k_1 = 0$ , in which case every function of the form (B) is a solution.

(d) If  $x_0 > 0$ , then  $a_1$  and  $a_2$  in (B) are uniquely determined by  $k_0$  and  $k_1$ , but  $b_1$  and  $b_2$  can be chosen arbitrarily. Therefore, (C) has a unique solution on  $(0, \infty)$ , but infinitely many solutions on  $(-\infty, \infty)$ . If  $x_0 < 0$ , then  $b_1$  and  $b_2$  in (B) are uniquely determined by  $k_0$  and  $k_1$ , but  $a_1$  and  $a_2$  can be chosen arbitrarily. Therefore, (C) has a unique solution on  $(-\infty, 0)$ , but infinitely many solutions on  $(-\infty, \infty)$ .

## 5.2 Constant Coefficient Homogeneous Equations

**5.2.2.**  $p(r) = r^2 - 4r + 5 = (r - 2)^2 + 1$ ;  $y = e^{2x}(c_1 \cos x + c_2 \sin x)$ .

**5.2.4.**  $p(r) = r^2 - 4r + 4 = (r - 2)^2$ ;  $y = e^{2x}(c_1 + c_2 x)$ .

**5.2.6.**  $p(r) = r^2 + 6r + 10 = (r + 3)^2 + 1$ ;  $y = e^{-3x}(c_1 \cos x + c_2 \sin x)$ .

**5.2.8.**  $p(r) = r^2 + r = r(r + 1)$ ;  $y = c_1 + c_2 e^{-x}$ .

**5.2.10.**  $p(r) = r^2 + 6r + 13 = (r + 3)^2 + 4$ ;  $y = e^{-3x}(c_1 \cos 2x + c_2 \sin 2x)$ .

**5.2.12.**  $p(r) = 10r^2 - 3r - 1 = (2r - 1)(5r + 1) = 10(r - 1/2)(r + 1/5)$ ;  $y = c_1 e^{-x/5} + c_2 e^{x/2}$ .

**5.2.14.**  $p(r) = 6r^2 - r - 1 = (2r - 1)(3r + 1) = 6(r - 1/2)(r + 1/3)$ ;  $y = c_1 e^{-x/3} + c_2 e^{x/2}$ ;  $y' = -\frac{c_1}{3} e^{-x/3} + \frac{c_2}{2} e^{x/2}$ ;  $y(0) = 10 \Rightarrow c_1 + c_2 = 10$ ,  $y'(0) = 0 \Rightarrow -\frac{c_1}{3} + \frac{c_2}{2} = 0$ ;  $c_1 = 6, c_2 = 4$ ;  $y = 4e^{x/2} + 6e^{-x/3}$ .

**5.2.16.**  $p(r) = 4r^2 - 4r - 3 = (2r - 3)(2r + 1) = 4(r - 3/2)(r + 1/2)$ ;  $y = c_1 e^{-x/2} + c_2 e^{3x/2}$ ;



$$y' = -\frac{c_1}{2}e^{-x/2} + \frac{3c_2}{2}e^{3x/2}; \quad y(0) = \frac{13}{12} \Rightarrow c_1 + c_2 = \frac{13}{12}, \quad y'(0) = \frac{23}{24} \Rightarrow -\frac{c_1}{2} + \frac{3c_2}{2} = \frac{23}{24};$$

$$c_1 = \frac{1}{3}, c_2 = \frac{3}{4}; \quad y = \frac{e^{-x/2}}{3} + \frac{3e^{3x/2}}{4}.$$

**5.2.18.**  $p(r) = r^2 + 7r + 12 = (r+3)(r+4)$ ;  $y = c_1e^{-4x} + c_2e^{-3x}$ ;  $y' = -4c_1e^{-4x} - 3c_2e^{-3x}$ ;  
 $y(0) = -1 \Rightarrow c_1 + c_2 = -1$ ,  $y'(0) = 0 \Rightarrow -4c_1 - 3c_2 = 0$ ;  $c_1 = 3$ ,  $c_2 = -4$ ;  $y = 3e^{-4x} - 4e^{-3x}$ .

**5.2.20.**  $p(r) = 36r^2 - 12r + 1 = (6r-1)^2 = 36(r-1/6)^2$ ;  $y = e^{x/6}(c_1 + c_2x)$ ;  $y' = \frac{e^{x/6}}{6}(c_1 + c_2x) + c_2e^{x/6}$ ;  
 $y(0) = 3 \Rightarrow c_1 = 3$ ,  $y'(0) = \frac{5}{2} \Rightarrow \frac{c_1}{6} + c_2 = \frac{5}{2} \Rightarrow c_2 = 2$ ;  $y = e^{x/6}(3 + 2x)$ .

**5.2.22.** (a) From (A),  $ay''(x) + by'(x) + cy(x) = 0$  for all  $x$ . Replacing  $x$  by  $x - x_0$  yields (C)  $ay''(x - x_0) + by'(x - x_0) + cy(x - x_0) = 0$ . If  $z(x) = y(x - x_0)$ , then the chain rule implies that  $z'(x) = y'(x - x_0)$  and  $z''(x) = y''(x - x_0)$ , so (C) is equivalent to  $az'' + bz' + cz = 0$ .

(b) If  $\{y_1, y_2\}$  is a fundamental set of solutions of (A) then Theorem 5.1.6 implies that  $y_2/y_1$  is nonconstant. Therefore,  $\frac{z_2(x)}{z_1(x)} = \frac{y_2(x - x_0)}{y_1(x - x_0)}$  is also nonconstant, so Theorem 5.1.6 implies that  $\{z_1, z_2\}$  is a fundamental set of solutions of (A).

(c) Let  $p(r) = ar^2 + br + c$  be the characteristic polynomial of (A). Then:

- If  $p(r) = 0$  has distinct real roots  $r_1$  and  $r_2$ , then the general solution of (A) is

$$y = c_1e^{r_1(x-x_0)} + c_2e^{r_2(x-x_0)}.$$

- If  $p(r) = 0$  has a repeated root  $r_1$ , then the general solution of (A) is

$$y = e^{r_1(x-x_0)}(c_1 + c_2(x - x_0)).$$

- If  $p(r) = 0$  has complex conjugate roots  $r_1 = \lambda + i\omega$  and  $r_2 = \lambda - i\omega$  (where  $\omega > 0$ ), then the general solution of (A) is

$$y = e^{\lambda(x-x_0)}(c_1 \cos \omega(x - x_0) + c_2 \sin \omega(x - x_0)).$$

**5.2.24.**  $p(r) = r^2 - 6r - 7 = (r-7)(r+1)$ ;

$$\begin{aligned} y &= c_1e^{-(x-2)} + c_2e^{7(x-2)}; \\ y' &= -c_1e^{-(x-2)} + 7c_2e^{7(x-2)}; \end{aligned}$$

$$y(2) = -\frac{1}{3} \Rightarrow c_1 + c_2 = -\frac{1}{3}, \quad y'(2) = -5 \Rightarrow -c_1 + 7c_2 = -5; \quad c_1 = \frac{1}{3}, c_2 = -\frac{2}{3}; \quad y = \frac{1}{3}e^{-(x-2)} - \frac{2}{3}e^{7(x-2)}.$$

**5.2.26.**  $p(r) = 9r^2 + 6r + 1 = (3r+1)^2 = 9(r+1/3)^2$ ;

$$\begin{aligned} y &= e^{-(x-2)/3}(c_1 + c_2(x-2)); \\ y' &= -\frac{1}{3}e^{-(x-2)/3}(c_1 + c_2(x-2)) + c_2e^{-(x-2)/3}; \end{aligned}$$

$$y(2) = 2 \Rightarrow c_1 = 2, \quad y'(2) = -\frac{14}{3} \Rightarrow -\frac{c_1}{3} + c_2 = -\frac{14}{3} \Rightarrow c_2 = -4; \quad y = e^{-(x-2)/3}(2 - 4(x-2)).$$

**5.2.28.**  $p(r) = r^2 + 3$ ;

$$\begin{aligned} y &= c_1 \cos \sqrt{3} \left( x - \frac{\pi}{3} \right) + c_2 \sin \sqrt{3} \left( x - \frac{\pi}{3} \right); \\ y' &= -\sqrt{3}c_1 \sin \sqrt{3} \left( x - \frac{\pi}{3} \right) + \sqrt{3}c_2 \cos \sqrt{3} \left( x - \frac{\pi}{3} \right); \end{aligned}$$

$$y(\pi/3) = 2 \Rightarrow c_1 = 2, \quad y'(\pi/3) = -1 \Rightarrow c_2 = -\frac{1}{\sqrt{3}};$$

$$y = 2 \cos \sqrt{3} \left( x - \frac{\pi}{3} \right) - \frac{1}{\sqrt{3}} \sin \sqrt{3} \left( x - \frac{\pi}{3} \right).$$

**5.2.30.**  $y$  is a solution of  $ay'' + by' + cy = 0$  if and only if

$$\begin{aligned} y &= c_1 e^{r_1(x-x_0)} + c_2 e^{r_2(x-x_0)} \\ y' &= r_1 c_1 e^{r_1(x-x_0)} + r_2 c_2 e^{r_2(x-x_0)}. \end{aligned}$$

Now  $y_1(x_0) = k_0$  and  $y'_1(x_0) = k_1 \Rightarrow c_1 + c_2 = k_0, r_1 c_1 + r_2 c_2 = k_1$ . Therefore,  $c_1 = \frac{r_2 k_0 - k_1}{r_2 - r_1}$  and  $c_2 = \frac{k_1 - r_1 k_0}{r_2 - r_1}$ . Substituting  $c_1$  and  $c_2$  into the above equations for  $y$  and  $y'$  yields

$$\begin{aligned} y &= \frac{r_2 k_0 - k_1}{r_2 - r_1} e^{r_1(x-x_0)} + \frac{k_1 - r_1 k_0}{r_2 - r_1} e^{r_2(x-x_0)} \\ &= \frac{k_0}{r_2 - r_1} (r_2 e^{r_1(x-x_0)} - r_1 e^{r_2(x-x_0)}) + \frac{k_1}{r_2 - r_1} (e^{r_2(x-x_0)} - e^{r_1(x-x_0)}). \end{aligned}$$

**5.2.32.**  $y$  is a solution of  $ay'' + by' + cy = 0$  if and only if

$$y = e^{\lambda(x-x_0)} (c_1 \cos \omega(x-x_0) + c_2 \sin \omega(x-x_0)) \quad (\text{A})$$

and

$$\begin{aligned} y' &= \lambda e^{\lambda(x-x_0)} (c_1 \cos \omega(x-x_0) + c_2 \sin \omega(x-x_0)) \\ &\quad + \omega e^{\lambda(x-x_0)} (-c_1 \sin \omega(x-x_0) + c_2 \cos \omega(x-x_0)). \end{aligned}$$

Now  $y_1(x_0) = k_0 \Rightarrow c_1 = k_0$  and  $y'_1(x_0) = k_1 \Rightarrow \lambda c_1 + \omega c_2 = k_1$ , so  $c_2 = \frac{k_1 - \lambda k_0}{\omega}$ . Substituting  $c_1$  and  $c_2$  into (A) yields

$$y = e^{\lambda(x-x_0)} \left[ k_0 \cos \omega(x-x_0) + \left( \frac{k_1 - \lambda k_0}{\omega} \right) \sin \omega(x-x_0) \right].$$

**5.2.34.** (b)

$$\begin{aligned} e^{i\theta_1} e^{i\theta_2} &= (\cos \theta_1 + i \sin \theta_1)(\cos \theta_2 + i \sin \theta_2) \\ &= (\cos \theta_1 \cos \theta_2 - \sin \theta_1 \sin \theta_2) + i(\sin \theta_1 \cos \theta_2 + \cos \theta_1 \sin \theta_2) \\ &= \cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2) = e^{i(\theta_1 + \theta_2)}. \end{aligned}$$

(c)

$$\begin{aligned}
e^{z_1+z_2} &= e^{(\alpha_1+i\beta_1)+(\alpha_2+i\beta_2)} = e^{(\alpha_1+\alpha_2)+i(\beta_1+\beta_2)} \\
&= e^{(\alpha_1+\alpha_2)} e^{i(\beta_1+\beta_2)} \text{ (from (F) with } \alpha = \alpha_1 + \alpha_2 \text{ and } \beta = \beta_1 + \beta_2) \\
&= e^{\alpha_1} e^{\alpha_2} e^{i(\beta_1+\beta_2)} \text{ (property of the real-valued exponential function)} \\
&= e^{\alpha_1} e^{\alpha_2} e^{i\beta_1} e^{i\beta_2} \text{ (from (b))} \\
&= e^{\alpha_1} e^{i\beta_1} e^{\alpha_2} e^{i\beta_2} = e^{\alpha_1+i\beta_1} e^{\alpha_2+i\beta_2} = e^{z_1} e^{z_2}.
\end{aligned}$$

(d) The real and imaginary parts of  $z_1 = e^{(\lambda+i\omega)x}$  are  $u_1 = e^{\lambda x} \cos \omega x$  and  $v_1 = e^{\lambda x} \sin \omega x$ , which are both solutions of  $ay'' + by' + cy = 0$ , by Theorem 5.2.1(c). Similarly, the real and imaginary parts of  $z_2 = e^{(\lambda-i\omega)x}$  are  $u_2 = e^{\lambda x} \cos(-\omega x) = e^{\lambda x} \cos \omega x$  and  $v_1 = e^{\lambda x} \sin(-\omega x) = -e^{\lambda x} \sin \omega x$ , which are both solutions of  $ay'' + by' + cy = 0$ , by Theorem 5.2.1(c).

### 5.3 Nonhomogeneous Linear Equations

**5.3.2.** The characteristic polynomial of the complementary equation is  $p(r) = r^2 - 4r + 5 = (r-2)^2 + 1$ , so  $\{e^{2x} \cos x, e^{2x} \sin x\}$  is a fundamental set of solutions for the complementary equation. Let  $y_p = A + Bx$ ; then  $y_p'' - 4y_p' + 5y_p = -4B + 5(A + Bx) = 1 + 5x$ . Therefore,  $5B = 5$ ,  $-4B + 5A = 1$ , so  $B = 1$ ,  $A = 1$ . Therefore,  $y_p = 1 + x$  is a particular solution and  $y = 1 + x + e^{2x}(c_1 \cos x + c_2 \sin x)$  is the general solution.

**5.3.4.** The characteristic polynomial of the complementary equation is  $p(r) = r^2 - 4r + 4 = (r-2)^2$ , so  $\{e^{2x}, xe^{2x}\}$  is a fundamental set of solutions for the complementary equation. Let  $y_p = A + Bx + Cx^2$ ; then  $y_p'' - 4y_p' + 4y_p = 2C - 4(B + 2Cx) + 4(A + Bx + Cx^2) = (2C - 4B + 4A) + (-8C + 4B)x + 4Cx^2 = 2 + 8x - 4x^2$ . Therefore,  $4C = -4$ ,  $-8C + 4B = 8$ ,  $2C - 4B + 4A = 2$ , so  $C = -1$ ,  $B = 0$ , and  $A = 1$ . Therefore,  $y_p = 1 - x^2$  is a particular solution and  $y = 1 - x^2 + e^{2x}(c_1 + c_2 x)$  is the general solution.

**5.3.6.** The characteristic polynomial of the complementary equation is  $p(r) = r^2 + 6r + 10 = (r+3)^2 + 1$ , so  $\{e^{-3x} \cos x, e^{-3x} \sin x\}$  is a fundamental set of solutions for the complementary equation. Let  $y_p = A + Bx$ ; then  $y_p'' + 6y_p' + 10y_p = 6B + 10(A + Bx) = 22 + 20x$ . Therefore,  $10B = 20$ ,  $6B + 10A = 22$ , so  $B = 2$ ,  $A = 1$ . Therefore,  $y_p = 1 + 2x$  is a particular solution and (A)  $y = 1 + 2x + e^{-3x}(c_1 \cos x + c_2 \sin x)$  is the general solution. Now  $y(0) = 2 \Rightarrow 2 = 1 + c_1 \Rightarrow c_1 = 1$ . Differentiating (A) yields  $y' = 2 - 3e^{-3x}(c_1 \cos x + c_2 \sin x) + e^{-3x}(-c_1 \sin x + c_2 \cos x)$ , so  $y'(0) = -2 \Rightarrow -2 = 2 - 3c_1 + c_2 \Rightarrow c_2 = -1$ .  $y = 1 + 2x + e^{-3x}(\cos x - \sin x)$  is the solution of the initial value problem.

**5.3.8.** If  $y_p = \frac{A}{x}$ , then  $x^2 y_p'' + 7xy_p' + 8y_p = A \left( x^2 \left( \frac{2}{x^3} \right) + 7x \left( \frac{-1}{x^2} \right) + \left( \frac{8}{x} \right) \right) = \frac{3A}{x} = \frac{6}{x}$  if  $A = 2$ . Therefore,  $y_p = \frac{2}{x}$  is a particular solution.

**5.3.10.** If  $y_p = Ax^3$ , then  $x^2 y_p'' - xy_p' + y_p = A(x^2(6x) - x(3x^2) + x^3) = 4Ax^3 = 2x^3$  if  $A = \frac{1}{2}$ . Therefore,  $y_p = \frac{x^3}{2}$  is a particular solution.

**5.3.12.** If  $y_p = Ax^{1/3}$ , then  $x^2 y_p'' + xy_p' + y_p = A \left( x^2 \left( \frac{-2x^{-5/3}}{9} \right) + x \left( \frac{x^{-2/3}}{3} \right) + x^{1/3} \right) = \frac{10A}{9} x^{1/3} = 10x^{1/3}$  if  $A = 9$ . Therefore,  $y_p = 9x^{1/3}$  is a particular solution.

**5.3.14.** If  $y_p = \frac{A}{x^3}$ , then  $x^2 y_p'' + 3x y_p' - 3y_p = A \left( x^2 \left( \frac{12}{x^5} \right) + 3x \left( \frac{-3}{x^4} \right) + \frac{3}{x^3} \right) = 0$ . Therefore,  $y_p$  is not a solution of the given equation for any choice of  $A$ .

**5.3.16.** The characteristic polynomial of the complementary equation is  $p(r) = r^2 + 5r - 6 = (r+6)(r-1)$ , so  $\{e^{-6x}, e^x\}$  is a fundamental set of solutions for the complementary equation. Let  $y_p = Ae^{3x}$ ; then  $y_p'' + 5y_p' - 6y_p = p(3)Ae^{3x} = 18Ae^{3x} = 6e^{3x}$  if  $A = \frac{1}{3}$ . Therefore,  $y_p = \frac{e^{3x}}{3}$  is a particular solution and  $y = \frac{e^{3x}}{3} + c_1 e^{-6x} + c_2 e^x$  is the general solution.

**5.3.18.** The characteristic polynomial of the complementary equation is  $p(r) = r^2 + 8r + 7 = (r+1)(r+7)$ , so  $\{e^{-7x}, e^{-x}\}$  is a fundamental set of solutions for the complementary equation. Let  $y_p = Ae^{-2x}$ ; then  $y_p'' + 8y_p' + 7y_p = p(-2)Ae^{-2x} = -5Ae^{-2x} = 10e^{-2x}$  if  $A = -2$ . Therefore,  $y_p = -2e^{-2x}$  is a particular solution and (A)  $y = -2e^{-2x} + c_1 e^{-7x} + c_2 e^{-x}$  is the general solution. Differentiating (A) yields  $y' = 4e^{-2x} - 7c_1 e^{-7x} - c_2 e^{-x}$ . Now  $y(0) = -2 \Rightarrow -2 = -2 + c_1 + c_2$  and  $y'(0) = 10 \Rightarrow 10 = 4 - 7c_1 - c_2$ . Therefore,  $c_1 = -1$  and  $c_2 = 1$ , so  $y = -2e^{-2x} - e^{-7x} + e^{-x}$  is the solution of the initial value problem.

**5.3.20.** The characteristic polynomial of the complementary equation is  $p(r) = r^2 + 2r + 10 = (r+1)^2 + 9$ , so  $\{e^{-x} \cos 3x, e^{-x} \sin 3x\}$  is a fundamental set of solutions for the complementary equation. If  $y_p = Ae^{x/2}$ , then  $y_p'' + 2y_p' + 10y_p = p(1/2)Ae^{x/2} = \frac{45}{4}Ae^{x/2} = e^{x/2}$  if  $A = \frac{4}{45}$ . Therefore,  $y_p = \frac{4}{45}e^{x/2}$  is a particular solution and  $y = \frac{4}{45}e^{x/2} + e^{-x}(c_1 \cos 3x + c_2 \sin 3x)$  is the general solution.

**5.3.22.** The characteristic polynomial of the complementary equation is  $p(r) = r^2 - 7r + 12 = (r-4)(r-3)$ . If  $y_p = Ae^{4x}$ , then  $y_p'' - 7y_p' + 12y_p = p(4)Ae^{4x} = 0 \cdot e^{4x} = 0$ , so  $y_p'' - 7y_p' + 12y_p \neq 5e^{4x}$  for any choice of  $A$ .

**5.3.24.** The characteristic polynomial of the complementary equation is  $p(r) = r^2 - 8r + 16 = (r-4)^2$ , so  $\{e^{4x}, xe^{4x}\}$  is a fundamental set of solutions for the complementary equation. If  $y_p = A \cos x + B \sin x$ , then  $y_p'' - 8y_p' + 16y_p = -(A \cos x + B \sin x) - 8(-A \sin x + B \cos x) + 16(A \cos x + B \sin x) = (15A - 8B) \cos x + (8A + 15B) \sin x$ , so  $15A - 8B = 23$ ,  $8A + 15B = -7$ , which implies that  $A = 1$  and  $B = -1$ . Hence  $y_p = \cos x - \sin x$  and  $y = \cos x - \sin x + e^{4x}(c_1 + c_2 x)$  is the general solution.

**5.3.26.** The characteristic polynomial of the complementary equation is  $p(r) = r^2 - 2r + 3 = (r-1)^2 + 2$ , so  $\{e^x \cos \sqrt{2}x, e^x \sin \sqrt{2}x\}$  is a fundamental set of solutions for the complementary equation. If  $y_p = A \cos 3x + B \sin 3x$ , then  $y_p'' - 2y_p' + 3y_p = -9(A \cos 3x + B \sin 3x) - 6(-A \sin 3x + B \cos 3x) + 3(A \cos 3x + B \sin 3x) = -(6A + 6B) \cos 3x + (6A - 6B) \sin 3x$ , so  $-6A - 6B = -6$ ,  $6A - 6B = 6$ , which implies that  $A = 1$  and  $B = 0$ . Hence  $y_p = \cos 3x$  is a particular solution and  $y = \cos 3x + e^x(c_1 \cos \sqrt{2}x + c_2 \sin \sqrt{2}x)$  is the general solution.

**5.3.28.** The characteristic polynomial of the complementary equation is  $p(r) = r^2 + 7r + 12 = (r+3)(r+4)$ , so  $\{e^{-4x}, e^{-3x}\}$  is a fundamental set of solutions for the complementary equation. If  $y_p = A \cos 2x + B \sin 2x$ , then  $y_p'' + 7y_p' + 12y_p = -4(A \cos 2x + B \sin 2x) + 14(-A \sin 2x + B \cos 2x) + 12(A \cos 2x + B \sin 2x) = (8A + 14B) \cos 2x + (8B - 14A) \sin 2x$ , so  $8A + 14B = -2$ ,  $-14A + 8B = 36$ , which implies that  $A = -2$  and  $B = 1$ . Hence  $y_p = -2 \cos 2x + \sin 2x$  is a particular solution and (A)  $y = -2 \cos 2x + \sin 2x + c_1 e^{-4x} + c_2 e^{-3x}$  is the general solution. Differentiating (A) yields  $y' = 2 \sin 2x + 2 \cos 2x - 4c_1 e^{-4x} - 3c_2 e^{-3x}$ . Now  $y(0) = -3 \Rightarrow -3 = -2 + c_1 + c_2$  and  $y'(0) = 3 \Rightarrow 3 = 2 - 4c_1 - 3c_2$ . Therefore,  $c_1 = 2$  and  $c_2 = -3$ , so  $y = -2 \cos 2x + \sin 2x + 2e^{-4x} - 3e^{-3x}$  is the solution of the initial value problem.

**5.3.30.**  $\{\cos \omega_0 x, \sin \omega_0 x\}$  is a fundamental set of solutions of the complementary equation. If  $y_p = A \cos \omega x + B \sin \omega x$ , then  $y_p'' + \omega_0^2 y_p = -\omega^2(A \cos \omega x + B \sin \omega x) + \omega_0^2(A \cos \omega x + B \sin \omega x) = (\omega_0^2 - \omega^2)(A \cos \omega x + B \sin \omega x) = M \cos \omega x + N \sin \omega x$  if  $A = \frac{M}{\omega_0^2 - \omega^2}$  and  $B = \frac{N}{\omega_0^2 - \omega^2}$ . Therefore,

$$y_p = \frac{1}{\omega_0^2 - \omega^2} (M \cos \omega x + N \sin \omega x)$$

is a particular solution of the given equation and

$$y = \frac{1}{\omega_0^2 - \omega^2} (M \cos \omega x + N \sin \omega x) + c_1 \cos \omega_0 x + c_2 \sin \omega_0 x$$

is the general solution.

**5.3.32.** If  $y_p = A \cos \omega x + B \sin \omega x$ , then  $ay_p'' + by_p' + cy_p = -a\omega^2(A \cos \omega x + B \sin \omega x) + b\omega(-A \sin \omega x + B \cos \omega x) + c(A \cos \omega x + B \sin \omega x) = [(c - a\omega^2)A + b\omega B] \cos \omega x + [-b\omega A + (c - a\omega^2)B] \sin \omega x$ . Therefore,  $y_p$  is a solution of (A) if and only if the set of equations (B)  $(c - a\omega^2)A + b\omega B = M$ ,  $-b\omega A + (c - a\omega^2)B = N$  has a solution. If  $(c - a\omega^2)^2 + (b\omega)^2 \neq 0$ , then (B) has the solution  $A = \frac{(c - a\omega^2)M - b\omega N}{(c - a\omega^2)^2 + (b\omega)^2}$ ,  $B = \frac{(c - a\omega^2)N + b\omega M}{(c - a\omega^2)^2 + (b\omega)^2}$ , and  $y_p = A \cos \omega x + B \sin \omega x$  is a solution of (A). If  $(c - a\omega^2)^2 + (b\omega)^2 = 0$  (which is true if and only if the left side of (A) is of the form  $a(y'' + \omega^2 y)$ ), then the coefficients of  $A$  and  $B$  in (B) are all zero, so (B) does not have a solution, so (A) does not have a solution of the form  $y_p = A \cos \omega x + B \sin \omega x$ .

**5.3.34.** From Exercises 5.3.2 and 5.3.17,  $y_{p_1} = 1 + x$  and  $y_{p_2} = e^{2x}$  are particular solutions of  $y'' - 4y' + 5y = 1 + 5x$  and  $y'' - 4y' + 5y = e^{2x}$  respectively, and  $\{e^{2x} \cos x, e^{2x} \sin x\}$  is a fundamental set of solutions of the complementary equation. Therefore,  $y_p = y_{p_1} + y_{p_2} = 1 + x + e^{2x}$  is a particular solution of the given equation, and  $y = 1 + x + e^{2x}(1 + c_1 \cos x + c_2 \sin x)$  is the general solution.

**5.3.36.** From Exercises 5.3.4 and 5.3.19,  $y_{p_1} = 1 - x^2$  and  $y_{p_2} = e^x$  are particular solutions of  $y'' - 4y' + 4y = 2 + 8x - 4x^2$  and  $y'' - 4y' + 4y = e^x$  respectively, and  $\{e^{2x}, xe^{2x}\}$  is a fundamental set of solutions of the complementary equation. Therefore,  $y_p = y_{p_1} + y_{p_2} = 1 - x^2 + e^x$  is a particular solution of the given equation, and  $y = 1 - x^2 + e^x + e^{2x}(c_1 + c_2 x)$  is the general solution.

**5.3.38.** From Exercises 5.3.6 and 5.3.21,  $y_{p_1} = 1 + 2x$  and  $y_{p_2} = e^{-3x}$  are particular solutions of  $y'' + 6y' + 10y = 22 + 20x$  and  $y'' + 6y' + 10y = e^{-3x}$  respectively, and  $\{e^{-3x} \cos x, e^{-3x} \sin x\}$  is a fundamental set of solutions of the complementary equation. Therefore,  $y_p = y_{p_1} + y_{p_2} = 1 + 2x + e^{-3x}$  is a particular solution of the given equation, and  $y = 1 + 2x + e^{-3x}(1 + c_1 \cos x + c_2 \sin x)$  is the general solution.

**5.3.40.** Letting  $c_1 = c_2 = 0$  shows that (A)  $y_p'' + p(x)y_p' + q(x)y_p = f$ . Letting  $c_1 = 1$  and  $c_2 = 0$  shows that (B)  $(y_1 + y_p)'' + p(x)(y_1 + y_p)' + q(x)(y_1 + y_p) = f$ . Now subtract (A) from (B) to see that  $y_1'' + p(x)y_1' + q(x)y_1 = 0$ . Letting  $c_1 = 0$  and  $c_2 = 1$  shows that (C)  $(y_2 + y_p)'' + p(x)(y_2 + y_p)' + q(x)(y_2 + y_p) = f$ . Now subtract (A) from (C) to see that  $y_2'' + p(x)y_2' + q(x)y_2 = 0$ .

## 5.4 The Method of Undetermined Coefficients I

**5.4.2.** If  $y = ue^{-3x}$ , then  $y'' - 6y' + 5y = e^{-3x}[(u'' - 6u' + 9u) - 6(u' - 3u) + 5u] = e^{-3x}(35 - 8x)$ , so  $u'' - 12u' + 32u = 35 - 8x$  and  $u_p = A + Bx$ , where  $-12B + 32(A + Bx) = 35 - 8x$ . Therefore,  $32B = -8$ ,

$32A - 12B = 35$ , so  $B = -\frac{1}{4}$ ,  $A = 1$ , and  $u_p = 1 - \frac{x}{4}$ . Therefore,  $y_p = e^{-3x} \left(1 - \frac{x}{4}\right)$ .

**5.4.4.** If  $y = ue^{2x}$ , then  $y'' + 2y' + y = e^{2x} [(u'' + 4u' + 4u) + 2(u' + 2u) + u] = e^{2x}(-7 - 15x + 9x^2)$  so  $u'' + 6u' + 9u = -7 - 15x + 9x^2$  and  $u_p = A + Bx + Cx^2$ , where  $2C + 6(B + 2Cx) + 9(A + Bx + Cx^2) = -7 - 15x + 9x^2$ . Therefore,  $9C = 9$ ,  $9B + 12C = -15$ ,  $9A + 6B + 2C = -7$ , so  $C = 1$ ,  $B = -3$ ,  $A = 1$ , and  $u_p = 1 - 3x + x^2$ . Therefore,  $y_p = e^{2x}(1 - 3x + x^2)$ .

**5.4.6.** If  $y = ue^x$ , then  $y'' - y' - 2y = e^x [(u'' + 2u' + u) - (u' + u) - 2u] = e^x(9 + 2x - 4x^2)$  so  $u'' + u' - 2u = 9 + 2x - 4x^2$ , and  $u_p = A + Bx + Cx^2$ , where  $2C + (B + 2Cx) - 2(A + Bx + Cx^2) = 9 + 2x - 4x^2$ . Therefore,  $-2C = -4$ ,  $-2B + 2C = 2$ ,  $-2A + B + 2C = 9$ , so  $C = 2$ ,  $B = 1$ ,  $A = -2$ , and  $u_p = -2 + x + 2x^2$ . Therefore,  $y_p = e^x(-2 + x + 2x^2)$ .

**5.4.8.** If  $y = ue^x$ , then  $y'' - 3y' + 2y = e^x [(u'' + 2u' + u) - 3(u' + u) + 2u] = e^x(3 - 4x)$ , so  $u'' - u' = 3 - 4x$  and  $u_p = Ax + Bx^2$ , where  $2B - (A + 2Bx) = 3 - 4x$ . Therefore,  $-2B = -4$ ,  $-A + 2B = 3$ , so  $B = 2$ ,  $A = 1$ , and  $u_p = x(1 + 2x)$ . Therefore,  $y_p = xe^x(1 + 2x)$ .

**5.4.10.** If  $y = ue^{2x}$ , then  $2y'' - 3y' - 2y = e^{2x} [2(u'' + 4u' + 4u) - 3(u' + 2u) - 2u] = e^{2x}(-6 + 10x)$ , so  $2u'' + 5u' = -6 + 10x$  and  $u_p = Ax + Bx^2$ , where  $2(2B) + 5(A + 2Bx) = -6 + 10x$ . Therefore,  $10B = 10$ ,  $5A + 4B = -6$ , so  $B = 1$ ,  $A = -2$ , and  $u_p = x(-2 + x)$ . Therefore,  $y_p = xe^{2x}(-2 + x)$ .

**5.4.12.** If  $y = ue^x$ , then  $y'' - 2y' + y = e^x [(u'' + 2u' + u) - 2(u' + u) + u] = e^x(1 - 6x)$ , so  $u'' = 1 - 6x$ . Integrating twice and taking the constants of integration to be zero yields  $u_p = x^2 \left(\frac{1}{2} - x\right)$ . Therefore,  $y_p = x^2 e^x \left(\frac{1}{2} - x\right)$ .

**5.4.14.** If  $y = ue^{-x/3}$ , then  $9y'' + 6y' + y = e^{-x/3} \left[ 9 \left( u'' - \frac{2u'}{3} + \frac{u}{9} \right) + 6 \left( u' - \frac{u}{3} \right) + u \right] = e^{-x/3} (2 - 4x + 4x^2)$ , so  $9u'' = 2 - 4x + 4x^2$ , or  $u'' = \frac{1}{9}(2 - 4x + 4x^2)$ . Integrating twice and taking the constants of integration to be zero yields  $u_p = \frac{x^2}{27}(3 - 2x + x^2)$ . Therefore,  $y_p = \frac{x^2 e^{-x/3}}{27}(3 - 2x + x^2)$ .

**5.4.16.** If  $y = ue^x$ , then  $y'' - 6y' + 8y = e^x [(u'' + 2u' + u) - 6(u' + u) + 8u] = e^x(11 - 6x)$ , so  $u'' - 4u' + 3u = 11 - 6x$  and  $u_p = A + Bx$ , where  $-4B + 3(A + Bx) = 11 - 6x$ . Therefore,  $3B = -6$ ,  $3A - 4B = 11$ , so  $B = -2$ ,  $A = 1$  and  $u_p = 1 - 2x$ . Therefore,  $y_p = e^x(1 - 2x)$ . The characteristic polynomial of the complementary equation is  $p(r) = r^2 - 6r + 8 = (r - 2)(r - 4)$ , so  $\{e^{2x}, e^{4x}\}$  is a fundamental set of solutions of the complementary equation. Therefore,  $y = e^x(1 - 2x) + c_1 e^{2x} + c_2 e^{4x}$  is the general solution of the nonhomogeneous equation.

**5.4.18.** If  $y = ue^x$ , then  $y'' + 2y' - 3y = e^x [(u'' + 2u' + u) + 2(u' + u) - 3u] = -16xe^x$ , so  $u'' + 4u' = -16x$  and  $u_p = Ax + Bx^2$ , where  $2B + 4(A + 2Bx) = -16x$ . Therefore,  $8B = -16$ ,  $4A + 2B = 0$ , so  $B = -2$ ,  $A = 1$ , and  $u_p = x(1 - 2x)$ . Therefore,  $y_p = xe^x(1 - 2x)$ . The characteristic polynomial of the complementary equation is  $p(r) = r^2 + 2r - 3 = (r + 3)(r - 1)$ , so  $\{e^x, e^{-3x}\}$  is a fundamental set of solutions of the complementary equation. Therefore,  $y = xe^x(1 - 2x) + c_1 e^x + c_2 e^{-3x}$  is the general solution of the nonhomogeneous equation.

**5.4.20.** If  $y = ue^{2x}$ , then  $y'' - 4y' - 5y = e^{2x} [(u'' + 4u' + 4u) - 4(u' + 2u) - 5u] = 9e^{2x}(1 + x)$ , so  $u'' - 9u = 9 + 9x$  and  $u_p = A + Bx$ , where  $-9(A + Bx) = 9 + 9x$ . Therefore,  $-9B = -9$ ,  $-9A = 9$ , so  $B = 1$ ,  $A = -1$ , and  $u_p = -1 + x$ . Therefore,  $y_p = -e^{2x}(1 + x)$ . The characteristic polynomial of the complementary equation is  $p(r) = r^2 - 4r - 5 = (r - 5)(r + 1)$ , so  $\{e^{-x}, e^{5x}\}$  is a fundamental set of solutions of the complementary equation. Therefore, (A)  $y = -e^{2x}(1 + x) + c_1 e^{-x} + c_2 e^{5x}$  is the general solution of the nonhomogeneous equation. Differentiating (A) yields  $y' = -2e^{2x}(1 + x) - e^{2x} - c_1 e^{-x} + 5c_2 e^{5x}$ . Now  $y(0) = 0$ ,  $y'(0) = -10 \Rightarrow 0 = -1 + c_1 + c_2$ ,  $-10 = -3 - c_1 + 5c_2$ , so

$c_1 = 2, c_2 = -1$ . Therefore,  $y = -e^{2x}(1+x) + 2e^{-x} - e^{5x}$  is the solution of the initial value problem.

**5.4.22.** If  $y = ue^{-x}$ , then  $y'' + 4y' + 3y = e^{-x}[(u'' - 2u' + u) + 4(u' - u) + 3u] = -e^{-x}(2 + 8x)$ , so  $u'' + 2u' = -2 - 8x$  and  $u_p = Ax + Bx^2$ , where  $2B + 2(A + 2Bx) = -2 - 8x$ . Therefore,  $4B = -8$ ,  $2A + 2B = -2$ , so  $B = -2$ ,  $A = 1$ , and  $u_p = x(1 - 2x)$ . Therefore,  $y_p = xe^{-x}(1 - 2x)$ . The characteristic polynomial of the complementary equation is  $p(r) = r^2 + 4r + 3 = (r + 3)(r + 1)$ , so  $\{e^{-x}, e^{-3x}\}$  is a fundamental set of solutions of the complementary equation. Therefore, (A)  $y = xe^{-x}(1 - 2x) + c_1e^{-x} + c_2e^{-3x}$  is the general solution of the nonhomogeneous equation. Differentiating (A) yields  $y' = -xe^{-x}(1 - 2x) + e^{-x}(1 - 4x) - c_1e^{-x} - 3c_2e^{-3x}$ . Now  $y(0) = 1, y'(0) = 2 \Rightarrow 1 = c_1 + c_2, 2 = 1 - c_1 - 3c_2$ , so  $c_1 = 2, c_2 = -1$ . Therefore,  $y = e^{-x}(2 + x - 2x^2) - e^{-3x}$  is the solution of the initial value problem.

**5.4.24.** We must find particular solutions  $y_{p_1}$  and  $y_{p_2}$  of (A)  $y'' + y' + y = xe^x$  and (B)  $y'' + y' + y = e^{-x}(1 + 2x)$ , respectively. To find a particular solution of (A) we write  $y = ue^x$ . Then  $y'' + y' + y = e^x[(u'' + 2u' + u) + (u' + u) + u] = xe^x$  so  $u'' + 3u' + 3u = x$  and  $u_p = A + Bx$ , where  $3B + 3(A + Bx) = x$ . Therefore,  $3B = 1, 3A + 3B = 0$ , so  $B = \frac{1}{3}, A = -\frac{1}{3}$ , and  $u_p = -\frac{1}{3}(1 - x)$ ,

so  $y_{p_1} = -\frac{e^x}{3}(1 - x)$ . To find a particular solution of (B) we write  $y = ue^{-x}$ . Then  $y'' + y' + y = e^{-x}[(u'' - 2u' + u) + (u' - u) + u] = e^{-x}(1 + 2x)$ , so  $u'' - u' + u = 1 + 2x$  and  $u_p = A + Bx$ , where  $-B + (A + Bx) = 1 + 2x$ . Therefore,  $B = 2, A - B = 1$ , so  $A = 3$ , and  $u_p = 2 + 3x$ , so  $y_{p_2} = e^{-x}(3 + 2x)$ . Now  $y_p = y_{p_1} + y_{p_2} = -\frac{e^x}{3}(1 - x) + e^{-x}(3 + 2x)$ .

**5.4.26.** We must find particular solutions  $y_{p_1}$  and  $y_{p_2}$  of (A)  $y'' - 8y' + 16y = 6xe^{4x}$  and (B)  $y'' - 8y' + 16y = 2 + 16x + 16x^2$ , respectively. To find a particular solution of (A) we write  $y = ue^{4x}$ . Then  $y'' - 8y' + 16y = e^x[(u'' + 8u' + 16u) - 8(u' + 4u) + 16u] = 6xe^{4x}$ , so  $u'' = 6x$ ,  $u_p = x^3$ , and  $y_{p_1} = x^3e^{4x}$ . To find a particular solution of (B) we write  $y_p = A + Bx + Cx^2$ . Then  $y_p'' - 8y_p' + 16y_p = 2C - 8(B + 2Cx) + 16(A + Bx + Cx^2) = (16A - 8B + 2C) + (16B - 16C)x + 16Cx^2 = 2 + 16x + 16x^2$  if  $16C = 16, 16B - 16C = 16, 16A - 8B + 2C = 2$ . Therefore,  $C = 1, B = 2, A = 1$ , and  $y_{p_2} = 1 + 2x + x^2$ . Now  $y_p = y_{p_1} + y_{p_2} = x^3e^{4x} + 1 + 2x + x^2$ .

**5.4.28.** We must find particular solutions  $y_{p_1}$  and  $y_{p_2}$  of (A)  $y'' - 2y' + 2y = e^x(1 + x)$  and (B)  $y'' - 2y' + 2y = e^{-x}(2 - 8x + 5x^2)$ , respectively. To find a particular solution of (A) we write  $y = ue^x$ . Then  $y'' - 2y' + 2y = e^x[(u'' + 2u' + u) - 2(u' + u) + 2u] = e^x(1 + x)$ , so  $u'' + u = 1 + x$  and  $u_p = 1 + x$ , so  $y_{p_1} = e^x(1 + x)$ . To find a particular solution of (B) we write  $y = ue^{-x}$ . Then  $y'' - 2y' + 2y = e^{-x}[(u'' - 2u' + u) - 2(u' - u) + 2u] = e^{-x}(2 - 8x + 5x^2)$ , so  $u'' - 4u' + 5u = 2 - 8x + 5x^2$  and  $u_p = A + Bx + Cx^2$ , where  $2C - 4(B + 2Cx) + 5(A + Bx + Cx^2) = 2 - 8x + 5x^2$ . Therefore,  $5C = 5, 5B - 8C = -8, 5A - 4B + 2C = 2$ , so  $C = 1, B = 0, A = 0$ , and  $u_p = x^2$ . Therefore,  $y_{p_2} = x^2e^{-x}$ . Now  $y_p = y_{p_1} + y_{p_2} = e^x(1 + x) + x^2e^{-x}$ .

**5.4.30.** (a) If  $y = ue^{\alpha x}$ , then  $ay'' + by' + cy = e^{\alpha x}[a(u'' + 2\alpha u' + \alpha^2 u) + b(u' + \alpha u) + cu] = e^{\alpha x}[au'' + (2a\alpha + b)u' + (a\alpha^2 + b\alpha + c)u] = e^{\alpha x}[au'' + p'(\alpha)u' + p(\alpha)u]$ . Therefore,  $ay'' + by' + cy = e^{\alpha x}G(x)$  if and only if  $au'' + p'(\alpha)u' + p(\alpha)u = G(x)$ .

(b) Substituting  $u_p = A + Bx + Cx^2 + Dx^3$  into (B) yields

$$\begin{aligned} & a(2C + 6Dx) + p'(\alpha)(B + 2Cx + 3Dx^2) + p(\alpha)(A + Bx + Cx^2 + Dx^3) \\ &= [p(\alpha)A + p'(\alpha)B + 2aC] + [p(\alpha)B + 2p'(\alpha)C + 6aD]x \\ & \quad + [p(\alpha)C + 3p'(\alpha)D]x^2 + p(\alpha)Dx^3 = g_0 + g_1x + g_2x^2 + g_3x^3 \end{aligned}$$

if

$$\begin{aligned} p(\alpha)D &= g_3 \\ p(\alpha)C + 3p'(\alpha)D &= g_2 \\ p(\alpha)B + 2p'(\alpha)C + 6aD &= g_1 \\ p(\alpha)A + p'(\alpha)B + 2aC &= g_0. \end{aligned} \tag{C}$$

Since  $e^{\alpha x}$  is not a solution of the complementary equation,  $p(\alpha) \neq 0$ . Therefore, the triangular system (C) can be solved successively for  $D$ ,  $C$ ,  $B$  and  $A$ .

(c) Since  $e^{\alpha x}$  is a solution of the complementary equation while  $xe^{\alpha x}$  is not,  $p(\alpha) = 0$  and  $p'(\alpha) \neq 0$ . Therefore, (B) reduces to (D)  $au'' + p'(\alpha)u = G(x)$ . Substituting  $u_p = Ax + Bx^2 + Cx^3 + Dx^4$  into (D) yields

$$\begin{aligned} &a(2B + 6Cx + 12Dx^2) + p'(\alpha)(A + 2Bx + 3Cx^2 + 4Dx^3) \\ &= (p'(\alpha)A + 2aB) + (2p'(\alpha)B + 6aC)x + (3p'(\alpha)C + 12aD)x^2 \\ &\quad + 4p'(\alpha)Dx^3 = g_0 + g_1x + g_2x^2 + g_3x^3 \end{aligned}$$

if

$$\begin{aligned} 4p'(\alpha)D &= g_3 \\ 3p'(\alpha)C + 12aD &= g_2 \\ 2p'(\alpha)B + 6aC &= g_1 \\ p'(\alpha)A + 2aB &= g_0. \end{aligned}$$

Since  $p'(\alpha) \neq 0$  this triangular system can be solved successively for  $D$ ,  $C$ ,  $B$  and  $A$ .

(d) Since  $e^{\alpha x}$  and  $xe^{\alpha x}$  are solutions of the complementary equation,  $p(\alpha) = 0$  and  $p'(\alpha) = 0$ . Therefore, (B) reduces to (D)  $au'' = G(x)$ , so  $u'' = \frac{G(x)}{a}$ . Integrating this twice and taking the constants of integration yields the particular solution  $u_p = x^2\left(\frac{g_0}{2} + \frac{g_1}{6}x + \frac{g_2}{12}x^2 + \frac{g_3}{20}x^3\right)$ .

**5.4.32.** If  $y_p = Axe^{4x}$ , then  $y_p'' - 7y_p' + 12y_p = [(8 + 16x) - 7(1 + 4x) + 12x]Ae^{4x} = Ae^{4x} = 5e^{4x}$  if  $A = 1$ , so  $y_p = 5xe^{4x}$ .

**5.4.34.** If  $y_p = e^{3x}(A + Bx + Cx^2)$ , then

$$\begin{aligned} y_p'' - 3y_p' + 2y_p &= e^{3x}[(9A + 6B + 2C) + (9B + 12C)x + 9Cx^2] \\ &\quad - 3e^{3x}[(3A + B) + (3B + 2C)x + 3Cx^2] \\ &\quad + 2e^{3x}(A + Bx + Cx^2) \\ &= e^{3x}[(2A + 3B + 2C) + (2B + 6C)x + 2Cx^2] \\ &= e^{3x}(-1 + 2x + x^2) \end{aligned}$$

if  $2C = 1$ ,  $2B + 6C = 2$ ,  $2A + 3B + 2C = -1$ . Therefore,  $C = \frac{1}{2}$ ,  $B = -\frac{1}{2}$ ,  $A = -\frac{1}{4}$ , and  $y_p = -\frac{e^{3x}}{4}(1 + 2x - 2x^2)$ .



**5.4.36.** If  $y_p = e^{-x/2}(Ax^2 + Bx^3 + Cx^4)$ , then

$$\begin{aligned} 4y_p'' + 4y_p' + y_p &= e^{-x/2}[8A - (8A - 24B)x + (A - 12B + 48C)x^2] \\ &\quad + e^{-x/2}[(B - 16C)x^3 + Cx^4] \\ &\quad + e^{-x/2}[8Ax - (2A - 12B)x^2 - (2B - 16C)x^3 - 2Cx^4] \\ &\quad + e^{-x/2}(Ax^2 + Bx^3 + Cx^4) \\ &= e^{-x/2}(8A + 24Bx + 48Cx^2) = e^{-x/2}(-8 + 48x + 144x^2) \end{aligned}$$

if  $48C = 144$ ,  $24B = 48$ , and  $8A = -8$ . Therefore,  $C = 3$ ,  $B = 2$ ,  $A = -1$ , and  $y_p = x^2e^{-x/2}(-1 + 2x + 3x^2)$ .

**5.4.38.** If  $y = \int e^{\alpha x} P(x) dx$ , then  $y' = e^{\alpha x} P(x)$ . Let  $y = ue^{\alpha x}$ ; then  $(u' + \alpha u)e^{\alpha x} = e^{\alpha x} P(x)$ , which implies (A). We must show that it is possible to choose  $A_0, \dots, A_k$  so that (B)  $(A_0 + A_1x \cdots + A_kx^k)' + \alpha(A_0 + A_1x \cdots + A_kx^k) = p_0 + p_1x + \cdots + p_kx^k$ . By equating the coefficients of  $x^k, x^{k-1}, \dots, 1$  (in that order) on the two sides of (B), we see that (B) holds if and only if  $\alpha A_k = p_k$  and  $(k - j + 1)A_{k-j+1} + \alpha A_k = p_{k-j}$ ,  $1 \leq j \leq k$ .

**5.4.40.** If  $y = \int x^k e^{\alpha x} dx$ , then  $y' = x^k e^{\alpha x}$ . Let  $y = ue^{\alpha x}$ ; then  $(u' + \alpha u)e^{\alpha x} = x^k e^{\alpha x}$ , so  $u' + \alpha u = x^k$ . This equation has a particular solution  $u_p = A_0 + A_1x \cdots + A_kx^k$ , where (A)  $(A_0 + A_1x \cdots + A_kx^k)' + \alpha(A_0 + A_1x \cdots + A_kx^k) = x^k$ . By equating the coefficients of  $x^k, x^{k-1}, \dots, 1$  on the two sides of (A), we see that (A) holds if and only if  $\alpha A_k = 1$  and  $(k - j + 1)A_{k-j+1} + \alpha A_k - j = 0$ ,  $1 \leq j \leq k$ . Therefore,  $A_k = \frac{1}{\alpha}$ ,  $A_{k-1} = -\frac{k}{\alpha^2}$ ,  $A_{k-2} = \frac{k(k-1)}{\alpha^3}$ , and, in general,  $A_{k-j} = (-1)^j \frac{k(k-1) \cdots (k-j+1)}{\alpha^{j+1}} = \frac{(-1)^j k!}{\alpha^{j+1}(k-j)!}$ ,  $1 \leq j \leq k$ . By introducing the index  $r = k - j$

we can rewrite this as  $A_r = \frac{(-1)^{k-r} k!}{\alpha^{k-r+1} r!}$ ,  $0 \leq r \leq k$ . Therefore,  $u_p = \frac{(-1)^k k!}{\alpha^{k+1}} \sum_{r=0}^k \frac{(-\alpha x)^r}{r!}$  and  $y = \frac{(-1)^k k! e^{\alpha x}}{\alpha^{k+1}} \sum_{r=0}^k \frac{(-\alpha x)^r}{r!} + c$ .

## 5.5 The Method of Undetermined Coefficients II

**5.5.2.** Let

$$\begin{aligned} y_p &= (A_0 + A_1x) \cos x + (B_0 + B_1x) \sin x; \text{ then} \\ y_p' &= (A_1 + B_0 + B_1x) \cos x + (B_1 - A_0 - A_1x) \sin x \\ y_p'' &= (2B_1 - A_0 - A_1x) \cos x - (2A_1 + B_0 + B_1x) \sin x, \text{ so} \end{aligned}$$

$$\begin{aligned} y_p'' + 3y_p' + y_p &= (3A_1 + 3B_0 + 2B_1 + 3B_1x) \cos x \\ &\quad + (3B_1 - 3A_0 - 2A_1 - 3A_1x) \sin x \\ &= (2 - 6x) \cos x - 9 \sin x \end{aligned}$$

if  $3B_1 = -6$ ,  $-3A_1 = 0$ ,  $3B_0 + 3A_1 + 2B_1 = 2$ ,  $-3A_0 + 3B_1 + 2A_1 = -9$ . Therefore,  $A_1 = 0$ ,  $B_1 = -2$ ,  $A_0 = 1$ ,  $B_0 = 2$ , and  $y_p = \cos x + (2 - 2x) \sin x$ .

**5.5.4.** Let  $y = ue^{2x}$ . Then

$$\begin{aligned} y'' + 3y' - 2y &= e^{2x} [(u'' + 4u' + 4u) + 3(u' + 2u) - 2u] \\ &= e^{2x}(u'' + 7u' + 8u) = -e^{2x}(5 \cos 2x + 9 \sin 2x) \end{aligned}$$

if  $u'' + 7u' + 8u = -5 \cos 2x - 9 \sin 2x$ . Now let  $u_p = A \cos 2x + B \sin 2x$ . Then

$$\begin{aligned} u_p'' + 7u_p' + 8u_p &= -4(A \cos 2x + B \sin 2x) + 14(-A \sin 2x + B \cos 2x) \\ &\quad + 8(A \cos 2x + B \sin 2x) \\ &= (4A + 14B) \cos 2x - (14A - 4B) \sin 2x \\ &= -5 \cos 2x - 9 \sin 2x \end{aligned}$$

if  $4A + 14B = -5$ ,  $-14A + 4B = -9$ . Therefore,  $A = \frac{1}{2}$ ,  $B = -\frac{1}{2}$ ,

$$u_p = \frac{1}{2}(\cos 2x - \sin 2x), \text{ and } y_p = \frac{e^{2x}}{2}(\cos 2x - \sin 2x).$$

**5.5.6.** Let  $y = ue^{-2x}$ . Then

$$\begin{aligned} y'' + 3y' - 2y &= e^{-2x} [(u'' - 4u' + 4u) + 3(u' - 2u) - 2u] \\ &= e^{-2x}(u'' - u' - 4u) \\ &= e^{-2x}[(4 + 20x) \cos 3x + (26 - 32x) \sin 3x] \end{aligned}$$

if  $u'' - u' - 4u = (4 + 20x) \cos 3x + (26 - 32x) \sin 3x$ . Let

$$\begin{aligned} u_p &= (A_0 + A_1 x) \cos 3x + (B_0 + B_1 x) \sin 3x; \text{ then} \\ u_p' &= (A_1 + 3B_0 + 3B_1 x) \cos 3x + (B_1 - 3A_0 - 3A_1 x) \sin 3x \\ u_p'' &= (6B_1 - 9A_0 - 9A_1 x) \cos 3x - (2A_1 + 9B_0 + 9B_1 x) \sin 3x, \text{ so} \end{aligned}$$

$$\begin{aligned} u_p'' - u_p' - 4u_p &= -[13A_0 + A_1 + 3B_0 - 6B_1 + (13A_1 + 3B_1)x] \cos 3x \\ &\quad - [13B_0 + B_1 - 3A_0 + 6A_1 + (13B_1 - 3A_1)x] \sin 3x \\ &= (4 + 20x) \cos 3x + (26 - 32x) \sin 3x \text{ if} \end{aligned}$$

$$\begin{aligned} -13A_1 - 3B_1 &= 20 & \text{and} & \quad -13A_0 - 3B_0 - A_1 + 6B_1 = 4 \\ 3A_1 - 13B_1 &= -32 & & \quad 3A_0 - 13B_0 - 6A_1 - B_1 = 26. \end{aligned}$$

From the first two equations,  $A_1 = -2$ ,  $B_1 = 2$ . Substituting these in the last two equations yields  $-13A_0 - 3B_0 = -10$ ,  $3A_0 - 13B_0 = 16$ . Solving this pair yields  $A_0 = 1$ ,  $B_0 = -1$ . Therefore,

$$u_p = (1 - 2x)(\cos 3x - \sin 3x) \text{ and } y_p = e^{-2x}(1 - 2x)(\cos 3x - \sin 3x).$$

**5.5.8.** Let

$$\begin{aligned} y_p &= (A_0 x + A_1 x^2) \cos x + (B_0 x + B_1 x^2) \sin x; \text{ then} \\ y_p' &= [A_0 + (2A_1 + B_0)x + B_1 x^2] \cos x + [B_0 + (2B_1 - A_0)x - B_1 x^2] \sin x \\ y_p'' &= [2A_1 + 2B_0 - (A_0 - 4B_1)x - A_1 x^2] \cos x \\ &\quad + [2B_1 - 2A_0 - (B_0 + 4A_1)x - B_1 x^2] \sin x, \text{ so} \end{aligned}$$

$$\begin{aligned} y_p'' + y_p &= (2A_1 + 2B_0 + 4B_1x) \cos x + (2B_1 - 2A_0 - 4A_1x) \sin x \\ &= (-4 + 8x) \cos x + (8 - 4x) \sin x \end{aligned}$$

if  $4B_1 = 8$ ,  $-4A_1 = -4$ ,  $2B_0 + 2A_1 = -4$ ,  $-2A_0 + 2B_1 = 8$ . Therefore,  $A_1 = 1$ ,  $B_1 = 2$ ,  $A_0 = -2$ ,  $B_0 = -3$ , and  $y_p = -x[(2-x)\cos x + (3-2x)\sin x]$ .

**5.5.10.** Let  $y = ue^{-x}$ . Then

$$\begin{aligned} y'' + 2y' + 2y &= e^{-x}[(u'' - 2u' + u) + 2(u' - u) + 2u] \\ &= e^{-x}(u'' + u) = e^{-x}(8\cos x - 6\sin x) \end{aligned}$$

if  $u'' + u = 8\cos x - 6\sin x$ . Now let

$$\begin{aligned} u_p &= Ax \cos x + Bx \sin x; \text{ then} \\ u_p' &= (A + Bx) \cos x + (B - Ax) \sin x \\ u_p'' &= (2B - Ax) \cos x - (2A + Bx) \sin x, \text{ so} \\ u_p'' + u_p &= 2B \cos x - 2A \sin x = 8 \cos x - 6 \sin x \end{aligned}$$

if  $2B = 8$ ,  $-2A = -6$ . Therefore,  $A = 3$ ,  $B = 4$ ,  $u_p = x(3\cos x + 4\sin x)$ , and  $y_p = xe^{-x}(3\cos x + 4\sin x)$ .

**5.5.12.** Let

$$\begin{aligned} y_p &= (A_0 + A_1x + A_2x^2) \cos x + (B_0 + B_1x + B_2x^2) \sin x; \text{ then} \\ y_p' &= [A_1 + B_0 + (2A_2 + B_1)x + B_2x^2] \cos x \\ &\quad + [B_1 - A_0 + (2B_2 - A_1)x - A_2x^2] \sin x, \\ y_p'' &= [-A_0 + 2A_2 + 2B_1 - (A_1 - 4B_2)x - A_2x^2] \cos x \\ &\quad + [-B_0 + 2B_2 - 2A_1 - (B_1 + 4A_2)x - B_2x^2] \sin x, \text{ so} \\ y_p'' + 2y_p' + y_p &= 2[A_1 + A_2 + B_0 + B_1 + (2A_2 + B_1 + 2B_2)x + B_2x^2] \cos x \\ &\quad + 2[B_1 + B_2 - A_0 - A_1 + (2B_2 - A_1 - 2A_2)x - A_2x^2] \sin x \\ &= 8x^2 \cos x - 4x \sin x \text{ if} \\ \text{(i)} \quad \begin{aligned} 2B_2 &= 8 \\ -2A_2 &= 0 \end{aligned}, \quad \text{(ii)} \quad \begin{aligned} 2B_1 + 4A_2 + 4B_2 &= 0 \\ -2A_1 - 4A_2 + 4B_2 &= -4 \end{aligned}, \\ \text{(iii)} \quad \begin{aligned} 2B_0 + 2A_1 + 2B_1 + 2A_2 &= 0 \\ -2A_0 - 2A_1 + 2B_1 + 2B_2 &= 0 \end{aligned}. \end{aligned}$$

From (i),  $A_2 = 0$ ,  $B_2 = 4$ . Substituting these into (ii) and solving for  $A_1$  and  $B_1$  yields  $A_1 = 10$ ,  $B_1 = -8$ . Substituting the known coefficients into (iii) and solving for  $A_0$  and  $B_0$  yields  $A_0 = -14$ ,  $B_0 = -2$ . Therefore,  $y_p = -(14 - 10x) \cos x - (2 + 8x - 4x^2) \sin x$ .

**5.5.14.** Let

$$\begin{aligned} y_p &= (A_0 + A_1x + A_2x^2) \cos 2x + (B_0 + B_1x + B_2x^2) \sin 2x; \text{ then} \\ y_p' &= [A_1 + 2B_0 + (2A_2 + 2B_1)x + 2B_2x^2] \cos 2x \\ &\quad + [B_1 - 2A_0 + (2B_2 - 2A_1)x - 2A_2x^2] \sin 2x \\ y_p'' &= [-4A_0 + 2A_2 + 4B_1 - (4A_1 - 8B_2)x - 4A_2x^2] \cos 2x \\ &\quad + [-4B_0 - 2B_2 - 4A_1 - (4B_1 + 8A_2)x - 4B_2x^2] \sin 2x, \text{ so} \end{aligned}$$

$$\begin{aligned}
y_p'' + 3y_p' + 2y_p &= [-2A_0 + 3A_1 + 4A_2 + 6B_0 + 4B_1 \\
&\quad -(2A_1 - 6A_2 - 6B_1 - 8B_2)x - (2A_2 - 6B_2)x^2] \cos 2x \\
&\quad + [-2B_0 + 3B_1 + 4B_2 - 6A_0 - 4A_1 \\
&\quad -(2B_1 - 6B_2 + 6A_1 + 8A_2)x - (2B_2 + 6A_2)x^2] \sin 2x \\
&= (1 - x - 4x^2) \cos 2x - (1 + 7x + 2x^2) \sin 2x \text{ if}
\end{aligned}$$

$$\begin{aligned}
\text{(i)} \quad \begin{aligned} -2A_2 + 6B_2 &= -4 \\ -6A_2 - 2B_2 &= -2 \end{aligned} \quad , \quad \text{(ii)} \quad \begin{aligned} -2A_1 + 6B_1 + 6A_2 + 8B_2 &= -1 \\ -6A_1 - 2B_1 - 8A_2 + 6B_2 &= -7 \end{aligned} ,
\end{aligned}$$

$$\begin{aligned}
\text{(iii)} \quad \begin{aligned} -2A_0 + 6B_0 + 3A_1 + 4B_1 + 2A_2 &= 1 \\ -6A_0 - 2B_0 - 4A_1 + 3B_1 + 2B_2 &= -1 \end{aligned} .
\end{aligned}$$

From (i),  $A_2 = \frac{1}{2}$ ,  $B_2 = -\frac{1}{2}$ . Substituting these into (ii) and solving for  $A_1$  and  $B_1$  yields  $A_1 = 0$ ,  $B_1 = 0$ . Substituting the known coefficients into (iii) and solving for  $A_0$  and  $B_0$  yields  $A_0 = 0$ ,  $B_0 = 0$ . Therefore,  $y_p = \frac{x^2}{2}(\cos 2x - \sin 2x)$ .

**5.5.16.** Let  $y = ue^x$ . Then

$$\begin{aligned}
y'' - 2y' + y &= e^x [(u'' + 2u' + u) - 2(u' + u) + u] = e^x u'' \\
&= -e^x [(3 + 4x - x^2) \cos x + (3 - 4x - x^2) \sin x]
\end{aligned}$$

if  $u'' = -(3 + 4x - x^2) \cos x - (3 - 4x - x^2) \sin x$ . Now let

$$\begin{aligned}
u_p &= (A_0 + A_1x + A_2x^2) \cos x + (B_0 + B_1x + B_2x^2) \sin x; \text{ then} \\
u_p' &= [A_1 + B_0 + (2A_2 + B_1)x + B_2x^2] \cos x \\
&\quad + [B_1 - A_0 + (2B_2 - A_1)x - A_2x^2] \sin x, \\
u_p'' &= [-A_0 + 2A_2 + 2B_1 - (A_1 - 4B_2)x - A_2x^2] \cos x \\
&\quad + [-B_0 + 2B_2 - 2A_1 - (B_1 + 4A_2)x - B_2x^2] \sin x \\
&= -(3 + 4x - x^2) \cos x - (3 - 4x - x^2) \sin x \text{ if}
\end{aligned}$$

$$\begin{aligned}
\text{(i)} \quad \begin{aligned} -A_2 &= 1 \\ -B_2 &= 1 \end{aligned} \quad , \quad \text{(ii)} \quad \begin{aligned} -A_1 + 4B_2 &= -4 \\ -B_1 - 4A_2 &= 4 \end{aligned} ,
\end{aligned}$$

$$\begin{aligned}
\text{(iii)} \quad \begin{aligned} -A_0 + 2B_1 + 2A_2 &= -3 \\ -B_0 - 2A_1 + 2B_2 &= -3 \end{aligned} .
\end{aligned}$$

From (i),  $A_2 = -1$ ,  $B_2 = -1$ . Substituting these into (ii) and solving for  $A_1$  and  $B_1$  yields  $A_1 = 0$ ,  $B_1 = 0$ . Substituting the known coefficients into (iii) and solving for  $A_0$  and  $B_0$  yields  $A_0 = 1$ ,  $B_0 = 1$ . Therefore,  $u_p = (1 - x^2)(\cos x + \sin x)$  and  $y_p = e^x(1 - x^2)(\cos x + \sin x)$ .

**5.5.18.** Let  $y = ue^{-x}$ . Then

$$\begin{aligned}
y'' + 2y' + y &= e^{-x} [(u'' - 2u' + u) + 2(u' - u) + u] \\
&= e^{-x} u'' = e^{-x} [(5 - 2x) \cos x - (3 + 3x) \sin x]
\end{aligned}$$

if  $u'' = (5 - 2x) \cos x - (3 + 3x) \sin x$ . Let

$$\begin{aligned} u_p &= (A_0 + A_1 x) \cos x + (B_0 + B_1 x) \sin x; \text{ then} \\ u'_p &= (A_1 + B_0 + B_1 x) \cos x + (B_1 - A_0 - A_1 x) \sin x \\ u''_p &= (2B_1 - A_0 - A_1 x) \cos x - (2A_1 + B_0 + B_1 x) \sin x \\ &= (5 - 2x) \cos x - (3 + 3x) \sin x \end{aligned}$$

if  $-A_1 = -2$ ,  $-B_1 = -3$ ,  $-A_0 + 2B_1 = 5$ ,  $-B_0 - 2A_1 = -3$ . Therefore,  $A_1 = 2$ ,  $B_1 = 3$ ,  $A_0 = 1$ ,  $B_0 = -1$ ,  $u_p = e^{-x} [(1 + 2x) \cos x - (1 - 3x) \sin x]$ , and  $y_p = e^{-x} [(1 + 2x) \cos x - (1 - 3x) \sin x]$ .

**5.5.20.** Let

$$\begin{aligned} y_p &= (A_0 x + A_1 x^2 + A_2 x^3) \cos x + (B_0 x + B_1 x^2 + B_2 x^3) \sin x; \text{ then} \\ y'_p &= [A_0 + (2A_1 + B_0)x + (3A_2 + B_1)x^2 + B_2 x^3] \cos x \\ &\quad + [B_0 + (2B_1 - A_0)x + (3B_2 - A_1)x^2 - A_2 x^3] \sin x \\ y''_p &= [2A_1 + 2B_0 - (A_0 - 6A_2 - 4B_1)x - (A_1 - 6B_2)x^2 - A_2 x^3] \cos x \\ &\quad + [2B_1 - 2A_0 - (B_0 + 6B_2 + 4A_1)x - (B_1 + 6A_2)x^2 - B_2 x^3] \sin x, \text{ so} \end{aligned}$$

$$\begin{aligned} y''_p + y_p &= [2A_1 + 2B_0 + (6A_2 + 4B_1)x + 6B_2 x^2] \cos x \\ &\quad + [2B_1 - 2A_0 + (6B_2 - 4A_1)x - 6A_2 x^2] \sin x \\ &= (2 + 2x) \cos x + (4 + 6x^2) \sin x \text{ if} \end{aligned}$$

$$\text{(i)} \quad \begin{matrix} 6B_2 &= & 0 \\ -6A_2 &= & 6 \end{matrix}, \text{ (ii)} \quad \begin{matrix} 4B_1 + 6A_2 &= & 2 \\ -4A_1 + 6B_2 &= & 0 \end{matrix}, \text{ (iii)} \quad \begin{matrix} 2B_0 + 2A_1 &= & 2 \\ -2A_0 + 2B_1 &= & 4 \end{matrix}.$$

From (i),  $A_2 = -1$ ,  $B_2 = 0$ . Substituting these into (ii) and solving for  $A_1$  and  $B_1$  yields  $A_1 = 0$ ,  $B_1 = 2$ . Substituting the known coefficients into (iii) and solving for  $A_0$  and  $B_0$  yields  $A_0 = 0$ ,  $B_0 = 2$ . Therefore,  $y_p = -x^3 \cos x + (x + 2x^2) \sin x$ .

**5.5.22.** Let  $y = ue^x$ . Then

$$\begin{aligned} y'' - 7y' + 6y &= e^x [(u'' + 2u' + u) - 7(u' + u) + 6u] \\ &= e^x (u'' - 5u') = -e^x (17 \cos x - 7 \sin x) \end{aligned}$$

if  $u'' - 5u' = -17 \cos x + 7 \sin x$ . Now let  $u_p = A \cos x + B \sin x$ . Then

$$\begin{aligned} u''_p - 5u'_p &= -(A \cos x + B \sin x) - 5(-A \sin x + B \cos x) \\ &= (-A - 5B) \cos x - (B - 5A) \sin x = -17 \cos x + 7 \sin x \end{aligned}$$

if  $-A - 5B = -17$ ,  $5A - B = 7$ . Therefore,  $A = 2$ ,  $B = 3$ ,  $u_p = 2 \cos x + 3 \sin x$ , and  $y_p = e^x (2 \cos x + 3 \sin x)$ . The characteristic polynomial of the complementary equation is  $p(r) = r^2 - 7r + 6 = (r - 1)(r - 6)$ , so  $\{e^x, e^{6x}\}$  is a fundamental set of solutions of the complementary equation. Therefore, (A)  $y = e^x (2 \cos x + 3 \sin x) + c_1 e^x + c_2 e^{6x}$  is the general solution of the nonhomogeneous equation. Differentiating (A) yields  $y' = e^x (2 \cos x + 3 \sin x) + e^x (-2 \sin x + 3 \cos x) + c_1 e^x + 6c_2 e^{6x}$ , so  $y(0) = 4$ ,  $y'(0) = 2 \Rightarrow 4 = 2 + c_1 + c_2$ ,  $2 = 2 + 3 + c_1 + 6c_2 \Rightarrow c_1 + c_2 = 2$ ,  $c_1 + 6c_2 = -3$ , so  $c_1 = 3$ ,  $c_2 = -1$ , and  $y = e^x (2 \cos x + 3 \sin x) + 3e^x - e^{6x}$ .

**5.5.24.** Let  $y = ue^x$ . Then

$$\begin{aligned} y'' + 6y' + 10y &= e^x [(u'' + 2u' + u) + 6(u' + u) + 10u] \\ &= e^x (u'' + 8u' + 17u) = -40e^x \sin x \end{aligned}$$

if  $u'' + 8u' + 17u = -40 \sin x$ . Let  $u_p = A \cos x + B \sin x$ . Then

$$\begin{aligned} u_p'' + 6u_p' + 17u_p &= -(A \cos x + B \sin x) + 8(-A \sin x + B \cos x) \\ &\quad + 17(A \cos x + B \sin x) \\ &= (16A + 8B) \cos x - (8A - 16B) \sin x = -40 \sin x \end{aligned}$$

if  $16A + 8B = 0$ ,  $-8A + 16B = -40$ . Therefore,  $A = 1$ ,  $B = -2$ , and  $y_p = e^x(\cos x - 2 \sin x)$ . The characteristic polynomial of the complementary equation is  $p(r) = r^2 + 6r + 10 = (r + 3)^2 + 1$ , so  $\{e^{-3x} \cos x, e^{-3x} \sin x\}$  is a fundamental set of solutions of the complementary equation, and (A)  $y = e^x(\cos x - 2 \sin x) + e^{-3x}(c_1 \cos x + c_2 \sin x)$  is the general solution of the nonhomogeneous equation. Therefore,  $y(0) = 2 \Rightarrow 2 = 1 + c_1$ , so  $c_1 = 1$ . Differentiating (A) yields  $y' = e^x(\cos x - 2 \sin x) - e^x(\sin x + 2 \cos x) - 3e^{-3x}(c_1 \cos x + c_2 \sin x) + e^{-3x}(-c_1 \sin x + c_2 \cos x)$ . Therefore,  $y'(0) = -3 \Rightarrow -3 = 1 - 2 - 3c_1 + c_2$ , so  $c_2 = 1$ , and  $y = e^x(\cos x - 2 \sin x) + e^{-3x}(\cos x + \sin x)$ .

**5.5.26.** Let  $y = ue^{3x}$ . Then

$$\begin{aligned} y'' - 3y' + 2y &= e^{3x} [(u'' + 6u' + 9u) - 3(u' + 3u) + 2u] \\ &= e^{3x} (u'' + 3u' + 2u) = e^{3x} [21 \cos x - (11 + 10x) \sin x] \end{aligned}$$

if  $u'' + 3u' + 2u = 21 \cos x - (11 + 10x) \sin x$ . Now let

$$\begin{aligned} u_p &= (A_0 + A_1 x) \cos x + (B_0 + B_1 x) \sin x; \text{ then} \\ u_p' &= (A_1 + B_0 + B_1 x) \cos x + (B_1 - A_0 - A_1 x) \sin x \\ u_p'' &= (2B_1 - A_0 - A_1 x) \cos x - (2A_1 + B_0 + B_1 x) \sin x, \text{ so} \end{aligned}$$

$$\begin{aligned} u'' + 3u' + 2u &= [A_0 + 3A_1 + 3B_0 + 2B_1 + (A_1 + 3B_1)x] \cos x \\ &\quad + [B_0 + 3B_1 - 3A_0 - 2A_1 + (B_1 - 3A_1)x] \sin x \\ &= 21 \cos x - (11 + 10x) \sin x \text{ if} \end{aligned}$$

$$\begin{aligned} A_1 + 3B_1 &= 0 & \text{and} & & A_0 + 3B_0 + 3A_1 + 2B_1 &= 21 \\ -3A_1 + B_1 &= -10 & & & -3A_0 + B_0 - 2A_1 + 3B_1 &= -11 \end{aligned}$$

From the first two equations  $A_1 = 3$ ,  $B_1 = -1$ . Substituting these in last two equations yields and solving for  $A_0$  and  $B_0$  yields  $A_0 = 2$ ,  $B_0 = 4$ . Therefore,  $u_p = (2 + 3x) \cos x + (4 - x) \sin x$  and  $y_p = e^{3x} [(2 + 3x) \cos x + (4 - x) \sin x]$ . The characteristic polynomial of the complementary equation is  $p(r) = r^2 - 3r + 2 = (r - 1)(r - 2)$ , so  $\{e^x, e^{2x}\}$  is a fundamental set of solutions of the complementary equation, and (A)  $y = e^{3x} [(2 + 3x) \cos x + (4 - x) \sin x] + c_1 e^x + c_2 e^{2x}$  is the general solution of the nonhomogeneous equation. Differentiating (A) yields

$$\begin{aligned} y' &= 3e^{3x} [(2 + 3x) \cos x + (4 - x) \sin x] \\ &\quad + e^{3x} [(7 - x) \cos x - (3 + 3x) \sin x] + c_1 e^x + 2c_2 e^{2x}. \end{aligned}$$

Therefore,  $y(0) = 0$ ,  $y'(0) = 6 \Rightarrow 0 = 2 + c_1 + c_2$ ,  $6 = 6 + 7 + c_1 + 2c_2$ , so  $c_1 + c_2 = -2$ ,  $c_1 + 2c_2 = -7$ . Therefore,  $c_1 = 3$ ,  $c_2 = -5$ , and  $y = e^{3x} [(2 + 3x) \cos x + (4 - x) \sin x] + 3e^x - 5e^{2x}$ .

**5.5.28.** We must find particular solutions  $y_{p_1}$ ,  $y_{p_2}$ , and  $y_{p_3}$  of (A)  $y'' + y = 4 \cos x - 2 \sin x$  and (B)  $y'' + y = xe^x$ , and (C)  $y'' + y = e^{-x}$ , respectively. To find a particular solution of (A) we write

$$\begin{aligned} y_{p_1} &= Ax \cos x + Bx \sin x; \text{ then} \\ y'_{p_1} &= (A + Bx) \cos x + (B - Ax) \sin x \\ y''_{p_1} &= (2B - Ax) \cos x - (2A + Bx) \sin x, \text{ so} \end{aligned}$$

$y''_{p_1} + y_{p_1} = 2B \cos x - 2A \sin x = 4 \cos x - 2 \sin x$  if  $2B = 4$ ,  $-2A = -2$ . Therefore,  $A = 1$ ,  $B = 2$ , and  $y_{p_1} = x(\cos x + 2 \sin x)$ . To find a particular solution of (B) we write  $y = ue^x$ . Then

$$\begin{aligned} y'' + y &= e^x [(u'' + 2u' + u) + u] \\ &= e^x (u'' + 2u' + 2u) = xe^x \end{aligned}$$

if  $u'' + 2u' + 2u = x$ . Now  $u_p = A + Bx$ , where  $2B + 2(A + Bx) = x$ . Therefore,  $2B = 1$ ,  $2A + 2B = 0$ , so  $B = \frac{1}{2}$ ,  $A = -\frac{1}{2}$ ,  $u_p = -\frac{1}{2}(1 - x)$ , and  $y_{p_2} = -\frac{e^x}{2}(1 - x)$ . To find a particular solution of (C) we write  $y_{p_3} = Ae^{-x}$ . Then  $y''_{p_3} + y_{p_3} = 2Ae^{-x} = e^{-x}$  if  $2A = 1$ , so  $A = \frac{1}{2}$  and  $y_{p_3} = \frac{e^{-x}}{2}$ . Now  $y_p = y_{p_1} + y_{p_2} + y_{p_3} = x(\cos x + 2 \sin x) - \frac{e^x}{2}(1 - x) + \frac{e^{-x}}{2}$ .

**5.5.30.** We must find particular solutions  $y_{p_1}$ ,  $y_{p_2}$  and  $y_{p_3}$  of (A)  $y'' - 2y' + 2y = 4xe^x \cos x$ , (B)  $y'' - 2y' + 2y = xe^{-x}$ , and (C)  $y'' - 2y' + 2y = 1 + x^2$ , respectively. To find a particular solution of (A) we write  $y = ue^x$ . Then  $y'' - 2y' + 2y = e^x [(u'' + 2u' + u) - 2(u' + u) + 2u] = e^x (u'' + u) = 4xe^x \cos x$  if  $u'' + u = 4x \cos x$ . Now let

$$\begin{aligned} u_p &= (A_0x + A_1x^2) \cos x + (B_0x + B_1x^2) \sin x; \text{ then} \\ u'_p &= [A_0 + (2A_1 + B_0)x + B_1x^2] \cos x + [B_0 + (2B_1 - A_0)x - B_1x^2] \sin x \\ u''_p &= [2A_1 + 2B_0 - (A_0 - 4B_1)x - A_1x^2] \cos x \\ &\quad + [2B_1 - 2A_0 - (B_0 + 4A_1)x - B_1x^2] \sin x, \text{ so} \end{aligned}$$

$$\begin{aligned} u''_p + u_p &= (2A_1 + 2B_0 + 4B_1x) \cos x + (2B_1 - 2A_0 - 4A_1x) \sin x \\ &= 4x \cos x \end{aligned}$$

if  $4B_1 = 4$ ,  $-4A_1 = 0$ ,  $2B_0 + 2A_1 = 0$ ,  $-2A_0 + 2B_1 = 0$ . Therefore,  $A_1 = 0$ ,  $B_1 = 1$ ,  $A_0 = 1$ ,  $B_0 = 0$ ,  $u_p = x(\cos x + x \sin x)$ , and  $y_{p_1} = xe^x(\cos x + x \sin x)$ . To find a particular solution of (B) we write  $y = ue^{-x}$ . Then

$$\begin{aligned} y'' - 2y' + 2y &= e^{-x} [(u'' - 2u' + u) - 2(u' - u) + 2u] \\ &= e^{-x} (u'' - 4u' + 5u) = xe^{-x} \end{aligned}$$

if  $u'' - 4u' + 5u = x$ . Now  $u_p = A + Bx$  where  $-4B + 5(A + Bx) = x$ . Therefore,  $5B = 1$ ,  $5A - 4B = 0$ ,  $B = \frac{1}{5}$ ,  $A = \frac{4}{25}$ ,  $u_p = \frac{1}{25}(4 + 5x)$ , and  $y_{p_2} = \frac{e^{-x}}{25}(4 + 5x)$ . To find a particular

solution of (C) we write  $y_{p_3} = A + Bx + Cx^2$ . Then

$$\begin{aligned} y_{p_3}'' - 2y_{p_3}' + 2y_{p_3} &= 2C - 2(B + 2Cx) + 2(A + Bx + Cx^2) \\ &= (2A - 2B + 2C) + (2B - 4C)x + 2Cx^2 = 1 + x^2 \end{aligned}$$

if  $2A - 2B + 2C = 1$ ,  $2B - 4C = 0$ ,  $2C = 1$ . Therefore,  $C = \frac{1}{2}$ ,  $B = 1$ ,  $A = 1$ , and  $y_{p_3} = 1 + x + \frac{x^2}{2}$ .

Now  $y_p = y_{p_1} + y_{p_2} + y_{p_3} = xe^x(\cos x + x \sin x) + \frac{e^{-x}}{25}(4 + 5x) + 1 + x + \frac{x^2}{2}$ .

**5.5.32.** We must find particular solutions  $y_{p_1}$  and  $y_{p_2}$  of (A)  $y'' - 4y' + 4y = 6e^{2x}$  and (B)  $y'' - 4y' + 4y = 25 \sin x$ , respectively. To find a particular solution of (A), let  $y = ue^{2x}$ . Then

$$\begin{aligned} y'' - 4y' + 4y &= e^{2x}[(u'' + 4u' + 4u) - 4(u' + 2u) + 4u] \\ &= e^{2x}u'' = 6e^{2x} \end{aligned}$$

if  $u'' = 6$ . Integrating twice and taking the constants of integration to be zero yields  $u_p = 3x^2$ , so  $y_{p_1} = 3x^2e^{2x}$ . To find a particular solution of (B), let  $y_{p_2} = A \cos x + B \sin x$ . Then

$$\begin{aligned} y_{p_2}'' - 4y_{p_2}' + 4y_{p_2} &= -(A \cos x + B \sin x) - 4(-A \sin x + B \cos x) \\ &\quad + 4(A \cos x + B \sin x) \\ &= (3A - 4B) \cos x + (4A + 3B) \sin x = 25 \sin x \end{aligned}$$

if  $3A - 4B = 0$ ,  $4A + 3B = 25$ . Therefore,  $A = 4$ ,  $B = 3$ , and  $y_{p_2} = 4 \cos x + 3 \sin x$ . Now  $y_p = y_{p_1} + y_{p_2} = 3x^2e^{2x} + 4 \cos x + 3 \sin x$ . The characteristic polynomial of the complementary equation is  $p(r) = r^2 - 4r + 4 = (r - 2)^2$ , so  $\{e^{2x}, xe^{2x}\}$  is a fundamental set of solutions of the complementary equation. Therefore, (C)  $y = 3x^2e^{2x} + 4 \cos x + 3 \sin x + e^{2x}(c_1 + c_2x)$  is the general solution of the nonhomogeneous equation. Now  $y(0) = 5 \Rightarrow 5 = 4 + c_1$ , so  $c_1 = 1$ . Differentiating (C) yields  $y' = 6e^{2x}(x + x^2) - 4 \sin x + 3 \cos x + 2e^{2x}(c_1 + c_2x) + c_2e^{2x}$ , so  $y'(0) = 3 \Rightarrow 3 = 3 + 2 + c_2$ . Therefore,  $c_2 = -2$ , and  $y = (1 - 2x + 3x^2)e^{2x} + 4 \cos x + 3 \sin x$ .

**5.5.34.** We must find particular solutions  $y_{p_1}$  and  $y_{p_2}$  of (A)  $y'' + 4y' + 4y = 2 \cos 2x + 3 \sin 2x$  and (B)  $y'' + 4y' + 4y = e^{-x}$ , respectively. To find a particular solution of (A) we write  $y_{p_1} = A \cos 2x + B \sin 2x$ . Then

$$\begin{aligned} y_{p_1}'' + 4y_{p_1}' + 4y_{p_1} &= -4(A \cos 2x + B \sin 2x) + 8(-A \sin 2x + B \cos 2x) \\ &\quad + 4(A \cos 2x + B \sin 2x) = -8A \sin 2x + 8B \cos 2x \\ &= 2 \cos 2x + 3 \sin 2x \end{aligned}$$

if  $8B = 2$ ,  $-8A = 3$ . Therefore,  $A = -\frac{3}{8}$ ,  $B = \frac{1}{4}$ , and  $y_{p_1} = -\frac{3}{8} \cos 2x + \frac{1}{4} \sin 2x$ . To find a particular solution of (B) we write  $y_{p_2} = Ae^{-x}$ . Then  $y_{p_2}'' + 4y_{p_2}' + 4y_{p_2} = A(1 - 4 + 4)e^{-x} = Ae^{-x} = e^{-x}$  if  $A = 1$ . Therefore,  $y_{p_2} = e^{-x}$ . Now  $y_p = y_{p_1} + y_{p_2} = -\frac{3}{8} \cos 2x + \frac{1}{4} \sin 2x + e^{-x}$ . The characteristic polynomial of the complementary equation is  $p(r) = r^2 + 4r + 4 = (r + 2)^2$ , so  $\{e^{-2x}, xe^{-2x}\}$  is a fundamental set of solutions of the complementary equation. Therefore, (C)  $y = -\frac{3}{8} \cos 2x + \frac{1}{4} \sin 2x + e^{-x} + e^{-2x}(c_1 + c_2x)$  is the general solution of the nonhomogeneous equation. Now  $y(0) = -1 \Rightarrow -1 = -\frac{3}{8} + 1 + c_1$ , so  $c_1 = -\frac{13}{8}$ . Differentiating (C) yields  $y' =$



$\frac{3}{4} \sin 2x + \frac{1}{2} \cos 2x - e^{-x} - 2e^{-2x}(c_1 + c_2x) + c_2e^{-2x}$ , so  $y'(0) = 2 \Rightarrow 2 = \frac{1}{2} - 1 - 2c_1 + c_2$ . Therefore,  $c_2 = -\frac{3}{4}$ , and  $y = -\frac{3}{8} \cos 2x + \frac{1}{4} \sin 2x + e^{-x} - \frac{13}{8}e^{-2x} - \frac{3}{4}xe^{-2x}$ .

**5.5.36.** (a), (b), and (c) require only routine manipulations. (d) The coefficients of  $\sin \omega x$  in  $y'_p$ ,  $y''_p$ ,  $ay'_p + by'_p + cy_p$ , and  $y''_p + \omega^2 y_p$  can be obtained by replacing  $A$  by  $B$  and  $B$  by  $-A$  in the corresponding coefficients of  $\cos \omega x$ .

**5.5.38.** Let  $y = ue^{\lambda x}$ . Then

$$\begin{aligned} ay'' + by' + cy &= e^{\lambda x} [a(u'' + 2\lambda u' + \lambda^2 u) + b(u' + \lambda u) + cu] \\ &= e^{\lambda x} [au'' + (2a\lambda + b)u' + (a\lambda^2 + b\lambda + c)u] \\ &= e^{\lambda x} [au'' + p'(\lambda)u' + p(\lambda)u] \\ &= e^{\lambda x} (P(x) \cos \omega x + Q(x) \sin \omega x) \text{ if} \end{aligned}$$

(A)  $au'' + p'(\lambda)u' + p(\lambda)uP(x) \cos \omega x + Q(x) \sin \omega x$ , where  $p(r) = ar^2 + br + c$  is that characteristic polynomial of the complementary equation (B)  $ay'' + by' + cy = 0$ . If  $e^{\lambda x} \cos \omega x$  and  $e^{\lambda x} \sin \omega x$  are not solutions of (B), then  $\cos \omega x$  and  $\sin \omega x$  are not solutions of the complementary equation for (A). Then Theorem 5.5.1 implies that (A) has a particular solution

$$u_p = (A_0 + A_1x + \cdots + A_kx^k) \cos \omega x + (B_0 + B_1x + \cdots + B_kx^k) \sin \omega x,$$

and  $y_p = u_p e^{\lambda x}$  is a particular solution of the stated form for the given equation. If  $e^{\lambda x} \cos \omega x$  and  $e^{\lambda x} \sin \omega x$  are solutions of (B), then  $\cos \omega x$  and  $\sin \omega x$  are solutions of the complementary equation for (A). Then Theorem 5.5.1 implies that (A) has a particular solution

$$u_p = (A_0x + A_1x^2 + \cdots + A_kx^{k+1}) \cos \omega x + (B_0x + B_1x^2 + \cdots + B_kx^{k+1}) \sin \omega x,$$

and  $y_p = u_p e^{\lambda x}$  is a particular solution of the stated form for the given equation.

**5.5.40.** (a) Let  $y = \int x^2 \cos x dx$ ; then  $y' = x^2 \cos x$ . Now let

$$\begin{aligned} y_p &= (A_0 + A_1x + A_2x^2) \cos x + (B_0 + B_1x + B_2x^2) \sin x; \text{ then} \\ y'_p &= [A_1 + B_0 + (2A_2 + B_1)x + B_2x^2] \cos x \\ &\quad + [B_1 - A_0 + (2B_2 - A_1)x - A_2x^2] \sin x = x^2 \cos x \text{ if} \end{aligned}$$

$$\begin{aligned} \text{(i)} \quad \frac{B_2}{-A_2} &= \frac{1}{0}, \quad \text{(ii)} \quad \frac{B_1 + 2A_2}{-A_1 + 2B_2} = \frac{0}{0}, \quad \text{(iii)} \quad \frac{B_0 + A_1}{-A_0 + B_1} = \frac{0}{0}. \end{aligned}$$

Solving these equations yields  $A_2 = 0$ ,  $B_2 = 1$ ,  $A_1 = 2$ ,  $B_1 = 0$ ,  $A_0 = 0$ ,  $B_0 = -2$ . Therefore,  $y_p = 2x \cos x - (2 - x^2) \sin x$  and  $y = 2x \cos x - (2 - x^2) \sin x + c$ .

(b) Let  $y = \int x^2 e^x \cos x dx = ue^x$ ; then  $y' = (u' + u)e^x = x^2 e^x \cos x$  if  $u' + u = x^2 \cos x$ . Now let

$$\begin{aligned} u_p &= (A_0 + A_1x + A_2x^2) \cos x + (B_0 + B_1x + B_2x^2) \sin x; \text{ then} \\ u'_p &= [A_1 + B_0 + (2A_2 + B_1)x + B_2x^2] \cos x \\ &\quad + [B_1 - A_0 + (2B_2 - A_1)x - A_2x^2] \sin x, \text{ so} \end{aligned}$$

$$\begin{aligned} u''_p + u_p &= [A_0 + A_1 + B_0 + (A_1 + 2A_2 + B_1)x + (A_2 + B_2)x^2] \cos x \\ &\quad + [B_0 + B_1 - A_0 + (B_1 + 2B_2 - A_1)x + (B_2 - A_2)x^2] \sin x \\ &= x^2 \cos x \text{ if} \end{aligned}$$

$$(i) \begin{array}{l} A_2 + B_2 = 1 \\ -A_2 + B_2 = 0 \end{array}, (ii) \begin{array}{l} A_1 + B_1 + 2A_2 = 0 \\ -A_1 + B_1 + 2B_2 = 0 \end{array},$$

$$(iii) \begin{array}{l} A_0 + B_0 + A_1 = 0 \\ -A_0 + B_0 + B_1 = 0 \end{array}.$$

From (i),  $A_2 = \frac{1}{2}$ ,  $B_2 = \frac{1}{2}$ . Substituting these into (ii) and solving for  $A_1$  and  $B_1$  yields  $A_1 = 0$ ,  $B_1 = -1$ . Substituting these into (iii) and solving for  $A_0$  and  $B_0$  yields  $A_0 = -\frac{1}{2}$ ,  $B_0 = \frac{1}{2}$ . Therefore,  $u_p = -\frac{1}{2} [(1-x^2) \cos x - (1-x)^2 \sin x]$  and  $y = -\frac{e^x}{2} [(1-x^2) \cos x - (1-x)^2 \sin x]$ .

(c) Let  $y = \int x e^{-x} \sin 2x dx = u e^{-x}$ ; then  $y' = (u' - u)e^{-x} = x e^{-x} \sin 2x$  if  $u' - u = x \sin 2x$ . Now let

$$\begin{aligned} u_p &= (A_0 + A_1 x) \cos 2x + (B_0 + B_1 x) \sin 2x; \text{ then} \\ u'_p &= [(A_1 + 2B_0) + 2B_1 x] \cos 2x + [(B_1 - 2A_0) - 2A_1 x] \sin 2x, \text{ so} \end{aligned}$$

$$\begin{aligned} u''_p - u_p &= [-A_0 + A_1 + 2B_0 - (A_1 - 2B_1)x] \cos 2x \\ &\quad + [-B_0 + B_1 - 2A_0 - (B_1 + 2A_1)x] \sin 2x = x \sin 2x \text{ if} \end{aligned}$$

$$(i) \begin{array}{l} -A_1 + 2B_1 = 0 \\ -2A_1 - B_1 = 1 \end{array}, (ii) \begin{array}{l} -A_0 + 2B_0 + A_1 = 0 \\ -2A_0 - B_0 + B_1 = 0 \end{array}.$$

From (i),  $A_1 = -\frac{2}{5}$ ,  $B_1 = -\frac{1}{5}$ . Substituting these into (ii) and solving for  $A_0$  and  $B_0$  yields  $A_0 = -\frac{4}{25}$ ,  $B_0 = \frac{3}{25}$ . Therefore,

$$u_p = -\frac{1}{25} [(4 + 10x) \cos 2x - (3 - 5x) \sin 2x] + c \text{ and}$$

$$y_p = -\frac{e^{-x}}{25} [(4 + 10x) \cos 2x - (3 - 5x) \sin 2x] + c.$$

(d) Let  $y = \int x^2 e^{-x} \sin x dx = u e^{-x}$ ; then  $y' = (u' - u)e^{-x} = x^2 e^{-x} \sin x$  if  $u' - u = x^2 \sin x$ . Now let

$$\begin{aligned} u_p &= (A_0 + A_1 x + A_2 x^2) \cos x + (B_0 + B_1 x + B_2 x^2) \sin x; \text{ then} \\ u'_p &= [A_1 + B_0 + (2A_2 + B_1)x + B_2 x^2] \cos x \\ &\quad + [B_1 - A_0 + (2B_2 - A_1)x - A_2 x^2] \sin x, \text{ so} \end{aligned}$$

$$\begin{aligned} u''_p - u_p &= [-A_0 + A_1 + B_0 - (A_1 - 2A_2 - B_1)x - (A_2 - B_2)x^2] \cos x \\ &\quad + [-B_0 + B_1 - A_0 - (B_1 - 2B_2 + A_1)x - (B_2 + A_2)x^2] \sin x \\ &= x^2 \sin x \text{ if} \end{aligned}$$

$$(i) \begin{array}{l} -A_2 + B_2 = 0 \\ -A_2 - B_2 = 1 \end{array}, (ii) \begin{array}{l} -A_1 + B_1 + 2A_2 = 0 \\ -A_1 - B_1 + 2B_2 = 0 \end{array},$$

$$(iii) \quad \begin{aligned} -A_0 + B_0 + A_1 &= 0 \\ -A_0 - B_0 + B_1 &= 0 \end{aligned}.$$

From (i),  $A_2 = -\frac{1}{2}$ ,  $B_2 = -\frac{1}{2}$ . Substituting these into (ii) and solving for  $A_1$  and  $B_1$  yields  $A_1 = -1$ ,  $B_1 = 0$ . Substituting these into (iii) and solving for  $A_0$  and  $B_0$  yields  $A_0 = -\frac{1}{2}$ ,  $B_0 = \frac{1}{2}$ . Therefore,

$$\begin{aligned} u_p &= -\frac{e^{-x}}{2} [(1+x)^2 \cos x - (1-x^2) \sin x] \text{ and} \\ y &= -\frac{e^{-x}}{2} [(1+x)^2 \cos x - (1-x^2) \sin x] + c. \end{aligned}$$

(e) Let  $y = \int x^3 e^x \sin x \, dx = u e^x$ ; then  $y' = (u' + u)e^x = x^3 e^x \sin x$  if  $u' + u = x^3 \sin x$ . Now let

$$\begin{aligned} u_p &= (A_0 + A_1 x + A_2 x^2 + A_3 x^3) \cos x + (B_0 + B_1 x + B_2 x^2 + B_3 x^3) \sin x; \text{ then} \\ u'_p &= [A_1 + B_0 + (2A_2 + B_1)x + (3A_3 + B_2)x^2 + B_3 x^3] \cos x \\ &\quad + [B_1 - A_0 + (2B_2 - A_1)x + (3B_3 - A_2)x^2 - A_3 x^3] \sin x, \text{ so} \end{aligned}$$

$$\begin{aligned} u''_p + u_p &= [A_0 + A_1 + B_0 + (A_1 + 2A_2 + B_1)x \\ &\quad + (A_2 + 3A_3 + B_2)x^2 + (A_3 + B_3)x^3] \cos x \\ &\quad + [B_0 + B_1 - A_0 + (B_1 + 2B_2 - A_1)x \\ &\quad + (B_2 + 3B_3 - A_2)x^2 + (B_3 - A_3)x^3] \sin x = x^3 \sin x \text{ if} \end{aligned}$$

$$(i) \quad \begin{aligned} A_3 + B_3 &= 0 \\ -A_3 + B_3 &= 1 \end{aligned}, \quad (ii) \quad \begin{aligned} A_2 + B_2 + 3A_3 &= 0 \\ -A_2 + B_2 + 3B_3 &= 0 \end{aligned},$$

$$(iii) \quad \begin{aligned} A_1 + B_1 + 2A_2 &= 0 \\ -A_1 + B_1 + 2B_2 &= 0 \end{aligned}, \quad (iv) \quad \begin{aligned} A_0 + B_0 + A_1 &= 0 \\ -A_0 + B_0 + B_1 &= 0 \end{aligned}.$$

From (i),  $A_3 = -\frac{1}{2}$ ,  $B_3 = \frac{1}{2}$ . Substituting these into (ii) and solving for  $A_2$  and  $B_2$  yields  $A_2 = \frac{3}{2}$ ,  $B_2 = 0$ . Substituting these into (iii) and solving for  $A_1$  and  $B_1$  yields  $A_1 = -\frac{3}{2}$ ,  $B_1 = -\frac{3}{2}$ . Substituting these into (iv) and solving for  $A_0$  and  $B_0$  yields  $A_0 = 0$ ,  $B_0 = \frac{3}{2}$ . Therefore,

$$\begin{aligned} u_p &= -\frac{1}{2} [x(3 - 3x + x^2) \cos x - (3 - 3x + x^3) \sin x] \text{ and} \\ y &= -\frac{e^x}{2} [x(3 - 3x + x^2) \cos x - (3 - 3x + x^3) \sin x] + c. \end{aligned}$$

(f) Let  $y = \int e^x [x \cos x - (1 + 3x) \sin x] \, dx = u e^x$ ; then  $y' = (u' + u)e^x = e^x [x \cos x - (1 + 3x) \sin x]$  if  $u' + u = x \cos x - (1 + 3x) \sin x$ . Now let

$$\begin{aligned} u_p &= (A_0 + A_1 x) \cos x + (B_0 + B_1 x) \sin x; \text{ then} \\ u'_p &= [A_1 + B_0 + B_1 x] \cos x + [B_1 - A_0 - A_1 x] \sin x, \text{ so} \end{aligned}$$

$$\begin{aligned}
u_p'' + u_p &= [A_0 + A_1 + B_0 + (A_1 + B_1)x] \cos x \\
&\quad + [B_0 + B_1 - A_0 + (B_1 - A_1)x] \sin x \\
&= x \cos x - (1 + 3x) \sin x \text{ if}
\end{aligned}$$

$$\begin{aligned}
\text{(i)} \quad \begin{aligned} A_1 + B_1 &= 1 \\ -A_1 + B_1 &= -3 \end{aligned} , \quad \text{(ii)} \quad \begin{aligned} A_0 + B_0 + A_1 &= 0 \\ -A_0 + B_0 + B_1 &= -1 \end{aligned} .
\end{aligned}$$

From (i),  $A_1 = 2$ ,  $B_1 = -1$ . Substituting these into (ii) and solving for  $A_0$  and  $B_0$  yields  $A_0 = -1$ ,  $B_0 = -1$ . Therefore,  $u_p = -[(1 - 2x) \cos x + (1 + x) \sin x]$  and  $y = -e^x [(1 - 2x) \cos x + (1 + x) \sin x] + c$ .

(g) Let  $y = \int e^{-x} [(1 + x^2) \cos x + (1 - x^2) \sin x] dx = ue^{-x}$ ; then

$$y' = (u' - u)e^{-x} = e^{-x} [(1 + x^2) \cos x + (1 - x^2) \sin x]$$

if  $u' - u = (1 + x^2) \cos x + (1 - x^2) \sin x$ . Now let

$$\begin{aligned}
u_p &= (A_0 + A_1x + A_2x^2) \cos x + (B_0 + B_1x + B_2x^2) \sin x; \text{ then} \\
u_p' &= [A_1 + B_0 + (2A_2 + B_1)x + B_2x^2] \cos x \\
&\quad + [B_1 - A_0 + (2B_2 - A_1)x - A_2x^2] \sin x, \text{ so}
\end{aligned}$$

$$\begin{aligned}
u_p'' - u_p &= [-A_0 + A_1 + B_0 - (A_1 - 2A_2 - B_1)x - (A_2 - B_2)x^2] \cos x \\
&\quad + [-B_0 + B_1 - A_0 - (B_1 - 2B_2 + A_1)x - (B_2 + A_2)x^2] \sin x \\
&= (1 + x^2) \cos x + (1 - x^2) \sin x \text{ if}
\end{aligned}$$

$$\begin{aligned}
\text{(i)} \quad \begin{aligned} -A_2 + B_2 &= 1 \\ -A_2 - B_2 &= -1 \end{aligned} , \quad \text{(ii)} \quad \begin{aligned} -A_1 + B_1 + 2A_2 &= 0 \\ -A_1 - B_1 + 2B_2 &= 0 \end{aligned} ,
\end{aligned}$$

$$\begin{aligned}
\text{(iii)} \quad \begin{aligned} -A_0 + B_0 + A_1 &= 1 \\ -A_0 - B_0 + B_1 &= 1 \end{aligned} .
\end{aligned}$$

From (i),  $A_2 = 0$ ,  $B_2 = 1$ . Substituting these into (ii) and solving for  $A_1$  and  $B_1$  yields  $A_1 = 1$ ,  $B_1 = 1$ . Substituting these into (iii) and solving for  $A_0$  and  $B_0$  yields  $A_0 = 0$ ,  $B_0 = 0$ . Therefore,  $u_p = x \cos x + x(1 + x) \sin x$  and  $y = e^{-x} [x \cos x + x(1 + x) \sin x] + c$ .

## 5.6 Reduction of Order

(NOTE: The term  $uy_1''$  is indicated by “...” in some of the following solutions, where  $y_1''$  is complicated. Since this term always drops out of the differential equation for  $u$ , it is not necessary to include it.)

**5.6.2.** If  $y = ux$ , then  $y' = u'x + u$  and  $y'' = u''x + 2u'$ , so  $x^2y'' + xy' - y = x^3u'' + 3x^2u' = \frac{4}{x^2}$  if  $u' = z$ , where (A)  $z' + \frac{3}{x}z = \frac{4}{x^5}$ . Since  $\int \frac{3}{x} dx = 3 \ln|x|$ ,  $z_1 = \frac{1}{x^3}$  is a solution of the complementary equation for (A). Therefore, the solutions of (A) are of the form (B)  $z = \frac{v}{x^3}$ , where  $\frac{v'}{x^3} = \frac{4}{x^5}$ , so  $v' = \frac{4}{x^2}$ . Hence,  $v = -\frac{4}{x} + C_1$ ;  $u' = z = -\frac{4}{x^4} + \frac{C_1}{x^3}$  (see (B));  $u = \frac{4}{3x^3} - \frac{C_1}{2x^2} + C_2$ ;

$y = ux = \frac{4}{3x^2} - \frac{C_1}{2x} + C_2x$ , or  $y = \frac{4}{3x^2} + c_1x + \frac{c_2}{x}$ . As a byproduct,  $\{x, 1/x\}$  is a fundamental set of solutions of the complementary equation.

**5.6.4.** If  $y = ue^{2x}$ , then  $y' = (u' + 2u)e^{2x}$  and  $y'' = (u'' + 4u' + 4u)e^{2x}$ , so  $y'' - 3y' + 2y = (u'' + u')e^{2x} = \frac{1}{1 + e^{-x}}$  if  $u' = z$ , where (A)  $z' + z = \frac{e^{-2x}}{1 + e^{-x}}$ . Since  $z_1 = e^{-x}$  is a solution of the complementary equation for (A), the solutions of (A) are of the form (B)  $z = ve^{-x}$ , where  $v'e^{-x} = \frac{e^{-2x}}{1 + e^{-x}}$ , so  $v' = \frac{e^{-x}}{1 + e^{-x}}$ . Hence,  $v = -\ln(1 + e^{-x}) + C_1$ ;  $u' = z = -e^{-x} \ln(1 + e^{-x}) + C_1e^{-x}$  (see (B));  $u = (1 + e^{-x}) \ln(1 + e^{-x}) - 1 - e^{-x} - C_1e^{-x} + C_2$ ;  $y = ue^{2x} = (e^{2x} + e^x) \ln(1 + e^{-x}) - (C_1 + 1)e^x + (C_2 - 1)e^{2x}$ , or  $y = (e^{2x} + e^x) \ln(1 + e^{-x}) + c_1e^{2x} + c_2e^x$ . As a byproduct,  $\{e^{2x}, e^x\}$  is a fundamental set of solutions of the complementary equation.

**5.6.6.** If  $y = ux^{1/2}e^x$ , then  $y' = u'x^{1/2}e^x + u\left(x^{1/2} + \frac{x^{-1/2}}{2}\right)e^x$  and  $y'' = u''x^{1/2}e^x + 2u'\left(x^{1/2} + \frac{x^{-1/2}}{2}\right)e^x + \dots$  so  $4x^2y'' + (4x - 8x^2)y' + (4x^2 - 4x - 1)y = e^x(4x^{5/2}u'' + 8x^{3/2}u') = 4x^{1/2}e^x(1 + 4x)$  if  $u' = z$ , where (A)  $z' + \frac{2}{x}z = \frac{1 + 4x}{x^2}$ . Since  $\int \frac{2}{x} dx = 2 \ln|x|$ ,  $z_1 = \frac{2}{x^2}$  is a solution of the complementary

equation for (A). Therefore, the solutions of (A) are of the form (B)  $z = \frac{v}{x^2}$ , where  $\frac{v'}{x^2} = \frac{1 + 4x}{x^2}$ ,

so  $v' = 1 + 4x$ . Hence,  $v = x + 2x^2 + C_1$ ;  $u' = z = \frac{1}{x^2} + 2 + \frac{C_1}{x^2}$  (see (B));  $u = \ln x + 2x - \frac{C_1}{x} + C_2$ ;  $y = ux^{1/2}e^x = e^x(2x^{3/2} + x^{1/2} \ln x - C_1x^{-1/2} + C_2x^{1/2})$ , or  $y = e^x(2x^{3/2} + x^{1/2} \ln x + c_1x^{1/2} + c_2x^{-1/2})$ . As a byproduct,  $\{x^{1/2}e^x, x^{-1/2}e^{-x}\}$  is a fundamental set of solutions of the complementary equation.

**5.6.8.** If  $y = ue^{-x^2}$ , then  $y' = u'e^{-x^2} - 2xue^{-x^2}$  and  $y'' = u''e^{-x^2} - 4xu'e^{-x^2} + \dots$ , so  $y'' + 4xy' + (4x^2 + 2)y = u''e^{-x^2} = 8e^{-x(x+2)} = 8e^{-x^2}e^{-2x}$  if  $u'' = 8e^{-2x}$ . Therefore,  $u' = -4e^{-2x} + C_1$ ;  $u = 2e^{-2x} + C_1x + C_2$ , and  $y = ue^{-x^2} = e^{-x^2}(2e^{-2x} + C_1x + C_2)$ , or  $y = e^{-x^2}(2e^{-2x} + c_1 + c_2x)$ . As a byproduct,  $\{e^{-x^2}, xe^{-x^2}\}$  is a fundamental set of solutions of the complementary equation.

**5.6.10.** If  $y = uxe^{-x}$ , then  $y' = u'xe^{-x} - ue^{-x}(x - 1)$  and  $y'' = u''xe^{-x} - 2u'e^{-x}(x - 1) + \dots$ , so  $x^2y'' + 2x(x - 1)y' + (x^2 - 2x + 2)y = x^3u'' = x^3e^{2x}$  if  $u'' = e^{3x}$ . Therefore,  $u' = \frac{e^{3x}}{3} + C_1$ ;

$u = \frac{e^{3x}}{9} + C_1x + C_2$ , and  $y = uxe^{-x} = \frac{xe^{2x}}{9} + xe^{-x}(C_1x + C_2)$ , or  $y = \frac{xe^{2x}}{9} + xe^{-x}(c_1 + c_2x)$ . As a byproduct,  $\{xe^{-x}, x^2e^{-x}\}$  is a fundamental set of solutions of the complementary equation.

**5.6.12.** If  $y = ue^x$ , then  $y' = (u' + u)e^x$  and  $y'' = (u'' + 2u' + u)e^x$ , so  $(1 - 2x)y'' + 2y' + (2x - 3)y = e^x[(1 - 2x)u'' + (4 - 4x)u'] = (1 - 4x + 4x^2)e^x$  if  $u' = z$ , where (A)  $z' + \frac{4 - 4x}{1 - 2x}z = 1 - 2x$ .

Since  $\int \frac{4 - 4x}{1 - 2x} dx = \int \left(2 + \frac{2}{1 - 2x}\right) dx = 2x - \ln|1 - 2x|$ ,  $z_1 = (1 - 2x)e^{-2x}$  is a solution of the complementary equation for (A). Therefore, the solutions of (A) are of the form (B)  $z =$

$v(1 - 2x)e^{-2x}$ , where  $v'(1 - 2x)e^{-2x} = (1 - 2x)$ , so  $v' = e^{2x}$ . Hence,  $v = \frac{e^{2x}}{2} + C_1$ ;  $u' = z =$

$\left(\frac{1}{2} + C_1e^{-2x}\right)(1 - 2x)$  (see (B));  $u = -\frac{(2x - 1)^2}{8} + C_1xe^{-2x} + C_2$ ;  $y = ue^x = -\frac{(2x - 1)^2e^x}{8} +$

$C_1xe^{-x} + C_2e^x$ , or  $y = -\frac{(2x - 1)^2e^x}{8} + c_1e^x + c_2xe^{-x}$ . As a byproduct,  $\{e^x, xe^{-x}\}$  is a fundamental set of solutions of the complementary equation.

**5.6.14.** If  $y = ue^{-x}$ , then  $y' = (u' - u)e^{-x}$  and  $y'' = (u'' - 2u' + u)e^{-x}$ , so  $2xy'' + (4x + 1)y' + (2x +$

1)  $y = e^{-x}(2xu'' + u') = 3x^{1/2}e^{-x}$  if  $u' = z$ , where (A)  $z' + \frac{1}{2x}z = \frac{3}{2}x^{-1/2}$ . Since  $\int \frac{1}{2x} dx = \frac{1}{2} \ln|x|$ ,  $z_1 = x^{-1/2}$  is a solution of the complementary equation for (A). Therefore, the solutions of (A) are of the form (B)  $z = vx^{-1/2}$ , where  $v'x^{-1/2} = \frac{3}{2}x^{-1/2}$ , so  $v' = \frac{3}{2}$ . Hence,  $v = \frac{3x}{2} + C_1$ ;  $u' = z = \frac{3}{2}x^{1/2} + C_1x^{-1/2}$  (see (B));  $u = x^{3/2} + 2C_1x^{1/2} + C_2$ ;  $y = ue^{-x} = e^{-x}(x^{3/2} + 2C_1x^{1/2} + C_2)$ , or  $y = e^{-x}(x^{3/2} + c_1 + c_2x^{1/2})$ . As a byproduct, is a  $\{e^{-x}, x^{1/2}e^{-x}\}$  fundamental set of solutions of the complementary equation.

**5.6.16.** If  $y = ux^{1/2}$ , then  $y' = u'x^{1/2} + \frac{u}{2x^{1/2}}$  and  $y'' = u''x^{1/2} + \frac{u'}{x^{1/2}} + \dots$  so  $4x^2y'' - 4x(x+1)y' + (2x+3)y = 4x^{5/2}(u'' - u') = 4x^{5/2}e^{2x}$  if  $u' = z$ , where (A)  $z' - z = e^{2x}$ . Since  $z_1 = e^x$  is a solution of the complementary equation for (A), the solutions of (A) are of the form (B)  $z = ve^x$ , where  $v'e^x = e^{2x}$ , so  $v' = e^x$ . Hence,  $v = e^x + C_1$ ;  $u' = z = e^{2x} + C_1e^x$  (see (B));  $u = \frac{e^{2x}}{2} + C_1e^x + C_2$ ;  $y = ux^{1/2} = x^{1/2}\left(\frac{e^{2x}}{2} + C_1e^x + C_2\right)$ , or  $y = x^{1/2}\left(\frac{e^{2x}}{2} + c_1 + c_2e^x\right)$ . As a byproduct,  $\{x^{1/2}, x^{1/2}e^x\}$  is a fundamental set of solutions of the complementary equation.

**5.6.18.** If  $y = ue^x$ , then  $y' = (u' + u)e^x$  and  $y'' = (u'' + 2u' + u)e^x$ , so  $xy'' + (2-2x)y' + (x-2)y = e^x(xu'' + 2u') = 0$  if  $\frac{u''}{u'} = -\frac{2}{x}$ ;  $\ln|u'| = -2\ln|x| + k$ ;  $u' = \frac{C_1}{x^2}$ ;  $u = -\frac{C_1}{x} + C_2$ . Therefore,  $y = ue^x = e^x\left(-\frac{C_1}{x} + C_2\right)$  is the general solution, and  $\{e^x, e^x/x\}$  is a fundamental set of solutions.

**5.6.20.** If  $y = u \ln|x|$ , then  $y' = u' \ln|x| + \frac{u}{x}$  and  $y'' = u'' \ln|x| + \frac{2u'}{x} \dots$ , so  $x^2(\ln|x|)^2y'' - (2x \ln|x|)y' + (2 + \ln|x|)y = x^2(\ln|x|)^3u'' = 0$  if  $u'' = 0$ ;  $u' = C_1$ ;  $u = C_1x + C_2$ . Therefore,  $y = u \ln|x| = (C_1x + C_2) \ln|x|$  is the general solution, and  $\{\ln|x|, x \ln|x|\}$  is a fundamental set of solutions.

**5.6.22.** If  $y = ue^x$ , then  $y' = u'e^x + ue^x$  and  $y'' = u''e^x + 2u'e^x + ue^x$ , so  $xy'' - (2x+2)y' + (x+2)y = e^x(xu'' - 2u') = 0$  if  $\frac{u''}{u'} = \frac{2}{x}$ ;  $\ln|u'| = 2\ln|x| + k$ ;  $u' = C_1x^2$ ;  $u = \frac{C_1x^3}{3} + C_2$ . Therefore,  $y = ue^x = \left(\frac{C_1x^3}{3} + C_2\right)e^x$  is the general solution, and  $\{e^x, x^3e^x\}$  is a fundamental set of solutions.

**5.6.24.** If  $y = ux \sin x$ , then  $y' = u'x \sin x + u(x \cos x + \sin x)$  and  $y'' = u''x \sin x + 2u'(x \cos x + \sin x) + \dots$ , so  $x^2y'' - 2xy' + (x^2 + 2)y = (x^3 \sin x)u'' + 2(x^3 \cos x)u' = 0$  if  $\frac{u''}{u'} = -\frac{2 \cos x}{\sin x}$ ;  $\ln|u'| = -2\ln|\sin x| + k$ ;  $u' = \frac{C_1}{\sin^2 x}$ ;  $u = -C_1 \cot x + C_2$ . Therefore,  $y = ux \sin x = x(-C_1 \cos x + C_2 \sin x)$  is the general solution, and  $\{x \sin x, x \cos x\}$  is a fundamental set of solutions.

**5.6.26.** If  $y = ux^{1/2}$ , then  $y' = u'x^{1/2} + \frac{u}{2x^{1/2}}$  and  $y'' = u''x^{1/2} + \frac{u'}{x^{1/2}} + \dots$  so  $4x^2(\sin x)y'' - 4x(x \cos x + \sin x)y' + (2x \cos x + 3 \sin x)y = 4x^{5/2}(u'' \sin x - u' \cos x) = 0$  if  $\frac{u''}{u'} = \frac{\cos x}{\sin x}$ ;  $\ln|u'| = \ln|\sin x| + k$ ;  $u' = C_1 \sin x$ ;  $u = -C_1 \cos x + C_2$ . Therefore,  $y = ux^{1/2} = (-C_1 \cos x + C_2)x^{1/2}$  is the general solution, and  $\{x^{1/2}, x^{1/2} \cos x\}$  is a fundamental set of solutions.

**5.6.28.** If  $y = \frac{u}{x}$ , then  $y' = \frac{u'}{x} - \frac{u}{x^2}$  and  $y'' = \frac{u''}{x} - \frac{2u'}{x^2} + \dots$ , so  $(2x+1)xy'' - 2(2x^2-1)y' -$

$4(x+1)y = (2x+1)u'' - (4x+4)u' = 0$  if  $\frac{u''}{u'} = \frac{4x+4}{2x+1} = 2 + \frac{2}{2x+1}$ ;  $\ln|u'| = 2x + \ln|2x+1| + k$ ;  $u' = C_1(2x+1)e^{2x}$ ;  $u = C_1xe^{2x} + C_2$ . Therefore,  $y = \frac{u}{x} = C_1e^{2x} + \frac{C_2}{x}$  is the general solution, and  $\{1/x, e^{2x}\}$  is a fundamental set of solutions.

**5.6.30.** If  $y = ue^{2x}$ , then  $y' = (u' + 2u)e^{2x}$  and  $y'' = (u'' + 4u' + 4u)e^{2x}$ , so  $xy'' - (4x+1)y' + (4x+2)y = e^{2x}(xu'' - u') = 0$  if  $\frac{u''}{u'} = \frac{1}{x}$ ;  $\ln|u'| = \ln|x| + k$ ;  $u' = C_1x$ ;  $u = \frac{C_1x^2}{2} + C_2$ . Therefore,  $y = ue^{2x} = e^{2x}\left(\frac{C_1x^2}{2} + C_2\right)$  is the general solution, and  $\{e^{2x}, x^2e^{2x}\}$  is a fundamental set of solutions.

**5.6.32.** If  $y = ue^{2x}$ , then  $y' = (u' + 2u)e^{2x}$  and  $y'' = (u'' + 4u' + 4u)e^{2x}$ , so  $(3x-1)y'' - (3x+2)y' - (6x-8)y = e^{2x}[(3x-1)u'' + (9x-6)u'] = 0$  if  $\frac{u''}{u'} = -\frac{9x-6}{3x-1} = -3 + \frac{3}{3x-1}$ . Therefore,  $\ln|u'| = -3x + \ln|3x-1| + k$ , so  $u' = C_1(3x-1)e^{-3x}$ ,  $u = -C_1xe^{-3x} + C_2$ . Therefore, the general solution is  $y = ue^{2x} = -C_1xe^{-x} + C_2e^{2x}$ , or (A)  $y = c_1e^{2x} + c_2xe^{-x}$ . Now  $y(0) = 2 \Rightarrow c_1 = 2$ . Differentiating (A) yields  $y' = 2c_1e^{2x} + c_2(e^{-x} - xe^{-x})$ . Now  $y'(0) = 3 \Rightarrow 3 = 2c_1 + c_2$ , so  $c_2 = -1$  and  $y = 2e^{2x} - xe^{-x}$ .

**5.6.34.** If  $y = ux$ , then  $y' = u'x + u$  and  $y'' = u''x + 2u'$ , so  $x^2y'' + 2xy' - 2y = x^3u'' + 4x^2u' = x^2$  if  $u' = z$ , where (A)  $z' + \frac{4}{x}z = \frac{1}{x}$ . Since  $\int \frac{4}{x} dx = 4 \ln|x|$ ,  $z_1 = \frac{1}{x^4}$  is a solution of the complementary equation for (A). Therefore, the solutions of (A) are of the form (B)  $z = \frac{v}{x^4}$ , where  $\frac{v'}{x^4} = \frac{1}{x}$ , so  $v' = x^3$ . Hence,  $v = \frac{x^4}{4} + C_1$ ;  $u' = z = \frac{1}{4} + \frac{C_1}{x^4}$  (see (B));  $u = \frac{x}{4} - \frac{C_1}{3x^3} + C_2$ . Therefore, the general solution is  $y = ux = \frac{x^2}{4} - \frac{C_1}{3x^2} + C_2x$ , or (C)  $y = \frac{x^2}{4} + c_1x + \frac{c_2}{x^2}$ . Differentiating (C) yields  $y' = \frac{x}{2} + c_1 - 2\frac{c_2}{x^3}$ . Now  $y(1) = \frac{5}{4}$ ,  $y'(1) = \frac{3}{2} \Rightarrow c_1 + c_2 = 1$ ,  $c_1 - 2c_2 = 1$ , so  $c_1 = 1$ ,  $c_2 = 0$  and  $y = \frac{x^2}{4} + x$ .

**5.6.36.** If  $y = uy_1$ , then  $y' = u'y_1 + uy_1'$  and  $y'' = u''y_1 + 2u'y_1' + uy_1''$ , so  $y'' + p_1(x)y' + p_2(x)y = y_1u'' + (2y_1' + p_1y_1)u' = 0$  if  $u$  is any function such that (B)  $\frac{u''}{u'} = -2\frac{y_1'}{y_1} - p_1$ . If  $\ln|u'(x)| = -2\ln|y_1(x)| - \int_{x_0}^x p_1(t) dt$ , then  $u$  satisfies (B); therefore, if (C)  $u'(x) = \frac{1}{y_1^2(x)} \exp\left(-\int_{x_0}^x p_1(s) ds\right)$ , then  $u$  satisfies (B). Since  $u(x) = \int_{x_0}^x \frac{1}{y_1^2(t)} \exp\left(-\int_{x_0}^t p_1(s) ds\right) dt$  satisfies (C),  $y_2 = uy_1$  is a solution of (A) on  $(a, b)$ . Since  $\frac{y_2}{y_1} = u$  is nonconstant, Theorem 5.1.6 implies that  $\{y_1, y_2\}$  is a fundamental set of solutions of (A) on  $(a, b)$ .

**5.6.38.** (a) The associated linear equation is (A)  $z'' + k^2z = 0$ , with characteristic polynomial  $p(r) = r^2 + k^2$ . The general solution of (A) is  $z = c_1 \cos kx + c_2 \sin kx$ . Since  $z' = -kc_1 \sin kx + kc_2 \cos kx$ ,  $y = \frac{z'}{z} = \frac{-kc_1 \sin kx + kc_2 \cos kx}{c_1 \cos kx + c_2 \sin kx}$ .

(b) The associated linear equation is (A)  $z'' - 3z' + 2z = 0$ , with characteristic polynomial  $p(r) = r^2 - 3r + 2 = (r-1)(r-2)$ . The general solution of (A) is  $z = c_1e^x + c_2e^{2x}$ . Since

$$z' = c_1 e^x + c_2 e^{2x}, y = \frac{z'}{z} = \frac{c_1 + 2c_2 e^x}{c_1 + c_2 e^x}.$$

(c) The associated linear equation is (A)  $z'' + 5z' - 6z = 0$ , with characteristic polynomial  $p(r) = r^2 + 5r - 6 = (r + 6)(r - 1)$ . The general solution of (A) is  $z = c_1 e^{-6x} + c_2 e^x$ . Since

$$z' = -6c_1 e^{-6x} + c_2 e^x, y = \frac{z'}{z} = \frac{-6c_1 + c_2 e^{7x}}{c_1 + c_2 e^{7x}}.$$

(d) The associated linear equation is (A)  $z'' + 8z' + 7z = 0$ , with characteristic polynomial  $p(r) = r^2 + 8r + 7 = (r + 7)(r + 1)$ . The general solution of (A) is  $z = c_1 e^{-7x} + c_2 e^{-x}$ . Since

$$z' = -7c_1 e^{-7x} - c_2 e^{-x}, y = \frac{z'}{z} = -\frac{7c_1 + c_2 e^{6x}}{c_1 + c_2 e^{6x}}.$$

(e) The associated linear equation is (A)  $z'' + 14z' + 50z = 0$ , with characteristic polynomial  $p(r) = r^2 + 14r + 50 = (r + 7)^2 + 1$ . The general solution of (A) is  $z = e^{-7x}(c_1 \cos x + c_2 \sin x)$ . Since  $z' = -7e^{-7x}(c_1 \cos x + c_2 \sin x) + e^{-7x}(-c_1 \sin x + c_2 \cos x) = -(7c_1 - c_2) \cos x - (c_1 + 7c_2) \sin x$ ,  $y = \frac{z'}{z} = -\frac{(7c_1 - c_2) \cos x + (c_1 + 7c_2) \sin x}{c_1 \cos x + c_2 \sin x}$ .

(f) The given equation is equivalent to (A)  $y' + y^2 - \frac{1}{6}y - \frac{1}{6} = 0$ . The associated linear equation is (B)  $z'' - \frac{1}{6}z' - \frac{1}{6}z = 0$ , with characteristic polynomial  $p(r) = r^2 - \frac{1}{6}r - \frac{1}{6} = \left(r + \frac{1}{3}\right)\left(r - \frac{1}{2}\right)$ . The general solution of (B) is  $z = c_1 e^{-x/3} + c_2 e^{x/2}$ . Since  $z' = -\frac{c_1}{3}e^{-x/3} + \frac{c_2}{2}e^{x/2}$ ,  $y = \frac{z'}{z} = \frac{-2c_1 + 3c_2 e^{5x/6}}{6(c_1 + c_2 e^{5x/6})}$ .

(g) The given equation is equivalent to (A)  $y' + y^2 - \frac{1}{3}y + \frac{1}{36} = 0$ . The associated linear equation is (B)  $z'' - \frac{1}{3}z' + \frac{1}{36}z = 0$ , with characteristic polynomial  $p(r) = r^2 - \frac{1}{3}r + \frac{1}{36} = \left(r - \frac{1}{6}\right)^2$ . The general solution of (B) is  $z = e^{x/6}(c_1 + c_2 x)$ . Since  $z' = \frac{e^{x/6}}{6}(c_1 + c_2 x) + c_2 e^{x/6} = \frac{e^{x/6}}{6}(c_1 + c_2(x + 6))$ ,  $y = \frac{z'}{z} = \frac{c_1 + c_2(x + 6)}{6(c_1 + c_2 x)}$ .

**5.6.40.** (a) Suppose that  $z$  is a solution of (B) and let  $y = \frac{z'}{rz}$ . Then (D)  $\frac{z''}{rz} + \left[p(x) - \frac{r'(x)}{r(x)}\right]y + q(x) = 0$  and  $y' = \frac{z''}{rz} - \frac{1}{r}\left(\frac{z'}{z}\right)^2 - \frac{r'z'}{r^2z} = \frac{z''}{rz} - ry^2 - \frac{r'}{r}y$ , so  $\frac{z''}{rz} = y' + ry^2 + \frac{r'}{r}y$ . Therefore, (D) implies that  $y$  satisfies (A). Now suppose that  $y$  is a solution of (A) and let  $z$  be any function such that  $z' = ryz$ . Then  $z'' = r'y z + ry'z + ryz' = \frac{r'}{r}z' + (y' + ry^2)rz = \frac{r'}{r}z' - (p(x)y + q(x))rz$ , so  $z'' - \frac{r'}{r}z' + p(x)ryz + q(x)rz = 0$ , which implies that  $z$  satisfies (B), since  $ryz = z'$ .

(b) If  $\{z_1, z_2\}$  is a fundamental set of solutions of (B) on  $(a, b)$ , then  $z = c_1 z_1 + c_2 z_2$  is the general solution of (B) on  $(a, b)$ . This and (a) imply that (C) is the general solution of (A) on  $(a, b)$ .



## 5.7 Variation of Parameters

**5.7.2.** (A)  $y_p = u_1 \cos 2x + u_2 \sin 2x$ ;

$$u'_1 \cos 2x + u'_2 \sin 2x = 0 \quad (\text{B})$$

$$-2u'_1 \sin 2x + 2u'_2 \cos 2x = \sin 2x \sec^2 x. \quad (\text{C})$$

Multiplying (B) by  $2 \sin 2x$  and (C) by  $\cos 2x$  and adding the resulting equations yields  $2u'_2 = \tan 2x$ , so  $u'_2 = \frac{\tan 2x}{2}$ . Then (B) implies that  $u'_1 = -u'_2 \tan(2x) = -\frac{\tan^2 2x}{2} = \frac{1 - \sec^2 2x}{2}$ . Therefore,  $u_1 = \frac{x}{2} - \frac{\tan 2x}{4}$  and  $u_2 = -\frac{\ln |\cos 2x|}{4}$ . Now (A) yields  $y_p = -\frac{\sin 2x \ln |\cos 2x|}{4} + \frac{x \cos 2x}{2} - \frac{\sin 2x}{4}$ . Since  $\sin 2x$  satisfies the complementary equation we redefine  $y_p = -\frac{\sin 2x \ln |\cos 2x|}{4} + \frac{x \cos 2x}{2}$ .

**5.7.4.** (A)  $y_p = u_1 e^x \cos x + u_2 e^x \sin x$ ;

$$u'_1 e^x \cos x + u'_2 e^x \sin x = 0 \quad (\text{B})$$

$$u'_1 (e^x \cos x - e^x \sin x) + u'_2 (e^x \sin x + e^x \cos x) = 3e^x \sec x. \quad (\text{C})$$

Subtracting (B) from (C) and cancelling  $e^x$  from the resulting equations yields

$$u'_1 \cos x + u'_2 \sin x = 0 \quad (\text{D})$$

$$-u'_1 \sin x + u'_2 \cos x = 3 \sec x. \quad (\text{E})$$

Multiplying (D) by  $\sin x$  and (E) by  $\cos x$  and adding the results yields  $u'_2 = 3$ . From (D),  $u'_1 = -u'_2 \tan x = -3 \tan x$ . Therefore  $u_1 = 3 \ln |\cos x|$ ,  $u_2 = 3x$ . Now (A) yields  $y_p = 3e^x (\cos x \ln |\cos x| + x \sin x)$ .

**5.7.6.** (A)  $y_p = u_1 e^x + u_2 e^{-x}$ ;

$$u'_1 e^x + u'_2 e^{-x} = 0 \quad (\text{B})$$

$$u'_1 e^x - u'_2 e^{-x} = \frac{4e^{-x}}{1 + e^{-2x}}. \quad (\text{C})$$

Adding (B) to (C) yields  $2u'_1 e^x = \frac{4e^{-x}}{1 + e^{-2x}}$ , so  $u'_1 = \frac{2e^{-2x}}{1 - e^{-2x}}$ . From (B),  $u'_2 = -e^{2x} u'_1 = -\frac{2}{1 - e^{-2x}} = \frac{2e^{2x}}{1 - e^{2x}}$ . Using the substitution  $v = e^{-2x}$  we integrate  $u'_1$  to obtain  $u_1 = \ln(1 - e^{-2x})$ . Using the substitution  $v = e^{2x}$  we integrate  $u'_2$  to obtain  $u_2 = \ln(1 - e^{2x})$ . Now (A) yields  $y_p = e^x \ln(1 - e^{-2x}) - e^{-x} \ln(e^{2x} - 1)$ .

**5.7.8.** (A)  $y_p = u_1 e^x + u_2 \frac{e^x}{x}$ ;

$$u'_1 e^x + u'_2 \frac{e^x}{x} = 0 \quad (\text{B})$$

$$u'_1 e^x + u'_2 \left( \frac{e^x}{x} - \frac{e^x}{x^2} \right) = \frac{e^{2x}}{x}. \quad (\text{C})$$

Subtracting (B) from (C) yields  $-\frac{u'_2 e^x}{x^2} = \frac{e^{2x}}{x}$ , so  $u'_2 = -xe^x$ . From (B),  $u'_1 = \frac{u'_2}{x} = e^x$ . Therefore

$u_1 = e^x$ ,  $u_2 = -xe^x + e^x$ . Now (A) yields  $y_p = \frac{e^{2x}}{x}$ .

**5.7.10.** (A)  $y_p = u_1 e^{-x^2} + u_2 x e^{-x^2}$ ;

$$u'_1 e^{-x^2} + u'_2 x e^{-x^2} = 0 \quad (\text{B})$$

$$-2xu'_1 e^{-x^2} + u'_2(e^{-x^2} - 2x^2 e^{-x^2}) = 4e^{-x(x+2)}. \quad (\text{C})$$

Multiplying (B) by  $2x$  and adding the result to (C) yields  $u'_2 e^{-x^2} = 4e^{-x(x+2)}$ , so  $u'_2 = 4e^{-2x}$ . From (B),  $u'_1 = -u'_2 x = -4xe^{-2x}$ . Therefore  $u_1 = (2x+1)e^{-2x}$ ,  $u_2 = -2e^{-2x}$ . Now (A) yields  $y_p = e^{-x(x+2)}$ .

**5.7.12.** (A)  $y_p = u_1 x + u_2 x^3$ ;

$$u'_1 x + u'_2 x^3 = 0 \quad (\text{B})$$

$$u'_1 + 3u'_2 x^2 = \frac{2x^4 \sin x}{x^2} = 2x^2 \sin x \quad (\text{C}).$$

Multiplying (B) by  $\frac{1}{x}$  and subtracting the result from (C) yields  $2x^2 u'_2 = 2x^2 \sin x$ , so  $u'_2 = \sin x$ . From (B),  $u'_1 = -u'_2 x^2 = -x^2 \sin x$ . Therefore  $u_1 = (x^2 - 2) \cos x - 2x \sin x$ ,  $u_2 = -\cos x$ . Now (A) yields  $y_p = -2x^2 \sin x - 2x \cos x$ .

**5.7.14.** (A)  $y_p = u_1 \cos \sqrt{x} + u_2 \sin \sqrt{x}$ ;

$$u'_1 \cos \sqrt{x} + u'_2 \sin \sqrt{x} = 0 \quad (\text{B})$$

$$-u'_1 \frac{\sin \sqrt{x}}{2\sqrt{x}} + u'_2 \frac{\cos \sqrt{x}}{2\sqrt{x}} = \frac{\sin \sqrt{x}}{4x} \quad (\text{C}).$$

Multiplying (B) by  $\frac{\sin \sqrt{x}}{2\sqrt{x}}$  and (C) by  $\cos \sqrt{x}$  and adding the resulting equations yields  $\frac{u'_2}{2\sqrt{x}} = \frac{\sin \sqrt{x} \cos \sqrt{x}}{4x}$ , so  $u'_2 = \frac{\sin \sqrt{x} \cos \sqrt{x}}{2\sqrt{x}}$ . From (B),  $u'_1 = -u'_2 \tan \sqrt{x} = -\frac{\sin^2 \sqrt{x}}{2\sqrt{x}}$ . Therefore,  $u_1 = \frac{\sin \sqrt{x} \cos \sqrt{x}}{2} - \frac{\sqrt{x}}{2}$ ,  $u_2 = \frac{\sin^2 \sqrt{x}}{2}$ . Now (A) yields  $y_p = \frac{\sin \sqrt{x}}{2} - \frac{\sqrt{x} \cos \sqrt{x}}{2}$ . Since  $\sin \sqrt{x}$  satisfies the complementary equation we redefine  $y_p = -\frac{\sqrt{x} \cos \sqrt{x}}{2}$ .

**5.7.16.** (A)  $y_p = u_1 x^a + u_2 x^a \ln x$ ;

$$u'_1 x^a + u'_2 x^a \ln x = 0 \quad (\text{B})$$

$$au'_1 x^{a-1} + u'_2(ax^{a-1} \ln x + x^{a-1}) = \frac{x^{a+1}}{x^2} = x^{a-1} \quad (\text{C}).$$

Multiplying (B) by  $\frac{a}{x}$  and subtracting the result from (C) yields  $u'_2 x^{a-1} = x^{a-1}$ , so  $u'_2 = 1$ . From (B),  $u'_1 = -u'_2 \ln x = -\ln x$ . Therefore,  $u_1 = x - \ln x$ ,  $u_2 = x$ . Now (A) yields  $y_p = x^{a+1}$ .

**5.7.18.**  $y_p = u_1 e^{x^2} + u_2 e^{-x^2}$ ;

$$u'_1 e^{x^2} + u'_2 e^{-x^2} = 0 \quad (\text{B})$$

$$2u'_1 x e^{x^2} - 2u'_2 x e^{-x^2} = \frac{8x^5}{x} = 8x^4. \quad (\text{B})$$

Multiplying (B) by  $2x$  and adding the result to (C) yields  $4u_1'xe^{x^2} = 8x^4$ , so  $u_1' = 2x^3e^{-x^2}$ . From (B),  $u_2' = -u_1'e^{2x^2} = -2x^3e^{x^2}$ . Therefore  $u_1 = -e^{-x^2}(x^2 + 1)$ ,  $u_2 = -e^{x^2}(x^2 - 1)$ . Now (A) yields  $y_p = -2x^2$ .

**5.7.20.** (A)  $y_p = u_1\sqrt{x}e^{2x} + u_2\sqrt{x}e^{-2x}$ ;

$$u_1'\sqrt{x}e^{2x} + u_2'\sqrt{x}e^{-2x} = 0 \quad (\text{B})$$

$$u_1'e^{2x}\left(2\sqrt{x} + \frac{1}{2\sqrt{x}}\right) - u_2'e^{-2x}\left(2\sqrt{x} - \frac{1}{2\sqrt{x}}\right) = \frac{8x^{5/2}}{4x^2} = 2\sqrt{x} \quad (\text{C}).$$

Multiplying (B) by  $\frac{1}{2x}$ , subtracting the result from (C), and cancelling common factors from the resulting equations yields

$$u_1'e^{2x} + u_2'e^{-2x} = 0 \quad (\text{D})$$

$$u_1'e^{2x} - u_2'e^{-2x} = 1. \quad (\text{E})$$

Adding (D) to (E) yields  $2u_1'e^{2x} = 1$ , so  $u_1' = \frac{e^{-2x}}{2}$ . From (D),  $u_2' = -u_1'e^{4x} = -\frac{e^{2x}}{2}$ . Therefore,  $u_1 = -\frac{e^{-2x}}{4}$ ,  $u_2 = -\frac{e^{2x}}{4}$ . Now (A) yields  $y_p = -\frac{\sqrt{x}}{2}$ .

**5.7.22.** (A)  $y_p = u_1xe^x + u_2xe^{-x}$ ;

$$u_1'xe^x + u_2'xe^{-x} = 0 \quad (\text{B})$$

$$u_1'(x+1)e^x - u_2'(x-1)e^{-x} = \frac{3x^4}{x^2} = 3x^2 \quad (\text{C}).$$

Multiplying (B) by  $\frac{1}{x}$ , subtracting the resulting equation from (C), and cancelling common factors yields

$$u_1'e^x + u_2'e^{-x} = 0 \quad (\text{D})$$

$$u_1'e^x - u_2'e^{-x} = 3x. \quad (\text{E})$$

Adding (D) to (E) yields  $2u_1'e^x = 3x$ , so  $u_1' = \frac{3xe^{-x}}{2}$ . From (D),  $u_2' = -u_1'e^{2x} = -\frac{3xe^x}{2}$ . Therefore  $u_1 = -\frac{3e^x(x+1)}{2}$ ,  $u_2 = -\frac{3e^x(x-1)}{2}$ . Now (A) yields  $y_p = -3x^2$ .

**5.7.24.** (A)  $y_p = \frac{u_1}{x} + u_2x^3$ ;

$$\frac{u_1'}{x} + u_2'x^3 = 0 \quad (\text{B})$$

$$-\frac{u_1'}{x^2} + 3u_2'x^2 = \frac{x^{3/2}}{x^2} = x^{-1/2}. \quad (\text{C})$$

Multiplying (B) by  $\frac{1}{x}$  and adding the result to (C) yields  $4u_2'x^2 = x^{-1/2}$ , so  $u_2' = \frac{x^{-5/2}}{4}$ . From (B),  $u_1' = -u_2'x^4 = -\frac{x^{3/2}}{4}$ . Therefore  $u_1 = -\frac{x^{5/2}}{10}$ ,  $u_2 = -\frac{x^{-3/2}}{6}$ . Now (A) yields  $y_p = -\frac{4x^{3/2}}{15}$ .

**5.7.26.** (A)  $y_p = u_1 x^2 e^x + u_2 x^3 e^x;$

$$u_1' x^2 e^x + u_2' x^3 e^x = 0 \quad (\text{B})$$

$$u_1' (x^2 e^x + 2x e^x) + u_2' (x^3 e^x + 3x^2 e^x) = \frac{2x e^x}{x^2} = \frac{2e^x}{x}. \quad (\text{C})$$

Subtracting (B) from (C) and cancelling common factors in the resulting equations yields

$$u_1' + u_2' x = 0 \quad (\text{D})$$

$$2u_1' x + 3u_2' x^2 = \frac{2}{x}. \quad (\text{E})$$

Multiplying (D) by  $2x$  and subtracting the result from (E) yields  $x^2 u_2' = \frac{2}{x}$ , so  $u_2' = \frac{2}{x^3}$ . From (D),  $u_1' = -u_2' x = -\frac{2}{x^2}$ . Therefore  $u_1 = \frac{2}{x}$ ,  $u_2 = -\frac{1}{x^2}$ . Now (A) yields  $y_p = x e^x$ .

**5.7.28.** (A)  $y_p = u_1 x + u_2 e^x;$

$$u_1' x + u_2' e^x = 0 \quad (\text{B})$$

$$u_1' + u_2' e^x = \frac{2(x-1)^2 e^x}{x-1} = 2(x-1)e^x. \quad (\text{C})$$

Subtracting (B) from (C) yields  $u_1' (1-x) = 2(x-1)e^x$ , so  $u_1' = -2e^x$ . From (B),  $u_2' = -u_1' x e^{-x} = 2x$ . Therefore,  $u_1 = -2e^x$ ,  $u_2 = x^2$ . Now (A) yields  $y_p = x e^x (x-2)$ .

**5.7.30.** (A)  $y_p = u_1 e^{2x} + u_2 x e^{-x};$

$$u_1' e^{2x} + u_2' x e^{-x} = 0 \quad (\text{B})$$

$$2u_1' e^{2x} + u_2' (e^{-x} - x e^{-x}) = \frac{(3x-1)^2 e^{2x}}{3x-1} = (3x-1)e^{2x}. \quad (\text{C})$$

Multiplying (B) by 2 and subtracting the result from (C) yields  $u_2' (1-3x)e^{-x} = (3x-1)e^{2x}$ , so  $u_2' = -e^{3x}$ . From (B),  $u_1' = -u_2' x e^{-3x} = x$ . Therefore  $u_1 = \frac{x^2}{2}$ ,  $u_2 = -\frac{e^{3x}}{3}$ . Now (A) yields  $y_p = \frac{x e^{2x} (3x-2)}{3}$ . The general solution of the given equation is  $y = \frac{x e^{2x} (3x-2)}{3} + c_1 e^{2x} + c_2 x e^{-x}$ .

Differentiating this yields  $y' = \frac{e^{2x} (3x^2 + x - 1)}{3} + 2c_1 e^{2x} + c_2 (1-x)e^{-x}$ . Now  $y(0) = 1$ ,  $y'(0) = 2 \Rightarrow c_1 = 1$ ,  $2 = -\frac{1}{3} + 2c_1 + c_2$ , so  $c_2 = \frac{1}{3}$ , and  $y = \frac{e^{2x} (3x^2 - 2x + 6)}{6} + \frac{x e^{-x}}{3}$ .

**5.7.32.** (A)  $y_p = u_1 (x-1)e^x + u_2 (x-1);$

$$u_1' (x-1)e^x + u_2' (x-1) = 0 \quad (\text{B})$$

$$u_1' x e^x + u_2' = \frac{(x-1)^3 e^x}{(x-1)^2} = (x-1)e^x. \quad (\text{C})$$

From (B),  $u_1' = -u_2' e^{-x}$ . Substituting this into (C) yields  $-u_2' (x-1) = (x-1)e^x$ , so  $u_2' = -e^x$ ,  $u_1' = 1$ . Therefore  $u_1 = x$ ,  $u_2 = e^x$ . Now (A) yields  $y_p = e^x (x-1)^2$ . The general solution of the given equation is  $y = (x-1)^2 e^x + c_1 (x-1)e^x + c_2 (x-1)$ . Differentiating this yields

$y' = (x^2 - 1)e^x + c_1xe^x + c_2$ . Now  $y(0) = 4$ ,  $y'(0) = -6 \Rightarrow 4 = 1 - c_1 - c_2$ ,  $-6 = -1 + c_2$ , so  $c_1 = 2$ ,  $c_2 = -5$  and  $y = (x^2 - 1)e^x - 5(x - 1)$ .

**5.7.34.** (A)  $y_p = u_1x + \frac{u_2}{x^2}$ ;

$$u_1'x + \frac{u_2'}{x^2} = 0 \quad (\text{B})$$

$$u_1' - \frac{2u_2'}{x^3} = -\frac{2x^2}{x^2} = -2. \quad (\text{C})$$

Multiplying (B) by  $\frac{2}{x}$  and adding the result to (C) yields  $3u_1' = -2$ , so  $u_1' = -\frac{2}{3}$ . From (B),  $u_2' = -u_1'x^3 = \frac{2x^3}{3}$ . Therefore  $u_1 = -\frac{2x}{3}$ ,  $u_2 = \frac{x^4}{6}$ . Now (A) yields  $y_p = -\frac{x^2}{2}$ . The general solution of the given equation is  $y = -\frac{x^2}{2} + c_1x + \frac{c_2}{x^2}$ . Differentiating this yields  $y' = -x + c_1 - \frac{2c_2}{x^3}$ . Now  $y(1) = 1$ ,  $y'(1) = -1 \Rightarrow 1 = -\frac{1}{2} + c_1 + c_2$ ,  $-1 = -1 + c_1 - 2c_2$ , so  $c_1 = 1$ ,  $c_2 = \frac{1}{2}$ , and

$$y = -\frac{x^2}{2} + x + \frac{1}{2x^2}.$$

**5.7.36.** Since  $\bar{y} = y_p - a_1y_1 - a_2y_2$ ,

$$\begin{aligned} P_0(x)\bar{y}'' + P_1(x)\bar{y}' + P_2(x)\bar{y} &= P_0(x)(y_p - a_1y_1 - a_2y_2)'' \\ &\quad + P_1(x)(y_p - a_1y_1 - a_2y_2)' \\ &\quad + P_2(x)(y_p - a_1y_1 - a_2y_2) \\ &= (P_0(x)y_p'' + P_1(x)y_p' + P_2(x)y_p) \\ &\quad - a_1[P_0(x)y_1'' + P_1(x)y_1' + P_2(x)y_1] \\ &\quad - a_2[P_0(x)y_2'' + P_1(x)y_2' + P_2(x)y_2] \\ &= F(x) - a_1 \cdot 0 - a_2 \cdot 0 = F(x); \end{aligned}$$

hence  $\bar{y}$  is a particular solution of (A).

**5.7.38.** (a)  $y_p = u_1e^x + u_2e^{-x}$  is a solution of (A) on  $(a, \infty)$  if  $u_1'e^x + u_2'e^{-x} = 0$  and  $u_1'e^x - u_2'e^{-x} = f(x)$ . Solving these two equations yields  $u_1' = \frac{e^{-x}f}{2}$ ,  $u_2' = -\frac{e^x f}{2}$ . The functions  $u_1(x) = \frac{1}{2} \int_0^x e^{-t} f(t) dt$  and  $u_2(x) = -\frac{1}{2} \int_0^x e^t f(t) dt$  satisfy these conditions. Therefore,

$$\begin{aligned} y_p(x) &= \frac{e^x}{2} \int_0^x e^{-t} f(t) dt - \frac{e^{-x}}{2} \int_0^x e^t f(t) dt \\ &= \frac{1}{2} \int_0^x f(t) (e^{(x-t)} - e^{-(x-t)}) dt = \int_0^x f(t) \sinh(x-t) dt. \end{aligned}$$

is a particular solution of  $y'' - y = f(x)$ . Differentiating  $y_p$  yields

$$\begin{aligned} y_p'(x) &= \frac{e^x}{2} \int_0^x e^{-t} f(t) dt + \frac{e^x}{2} e^{-x} + \frac{e^{-x}}{2} \int_0^x e^t f(t) dt - \frac{e^{-x}}{2} e^x \\ &= \frac{e^x}{2} \int_0^x e^{-t} f(t) dt + \frac{e^{-x}}{2} \int_0^x e^t f(t) dt \\ &= \frac{1}{2} \int_0^x f(t) (e^{(x-t)} + e^{-(x-t)}) dt = \int_0^x f(t) \cosh(x-t) dt. \end{aligned}$$

Since  $y_p(x_0) = y_p'(x_0) = 0$ , the solution of the initial value problem is

$$\begin{aligned} y &= y_p + k_0 \cosh x + k_1 \sinh x \\ &= k_0 \cosh x + k_1 \sinh x + \int_0^x \sinh(x-t) f(t) dt. \end{aligned}$$

The derivative of the solution is

$$\begin{aligned} y' &= y_p' + k_0 \sinh x + k_1 \cosh x \\ &= k_0 \sinh x + k_1 \cosh x + \int_0^x \cosh(x-t) f(t) dt. \end{aligned}$$

## Chapter 6

# Applications of Linear Second Order Equations

### 6.1 Spring Problems I

**6.1.2.** Since  $\frac{k}{m} = \frac{g}{\Delta l} = \frac{32}{.1} = 320$  the equation of motion is (A)  $y'' + 320y = 0$ . The general solution of (A) is  $y = c_1 \cos 8\sqrt{5}t + c_2 \sin 8\sqrt{5}t$ , so  $y' = 8\sqrt{5}(-c_1 \sin 8\sqrt{5}t + c_2 \cos 8\sqrt{5}t)$ . Now  $y(0) = -\frac{1}{4} \Rightarrow c_1 = -\frac{1}{4}$  and  $y'(0) = -2 \Rightarrow c_2 = -\frac{1}{4\sqrt{5}}$ , so  $y = -\frac{1}{4} \cos 8\sqrt{5}t - \frac{1}{4\sqrt{5}} \sin 8\sqrt{5}t$  ft.

**6.1.4.** Since  $\frac{k}{m} = \frac{g}{\Delta l} = \frac{32}{.5} = 64$  the equation of motion is (A)  $y'' + 64y = 0$ . The general solution of (A) is  $y = c_1 \cos 8t + c_2 \sin 8t$ , so  $y' = 8(-c_1 \sin 8t + c_2 \cos 8t)$ . Now  $y(0) = \frac{1}{4} \Rightarrow c_1 = \frac{1}{4}$  and  $y'(0) = -\frac{1}{2} \Rightarrow c_2 = -\frac{1}{16}$ , so  $y = \frac{1}{4} \cos 8t - \frac{1}{16} \sin 8t$  ft;  $R = \frac{\sqrt{17}}{16}$  ft;  $\omega_0 = 8$  rad/s;  $T = \pi/4$  s;  $\phi \approx -.245$  rad  $\approx -14.04^\circ$ .

**6.1.6.** Since  $k = \frac{mg}{\Delta l} = \frac{(9.8)10}{.7} = 140$ , the equation of motion of the 2 kg mass is (A)  $y'' + 70y = 0$ . The general solution of (A) is  $y = c_1 \cos \sqrt{70}t + c_2 \sin \sqrt{70}t$ , so  $y' = \sqrt{70}(-c_1 \sin \sqrt{70}t + c_2 \cos \sqrt{70}t)$ . Now  $y(0) = -\frac{1}{4} \Rightarrow c_1 = -\frac{1}{4}$  and  $y'(0) = 2 \Rightarrow c_2 = \frac{2}{\sqrt{70}}$ , so  $y = -\frac{1}{4} \cos \sqrt{70}t + \frac{2}{\sqrt{70}} \sin \sqrt{70}t$  m;  $R = \frac{1}{4} \sqrt{\frac{67}{35}}$  m;  $\omega_0 = \sqrt{70}$  rad/s;  $T = 2\pi/\sqrt{70}$  s;  $\phi \approx 2.38$  rad  $\approx 136.28^\circ$ .

**6.1.8.** Since  $\frac{k}{m} = \frac{g}{\Delta l} = \frac{32}{1/2} = 64$  the equation of motion is (A)  $y'' + 64y = 0$ . The general solution of (A) is  $y = c_1 \cos 8t + c_2 \sin 8t$ , so  $y' = 8(-c_1 \sin 8t + c_2 \cos 8t)$ . Now  $y(0) = \frac{1}{2} \Rightarrow c_1 = \frac{1}{2}$  and  $y'(0) = -3 \Rightarrow c_2 = -\frac{3}{8}$ , so  $y = \frac{1}{2} \cos 8t - \frac{3}{8} \sin 8t$  ft.

**6.1.10.**  $m = \frac{64}{32} = 2$ , so the equation of motion is  $2y'' + 8y = 2 \sin t$ , or (A)  $y'' + 4y = \sin t$ . Let  $y_p = A \cos t + B \sin t$ ; then  $y_p'' = -A \cos t - B \sin t$ , so  $y_p'' + 4y_p = 3A \cos t + 3B \sin t = \sin t$  if  $3A = 0$ ,  $3B = 1$ . Therefore,  $A = 0$ ,  $B = \frac{1}{3}$ , and  $y_p = \frac{1}{3} \sin t$ . The general solution of

(A) is (B)  $y = \frac{1}{3} \sin t + c_1 \cos 2t + c_2 \sin 2t$ , so  $y(0) = \frac{1}{2} \Rightarrow c_1 = \frac{1}{2}$ . Differentiating (B) yields  $y' = \frac{1}{3} \cos t - 2c_1 \sin 2t + 2c_2 \cos 2t$ , so  $y'(0) = 2 \Rightarrow 2 = \frac{1}{3} + 2c_2$ , so  $c_2 = \frac{5}{6}$ . Therefore,  $y = \frac{1}{3} \sin t + \frac{1}{2} \cos 2t + \frac{5}{6} \sin 2t$  ft.

**6.1.12.**  $m = \frac{4}{32} = \frac{1}{8}$  and  $k = \frac{mg}{\Delta l} = 4$ , so the equation of motion is  $\frac{1}{8}y'' + 4y = \frac{1}{4} \sin 8t$ , or (A)  $y'' + 32y = 2 \sin 8t$ . Let  $y_p = A \cos 8t + B \sin 8t$ ; then  $y_p'' = -64A \cos 8t - 64B \sin 8t$ , so  $y_p'' + 32y_p = -32A \cos 8t - 32B \sin 8t = 2 \sin 8t$  if  $-32A = 0$ ,  $-32B = 2$ . Therefore,  $A = 0$ ,  $B = -\frac{1}{16}$ , and  $y_p = -\frac{1}{16} \sin 8t$ . The general solution of (A) is (B)  $y = -\frac{1}{16} \sin 8t + c_1 \cos 4\sqrt{2}t + c_2 \sin 4\sqrt{2}t$ , so  $y(0) = \frac{1}{3} \Rightarrow c_1 = \frac{1}{3}$ . Differentiating (B) yields  $y' = -\frac{1}{2} \cos 8t + 4\sqrt{2}(-c_1 \sin 4\sqrt{2}t + c_2 \cos 4\sqrt{2}t)$ , so  $y'(0) = -1 \Rightarrow -1 = -\frac{1}{2} + 4\sqrt{2}c_2$ , so  $c_2 = -\frac{1}{8\sqrt{2}}$ . Therefore,  $y = -\frac{1}{16} \sin 8t + \frac{1}{3} \cos 4\sqrt{2}t - \frac{1}{8\sqrt{2}} \sin 4\sqrt{2}t$  ft.

**6.1.14.** Since  $T = \frac{2\pi}{\omega_0} = 2\pi\sqrt{\frac{m}{k}}$  the period is proportional to the square root of the mass. Therefore, doubling the mass multiplies the period by  $\sqrt{2}$ ; hence the period of the system with the 20 gm mass is  $T = 4\sqrt{2}$  s.

**6.1.16.**  $m = \frac{6}{32} = \frac{3}{16}$  and  $k = \frac{mg}{\Delta l} = \frac{6}{1/3} = 18$  so the equation of motion is  $\frac{3}{16}y'' + 18y = 4 \sin \omega t - 6 \cos \omega t$ , or (A)  $y'' + 96y = \frac{64}{3} \sin \omega t - 32 \cos \omega t$ . The displacement will be unbounded if  $\omega = \sqrt{96} = 4\sqrt{6}$ , in which case (A) becomes (B)  $y'' + 96y = \frac{64}{3} \sin 4\sqrt{6}t - 32 \cos 4\sqrt{6}t$ . Let

$$\begin{aligned} y_p &= At \cos 4\sqrt{6}t + Bt \sin 4\sqrt{6}t; \text{ then} \\ y_p' &= (A + 4\sqrt{6}Bt) \cos 4\sqrt{6}t + (B - 4\sqrt{6}At) \sin 4\sqrt{6}t \\ y_p'' &= (8\sqrt{6}B - 96At) \cos 4\sqrt{6}t - (8\sqrt{6}A + 96Bt) \sin 4\sqrt{6}t, \text{ so} \end{aligned}$$

$$y_p'' + 96y_p = 8\sqrt{6}B \cos 4\sqrt{6}t - 8\sqrt{6}A \sin 4\sqrt{6}t = \frac{64}{3} \sin 4\sqrt{6}t - 32 \cos 4\sqrt{6}t$$

if  $8\sqrt{6}B = -32$ ,  $-8\sqrt{6}A = \frac{64}{3}$ . Therefore,  $A = -\frac{8}{3\sqrt{6}}$ ,  $B = -\frac{4}{\sqrt{6}}$ , and  $y_p = -\frac{t}{\sqrt{6}} \left( \frac{8}{3} \cos 4\sqrt{6}t + 4 \sin 4\sqrt{6}t \right)$ . The general solution of (B) is

$$y = -\frac{t}{\sqrt{6}} \left( \frac{8}{3} \cos 4\sqrt{6}t + 4 \sin 4\sqrt{6}t \right) + c_1 \cos 4\sqrt{6}t + c_2 \sin 4\sqrt{6}t, \quad (\text{C})$$

so  $y(0) = 0 \Rightarrow c_1 = 0$ . Differentiating (C) yields

$$\begin{aligned} y' &= -\left( \frac{8}{3\sqrt{6}} \cos 4\sqrt{6}t + \frac{4}{\sqrt{6}} \sin 4\sqrt{6}t \right) - 4t \left( -\frac{8}{3} \sin 4\sqrt{6}t + 4 \cos 4\sqrt{6}t \right) \\ &\quad + 4\sqrt{6}(-c_1 \sin 4\sqrt{6}t + c_2 \cos 4\sqrt{6}t), \end{aligned}$$



so  $y'(0) = 0 \Rightarrow 0 = -\frac{8}{3\sqrt{6}} + 4\sqrt{6}c_2$ , and  $c_2 = \frac{1}{9}$ . Therefore,

$$y = -\frac{t}{\sqrt{6}} \left( \frac{8}{3} \cos 4\sqrt{6}t + 4 \sin 4\sqrt{6}t \right) + \frac{1}{9} \sin 4\sqrt{6}t \text{ ft.}$$

**6.1.18.** The equation of motion is (A)  $y'' + \omega_0^2 y = 0$ . The general solution of (A) is  $y = c_1 \cos \omega_0 t + c_2 \sin \omega_0 t$ . Now  $y(0) = y_0 \Rightarrow c_1 = y_0$ . Since  $y' = \omega_0(-c_1 \sin \omega_0 t + c_2 \cos \omega_0 t)$ ,  $y'(0) = v_0 \Rightarrow c_2 = \frac{v_0}{\omega_0}$ .

Therefore,  $y = y_0 \cos \omega_0 t + \frac{v_0}{\omega_0} \sin \omega_0 t$ ;

$$R = \frac{1}{\omega_0} \sqrt{(\omega_0 y_0)^2 + (v_0)^2}; \cos \phi = \frac{y_0 \omega_0}{\sqrt{(\omega_0 y_0)^2 + (v_0)^2}}; \sin \phi = \frac{v_0}{\sqrt{(\omega_0 y_0)^2 + (v_0)^2}}.$$

**Discussion 6.1.1** In Exercises 19, 20, and 21 we use the fact that in a spring-mass system with mass  $m$  and spring constant  $k$  the period of the motion is  $T = 2\pi\sqrt{\frac{m}{k}}$ . Therefore, if we have two systems with masses  $m_1$  and  $m_2$  and spring constants  $k_1$  and  $k_2$ , then the periods are related by  $\frac{T_2}{T_1} = \sqrt{\frac{m_2 k_1}{m_1 k_2}}$ . We will use this formula in the solutions of these exercises.

**6.1.20.** Let  $m_2 = 2m_1$ . Since  $k_1 = k_2$ ,  $\frac{T_2}{T_1} = \sqrt{\frac{2m_1}{m_1}} = \sqrt{2}$ , so  $T_2 = \sqrt{2}T_1$ .

**6.1.21.** Suppose that  $T_2 = 3T_1$ . Since  $m_1 = m_2$ ,  $\sqrt{\frac{k_1}{k_2}} = 3$ ,  $k_1 = 9k_2$ .

## 6.2 Spring Problems II

**6.2.2.** Since  $k = \frac{mg}{\Delta l} = \frac{16}{3.2} = 5$  the equation of motion is  $\frac{1}{2}y'' + y' + 5y = 0$ , or (A)  $y'' + 2y' + 10y = 0$ . The characteristic polynomial of (A) is  $p(r) = r^2 + 2r + 10 = (r + 1)^2 + 9$ . Therefore, the general solution of (A) is  $y = e^{-t}(c_1 \cos 3t + c_2 \sin 3t)$ , so  $y' = -y + 3e^{-t}(-c_1 \sin 3t + c_2 \cos 3t)$ . Now  $y(0) = -3$  and  $y'(0) = 2 \Rightarrow c_1 = -3$  and  $2 = 3 + 3c_2$ , or  $c_2 = -\frac{1}{3}$ . Therefore,  $y = -e^{-t} \left( 3 \cos 3t + \frac{1}{3} \sin 3t \right)$  ft.

The time-varying amplitude is  $\frac{\sqrt{82}}{3}e^{-t}$  ft.

**6.2.4.** Since  $k = \frac{mg}{\Delta l} = \frac{96}{3.2} = 30$  the equation of motion is  $3y'' + 18y' + 30y = 0$ , or (A)  $y'' + 6y' + 10y = 0$ . The characteristic polynomial of (A) is  $p(r) = r^2 + 6r + 10 = (r + 3)^2 + 1$ . Therefore, the general solution of (A) is  $y = e^{-3t}(c_1 \cos t + c_2 \sin t)$ , so  $y' = -3y + e^{-3t}(-c_1 \sin t + c_2 \cos t)$ . Now  $y(0) = -\frac{5}{4}$  and  $y'(0) = -12 \Rightarrow c_1 = -\frac{5}{4}$  and  $-12 = \frac{15}{4} + c_2$ , or  $c_2 = -\frac{63}{4}$ . Therefore,  $y = -\frac{e^{-3t}}{4}(5 \cos t + 63 \sin t)$  ft.

**6.2.6.** Since  $k = \frac{mg}{\Delta l} = \frac{8}{.32} = 25$  the equation of motion is  $\frac{1}{4}y'' + \frac{3}{2}y' + 25y = 0$ , or (A)  $y'' + 6y' + 100y = 0$ . The characteristic polynomial of (A) is  $p(r) = r^2 + 6r + 100 = (r + 3)^2 + 91$ . Therefore, the

general solution of (A) is  $y = e^{-3t}(c_1 \cos \sqrt{91}t + c_2 \sin \sqrt{91}t)$ , so  $y' = -3y + \sqrt{91}e^{-3t}(-c_1 \sin \sqrt{91}t + c_2 \cos \sqrt{91}t)$ . Now  $y(0) = \frac{1}{2}$  and  $y'(0) = 4 \Rightarrow c_1 = \frac{1}{2}$  and  $4 = -\frac{3}{2} + \sqrt{91}c_2$ , or  $c_2 = \frac{11}{2\sqrt{91}}$ .

Therefore,  $y = \frac{1}{2}e^{-3t} \left( \cos \sqrt{91}t + \frac{11}{\sqrt{91}} \sin \sqrt{91}t \right)$  ft.

**6.2.8.** Since  $k = \frac{mg}{\Delta l} = \frac{20 \cdot 980}{5} = 3920$  the equation of motion is  $20y'' + 400y' + 3920y = 0$ , or (A)  $y'' + 20y' + 196y = 0$ . The characteristic polynomial of (A) is  $p(r) = r^2 + 20r + 196 = (r + 10)^2 + 96$ . Therefore, the general solution of (A) is  $y = e^{-10t}(c_1 \cos 4\sqrt{6}t + c_2 \sin 4\sqrt{6}t)$ , so  $y' = -10y + 4\sqrt{6}e^{-10t}(-c_1 \sin 4\sqrt{6}t + c_2 \cos 4\sqrt{6}t)$ . Now  $y(0) = 9$  and  $y'(0) = 0 \Rightarrow c_1 = 9$  and  $0 = -90 + 4\sqrt{6}c_2$ , or  $c_2 = \frac{45}{2\sqrt{6}}$ . Therefore,  $y = e^{-10t} \left( 9 \cos 4\sqrt{6}t + \frac{45}{2\sqrt{6}} \sin 4\sqrt{6}t \right)$  cm.

**6.2.10.** Since  $k = \frac{mg}{\Delta l} = \frac{32}{1} = 32$  the equation of motion is (A)  $y'' + 3y' + 32y = 0$ . The characteristic polynomial of (A) is  $p(r) = r^2 + 3r + 32 = \left(r + \frac{3}{2}\right)^2 + \frac{119}{4}$ . Therefore, the general solution of (A) is  $y = e^{-3t/2} \left( c_1 \cos \frac{\sqrt{119}}{2}t + c_2 \sin \frac{\sqrt{119}}{2}t \right)$ , so  $y' = -\frac{3}{2}y + \frac{\sqrt{119}}{2}e^{-3t/2} \left( -c_1 \sin \frac{\sqrt{119}}{2}t + c_2 \cos \frac{\sqrt{119}}{2}t \right)$ . Now  $y(0) = \frac{1}{2}$  and  $y'(0) = -3 \Rightarrow c_1 = \frac{1}{2}$  and  $-3 = -\frac{3}{4} + \frac{\sqrt{119}}{2}c_2$ , or  $c_2 = -\frac{9}{2\sqrt{119}}$ . Therefore,  $y = e^{-\frac{3}{2}t} \left( \frac{1}{2} \cos \frac{\sqrt{119}}{2}t - \frac{9}{2\sqrt{119}} \sin \frac{\sqrt{119}}{2}t \right)$  ft.

**6.2.12.** Since  $k = \frac{mg}{\Delta l} = \frac{2}{.32} = \frac{25}{4}$  the equation of motion is  $\frac{1}{16}y'' + \frac{1}{8}y' + \frac{25}{4}y = 0$ , or (A)  $y'' + 2y' + 100y = 0$ . The characteristic polynomial of (A) is  $p(r) = r^2 + 2r + 100 = (r + 1)^2 + 99$ . Therefore, the general solution of (A) is  $y = e^{-t}(c_1 \cos 3\sqrt{11}t + c_2 \sin 3\sqrt{11}t)$ , so  $y' = -y + 3\sqrt{11}e^{-t}(-c_1 \sin 3\sqrt{11}t + c_2 \cos 3\sqrt{11}t)$ . Now  $y(0) = -\frac{1}{3}$  and  $y'(0) = 5 \Rightarrow c_1 = -\frac{1}{3}$  and  $5 = \frac{1}{3} + 3\sqrt{11}c_2$ , or  $c_2 = \frac{14}{9\sqrt{11}}$ . Therefore,  $y = e^{-t} \left( -\frac{1}{3} \cos 3\sqrt{11}t + \frac{14}{9\sqrt{11}} \sin 3\sqrt{11}t \right)$  ft.

**6.2.14.** Since  $k = \frac{mg}{\Delta l} = 32$  the equation of motion is (A)  $y'' + 12y' + 32y = 0$ . The characteristic polynomial of (A) is  $p(r) = r^2 + 12r + 32 = (r + 8)(r + 4)$ . Therefore, the general solution of (A) is  $y = c_1e^{-8t} + c_2e^{-4t}$ , so  $y' = -8c_1e^{-8t} - 4c_2e^{-4t}$ . Now  $y(0) = -\frac{2}{3}$  and  $y'(0) = 0 \Rightarrow c_1 + c_2 = \frac{2}{3}$ ,  $8c_1 - 4c_2 = 0$ , so  $c_1 = -\frac{2}{3}$ ,  $c_2 = \frac{4}{3}$ , and  $y = -\frac{2}{3}(e^{-8t} - 2e^{-4t})$ .

**6.2.16.** Since  $k = \frac{mg}{\Delta l} = \frac{100 \cdot 980}{98} = 100$  the equation of motion is  $100y'' + 600y' + 1000y = 0$ , or (A)  $y'' + 6y' + 10y = 0$ . The characteristic polynomial of (A) is  $p(r) = r^2 + 6r + 10 = (r + 3)^2 + 1$ . Therefore, the general solution of (A) is  $y = e^{-3t}(c_1 \cos t + c_2 \sin t)$ , so  $y' = -3y + e^{-t}(-c_1 \sin t + c_2 \cos t)$ . Now  $y(0) = 10$  and  $y'(0) = -100 \Rightarrow c_1 = 10$  and  $-100 = -30 + c_2$ , or  $c_2 = -70$ . Therefore,  $y = e^{-3t}(10 \cos t - 70 \sin t)$  cm.

**6.2.18.** The equation of motion is (A)  $2y'' + 4y' + 20y = 3 \cos 4t - 5 \sin 4t$ . The steady state component of the solution of (A) is of the form  $y_p = A \cos 4t + B \sin 4t$ ; therefore  $y_p' = -4A \sin 4t + 4B \cos 4t$  and  $y_p'' = -16A \cos 4t - 16B \sin 4t$ , so  $2y_p'' + 4y_p' + 20y_p = (-12A + 16B) \cos 4t - (16A +$

$12B) \sin 4t = 3 \cos 4t - 5 \sin 4t$  if  $-12A + 16B = 3$ ,  $-16A - 12B = -5$ ; therefore  $A = \frac{11}{100}$ ,  $B = \frac{27}{100}$ ,

and  $y_p = \frac{11}{100} \cos 4t + \frac{27}{100} \sin 4t$  cm.

**6.2.20.** Since  $k = \frac{mg}{\Delta l} = \frac{9.8}{.49} = 20$  the equation of motion is (A)  $y'' + 4y' + 20y = 8 \sin 2t - 6 \cos 2t$ .

The steady state component of the solution of (A) is of the form  $y_p = A \cos 2t + B \sin 2t$ ; therefore  $y'_p = -2A \sin 2t + 2B \cos 2t$  and  $y''_p = -4A \cos 2t - 4B \sin 2t$ , so  $y''_p + 4y'_p + 20y_p = (16A + 8B) \cos 2t - (8A - 16B) \sin 2t = 8 \sin 2t - 6 \cos 2t$  if  $16A + 8B = -6$ ,  $-8A + 16B = 8$ ; therefore  $A = -\frac{1}{2}$ ,  $B = \frac{1}{4}$ ,

and  $y = -\frac{1}{2} \cos 2t + \frac{1}{4} \sin 2t$  m.

**6.2.22.** If  $e^{r_1 t}(c_1 + c_2 t) = 0$ , then (A)  $c_1 + c_2 t = 0$ . If  $c_2 = 0$ , then  $c_1 \neq 0$  (by assumption), so (A) is impossible. If  $c_1 \neq 0$ , then the left side of (A) is strictly monotonic and therefore cannot have the same value for two distinct values of  $t$ .

**6.2.24.** If  $y = e^{-ct/2m}(c_1 \cos \omega_1 t + c_2 \sin \omega_1 t)$ , then  $y' = -\frac{c}{2m}y + \omega_1 e^{-ct/2m}(-c_1 \sin \omega_1 t + c_2 \cos \omega_1 t)$ ,

so  $y(0) = y_0$  and  $y'(0) = v_0 \Rightarrow c_1 = y_0$  and  $v_0 = -\frac{cy_0}{2m} + c_2 \omega_1$ , so  $c_2 = \frac{1}{\omega_1} \left( v_0 + \frac{cy_0}{2m} \right)$  and

$y = e^{-ct/2m} \left( y_0 \cos \omega_1 t + \frac{1}{\omega_1} \left( v_0 + \frac{cy_0}{2m} \right) \sin \omega_1 t \right)$ .

**6.2.26.** If  $y = e^{r_1 t}(c_1 + c_2 t)$ , then  $y' = r_1 y + c_2 e^{r_1 t}$ , so  $y(0) = y_0$  and  $y'(0) = v_0 \Rightarrow c_1 = y_0$  and  $v_0 = r_1 y_0 + c_2$ , so  $c_2 = v_0 - r_1 y_0$ . Therefore,  $y = e^{r_1 t}(y_0 + (v_0 - r_1 y_0)t)$ .

### 6.3 The RLC Circuit

**6.3.2.**  $\frac{1}{20}Q'' + 2Q' + 100Q = 0$ ;  $Q'' + 40Q' + 2000Q = 0$ ;  $r^2 + 40r + 2000 = (r + 20)^2 + 1600 = 0$ ;  $r = -20 \pm 40i$ ;  $Q = e^{-20t}(2 \cos 40t + c_2 \sin 40t)$  (since  $Q_0 = 2$ );  $I = Q' = e^{-20t}((40c_2 - 40) \cos 40t - (20c_2 + 80) \sin 40t)$ ;  $I_0 = 2 \Rightarrow 40c_2 - 40 = 2 \Rightarrow c_2 = \frac{21}{20}$ , so  $20c_2 + 80 = 101$ ;  $I = e^{-20t}(2 \cos 40t - 101 \sin 40t)$ .

**6.3.4.**  $\frac{1}{10}Q'' + 6Q' + 250Q = 0$ ;  $Q'' + 60Q' + 2500Q = 0$ ;  $r^2 + 60r + 2500 = (r + 30)^2 + 1600 = 0$ ;  $r = -30 \pm 40i$ ;  $Q = e^{-30t}(3 \cos 40t + c_2 \sin 40t)$  (since  $Q_0 = 3$ );  $I = Q' = e^{-30t}((40c_2 - 90) \cos 40t - (30c_2 + 120) \sin 40t)$ ;  $I_0 = -10 \Rightarrow 40c_2 - 90 = -10 \Rightarrow c_2 = 2$ , so  $-30c_2 - 120 = -180$ ;  $I = -10e^{-30t}(\cos 40t + 18 \sin 40t)$ .

**6.3.6.**  $Q_p = A \cos 10t + B \sin 10t$ ;  $Q'_p = 10B \cos 10t - 10A \sin 10t$ ;  $Q''_p = -100A \cos 10t - 100B \sin 10t$ ;

$\frac{1}{10}Q''_p + 3Q'_p + 100Q_p = (90A + 30B) \cos 10t - (30A - 90B) \sin 10t = 5 \cos 10t - 5 \sin 10t$ , so  $90A + 30B = 5$ ,  $-30A + 90B = -5$ . Therefore,  $A = 1/15$ ,  $B = -1/30$ ,  $Q_p = \frac{\cos 10t}{15} - \frac{\sin 10t}{30}$ , and

$I_p = -\frac{1}{3}(\cos 10t + 2 \sin 10t)$ .

**6.3.8.**  $Q_p = A \cos 50t + B \sin 50t$ ;  $Q'_p = 50B \cos 50t - 50A \sin 50t$ ;  $Q''_p = -2500A \cos 50t - 2500B \sin 50t$ ;  $\frac{1}{10}Q''_p + 2Q'_p + 100Q_p = (-150A + 100B) \cos 50t - (100A + 150B) \sin 50t = 3 \cos 50t - 6 \sin 50t$ , so  $-150A + 100B = 3$ ,  $-100A + 150B = -6$ . Therefore,  $A = 3/650$ ,  $B = 12/325$ ,

$Q_p = \frac{3}{650}(\cos 50t + 8 \sin 50t)$ , and  $I_p = \frac{3}{13}(8 \cos 50t - \sin 50t)$ .

**6.3.10.**  $Q_p = A \cos 30t + B \sin 30t$ ;  $Q'_p = 30B \cos 30t - 30A \sin 30t$ ;  $Q''_p = -900A \cos 30t -$

$900B \sin 30t$ ;  $\frac{1}{20}Q_p'' + 4Q_p' + 125Q_p = (80A + 120B) \cos 30t - (120A - 80B) \sin 30t = 15 \cos 30t - 30 \sin 30t$ , so  $80A + 120B = 15$ ,  $-120A + 80B = -30$ ,  $A = 3/13$ ,  $B = -3/104$ ,  $Q_p = \frac{3}{104}(8 \cos 30t - \sin 30t)$ , and  $I_p = -\frac{45}{52}(\cos 30t + 8 \sin 30t)$ .

**6.3.12.** Let  $\sigma = \sigma(\omega)$  be the amplitude of  $I_p$ . From the solution of Exercise 6.3.11,  $Q_p = A \cos \omega t + B \sin \omega t$ , where  $A = \frac{(1/C - L\omega^2)U - R\omega V}{\Delta}$ ,  $B = \frac{R\omega U + (1/C - L\omega^2)V}{\Delta}$ , and  $\Delta = (1/C - L\omega^2)^2 + R^2\omega^2$ . Since  $I_p = Q_p' = \omega(-A \sin \omega t + B \cos \omega t)$ , it follows that  $\sigma^2(\omega) = \omega^2(A^2 + B^2) = \frac{U^2 + V^2}{\rho(\omega)}$ , with  $\rho(\omega) = \frac{\Delta}{\omega^2} = (1/C\omega - L\omega)^2 + R^2$ , which attains its minimum value  $R^2$  when  $\omega = \omega_0 = \frac{1}{\sqrt{LC}}$ .

The maximum amplitude of  $I_p$  is  $\sigma(\omega) = \frac{\sqrt{U^2 + V^2}}{R}$ .

## 6.4 Motion Under a Central Force

**6.4.2.** Let  $h = r_0^2\theta_0'$ ; then  $\rho = \frac{h^2}{k}$ . Since  $r = \frac{\rho}{1 + e \cos(\theta - \phi)}$ , it follows that (A)  $e \cos(\theta - \phi) = \frac{\rho}{r} - 1$ .

Differentiating this with respect to  $t$  yields  $-e \sin(\theta - \phi)\theta' = -\frac{\rho r'}{r^2}$ , so (B)  $e \sin(\theta - \phi) = \frac{\rho r'}{h}$ , since  $r^2\theta' \equiv h$ . Squaring and adding (A) and (B) and setting  $t = 0$  in the result yields  $e = \left[ \left( \frac{\rho}{r_0} - 1 \right)^2 + \left( \frac{\rho r_0'}{h} \right)^2 \right]^{1/2}$ . If  $e = 0$ , then  $\theta_0$  is undefined, but also irrelevant; if  $e \neq 0$ , then

set  $t = 0$  in (A) and (B) to see that  $\phi = \theta_0 - \alpha$ , where  $-\pi \leq \alpha < \pi$ ,  $\cos \alpha = \frac{1}{e} \left( \frac{\rho}{r_0} - 1 \right)$  and  $\sin \alpha = \frac{\rho r_0'}{eh}$ .

**6.4.4.** Recall that (A)  $\frac{d^2u}{d\theta^2} = -\frac{1}{mh^2u^2}f(1/u)$ . Let  $u = \frac{1}{r} = \frac{1}{c\theta^2}$ ; then  $\frac{d^2u}{d\theta^2} = \frac{6}{c\theta^4} = 6cu^2$ .  $6cu^2 + u = -\frac{1}{mh^2u^2}f(1/u)$ , so  $f(1/u) = -mh^2(6cu^4 + u^3)$  and  $f(r) = -mh^2 \left( \frac{6c}{r^4} + \frac{1}{r^3} \right)$ .

**6.4.6. (a)** With  $f(r) = -\frac{mk}{r^3}$ , Eqn. 6.4.11 becomes (A)  $\frac{d^2u}{d\theta^2} + \left( 1 - \frac{k}{h^2} \right)u = 0$ . The initial conditions imply that  $u(\theta_0) = \frac{1}{r_0}$  and  $\frac{du}{d\theta}(\theta_0) = -\frac{r_0'}{h}$  (see Eqn. (6.4.)

(b) Let  $\gamma = \left| 1 - \frac{k}{h^2} \right|^{1/2}$ . (i) If  $h^2 < k$ , then (A) becomes  $\frac{d^2u}{d\theta^2} - \gamma^2u = 0$ , and the solution of the initial value problem for  $u$  is  $u = \frac{1}{r_0} \cosh \gamma(\theta - \theta_0) - \frac{r_0'}{\gamma h} \sinh \gamma(\theta - \theta_0)$ ; therefore  $r = r_0 \left( \cosh \gamma(\theta - \theta_0) - \frac{r_0 r_0'}{\gamma h} \sinh \gamma(\theta - \theta_0) \right)^{-1}$ . (ii) If  $h^2 = k$ , then (A) becomes  $\frac{d^2u}{d\theta^2} = 0$ , and the solution of the initial value problem for  $u$  is  $u = \frac{1}{r_0} - \frac{r_0'}{h}(\theta - \theta_0)$ ; therefore  $r = r_0 \left( 1 - \frac{r_0 r_0'}{h}(\theta - \theta_0) \right)^{-1}$ .

(iii) If  $h^2 > k$ , then (A) becomes  $\frac{d^2u}{d\theta^2} + \gamma^2 u = 0$ , and the solution of the initial value problem for  $u$  is  $u = \frac{1}{r_0} \cos \gamma(\theta - \theta_0) - \frac{r'_0}{\gamma h} \sin \gamma(\theta - \theta_0)$ ; therefore  $r = r_0 \left( \cos \gamma(\theta - \theta_0) - \frac{r_0 r'_0}{\gamma h} \sin \gamma(\theta - \theta_0) \right)^{-1}$ .

## Chapter 7

# Series Solutions of Linear Second Equations

### 7.1 Review of Power Series

**7.1.2.** From Theorem 7.1.3,  $\sum_{m=0}^{\infty} b_m z^m$  converges if  $|z| < 1/L$  and diverges if  $|z| > 1/L$ . Therefore,  $\sum_{m=0}^{\infty} b_m (x - x_0)^2$  converges if  $|x - x_0| < 1/\sqrt{L}$  and diverges if  $|x - x_0| > 1/\sqrt{L}$ .

**7.1.4.** From Theorem 7.1.3,  $\sum_{m=0}^{\infty} b_m z^m$  converges if  $|z| < 1/L$  and diverges if  $|z| > 1/L$ . Therefore,  $\sum_{m=0}^{\infty} b_m (x - x_0)^{km}$  converges if  $|x - x_0| < 1/\sqrt[k]{L}$  and diverges if  $|x - x_0| > 1/\sqrt[k]{L}$ .

$$\begin{aligned} \text{7.1.12. } (1 + 3x^2)y'' + 3x^2y' - 2y &= \sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} + 3 \sum_{n=2}^{\infty} n(n-1)a_n x^n + 3 \sum_{n=1}^{\infty} na_n x^{n+1} - \\ 2 \sum_{n=0}^{\infty} a_n x^n &= \sum_{n=0}^{\infty} (n+2)(n+1)a_{n+2} x^n + 3 \sum_{n=1}^{\infty} n(n-1)na_n x^n + 3 \sum_{n=1}^{\infty} (n-1)a_{n-1} x^n - 2 \sum_{n=0}^{\infty} a_n x^n = \\ 2a_2 - 2a_0 + \sum_{n=1}^{\infty} [(n+2)(n+1)a_{n+2} + (3n(n-1) - 2)a_n + 3(n-1)a_{n-1}] x^n. \end{aligned}$$

$$\begin{aligned} \text{7.1.13. } (1 + 2x^2)y'' + (2 - 3x)y' + 4y &= \sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} + 2 \sum_{n=2}^{\infty} n(n-1)a_n x^n + 2 \sum_{n=1}^{\infty} na_n x^{n-1} - \\ 2 \sum_{n=0}^{\infty} a_n x^n + 4 \sum_{n=0}^{\infty} a_n x^n &= \sum_{n=0}^{\infty} (n+2)(n+1)a_{n+2} x^n + 2 \sum_{n=0}^{\infty} n(n-1)a_n x^n + 2 \sum_{n=0}^{\infty} (n+1)a_{n+1} x^n - \\ 3 \sum_{n=0}^{\infty} a_n x^n + 4 \sum_{n=0}^{\infty} a_n x^n &\sum_{n=0}^{\infty} [(n+2)(n+1)a_{n+2} + 2(n+1)a_{n+1} + (2n^2 - 5n + 4)a_n] x^n. \end{aligned}$$

$$\begin{aligned} \text{7.1.14. } (1 + x^2)y'' + (2 - x)y' + 3y &= \sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} + \sum_{n=2}^{\infty} n(n-1)a_n x^n + 2 \sum_{n=1}^{\infty} na_n x^{n-1} - \\ \sum_{n=1}^{\infty} na_n x^n + 3 \sum_{n=0}^{\infty} a_n x^n &= \sum_{n=0}^{\infty} (n+2)(n+1)a_{n+2} x^n \\ + \sum_{n=0}^{\infty} n(n-1)a_n x^n + 2 \sum_{n=0}^{\infty} (n+1)a_{n+1} x^n - \sum_{n=0}^{\infty} na_n x^n + 3 \sum_{n=0}^{\infty} a_n x^n &= \\ = \sum_{n=0}^{\infty} [(n+2)(n+1)a_{n+2} + 2(n+1)a_{n+1} + (n^2 - 2n + 3)a_n] x^n. \end{aligned}$$

**7.1.16.** Let  $t = x + 1$ ; then  $xy'' + (4 + 2x)y' + (2 + x)y = (-1 + t)y'' + (2 + 2t)y' + (1 + t)y = -\sum_{n=2}^{\infty} n(n-1)a_n t^{n-2} + \sum_{n=2}^{\infty} n(n-1)a_n t^{n-1} + 2\sum_{n=1}^{\infty} na_n t^{n-1} + 2\sum_{n=1}^{\infty} na_n t^n + \sum_{n=0}^{\infty} a_n t^n + \sum_{n=0}^{\infty} a_n t^{n+1} = -\sum_{n=0}^{\infty} (n+2)(n+1)a_{n+2}t^n + \sum_{n=0}^{\infty} (n+1)na_{n+1}t^n + 2\sum_{n=0}^{\infty} (n+1)a_{n+1}t^n + 2\sum_{n=0}^{\infty} na_n t^n + \sum_{n=0}^{\infty} a_n t^n + \sum_{n=1}^{\infty} a_{n-1}t^n = (-2a_2 + 2a_1 + a_0) + \sum_{n=1}^{\infty} [-(n+2)(n+1)a_{n+2} + (n+1)(n+2)a_{n+1} + (2n+1)a_n + a_{n-1}](x+2)^n.$

**7.1.20.**  $y'(x) = x^r \sum_{n=0}^{\infty} na_n x^{n-1} + rx^{r-1} \sum_{n=0}^{\infty} a_n x^n = \sum_{n=0}^{\infty} (n+r)x^{n+r-1}$   
 $y'' = \frac{d}{dx}y'(x) = \frac{d}{dx} \left[ x^{r-1} \sum_{n=0}^{\infty} (n+r)a_n x^n \right] = x^{r-1} \sum_{n=0}^{\infty} (n+r)na_n x^{n-1} + (r-1)x^{r-2} \sum_{n=0}^{\infty} (n+r)a_n x^n = \sum_{n=0}^{\infty} (n+r)(n+r-1)a_n x^{n+r-2}.$

**7.1.22.**  $x^2(1+x)y'' + x(1+2x)y' - (4+6x)y = (x^2y'' + xy' - 4y) + x(x^2y'' + 2xy' - 6y) = \sum_{n=0}^{\infty} [(n+r)(n+r-1) + (n+r) - 4]a_n x^{n+r} + \sum_{n=0}^{\infty} [(n+r)(n+r-1) + 2(n+r) - 6]a_n x^{n+r+1} = \sum_{n=0}^{\infty} (n+r-2)(n+r+2)a_n x^{n+r} + \sum_{n=0}^{\infty} (n+r+3)(n+r-2)a_n x^{n+r+1} = \sum_{n=0}^{\infty} (n+r-2)(n+r+2)a_n x^{n+r} + \sum_{n=1}^{\infty} (n+r+2)(n+r-3)a_{n-1}x^{n+r} = x^r \sum_{n=0}^{\infty} b_n x^n$  with  $b_0 = (r-2)(r+2)a_0$  and  $b_n = (n+r-2)(n+r+2)a_n + (n+r+2)(n+r-3)a_{n-1}$ ,  $n \geq 1$ .

**7.1.24.**  $x^2(1+3x)y'' + x(2+12x+x^2)y' + 2x(3+x)y = (x^2y'' + 2xy') + x(3x^2y'' + 12xy' + 6y) + x^2(xy' + 2y) = \sum_{n=0}^{\infty} [(n+r)(n+r-1) + 2(n+r)]a_n x^{n+r} + \sum_{n=0}^{\infty} [3(n+r)(n+r-1) + 12(n+r) + 6]a_n x^{n+r+1} + \sum_{n=0}^{\infty} [(n+r) + 2]a_n x^{n+r+2} = \sum_{n=0}^{\infty} (n+r)(n+r+1)a_n x^{n+r} + 3\sum_{n=0}^{\infty} (n+r+1)(n+r+2)a_n x^{n+r+1} + \sum_{n=0}^{\infty} (n+r+2)a_n x^{n+r+2} = \sum_{n=0}^{\infty} (n+r)(n+r+1)a_n x^{n+r} + 3\sum_{n=1}^{\infty} (n+r)(n+r+1)a_{n-1}x^{n+r} + \sum_{n=2}^{\infty} (n+r)a_{n-2}x^{n+r} = x^r \sum_{n=0}^{\infty} b_n x^n$  with  $b_0 = r(r+1)a_0$ ,  $b_1 = (r+1)(r+2)a_1 + 3(r+1)(r+2)a_0$ ,  $b_n = (n+r)(n+r+1)a_n + 3(n+r)(n+r+1)a_{n-1} + (n+r)a_{n-2}$ ,  $n \geq 2$ .

**7.1.26.**  $x^2(2+x^2)y'' + 2x(5+x^2)y' + 2(3-x^2)y = (2x^2y'' + 10xy' + 6y) + x^2(x^2y'' + 2xy' - 2y) = \sum_{n=0}^{\infty} [2(n+r)(n+r-1) + 10(n+r) + 6]a_n x^{n+r} + \sum_{n=0}^{\infty} [(n+r)(n+r-1) + 2(n+r) - 2]a_n x^{n+r+2} = 2\sum_{n=0}^{\infty} (n+r+1)(n+r+3)a_n x^{n+r} + \sum_{n=0}^{\infty} (n+r-1)(n+r+2)a_n x^{n+r+2} = 2\sum_{n=0}^{\infty} (n+r+1)(n+r+3)a_n x^{n+r} + \sum_{n=2}^{\infty} (n+r-3)(n+r)a_{n-2}x^{n+r} = x^r \sum_{n=0}^{\infty} b_n x^n$  with  $b_0 = 2(r+1)(r+3)a_0$ ,  $b_1 = 2(r+2)(r+4)a_1$ ,  $b_n = 2(n+r+1)(n+r+3)a_n + (n+r-3)(n+r)a_{n-2}$ ,  $n \geq 2$ .

## 7.2 Series Solutions Near an Ordinary Point I

**7.2.2.**  $p(n) = n(n-1) + 2n - 2 = (n+2)(n-1)$ ;  $a_{n+2} = -\frac{n-1}{n+1}a_n$ ;  $a_{2m+2} = -\frac{2m-1}{2m+1}a_{2m}$ , so  $a_{2m} = \frac{(-1)^m}{2m-1}a_0$ ;  $a_{2m+3} = -\frac{m}{m+2}a_{2m+1} = 0$  if  $m \geq 0$ ;  $y = a_0 \sum_{m=0}^{\infty} (-1)^{m+1} \frac{x^{2m}}{2m-1} + a_1 x$ .

**7.2.4.**  $p(n) = -n(n-1) - 8n - 12 = -(n+3)(n+4)$ ;  $a_{n+2} = -\frac{(n+3)(n+4)}{(n+2)(n+1)}a_n$ ;  $a_{2m+2} = -\frac{(m+2)(2m+3)}{(m+1)(2m+1)}a_{2m}$ , so  $a_{2m} = (m+1)(2m+1)a_0$ ;  $a_{2m+3} = \frac{(m+2)(2m+5)}{(m+1)(2m+3)}a_{2m+1}$  so  $a_{2m+1} = \frac{(m+1)(2m+3)}{3}a_1$ ;  $y = a_0 \sum_{m=0}^{\infty} (m+1)(2m+1)x^{2m} + \frac{a_1}{3} \sum_{m=0}^{\infty} (m+1)(2m+3)x^{2m+1}$ .

**7.2.6.**  $p(n) = n(n-1) + 2n + \frac{1}{4} = \frac{(2n+1)^2}{4}$ ;  $a_{n+2} = -\frac{(2n+1)^2}{4(n+2)(n+1)}a_n$ ;  $a_{2m+2} = -\frac{(4m+1)^2}{8(m+1)(2m+1)}a_{2m}$ , so  $a_{2m} = (-1)^m \left[ \prod_{j=0}^{m-1} \frac{(4j+1)^2}{2j+1} \right] \frac{1}{8^m m!} a_0$ ;  $a_{2m+3} = -\frac{(4m+3)^2}{8(2m+3)(m+1)}a_{2m+1}$  so  $a_{2m+1} = (-1)^m \left[ \prod_{j=0}^{m-1} \frac{(4j+3)^2}{2j+3} \right] \frac{1}{8^m m!}$ ;  $y = a_0 \sum_{m=0}^{\infty} (-1)^m \left[ \prod_{j=0}^{m-1} \frac{(4j+1)^2}{2j+1} \right] \frac{x^{2m}}{8^m m!} + a_1 \sum_{m=0}^{\infty} (-1)^m \left[ \prod_{j=0}^{m-1} \frac{(4j+3)^2}{2j+3} \right] \frac{x^{2m+1}}{8^m m!}$ .

**7.2.8.**  $p(n) = n(n-1) - 10n + 28 = (n-7)(n-4)$ ;  $a_{n+2} = -\frac{(n-7)(n-4)}{(n+2)(n+1)}a_n$ ;  $a_{2m+2} = -\frac{2(2m-7)(m-2)}{2(m+1)(2m+1)}a_{2m}$ , so  $a_2 = -14a_0$ ,  $a_4 = -\frac{5}{6}a_2 = \frac{35}{3}a_0$ ,  $a_{2m} = 0$  if  $m \geq 3$ ;  $a_{2m+3} = -\frac{(m-3)(2m-3)}{(2m+3)(m+1)}a_{2m+1}$ , so  $a_3 = -3a_1$ ,  $a_5 = -\frac{1}{5}a_3 = \frac{3}{5}a_1$ ,  $a_7 = \frac{1}{21}a_5 = \frac{1}{35}a_1$ ;  $y = a_0 \left( 1 - 14x^2 + \frac{35}{3}x^4 \right) + a_1 \left( x - 3x^3 + \frac{3}{5}x^5 + \frac{1}{35}x^7 \right)$ .

**7.2.10.**  $p(n) = 2n + 3$ ;  $a_{n+2} = -\frac{2n+3}{(n+2)(n+1)}a_n$ ;  $a_{2m+2} = -\frac{4m+3}{2(m+1)(2m+1)}a_{2m}$ , so  $a_{2m} = \left[ \prod_{j=0}^{m-1} \frac{4j+3}{2j+1} \right] \frac{(-1)^m}{2^m m!} a_0$ ;  $a_{2m+3} = -\frac{4m+5}{2(2m+3)(m+1)}a_{2m+1}$  so  $a_{2m+1} = \left[ \prod_{j=0}^{m-1} \frac{4j+5}{2j+3} \right] \frac{(-1)^m}{2^m m!} a_1$ ;  $y = a_0 \sum_{m=0}^{\infty} (-1)^m \left[ \prod_{j=0}^{m-1} \frac{4j+3}{2j+1} \right] \frac{x^{2m}}{2^m m!} + a_1 \sum_{m=0}^{\infty} (-1)^m \left[ \prod_{j=0}^{m-1} \frac{4j+5}{2j+3} \right] \frac{x^{2m+1}}{2^m m!}$ .

**7.2.12.**  $p(n) = 2n(n-1) - 9n - 6 = (n-6)(2n+1)$ ;  $a_{n+2} = -\frac{(n-6)(2n+1)}{(n+2)(n+1)}a_n$ ;  $a_0 = y(0) = 1$ ;  $a_1 = y'(0) = -1$ .

**7.2.13.**  $p(n) = 8n(n-1) + 2 = 2(2n-1)^2$ ;  $a_{n+2} = -\frac{2(2n-1)^2}{(n+2)(n+1)}a_n$ ;  $a_0 = y(0) = 2$ ;  $a_1 = y'(0) = -1$ .

**7.2.16.**  $p(n) = -1$ ;  $a_{n+2} = \frac{1}{(n+2)(n+1)}a_n$ ;  $a_{2m+2} = \frac{1}{(2m+2)(2m+1)}a_{2m}$ , so  $a_{2m} = \frac{1}{(2m)!}a_0$ ;  $a_{2m+3} = \frac{1}{(2m+3)(2m+1)}a_{2m+1}$ , so  $a_{2m+1} = \frac{1}{(2m+1)!}a_1$ ;  $y = a_0 \sum_{m=0}^{\infty} \frac{(x-3)^{2m}}{(2m)!} + a_1 \sum_{m=0}^{\infty} \frac{(x-3)^{2m+1}}{(2m+1)!}$ .

**7.2.18.** Let  $t = x - 1$ ; then  $(1 - 2t^2)y'' - 10ty' - 6y = 0$ ;  $p(n) = -2n(n-1) - 10n - 6 =$



$$-2(n+1)(n+3); a_{n+2} = \frac{2(n+3)}{n+2}a_n; a_{2m+2} = \frac{2m+3}{m+1}a_{2m}, \text{ so } a_{2m} = \frac{1}{m!} \left[ \prod_{j=0}^{m-1} (2j+3) \right] a_0;$$

$$a_{2m+3} = \frac{4(m+2)}{2m+3}a_{2m+1}, \text{ so } a_{2m+1} = \frac{4^m(m+1)!}{\prod_{j=0}^{m-1} (2j+3)} a_1; y = a_0 \sum_{m=0}^{\infty} \left[ \prod_{j=0}^{m-1} (2j+3) \right] \frac{(x-1)^{2m}}{m!}$$

$$+ a_1 \sum_{m=0}^{\infty} \frac{4^m(m+1)!}{\prod_{j=0}^{m-1} (2j+3)} (x-1)^{2m+1}.$$

**7.2.20.** Let  $t = x + 1$ ; then  $\left(1 + \frac{3t^2}{3}\right)y'' - \frac{9t}{2}y' + \frac{3}{2}y = 0$ ;  $p(n) = \frac{3}{2}n(n-1) + \frac{9}{2}n + \frac{3}{2} = \frac{3}{2}(n+1)^2$ ;

$$a_{n+2} = -\frac{3(n+1)}{2(n+2)}a_n; a_{2m+2} = -\frac{3(2m+1)}{4(m+1)}a_{2m}, \text{ so } a_{2m} = (-1)^m \left[ \prod_{j=0}^{m-1} (2j+1) \right] \frac{3^m}{4^m m!} a_0; a_{2m+3} =$$

$$-\frac{3(m+1)}{2m+3}a_{2m+1}, \text{ so } a_{2m+1} = (-1)^m \frac{3^m m!}{\prod_{j=0}^{m-1} (2j+3)} a_1;$$

$$y = a_0 \sum_{m=0}^{\infty} (-1)^m \left[ \prod_{j=0}^{m-1} (2j+1) \right] \frac{3^m}{4^m m!} (x+1)^{2m} + a_1 \sum_{m=0}^{\infty} (-1)^m \frac{3^m m!}{\prod_{j=0}^{m-1} (2j+3)} (x+1)^{2m+1}.$$

**7.2.22.**  $p(n) = n + 3$ ;  $a_{n+2} = -\frac{n+3}{(n+2)(n+1)}a_n$ ;  $a_0 = y(3) = -2$ ;  $a_1 = y'(3) = 3$ .

**7.2.24.** Let  $t = x - 3$ ;  $(1 + 4t^2)y'' + y = 0$ ;  $p(n) = (4n(n-1) + 1) = 2n(n-1)^2$ ;  $a_{n+2} = -\frac{(2n-1)^2}{(n+2)(n+1)}a_n$ ;  $a_0 = y(3) = 4$ ;  $a_1 = y'(3) = -6$ .

**7.2.26.** Let  $t = x + 1$ ;  $\left(1 + \frac{2t^2}{3}\right)y'' - \frac{20}{3}ty' + 20y = 0$ ;  $p(n) = \frac{2}{3}n(n-1) - \frac{20}{3}n + 20 = \frac{2(n-6)(n-5)}{3}$ ;  $a_{n+2} = -\frac{2(n-6)(n-5)}{3(n+2)(n+1)}a_n$ ;  $a_0 = y(-1) = 3$ ;  $a_1 = y'(-1) = -3$ .

**7.2.28.** From Theorem 7.2.2,  $a_{n+2} = -\frac{p(n)}{(n+2)(n+1)}a_n$ ;  $a_{2m+2} = -\frac{p(2m)}{(2m+2)(2m+1)}a_{2m}$ , so

$$a_{2m} = \left[ \prod_{j=0}^{m-1} p(2j) \right] \frac{(-1)^m}{(2m)!} a_0; a_{2m+3} = -\frac{p(2m+1)}{(2m+3)(2m+2)}a_{2m}, \text{ so } a_{2m+1} = \left[ \prod_{j=0}^{m-1} p(2j+1) \right] \frac{(-1)^m}{(2m+1)!} a_1.$$

**7.2.30.** (a) Here  $p(n) = -[n(n-1) + 2bn - \alpha(\alpha + 2b - 1)] = -(n-\alpha)(n+\alpha+2b-1)$ , so Exercise 7.2.28 implies that  $y_1$  and  $y_2$  have the stated forms. If  $\alpha = 2k$ , then

$$y_1 = \sum_{m=0}^{\infty} \left[ \prod_{j=0}^{m-1} (2j-2k)(2j+2k+2b-1) \right] \frac{x^{2m}}{(2m)!}. \quad (\text{C})$$

If  $\alpha = 2k + 1$ , then

$$y_2 = \sum_{m=0}^{\infty} \left[ \prod_{j=0}^{m-1} (2j-2k)(2j+2k+2b) \right] \frac{x^{2m+1}}{(2m+1)!}. \quad (\text{D})$$

Since  $2b$  is not a negative integer and  $\prod_{j=0}^{m-1} (2j-2k) = 0$  if  $m > k$ ,  $y_1$  in (C) and  $y_2$  in (D) have the stated properties. This implies the conclusions regarding  $P_n$ .

(b) Multiplying (A) through by  $(1-x^2)^{b-1}$  yields

$$[(1-x^2)^b P_n']' = -n(n+2b-1)(1-x^2)^{b-1} P_n. \quad (\text{E})$$

(c) Therefore,

$$[(1-x^2)^b P'_m]' = -m(m+2b-1)(1-x^2)^{b-1} P_m. \quad (\text{F})$$

Subtract  $P_n$  times (F) from  $P_m$  times (E) to obtain (B).

(d) Integrating the left side of (B) by parts over  $[-1, 1]$  yields zero, which implies the conclusion.

**7.2.32.** (a) Let  $Ly = (1 + \alpha x^3)y'' + \beta x^2 y' + \gamma xy$ . If  $y = \sum_{n=0}^{\infty} a_n x^n$ , then  $Ly = \sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} + \sum_{n=0}^{\infty} p(n)a_n x^{n+1} = 2a_2 + \sum_{n=0}^{\infty} [(n+3)(n+2)a_{n+3} + p(n)a_n]x^{n+1} = 0$  if and only if  $a_2 = 0$  and  $a_{n+3} = -\frac{p(n)}{(n+3)(n+2)}a_n$  for  $n \geq 0$ .

**7.2.34.**  $p(r) = -2r(r-1) - 10r - 8 = -2(r+2)^2$ ; (A)  $\prod_{j=0}^{m-1} \frac{p(3j)}{3j+2} = \prod_{j=0}^{m-1} \frac{(-2)(3j+2)^2}{3j+2} = (-1)^m 2^m \prod_{j=0}^{m-1} (3j+2)$ ; (B)  $\prod_{j=0}^{m-1} \frac{p(3j+1)}{3j+4} = \prod_{j=0}^{m-1} \frac{(-2)(3j+3)^2}{3j+4} = \frac{(-1)^m 2^m (m!)^2}{\prod_{j=0}^{m-1} (3j+4)}$ . Substituting (A) and (B) into the result of Exercise 7.2.32(c) yields

$$y = a_0 \sum_{m=0}^{\infty} \left(\frac{2}{3}\right)^m \left[ \prod_{j=0}^{m-1} (3j+2) \right] \frac{x^{3m}}{m!} + a_1 \sum_{m=0}^{\infty} \frac{6^m m!}{\prod_{j=0}^{m-1} (3j+4)} x^{3m+1}.$$

**7.2.36.**  $p(r) = -2r(r-1) + 6r + 24 = -2(r-6)(r+2)$ ; (A)  $\prod_{j=0}^{m-1} \frac{p(3j)}{3j+2} = \prod_{j=0}^{m-1} (-6)(j-2)$ .

(B)  $\prod_{j=0}^{m-1} \frac{p(3j+1)}{3j+4} = \prod_{j=0}^{m-1} \frac{(-6)(j+1)(3j-5)}{3j+4} = (-1)^m 6^m m \prod_{j=0}^{m-1} \frac{3j-5}{3j+4}$ . Substituting (A) and (B) into the result of Exercise 7.2.32(c) yields

$$y = a_0(1 - 4x^3 + 4x^6) + a_1 \sum_{m=0}^{\infty} 2^m \left[ \prod_{j=0}^{m-1} \frac{3j-5}{3j+4} \right] x^{3m+1}.$$

**7.2.38.** (a) Let  $Ly = (1 + \alpha x^{k+2})y'' + \beta x^{k+1}y' + \gamma x^k y$ . If  $y = \sum_{n=0}^{\infty} a_n x^n$ , then  $Ly = \sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} + \sum_{n=0}^{\infty} p(n)a_n x^{n+k} = \sum_{n=-k}^{-1} (n+k+2)(n+k-1)a_{n+k+2} x^{n+k} + \sum_{n=0}^{\infty} [(n+k+2)(n+k+1)a_{n+k+2} + p(n)a_n]x^{n+k} = 0$  if and only if  $a_k = 0$  for  $2 \leq n \leq k+1$  and (A)  $a_{n+k+1} = -\frac{p(n)}{(n+k+2)(n+k+1)}a_n$  for  $n \geq 0$ .

(b) If  $a_n = 0$  the  $a_{n+(k+2)m} = 0$  for all  $m \geq 0$ , from (A).

**7.2.40.**  $k = 2$  and  $p(r) = 1$ ; (A)  $\prod_{j=0}^{m-1} \frac{p(4j)}{4j+3} = \frac{1}{\prod_{j=0}^{m-1} (4j+3)}$ ; (B)  $\prod_{j=0}^{m-1} \frac{p(4j+1)}{(4j+5)} = \frac{1}{\prod_{j=0}^{m-1} (4j+5)}$ .

Substituting (A) and (B) into the result of Exercise 7.2.38(c) yields

$$y = a_0 \sum_{m=0}^{\infty} (-1)^m \frac{x^{4m}}{4^m m! \prod_{j=0}^{m-1} (4j+3)} + a_1 \sum_{m=0}^{\infty} (-1)^m \frac{x^{4m+1}}{4^m m! \prod_{j=0}^{m-1} (4j+5)}.$$

**7.2.42.**  $k = 6$  and  $p(r) = r(r-1) - 16r + 72 = (r-9)(r-8)$ ; (A)  $\prod_{j=0}^{m-1} \frac{p(8j)}{8j+7} = \prod_{j=0}^{m-1} \frac{8(j-1)(8j-9)}{8j+7}$ ;

$$(B) \prod_{j=0}^{m-1} \frac{p(8j+1)}{(8j+9)} = \prod_{j=0}^{m-1} \frac{8(j-1)(8j-7)}{8j+9};$$

Substituting (A) and (B) into the result of Exercise 7.2.38(c) yields

$$y = a_0 \left(1 - \frac{9}{7}x^8\right) + a_1 \left(x - \frac{7}{9}x^9\right).$$

$$7.2.44. \quad k = 4 \text{ and } p(r) = r + 6; (A) \prod_{j=0}^{m-1} \frac{p(6j)}{6j+5} = \prod_{j=0}^{m-1} \frac{6(j+1)}{6j+5} = \frac{6^m m!}{\prod_{j=0}^{m-1} (6j+5)};$$

$$(B) \prod_{j=0}^{m-1} \frac{p(6j+1)}{(6j+7)} = 1;$$

Substituting (A) and (B) into the result of Exercise 7.2.38(c) yields

$$y = a_0 \sum_{m=0}^{\infty} (-1)^m \frac{x^{6m}}{\prod_{j=0}^{m-1} (6j+5)} + a_1 \sum_{m=0}^{\infty} (-1)^m \frac{x^{6m+1}}{6^m m!}.$$

### 7.3 Series Solutions Near an Ordinary Point II

**7.3.2.** If  $y = \sum_{n=0}^{\infty} a_n x^n$ , then  $(1+x+2x^2)y'' + (2+8x)y' + 4y = \sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} + \sum_{n=2}^{\infty} n(n-1)a_n x^{n-1} + 2 \sum_{n=2}^{\infty} n(n-1)a_n x^n + 2 \sum_{n=1}^{\infty} n a_n x^{n-1} + 8 \sum_{n=1}^{\infty} n a_n x^n + 4 \sum_{n=0}^{\infty} a_n x^n = \sum_{n=0}^{\infty} (n+2)(n+1)(a_{n+2} + a_{n+1} + 2a_n)x^n = 0$  if  $a_{n+2} = -a_{n+1} - 2a_n$ ,  $a_n \geq 0$ . Starting with  $a_0 = -1$  and  $a_1 = 2$  yields  $y = -1 + 2x - 4x^3 + 4x^4 + 4x^5 - 12x^6 + 4x^7 + \dots$ .

**7.3.4.** If  $y = \sum_{n=0}^{\infty} a_n x^n$ , then  $(1+x+3x^2)y'' + (2+15x)y' + 12y = \sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} + \sum_{n=2}^{\infty} n(n-1)a_n x^{n-1} + 3 \sum_{n=2}^{\infty} n(n-1)a_n x^n + 2 \sum_{n=1}^{\infty} n a_n x^{n-1} + 15 \sum_{n=1}^{\infty} n a_n x^n + 12 \sum_{n=0}^{\infty} a_n x^n = \sum_{n=0}^{\infty} [(n+2)(n+1)a_{n+2} + (n+1)(n+2)a_{n+1} + 3(n+2)^2 a_n]x^n = 0$  if  $a_{n+2} = -a_{n+1} - \frac{3(n+2)}{n+1}a_n$ ,  $a_n \geq 0$ . Starting with  $a_0 = 0$  and  $a_1 = 1$  yields  $y = x - x^2 - \frac{7}{2}x^3 + \frac{15}{2}x^4 + \frac{45}{8}x^5 - \frac{261}{8}x^6 + \frac{207}{16}x^7 + \dots$ .

**7.3.6.** If  $y = \sum_{n=0}^{\infty} a_n x^n$ , then  $(3+3x+x^2)y'' + (6+4x)y' + 2y = 3 \sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} + 3 \sum_{n=2}^{\infty} n(n-1)a_n x^{n-1} + \sum_{n=2}^{\infty} n(n-1)a_n x^n + 6 \sum_{n=1}^{\infty} n a_n x^{n-1} + 4 \sum_{n=1}^{\infty} n a_n x^n + 2 \sum_{n=0}^{\infty} a_n x^n = \sum_{n=0}^{\infty} (n+2)(n+1)[3a_{n+2} + 3a_{n+1} + a_n]x^n = 0$  if  $a_{n+2} = -a_{n+1} - a_n/3$ ,  $a_n \geq 0$ . Starting with  $a_0 = 7$  and  $a_1 = 3$  yields  $y = 7 + 3x - \frac{16}{3}x^2 + \frac{13}{3}x^3 - \frac{23}{9}x^4 + \frac{10}{9}x^5 - \frac{7}{27}x^6 - \frac{1}{9}x^7 + \dots$ .

**7.3.8.** The equation is equivalent to  $(1+t+2t^2)y'' + (2+6t)y' + 2y = 0$  with  $t = x - 1$ . If  $y = \sum_{n=0}^{\infty} a_n t^n$ , then  $(1+t+2t^2)y'' + (2+6t)y' + 2y = \sum_{n=2}^{\infty} n(n-1)a_n t^{n-2} + \sum_{n=2}^{\infty} n(n-1)a_n t^{n-1} + 2 \sum_{n=2}^{\infty} n(n-1)a_n t^n + 2 \sum_{n=1}^{\infty} n a_n t^{n-1} + 6 \sum_{n=1}^{\infty} n a_n t^n + 2 \sum_{n=0}^{\infty} a_n t^n = \sum_{n=0}^{\infty} (n+1)[(n+2)a_{n+2} + (n+2)a_{n+1} + 2(n+1)a_n]t^n = 0$  if  $a_{n+2} = -a_{n+1} - \frac{2(n+1)}{n+2}a_n$ ,  $a_n \geq 0$ . Starting with  $a_0 = 1$  and  $a_1 = -1$  yields

$$y = 1 - (x-1) + \frac{4}{3}(x-1)^3 - \frac{4}{3}(x-1)^4 - \frac{4}{5}(x-1)^5 + \frac{136}{45}(x-1)^6 - \frac{104}{63}(x-1)^7 + \dots$$

**7.3.10.** The equation is equivalent to  $(1+t+t^2)y'' + (3+4t)y' + 2y = 0$  with  $t = x-1$ . If  $y = \sum_{n=0}^{\infty} a_n t^n$ , then  $(1+t+t^2)y'' + (3+4t)y' + 2y = \sum_{n=2}^{\infty} n(n-1)a_n t^{n-2} + \sum_{n=2}^{\infty} n(n-1)a_n t^{n-1} + \sum_{n=2}^{\infty} n(n-1)a_n t^n + 3 \sum_{n=1}^{\infty} n a_n t^{n-1} + 4 \sum_{n=1}^{\infty} n a_n t^n + 2 \sum_{n=0}^{\infty} a_n t^n = \sum_{n=0}^{\infty} (n+1)[(n+2)a_{n+2} + (n+3)a_{n+1} + (n+2)a_n]t^n = 0$  if  $a_{n+2} = -\frac{n+3}{n+2}a_{n+1} - a_n$ ,  $a_n \geq 0$ . Starting with  $a_0 = 2$  and  $a_1 = -1$  yields  $y =$

$$2 - (x-1) - \frac{1}{2}(x-1)^2 + \frac{5}{3}(x-1)^3 - \frac{19}{12}(x-1)^4 + \frac{7}{30}(x-1)^5 + \frac{59}{45}(x-1)^6 - \frac{1091}{630}(x-1)^7 + \dots$$

**7.3.12.** The equation is equivalent to  $(1+2t+t^2)y'' + (1+7t)y' + 8y = 0$  with  $t = x-1$ . If  $y = \sum_{n=0}^{\infty} a_n t^n$ , then  $(1+2t+t^2)y'' + (1+7t)y' + 8y = \sum_{n=2}^{\infty} n(n-1)a_n t^{n-2} + 2 \sum_{n=2}^{\infty} n(n-1)a_n t^{n-1} + \sum_{n=2}^{\infty} n(n-1)a_n t^n + \sum_{n=1}^{\infty} n a_n t^{n-1} + 7 \sum_{n=1}^{\infty} n a_n t^n + 8 \sum_{n=0}^{\infty} a_n t^n = \sum_{n=0}^{\infty} [(n+2)(n+1)a_{n+2} + (n+1)(2n+1)a_{n+1} + (n+2)(n+4)a_n]t^n = 0$  if  $a_{n+2} = -\frac{2n+1}{n+2}a_{n+1} - \frac{n+4}{n+1}a_n$ ,  $a_n \geq 0$ . Starting with  $a_0 = 1$  and  $a_1 = -2$

$$\text{yields } y = 1 - 2(x-1) - 3(x-1)^2 + 8(x-1)^3 - 4(x-1)^4 - \frac{42}{5}(x-1)^5 + 19(x-1)^6 - \frac{604}{35}(x-1)^7 + \dots$$

**7.3.16.** If  $y = \sum_{n=0}^{\infty} a_n x^n$ , then  $(1-x)y'' - (2-x)y' + y = \sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} - \sum_{n=2}^{\infty} n(n-1)a_n x^{n-1} - 2 \sum_{n=1}^{\infty} n a_n x^{n-1} + \sum_{n=1}^{\infty} n a_n x^n + \sum_{n=0}^{\infty} a_n x^n = \sum_{n=0}^{\infty} [(n+2)(n+1)a_{n+2} - (n+2)(n+1)a_{n+1} + (n+1)a_n]x^n = 0$  if  $a_{n+2} = a_{n+1} - \frac{a_n}{n+2}$ ,  $a_n \geq 0$ .

**7.3.18.** If  $y = \sum_{n=0}^{\infty} a_n x^n$ , then  $(1+x^2)y'' + y' + 2y = \sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} + \sum_{n=2}^{\infty} n(n-1)a_n x^n + \sum_{n=1}^{\infty} n a_n x^{n-1} + 2 \sum_{n=0}^{\infty} a_n x^n = \sum_{n=0}^{\infty} [(n+2)(n+1)a_{n+2} + (n+1)a_{n+1} + (n^2 - n + 2)a_n]x^n = 0$  if  $a_{n+2} = -\frac{1}{n+2}a_{n+1} - \frac{n^2 - n + 2}{(n+2)(n+1)}a_n$ .

**7.3.20.** The equation is equivalent to  $(3+2t)y'' + (1+2t)y' - (1-2t)y = 0$  with  $t = x-1$ . If  $y = \sum_{n=0}^{\infty} a_n t^n$ , then  $(3+2t)y'' + (1+2t)y' - (1-2t)y = 3 \sum_{n=2}^{\infty} n(n-1)a_n t^{n-2} + 2 \sum_{n=2}^{\infty} n(n-1)a_n t^{n-1} + \sum_{n=1}^{\infty} n a_n t^{n-1} + 2 \sum_{n=1}^{\infty} n a_n t^n - \sum_{n=0}^{\infty} a_n t^n + 2 \sum_{n=0}^{\infty} a_n t^{n+1} = (6a_2 + a_1 - a_0) + \sum_{n=1}^{\infty} [3(n+2)(n+1)a_{n+2} + (n+1)(2n+1)a_{n+1} + (2n-1)a_n + 2a_{n-1}]t^n = 0$  if  $a_2 = -\frac{a_1 - a_0}{6}$  and  $a_{n+2} = -\frac{2n+1}{3(n+2)}a_{n+1} - \frac{2n-1}{3(n+2)(n+1)}a_n - \frac{2}{3(n+2)(n+1)}a_{n-1}$ ,  $n \geq 1$ . Starting with  $a_0 = 1$  and  $a_1 = -2$  yields  $y = 1 - 2(x-1) + \frac{1}{2}(x-1)^2 - \frac{1}{6}(x-1)^3 + \frac{5}{36}(x-1)^4 - \frac{73}{1080}(x-1)^5 + \dots$

**7.3.22.** The equation is equivalent to  $(1+t)y'' + (2-2t)y' + (3+t)y = 0$  with  $t = x+3$ . If  $y = \sum_{n=0}^{\infty} a_n t^n$ , then  $(1+t)y'' + (2-2t)y' + (3+t)y = \sum_{n=2}^{\infty} n(n-1)a_n t^{n-2} + \sum_{n=2}^{\infty} n(n-1)a_n t^{n-1} +$

$$2 \sum_{n=1}^{\infty} n a_n t^{n-1} - 2 \sum_{n=1}^{\infty} n a_n t^n + 3 \sum_{n=0}^{\infty} a_n t^n + \sum_{n=0}^{\infty} a_n t^{n+1} = (2a_2 + 2a_1 + 3a_0) + \sum_{n=1}^{\infty} [(n+2)(n+1)a_{n+2} + (n+2)(n+1)a_{n+1} - (2n-3)a_n + a_{n-1}] t^n = 0$$

if  $a_2 = -\frac{2a_1 + 3a_0}{2}$  and  $a_{n+2} = -a_{n+1} + \frac{(2n-3)a_n - a_{n-1}}{(n+2)(n+1)}$ ,  $n \geq 1$ . Starting with  $a_0 = 2$  and  $a_1 = -2$  yields

$$y = 2 - 2(x+3) - (x+3)^2 + (x+3)^3 - \frac{11}{12}(x+3)^4 + \frac{67}{60}(x+3)^5 + \dots$$

**7.3.24.** The equation is equivalent to  $(1+2t)y'' + 3y' + (1-t)y = 0$  with  $t = x+1$ . If  $y = \sum_{n=0}^{\infty} a_n t^n$ , then  $(1+2t)y'' + 3y' + (1-t)y = \sum_{n=2}^{\infty} n(n-1)a_n t^{n-2} + 2 \sum_{n=2}^{\infty} n(n-1)a_n t^{n-1} + 3 \sum_{n=1}^{\infty} n a_n t^{n-1} + \sum_{n=0}^{\infty} a_n t^n - \sum_{n=0}^{\infty} a_n t^{n+1} = (2a_2 + 3a_1 + a_0) + \sum_{n=1}^{\infty} [(n+2)(n+1)a_{n+2} + (2n+3)(n+1)a_{n+1} + a_n - a_{n-1}] t^n = 0$  if  $a_2 = -\frac{3a_1 + a_0}{2}$  and  $a_{n+2} = -\frac{2n+3}{n+2}a_{n+1} - \frac{a_n - a_{n-1}}{(n+2)(n+1)}$ ,  $n \geq 1$ . Starting with  $a_0 = 2$  and  $a_1 = -3$  yields  $y = 2 - 3(x+1) + \frac{7}{2}(x+1)^2 - 5(x+1)^3 + \frac{197}{24}(x+1)^4 - \frac{287}{20}(x+1)^5 + \dots$

**7.3.26.** The equation is equivalent to  $(6-2t)y'' + (3+t)y = 0$  with  $t = x-2$ . If  $y = \sum_{n=0}^{\infty} a_n t^n$ , then  $(6-2t)y'' + (3+t)y = 6 \sum_{n=2}^{\infty} n(n-1)a_n t^{n-2} - 2 \sum_{n=2}^{\infty} n(n-1)a_n t^{n-1} + 3 \sum_{n=0}^{\infty} a_n t^n + \sum_{n=0}^{\infty} a_n t^{n+1} = (12a_2 + 3a_0) + \sum_{n=1}^{\infty} [6(n+2)(n+1)a_{n+2} - 2(n+1)na_{n+1} + 3a_n + a_{n-1}] t^n = 0$  if  $a_2 = -\frac{a_0}{4}$  and  $a_{n+2} = \frac{n}{3(n+2)}a_{n+1} - \frac{3a_n + a_{n-1}}{6(n+2)(n+1)}$ ,  $n \geq 1$ . Starting with  $a_0 = 2$  and  $a_1 = -4$  yields  $y = 2 - 4(x-2) - \frac{1}{2}(x-2)^2 + \frac{2}{9}(x-2)^3 + \frac{49}{432}(x-2)^4 + \frac{23}{1080}(x-2)^5 + \dots$

**7.3.28.** The equation is equivalent to  $(2+4t)y'' - (1-2t)y = 0$  with  $t = x+4$ . If  $y = \sum_{n=0}^{\infty} a_n t^n$ , then  $(2+4t)y'' - (1-2t)y = 2 \sum_{n=2}^{\infty} n(n-1)a_n t^{n-2} + 4 \sum_{n=2}^{\infty} n(n-1)a_n t^{n-1} - \sum_{n=0}^{\infty} a_n t^n + 2 \sum_{n=0}^{\infty} a_n t^{n+1} = (4a_2 - a_0) + \sum_{n=1}^{\infty} [2(n+2)(n+1)a_{n+2} + 4(n+1)na_{n+1} - a_n + 2a_{n-1}] t^n = 0$  if  $a_2 = \frac{a_0}{4}$  and  $a_{n+2} = -\frac{2n}{n+2}a_{n+1} + \frac{a_n - 2a_{n-1}}{2(n+2)(n+1)}$ ,  $n \geq 1$ . Starting with  $a_0 = -1$  and  $a_1 = 2$  yields  $y = -1 + 2(x+4) - \frac{1}{4}(x+4)^2 + \frac{1}{2}(x+4)^3 - \frac{65}{96}(x+4)^4 + \frac{67}{80}(x+4)^5 + \dots$

N=5; b=zeros(N,1); b(1)=-1;b(2)=2; b(3)=b(1)/4; for n=1:N-2 b(n+3)=-2\*n\*b(n+2)/(n+2)+(b(n+1)-2\*b(n))/(2\*(n+2)\*(n+1)); end

**7.3.29.** Let  $Ly = (1+\alpha x + \beta x^2)y'' + (\gamma + \delta x)y' + \epsilon y$ . If  $y = \sum_{n=0}^{\infty} a_n x^n$ , then  $Ly = \sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} + \alpha \sum_{n=2}^{\infty} n(n-1)a_n x^{n-1} + \beta \sum_{n=2}^{\infty} n(n-1)a_n x^n + \gamma \sum_{n=1}^{\infty} n a_n x^{n-1} + \delta \sum_{n=1}^{\infty} n a_n x^n + \epsilon \sum_{n=0}^{\infty} a_n x^n = \sum_{n=0}^{\infty} (n+2)(n+1)a_{n+2} x^n + \alpha \sum_{n=0}^{\infty} (n+1)na_{n+1} x^n + \beta \sum_{n=0}^{\infty} n(n-1)a_n x^n + \gamma \sum_{n=0}^{\infty} (n+1)a_{n+1} x^n + \delta \sum_{n=0}^{\infty} n a_n x^n + \epsilon \sum_{n=0}^{\infty} a_n x^n =$

$\sum_{n=0}^{\infty} b_n x^n$ , where  $b_n = (n+1)(n+2)a_{n+2} + (n+1)(\alpha n + \gamma)a_{n+1} + [\beta n(n-1) + \delta n + \epsilon]a_n$ , which implies the conclusion.

**7.3.30.** (a) Let  $\gamma = 2\alpha$ ,  $\delta = 4\beta$ , and  $\epsilon = 2\beta$  in Exercise 7.3.29 to obtain (B).

(b) If  $a_n = c_1 r_1^n + c_2 r_2^n$ , then  $a_{n+2} + \alpha a_{n+1} + \beta a_n = c_1 r_1^n (r_1^2 + \alpha r_1 + \beta) + c_2 r_2^n (r_2^2 + \alpha r_2 + \beta) = c_1 r_1^n P_0(r_1) + c_2 r_2^n P_0(r_2) = 0$ , so  $\{a_n\}$  satisfies (B). Since  $1/r_1$  and  $1/r_2$  are the zeros of  $P_0$ , Theorem 7.2.1 implies that  $\sum_{n=0}^{\infty} (c_1 r_1^n + c_2 r_2^n) x^n$  is a solution of (A) on  $(-\rho, \rho)$ .

(c) If  $|x| < \rho$ , then  $|r_1 x| < \rho$  and  $|r_2 x| < 1$ , so  $\sum_{n=0}^{\infty} r_i^n x^n = \frac{1}{1 - r_i x} = y_i$ ,  $i = 1, 2$ . Therefore, (b) implies that  $\{y_1, y_2\}$  is a fundamental set of solutions of (A) on  $(-\rho, \rho)$ .

(d) (A) can be written as  $P_0 y'' + 2P_0' y' + P_0'' y = (P_0 y)'' = 0$ . Therefore,  $P_0 y = a + bx$  where  $a$  and  $b$  are arbitrary constants, and a partial fraction expansion shows that the general solution of (A) on any interval not containing  $1/r_1$  or  $1/r_2$  is  $y = \frac{a + bx}{P_0(x)} = \frac{c_1}{1 - r_1 x} + \frac{c_2}{1 - r_2 x} = c_1 y_1 + c_2 y_2$ .

(e) If  $a_n = c_1 r_1^n + c_2 r_2^n$ , then  $a_{n+2} + \alpha a_{n+1} + \beta a_n = c_1 r_1^n (r_1^2 + \alpha r_1 + \beta) + c_2 r_2^n [(n+2)r_2^2 + \alpha(n+1)r_2 + \beta n] = (c_1 + nc_2)r_1^n P_0(r_1) + c_2 r_2^n P_0'(r_1) = 0$ , so  $\{a_n\}$  satisfies (B). Since  $1/r_1$  is the only zero of  $P_0$ , Theorem 7.2.1 implies that  $\sum_{n=0}^{\infty} (c_1 + c_2 n) r_1^n x^n$  is a solution of (A) on  $(-\rho, \rho)$ .

(f) If  $|x| < \rho$ , then  $|r_1 x| < \rho$ , so  $\sum_{n=0}^{\infty} r_1^n x^n = \frac{1}{1 - r_1 x} = y_1$ . Differentiating this and multiplying the result by  $x$  shows that  $\sum_{n=0}^{\infty} n r_1^n x^n = \frac{r_1 x}{(1 - r_1 x)^2} = r_1 y_2$ . Therefore, (e) implies that  $\{y_1, y_2\}$  is a fundamental set of solutions of (A) on  $(-\rho, \rho)$ .

(g) The argument is the same as in (c), but now the partial fraction expansion can be written as  $y = \frac{a + bx}{P_0(x)} = \frac{c_1}{1 - r_1 x} + \frac{c_2 x}{(1 - r_2 x)^2} = c_1 y_1 + c_2 y_2$ .

**7.3.32.** If  $y = \sum_{n=0}^{\infty} a_n x^n$ , then  $y'' + 2xy' + (3+2x^2)y = \sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} + 2 \sum_{n=1}^{\infty} n a_n x^n + 3 \sum_{n=0}^{\infty} a_n x^n + 2 \sum_{n=0}^{\infty} a_n x^{n+2} = (2a_2 + 3a_0) + (6a_3 + 5a_1)x + \sum_{n=2}^{\infty} [(n+2)(n+1)a_{n+2} + (2n+3)a_n + 2a_{n-2}]x^n = 0$  if  $a_2 = -3a_0/2$ ,  $a_3 = -5a_1/6$ , and  $a_{n+2} = -\frac{(2n+3)a_n + 2a_{n-2}}{(n+2)(n+1)}$ ,  $n \geq 2$ . Starting with  $a_0 = 1$  and  $a_1 = -2$  yields  $y = 1 - 2x - \frac{3}{2}x^2 + \frac{5}{3}x^3 + \frac{17}{24}x^4 - \frac{11}{20}x^5 + \dots$ .

**7.3.34.** If  $y = \sum_{n=0}^{\infty} a_n x^n$ , then  $y'' + 5xy' - (3-x^2)y = \sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} + 5 \sum_{n=1}^{\infty} n a_n x^n - 3 \sum_{n=0}^{\infty} a_n x^n + \sum_{n=0}^{\infty} a_n x^{n+2} = (2a_2 - 3a_0) + (6a_3 + 2a_1)x + \sum_{n=2}^{\infty} [(n+2)(n+1)a_{n+2} + (5n-3)a_n + a_{n-2}]x^n = 0$  if  $a_2 = 3a_0/2$ ,  $a_3 = -a_1/3$ , and  $a_{n+2} = -\frac{(5n-3)a_n + a_{n-2}}{(n+2)(n+1)}$ ,  $n \geq 2$ . Starting with  $a_0 = 6$  and  $a_1 = -2$  yields  $y = 6 - 2x + 9x^2 + \frac{2}{3}x^3 - \frac{23}{4}x^4 - \frac{3}{10}x^5 + \dots$ .

**7.3.36.** If  $y = \sum_{n=0}^{\infty} a_n x^n$ , then  $y'' - 3xy' + (2+4x^2)y = \sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} - 3 \sum_{n=1}^{\infty} n a_n x^n + 2 \sum_{n=0}^{\infty} a_n x^n +$

$4 \sum_{n=0}^{\infty} a_n x^{n+2} = (2a_2 + 2a_0) + (6a_3 - a_1)x + \sum_{n=2}^{\infty} [(n+2)(n+1)a_{n+2} - (3n-2)a_n + 4a_{n-2}]x^n = 0$  if  $a_2 = -a_0$ ,  $a_3 = a_1/6$ , and  $a_{n+2} = \frac{(3n-2)a_n - 4a_{n-2}}{(n+2)(n+1)}$ ,  $n \geq 2$ . Starting with  $a_0 = 3$  and  $a_1 = 6$  yields  $y = 3 + 6x - 3x^2 + x^3 - 2x^4 - \frac{17}{20}x^5 + \dots$ .

**7.3.38.** If  $y = \sum_{n=0}^{\infty} a_n x^n$ , then  $3y'' + 2xy' + (4 - x^2)y = 3 \sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} + 2 \sum_{n=1}^{\infty} na_n x^n + 4 \sum_{n=0}^{\infty} a_n x^n - \sum_{n=0}^{\infty} a_n x^{n+2} = (6a_2 + 4a_0) + (18a_3 + 6a_1)x + \sum_{n=2}^{\infty} [3(n+2)(n+1)a_{n+2} + (2n+4)a_n - a_{n-2}]x^n = 0$  if  $a_2 = -2a_0/3$ ,  $a_3 = -a_1/3$ , and  $a_{n+2} = -\frac{(2n+4)a_n - a_{n-2}}{3(n+2)(n+1)}$ ,  $n \geq 2$ . Starting with  $a_0 = -2$  and  $a_1 = 3$  yields  $y = -2 + 3x + \frac{4}{3}x^2 - x^3 - \frac{19}{54}x^4 + \frac{13}{60}x^5 + \dots$ .

**7.3.40.** If  $y = \sum_{n=0}^{\infty} a_n x^n$ , then  $(1+x)y'' + x^2y' + (1+2x)y = \sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} + \sum_{n=2}^{\infty} n(n-1)a_n x^{n-1} + \sum_{n=1}^{\infty} na_n x^{n+1} + \sum_{n=0}^{\infty} a_n x^n + 2 \sum_{n=0}^{\infty} a_n x^{n+1} = (2a_2 + a_0) + \sum_{n=1}^{\infty} [(n+2)(n+1)a_{n+2} + (n+1)na_{n+1} + a_n + (n+1)a_{n-1}]x^n = 0$  if  $a_2 = -a_0/2$  and  $a_{n+2} = -\frac{(n+1)na_{n+1} + a_n + (n+1)a_{n-1}}{(n+2)(n+1)}$ ,  $n \geq 1$ . Starting with  $a_0 = -2$  and  $a_1 = 3$  yields  $y = -2 + 3x + x^2 - \frac{1}{6}x^3 - \frac{3}{4}x^4 + \frac{31}{120}x^5 + \dots$ .

**7.3.42.** If  $y = \sum_{n=0}^{\infty} a_n x^n$ , then  $(1+x^2)y'' + (2+x^2)y' + xy = \sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} + \sum_{n=2}^{\infty} n(n-1)a_n x^n + 2 \sum_{n=1}^{\infty} na_n x^{n-1} + \sum_{n=1}^{\infty} na_n x^{n+1} + \sum_{n=1}^{\infty} a_n x^{n+1} = (2a_2 + 2a_1) + \sum_{n=1}^{\infty} [(n+2)(n+1)a_{n+2} + 2(n+1)a_{n+1} + n(n-1)a_n + na_{n-1}]x^n = 0$  if  $a_2 = -a_1$  and  $a_{n+2} = -\frac{[2(n+1)a_{n+1} + n(n-1)a_n + na_{n-1}]}{(n+2)(n+1)}$ ,  $n \geq 1$ . Starting with  $a_0 = -3$  and  $a_1 = 5$  yields  $y = -3 + 5x - 5x^2 + \frac{23}{6}x^3 - \frac{23}{12}x^4 + \frac{11}{30}x^5 + \dots$ .

**7.3.44.** The equation is equivalent to  $y'' + (1+3t^2)y' + (1+2t)y = 0$  with  $t = x-2$ . If  $y = \sum_{n=0}^{\infty} a_n t^n$ , then  $y'' + (1+3t^2)y' + (1+2t)y = \sum_{n=2}^{\infty} n(n-1)a_n t^{n-2} + \sum_{n=1}^{\infty} na_n t^{n-1} + 3 \sum_{n=1}^{\infty} na_n t^{n+1} + \sum_{n=0}^{\infty} a_n t^n + 2 \sum_{n=0}^{\infty} a_n t^{n+1} = (2a_2 + a_1 + a_0) + \sum_{n=1}^{\infty} [(n+2)(n+1)a_{n+2} + (n+1)a_{n+1} + a_n + (3n-1)a_{n-1}]t^n = 0$  if  $a_2 = -(a_1 + a_0)/2$  and  $a_{n+2} = -\frac{[(n+1)a_{n+1} + a_n + (3n-1)a_{n-1}]}{(n+2)(n+1)}$ ,  $n \geq 1$ . Starting with  $a_0 = 2$  and  $a_1 = -3$  yields

$$y = 2 - 3(x+2) + \frac{1}{2}(x+2)^2 - \frac{1}{3}(x+2)^3 + \frac{31}{24}(x+2)^4 - \frac{53}{120}(x+2)^5 + \dots$$

**7.3.46.** The equation is equivalent to  $(1-t^2)y'' - (7-8t+t^2)y' + ty = 0$  with  $t = x+2$ . If  $y = \sum_{n=0}^{\infty} a_n t^n$ , then  $(1-t^2)y'' - (7-8t+t^2)y' + ty = \sum_{n=2}^{\infty} n(n-1)a_n t^{n-2} - \sum_{n=2}^{\infty} n(n-1)a_n t^n - 7 \sum_{n=1}^{\infty} na_n t^{n-1} +$

$$8 \sum_{n=1}^{\infty} n a_n t^n - \sum_{n=1}^{\infty} n a_n t^{n+1} + \sum_{n=0}^{\infty} a_n t^{n+1} = (2a_2 - 7a_1) + \sum_{n=1}^{\infty} [(n+2)(n+1)a_{n+2} - 7(n+1)a_{n+1} - n(n-9)a_n - (n-2)a_{n-1}] t^n = 0$$

if  $a_2 = 7a_1/2$  and  $a_{n+2} = \frac{[7(n+1)a_{n+1} + n(n-9)a_n + (n-2)a_{n-1}]}{(n+2)(n+1)}$ ,  $n \geq 1$ .

Starting with  $a_0 = 2$  and  $a_1 = -1$  yields

$$y = 2 - (x+2) - \frac{7}{2}(x+2)^2 - \frac{43}{6}(x+2)^3 - \frac{203}{24}(x+2)^4 - \frac{167}{30}(x+2)^5 + \dots$$

**7.3.48.** The equation is equivalent to  $(1+3t+2t^2)y'' - (3+t-t^2)y' - (3+t)y = 0$  with  $t = x-1$ .

If  $y = \sum_{n=0}^{\infty} a_n t^n$ , then  $(1+3t+2t^2)y'' - (3+t-t^2)y' - (3+t)y = \sum_{n=2}^{\infty} n(n-1)a_n t^{n-2} + 3 \sum_{n=2}^{\infty} n(n-1)a_n t^{n-1} + 2 \sum_{n=2}^{\infty} n(n-1)a_n t^n - 3 \sum_{n=1}^{\infty} n a_n t^{n-1} - \sum_{n=1}^{\infty} n a_n t^n + \sum_{n=1}^{\infty} n a_n t^{n+1} - 3 \sum_{n=0}^{\infty} a_n t^n - \sum_{n=0}^{\infty} a_n t^{n+1} =$

$$(2a_2 - 3a_1 - 3a_0) + \sum_{n=1}^{\infty} [(n+2)(n+1)a_{n+2} + 3(n^2-1)a_{n+1} + (2n^2-3n-3)(n+1)a_n + (n-2)a_{n-1}] t^n = 0$$

if  $a_2 = 3(a_1 + a_0)/2$  and  $a_{n+2} = -\frac{[3(n^2-1)a_{n+1} + (2n^2-3n-3)(n+1)a_n + (n-2)a_{n-1}]}{(n+2)(n+1)}$ ,  $n \geq 1$ .

Starting with  $a_0 = 1$  and  $a_1 = 0$  yields  $y = 1 + \frac{3}{2}(x-1)^2 + \frac{1}{6}(x-1)^3 - \frac{1}{8}(x-1)^5 + \dots$

## 7.4 Regular Singular Points; Euler Equations

**7.4.2.**  $p(r) = r(r-1) - 7r + 7 = (r-7)(r-1)$ ;  $y = c_1 x + c_2 x^7$ .

**7.4.4.**  $p(r) = r(r-1) + 5r + 4 = (r+2)^2$ ;  $y = x^{-2}(c_1 + c_2 \ln x)$

**7.4.6.**  $p(r) = r(r-1) - 3r + 13 = (r-2)^2 + 9$ ;  $y = x^2[c_1 \cos(3 \ln x) + c_2 \sin(3 \ln x)]$ .

**7.4.8.**  $p(r) = 12r(r-1) - 5r + 6 = (3r-2)(4r-3)$ ;  $y = c_1 x^{2/3} + c_2 x^{3/4}$ .

**7.4.10.**  $p(r) = 3r(r-1) - r + 1 = (r-1)(3r-1)$ ;  $y = c_1 x + c_2 x^{1/3}$ .

**7.4.12.**  $p(r) = r(r-1) + 3r + 5 = (r+1)^2 + 4$ ;  $y = \frac{1}{x} [c_1 \cos(2 \ln x) + c_2 \sin(2 \ln x)]$

**7.4.14.**  $p(r) = r(r-1) - r + 10 = (r-1)^2 + 9$ ;  $y = x [c_1 \cos(3 \ln x) + c_2 \sin(3 \ln x)]$ .

**7.4.16.**  $p(r) = 2r(r-1) + 3r - 1 = (r+1)(2r-1)$ ;  $y = \frac{c_1}{x} + c_2 x^{1/2}$ .

**7.4.18.**  $p(r) = 2r(r-1) + 10r + 9 = 2(r+2)^2 + 1$ ;  $y = \frac{1}{x^2} \left[ c_1 \cos \left( \frac{1}{\sqrt{2}} \ln x \right) + c_2 \sin \left( \frac{1}{\sqrt{2}} \ln x \right) \right]$ .

**7.4.20.** If  $p(r) = ar(r-1) + br + c = a(r-r_1)^2$ , then (A)  $p(r_1) = p'(r_1) = 0$ . If  $y = ux^{r_1}$ , then  $y' = u'x^{r_1} + r_1 ux^{r_1-1}$  and  $y'' = u''x^{r_1} + 2r_1 u'x^{r_1-1} + r_1(r_1-1)ux^{r_1-2}$ , so

$$\begin{aligned} ax^2 y'' + bxy' + cy &= ax^{r_1+2}u'' + (2ar_1 + b)x^{r_1+1}u' + (ar_1(r_1-1) + br_1 + c)x^{r_1}u \\ &= ax^{r_1+2}u'' + p'(r_1)x^{r_1+1}u' + p(r)x^{r_1}u = ax^{r_1+2}u'', \end{aligned}$$

from (A). Therefore,  $u'' = 0$ , so  $u = c_1 + c_2 x$  and  $y = x^{r_1}(c_1 + c_2 x)$ .

**7.4.22.** (a) If  $t = x-1$  and  $Y(t) = y(t+1) = y(x)$ , then  $(1-x^2)y'' - 2xy' + \alpha(\alpha+1)y = -t(2+t)\frac{d^2Y}{dt^2} - 2(1+t)\frac{dY}{dt} + \alpha(\alpha+1)Y = 0$ , so  $y$  satisfies Legendre's equation if and only

if  $Y$  satisfies (A)  $t(2+t)\frac{d^2Y}{dt^2} + 2(1+t)\frac{dY}{dt} - \alpha(\alpha+1)Y = 0$ . Since (A) can be rewritten as  $t^2(2+t)\frac{d^2Y}{dt^2} + 2t(1+t)\frac{dY}{dt} - \alpha(\alpha+1)tY = 0$ , (A) has a regular singular point at  $t = 0_0$ .



(b) If  $t = x + 1$  and  $Y(t) = y(t - 1) = y(x)$ , then  $(1 - x^2)y'' - 2xy' + \alpha(\alpha + 1)y = t(2 - t)\frac{d^2Y}{dt^2} + 2(1 - t)\frac{dY}{dt} + \alpha(\alpha + 1)Y$ , so  $y$  satisfies Legendre's equation if and only if  $Y$  satisfies (B)  $t(2 - t)\frac{d^2Y}{dt^2} + 2(1 - t)\frac{dY}{dt} + \alpha(\alpha + 1)Y$ , Since (B) can be rewritten as (B)  $t^2(2 - t)\frac{d^2Y}{dt^2} + 2t(1 - t)\frac{dY}{dt} + \alpha(\alpha + 1)tY$ , (B) has a regular singular point at  $t = 0_0$ .

## 7.5 The Method of Frobenius I

**7.5.2.**  $p_0(r) = r(3r - 1)$ ;  $p_1(r) = 2(r + 1)$ ;  $p_2(r) = -4(r + 2)$ .

$$a_1(r) = -\frac{2}{3r + 2}; a_n(r) = -\frac{2a_{n-1}(r) - 4a_{n-2}(r)}{3n + 3r - 1}, n \geq 1.$$

$$r_1 = 1/3; a_1(1/3) = -2/3; a_n(1/3) = -\frac{2a_{n-1}(1/3) - 4a_{n-2}(1/3)}{3n}, n \geq 1;$$

$$y_1 = x^{1/3} \left( 1 - \frac{2}{3}x + \frac{8}{9}x^2 - \frac{40}{81}x^3 + \dots \right).$$

$$r_2 = 0; a_1(0) = -1; a_n(0) = -\frac{2a_{n-1}(0) - 4a_{n-2}(0)}{3n - 1}, n \geq 1;$$

$$y_2 = 1 - x + \frac{6}{5}x^2 - \frac{4}{5}x^3 + \dots.$$

**7.5.4.**  $p_0(r) = (r + 1)(4r - 1)$ ;  $p_1(r) = 2(r + 2)$ ;  $p_2(r) = 4r + 7$ .

$$a_1(r) = -\frac{2}{4r + 3}; a_n(r) = -\frac{2}{4n + 4r - 1}a_{n-1}(r) - \frac{1}{n + r + 1}a_{n-2}(r), n \geq 1.$$

$$r_1 = 1/4; a_1(1/4) = -1/2; a_n(1/4) = -\frac{1}{2n}a_{n-1}(1/4) - \frac{4}{4n + 5}a_{n-2}(1/4), n \geq 1;$$

$$y_1 = x^{1/4} \left( 1 - \frac{1}{2}x - \frac{19}{104}x^2 + \frac{1571}{10608}x^3 + \dots \right).$$

$$r_2 = -1; a_1(-1) = 2; a_n(-1) = -\frac{2}{4n - 5}a_{n-1}(-1) - \frac{1}{n}a_{n-2}(-1), n \geq 1;$$

$$y_2 = x^{-1} \left( 1 + 2x - \frac{11}{6}x^2 - \frac{1}{7}x^3 + \dots \right).$$

**7.5.6.**  $p_0(r) = r(5r - 1)$ ;  $p_1(r) = (r + 1)^2$ ;  $p_2(r) = 2(r + 2)(5r + 9)$ .

$$a_1(r) = -\frac{r + 1}{5r + 4}; a_n(r) = -\frac{n + r}{5n + 5r - 1}a_{n-1}(r) - \frac{2}{n + r + 1}a_{n-2}(r), n \geq 1.$$

$$r_1 = 1/5; a_1(1/5) = -6/25; a_n(1/5) = -\frac{5n + 1}{25n}a_{n-1}(1/5) - \frac{2}{n + 1}a_{n-2}(1/5), n \geq 1;$$

$$y_1 = x^{1/5} \left( 1 - \frac{6}{25}x - \frac{1217}{625}x^2 + \frac{41972}{46875}x^3 + \dots \right).$$

$$r_2 = 0; a_1(0) = -1/4; a_n(0) = -\frac{n}{5n - 1}a_{n-1}(0) - \frac{2}{n}a_{n-2}(0), n \geq 1;$$

$$y_2 = x - \frac{1}{4}x^2 - \frac{35}{18}x^3 + \frac{11}{12}x^4 + \dots.$$

**7.5.8.**  $p_0(r) = (3r - 1)(6r + 1)$ ;  $p_1(r) = (3r + 2)(6r + 1)$ ;  $p_2(r) = 3r + 5$ .

$$a_1(r) = -\frac{6r + 1}{6r + 7}; a_n(r) = -\frac{6n + 6r - 5}{6n + 6r + 1}a_{n-1}(r) - \frac{1}{6n + 6r + 1}a_{n-2}(r), n \geq 1.$$

$$r_1 = 1/3; a_1(1/3) = -1/3; a_n(1/3) = -\frac{2n - 1}{2n + 1}a_{n-1}(1/3) - \frac{1}{6n + 3}a_{n-2}(1/3), n \geq 1;$$

$$y_1 = x^{1/3} \left( 1 - \frac{1}{3}x + \frac{2}{15}x^2 - \frac{5}{63}x^3 + \dots \right).$$

$$r_2 = -1/6; a_1(-1/6) = 0; a_n(-1/6) = -\frac{n-1}{n}a_{n-1}(-1/6) - \frac{1}{6n}a_{n-2}(-1/6), n \geq 1;$$

$$y_2 = x^{-1/6} \left( 1 - \frac{1}{12}x^2 + \frac{1}{18}x^3 + \dots \right).$$

$$\mathbf{7.5.10.} \quad p_0(r) = (2r+1)(5r-1); p_1(r) = (2r-1)(5r+4); p_2(r) = 2(2r+5)(5r-1).$$

$$a_1(r) = -\frac{2r-1}{2r+3}; a_n(r) = -\frac{2n+2r-3}{2n+2r+1}a_{n-1}(r) - \frac{10n+10r-22}{5n+5r-1}a_{n-2}(r), n \geq 1.$$

$$r_1 = 1/5; a_1(1/5) = 3/17; a_n(1/5) = -\frac{10n-13}{10n+7}a_{n-1}(1/5) - \frac{2n-4}{n}a_{n-2}(1/5), n \geq 1;$$

$$y_1 = x^{1/5} \left( 1 + \frac{3}{17}x - \frac{7}{153}x^2 - \frac{547}{5661}x^3 + \dots \right).$$

$$r_2 = -1/2; a_1(-1/2) = 1; a_n(-1/2) = -\frac{n-2}{n}a_{n-1}(-1/2) - \frac{20n-54}{10n-7}a_{n-2}(-1/2), n \geq 1;$$

$$y_2 = x^{-1/2} \left( 1 + x + \frac{14}{13}x^2 - \frac{556}{897}x^3 + \dots \right).$$

$$\mathbf{7.5.14.} \quad p_0(r) = (r+1)(2r-1); p_1(r) = 2r+1; a_n(r) = -\frac{1}{n+r+1}a_{n-1}(r).$$

$$r_1 = 1/2; a_n(1/2) = -\frac{2}{2n+3}a_{n-1}(1/2); y_1 = x^{1/2} \sum_{n=0}^{\infty} \frac{(-2)^n}{\prod_{j=1}^n (2j+3)} x^n.$$

$$r_2 = -1; a_n(-1) = -\frac{1}{n}a_{n-1}(-1); y_2 = x^{-1} \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} x^n.$$

$$\mathbf{7.5.16.} \quad p_0(r) = (r+2)(2r-1); p_1(r) = r+3; a_n(r) = -\frac{1}{2n+2r-1}a_{n-1}(r).$$

$$r_1 = 1/2; a_n(1/2) = -\frac{1}{2n}a_{n-1}(1/2); y_1 = x^{1/2} \sum_{n=0}^{\infty} \frac{(-1)^n}{2^n n!} x^n.$$

$$r_2 = -2; a_n(-2) = -\frac{1}{2n-5}a_{n-1}(-2); y_2 = \frac{1}{x^2} \sum_{n=0}^{\infty} \frac{(-1)^n}{\prod_{j=1}^n (2j-5)} x^n.$$

$$\mathbf{7.5.18.} \quad p_0(r) = (r-1)(2r-1); p_1(r) = -2; a_n(r) = \frac{2}{(n+r-1)(2n+2r-1)}a_{n-1}(r).$$

$$r_1 = 1; a_n(1) = \frac{2}{n(2n+1)}a_{n-1}(1); y_1 = x \sum_{n=0}^{\infty} \frac{2^n}{n! \prod_{j=1}^n (2j+1)} x^n.$$

$$r_2 = 1/2; a_n(1/2) = \frac{2}{n(2n-1)}a_{n-1}(1/2); y_2 = x^{1/2} \sum_{n=0}^{\infty} \frac{2^n}{n! \prod_{j=1}^n (2j-1)} x^n.$$

$$\mathbf{7.5.20.} \quad p_0(r) = (r-1)(3r+1); p_1(r) = r-3; a_n(r) = -\frac{n+r-4}{(n+r-1)(3n+3r+1)}a_{n-1}(r).$$

$$r_1 = ; a_n(1) = -\frac{n-3}{n(3n+4)}a_{n-1}(1); y_1 = x \left( 1 + \frac{2}{7}x + \frac{1}{70}x^2 \right).$$

$$r_2 = -1/3; a_n(-1/3) = -\frac{3n-13}{3n(3n-4)}a_{n-1}(-1/3); y_2 = x^{-1/3} \sum_{n=0}^{\infty} \frac{(-1)^n}{3^n n!} \left( \prod_{j=1}^n \frac{3j-13}{3j-4} \right) x^n.$$

$$\mathbf{7.5.22.} \quad p_0(r) = (r-1)(4r-1); p_1(r) = r(r+2); a_n(r) = -\frac{n+r+1}{4n+4r-1}a_{n-1}(r).$$

$$r_1 = 1; a_n(1) = -\frac{n+2}{4n+3}a_{n-1}(1); y_1 = x \sum_{n=0}^{\infty} \frac{(-1)^n (n+2)!}{2 \prod_{j=1}^n (4j+3)} x^n.$$

$$r_2 = 1/4; a_n(1/4) = -\frac{4n+5}{16n}a_{n-1}(1/4); y_2 = x^{1/4} \sum_{n=0}^{\infty} \frac{(-1)^n}{16^n n!} \prod_{j=1}^n (4j+5) x^n$$

$$\mathbf{7.5.24.} \quad p_0(r) = (r+1)(3r-1); p_1(r) = 2(r+2)(2r+3); a_n(r) = -2 \frac{2n+2r+1}{3n+3r-1} a_{n-1}(r).$$

$$r_1 = 1/3; a_n(1/3) = -2 \frac{6n+5}{9n} a_{n-1}(1/3); y_1 = x^{1/3} \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} \left(\frac{2}{9}\right)^n \left(\prod_{j=1}^n (6j+5)\right) x^n;$$

$$r_2 = -1; a_n(-1) = -2 \frac{2n-1}{3n-4} a_{n-1}(-1); y_2 = x^{-1} \sum_{n=0}^{\infty} (-1)^n 2^n \left(\prod_{j=1}^n \frac{2j-1}{3j-4}\right) x^n$$

$$\mathbf{7.5.28.} \quad p_0(r) = (2r-1)(4r-1); p_1(r) = (r+1)^2; a_n(r) = -\frac{(n+r)^2}{(2n+2r-1)(4n+4r-1)} a_{n-1}(r).$$

$$r_1 = 1/2; a_n(1/2) = -\frac{4n^2+4n+1}{8n(4n+1)} a_{n-1}(1/2); y_1 = x^{1/2} \left(1 - \frac{9}{40}x + \frac{5}{128}x^2 - \frac{245}{39936}x^3 + \dots\right).$$

$$r_2 = 1/4; a_n(1/4) = -\frac{16n^2+8n+1}{32n(4n-1)} a_{n-1}(1/4); y_2 = x^{1/4} \left(1 - \frac{25}{96}x + \frac{675}{14336}x^2 - \frac{38025}{5046272}x^3 + \dots\right).$$

$$\mathbf{7.5.30.} \quad p_0(r) = (2r-1)(2r+1); p_1(r) = (2r+1)(3r+1); a_n(r) = -\frac{(3n+3r-2)}{(2n+2r+1)} a_{n-1}(r).$$

$$r_1 = 1/2; a_n(1/2) = -\frac{6n-1}{4(n+1)} a_{n-1}(1/2); y_1 = x^{1/2} \left(1 - \frac{5}{8}x + \frac{55}{96}x^2 - \frac{935}{1536}x^3 + \dots\right).$$

$$r_2 = -1/2; a_n(-1/2) = -\frac{6n-7}{4n} a_{n-1}(-1/2); y_2 = x^{-1/2} \left(1 + \frac{1}{4}x - \frac{5}{32}x^2 - \frac{55}{384}x^3 + \dots\right).$$

$$\mathbf{7.5.32.} \quad p_0(r) = (2r+1)(3r+1); p_1(r) = (r+1)(r+2); a_n(r) = \frac{(n+r)(n+r+1)}{(2n+2r+1)(3n+3r+1)} a_n(r).$$

$$r_1 = -1/3; a_n(-1/3) = -\frac{(3n-1)(3n+2)}{9n(6n+1)} a_{n-1}(-1/3); y_1 = x^{-1/3} \left(1 - \frac{10}{63}x + \frac{200}{7371}x^2 - \frac{17600}{3781323}x^3 + \dots\right).$$

$$r_2 = -1/2; a_n(-1/2) = -\frac{(2n-1)(2n+1)}{4n(6n-1)} a_{n-1}(-1/2); y_2 = x^{-1/2} \left(1 - \frac{3}{20}x + \frac{9}{352}x^2 - \frac{105}{23936}x^3 + \dots\right).$$

$$\mathbf{7.5.34.} \quad p_0(r) = (2r-1)(4r-1); p_2(r) = -(2r+3)(4r+3); a_{2m}(r) = \frac{8m+4r-5}{8m+4r-1} a_{2m-2}(r).$$

$$r_1 = 1/2; a_{2m}(1/2) = \frac{8m-3}{8m+1} a_{2m-2}(1/2); y_1 = x^{1/2} \sum_{m=0}^{\infty} \left(\prod_{j=1}^m \frac{8j-3}{8j+1}\right) x^{2m}.$$

$$r_2 = 1/4; a_{2m}(1/4) = \frac{2m-1}{2m} a_{2m-2}(1/4); y_2 = x^{1/4} \sum_{m=0}^{\infty} \frac{1}{2^m m!} \left(\prod_{j=1}^m (2j-1)\right) x^{2m}$$

$$\mathbf{7.5.36.} \quad p_0(r) = r(3r-1); p_2(r) = (r-4)(r+2); a_{2m}(r) = -\frac{2m+r-6}{6m+3r-1} a_{2m-2}(r).$$

$$r_1 = 1/3; a_{2m}(1/3) = -\frac{6m-17}{18m} a_{2m-2}(1/3); y_1 = x^{1/3} \sum_{m=0}^{\infty} \frac{(-1)^m}{18^m m!} \left(\prod_{j=1}^m (6j-17)\right) x^{2m}.$$

$$r_2 = 0; a_{2m}(0) = -\frac{2m-6}{6m-1} a_{2m-2}(0); y_2 = 1 + \frac{4}{5}x^2 + \frac{8}{55}x^4$$

$$\mathbf{7.5.38.} \quad p_0(r) = (2r-1)(3r-1); p_2(r) = -(r+1)(3r+5); a_{2m}(r) = \frac{2m+r-1}{4m+2r-1} a_{2m-2}(r).$$

$$r_1 = 1/2; a_{2m}(1/2) = \frac{4m-1}{8m} a_{2m-2}(1/2); y_1 = x^{1/2} \sum_{m=0}^{\infty} \frac{1}{8^m m!} \left(\prod_{j=1}^m (4j-1)\right) x^{2m}.$$

$$r_2 = 1/3; a_{2m}(1/3) = \frac{6m-2}{12m-1}a_{2m-2}(1/3); y_2 = x^{1/3} \sum_{m=0}^{\infty} 2^m \left( \prod_{j=1}^m \frac{3j-1}{12j-1} \right) x^{2m}.$$

$$\mathbf{7.5.40.} \quad p_0(r) = (2r-1)(2r+1); p_1(r) = (r+1)(2r+3); a_{2m}(r) = -\frac{2m+r-1}{4m+2r+1}a_{2m-2}(r).$$

$$r_1 = 1/2; a_{2m}(1/2) = -\frac{4m-1}{4(2m+1)}a_{2m-2}(1/2); y_1 = x^{1/2} \sum_{m=0}^{\infty} \frac{(-1)^m}{4^m} \left( \prod_{j=1}^m \frac{4j-1}{2j+1} \right) x^{2m}.$$

$$r_2 = -1/2; a_{2m}(-1/2) = -\frac{4m-3}{8m}a_{2m-2}(-1/2); y_2 = x^{-1/2} \sum_{m=0}^{\infty} \frac{(-1)^m}{8^m m!} \left( \prod_{j=1}^m (4j-3) \right) x^{2m}$$

$$\mathbf{7.5.42.} \quad p_0(r) = (r+1)(3r-1); p_1(r) = (r-1)(3r+5); a_{2m}(r) = -\frac{2m+r-3}{2m+r+1}a_{2m-2}(r).$$

$$r_1 = 1/3; a_{2m}(1/3) = -\frac{3m-4}{3m+2}a_{2m-2}(1/3); y_1 = x^{1/3} \sum_{m=0}^{\infty} (-1)^m \left( \prod_{j=1}^m \frac{3j-4}{3j+2} \right) x^{2m}.$$

$$r_2 = -1; a_{2m}(-1) = -\frac{m-2}{m}a_{2m-2}(-1); y_2 = x^{-1}(1+x^2)$$

$$\mathbf{7.5.44.} \quad p_0(r) = (r+1)(2r-1); p_1(r) = r^2; a_{2m}(r) = -\frac{(2m+r-2)^2}{(2m+r+1)(4m+2r-1)}a_{2m-2}(r).$$

$$r_1 = 1/2; a_{2m}(1/2) = -\frac{(4m-3)^2}{8m(4m+3)}a_{2m-2}(1/2); y_1 = x^{1/2} \sum_{m=0}^{\infty} \frac{(-1)^m}{8^m m!} \left( \prod_{j=1}^m \frac{(4j-3)^2}{4j+3} \right) x^{2m}.$$

$$r_2 = -1; a_{2m}(-1) = -\frac{(2m-3)^2}{2m(4m-3)}a_{2m-2}(-1); y_2 = x^{-1} \sum_{m=0}^{\infty} \frac{(-1)^m}{2^m m!} \left( \prod_{j=1}^m \frac{(2j-3)^2}{4j-3} \right) x^{2m}.$$

$$\mathbf{7.5.46.} \quad p_0(r) = (3r-1)(3r+1); p_1(r) = 3r+5; a_{2m}(r) = -\frac{1}{6m+3r+1}a_{2m-2}(r).$$

$$r_1 = 1/3; a_{2m}(1/3) = -\frac{1}{2(3m+1)}a_{2m-2}(1/3); y_1 = x^{1/3} \sum_{m=0}^{\infty} \frac{(-1)^m}{2^m \prod_{j=1}^m (3j+1)} x^{2m}.$$

$$r_2 = -1/3; a_{2m}(-1/3) = -\frac{1}{6m}a_{2m-2}(-1/3); y_2 = x^{-1/3} \sum_{m=0}^{\infty} \frac{(-1)^m}{6^m m!} x^{2m}$$

$$\mathbf{7.5.48.} \quad p_0(r) = 2(r+1)(4r-1); p_2(r) = (r+3)^2; a_{2m}(r) = -\frac{2m+r+1}{2(8m+4r-1)}a_{2m-2}(r).$$

$$r_1 = 1/4; a_{2m}(1/4) = -\frac{8m+5}{64m}a_{2m-2}(1/4); y_1 = x^{1/4} \left( 1 - \frac{13}{64}x^2 + \frac{273}{8192}x^4 - \frac{2639}{524288}x^6 + \dots \right).$$

$$r_2 = -1; a_{2m}(-1) = -\frac{m}{8m-5}a_{2m-2}(-1); y_2 = x^{-1} \left( 1 - \frac{1}{3}x^2 + \frac{2}{33}x^4 - \frac{2}{209}x^6 + \dots \right).$$

$$\mathbf{7.5.50.} \quad p_0(r) = (2r-1)(2r+1); p_2(r) = (2r+5)^2; a_{2m}(r) = -\frac{4m+2r+1}{4m+2r-1}a_{2m-2}(r).$$

$$r_1 = 1/2; a_{2m}(1/2) = -\frac{2m+1}{2m}a_{2m-2}(1/2); y_1 = x^{1/2} \left( 1 - \frac{3}{2}x^2 + \frac{15}{8}x^4 - \frac{35}{16}x^6 + \dots \right).$$

$$r_2 = -1/2; a_{2m}(-1/2) = -\frac{2m}{2m-1}a_{2m-2}(-1/2); y_2 = x^{-1/2} \left( 1 - 2x^2 + \frac{8}{3}x^4 - \frac{16}{5}x^6 + \dots \right).$$

**7.5.52.** (a) Multiplying (A)  $c_1 y_1 + c_2 y_2 \equiv 0$  by  $x^{-r_2}$  yields  $c_1 x^{r_1-r_2} \sum_{n=0}^{\infty} a_n x^n + c_2 \sum_{n=0}^{\infty} b_n x^n = 0$ ,  $0 < x < \rho$ . Letting  $x \rightarrow 0+$  shows that  $c_2 = 0$ , since  $b_0 = 1$ . Now (A) reduces to  $c_1 y_1 \equiv 0$ , so  $c_1 = 0$ . Therefore,  $y_1$  and  $y_2$  are linearly independent on  $(0, \rho)$ .

(b) Since  $y_1 = \sum_{n=0}^{\infty} a_n(r_1)x^n$  and  $y_2 = \sum_{n=0}^{\infty} a_n(r_2)x^n$  are linearly independent solutions of  $Ly = 0$   $(0, \rho)$ ,  $\{y_1, y_2\}$  is a fundamental set of solutions of  $Ly = 0$  on  $(0, \rho)$ , by Theorem 5.1.6.

**7.5.54.** (a) If  $x > 0$ , then  $|x|^r x^n = x^{n+r}$ , so the assertions are obvious. If  $x < 0$ , then  $|x|^r = (-x)^r$ , so  $\frac{d}{dx}|x|^r = -r(-x)^{r-1} = \frac{r(-x)^r}{x} = \frac{r|x|^r}{x}$ . Therefore, (A)  $\frac{d}{dx}(|x|^r x^n) = \frac{r|x|^r}{x} x^n + |x|^r (nx^{n-1}) = (n+r)|x|^r x^{n-1}$  and  $\frac{d^2}{dx^2}(|x|^r x^n) = (n+r) \frac{d}{dx}(|x|^r x^{n-1}) = (n+r)(n+r-1)|x|^r x^{n-2}$ , from (A) with  $n$  replaced by  $n-1$ .

**7.5.56.** (a) Here  $p_1 \equiv 0$ , so Eqn. (7.5.12) reduces to  $a_0(r) = 1$ ,  $a_1(r) = 0$ ,  $a_n(r) = -\frac{p_2(n+r-2)}{p_0(n+r)} a_{n-2}(r)$ ,  $r \geq 0$ , which implies that  $a_{2m+1}(r) = 0$  for  $m = 1, 2, 3, \dots$ . Therefore, Eqn. (7.5.12) actually reduces to  $a_0(r) = 1$ ,  $a_{2m}(r) = -\frac{p_2(2m+r-2)}{p_0(2m+r)}$ , which holds because of condition (A).

(b) Similar to the proof of Exercise 7.5.55(a).

(c)  $p_0(2m+r_1) = 2m\alpha_0(2m+r_1-r_2)$ , which is nonzero if  $m > 0$ , since  $r_1 - r_2 \geq 0$ . Therefore, the assumptions of Theorem 7.5.2 hold with  $r = r_1$ , and  $Ly_1 = p_0(r_1)x^{r_1} = 0$ . If  $r_1 - r_2$  is not an even integer, then  $p_0(2m+r_2) = 2m\alpha_0(2m-r_1+r_2) \neq 0$ ,  $m = 1, 2, \dots$ . Hence, the assumptions of Theorem 7.5.2 hold with  $r = r_2$  and  $Ly_2 = p_0(r_2)x^{r_2} = 0$ . From Exercise 7.5.52,  $\{y_1, y_2\}$  is a fundamental set of solutions.

(d) Similar to the proof of Exercise 7.5.55(c).

**7.5.58.** (a) From Exercise 7.5.57,  $b_n = 0$  for  $n \geq 1$ .

**7.5.60.** (a)  $(\alpha_0 + \alpha_1 x + \alpha_2 x^2) \sum_{n=0}^{\infty} a_n x^n = \alpha_0 a_0 + (\alpha_0 a_1 + \alpha_1 a_0)x + \sum_{n=2}^{\infty} (\alpha_0 a_n + \alpha_1 a_{n-1} + \alpha_2 a_{n-2})x^n = 1$ , so  $\sum_{n=0}^{\infty} a_n x^n = \frac{\alpha_0 a_0}{\alpha_0 + \alpha_1 x + \alpha_2 x^2}$ .

(b) If  $\frac{p_1(r-1)}{p_0(r)} = \frac{\alpha_1}{\alpha_0}$  and  $\frac{p_2(r-2)}{p_0(r)} = \frac{\alpha_2}{\alpha_0}$ , then Eqn. (7.5.12) is equivalent to  $a_0(r) = 1$ ,  $\alpha_0 a_1(r) + \alpha_1 a_0(r) = 0$ ,  $\alpha_0 a_n(r) + \alpha_1 a_{n-1}(r) + \alpha_2 a_{n-2}(r) = 0$ ,  $n \geq 2$ . Therefore, Theorem 7.5.2 implies the conclusion.

**7.5.62.**  $p_0(r) = (2r-1)(3r-1)$ ;  $p_1(r) = 0$ ;  $p_2(r) = 2(2r+3)(3r+5)$ ;  $\frac{p_1(r-1)}{p_0(r)} = 0 = \frac{\alpha_1}{\alpha_0}$ ;  $\frac{p_2(r-2)}{p_0(r)} = 2 = \frac{\alpha_2}{\alpha_0}$ ;  $y_1 = \frac{x^{1/3}}{1+2x^2}$ ;  $y_2 = \frac{x^{1/2}}{1+2x^2}$ .

**7.5.64.**  $p_0(r) = 5(3r-1)(3r+1)$ ;  $p_1(r) = (3r+2)(3r+4)$ ;  $p_2(r) = 0$ ;  $\frac{p_1(r-1)}{p_0(r)} = \frac{1}{5} = \frac{\alpha_1}{\alpha_0}$ ;  $\frac{p_2(r-2)}{p_0(r)} = 0 = \frac{\alpha_2}{\alpha_0}$ ;  $y_1 = \frac{x^{1/3}}{5+x}$ ;  $y_2 = \frac{x^{-1/3}}{5+x}$ .

**7.5.66.**  $p_0(r) = (2r-3)(2r-1)$ ;  $p_1(r) = 3(2r-1)(2r+1)$ ;  $p_2(r) = (2r+1)(2r+3)$ ;  $\frac{p_1(r-1)}{p_0(r)} = 3 = \frac{\alpha_1}{\alpha_0}$ ;  $\frac{p_2(r-2)}{p_0(r)} = 1 = \frac{\alpha_2}{\alpha_0}$ ;  $y_1 = \frac{x^{1/2}}{1+3x+x^2}$ ;  $y_2 = \frac{x^{3/2}}{1+3x+x^2}$ .

**7.5.68.**  $p_0(r) = 3(r-1)(4r-1)$ ;  $p_1(r) = 2r(4r+3)$ ;  $p_2(r) = (r+1)(4r+7)$ ;  $\frac{p_1(r-1)}{p_0(r)} = \frac{2}{3} = \frac{\alpha_1}{\alpha_0}$ ;  $\frac{p_2(r-2)}{p_0(r)} = \frac{1}{3} = \frac{\alpha_2}{\alpha_0}$ ;  $y_1 = \frac{x}{3+2x+x^2}$ ;  $y_2 = \frac{x^{1/4}}{3+2x+x^2}$ .

## 7.6 The Method of Frobenius II

**7.6.2.**  $p_0(r) = (r+1)^2$ ;  $p_1(r) = (r+2)(r+3)$ ;  $p_2(r) = (r+3)(2r-1)$ ;

$$a_1(r) = -\frac{r+3}{r+2}; a_n(r) = -\frac{n+r+2}{n+r+1}a_{n-1}(r) - \frac{2n+2r-5}{n+r+1}a_{n-2}(r), n \geq 2.$$

$$a'_1(r) = \frac{1}{(r+2)^2}; a'_n(r) = -\frac{n+r+2}{n+r+1}a'_{n-1}(r) - \frac{2n+2r-5}{n+r+1}a'_{n-2}(r) + \frac{1}{(n+r+1)^2}a_{n-1}(r) - \frac{7}{(n+r+1)^2}a_{n-2}(r), n \geq 2.$$

$$r_1 = -1; a_1(-1) = -2; a_n(-1) = -\frac{n+1}{n}a_{n-1}(-1) - \frac{2n-7}{n}a_{n-2}(-1), n \geq 2;$$

$$y_1 = x^{-1} \left( 1 - 2x + \frac{9}{2}x^2 - \frac{20}{3}x^3 + \dots \right);$$

$$a'_1(-1) = 1; a'_n(-1) = -\frac{n+1}{n}a'_{n-1}(-1) - \frac{2n-7}{n}a'_{n-2}(-1) + \frac{1}{n^2}a_{n-1}(-1) - \frac{7}{n^2}a_{n-2}(-1), n \geq 2;$$

$$y_2 = y_1 \ln x + 1 - \frac{15}{4}x + \frac{133}{18}x^2 + \dots.$$

**7.6.4.**  $p_0(r) = (2r-1)^2$ ;  $p_1(r) = (2r+1)(2r+3)$ ;  $p_2(r) = (2r+1)(2r+3)$ ;

$$a_1(r) = -\frac{2r+3}{2r+1}; a_n(r) = -\frac{(2n+2r+1)a_{n-1}(r) - (2n+2r-3)a_{n-2}(r)}{2n+2r-1}, n \geq 2.$$

$$a'_1(r) = \frac{4}{(2r+1)^2}; a'_n(r) = -\frac{(2n+2r+1)a'_{n-1}(r) - (2n+2r-3)a'_{n-2}(r)}{2n+2r-1} + \frac{4(a_{n-1}(r) - a_{n-2}(r))}{(2n+2r-1)^2};$$

$n \geq 2.$

$$r_1 = 1/2; a_1(1/2) = -2; a_n(1/2) = -\frac{(n+1)a_{n-1}(1/2) + (n-1)a_{n-2}(1/2)}{n}; n \geq 2;$$

$$y_1 = x^{1/2} \left( 1 - 2x + \frac{5}{2}x^2 - 2x^3 + \dots \right);$$

$$a'_1(1/2) = 1; a'_n(1/2) = -\frac{(n+1)a'_{n-1}(1/2) + (n-1)a'_{n-2}(1/2)}{n} + \frac{a_{n-1}(1/2) - a_{n-2}(1/2)}{n^2}, n \geq 2;$$

$$y_2 = y_1 \ln x + x^{3/2} \left( 1 - \frac{9}{4}x + \frac{17}{6}x^2 + \dots \right).$$

**7.6.6.**  $p_0(r) = (3r+1)^2$ ;  $p_1(r) = 3(3r+4)$ ;  $p_2(r) = -2(3r+7)$ ;

$$a_1(r) = -\frac{3}{3r+4}; a_n(r) = \frac{-3a_{n-1}(r) + 2a_{n-2}(r)}{3n+3r+1}; n \geq 2;$$

$$a'_1(r) = \frac{9}{(3r+4)^2}; a'_n(r) = \frac{-3a'_{n-1}(r) + 2a'_{n-2}(r)}{3n+3r+1} + \frac{9a_{n-1}(r) - 6a_{n-2}(r)}{(3n+3r+1)^2}; n \geq 2.$$

$$r_1 = -1/3; a_1(-1/3) = -1; a_n(-1/3) = \frac{-3a_{n-1}(-1/3) + 2a_{n-2}(-1/3)}{3n}, n \geq 2;$$

$$y_1 = x^{-1/3} \left( 1 - x + \frac{5}{6}x^2 - \frac{1}{2}x^3 + \dots \right);$$

$$a'_1(-1/3) = 1; a'_n(-1/3) = \frac{-3a'_{n-1}(r) + 2a'_{n-2}(r)}{3n} + \frac{3a_{n-1}(r) - 2a_{n-2}(r)}{3n^2}; n \geq 2;$$

$$y_2 = y_1 \ln x + x^{2/3} \left( 1 - \frac{11}{12}x + \frac{25}{36}x^2 + \dots \right).$$

**7.6.8.**  $p_0(r) = (r+2)^2$ ;  $p_1(r) = 2(r+3)^2$ ;  $p_2(r) = 3(r+4)$ ;

$$a_1(r) = -2; a_n(r) = -2a_{n-1}(r) - \frac{3a_{n-2}(r)}{n+r+2}; n \geq 2;$$

$$a'_1(r) = 0; a'_n(r) = -2a'_{n-1}(r) - \frac{3a'_{n-2}(r)}{n+r+2} + \frac{3a_{n-2}(r)}{(n+r+2)^2}, n \geq 2.$$

$$r_1 = -2; a_1(-2) = -2; a_n(-2) = -2a_{n-1}(-2) - \frac{3a_{n-2}(-2)}{n}, n \geq 2;$$

$$y_1 = x^{-2} \left( 1 - 2x + \frac{5}{2}x^2 - 3x^3 + \dots \right);$$

$$a'_1(-2) = 0; a'_n(-2) = -2a_{n-1}(-2) - \frac{3a_{n-2}(-2)}{n} + \frac{3a_{n-2}(-2)}{n^2}; n \geq 2;$$

$$y_2 = y_1 \ln x + \frac{3}{4} - \frac{13}{6}x + \dots.$$

$$\mathbf{7.6.10.} \quad p_0(r) = (4r+1)^2; p_1(r) = 4r+5; p_2(r) = 2(4r+9);$$

$$a_1(r) = -\frac{1}{4r+5}; a_n(r) = -\frac{a_{n-1}(r) + 2a_{n-2}(r)}{4n+4r+1}; n \geq 2;$$

$$a'_1(r) = \frac{4}{(4r+5)^2}; a'_n(r) = -\frac{a'_{n-1}(r) + 2a'_{n-2}(r)}{4n+4r+1} + \frac{4a_{n-1}(r) + 8a_{n-2}(r)}{(4n+4r+1)^2}; n \geq 2.$$

$$r_1 = -1/4; a_1(-1/4) = -1/4; a_n(-1/4) = -\frac{a_{n-1}(-1/4) + 2a_{n-2}(-1/4)}{4n}; n \geq 2;$$

$$y_1 = x^{-1/4} \left( 1 - \frac{1}{4}x - \frac{7}{32}x^2 + \frac{23}{384}x^3 + \dots \right);$$

$$a'_1(-1/4) = 1/4; a'_n(-1/4) = -\frac{a'_{n-1}(-1/4) + 2a'_{n-2}(-1/4)}{4n} + \frac{a_{n-1}(-1/4) + 2a_{n-2}(-1/4)}{4n^2}; n \geq 2;$$

$$y_2 = y_1 \ln x + x^{3/4} \left( \frac{1}{4} + \frac{5}{64}x - \frac{157}{2304}x^2 + \dots \right).$$

$$\mathbf{7.6.12.} \quad p_0(r) = (2r-1)^2; p_1(r) = 4;$$

$$a_n(r) = -\frac{4}{(2n+2r-1)^2} a_{n-1}(r);$$

$$a_n(r) = \frac{(-4)^n}{\prod_{j=1}^n (2j+2r-1)}.$$

By logarithmic differentiation,  $a'_n(r) = a_n(r) \sum_{j=1}^n \frac{2}{2j+2r-1};$

$$r_1 = 1/2; a_n(1/2) = \frac{(-1)^n}{(n!)^2};$$

$$a'_n(1/2) = a_n(1/2) \left( -2 \sum_{j=1}^n \frac{1}{j} \right);$$

$$y_1 = x^{1/2} \sum_{n=0}^{\infty} \frac{(-1)^n}{(n!)^2} x^n;$$

$$y_2 = y_1 \ln x - 2x^{1/2} \sum_{n=1}^{\infty} \frac{(-1)^n}{(n!)^2} \left( \sum_{j=1}^n \frac{1}{j} \right) x^n;$$

$$\mathbf{7.6.14.} \quad p_0(r) = (r-2)^2; p_1(r) = r^2; a_n(r) = -\frac{(n+r-1)^2}{(n+r-2)^2} a_{n-1}(r); a_n(r) = (-1)^n \frac{(n+r-1)^2}{(r-1)^2};$$

$$a'_n(r) = (-1)^{n+1} \frac{2n(r+n-1)}{(r-1)^3}; r_1 = 2; a_n(2) = (-1)^n (n+1)^2; a'_n(2) = (-1)^{n+1} 2n(n+1);$$

$$y_1 = x^2 \sum_{n=0}^{\infty} (-1)^n (n+1)^2 x^n; y_2 = y_1 \ln x - 2x^2 \sum_{n=1}^{\infty} (-1)^n n(n+1) x^n.$$

$$\mathbf{7.6.16.} \quad p_0(r) = (5r-1)^2; p_1(r) = r+1;$$

$$a_n(r) = -\frac{(n+r)}{(5n+5r-1)^2} a_{n-1}(r);$$

$$a_n(r) = (-1)^n \prod_{j=1}^n \frac{(j+r)}{(5j+5r-1)^2};$$

By logarithmic differentiation,

$$a'_n(r) = -a_n(r) \sum_{j=1}^n \frac{(5j+5r+1)}{(j+r)(5j+5r-1)};$$

$$r_1 = 1/5; a_n(1/5) = (-1)^n \prod_{j=1}^n \frac{(5j+1)}{125^n (n!)^2};$$

$$a'_n(1/5) = a_n(1/5) \sum_{j=1}^n \frac{5j+2}{j(5j+1)};$$

$$y_1 = x^{1/5} \sum_{n=0}^{\infty} \frac{(-1)^n \prod_{j=1}^n (5j+1)}{125^n (n!)^2} x^n;$$

$$y_2 = y_1 \ln x - x^{1/5} \sum_{n=1}^{\infty} \frac{(-1)^n \prod_{j=1}^n (5j+1)}{125^n (n!)^2} \left( \sum_{j=1}^n \frac{5j+2}{j(5j+1)} \right) x^n.$$

**7.6.18.**  $p_0(r) = (3r-1)^2$ ;  $p_1(r) = (2r-1)^2$ ;

$$a_n(r) = -\frac{(2n+2r-3)^2}{(3n+3r-1)^2} a_{n-1}(r);$$

$$a_n(r) = (-1)^n \prod_{j=1}^n \frac{(2j+2r-3)^2}{(3j+3r-1)^2};$$

By logarithmic differentiation,

$$a'_n(r) = 14a_n(r) \sum_{j=1}^n \frac{1}{(2j+2r-3)(3j+3r-1)};$$

$$r_1 = 1/3; a_n(1/3) = \frac{(-1)^n \prod_{j=1}^n (6j-7)^2}{81^n (n!)^2};$$

$$a'_n(1/3) = 14a_n(1/3) \sum_{j=1}^n \frac{1}{j(6j-7)};$$

$$y_1 = x^{1/3} \sum_{n=0}^{\infty} \frac{(-1)^n \prod_{j=1}^n (6j-7)^2}{81^n (n!)^2} x^n;$$

$$y_2 = y_1 \ln x + 14x^{1/3} \sum_{n=1}^{\infty} \frac{(-1)^n \prod_{j=1}^n (6j-7)^2}{81^n (n!)^2} \left( \sum_{j=1}^n \frac{1}{j(6j-7)} \right) x^n.$$

**7.6.20.**  $p_0(r) = (r+1)^2$ ;  $p_1(r) = -2(r+2)(2r+3)$ ;

$$a_n(r) = \frac{2(2n+2r+1)}{n+r+1} a_{n-1}(r), n \geq 1; a_n(r) = 2^n \prod_{j=1}^n \frac{2j+2r+1}{j+r+1};$$

By logarithmic differentiation,

$$a'_n(r) = a_n(r) \sum_{j=1}^n \frac{1}{(j+r+1)(2j+2r+1)};$$

$$r_1 = -1; a_n(-1) = \frac{2^n \prod_{j=1}^n (2j-1)}{n!};$$

$$a'_n(-1) = a_n(-1) \sum_{j=1}^n \frac{1}{j(2j-1)};$$



$$y_1 = \frac{1}{x} \sum_{n=0}^{\infty} \frac{2^n \prod_{j=1}^n (2j-1)}{n!} x^n;$$

$$y_2 = y_1 \ln x + \frac{1}{x} \sum_{n=1}^{\infty} \frac{2^n \prod_{j=1}^n (2j-1)}{n!} \left( \sum_{j=1}^n \frac{1}{j(2j-1)} \right) x^n.$$

**7.6.22.**  $p_0(r) = 2(r-2)^2$ ;  $p_1(r) = (r-1)(2r+1)$ ;

$$a_n(r) = -\frac{2n+2r-1}{2(n+r-2)} a_{n-1}(r);$$

$$a_n(r) = \frac{(-1)^n}{2^n} \prod_{j=1}^n \frac{2j+2r-1}{j+r-2};$$

By logarithmic differentiation,

$$a'_n(r) = -3a_n(r) \sum_{j=1}^n \frac{1}{(j+r-2)(2j+2r-1)};$$

$$r_1 = 2; a_n(2) = \frac{(-1)^n \prod_{j=1}^n (2j+3)}{2^n n!};$$

$$a'_n(2) = -3a_n(2) \sum_{j=1}^n \frac{1}{j(2j+3)};$$

$$y_1 = x^2 \sum_{n=0}^{\infty} \frac{(-1)^n \prod_{j=1}^n (2j+3)}{2^n n!} x^n;$$

$$y_2 = y_1 \ln x - 3x^2 \sum_{n=0}^{\infty} \frac{(-1)^n \prod_{j=1}^n (2j+3)}{2^n n!} \left( \sum_{j=1}^n \frac{1}{j(2j+3)} \right) x^n.$$

**7.6.24.**  $p_0(r) = (r-3)^2$ ;  $p_1(r) = -2(r-1)(r+2)$ ;

$$a_n(r) = \frac{2(n+r-2)(n+r+1)}{(n+r-3)^2} a_{n-1}(r);$$

$$a'_n(r) = \frac{2(n+r-2)(n+r+1)}{(n+r-3)^2} a'_{n-1}(r) - \frac{2(5n+5r-7)}{(n+r-3)^3} a_{n-1}(r);$$

$$r_1 = 3; a_n(3) = \frac{2(n+1)(n+4)}{n^2} a_{n-1}(3);$$

$$y_1 = x^3(1+20x+180x^2+1120x^3+\cdots);$$

$$a'_n(3) = \frac{2(n+1)(n+4)}{n^2} a'_{n-1}(3) - \frac{2(5n+8)}{n^3} a_{n-1}(3);$$

$$y_2 = y_1 \ln x - x^4 \left( 26 + 324x + \frac{6968}{3}x^2 + \cdots \right)$$

**7.6.26.**  $p_0(r) = r^2$ ;  $p_1(r) = r^2 + r + 1$ ;

$$a_n(r) = -\frac{(n^2 + n(2r-1) + r^2 - r + 1)}{(n+r)^2} a_{n-1}(r);$$

$$a'_n(r) = -\frac{(n^2 + n(2r-1) + r^2 - r + 1)}{(n+r)^2} a'_{n-1}(r) - \frac{(n+r-2)}{(n+r)^3} a_{n-1}(r);$$

$$r_1 = 0; a_n(0) = -\frac{(n^2 - n + 1)}{n^2} a_{n-1}(0);$$

$$y_1 = 1 - x + \frac{3}{4}x^2 - \frac{7}{12}x^3 + \cdots;$$

$$a'_n(0) = -\frac{(n^2 - n + 1)}{n^2} a'_{n-1}(0) - \frac{(n-2)}{n^3} a_{n-1}(0);$$

$$y_2 = y_1 \ln x + x \left( 1 - \frac{3}{4}x + \frac{5}{9}x^2 + \cdots \right).$$

**7.6.28.**  $p_0(r) = (r-1)^2$ ;  $p_2(r) = r+1$ ;

$$a_{2m}(r) = -\frac{1}{2m+r-1}a_{2m-2}(r), \quad n \geq 1; \quad a_{2m}(r) = \frac{(-1)^m}{\prod_{j=1}^m (2j+r-1)};$$

By logarithmic differentiation,

$$a'_{2m}(r) = a_{2m}(r) \sum_{j=1}^m \frac{(-1)^m}{2^m m!};$$

$$r_1 = 1; \quad a_{2m}(1) = \frac{(-1)^m}{2^m m!};$$

$$a'_{2m}(1) = -\frac{1}{2}a_{2m}(1) \sum_{j=1}^m \frac{1}{j};$$

$$y_1 = x \sum_{m=0}^{\infty} \frac{(-1)^m}{2^m m!} x^{2m};$$

$$y_2 = y_1 \ln x - \frac{x}{2} \sum_{m=1}^{\infty} \frac{(-1)^m}{2^m m!} \left( \sum_{j=1}^m \frac{1}{j} \right) x^{2m}.$$

**7.6.30.**  $p_0(r) = (2r-1)^2$ ;  $p_2(r) = 2r+3$ ;

$$a_{2m}(r) = -\frac{1}{4m+2r-1}a_{2m-2}(r);$$

$$a_{2m}(r) = \frac{(-1)^m}{\prod_{j=1}^m (4j+2r-1)};$$

By logarithmic differentiation,

$$a'_{2m}(r) = -2a_{2m}(r) \sum_{j=1}^m \frac{1}{4j+2r-1};$$

$$r_1 = 1/2; \quad a_{2m}(1/2) = \frac{(-1)^m}{4^m m!};$$

$$a'_{2m}(1/2) = -\frac{1}{2}a_{2m}(1/2) \sum_{j=1}^m \frac{1}{j};$$

$$y_1 = x^{1/2} \sum_{m=0}^{\infty} \frac{(-1)^m}{4^m m!} x^{2m};$$

$$y_2 = y_1 \ln x - \frac{x^{1/2}}{2} \sum_{m=1}^{\infty} \frac{(-1)^m}{4^m m!} \left( \sum_{j=1}^m \frac{1}{j} \right) x^{2m}.$$

**7.6.32.**  $p_0(r) = (2r-1)^2$ ;  $p_2(r) = (r+1)(2r+3)$ ;  $a_{2m}(r) = -\frac{2m+r-1}{4m+2r-1}a_{2m-2}(r)$ ;

$$a_{2m}(r) = (-1)^m \prod_{j=1}^m \frac{2j+r-1}{4j+2r-1};$$

By logarithmic differentiation,

$$a'_{2m}(r) = a_{2m}(r) \sum_{j=1}^m \frac{1}{(2j+r-1)(4j+2r-1)};$$

$$r_1 = 1/2; \quad a_{2m}(1/2) = \frac{(-1)^m \prod_{j=1}^m (4j-1)}{8^m m!};$$

$$a'_{2m}(1/2) = a_{2m}(1/2) \sum_{j=1}^m \frac{1}{2j(4j-1)};$$

$$y_1 = x^{1/2} \sum_{m=0}^{\infty} \frac{(-1)^m \prod_{j=1}^m (4j-1)}{8^m m!} x^{2m};$$

$$y_2 = y_1 \ln x + \frac{x^{1/2}}{2} \sum_{m=1}^{\infty} \frac{(-1)^m \prod_{j=1}^m (4j-1)}{8^m m!} \left( \sum_{j=1}^m \frac{1}{j(4j-1)} \right) x^{2m}.$$

**7.6.34.**  $p_0(r) = (4r+1)^2$ ;  $p_2(r) = (r-1)(4r+9)$ ;

$$a_{2m}(r) = -\frac{2m+r-3}{8m+4r+1} a_{2m-2}(r);$$

$$a_{2m}(r) = (-1)^m \prod_{j=1}^m \frac{2j+r-3}{8j+4r+1};$$

By logarithmic differentiation,

$$a'_{2m}(r) = a_{2m}(r) \sum_{j=1}^m \frac{13}{(2j+r-3)(8j+4r+1)};$$

$$r_1 = -1/4; a_{2m}(-1/4) = \frac{(-1)^m \prod_{j=1}^m (8j-13)}{(32)^m m!};$$

$$a'_{2m}(-1/4) = a_{2m}(-1/4) \sum_{j=1}^m \frac{13}{2j(8j-13)};$$

$$y_1 = x^{-1/4} \sum_{m=0}^{\infty} \frac{(-1)^m \prod_{j=1}^m (8j-13)}{(32)^m m!} x^{2m};$$

$$y_2 = y_1 \ln x + \frac{13}{2} x^{-1/4} \sum_{m=1}^{\infty} \frac{(-1)^m \prod_{j=1}^m (8j-13)}{(32)^m m!} \left( \sum_{j=1}^m \frac{1}{j(8j-13)} \right) x^{2m}.$$

**7.6.36.**  $p_0(r) = (2r-1)^2$ ;  $p_2(r) = 16r(r+1)$ ;

$$a_{2m}(r) = -\frac{16(2m+r-2)(2m+r-1)}{(4m+2r-1)^2} a_{2m-2}(r);$$

$$a_{2m}(r) = (-16)^m \prod_{j=1}^m \frac{(2j+r-2)(2j+r-1)}{(4j+2r-1)^2};$$

By logarithmic differentiation,

$$a'_{2m}(r) = a_{2m}(r) \sum_{j=1}^m \frac{8j+4r-5}{(2j+r-2)(2j+r-1)(4j+2r-1)};$$

$$r_1 = 1/2; a_{2m}(1/2) = \frac{(-1)^m \prod_{j=1}^m (4j-3)(4j-1)}{4^m (m!)^2};$$

$$a'_{2m}(1/2) = a_{2m}(1/2) \sum_{j=1}^m \frac{8j-3}{j(4j-3)(4j-1)};$$

$$y_1 = x^{1/2} \sum_{m=0}^{\infty} \frac{(-1)^m \prod_{j=1}^m (4j-3)(4j-1)}{4^m (m!)^2} x^{2m};$$

$$y_2 = y_1 \ln x + x^{1/2} \sum_{m=1}^{\infty} \frac{(-1)^m \prod_{j=1}^m (4j-3)(4j-1)}{4^m (m!)^2} \left( \sum_{j=1}^m \frac{8j-3}{j(4j-3)(4j-1)} \right) x^{2m}.$$

**7.6.38.**  $p_0(r) = (r+1)^2$ ;  $p_2(r) = (r+3)(2r-1)$ ;

$$a_{2m}(r) = -\frac{4m+2r-5}{2m+r+1} a_{2m-2}(r);$$

$$a_{2m}(r) = (-1)^m \prod_{j=1}^m \frac{4j+2r-5}{2j+r+1};$$

By logarithmic differentiation,

$$\begin{aligned} a'_{2m}(r) &= a_{2m}(r) \sum_{j=1}^m \frac{7}{(2j+r+1)(4j+2r-5)}; \\ r_1 &= -1; a_{2m}(-1) = \frac{(-1)^m \prod_{j=1}^m (4j-7)}{2^m m!}; \\ a'_{2m}(-1) &= a_{2m}(-1) \sum_{j=1}^m \frac{7}{2j(4j-7)}; \\ y_1 &= \frac{1}{x} \sum_{m=0}^{\infty} \frac{(-1)^m \prod_{j=1}^m (4j-7)}{2^m m!} x^{2m}; \\ y_2 &= y_1 \ln x + \frac{7}{2x} \sum_{m=1}^{\infty} \frac{(-1)^m \prod_{j=1}^m (4j-7)}{2^m m!} \left( \sum_{j=1}^m \frac{1}{j(4j-7)} \right) x^{2m}. \end{aligned}$$

**7.6.40.**  $p_0(r) = (r-1)^2$ ;  $p_2(r) = r+1$ ;

$$\begin{aligned} a_{2m}(r) &= -\frac{1}{2m+r-1} a_{2m-2}(r); \\ a'_{2m}(r) &= -\frac{1}{2m+r-1} a'_{2m-2}(r) + \frac{1}{(2m+r-1)^2} a_{2m-2}(r); \\ r_1 &= 1; a_{2m}(1) = -\frac{1}{2m} a_{2m-2}(1); \\ y_1 &= x \left( 1 - \frac{1}{2}x^2 + \frac{1}{8}x^4 - \frac{1}{48}x^6 + \dots \right); \\ a'_{2m}(1) &= -\frac{1}{2m} a'_{2m-2}(1) + \frac{1}{4m^2} a_{2m-2}(1), \quad m \geq 1; \\ y_2 &= y_1 \ln x + x^3 \left( \frac{1}{4} - \frac{3}{32}x^2 + \frac{11}{576}x^4 + \dots \right). \end{aligned}$$

**7.6.42.**  $p_0(r) = 2(r+3)^2$ ;  $p_2(r) = r^2 - 2r + 2$ ;

$$\begin{aligned} a_{2m}(r) &= -\frac{4m^2 + 4m(r-3) + r^2 - 6r + 10}{2(2m+r+3)^2} a_{2m-2}(r); \\ a'_{2m}(r) &= -\frac{4m^2 + 4m(r-3) + r^2 - 6r + 10}{2(2m+r+3)^2} a'_{2m-2}(r) - \frac{12m+6r-19}{(2m+r+3)^3} a_{2m-2}(r); \\ r_1 &= -3; a_{2m}(-3) = -\frac{4m^2 - 24m + 37}{8m^2} a_{2m-2}(-3); \\ y_1 &= x^{-3} \left( 1 - \frac{17}{8}x^2 + \frac{85}{256}x^4 - \frac{85}{18432}x^6 + \dots \right); \\ a'_{2m}(-3) &= -\frac{4m^2 - 24m + 37}{8m^2} a'_{2m-2}(-3) + \frac{37-12m}{8m^3} a_{2m-2}(-3), \quad m \geq 1; \\ y_2 &= y_1 \ln x + x^{-1} \left( \frac{25}{8} - \frac{471}{512}x^2 + \frac{1583}{110592}x^4 + \dots \right). \end{aligned}$$

**7.6.44.**  $p_0(r) = (r+1)^2$ ;  $p_1(r) = 2(2-r)(r+1)$ ;  $r_1 = -1$ .

$a_n(r) = \frac{2(n+r)(n+r-3)}{(n+r+1)^2} a_{n-1}(r)$ ;  $a_n(r) = 2^n \prod_{j=1}^n \frac{(j+r)(j+r-3)}{(j+r+1)^2}$ ,  $n \geq 0$ . Therefore,  $a_n(-1) = 0$  if  $n \geq 1$  and  $y_1 = 1/x$ . If  $n \geq 4$ , then  $a_n(r) = (r+1)^2 b_n(r)$ , where  $b'_n(-1)$  exists; therefore  $a'_n(-1) = 0$  if  $n \geq 4$ . For  $r = 1, 2, 3$ ,  $a_n(r) = (r+1)c_n(r)$ , where  $c_1(r) = \frac{2(r-2)}{(r+2)^2}$ ,  $c_2(r) = \frac{4(r-2)(r-1)}{(r+2)(r+3)^2}$ ,  $c_3(r) = \frac{8r(r-2)(r-1)}{(r+2)(r+3)(r+4)^2}$ . Hence,  $a'_1(-1) = c_1(-1) = -6$ ,  $a'_2(-1) =$

$c_2(-1) = 6$ ,  $a'_3(-1) = c_3(-1) = -8/3$ , and  $y_2 = y_1 \ln x - 6 + 6x - \frac{8}{3}x^2$ .

**7.6.46.**  $p_0(r) = (r+1)^2$ ;  $p_1(r) = -(r-1)(r+2)$ ;  $r_1 = -1$ .

$a_n(r) = \frac{n+r-2}{n+r+1}a_{n-1}(r)$ ;  $a_n(r) = \prod_{j=1}^m \frac{j+r-2}{j+r+1}$ ,  $n \geq 0$ . Therefore,  $a_1(-1) = -2$ ,  $a_2(-1) = 1$ ,

and  $a_n(-1) = 0$  if  $n \geq 3$ , so  $y_1 = \frac{(x-1)^2}{x}$ .

$a_1(r) = \frac{r-1}{r+2}$ ,  $a'_1(r) = \frac{3}{(r+2)^2}$ ,  $a'_1(-1) = 3$ ;  $a_2(r) = \frac{r(r-1)}{(r+2)(r+3)}$ ,  $a'_2(r) = \frac{6(r^2+2r-1)}{(r+2)^2(r+3)^2}$ ,

$a'_2(-1) = -3$ ; if  $n \geq 3$   $a_n(r) = (r+1)c_n(r)$  where  $c_n(r) = \frac{r(r-1)}{(n+r)(n+r-1)(n+r+1)}$ , so

$a'_n(-1) = c_n(-1) = \frac{2}{n(n-2)(n-1)}$  and  $y_2 = y_1 \ln x + 3 - 3x + 2 \sum_{n=2}^{\infty} \frac{1}{n(n^2-1)}x^n$ .

**7.6.48.**  $p_0(r) = (r-2)^2$ ;  $p_1(r) = -(r-5)(r-1)$ ;  $r_1 = 2$ .

$a_n(r) = \frac{n+r-6}{n+r-2}a_{n-1}(r)$ ;

$a_n(r) = \prod_{j=1}^m \frac{j+r-6}{j+r-2}$ ,  $n \geq 0$ . Therefore,  $a_1(2) = -3$ ,  $a_2(2) = 3$ ,  $a_3(2) = -1$ , and  $a_n(2) = 0$  if

$n \geq 4$ , so  $y_1 = x^2(1-x)^3$ .

$a_1(r) = \frac{r-5}{r-1}$ ,  $a'_1(r) = \frac{4}{(r-1)^2}$ ,  $a'_1(2) = 4$ ;

$a_2(r) = \frac{(r-5)(r-4)}{r(r-1)}$ ,  $a'_2(r) = \frac{4(2r^2-10r+5)}{r^2(r-1)^2}$ ,  $a'_2(2) = -7$ ;

$a_3(r) = \frac{(r-5)(r-4)(r-3)}{r(r-1)(r+1)}$ ,  $a'_3(r) = \frac{12(r^4-8r^3+16r^2-5)}{r^2(r-1)^2(r+1)^2}$ ,  $a'_3(2) = 11/3$ ; if  $n \geq 4$ , then

$a_n(r) = (r-2)c_n(r)$  where  $c_n(r) = \frac{(r-5)(r-4)(r-3)}{(n+r-5)(n+r-4)(n+r-3)(n+r-2)}$ , so  $a'_n(2) = c_n(2) =$

$-\frac{6}{n(n-2)(n^2-1)}$  and

$y_2 = y_1 \ln x + x^3 \left( 4 - 7x + \frac{11}{3}x^2 - 6 \sum_{n=3}^{\infty} \frac{1}{n(n-2)(n^2-1)}x^n \right)$ .

**7.6.50.**  $p_0(r) = (3r-1)^2$ ;  $p_2(r) = 7-3r$ ;  $r_1 = 1/3$ .

$a_{2m}(r) = \frac{6m+3r-13}{(6m+3r-1)^2}a_{2m-2}(r)$ ;

$a_{2m}(r) = \prod_{j=1}^m \frac{6j+3r-13}{(6j+3r-1)^2}$ ,  $m \geq 0$ . Therefore,  $a_2(1/3) = 1/6$  and  $a_{2m}(1/3) = 0$  if  $m \geq 2$ , so

$y_1 = x^{1/3} \left( 1 - \frac{1}{6}x^2 \right)$ .

$a_2(r) = \frac{3r-7}{(3r+5)^2}$ ;  $a'_2(r) = \frac{3(19-3r)}{(3r+5)^3}$ ;  $a'_2(1/3) = 1/4$ . If  $m \geq 2$ , then  $a_{2m}(r) = (r-1/3)c_{2m}(r)$

where  $c_{2m}(r) = \frac{3(3r-7)}{(6m+3r-7)(6m+3r-1) \prod_{j=1}^m (6j+3r-1)}$ , so  $a'_{2m}(1/3) = c_{2m}(1/3) = -\frac{1}{12} \frac{1}{6^{m-1}(m-1)m!}$ ,

and

$$y_2 = y_1 \ln x + x^{7/3} \left( \frac{1}{4} - \frac{1}{12} \sum_{m=1}^{\infty} \frac{1}{6^m m(m+1)(m+1)!} x^{2m} \right).$$

**7.6.52.**  $p_0(r) = (2r+1)^2$ ;  $p_2(r) = 7-2r$ ;  $r_1 = -1/2$ .

$$a_{2m}(r) = \frac{4m+2r-11}{(4m+2r+1)^2} a_{2m-2}(r);$$

$$a_{2m}(r) = \prod_{j=1}^m \frac{4j+2r-11}{(4j+2r+1)^2}, \quad m \geq 0. \quad \text{Therefore, } a_2(-1/2) = -1/2, \quad a_4(-1/2) = 1/32, \quad \text{and}$$

$$a_{2m}(-1/2) = 0 \text{ if } m \geq 3, \text{ so } y_1 = x^{-1/2} \left( 1 - \frac{1}{2}x^2 + \frac{1}{32}x^4 \right).$$

$$a_2(r) = \frac{2r-7}{(2r+5)^2}, \quad a'_2(r) = \frac{2(19-2r)}{(2r+5)^3}, \quad a'_2(-1/2) = 5/8,$$

$$a_4(r) = \frac{(2r-7)(2r-3)}{(2r+5)^2(2r+9)^2}, \quad a'_4(r) = -\frac{4(8r^3-60r^2-146r+519)}{(2r+5)^3(2r+9)^3}, \quad a'_4(-1/2) = -9/128; \text{ if } m \geq 3,$$

then  $a_{2m}(r) = (r+1/2)c_{2m}(r)$  where

$$c_{2m}(r) = \frac{2(2r-7)(2r-3)}{(4m+2r-7)(4m+2r-3)(4m+2r+1) \prod_{j=1}^m (4j+2r+1)}, \text{ so } a'_{2m}(-1/2) = c_{2m}(-1/2) =$$

$$\frac{1}{4^m(m-2)(m-1)m!}, \text{ and}$$

$$y_2 = y_1 \ln x + x^{3/2} \left( \frac{5}{8} - \frac{9}{128}x^2 + \sum_{m=2}^{\infty} \frac{1}{4^{m+1}(m-1)m(m+1)(m+1)!} x^{2m} \right).$$

**7.6.54.** (a) If  $p_0(r) = \alpha_0(r-r_1)^2$ , then (A)  $a_n(r) = \frac{(-1)^n}{\alpha_0^n} \prod_{j=1}^n \frac{p_1(j+r-1)}{(j+r-r_1)^2}$ . Therefore,  $a_n(r_1) =$

$$\frac{(-1)^n}{\alpha_0^n(n!)^2} \prod_{j=1}^n p_1(j+r_1-1). \text{ Theorem 7.6.2 implies } Ly_1 = 0.$$

$$(b) \text{ From (A), } \ln |a_n(r)| = -n \ln |\alpha_0| + \sum_{j=1}^n (\ln |p_1(j+r-1)| - 2 \ln |j+r-r_1|), \text{ so } a'_n(r) = a_n(r) \sum_{j=1}^n \left( \frac{p'_1(j+r-1)}{p_1(j+r-1)} - \frac{2}{j+r-r_1} \right).$$

$$\text{and } a'_n(r_1) = a_n(r_1) \sum_{j=1}^n \left( \frac{p'_1(j+r_1-1)}{p_1(j+r_1-1)} - \frac{2}{j} \right). \text{ Theorem 7.6.2 implies that } Ly_2 = 0.$$

(c) Since  $p_1(r) = \gamma_1$ ,  $y_1$  and  $y_2$  reduce to the stated forms. If  $\gamma_1 = 0$ , then  $y_1 = x^{r_1}$  and  $y_2 = x^{r_1} \ln x$ , which are solutions of the Euler equation  $\alpha_0 x^2 y'' + \beta_0 x y' + \gamma_0 y = 0$ .

**7.6.54.** (a)  $Ly_1 = p_0(r_1)x^{r_1} = 0$ . Now use the fact that  $p_0(j+r_1) = \alpha_0 j^2$ , so  $\prod_{j=1}^n p_0(j+r_1) = \alpha_0^n (n!)^2$ .

(b) From Theorem 7.6.2,  $y_2 = y_1 \ln x + x^{r_1} \sum_{n=1}^{\infty} a'_n(r_1)x^n$  is a second solution of  $Ly = 0$ . Since

$$a_n(r) = \frac{(-1)^n}{\alpha_0^n} \prod_{j=1}^n \frac{p_1(j+r-1)}{(j+r-r_1)^2}, \quad (A) \quad \ln |a_n(r)| = -n \ln |\alpha_0| + \sum_{j=1}^n \ln |p_1(j+r-1)| - 2 \sum_{j=1}^n \ln |j+r-r_1|,$$

provided that  $p_1(j+r-1)$  and  $j+r-r_1$  are nonzero for all positive integers  $j$ . Differentiating (A)

$$\text{and then setting } r = r_1 \text{ yields } \frac{a'_n(r_1)}{a_n(r_1)} = \sum_{j=1}^n \frac{p'_1(j+r_1-1)}{p_1(j+r_1-1)} - 2 \sum_{j=1}^n \frac{1}{j}, \text{ which implies the conclusion.}$$

(c) In this case  $p_1(r) = \gamma_1$  and  $p'_1(r) = 0$ , so  $a_n(r_1) = \frac{(-1)^n}{(n!)^2} \left( \frac{\gamma_1}{\alpha_0} \right)^n$  and  $J_n = -2 \sum_{j=1}^n \frac{1}{j}$ . If  $\gamma_1 = 0$ , then  $y_1 = x^{r_1}$  and  $y_2 = x^{r_1} \ln x$ , while the differential equation is an Euler equation with

indicial polynomial  $\alpha_0(r - r_1^2)$ . See Theorem 7.4.3.

**7.6.56.**  $p_0(r) = r^2$ ;  $p_1(r) = 1$ ;  $r_1 = 0$ .  $a_{2m}(r) = -\frac{a_{2m-1}(r)}{(2m+r)^2}$ ,  $m \geq 1$ ;  $a_{2m}(r) = \frac{(-1)^m}{\prod_{j=1}^m (2j+r)^2}$ ,

$m \geq 0$ . Therefore,  $a_{2m}(0) = \frac{(-1)^m}{4^m(m!)^2}$ , so  $y_1 = \sum_{m=0}^{\infty} \frac{(-1)^m}{4^m(m!)^2} x^{2m}$ .

By logarithmic differentiation,  $a'_{2m}(r) = -2a_{2m}(r) \sum_{j=1}^m \frac{1}{2m+r}$ , so  $a'_{2m}(0) = -a_{2m}(0) \sum_{j=1}^m \frac{1}{j}$  and

$$y_2 = y_1 \ln x - \sum_{m=1}^{\infty} \frac{(-1)^m}{4^m(m!)^2} \left( \sum_{j=1}^m \frac{1}{j} \right) x^{2m}.$$

**7.6.58.**  $p_0(r) = (2r-1)^2$ ;  $p_1(r) = (2r+1)^2$ ;  $p_2(r) = 0$ ;  $\frac{p_1(r-1)}{p_0(r)} = 1 = \frac{\alpha_1}{\alpha_0}$ ;  $\frac{p_2(r-2)}{p_0(r)} = 0 = \frac{\alpha_2}{\alpha_0}$ ;

$$y_1 = \frac{x^{1/2}}{1+x}; y_2 = \frac{x^{1/2} \ln x}{1+x}.$$

**7.6.60.**  $p_0(r) = 2(r-1)^2$ ;  $p_1(r) = 0$ ;  $p_2(r) = -(r+1)^2$ ;  $\frac{p_1(r-1)}{p_0(r)} = 0 = \frac{\alpha_1}{\alpha_0}$ ;  $\frac{p_2(r-2)}{p_0(r)} = -1/2 =$

$$\frac{\alpha_2}{\alpha_0}; y_1 = \frac{x}{2-x^2}; y_2 = \frac{x \ln x}{2-x^2}.$$

**7.6.62.**  $p_0(r) = 4(r-1)^2$ ;  $p_1(r) = 3r^2$ ;  $p_2(r) = 0$ ;  $\frac{p_1(r-1)}{p_0(r)} = 3/4 = \frac{\alpha_1}{\alpha_0}$ ;  $\frac{p_2(r-2)}{p_0(r)} = 0 = \frac{\alpha_2}{\alpha_0}$ ;

$$y_1 = \frac{x}{4+3x}; y_2 = \frac{x \ln x}{4+3x}.$$

**7.6.64.**  $p_0(r) = (r-1)^2$ ;  $p_1(r) = -2r^2$ ;  $p_2(r) = (r+1)^2$ ;  $\frac{p_1(r-1)}{p_0(r)} = -2 = \frac{\alpha_1}{\alpha_0}$ ;  $\frac{p_2(r-2)}{p_0(r)} = 1 = \frac{\alpha_2}{\alpha_0}$ ;

$$y_1 = \frac{x}{(1-x)^2}; y_2 = \frac{x \ln x}{(1-x)^2}.$$

**7.6.66.** See the proofs of Theorems 7.6.1 and 7.6.2.

## 7.7 The Method of Frobenius III

**7.7.2.**  $p_0(r) = r(r-1)$ ;  $p_1(r) = 1$ ;  $r_1 = 1$ ;  $r_2 = 0$ ;  $k = r_1 - r_2 = 1$ ;

$$a_n(r) = -\frac{1}{(n+r)(n+r-1)} a_{n-1}(r);$$

$$a_n(r) = \frac{(-1)^n}{\prod_{j=1}^n (j+r)(j+r-1)};$$

$$a_n(1) = \frac{(-1)^n}{n!(n+1)!};$$

$$y_1 = x \sum_{n=0}^{\infty} \frac{(-1)^n}{n!(n+1)!} x^n;$$

$$z = 1; C = -p_1(0)a_0(0) = -1.$$

By logarithmic differentiation,

$$a'_n(r) = -a_n(r) \sum_{j=1}^n \frac{2n+2r-1}{(n+r)(n+r-1)};$$

$$a'_n(1) = -a_n(1) \sum_{j=1}^n \frac{2j+1}{j(j+1)};$$

$$y_2 = 1 - y_1 \ln x + x \sum_{n=1}^{\infty} \frac{(-1)^n}{n!(n+1)!} \left( \sum_{j=1}^n \frac{2j+1}{j(j+1)} \right) x^n.$$

**7.7.4.**  $p_0(r) = r(r-1)$ ;  $p_1(r) = r+1$ ;  $r_1 = 1$ ;  $r_2 = 0$ ;  $k = r_1 - r_2 = 1$ ;  $a_n(r) = -\frac{a_{n-1}(r)}{n+r-1}$ ;  $a_n(r) = \frac{(-1)^n}{\prod_{j=1}^n (j+r-1)}$ ;  $a_n(1) = \frac{(-1)^n}{n!}$ ;  $y_1 = x \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} x^n = xe^{-x}$ ;  $z = 1$ ;  $C = -p_1(0)a_0(0) = -1$ . By

logarithmic differentiation,  $a'_n(r) = -a_n(r) \sum_{j=1}^n \frac{1}{j+r-1}$ ;  $a'_n(1) = -a_n(1) \sum_{j=1}^n \frac{1}{j}$ ;  $y_2 = 1 - y_1 \ln x + x \sum_{n=1}^{\infty} \frac{(-1)^n}{n!} \left( \sum_{j=1}^n \frac{1}{j} \right) x^n$ .

**7.7.6.**  $p_0(r) = (r-1)(r+2)$ ;  $p_1(r) = r+3$ ;  $r_1 = 1$ ;  $r_2 = -2$ ;  $k = r_1 - r_2 = 3$ .  $a_n(r) = -\frac{1}{n+r-1} a_{n-1}(r)$ ;  $a_n(r) = \frac{(-1)^n}{\prod_{j=1}^n (j+r-1)}$ ;  $a_n(1) = \frac{(-1)^n}{n!}$ ;  $y_1 = x \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} x^n = xe^{-x}$ ;  $z =$

$x^{-2} \left( 1 + \frac{1}{2}x + \frac{1}{2}x^2 \right)$ ;  $C = -\frac{p_1(0)}{3} a_2(-2) = -1/2$ . By logarithmic differentiation,  $a'_n(r) = -a_n(r) \sum_{j=1}^n \frac{1}{j+r-1}$ ;

$$a'_n(1) = a_n(1) \sum_{j=1}^n \frac{1}{j}; y_2 = x^{-2} \left( 1 + \frac{1}{2}x + \frac{1}{2}x^2 \right) - \frac{1}{2} \left( y_1 \ln x - x \sum_{n=1}^{\infty} \frac{(-1)^n}{n!} \left( \sum_{j=1}^n \frac{1}{j} \right) x^n \right);$$

**7.7.8.**  $p_0(r) = (r+2)(r+7)$ ;  $p_1(r) = 1$ ;  $r_1 = -2$ ;  $r_2 = -7$ ;  $k = r_1 - r_2 = 5$ ;  $a_n(r) = -\frac{a_{n-1}(r)}{(n+r+2)(n+r+7)}$ ;  $a_n(r) = \frac{(-1)^n}{\prod_{j=1}^n (j+r+2)(j+r+7)}$ ;  $a_n(-2) = 120 \prod_{j=1}^n \frac{(-1)^n}{n!(n+5)!}$ ;  $y_1 = \frac{120}{x^2} \sum_{n=0}^{\infty} \frac{(-1)^n}{n!(n+5)!} x^n$ ;

$z = x^{-7} \left( 1 + \frac{1}{4}x + \frac{1}{24}x^2 + \frac{1}{144}x^3 + \frac{1}{576}x^4 \right)$ ;  $C = -\frac{p_1(-3)}{5} a_4(-7) = -1/2880$ . By logarithmic

differentiation,  $a'_n(r) = -a_n(r) \sum_{j=1}^n \frac{2j+2r+9}{(j+r+2)(j+r+7)}$ ;  $a'_n(-2) = -a_n(-2) \sum_{j=1}^n \frac{2j+5}{j(j+5)}$ ;

$$y_2 = x^{-7} \left( 1 + \frac{1}{4}x + \frac{1}{24}x^2 + \frac{1}{144}x^3 + \frac{1}{576}x^4 \right) - \frac{1}{2880} \left( y_1 \ln x - \frac{120}{x^2} \sum_{n=1}^{\infty} \frac{(-1)^n}{n!(n+5)!} \left( \sum_{j=1}^n \frac{2j+5}{j(j+5)} \right) x^n \right).$$

**7.7.10.**  $p_0(r) = r(r-4)$ ;  $p_1(r) = (r-6)(r-5)$ ;  $r_1 = 4$ ;  $r_2 = 0$ ;  $k = r_1 - r_2 = 4$ ;  $a_n(r) = -\frac{(n+r-7)(n+r-6)}{(n+r)(n+r-4)} a_{n-1}(r)$ ;  $a_n(r) = (-1)^n \prod_{j=1}^n \frac{(j+r-7)(j+r-6)}{(j+r)(j+r-4)}$ . Setting  $r = 4$  yields

$$y_1 = x^4 \left( 1 - \frac{2}{5}x \right). z = 1 + 10x + 50x^2 + 200x^3; C = -\frac{p_1(3)}{4} a_3(0) = -300. a_1(r) = -\frac{(r-6)(r-5)}{(r-3)(r+1)};$$

$$a'_1(r) = -\frac{3(3r^2 - 22r + 31)}{(r-3)^2(r+1)^2}; a'_1(4) = 27/25. a_2(r) = (r-4)c_2(r), \text{ with } c_2(r) = \frac{(r-6)(r-5)^2}{(r-3)(r-2)(r+1)(r+2)},$$

so  $a'_2(4) = c_2(4) = -1/30$ . If  $n \geq 3$ , then  $a_n(r) = (r-4)^2 b_n(r)$  where  $b'_n(4)$  exists, so  $a'_n(4) = 0$

and  $y_2 = 1 + 10x + 50x^2 + 200x^3 - 300 \left( y_1 \ln x + \frac{27}{25}x^5 - \frac{1}{30}x^6 \right)$ .

**7.7.12.**  $p_0(r) = (r-2)(r+2)$ ;  $p_1(r) = -2r-1$ ;  $r_1 = 2$ ;  $r_2 = -2$ ;  $k = r_1 - r_2 = 4$ ;  $a_n(r) =$

$$\frac{2j+2r-1}{(j+r-2)(j+r+2)} a_{n-1}(r); a_n(r) = \prod_{j=1}^n \frac{2n+2r-1}{(n+r-2)(n+r+2)}; a_n(2) = \frac{1}{n!} \left( \prod_{j=1}^n \frac{2j+3}{j+4} \right); y_1 = x^2 \sum_{n=0}^{\infty} \frac{1}{n!} \left( \prod_{j=1}^n \frac{2j+3}{j+4} \right) x^n$$

$z = x^{-2} \left( 1 + x + \frac{1}{4}x^2 - \frac{1}{12}x^3 \right)$ ;  $C = -\frac{p_1(1)}{4} a_3(-2) = -1/16$ . By logarithmic differentiation,



$$\begin{aligned}
a'_n(r) &= -2a_n(r) \sum_{j=1}^n \frac{j^2 + j(2r-1) + r^2 - r + 4}{(j+r-2)(j+r+2)(2j+2r-1)}; \quad a'_n(2) = -2a_n(2) \sum_{j=1}^n \frac{(j^2 + 3j + 6)}{j(j+4)(2j+3)}; \\
y_2 &= x^{-2} \left( 1 + x + \frac{1}{4}x^2 - \frac{1}{12}x^3 \right) - \frac{1}{16}y_1 \ln x + \frac{x^2}{8} \sum_{n=1}^{\infty} \frac{1}{n!} \left( \prod_{j=1}^n \frac{2j+3}{j+4} \right) \left( \sum_{j=1}^n \frac{(j^2 + 3j + 6)}{j(j+4)(2j+3)} \right) x^n. \\
\mathbf{7.7.14.} \quad &p_0(r) = (r+1)(r+7); \quad p_1(r) = (r+5)(2r+1); \quad r_1 = -1; \quad r_2 = -7; \quad k = r_1 - r_2 = 6; \\
a_n(r) &= -\frac{(n+r+4)(2n+2r-1)}{(n+r+1)(n+r+7)} a_{n-1}(r); \quad a_n(r) = (-1)^n \prod_{j=1}^n \frac{(j+r+4)(2j+2r-1)}{(j+r+1)(j+r+7)}; \quad a_n(-1) = \\
&\frac{(-1)^n}{n!} \left( \prod_{j=1}^n \frac{(j+3)(2j-3)}{j+6} \right); \quad y_1 = \frac{1}{x} \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} \left( \prod_{j=1}^n \frac{(j+3)(2j-3)}{j+6} \right) x^n; \quad z = x^{-7} \left( 1 + \frac{26}{5}x + \frac{143}{20}x^2 \right); \\
C &= -\frac{p_1(-2)}{6} a_5(-7) = 0; \quad y_2 = x^{-7} \left( 1 + \frac{26}{5}x + \frac{143}{20}x^2 \right). \\
\mathbf{7.7.16.} \quad &p_0(r) = (3r-10)(3r+2); \quad p_1(r) = r(3r-4); \quad r_1 = 10/3; \quad r_2 = -2/3; \quad k = r_1 - r_2 = 4; \quad a_n(r) = \\
&-\frac{(n+r-1)(3n+3r-7)}{(3n+3r-10)(3n+3r+2)} a_{n-1}(r); \quad a_n(r) = (-1)^n \prod_{j=1}^n \frac{(j+r-1)(3j+3r-7)}{(3j+3r-10)(3j+3r+2)}; \quad a_n(10/3) = \\
&\frac{(-1)^n(n+1)}{9^n} \left( \prod_{j=1}^n \frac{3j+7}{j+4} \right); \quad y_1 = x^{10/3} \sum_{n=0}^{\infty} \frac{(-1)^n(n+1)}{9^n} \left( \prod_{j=1}^n \frac{3j+7}{j+4} \right) x^n; \quad z = x^{-2/3} \left( 1 + \frac{4}{27}x - \frac{1}{243}x^2 \right); \\
C &= -\frac{p_1(7/3)}{36} a_3(-2/3) = 0; \quad y_2 = x^{-2/3} \left( 1 + \frac{4}{27}x - \frac{1}{243}x^2 \right). \\
\mathbf{7.7.18.} \quad &p_0(r) = (r-3)(r+2); \quad p_1(r) = (r+1)^2; \quad r_1 = 3; \quad r_2 = -2; \quad k = r_1 - r_2 = 5; \quad a_n(r) = \\
&-\frac{(n+r)^2}{(n+r-3)(n+r+2)} a_{n-1}(r); \quad a_n(r) = (-1)^n \prod_{j=1}^n \frac{(j+r)^2}{(j+r-3)(j+r+2)}; \quad a_n(3) = \frac{(-1)^n}{n!} \left( \prod_{j=1}^n \frac{(j+3)^2}{j+5} \right); \\
y_1 &= x^3 \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} \left( \prod_{j=1}^n \frac{(j+3)^2}{j+5} \right) x^n; \quad z = x^{-2} \left( 1 + \frac{1}{4}x \right); \quad C = -\frac{p_1(2)}{5} a_4(-2) = 0; \quad y_2 = x^{-2} \left( 1 + \frac{1}{4}x \right). \\
\mathbf{7.7.20.} \quad &p_0(r) = (r-6)(r-1); \quad p_1(r) = (r-8)(r-4); \quad r_1 = 6; \quad r_2 = 1; \quad k = r_1 - r_2 = 5; \quad a_n(r) = \\
&-\frac{(n+r-9)(n+r-5)}{(n+r-6)(n+r-1)} a_{n-1}(r); \quad y_1 = x^6 \left( 1 + \frac{2}{3}x + \frac{1}{7}x^2 \right); \quad z = x \left( 1 + \frac{21}{4}x + \frac{21}{2}x^2 + \frac{35}{4}x^3 \right); \quad C = \\
&-\frac{p_1(5)}{6} a_5(1) = 0; \quad y_2 = x \left( 1 + \frac{21}{4}x + \frac{21}{2}x^2 + \frac{35}{4}x^3 \right). \\
\mathbf{7.7.22.} \quad &p_0(r) = r(r-10); \quad p_1(r) = 2(r-6)(r+1); \quad r_1 = 10; \quad r_2 = 0; \quad k = r_1 - r_2 = 10; \quad a_n(r) = \\
&-\frac{2(n+r-7)}{n+r-10} a_{n-1}(r); \quad a_n(r) = (-2)^n \frac{(n+r-9)(n+r-8)(n+r-7)}{(r-9)(r-8)(r-7)}; \quad a_n(10) = \frac{(-1)^n 2^n (n+1)(n+2)(n+3)}{6}; \\
y_1 &= \frac{x^{10}}{6} \sum_{n=0}^{\infty} (-1)^n 2^n (n+1)(n+2)(n+3) x^n; \quad z = \left( 1 - \frac{4}{3}x + \frac{5}{3}x^2 - \frac{40}{21}x^3 + \frac{40}{21}x^4 - \frac{32}{21}x^5 + \frac{16}{21}x^6 \right); \\
C &= -\frac{p_1(9)}{10} a_9(0) = 0; \quad y_2 = \left( 1 - \frac{4}{3}x + \frac{5}{3}x^2 - \frac{40}{21}x^3 + \frac{40}{21}x^4 - \frac{32}{21}x^5 + \frac{16}{21}x^6 \right). \\
\text{Note: in the solutions to Exercises 7.7.23--7.7.40, } &z = x^{r_2} \sum_{m=0}^{k-1} a_{2m}(r_2) x^{2m}. \\
\mathbf{7.7.24.} \quad &p_0(r) = (r-6)(r-2); \quad p_2(r) = r; \quad r_1 = 6; \quad r_2 = 2; \quad k = (r_1 - r_2)/2 = 2; \quad a_{2m}(r) = \\
&-\frac{a_{2m-2}(r)}{2m+r-6}; \quad a_{2m}(r) = \frac{(-1)^m}{\prod_{j=1}^m (2j+r-6)}; \quad a_{2m}(6) = \frac{(-1)^m}{2^m m!}; \quad y_1 = x^6 \sum_{m=0}^{\infty} \frac{(-1)^m}{2^m m!} x^{2m} = x^6 e^{-x^2/2}; \\
z &= x^2 \left( 1 + \frac{1}{2}x^2 \right); \quad C = -\frac{p_2(4)}{4} a_2(2) = -1/2. \quad \text{By logarithmic differentiation, } a'_{2m}(r) = -a_{2m}(r) \sum_{j=1}^m \frac{1}{2j+r-6};
\end{aligned}$$

$$a'_{2m}(6) = -a_{2m}(6) \sum_{j=1}^m \frac{1}{2j}; y_2 = x^2 \left(1 + \frac{1}{2}x^2\right) - \frac{1}{2}y_1 \ln x + \frac{x^6}{4} \sum_{m=1}^{\infty} \frac{(-1)^m}{2^m m!} \left(\sum_{j=1}^m \frac{1}{j}\right) x^{2m}.$$

**7.7.26.**  $p_0(r) = (r-1)(r+1)$ ;  $p_2(r) = 2r+10$ ;  $r_1 = 1$ ;  $r_2 = -1$ ;  $k = (r_1 - r_2)/2 = 1$ ;  $a_{2m}(r) = -\frac{2(2m+r+3)}{(2m+r-1)(2m+r+1)} a_{2m-2}(r)$ ;  $a_{2m}(r) = (-2)^m \prod_{j=1}^m \frac{2j+r+3}{(2j+r-1)(2j+r+1)}$ ;  
 $a_{2m}(1) = \frac{(-1)^m(m+2)}{2m!}$ ;  $y_1 = \frac{x}{2} \sum_{m=0}^{\infty} \frac{(-1)^m(m+2)}{m!} x^{2m}$ ;  $z = x^{-1}$ ;  $C = -\frac{p_2(-1)}{2} a_0(-1) = -4$ . By

logarithmic differentiation,

$$a'_{2m}(r) = -a_{2m}(r) \sum_{j=1}^m \frac{(4j^2 + 4j(r+3) + r^2 + 6r + 1)}{(2j+r-1)(2j+r+1)(2j+r+3)};$$

$$a'_{2m}(1) = -a_{2m}(1) \sum_{j=1}^m \frac{j^2 + 4j + 2}{2j(j+1)(j+2)};$$

$$y_2 = x^{-1} - 4y_1 \ln x + x \sum_{m=1}^{\infty} \frac{(-1)^m(m+2)}{m!} \left(\sum_{j=1}^m \frac{j^2 + 4j + 2}{j(j+1)(j+2)}\right) x^{2m}.$$

**7.7.28.**  $p_0(r) = (2r+1)(2r+5)$ ;  $p_2(r) = 2r+3$ ;  $r_1 = -1/2$ ;  $r_2 = -5/2$ ;  $k = (r_1 - r_2)/2 = 1$ ;  
 $a_{2m}(r) = -\frac{(4m+2r-1)}{(4m+2r+1)(4m+2r+5)} a_{2m-2}(r)$ ;  $a_{2m}(r) = (-1)^m \prod_{j=1}^m \frac{(4j+2r-1)}{(4j+2r+1)(4j+2r+5)}$ ;

$$a_{2m}(-1/2) = \frac{(-1)^m \prod_{j=1}^m (2j-1)}{8^m m! (m+1)!}; y_1 = x^{-1/2} \sum_{m=0}^{\infty} \frac{(-1)^m \prod_{j=1}^m (2j-1)}{8^m m! (m+1)!} x^{2m}; z = x^{-5/2}; C = -\frac{p_2(-5/2)}{8} a_0(-5/2) =$$

1/4. By logarithmic differentiation,

$$a'_{2m}(r) = -a_{2m}(r) \sum_{j=1}^m \frac{(16j^2 + 8j(2r-1) + 4r^2 - 4r - 11)}{(4j+2r-1)(4j+2r+1)(4j+2r+5)}; a'_{2m}(-1/2) = -a_{2m}(-1/2) \sum_{j=1}^m \frac{2j^2 - 2j - 1}{2j(j+1)(2j-1)};$$

$$y_2 = x^{-5/2} + \frac{1}{4}y_1 \ln x - x^{-1/2} \sum_{m=1}^{\infty} \frac{(-1)^m \prod_{j=1}^m (2j-1)}{8^{m+1} m! (m+1)!} \left(\sum_{j=1}^m \frac{2j^2 - 2j - 1}{j(j+1)(2j-1)}\right) x^{2m}.$$

**7.7.30.**  $p_0(r) = (r-2)(r+2)$ ;  $p_2(r) = -2(r+4)$ ;  $r_1 = 2$ ;  $r_2 = -2$ ;  $k = (r_1 - r_2)/2 = 2$ ;  $a_{2m}(r) = \frac{2}{2m+r-2} a_{2m-2}(r)$ ;  $a_{2m}(r) = \frac{2^m}{\prod_{j=1}^m (2j+r-2)}$ ;  $a_{2m}(2) = \frac{1}{m!}$ ;  $y_1 = x^2 \sum_{m=0}^{\infty} \frac{1}{m!} x^{2m} = x^2 e^{x^2}$ ;  $z =$

$$x^{-2}(1-x^2); C = -\frac{p_2(0)}{4} a_2(-2) = -2. \text{ By logarithmic differentiation, } a'_{2m}(r) = -a_{2m}(r) \sum_{j=1}^m \frac{1}{2j+r-2};$$

$$a'_{2m}(2) = -a_{2m}(2) \sum_{j=1}^m \frac{1}{2j}; y_2 = x^{-2}(1-x^2) - 2y_1 \ln x + x^2 \sum_{m=1}^{\infty} \frac{1}{m!} \left(\sum_{j=1}^m \frac{1}{j}\right) x^{2m}.$$

**7.7.32.**  $p_0(r) = (3r-13)(3r-1)$ ;  $p_2(r) = 2(5-3r)$ ;  $r_1 = 13/3$ ;  $r_2 = 1/3$ ;  $k = (r_1 - r_2)/2 = 2$ ;  $a_{2m}(r) = \frac{2(6m+3r-11)}{(6m+3r-13)(6m+3r-1)} a_{2m-2}(r)$ ;  $a_{2m}(r) = 2^m \prod_{j=1}^m \frac{(6j+3r-11)}{(6j+3r-13)(6j+3r-1)}$ ;

$$a_{2m}(13/3) = \frac{\prod_{j=1}^m (3j+1)}{9^m m! (m+2)!}; y_1 = 2x^{13/3} \sum_{m=0}^{\infty} \frac{\prod_{j=1}^m (3j+1)}{9^m m! (m+2)!} x^{2m}; z = x^{1/3} \left(1 + \frac{2}{9}x^2\right); C = -\frac{p_2(7/3)}{36} a_2(1/3) =$$

$$2/81. \text{ By logarithmic differentiation, } a'_{2m}(r) = -9a_{2m}(r) \sum_{j=1}^m \frac{(12j^2 + 4j(3r-11) + 3r^2 - 22r + 47)}{(6j+3r-13)(6j+3r-11)(6j+3r-1)};$$

$$a'_{2m}(13/3) = -a_{2m}(13/3) \sum_{j=1}^m \frac{3j^2 + 2j + 2}{2j(j+2)(3j+1)};$$

$$\begin{aligned}
y_2 &= x^{1/3} \left( 1 + \frac{2}{9}x^2 \right) + \frac{2}{81} \left( y_1 \ln x - x^{13/3} \sum_{m=0}^{\infty} \frac{\prod_{j=1}^m (3j+1)}{9^m m! (m+2)!} \left( \sum_{j=1}^m \frac{3j^2+2j+2}{j(j+2)(3j+1)} \right) x^{2m} \right). \\
\mathbf{7.7.34.} \quad &p_0(r) = (r-2)(r+2); p_2(r) = -3(r-4); r_1 = 2; r_2 = -2; k = (r_1 - r_2)/2 = 2; a_{2m}(r) = \\
&\frac{3(2m+r-6)}{(2m+r-2)(2m+r+2)} a_{2m-2}(r); y_1 = x^2 \left( 1 - \frac{1}{2}x^2 \right); z = x^{-2} \left( 1 + \frac{9}{2}x^2 \right); C = -\frac{p_2(0)}{4} a_2(-2) = \\
&-27/2; a_2(r) = \frac{3(r-4)}{r(r+4)}, a'_2(r) = -\frac{3(r^2-8r-16)}{r^2(r+4)^2}, a'_2(2) = 7/12. \text{ If } m \geq 2, \text{ then } a_{2m}(r) = \\
&(r-2)c_{2m}(r) \text{ where } c_{2m}(r) = \frac{3^m(r-4)}{(2m+r-4)(2m+r-2)\prod_{j=1}^m (2j+r+2)}, \text{ so } a'_{2m}(2) = c_{2m}(2) = \\
&-\frac{\left(\frac{3}{2}\right)^m}{m(m-1)(m+2)!}; y_2 = x^{-2} \left( 1 + \frac{9}{2}x^2 \right) - \frac{27}{2} \left( y_1 \ln x + \frac{7}{12}x^4 - x^2 \sum_{m=2}^{\infty} \frac{\left(\frac{3}{2}\right)^m}{m(m-1)(m+2)!} x^{2m} \right). \\
\mathbf{7.7.36.} \quad &p_0(r) = (2r-5)(2r+7); p_2(r) = (2r-1)^2; r_1 = 5/2; r_2 = -7/2; k = (r_1 - r_2)/2 = 3; \\
&a_{2m}(r) = -\frac{4m+2r-5}{4m+2r+7} a_{2m-2}(r); a_{2m}(r) = \frac{(2r-1)(2r+3)(2r+7)}{(4m+2r-1)(4m+2r+3)(4m+2r+7)}; a_{2m}(5/2) = \\
&\frac{(-1)^m}{(m+1)(m+2)(m+3)}; y_1 = x^{5/2} \sum_{m=0}^{\infty} \frac{(-1)^m}{(m+1)(m+2)(m+3)} x^{2m}; z = x^{-7/2}(1+x^2)^2 C = -\frac{p_2(1/2)}{24} a_4(-7/2) = \\
&0; y_2 = x^{-7/2}(1+x^2)^2. \\
\mathbf{7.7.38.} \quad &p_0(r) = (r-3)(r+7); p_2(r) = r(r+1); r_1 = 3; r_2 = -7; k = (r_1 - r_2)/2 = \\
&5; a_{2m}(r) = -\frac{(2m+r-2)(2m+r-1)}{(2m+r-3)(2m+r+7)} a_{2m-2}(r); a_{2m}(r) = (-1)^m \prod_{j=1}^m \frac{(2j+r-2)(2j+r-1)}{(2j+r-3)(2j+r+7)}; \\
&a_{2m}(3) = (-1)^m \frac{m+1}{2^m} \left( \prod_{j=1}^m \frac{2j+1}{j+5} \right); y_1 = x^3 \sum_{m=0}^{\infty} (-1)^m \frac{m+1}{2^m} \left( \prod_{j=1}^m \frac{2j+1}{j+5} \right) x^{2m}; z = x^{-7} \left( 1 + \frac{21}{8}x^2 + \frac{35}{16}x^4 + \frac{35}{64}x^6 \right) \\
&C = -\frac{p_2(1)}{10} a_8(-7) = 0; y_2 = x^{-7} \left( 1 + \frac{21}{8}x^2 + \frac{35}{16}x^4 + \frac{35}{64}x^6 \right). \\
\mathbf{7.7.40.} \quad &p_0(r) = (2r-3)(2r+5); p_2(r) = (2r-1)(2r+1); r_1 = 3/2; r_2 = -5/2; k = \\
&(r_1 - r_2)/2 = 2; a_{2m}(r) = -\frac{4m+2r-5}{4m+2r+5} a_{2m-2}(r); a_{2m}(r) = (-1)^m \prod_{j=1}^m \frac{4j+2r-5}{4j+2r+5}; a_{2m}(3/2) = \\
&\frac{(-1)^m \prod_{j=1}^m (2j-1)}{2^{m-1}(m+2)!}; y_1 = x^{3/2} \sum_{m=0}^{\infty} \frac{(-1)^m \prod_{j=1}^m (2j-1)}{2^{m-1}(m+2)!} x^{2m}; z = x^{-5/2} \left( 1 + \frac{3}{2}x^2 \right) C = -\frac{p_2(-1/2)}{16} a_2(-5/2) = \\
&0; y_2 = x^{-5/2} \left( 1 + \frac{3}{2}x^2 \right). \\
\mathbf{7.7.42.} \quad &p_0(r) = r^2 - \nu^2; p_2(r) = 1; r_1 = \nu; r_2 = -\nu; k = (r_1 - r_2)/2 = \nu; a_{2m}(r) = \\
&-\frac{a_{2m-2}(r)}{(2m+r+\nu)(2m+r-\nu)}; a_{2m}(r) = \frac{(-1)^m}{\prod_{j=1}^m (2j+r+\nu)(2j+r-\nu)}; a_{2m}(\nu) = \frac{(-1)^m}{4^m m! \prod_{j=1}^m (j+\nu)}; \\
&a_{2m}(-\nu) = \frac{(-1)^m}{4^m m! \prod_{j=1}^m (j-\nu)}, j = 0, \dots, \nu-1; y_1 = x^\nu \sum_{m=0}^{\infty} \frac{(-1)^m}{4^m m! \prod_{j=1}^m (j+\nu)} x^{2m}; z = x^{-\nu} \sum_{m=0}^{\nu-1} \frac{(-1)^m}{4^m m! \prod_{j=1}^m (j-\nu)} x^{2m}; \\
&C = -\frac{p_2(\nu-2)}{2\nu} a_{2\nu-2}(-\nu) = -\frac{a_{2\nu-2}(-\nu)}{2\nu} = -\frac{2}{4^\nu \nu! (\nu-1)!}. \text{ By logarithmic differentiation,} \\
&a'_{2m}(r) = -2a_{2m}(r) \sum_{j=1}^m \frac{2j+\nu}{(2j+r+\nu)(2j+r-\nu)}; a'_{2m}(\nu) = -a_{2m}(\nu) \sum_{j=1}^m \frac{2j+\nu}{2j(j+\nu)}; y_2 = x^{-\nu} \sum_{m=0}^{\nu-1} \frac{(-1)^m}{4^m m! \prod_{j=1}^m (j-\nu)} x^{2m} -
\end{aligned}$$

**7.7.44.** Since  $a_n(r_2) = -\frac{p_1(n+r_2-1)}{p_0(n+r_2)}a_{n-1}(r_2)$ ,  $1 \leq n \leq k-1$ ,  $a_{k-1}(r_2) = (-1)^{k-1} \prod_{j=1}^{k-1} \frac{p_1(r_2+j-1)}{p_0(r_2+j)}$ .

But  $C = -\frac{p_1(r_1-1)}{k\alpha_0}a_{k-1}(r_2) = -\frac{p_1(r_2+k-1)}{k\alpha_0}a_{k-1}(r_2) = (-1)^k \frac{\prod_{j=1}^k p_1(r_2+j-1)}{k\alpha_0 \prod_{j=1}^{k-1} p_0(r_2+j)} = 0$  if and

only if  $\prod_{j=1}^k p_1(r_2+j-1) = 0$ .

**7.7.46.** Since  $p_1(r) = \gamma_1$ ,  $a_n(r) = -\frac{\gamma_1}{\alpha_0(n+r-r_1)(n+r-r_2)}a_{n-1}(r)$  and (A)  $a_n(r) = (-1)^n \left(\frac{\gamma_1}{\alpha_0}\right)^n \frac{1}{\prod_{j=1}^n (j+r-r_1)(j+r-r_2)}$ .

Therefore,  $a_n(r_1) = \frac{(-1)^n}{n!} \left(\frac{\gamma_1}{\alpha_0}\right)^n \frac{1}{\prod_{j=1}^n (j+k)}$  for  $n \geq 0$  (so  $Ly_1 = 0$ ) and  $a_n(r_2) = \frac{(-1)^n}{n!} \left(\frac{\gamma_1}{\alpha_0}\right)^n \frac{1}{\prod_{j=1}^n (j-k)}$

for  $n = 0, \dots, k-1$ .  $Ly_2 = 0$  if  $y_2 = x^{r_2} \sum_{n=0}^{k-1} a_n(r_2)x^n + C \left( y_1 \ln x + x^{r_1} \sum_{n=1}^{\infty} a'_n(r_1)x^n \right)$  if  $C =$

$-\frac{\gamma_1}{k\alpha_0}a_{k-1}(r_2) = -\frac{\gamma_1}{k\alpha_0} \frac{(-1)^{k-1}}{(k-1)!} \left(\frac{\gamma_1}{\alpha_0}\right)^{k-1} \frac{(-1)^{k-1}}{(k-1)!} = -\frac{1}{k!(k-1)!} \left(\frac{\gamma_1}{\alpha_0}\right)^k$ . From (A),  $\ln |a_n(r)| = -n \ln \left| \frac{\gamma_1}{\alpha_0} \right| - \sum_{j=1}^n ((\ln |j+r-r_1| + \ln |j+r-r_2|))$ , so  $a'_n(r) = -a_n(r) \sum_{j=1}^n \left( \frac{1}{j+r-r_1} + \frac{1}{j+r-r_2} \right)$

and  $a'_n(r_1) = -a_n(r_1) \sum_{j=1}^n \frac{2j+k}{j(j+k)}$ .

**7.7.48.** (a) From Exercise 7.6.66(a) of Section 7.6,  $L \left( \frac{\partial y}{\partial r}(x, r) \right) = p'_0(r)x^r + x^r p_0(r) \ln x$ . Setting  $r = r_1$  yields  $L \left( y_1 \ln x + x^{r_1} \sum_{n=1}^{\infty} a'_n(r_1)x^n \right) = p'_0(r_1)x^{r_1}$ . Since  $p'_0(r) = \alpha_0(2r - r_1 - r_2)$ ,  $p'_0(r_1) = k\alpha_0$ .

(b) From Exercise 7.5.57 of Section 7.5,  $L \left( x^{r_2} \sum_{n=0}^{\infty} a_n(r_2)x^n \right) = x^{r_2} \sum_{n=0}^{\infty} b_n x^n$ , where  $b_0 = p_0(r_2) = 0$  and  $b_n = \sum_{j=0}^n p_j(n+r_2-j)a_{n-j}(r_2)$  if  $n \geq 1$ . From the definition of  $\{a_n(r_2)\}$ ,  $b_n = 0$  if  $n \neq k$ , while  $b_k = \sum_{j=0}^k p_k(k+r_2-j)a_{k-j}(r_2) = \sum_{j=1}^k p_j(r_1-j)a_{k-j}(r_2)$ .

(d) Let  $\{\tilde{a}_n(r_2)\}$  be the coefficients that would be obtained if  $\tilde{a}_k(r_2) = 0$ . Then  $a_n(r_2) = \tilde{a}_n(r_2)$  if  $n = 0, \dots, k-1$ , and (A)  $a_n(r_2) - \tilde{a}_n(r_2) = -\frac{1}{p_0(n+r_2)} \sum_{j=0}^{n-k} p_j(n+r_2-j)(a_{n-j}(r_2) - \tilde{a}_{n-j}(r_2))$  if  $n > k$ . Now let  $c_m = a_{k+m}(r_2) - \tilde{a}_{k+m}(r_2)$ . Setting  $n = m+k$  in (A) and recalling the  $k+r_2 = r_1$  yields (B)  $c_m = -\frac{1}{p_0(m+r_1)} \sum_{j=0}^m p_j(m+r_1-j)c_{m-j}$ . Since  $c_k = a_k(r_2)$ , (B) implies that  $c_m = a_k(r_2)a_m(r_1)$  for all  $m \geq 0$ , which implies the conclusion.

## Chapter 8

# Laplace Transforms

### 8.1 Introduction to the Laplace Transform

$$8.1.2. \text{ (a) } \cosh t \sin t = \frac{1}{2} (e^t \sin t + e^{-t} \sin t) \leftrightarrow \frac{1}{2} \left[ \frac{1}{(s-1)^2 + 1} + \frac{1}{(s+1)^2 + 1} \right] = \frac{s^2 + 2}{[(s-1)^2 + 1][(s+1)^2 + 1]}.$$

$$\text{(b) } \sin^2 t = \frac{1 - \cos 2t}{2} \leftrightarrow \frac{1}{2} \left[ \frac{1}{s} - \frac{s}{s^2 + 4} \right] = \frac{2}{s(s^2 + 4)}.$$

$$\text{(c) } \cos^2 2t = \frac{1}{2} \left[ \frac{1}{s} + \frac{s}{s^2 + 16} \right] = \frac{s^2 + 8}{s(s^2 + 16)}.$$

$$\text{(d) } \cosh^2 t = \frac{(e^t + e^{-t})^2}{4} = \frac{(e^{2t} + 2 + e^{-2t})}{4} \leftrightarrow \frac{1}{4} \left( \frac{1}{s-2} + \frac{2}{s} + \frac{1}{s+2} \right) = \frac{s^2 - 2}{s(s^2 - 4)}.$$

$$\text{(e) } t \sinh 2t = \frac{te^{2t} - te^{-2t}}{2} \leftrightarrow \frac{1}{2} \left( \frac{1}{(s-2)^2} - \frac{1}{(s+2)^2} \right) = \frac{4s}{(s^2 - 4)^2}.$$

$$\text{(f) } \sin t \cos t = \frac{\sin 2t}{2} \leftrightarrow \frac{1}{s^2 + 4}.$$

$$\text{(g) } \sin(t + \pi/4) = \sin t \cos(\pi/4) + \cos t \sin(\pi/4) \leftrightarrow \frac{1}{\sqrt{2}} \frac{s+1}{s^2 + 1}.$$

$$\text{(h) } \cos 2t - \cos 3t \leftrightarrow \frac{s}{s^2 + 4} - \frac{s}{s^2 + 9} = \frac{5s}{(s^2 + 4)(s^2 + 9)}.$$

$$\text{(i) } \sin 2t + \cos 4t \leftrightarrow \frac{2}{s^2 + 4} + \frac{s}{s^2 + 16} = \frac{s^3 + 2s^2 + 4s + 32}{(s^2 + 4)(s^2 + 16)}.$$

8.1.6. If  $F(s) = \int_0^\infty e^{-st} f(t) dt$ , then  $F'(s) = \int_0^\infty (-te^{-st}) f(t) dt = - \int_0^\infty e^{-st} (tf(t)) dt$ . Applying this argument repeatedly yields the assertion.

8.1.8. Let  $f(t) = 1$  and  $F(s) = 1/s$ . From Exercise 8.1.6,  $t^n \leftrightarrow (-1)^n F^{(n)}(s) = n!/s^{n+1}$ .

8.1.10. If  $|f(t)| \leq Me^{s_0 t}$  for  $t \geq t_0$ , then  $|f(t)e^{-st}| \leq Me^{-(s-s_0)t}$  for  $t \geq t_0$ . Let  $g(t) = e^{-st} f(t)$ ,  $w(t) = Me^{-(s-s_0)t}$ , and  $\tau = t - t_0$ . Since  $\int_{t_0}^\infty w(t) dt$  converges if  $s > s_0$ ,  $F(s)$  is defined for  $s > s_0$ .

$$8.1.12. \int_0^T e^{-st} \left( \int_0^t f(\tau) d\tau \right) dt = -\frac{e^{-st}}{s} \int_0^t f(\tau) d\tau \Big|_0^T + \frac{1}{s} \int_0^T e^{-st} f(t) dt = -\frac{e^{-sT}}{s} \int_0^T f(\tau) d\tau +$$

$\frac{1}{s} \int_0^T e^{-st} f(t) dt$ . Since  $f$  is of exponential order  $s_0$ , the second integral on the right converges to

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$\frac{1}{s} L(f)$  as  $T \rightarrow \infty$  (Exercise 8.1.10). Now it suffices to show that (A)  $\lim_{T \rightarrow \infty} e^{-sT} \int_0^T f(\tau) d\tau = 0$  if  $s > s_0$ . Suppose that  $|f(t)| \leq M e^{s_0 t}$  if  $t \geq t_0$  and  $|f(t)| \leq K$  if  $0 \leq t \leq t_0$ , and let  $T > t_0$ . Then  $\left| \int_0^T f(\tau) d\tau \right| \leq \left| \int_0^{t_0} f(\tau) d\tau \right| + \left| \int_{t_0}^T f(\tau) d\tau \right| < K t_0 + M \int_{t_0}^T e^{s_0 \tau} d\tau < K t_0 + \frac{M e^{s_0 T}}{s_0}$ , which proves (A).

**8.1.14.** (a) If  $T > 0$ , then  $\int_0^T e^{-st} f(t) dt = \int_0^T e^{-(s-s_0)t} (e^{-s_0 t} f(t)) dt$ . Use integration by parts with  $u = e^{-(s-s_0)t}$ ,  $dv = e^{-s_0 t} f(t) dt$ ,  $du = -(s-s_0)e^{-(s-s_0)t}$ , and  $v = g(t)$  obtain  $\int_0^T e^{-st} f(t) dt = e^{-(s-s_0)t} g(t) \Big|_0^T + (s-s_0) \int_0^T e^{-(s-s_0)t} g(t) dt$ . Since  $g(0) = 0$  this reduces to  $\int_0^T e^{-st} f(t) dt = e^{-(s-s_0)T} g(T) + (s-s_0) \int_0^T e^{-(s-s_0)t} g(t) dt$ . Since  $|g(t)| \leq M$  for all  $t \geq 0$ , we can let  $t \rightarrow \infty$  to conclude that  $\int_0^\infty e^{-st} f(t) dt = (s-s_0) \int_0^\infty e^{-(s-s_0)t} g(t) dt$  if  $s > s_0$ .

(b) If  $F(s_0)$  exists, then  $g(t)$  is bounded on  $[0, \infty)$ . Now apply (a).

(c) Since  $f(t) = \frac{1}{2} \frac{d}{dt} \sin(e^{t^2})$ ,  $\left| \int_0^t f(\tau) d\tau \right| = \frac{|\sin(e^{t^2}) - \sin(1)|}{2} \leq 1$  for all  $t \geq 0$ . Now apply (a) with  $s_0 = 0$ .

**8.1.16.** (a)  $\Gamma(\alpha) = \int_0^\infty x^{\alpha-1} e^{-x} dx = \frac{x^\alpha e^{-x}}{\alpha} \Big|_0^\infty + \frac{1}{\alpha} \int_0^\infty x^\alpha e^{-x} dx = \frac{\Gamma(\alpha+1)}{\alpha}$ .

(b) Use induction.  $\Gamma(1) = \int_0^\infty e^{-x} dx = 1$ . If (A)  $\Gamma(n+1) = n!$ , then  $\Gamma(n+2) = (n+1)\Gamma(n+1)$  (from (a))  $= (n+1)n!$  (from (A))  $= (n+1)!$ .

(c)  $\Gamma(\alpha+1) = \int_0^\infty x^\alpha e^{-x} dx$ . Let  $x = st$ . Then  $\Gamma(\alpha+1) = \int_0^\infty (st)^\alpha e^{-st} s dt$ , so  $\int_0^\infty e^{-st} t^\alpha dt = \frac{\Gamma(\alpha+1)}{\alpha}$ .

**8.1.18.** (a)  $\int_0^2 e^{-st} f(t) dt = \int_0^1 e^{-st} t dt + \int_1^2 e^{-st} (2-t) dt = \left( \frac{1}{s^2} - \frac{e^{-s}(s+1)}{s^2} \right) + \left( \frac{e^{-s}(s-1)}{s^2} + \frac{e^{-2s}}{s^2} \right) = -\frac{2e^{-s}}{s^2} + \frac{e^{-2s}}{s^2} + \frac{1}{s^2} = \frac{(1-e^{-s})^2}{s^2}$ . Therefore,  $F(s) = \frac{(1-e^{-s})^2}{s^2(1-e^{-2s})} = \frac{1-e^{-s}}{s^2(1+e^{-s})} = \frac{1}{s^2} \tanh \frac{s}{2}$ .

(b)  $\int_0^1 e^{-st} f(t) dt = \int_0^{1/2} e^{-st} dt - \int_{1/2}^1 e^{-st} dt = \frac{1}{s} - \frac{e^{-s/2}}{s} + \frac{e^{-s}}{s} - \frac{e^{-s/2}}{s} = -\frac{2e^{-s/2}}{s} + \frac{e^{-s}}{s} + \frac{1}{s} = \frac{(1-e^{-s/2})^2}{s}$ . Therefore,  $F(s) = \frac{(1-e^{-s/2})^2}{s(1-e^{-s})} = \frac{1-e^{-s/2}}{s(1+e^{-s/2})} = \frac{1}{s} \tanh \frac{s}{4}$ .

(c)  $\int_0^\pi e^{-st} f(t) dt = \int_0^\pi e^{-st} \sin t dt = \frac{1+e^{-\pi s}}{(s^2+1)}$ . Therefore,  $F(s) = \frac{1+e^{-\pi s}}{(s^2+1)(1-e^{-\pi s})} \frac{1}{s^2+1} \coth \frac{\pi s}{2}$ .

$$(d) \int_0^{2\pi} e^{-st} f(t) dt = \int_0^{\pi} e^{-st} \sin t dt = \frac{1 + e^{-\pi s}}{(s^2 + 1)}. \text{ Therefore, } F(s) = \frac{1 + e^{-\pi s}}{(s^2 + 1)1 + e^{-2\pi s}} = \frac{1}{(s^2 + 1)(1 - e^{-\pi s})}.$$

## 8.2 The Inverse Laplace Transform

$$8.2.2. (a) \frac{2s+3}{(s-7)^4} = \frac{2(s-7)+17}{(s-7)^4} = \frac{2}{(s-7)^3} + \frac{17}{(s-7)^4} = \frac{2!}{(s-7)^3} + \frac{17}{6} \frac{3!}{(s-7)^4} \leftrightarrow e^{7t} \left( t^2 + \frac{17}{6} t^3 \right).$$

$$(b) \frac{s^2-1}{(s-2)^6} = \frac{[(s-2)+2]^2-1}{(s-2)^6} = \frac{(s-2)^2+4(s-2)+3}{(s-2)^6} = \frac{1}{(s-2)^4} + \frac{4}{(s-2)^5} + \frac{3}{(s-2)^6} = \frac{1}{6} \frac{3!}{(s-2)^4} + \frac{1}{6} \frac{4!}{(s-2)^5} + \frac{1}{40} \frac{5!}{(s-2)^6} \leftrightarrow \left( \frac{1}{6} t^3 + \frac{1}{6} t^4 + \frac{1}{40} t^5 \right) e^{2t}.$$

$$(c) \frac{s+5}{s^2+6s+18} = \frac{(s+3)}{(s+3)^2+9} + \frac{2}{3} \frac{3}{(s+3)^2+9} \leftrightarrow e^{-3t} \left( \cos 3t + \frac{2}{3} \sin 3t \right).$$

$$(d) \frac{2s+1}{s^2+9} = 2 \frac{s}{s^2+9} + \frac{1}{3} \frac{3}{s^2+9} \leftrightarrow 2 \cos 3t + \frac{1}{3} \sin 3t.$$

$$(e) \frac{s}{s^2+2s+1} = \frac{(s+1)-1}{(s+1)^2} = \frac{1}{s+1} - \frac{1}{(s+1)^2} \leftrightarrow (1-t)e^{-t}.$$

$$(f) \frac{s+1}{s^2-9} = \frac{s}{s^2-9} + \frac{1}{3} \frac{3}{s^2-9} \leftrightarrow \cosh 3t + \frac{1}{3} \sinh 3t.$$

$$(g) \text{ Expand the numerator in powers of } s+1: s^3+2s^2-s-3 = [(s+1)-1]^3+2[(s+1)-1]^2-[(s+1)-1]-3 = (s+1)^3-(s+1)^2-2(s+1)-1; \text{ therefore } \frac{s^3+2s^2-s-3}{(s+1)^4} =$$

$$\frac{1}{s+1} - \frac{1}{(s+1)^2} - \frac{2}{(s+1)^3} - \frac{1}{6} \frac{6}{(s+1)^4} \leftrightarrow \left( 1-t-t^2-\frac{1}{6}t^3 \right) e^{-t}.$$

$$(h) \frac{2s+3}{(s-1)^2+4} = 2 \frac{(s-1)}{(s-1)^2+4} + \frac{5}{2} \frac{2}{(s-1)^2+4} \leftrightarrow e^t \left( 2 \cos 2t + \frac{5}{2} \sin 2t \right).$$

$$(i) \frac{1}{s} - \frac{s}{s^2+1} \leftrightarrow 1 - \cos t.$$

$$(j) \frac{3s+4}{s^2-1} = \frac{3s}{s^2-1} + \frac{4}{s^2-1} \leftrightarrow 3 \cosh t + 4 \sinh t. \text{ Alternatively, } \frac{3s+4}{s^2-1} = \frac{3s+4}{(s-1)(s+1)} =$$

$$\frac{1}{2} \left[ \frac{7}{s-1} - \frac{1}{s+1} \right] \leftrightarrow \frac{7e^t - e^{-t}}{2}.$$

$$(k) \frac{3}{s-1} + \frac{4s+1}{s^2+9} = 3 \frac{1}{s-1} + 4 \frac{s}{s^2+9} + \frac{1}{3} \frac{3}{s^2+9} \leftrightarrow 3e^t + 4 \cos 3t + \frac{1}{3} \sin 3t.$$

$$(l) \frac{3}{(s+2)^2} - \frac{2s+6}{s^2+4} = 3 \frac{1}{(s+2)^2} - 2 \frac{s}{s^2+4} - 3 \frac{2}{s^2+4} \leftrightarrow 3te^{-2t} - 2 \cos 2t - 3 \sin 2t.$$

8.2.4. (a)

$$\frac{2+3s}{(s^2+1)(s+2)(s+1)} = \frac{A}{s+2} + \frac{B}{s+1} + \frac{Cs+D}{s^2+1},$$

where

$$A(s^2+1)(s+1) + B(s^2+1)(s+2) + (Cs+D)(s+2)(s+1) = 2+3s.$$

$$\begin{aligned} -5A &= -4 & (\text{set } s = -2); \\ 2B &= -1 & (\text{set } s = -1); \\ A+2B+2D &= 2 & (\text{set } s = 0); \\ A+B+C &= 0 & (\text{equate coefficients of } s^3). \end{aligned}$$

Solving this system yields  $A = \frac{4}{5}$ ,  $B = -\frac{1}{2}$ ,  $C = -\frac{3}{10}$ ,  $D = \frac{11}{10}$ . Therefore,

$$\begin{aligned}\frac{2+3s}{(s^2+1)(s+2)(s+1)} &= \frac{4}{5} \frac{1}{s+2} - \frac{1}{2} \frac{1}{s+1} - \frac{1}{10} \frac{3s-11}{s^2+1} \\ &\leftrightarrow \frac{4}{5} e^{-2t} - \frac{1}{2} e^{-t} - \frac{3}{10} \cos t + \frac{11}{10} \sin t.\end{aligned}$$

(b)

$$\frac{3s^2+2s+1}{(s^2+1)(s^2+2s+2)} = \frac{As+B}{s^2+1} + \frac{C(s+1)+D}{(s+1)^2+1},$$

where

$$(As+B)((s+1)^2+1) + (C(s+1)+D)(s^2+1) = 3s^2+2s+1.$$

$$\begin{aligned}2B+C+D &= 1 & (\text{set } s=0); \\ -A+B+2D &= 2 & (\text{set } s=-1); \\ 2B+C+D &= 1 & (\text{set } s=0); \\ A+C &= 0 & (\text{equate coefficients of } s^3).\end{aligned}$$

Solving this system yields  $A = 6/5$ ,  $B = 2/5$ ,  $C = -6/5$ ,  $D = 7/5$ . Therefore,

$$\begin{aligned}\frac{3s^2+2s+1}{(s^2+1)(s^2+2s+2)} &= \frac{1}{5} \left[ \frac{6s+2}{s^2+1} - \frac{6(s+1)-7}{(s+1)^2+1} \right] \\ &\leftrightarrow \frac{6}{5} \cos t + \frac{2}{5} \sin t - \frac{6}{5} e^{-t} \cos t + \frac{7}{5} e^{-t} \sin t.\end{aligned}$$

(c)  $s^2+2s+5 = (s+1)^2+4$ ;

$$\frac{3s+2}{(s-2)((s+1)^2+4)} = \frac{A}{s-2} + \frac{B(s+1)+C}{(s+1)^2+4},$$

where

$$A((s+1)^2+4) + (B(s+1)+C)(s-2) = 3s+2.$$

$$\begin{aligned}13A &= 8 & (\text{set } s=2); \\ 4A-3C &= -1 & (\text{set } s=-1); \\ A+B &= 0 & (\text{equate coefficients of } s^2).\end{aligned}$$

Solving this system yields  $A = \frac{8}{13}$ ,  $B = -\frac{8}{13}$ ,  $C = \frac{15}{13}$ . Therefore,

$$\begin{aligned}\frac{3s+2}{(s-2)((s+1)^2+4)} &= \frac{1}{13} \left[ \frac{8}{s-2} - \frac{8(s-1)-15}{(s+1)^2+4} \right] \\ &\leftrightarrow \frac{8}{13} e^{2t} - \frac{8}{13} e^{-t} \cos 2t + \frac{15}{26} e^{-t} \sin 2t.\end{aligned}$$

(d)

$$\frac{3s^2+2s+1}{(s-1)^2(s+2)(s+3)} = \frac{A}{s-1} + \frac{B}{(s-1)^2} + \frac{C}{s+2} + \frac{D}{s+3},$$



where

$$(A(s-1) + B)(s+2)(s+3) + (C(s+3) + D(s+2))(s-1)^2 = 3s^2 + 2s + 1.$$

$$\begin{aligned} 12B &= 6 & (\text{set } s = 1); \\ 9C &= 9 & (\text{set } s = -2); \\ -16D &= 22 & (\text{set } s = -3); \\ A + C + D &= 0 & (\text{equate coefficients of } s^3). \end{aligned}$$

Solving this system yields  $A = 3/8$ ,  $B = 1/2$ ,  $C = 1$ ,  $D = -11/8$ . Therefore,

$$\begin{aligned} \frac{3s^2 + 2s + 1}{(s-1)^2(s+2)(s+3)} &= \frac{3}{8} \frac{1}{s-1} + \frac{1}{2} \frac{1}{(s-1)^2} + \frac{1}{s+2} - \frac{11}{8} \frac{1}{s+3} \\ &\leftrightarrow \frac{3}{8}e^t + \frac{1}{2}te^t + e^{-2t} - \frac{11}{8}e^{-3t}. \end{aligned}$$

(e)

$$\frac{2s^2 + s + 3}{(s-1)^2(s+2)^2} = \frac{A}{s-1} + \frac{B}{(s-1)^2} + \frac{C}{s+2} + \frac{D}{(s+2)^2},$$

where

$$(A(s-1) + B)(s+2)^2 + (C(s+2) + D)(s-1)^2 = 2s^2 + s + 3.$$

$$\begin{aligned} 9B &= 6 & (\text{set } s = 1); \\ 9D &= 9 & (\text{set } s = -2); \\ -4A + 4B + 2C + D &= 3 & (\text{set } s = 0); \\ A + C &= 0 & (\text{equate coefficients of } s^3). \end{aligned}$$

Solving this system yields  $A = 1/9$ ,  $B = 2/3$ ,  $C = -1/9$ ,  $D = 1$ . Therefore,

$$\begin{aligned} \frac{2s^2 + s + 3}{(s-1)^2(s+2)^2} &= \frac{1}{9} \frac{1}{s-1} + \frac{2}{3} \frac{1}{(s-1)^2} - \frac{1}{9} \frac{1}{s+2} + \frac{1}{(s+2)^2} \\ &\leftrightarrow \frac{1}{9}e^t + \frac{2}{3}te^t - \frac{1}{9}e^{-2t} + te^{-2t}. \end{aligned}$$

(f)

$$\frac{3s+2}{(s^2+1)(s-1)^2} = \frac{A}{s-1} + \frac{B}{(s-1)^2} + \frac{Cs+D}{s^2+1},$$

where

$$A(s-1)(s^2+1) + B(s^2+1) + (Cs+D)(s-1)^2 = 3s+2. \quad (\text{A})$$

Setting  $s = 1$  yields  $2B = 5$ , so  $B = \frac{5}{2}$ . Substituting this into (A) shows that

$$\begin{aligned} A(s-1)(s^2+1) + (Cs+D)(s-1)^2 &= 3s+2 - \frac{5}{2}(s^2+1) \\ &= -\frac{5s^2-6s+1}{2} = -\frac{(s-1)(5s-1)}{2}. \end{aligned}$$

Therefore,

$$A(s^2+1) + (Cs+D)(s-1) = \frac{1-5s}{2}.$$

$$\begin{aligned} 2A &= -2 & (\text{set } s = 1); \\ A - D &= 1/2 & (\text{set } s = 0); \\ A + C &= 0 & (\text{equate coefficients of } s^2). \end{aligned}$$

Solving this system yields  $A = -1$ ,  $C = 1$ ,  $D = -\frac{3}{2}$ . Therefore,

$$\begin{aligned} \frac{3s+2}{(s^2+1)(s-1)^2} &= -\frac{1}{s-1} + \frac{5}{2} \frac{1}{(s-1)^2} + \frac{s-3/2}{s^2+1} \\ &\leftrightarrow -e^t + \frac{5}{2}te^t + \cos t - \frac{3}{2}\sin t. \end{aligned}$$

**8.2.6. (a)**

$$\frac{17s-15}{(s^2-2s+5)(s^2+2s+10)} = \frac{A(s-1)+B}{(s-1)^2+4} + \frac{C(s+1)+D}{(s+1)^2+9}$$

where

$$(A(s-1)+B)((s+1)^2+9) + (C(s+1)+D)((s-1)^2+4) = 17s-15.$$

$$\begin{aligned} 13B+8C+4D &= 2 & (\text{set } s = 1); \\ -18A+9B+8D &= -32 & (\text{set } s = -1); \\ -10A+10B+5C+5D &= -15 & (\text{set } s = 0); \\ A+C &= 0 & (\text{equate coefficients of } s^3). \end{aligned}$$

Solving this system yields  $A = 1$ ,  $B = 2$ ,  $C = -1$ ,  $D = -4$ . Therefore,

$$\begin{aligned} \frac{17s-15}{(s^2-2s+5)(s^2+2s+10)} &= \frac{(s-1)+2}{(s-1)^2+4} - \frac{(s+1)+4}{(s+1)^2+9} \\ &\leftrightarrow e^t(\cos 2t + \sin 2t) - e^{-t} \left( \cos 3t + \frac{4}{3}\sin 3t \right). \end{aligned}$$

(b)

$$\frac{8s+56}{(s^2-6s+13)(s^2+2s+5)} = \frac{A(s-3)+B}{(s-3)^2+4} + \frac{C(s+1)+D}{(s+1)^2+4}$$

where

$$(A(s-3)+B)((s+1)^2+4) + (C(s+1)+D)((s-3)^2+4) = 8s+56.$$

$$\begin{aligned} 20B+16C+4D &= 80 & (\text{set } s = 3); \\ -16A+4B+20D &= 48 & (\text{set } s = -1); \\ -15A+5B+13C+13D &= 56 & (\text{set } s = 0); \\ A+C &= 0 & (\text{equate coefficients of } s^3). \end{aligned}$$

Solving this system yields  $A = -1$ ,  $B = 3$ ,  $C = 1$ ,  $D = 1$ . Therefore,

$$\begin{aligned} \frac{8s+56}{(s^2-6s+13)(s^2+2s+5)} &= \frac{-(s-3)+3}{(s-3)^2+4} + \frac{(s+1)+1}{(s+1)^2+4} \\ &\leftrightarrow e^{3t} \left( -\cos 2t + \frac{3}{2}\sin 2t \right) + e^{-t} \left( \cos 2t + \frac{1}{2}\sin 2t \right). \end{aligned}$$

(c)

$$\frac{s+9}{(s^2+4s+5)(s^2-4s+13)} = \frac{A(s+2)+B}{(s+2)^2+1} + \frac{C(s-2)+D}{(s-2)^2+9}$$

where

$$\begin{aligned}
 (A(s+2) + B)((s-2)^2 + 9) + (C(s-2) + D)((s+2)^2 + 1) &= s + 9. \\
 25B - 4C + D &= 7 \quad (\text{set } s = -2); \\
 36A + 9B + 17D &= 11 \quad (\text{set } s = 2); \\
 26A + 13B - 10C + 5D &= 9 \quad (\text{set } s = 0); \\
 A + C &= 0 \quad (\text{equate coefficients of } s^3).
 \end{aligned}$$

Solving this system yields  $A = 1/8$ ,  $B = 1/4$ ,  $C = -1/8$ ,  $D = 1/4$ . Therefore,

$$\begin{aligned}
 \frac{s+9}{(s^2+4s+5)(s^2-4s+13)} &= \left[ \frac{1}{8} \frac{(s+2)+2}{(s+2)^2+1} - \frac{(s-2)-2}{(s-2)^2+3} \right] \\
 &\leftrightarrow e^{-2t} \left( \frac{1}{8} \cos t + \frac{1}{4} \sin t \right) - e^{2t} \left( \frac{1}{8} \cos 3t - \frac{1}{12} \sin 3t \right).
 \end{aligned}$$

(d)

$$\frac{3s-2}{(s^2-4s+5)(s^2-6s+13)} = \frac{A(s-2)+B}{(s-2)^2+1} + \frac{C(s-3)+D}{(s-3)^2+4}$$

where

$$\begin{aligned}
 (A(s-2) + B)((s-3)^2 + 4) + (C(s-3) + D)((s-2)^2 + 1) &= 3s - 2. \\
 5B - C + D &= 4 \quad (\text{set } s = 2); \\
 4A + 4B + 2D &= 7 \quad (\text{set } s = 3); \\
 -26A + 13B - 15C + 5D &= -2 \quad (\text{set } s = 0); \\
 A + C &= 0 \quad (\text{equate coefficients of } s^3).
 \end{aligned}$$

Solving this system yields  $A = 1$ ,  $B = 1/2$ ,  $C = -1$ ,  $D = 1/2$ . Therefore,

$$\begin{aligned}
 \frac{3s-2}{(s^2-4s+5)(s^2-6s+13)} &= \frac{1}{2} \left[ \frac{2(s-2)+1}{(s-2)^2+1} - \frac{2(s-3)-1}{(s-3)^2+4} \right] \\
 &\leftrightarrow e^{2t} \left( \cos t + \frac{1}{2} \sin t \right) - e^{3t} \left( \cos 2t - \frac{1}{4} \sin 2t \right).
 \end{aligned}$$

(e)

$$\frac{3s-1}{(s^2-2s+2)(s^2+2s+5)} = \frac{A(s-1)+B}{(s-1)^2+1} + \frac{C(s+1)+D}{(s+1)^2+4}$$

where

$$\begin{aligned}
 (A(s-1) + B)((s+1)^2 + 4) + (C(s+1) + D)((s-1)^2 + 1) &= 3s - 1. \\
 8B + 2C + D &= 2 \quad (\text{set } s = 1); \\
 -8A + 4B + 5D &= -4 \quad (\text{set } s = -1); \\
 -5A + 5B + 2C + 2D &= -1 \quad (\text{set } s = 0); \\
 A + 5B + C &= 0 \quad (\text{equate coefficients of } s^3).
 \end{aligned}$$

Solving this system yields  $A = 1/5$ ,  $B = 2/5$ ,  $C = -1/5$ ,  $D = -4/5$ . Therefore,

$$\begin{aligned}
 \frac{3s-1}{(s^2-2s+2)(s^2+2s+5)} &= \frac{1}{5} \left[ \frac{(s-1)+2}{(s-1)^2+1} - \frac{(s+1)+4}{(s+1)^2+4} \right] \\
 &\leftrightarrow e^t \left( \frac{1}{5} \cos t + \frac{2}{5} \sin t \right) - e^{-t} \left( \frac{1}{5} \cos 2t + \frac{2}{5} \sin 2t \right).
 \end{aligned}$$

(f)

$$\frac{20s + 40}{(4s^2 - 4s + 5)(4s^2 + 4s + 5)} = \left[ \frac{A(s - 1/2) + B}{(s - 1/2)^2 + 1} + \frac{C(s + 1/2) + D}{(s + 1/2)^2 + 1} \right]$$

where

$$(A(s - 1/2) + B)((s + 1/2)^2 + 1) + (C(s + 1/2) + D)((s - 1/2)^2 + 1) = \frac{5s + 10}{4}.$$

$$\begin{aligned} 2B + C + D &= 25/8 & (\text{set } s = 1/2); \\ -A + B + 2D &= 15/8 & (\text{set } s = -1/2); \\ -5A + 10B + 5C + 10D &= 20 & (\text{set } s = 0); \\ A + C &= 0 & (\text{equate coefficients of } s^3). \end{aligned}$$

Solving this system yields  $A = -1$ ,  $B = 9/8$ ,  $C = 1$ ,  $D = -1/8$ . Therefore,

$$\begin{aligned} \frac{20s + 40}{(4s^2 - 4s + 5)(4s^2 + 4s + 5)} &= \frac{1}{8} \left[ \frac{-8(s - 1/2) + 9}{(s - 1/2)^2 + 1} + \frac{8(s + 1/2) - 1}{(s + 1/2)^2 + 1} \right] \\ &\leftrightarrow e^{t/2} \left( -\cos t + \frac{9}{8} \sin t \right) + e^{-t/2} \left( \cos t - \frac{1}{8} \sin t \right). \end{aligned}$$

**8.2.8.** (a)

$$\frac{2s + 1}{(s^2 + 1)(s - 1)(s - 3)} = \frac{A}{s - 1} + \frac{B}{s - 3} + \frac{Cs + D}{s^2 + 1}$$

where

$$(A(s - 3) + B(s - 1))(s^2 + 1) + (Cs + D)(s - 1)(s - 3) = 2s + 1.$$

$$\begin{aligned} -4A &= 3 & (\text{set } s = 1); \\ 20B &= 7 & (\text{set } s = 3); \\ -3A - B + 3D &= 1 & (\text{set } s = 0); \\ A + B + C &= 0 & (\text{equate coefficients of } s^3). \end{aligned}$$

Solving this system yields  $A = -3/4$ ,  $B = 7/20$ ,  $C = 2/5$ ,  $D = -3/10$ . Therefore,

$$\begin{aligned} \frac{2s + 1}{(s^2 + 1)(s - 1)(s - 3)} &= -\frac{3}{4} \frac{1}{s - 1} + \frac{7}{20} \frac{1}{s - 3} + \frac{2}{5} \frac{s}{s^2 + 1} - \frac{3}{10} \frac{1}{s^2 + 1} \\ &\leftrightarrow -\frac{3}{4} e^t + \frac{7}{20} e^{3t} + \frac{2}{5} \cos t - \frac{3}{10} \sin t. \end{aligned}$$

(b)

$$\frac{s + 2}{(s^2 + 2s + 2)(s^2 - 1)} = \frac{A}{s - 1} + \frac{B}{s + 1} + \frac{C(s + 1) + D}{(s + 1)^2 + 1}$$

where

$$(A(s + 1) + B(s - 1))((s + 1)^2 + 1) + (C(s + 1) + D)(s^2 - 1) = s + 2.$$

$$\begin{aligned} 10A &= 3 & (\text{set } s = 1); \\ -2B &= 1 & (\text{set } s = -1); \\ 2A - 2B - C - D &= 2 & (\text{set } s = 0); \\ A + B + C &= 0 & (\text{equate coefficients of } s^3). \end{aligned}$$

Solving this system yields  $A = 3/10$ ,  $B = -1/2$ ,  $C = 1/5$ ,  $D = -3/5$ . Therefore,

$$\begin{aligned}\frac{s+2}{(s^2+2s+2)(s^2-1)} &= \frac{3}{10} \frac{1}{s-1} - \frac{1}{2} \frac{1}{s+1} + \frac{1}{5} \frac{s+1}{(s+1)^2+1} - \frac{3}{5} \frac{1}{(s+1)^2+1} \\ &\leftrightarrow \frac{3}{10} e^t - \frac{1}{2} e^{-t} + \frac{1}{5} e^{-t} \cos t e^{-t} \sin t.\end{aligned}$$

(c)

$$\frac{2s-1}{(s^2-2s+2)(s+1)(s-2)} = \frac{A}{s-2} + \frac{B}{s+1} + \frac{C(s-1)+D}{(s-1)^2+1}$$

where

$$(A(s+1) + B(s-2))((s-1)^2+1) + (C(s-1)+D)(s-2)(s+1) = 2s-1.$$

$$\begin{aligned}6A &= 3 & (\text{set } s=2); \\ -15B &= -3 & (\text{set } s=-1); \\ 2A-4B+2C-2D &= -1 & (\text{set } s=0); \\ A+B+C &= 0 & (\text{equate coefficients of } s^3).\end{aligned}$$

Solving this system yields  $A = 1/2$ ,  $B = 1/5$ ,  $C = -7/10$ ,  $D = -1/10$ . Therefore,

$$\begin{aligned}\frac{2s-1}{(s^2-2s+2)(s+1)(s-2)} &= \frac{1}{2} \frac{1}{s-2} + \frac{1}{5} \frac{1}{s+1} - \frac{7}{10} \frac{s-1}{(s-1)^2+1} - \frac{1}{10} \frac{1}{(s-1)^2+1} \\ &\leftrightarrow \frac{1}{2} e^{2t} + \frac{1}{5} e^{-t} - \frac{7}{10} e^t \cos t - \frac{1}{10} e^t \sin t.\end{aligned}$$

(d)

$$\frac{s-6}{(s^2-1)(s^2+4)} = \frac{A}{s-1} + \frac{B}{s+1} + \frac{Cs+D}{s^2+4}$$

where

$$(A(s+1) + B(s-1))(s^2+4) + (Cs+D)(s^2-1) = s-6.$$

$$\begin{aligned}10A &= -5 & (\text{set } s=1); \\ -10B &= -7 & (\text{set } s=-1); \\ 4A-4B-D &= -6 & (\text{set } s=0); \\ A+B+C &= 0 & (\text{equate coefficients of } s^3).\end{aligned}$$

Solving this system yields  $A = -1/2$ ,  $B = 7/10$ ,  $C = -1/5$ ,  $D = 6/5$ . Therefore,

$$\begin{aligned}\frac{s-6}{(s^2-1)(s^2+4)} &= -\frac{1}{2} \frac{1}{s-1} + \frac{7}{10} \frac{1}{s+1} - \frac{1}{5} \frac{s}{s^2+4} + \frac{3}{5} + \frac{1}{s^2+4} \\ &\leftrightarrow -\frac{1}{2} e^t + \frac{7}{10} e^{-t} - \frac{1}{5} \cos 2t + \frac{3}{5} \sin 2t.\end{aligned}$$

(e)

$$\frac{2s-3}{s(s-2)(s^2-2s+5)} = \frac{A}{s} + \frac{B}{s-2} + \frac{C(s-1)+D}{(s-1)^2+4}$$

where

$$(A(s-2) + Bs)((s-1)^2+4) + (C(s-1)+D)s(s-2) = 2s-3.$$

$$\begin{aligned}
-10A &= -3 & (\text{set } s = 0); \\
10B &= 1 & (\text{set } s = 2); \\
-4A + 4B - D &= -1 & (\text{set } s = 1); \\
A + B + C &= 0 & (\text{equate coefficients of } s^3).
\end{aligned}$$

Solving this system yields  $A = 3/10$ ,  $B = 1/10$ ,  $C = -2/5$ ,  $D = 1/5$ . Therefore,

$$\begin{aligned}
\frac{2s-3}{s(s-2)(s^2-2s+5)} &= \frac{3}{10s} + \frac{1}{10} \frac{1}{s-2} - \frac{2}{5} \frac{s-1}{(s-1)^2+4} + \frac{1}{5} \frac{1}{(s-1)^2+4} \\
&\leftrightarrow \frac{3}{10} + \frac{1}{10}e^{2t} - \frac{2}{5}e^t \cos 2t + \frac{1}{10}e^t \sin 2t.
\end{aligned}$$

(f)

$$\frac{5s-15}{(s^2-4s+13)(s-2)(s-1)} = \frac{A}{s-1} + \frac{B}{s-2} + \frac{C(s-2)+D}{(s-2)^2+9}$$

where

$$(A(s-2) + B(s-1))((s-2)^2+9) + (C(s-2)+D)(s-1)(s-2) = 5s-15.$$

$$\begin{aligned}
-10A &= -10 & (\text{set } s = 1); \\
9B &= -5 & (\text{set } s = 2); \\
-26A - 13B - 4C + 2D &= -15 & (\text{set } s = 0); \\
A + B + C &= 0 & (\text{equate coefficients of } s^3).
\end{aligned}$$

Solving this system yields  $A = 1$ ,  $B = -5/9$ ,  $C = -4/9$ ,  $D = 1$ . Therefore,

$$\begin{aligned}
\frac{5s-15}{(s^2-4s+13)(s-2)(s-1)} &= \frac{1}{s-1} - \frac{5}{9} \frac{1}{s-2} - \frac{4}{9} \frac{s-2}{(s-2)^2+9} + \frac{1}{(s-2)^2+9} \\
&\leftrightarrow e^t - \frac{5}{9}e^{2t} - \frac{4}{9}e^{2t} \cos 3t + \frac{1}{3}e^{2t} \sin 3t.
\end{aligned}$$

**8.2.10.** (a) Let  $i = 1$ . (The proof for  $i = 2, \dots, n$ ) is similar. Multiplying the given equation through by  $s - s_1$  yields

$$\frac{P(s)}{(s-s_2)\cdots(s-s_n)} = A_1 + (s-s_1) \left[ \frac{A_2}{s-s_2} + \cdots + \frac{A_n}{s-s_n} \right],$$

and setting  $s = s_1$  yields  $A_1 = \frac{P(s_1)}{(s_1-s_2)\cdots(s_1-s_n)}$ .

(b) From calculus we know that  $F$  has a partial fraction expansion of the form  $\frac{P(s)}{(s-s_1)Q_1(s)} = \frac{A}{s-s_1} + G(s)$  where  $G$  is continuous at  $s_1$ . Multiplying through by  $s - s_1$  shows that  $\frac{P(s)}{Q_1(s)} = A + (s-s_1)G(s)$ . Now set  $s = s_1$  to obtain  $A = \frac{P(s_1)}{Q(s_1)}$ .

(c) The result in (b) is generalization of the result in (a), since it shows that if  $s_1$  is a simple zero of the denominator of the rational function, then Heaviside's method can be used to determine the coefficient of  $1/(s-s_1)$  in the partial fraction expansion even if some of the other zeros of the denominator are repeated or complex.

### 8.3 Solution of Initial Value Problems

**8.3.2.**

$$(s^2 - s - 6)Y(s) = \frac{2}{s} + s - 1 = \frac{2 + s(s-1)}{s}.$$

Since  $(s^2 - s - 6) = (s-3)(s+2)$ ,

$$Y(s) = \frac{2 + s(s-1)}{s(s-3)(s+2)} = -\frac{1}{3s} + \frac{8}{15} \frac{1}{s-3} + \frac{4}{5} \frac{1}{s+2}$$

and  $y = -\frac{1}{3} + \frac{8}{15}e^{3t} + \frac{4}{5}e^{-2t}$ .

**8.3.4.**

$$(s^2 - 4)Y(s) = \frac{2}{s-3} + (-1 + s) = \frac{2 + (s-1)(s-3)}{s-3}.$$

Since  $s^2 - 4 = (s-2)(s+2)$ ,

$$Y(s) = \frac{2 + (s-1)(s-3)}{(s-2)(s+2)(s-3)} = -\frac{1}{4} \frac{1}{s-2} + \frac{17}{20} \frac{1}{s+2} + \frac{2}{5} \frac{1}{s-3}$$

and  $y = -\frac{1}{4}e^{2t} + \frac{17}{20}e^{-2t} + \frac{2}{5}e^{3t}$ .

**8.3.6.**

$$(s^2 + 3s + 2)Y(s) = \frac{6}{s-1} + (-1 + s) + 3 = \frac{6 + (s-1)(s+2)}{s-1}.$$

Since  $s^2 + 3s + 2 = (s+2)(s+1)$ ,

$$Y(s) = \frac{6 + (s-1)(s+2)}{(s-1)(s+2)(s+1)} = \frac{1}{s-1} + \frac{2}{s+2} - \frac{2}{s+1}$$

and  $y = e^t + 2e^{-2t} - 2e^{-t}$ .

**8.3.8.**

$$(s^2 - 3s + 2)Y(s) = \frac{2}{s-3} + (-1 + s) - 3 = \frac{2 + (s-3)(s-4)}{s-3}.$$

Since  $s^2 - 3s + 2 = (s-1)(s-2)$ ,

$$Y(s) = \frac{2 + (s-3)(s-4)}{(s-1)(s-2)(s-3)} = \frac{4}{s-1} - \frac{4}{s-2} + \frac{1}{s-3}$$

and  $y = 4e^t - 4e^{2t} + e^{3t}$ .

**8.3.10.**

$$(s^2 - 3s + 2)Y(s) = \frac{1}{s-3} + (-4 - s) + 3 = \frac{1 - (s-3)(s+1)}{s-3}.$$

Since  $s^2 - 3s + 2 = (s-1)(s-2)$ ,

$$Y(s) = \frac{1 - (s-3)(s+1)}{(s-1)(s-2)(s-3)} = \frac{5}{2} \frac{1}{s-1} - \frac{4}{s-2} + \frac{1}{2} \frac{1}{s-3}$$

and  $y = \frac{5}{2}e^t - 4e^{2t} + \frac{1}{2}e^{3t}$ .

**8.3.12.**

$$(s^2 + s - 2)Y(s) = -\frac{4}{s} + (3 + 2s) + 2 = \frac{-4 + s(5 + 2s)}{s}.$$

Since  $(s^2 + s - 2) = (s + 2)(s - 1)$ ,

$$Y(s) = \frac{-4 + s(5 + 2s)}{s(s + 2)(s - 1)} = \frac{2}{s} - \frac{1}{s + 2} + \frac{1}{s - 1},$$

and  $y = 2 - e^{-2t} + e^t$ .

**8.3.14.**

$$(s^2 - s - 6)Y(s) = \frac{2}{s} + s - 1 = \frac{2 + s(s - 1)}{s}.$$

Since  $s^2 - s - 6 = (s - 3)(s + 2)$ ,

$$Y(s) = \frac{2 + s(s - 1)}{s(s - 3)(s + 2)} = -\frac{1}{3s} + \frac{8}{15} \frac{1}{s - 3} + \frac{4}{5} \frac{1}{s + 2}$$

and  $y = -\frac{1}{3} + \frac{8}{15}e^{3t} + \frac{4}{5}e^{-2t}$ .

**8.3.16.**

$$(s^2 - 1)Y(s) = \frac{1}{s} + s = \frac{1 + s^2}{s}.$$

Since  $s^2 - 1 = (s - 1)(s + 1)$ ,

$$Y(s) = \frac{1 + s^2}{s(s - 1)(s + 1)} = -\frac{1}{s} + \frac{1}{s - 1} + \frac{1}{s + 1}$$

and  $y = -1 + e^t + e^{-t}$ .

**8.3.18.**

$$(s^2 + s)Y(s) = \frac{2}{s - 3} + (4 - s) - 1 = \frac{2 - (s - 3)^2}{s - 3}.$$

Since  $s^2 + s = s(s + 1)$ ,

$$Y(s) = \frac{2 - (s - 3)^2}{s(s + 1)(s - 3)} = \frac{7}{3s} - \frac{7}{2} \frac{1}{s + 1} + \frac{1}{6} \frac{1}{s - 3}$$

and  $y = \frac{7}{3} - \frac{7}{2}e^{-t} + \frac{1}{6}e^{3t}$ .

**8.3.20.**

$$(s^2 + 1)Y(s) = \frac{1}{s^2} + 2, \quad \text{so} \quad Y(s) = \frac{1}{(s^2 + 1)s^2} + \frac{2}{s^2 + 1}.$$

Substituting  $x = s^2$  into

$$\frac{1}{(x + 1)x} = \frac{1}{x + 1} - \frac{1}{x} \quad \text{yields} \quad \frac{1}{(s^2 + 1)s^2} = \frac{1}{s^2} - \frac{1}{s^2 + 1},$$

so  $Y(s) = \frac{1}{s^2} + \frac{1}{s^2 + 1}$  and  $y = t + \sin t$ .



**8.3.22.**

$$(s^2 + 5s + 6)Y(s) = \frac{2}{s+1} + (3+s) + 5 = \frac{2 + (s+1)(s+8)}{s+1}.$$

Since  $s^2 + 5s + 6 = (s+2)(s+3)$ ,

$$Y(s) = \frac{2 + (s+1)(s+8)}{(s+1)(s+2)(s+3)} = \frac{1}{s+1} + \frac{4}{s+2} - \frac{4}{s+3}$$

and  $y = e^{-t} + 4e^{-2t} - 4e^{-3t}$ .

**8.3.24.**

$$(s^2 - 2s - 3)Y(s) = \frac{10s}{s^2 + 1} + (7 + 2s) - 4 = \frac{10s}{s^2 + 1} + (2s + 3).$$

Since  $s^2 - 2s - 3 = (s-3)(s+1)$ ,

$$Y(s) = \frac{10s}{(s-3)(s+1)(s^2+1)} + \frac{2s+3}{(s-3)(s+1)}. \quad (\text{A})$$

$$\frac{2s+3}{(s-3)(s+1)} = \frac{9}{4} \frac{1}{s-3} - \frac{1}{4} \frac{1}{s+1} \leftrightarrow \frac{9}{4} e^{3t} - \frac{1}{4} e^{-t}. \quad (\text{B})$$

$$\frac{10s}{(s-3)(s+1)(s^2+1)} = \frac{A}{s-3} + \frac{B}{s+1} + \frac{Cs+D}{s^2+1}$$

where

$$(A(s+1) + B(s-3))(s^2+1) + (Cs+D)(s-3)(s+1) = 10s.$$

$$\begin{aligned} 40A &= 30 & (\text{set } s=3); \\ -8B &= -10 & (\text{set } s=-1); \\ A-3B-3D &= 0 & (\text{set } s=0); \\ A+B+C &= 0 & (\text{equate coefficients of } s^3). \end{aligned}$$

Solving this system yields  $A = 3/4$ ,  $B = 5/4$ ,  $C = -2$ ,  $D = -1$ . Therefore,

$$\begin{aligned} \frac{10s}{(s-3)(s+1)(s^2+1)} &= \frac{3}{4} \frac{1}{s-3} + \frac{5}{4} \frac{1}{s+1} - \frac{2s+1}{s^2+1} \\ &\leftrightarrow \frac{3}{4} e^{3t} + \frac{5}{4} e^{-t} - 2 \cos t - \sin t. \end{aligned}$$

From this, (A), and (B),  $y = -\sin t - 2 \cos t + 3e^{3t} + e^{-t}$ .

**8.3.26.**

$$(s^2 + 4)Y(s) = \frac{16}{s^2 + 4} + \frac{9s}{s^2 + 1} + s, \quad \text{so}$$

$$Y(s) = \frac{16}{(s^2 + 4)^2} + \frac{9s}{(s^2 + 4)(s^2 + 1)} + \frac{s}{s^2 + 4}.$$

From the table of Laplace transforms,

$$\begin{aligned} t \cos 2t &\leftrightarrow \frac{s^2 - 4}{(s^2 + 4)^2} = \frac{s^2 + 4}{(s^2 + 4)^2} - \frac{8}{(s^2 + 4)^2} \\ &= \frac{1}{s^2 + 4} - \frac{8}{(s^2 + 4)^2}. \end{aligned}$$

Therefore,

$$\frac{8}{(s^2 + 4)^2} = \frac{1}{s^2 + 4} - L(t \cos 2t), \text{ so } \frac{16}{(s^2 + 4)^2} \leftrightarrow \sin 2t - 2t \cos 2t. \quad (\text{A})$$

Substituting  $x = s^2$  into

$$\frac{9}{(x+4)(x+1)} = \frac{3}{x+1} - \frac{3}{x+4}$$

and multiplying by  $s$  yields

$$\frac{9s}{(s^2 + 4)(s^2 + 1)} = \frac{3s}{s^2 + 1} - \frac{3s}{s^2 + 4} \leftrightarrow 3 \cos t - 3 \cos 2t. \quad (\text{B})$$

Finally,

$$\frac{s}{s^2 + 4} \leftrightarrow \cos 2t. \quad (\text{C})$$

Adding (A), (B), and (C) yields  $y = -(2t + 2) \cos 2t + \sin 2t + 3 \cos t$ .

**28.**

$$(s^2 + 2s + 2)Y(s) = \frac{2}{s^2} + (-7 + 2s) + 4.$$

Since  $(s^2 + 2s + 2) = (s + 1)^2 + 1$ ,

$$Y(s) = \frac{2}{s^2((s+1)^2 + 1)} + \frac{2s-3}{(s+1)^2 + 1}. \quad (\text{A})$$

$$\frac{2s-3}{(s+1)^2 + 1} = \frac{2(s+1)-5}{(s+1)^2 + 1} \leftrightarrow e^{-t}(2 \cos t - 5 \sin t). \quad (\text{B})$$

$$\frac{2}{s^2((s+1)^2 + 1)} = \frac{A}{s} + \frac{B}{s^2} + \frac{C(s+1) + D}{(s+1)^2 + 1},$$

where  $(As + B)((s + 1)^2 + 1) + s^2(C(s + 1) + D) = 2$ .

$$\begin{aligned} 2B &= 2 && (\text{set } s = 0); \\ -A + B + D &= 2 && (\text{set } s = -1); \\ A + C &= 0 && (\text{equate coefficients of } s^3); \\ 2A + B + C + D &= 0 && (\text{equate coefficients of } s^2). \end{aligned}$$

Solving this system yields  $A = -1$ ,  $B = 1$ ,  $C = 1$ ,  $D = 0$ . Therefore,

$$\frac{2}{s^2((s+1)^2 + 1)} = -\frac{1}{s} + \frac{1}{s^2} + \frac{(s+1)}{(s+1)^2 + 1} \leftrightarrow -1 + t + e^{-t} \cos t.$$

From this, (A), and (B),  $y = -1 + t + e^{-t}(\cos t - 5 \sin t)$ .

**8.3.30.**  $(s^2 + 4s + 5)Y(s) = \frac{(s+1)+3}{(s+1)^2 + 1} + 4$ . Since  $(s^2 + 4s + 5) = (s + 2)^2 + 1$ ,

$$Y(s) = \frac{s+4}{((s+1)^2 + 1)((s+2)^2 + 1)} + \frac{4}{(s+2)^2 + 1}. \quad (\text{A})$$

$$\frac{4}{(s+2)^2 + 1} \leftrightarrow 4e^{-2t} \sin t. \quad (\text{B})$$

$$\frac{s+4}{((s+1)^2 + 1)((s+2)^2 + 1)} = \frac{A(s+1) + B}{(s+1)^2 + 1} + \frac{C(s+2) + D}{(s+2)^2 + 1},$$

where  $(A(s+1) + B)((s+2)^2 + 1) + (C(s+2) + D)((s+1)^2 + 1) = 4 + s$ .

$$\begin{aligned} 5A + 5B + 4C + 2D &= 4 & (\text{set } s = 0); \\ 2B + C + D &= 3 & (\text{set } s = -1); \\ -A + B + 2D &= 2 & (\text{set } s = -2); \\ A + C &= 0 & (\text{equate coefficients of } s^3). \end{aligned}$$

Solving this system yields  $A = -1$ ,  $B = 1$ ,  $C = 1$ ,  $D = 0$ . Therefore,

$$\frac{s+4}{((s+1)^2+1)((s+2)^2+1)} = \frac{-(s+1)+1}{(s+1)^2+1} + \frac{s+2}{(s+2)^2+1}, \leftrightarrow e^{-t}(-\cos t + \sin t) + e^{-2t} \cos t.$$

From this, (A), and (B),  $y = e^t(-\cos t + \sin t) + e^{-2t}(\cos t + 4 \sin t)$ .

**8.3.32.**

$$(2s^2 - 3s - 2)Y(s) = \frac{4}{s-1} + 2(-2+s) - 3 = \frac{4 + (2s-7)(s-1)}{s-1}$$

Since  $2s^2 - 3s - 2 = (s-2)(2s+1)$ ,

$$Y(s) = \frac{4 + (2s-7)(s-1)}{2(s-2)(s-1)(s+1/2)} = \frac{1}{5} \frac{1}{s-2} - \frac{4}{3} \frac{1}{s-1} + \frac{32}{15} \frac{1}{s+1/2}$$

and  $y = \frac{1}{5}e^{2t} - \frac{4}{3}e^t + \frac{32}{15}e^{-t/2}$ .

**8.3.34.**

$$(2s^2 + 2s + 1)Y(s) = \frac{2}{s^2} + 2(-1+s) + 2 = \frac{2}{s^2} + 2s.$$

Since  $2s^2 + 2s + 1 = 2((s+1/2)^2 + 1/4)$ ,

$$Y(s) = \frac{1}{s^2((s+1/2)^2 + 1/4)} + \frac{s}{((s+1/2)^2 + 1/4)}. \quad (\text{A})$$

$$\frac{s}{((s+1/2)^2 + 1/4)} \leftrightarrow e^{-t/2}(\cos(t/2) - \sin(t/2)). \quad (\text{B})$$

$$\frac{1}{s^2((s+1/2)^2 + 1/4)} = \frac{A}{s} + \frac{B}{s^2} + \frac{C(s+1/2) + D}{((s+1/2)^2 + 1/4)}$$

where

$$(As + B)((s+1/2)^2 + 1/4) + (C(s+1/2) + D)s^2 = 1.$$

$$\begin{aligned} B &= 2 & (\text{set } s = 0); \\ -A + 2B + 2D &= 8 & (\text{set } s = -1/2); \\ 5A + 10B + 2C + 2D &= 8 & (\text{set } s = 1/2); \\ A + C &= 0 & (\text{equate coefficients of } s^3). \end{aligned}$$

Solving this system yields  $A = -4$ ,  $B = 2$ ,  $C = 4$ ,  $D = 0$ . Therefore,

$$\begin{aligned} \frac{1}{s^2((s+1/2)^2 + 1/4)} &= -\frac{4}{s} + \frac{2}{s^2} + \frac{4(s+1/2)}{(s+1/2)^2 + 1/4} \\ &\leftrightarrow -4 + 2t + 4e^{-t/2} \cos(t/2). \end{aligned}$$

This, (A), and (B) imply that  $y = e^{-t/2}(5 \cos(t/2) - \sin(t/2)) + 2t - 4$ .

**8.3.36.**

$$(4s^2 + 4s + 1)Y(s) = \frac{3+s}{s^2+1} + 4(-1+2s) + 8 = \frac{3+s}{s^2+1} + 4(-1+2s) + 8s + 4.$$

Since  $4s^2 + 4s + 1 = 4(s + 1/2)^2$ ,

$$\begin{aligned} Y(s) &= \frac{3+s}{4(s+1/2)^2(s^2+1)} + \frac{2}{s+1/2}. \\ \frac{3+s}{4(s+1/2)^2(s^2+1)} &= \frac{A}{s+1/2} + \frac{B}{(s+1/2)^2} + \frac{Cs+D}{s^2+1} \end{aligned} \quad (\text{A})$$

where

$$\begin{aligned} (A(s+1/2) + B)(s^2+1) + (Cs+D)(s+1/2)^2 &= \frac{3+s}{4}. \\ 10B &= 5 \quad (\text{set } s = -1/2); \\ 2A + 4B + D &= 3 \quad (\text{set } s = 0); \\ 12A + 8B + 9C + 9D &= 4 \quad (\text{set } s = 1); \\ A + C &= 0 \quad (\text{equate coefficients of } s^3). \end{aligned}$$

Solving this system yields  $A = 3/5$ ,  $B = 1/2$ ,  $C = -3/5$ ,  $D = -1/5$ . Therefore,

$$\begin{aligned} \frac{3+s}{4(s+1/2)^2(s^2+1)} &= \frac{3}{5} \frac{1}{s+1/2} + \frac{1}{2} \frac{1}{(s+1/2)^2} - \frac{1}{5} \frac{3s+1}{s^2+1}. \\ \Leftrightarrow &\frac{3}{5} e^{-t/2} + \frac{1}{2} t e^{-t/2} - \frac{1}{5} (3 \cos t + \sin t). \end{aligned}$$

Since  $\frac{2}{s+1/2} \Leftrightarrow 2e^{-t/2}$ , this and (A) imply that  $y = \frac{e^{-t/2}}{10}(5t + 26) - \frac{1}{5}(3 \cos t + \sin t)$ .

**8.3.38.** Transforming the initial value problem

$$ay'' + by' + cy = 0, \quad y(0) = 1, \quad y'(0) = 0$$

yields  $(as^2 + bs + c)Y(s) = as + b$ , so  $Y(s) = \frac{as+b}{as^2+bs+c}$ . Therefore,  $y_1 = L^{-1}\left(\frac{as+b}{as^2+bs+c}\right)$  satisfies the initial conditions  $y_1(0) = 1$ ,  $y_1'(0) = 0$ .

Transforming the initial value problem

$$ay'' + by' + cy = 0, \quad y(0) = 0, \quad y'(0) = 1$$

yields  $(as^2 + bs + c)Y(s) = a$ , so  $Y(s) = \frac{a}{as^2+bs+c}$ . Therefore,  $y_2 = L^{-1}\left(\frac{a}{as^2+bs+c}\right)$  satisfies the initial conditions  $y_2(0) = 0$ ,  $y_2'(0) = 1$ .

## 8.4 The Unit Step Function

**8.4.2.**

$$L(f) = \int_0^\infty e^{-st} f(t) dt = \int_0^1 e^{-st} t dt + \int_1^\infty e^{-st} dt. \quad (\text{A})$$

To relate the first term to a Laplace transform we add and subtract  $\int_1^\infty e^{-st} t dt$  in (A) to obtain

$$L(f) = \int_0^\infty e^{-st} t dt + \int_1^\infty e^{-st} (1-t) dt = L(t) - \int_1^\infty e^{-st} (t-1) dt. \quad (\text{B})$$

Letting  $t = x + 1$  in the last integral yields

$$\int_1^\infty e^{-st} (t-1) dt = - \int_0^\infty e^{-s(x+1)} x dx = e^{-s} L(t).$$

This and (B) imply that  $L(f) = (1 - e^{-s})L(t) = \frac{1 - e^{-s}}{s^2}$ .

Alternatively,  $f(t) = t - u(t-1)(t-1) \leftrightarrow (1 - e^{-s})L(t) = \frac{1 - e^{-s}}{s^2}$ .

**8.4.4.**

$$L(f) = \int_0^\infty e^{-st} f(t) dt = \int_0^1 e^{-st} dt + \int_1^\infty e^{-st} (t+2) dt. \quad (\text{A})$$

To relate the first term to a Laplace transform we add and subtract  $\int_1^\infty e^{-st} dt$  in (A) to obtain

$$L(f) = \int_0^\infty e^{-st} dt + \int_1^\infty e^{-st} (t+1) dt = L(t) + \int_1^\infty e^{-st} (t+1) dt. \quad (\text{B})$$

Letting  $t = x + 1$  in the last integral yields

$$\int_1^\infty e^{-st} (t+1) dt = \int_0^\infty e^{-s(x+1)} (x+2) dx = e^{-s} L(t+2).$$

This and (B) imply that  $L(f) = L(1) + e^{-s} L(t+2) = \frac{1}{s} + e^{-s} \left( \frac{1}{s^2} + \frac{2}{s} \right)$ .

Alternatively,

$$f(t) = 1 + u(t-1)(t+1) \leftrightarrow L(1) + e^{-s} L(t+2) = \frac{1}{s} + e^{-s} \left( \frac{1}{s^2} + \frac{2}{s} \right).$$

**8.4.6.**

$$L(f) = \int_0^\infty e^{-st} f(t) dt = \int_0^1 e^{-st} t^2 dt = L(t^2) - \int_1^\infty t^2 dt. \quad (\text{A})$$

Letting  $t = x + 1$  in the last integral yields

$$\int_1^\infty e^{-st} t^2 dt = \int_0^\infty e^{-s(x+1)} (t^2 + 2t + 1) dx = e^{-s} L(t^2 + 2t + 1).$$

This and (A) imply that

$$L(f) = L(t^2) + e^{-s} L(t^2 + 2t + 1) = \frac{2}{s^3} - e^{-s} \left( \frac{2}{s^3} + \frac{2}{s^2} + \frac{1}{s} \right).$$

Alternatively,

$$f(t) = t^2 (1 - u(t-1)) \leftrightarrow L(t^2) + e^{-s} L(t^2 + 2t + 1) = \frac{2}{s^3} - e^{-s} \left( \frac{2}{s^3} + \frac{2}{s^2} + \frac{1}{s} \right)$$

**8.4.8.**  $f(t) = t^2 + 2 + u(t-1)(t-t^2-2)$ . Since  $t^2 + 2 \leftrightarrow \frac{2}{s^3} + \frac{2}{s}$  and

$$\begin{aligned} L(u(t-1)(t-t^2-2)) &= e^{-s}L((t+1)-(t+1)^2-2) \\ &= -e^{-s}L(t^2+t+2) = -e^{-s}\left(\frac{2}{s^3} + \frac{1}{s^2} + \frac{2}{s}\right), \end{aligned}$$

it follows that  $F(s) = \frac{2}{s^3} + \frac{2}{s} - e^{-s}\left(\frac{2}{s^3} + \frac{1}{s^2} + \frac{2}{s}\right)$ .

**8.4.10.**  $f(t) = e^{-t} + u(t-1)(e^{-2t} - e^{-t}) \leftrightarrow L(e^{-t}) + e^{-s}L(e^{-2(t+1)}) - e^{-s}L(e^{-t-1}) = L(e^{-t}) + e^{-(s+2)}L(e^{-2t}) - e^{-(s+1)}L(e^{-t}) = \frac{1-e^{-(s+1)}}{s+1} + \frac{e^{-(s+2)}}{s+2}$ .

**8.4.12.**  $f(t) = [u(t-1) - u(t-2)]t \leftrightarrow e^{-s}L(t+1) - e^{-2s}L(t+2) = e^{-s}\left(\frac{1}{s^2} + \frac{1}{s}\right) - e^{-2s}\left(\frac{1}{s^2} + \frac{2}{s}\right)$ .

**8.4.14.**

$$\begin{aligned} f(t) &= t - 2u(t-1)(t-1) + u(t-2)(t+4) \leftrightarrow \frac{1}{s^2} - 2e^{-s}L(t) + e^{-2s}L(t+6) \\ &= \frac{1}{s^2} - \frac{2e^{-s}}{s^2} + e^{-2s}\left(\frac{1}{s^2} + \frac{6}{s}\right). \end{aligned}$$

**8.4.16.**  $f(t) = 2 - 2u(t-1)t + u(t-3)(5t-2) \leftrightarrow L(2) - 2e^{-s}L(t+1) + e^{-3s}L(5t+13) = \frac{2}{s} - e^{-s}\left(\frac{2}{s^2} + \frac{2}{s}\right) + e^{-3s}\left(\frac{5}{s^2} + \frac{13}{s}\right)$ .

**8.4.18.**  $f(t) = (t+1)^2 + u(t-1)((t+2)^2 - (t+1)^2) = t^2 + 2t + 1 + u(t-1)(2t+3) \leftrightarrow L(t^2 + 2t + 1) + e^{-s}L(2t+5) = \frac{2}{s^3} + \frac{2}{s^2} + \frac{1}{s} + e^{-s}\left(\frac{2}{s^2} + \frac{5}{s}\right)$ .

**8.4.20.**  $\frac{1}{s(s+1)} = \frac{1}{s} - \frac{1}{s+1} \leftrightarrow 1 - e^{-t} \Rightarrow e^{-s}\frac{1}{s(s+1)} \leftrightarrow u(t-1)(1 - e^{-(t-1)}) = \begin{cases} 0, & 0 \leq t < 1, \\ 1 - e^{-(t-1)}, & t \geq 1. \end{cases}$

**8.4.22.**

$$\frac{3}{s} - \frac{1}{s^2} \leftrightarrow 3 - t \Rightarrow e^{-s}\left(\frac{3}{s} - \frac{1}{s^2}\right) \leftrightarrow u(t-1)(3 - (t-1)) = u(t-1)(4-t);$$

$$\frac{1}{s} + \frac{1}{s^2} \leftrightarrow 1 + t \Rightarrow e^{-3s}\left(\frac{1}{s} + \frac{1}{s^2}\right) \leftrightarrow u(t-3)(1 + (t-3)) = u(t-3)(t-2);$$

therefore

$$h(t) = 2 + t + u(t-1)(4-t) + u(t-3)(t-2) = \begin{cases} 2+t, & 0 \leq t < 1, \\ 6, & 1 \leq t < 3, \\ t+4, & t \geq 3. \end{cases}$$

**8.4.24.**

$$\frac{1-2s}{s^2+4s+5} = \frac{5-2(s+2)}{(s+2)^2+1} \leftrightarrow e^{-2t}(5\sin t - 2\cos t);$$

therefore,

$$\begin{aligned} h(t) &= u(t-\pi)e^{-2(t-\pi)}(5\sin(t-\pi)-2\cos(t-\pi)) \\ &= u(t-\pi)e^{-2(t-\pi)}(2\cos t-5\sin t) \\ &= \begin{cases} 0, & 0 \leq t < \pi, \\ e^{-2(t-\pi)}(2\cos t-5\sin t), & t \geq \pi. \end{cases} \end{aligned}$$

**8.4.26.** Denote  $F(s) = \frac{3(s-3)}{(s+1)(s-2)} - \frac{s+1}{(s-1)(s-2)}$ . Since  $\frac{3(s-3)}{(s+1)(s-2)} = \frac{4}{s+1} - \frac{1}{s-2}$  and  $\frac{s+1}{(s-1)(s-2)} = \frac{3}{s-2} - \frac{2}{s-1}$ ,  $F(s) = \frac{4}{s+1} - \frac{4}{s-2} + \frac{2}{s-1} \leftrightarrow 4e^{-t} - 4e^{2t} + 2e^t$ . Therefore,  $e^{-2s}F(s) \leftrightarrow u(t-2)(4e^{-(t-2)} - 4e^{2(t-2)} + 2e^{(t-2)}) = \begin{cases} 0, & 0 \leq t < 2, \\ 4e^{-(t-2)} - 4e^{2(t-2)} + 2e^{(t-2)}, & t \geq 2. \end{cases}$

**8.4.28.**

$$\frac{3}{s} - \frac{1}{s^3} \leftrightarrow 3 - \frac{t^2}{2} \Rightarrow e^{-2s} \left( \frac{3}{s} - \frac{1}{s^3} \right) \leftrightarrow u(t-2) \left( 3 - \frac{(t-2)^2}{2} \right) = u(t-2) \left( -\frac{t^2}{2} + 2t + 1 \right);$$

$$\frac{1}{s^2} \leftrightarrow t \Rightarrow \frac{e^{-4s}}{s^2} \leftrightarrow u(t-4)(t-4);$$

therefore

$$\begin{aligned} h(t) &= 1 - t^2 + u(t-2) \left( -\frac{t^2}{2} + 2t + 1 \right) + u(t-4)(t-4) \\ &= \begin{cases} 1 - t^2, & 0 \leq t < 2 \\ -\frac{3t^2}{2} + 2t + 2, & 2 \leq t < 4, \\ -\frac{3t^2}{2} + 3t - 2, & t \geq 4. \end{cases} \end{aligned}$$

**8.4.30.** Let  $T$  be an arbitrary positive number. Since  $\lim_{m \rightarrow \infty} t_m = \infty$ , only finitely many members of  $\{t_m\}$  are in  $[0, T]$ . Since  $f_m$  is continuous on  $[t_m, \infty)$  for each  $m$ ,  $f$  is piecewise continuous on  $[0, T]$ . If  $t_M \leq t < t_{M+1}$ , then  $u(t-t_m) = 1$  if  $m \leq M$ , while  $u(t-t_m) = 0$  if  $m > M$ . Therefore,

$$f(t) = f_0(t) + \sum_{m=1}^M (f_m(t) - f_{m-1}(t)) = f_M(t)$$

**8.4.32.** Since  $\sum_{m=0}^{\infty} e^{-\rho K m}$  converges if  $\rho > 0$ ,  $\sum_{m=0}^{\infty} e^{-\rho t_m}$  converges if  $\rho > 0$ , by the comparison test. Therefore, (C) of Exercise 8.3.31 holds if  $s > s_0 + \rho$  if  $\rho$  is any positive number. This implies that it holds if  $s > s_0$ .

**8.4.34.** Let  $t_m = m$  and  $f_m(t) = (-1)^m$ ,  $m = 0, 1, 2, \dots$ . Then  $f_m(t) - f_{m-1}(t) = (-1)^m$ , so  $f(t) = 1 + 2 \sum_{m=1}^{\infty} (-1)^m u(t-m)$  and  $F(s) = \frac{1}{s} \left( 1 + 2 \sum_{m=1}^{\infty} (-1)^m e^{-ms} \right)$ . Substituting  $x = e^{-s}$  in the identity  $\sum_{m=1}^{\infty} (-1)^m x^m = -\frac{x}{1+x}$  ( $|x| < 1$ ) yields  $F(s) = \frac{1}{s} \left( 1 - \frac{2e^{-s}}{1+e^{-s}} \right) = \frac{1}{s} \frac{1-e^{-s}}{1+e^{-s}}$ .

**8.4.36.** Let  $t_m = m$  and  $f_m(t) = (-1)^m m$ ,  $m = 0, 1, 2, \dots$ . Then  $f_m(t) - f_{m-1}(t) = (-1)^m (2m-1)$ , so  $f(t) = \sum_{m=1}^{\infty} (-1)^m (2m-1) u(t-m)$  and  $F(s) = \frac{1}{s} \sum_{m=1}^{\infty} (-1)^m (2m-1) e^{-ms}$ . Substituting  $x = e^{-s}$  in the identities  $\sum_{m=1}^{\infty} (-1)^m x^m = -\frac{x}{1+x}$  and  $\sum_{m=1}^{\infty} (-1)^m m x^m = -\frac{x}{(1+x)^2}$  ( $|x| < 1$ ) yields  $F(s) = \frac{1}{s} \left[ \frac{e^{-s}}{1+e^{-s}} - \frac{2e^{-s}}{(1+e^{-s})^2} \right] = \frac{1}{s} \frac{(1-e^s)}{(1+e^s)^2}$ .

## 8.5 Constant Coefficient Equations with Piecewise Continuous Forcing Functions

**8.5.2.**  $y'' + y = 3 + u(t-4)(2t-8)$ ,  $y(0) = 1$ ,  $y'(0) = 0$ . Since

$$L(u(t-4)(2t-8)) = e^{-4s} L(2(t+4)-8) = e^{-4s} L(2t) = \frac{2e^{-4s}}{s^2},$$

$$(s^2 + 1)Y(s) = \frac{3}{s} + \frac{2e^{-4s}}{s^2} + s.$$

$$\begin{aligned} Y(s) &= \frac{3}{s(s^2 + 1)} + \frac{2e^{-4s}}{s^2(s^2 + 1)} + \frac{s}{s^2 + 1} \\ &= 3 \left( \frac{1}{s} - \frac{s}{s^2 + 1} \right) + 2e^{-4s} \left( \frac{1}{s^2} - \frac{1}{s^2 + 1} \right) + \frac{s}{s^2 + 1} \\ &= \frac{3}{s} - \frac{2s}{s^2 + 1} + 2e^{-4s} \left( \frac{1}{s^2} - \frac{1}{s^2 + 1} \right). \end{aligned}$$

Since

$$\frac{1}{s^2} - \frac{1}{s^2 + 1} \leftrightarrow t - \sin t \Rightarrow e^{-4s} \left( \frac{1}{s^2} - \frac{1}{s^2 + 1} \right) \leftrightarrow u(t-4)(t-4 - \sin(t-4)),$$

$$y = 3 - 2 \cos t + 2u(t-4)(t-4 - \sin(t-4)).$$

**8.5.4.**  $y'' - y = e^{2t} + u(t-2)(1 - e^{2t})$ ,  $y(0) = 3$ ,  $y'(0) = -1$ . Since

$$L(u(t-2)(1 - e^{2t})) = e^{-2s} L(1 - e^{2(t+2)}) = e^{-2s} \left( \frac{1}{s} - \frac{e^4}{s-2} \right),$$

$$(s^2 - 1)Y(s) = \frac{1}{s-2} + e^{-2s} \left( \frac{1}{s} - \frac{e^4}{s-2} \right) + (-1 + 3s).$$

Therefore,

$$\begin{aligned} Y(s) &= \frac{1}{(s-1)(s+1)(s-2)} + \frac{3s-1}{(s-1)(s+1)} \\ &\quad + e^{-2s} \left( \frac{1}{s(s-1)(s+1)} - \frac{e^4}{(s-1)(s+1)(s-2)} \right). \\ \frac{1}{(s-1)(s+1)(s-2)} &= -\frac{1}{2} \frac{1}{s-1} + \frac{1}{6} \frac{1}{s+1} + \frac{1}{3} \frac{1}{s-2} \\ &\leftrightarrow -\frac{1}{2} e^t + \frac{1}{6} e^{-t} + \frac{1}{3} e^{2t}; \end{aligned}$$



$$\begin{aligned}
\frac{e^{-2s}e^4}{(s-1)(s+1)(s-2)} &\leftrightarrow u(t-2) \left( -\frac{1}{2}e^{t+2} + \frac{1}{6}e^{-(t-6)} + \frac{1}{3}e^{2t} \right); \\
\frac{1}{s(s-1)(s+1)} &= -\frac{1}{s} + \frac{1}{2} \frac{1}{s-1} + \frac{1}{2} \frac{1}{s+1} \leftrightarrow -1 + \frac{1}{2}e^t + \frac{1}{2}e^{-t}; \\
\frac{e^{-2s}}{s(s-1)(s+1)} &\leftrightarrow u(t-2) \left( -1 + \frac{1}{2}e^{t-2} + \frac{1}{2}e^{-(t-2)} \right); \\
\frac{3s-1}{(s-1)(s+1)} &= \frac{1}{s-1} + \frac{2}{s+1} \leftrightarrow e^t + 2e^{-t}.
\end{aligned}$$

Therefore,

$$y = \frac{1}{2}e^t + \frac{13}{6}e^{-t} + \frac{1}{3}e^{2t} + u(t-2) \left( -1 + \frac{1}{2}e^{t-2} + \frac{1}{2}e^{-(t-2)} + \frac{1}{2}e^{t+2} - \frac{1}{6}e^{-(t-6)} - \frac{1}{3}e^{2t} \right).$$

**8.5.6.** Note that  $|\sin t| = \sin t$  if  $0 \leq t < \pi$ , while  $|\sin t| = -\sin t$  if  $\pi \leq t < 2\pi$ . Rewrite the initial value problem as

$$y'' + 4y = \sin t - 2u(t-\pi)\sin t + u(t-2\pi)\sin t, \quad y(0) = -3, \quad y'(0) = 1.$$

Since

$$L(u(t-\pi)\sin t) = e^{-\pi s}L(\sin(t+\pi)) = -e^{-\pi s}L(\sin t)$$

and

$$\begin{aligned}
L(u(t-2\pi)\sin t) &= e^{-2\pi s}L(\sin(t+2\pi)) = e^{-2\pi s}L(\sin t), \\
(s^2+4)Y(s) &= \frac{1+2e^{-\pi s}+e^{-2\pi s}}{(s^2+1)} + 1-3s, \text{ so } Y(s) = \frac{1+2e^{-\pi s}+e^{-2\pi s}}{(s^2+1)(s^2+4)} + \frac{1-3s}{s^2+4}. \\
\frac{1}{(s^2+1)(s^2+4)} &= \left( \frac{1}{s^2+1} - \frac{1}{s^2+4} \right) \leftrightarrow \frac{1}{3}\sin t - \frac{1}{6}\sin 2t;
\end{aligned}$$

therefore

$$\begin{aligned}
\frac{e^{-\pi s}}{(s^2+1)(s^2+4)} &\leftrightarrow u(t-\pi) \left( \frac{1}{3}\sin(t-\pi) - \frac{1}{6}\sin 2(t-\pi) \right) \\
&= -u(t-\pi) \left( \frac{1}{3}\sin t + \frac{1}{6}\sin 2t \right)
\end{aligned}$$

and

$$\begin{aligned}
\frac{e^{-2\pi s}}{(s^2+1)(s^2+4)} &\leftrightarrow u(t-2\pi) \left( \frac{1}{3}\sin(t-2\pi) - \frac{1}{6}\sin 2(t-2\pi) \right) \\
&= u(t-2\pi) \left( \frac{1}{3}\sin t - \frac{1}{6}\sin 2t \right);
\end{aligned}$$

therefore

$$y = \frac{1}{3}\sin 2t - 3\cos 2t + \frac{1}{3}\sin t - 2u(t-\pi) \left( \frac{1}{3}\sin t + \frac{1}{6}\sin 2t \right) + u(t-2\pi) \left( \frac{1}{3}\sin t - \frac{1}{6}\sin 2t \right).$$

**8.5.8.**  $y'' + 9y = \cos t + u(t - 3\pi/2)(\sin t - \cos t)$ ,  $y(0) = 0$ ,  $y'(0) = 0$ . Since

$$\begin{aligned} L(u(t - 3\pi/2)(\sin t - \cos t)) &= e^{-3\pi s/2} L(\sin(t + 3\pi/2) - \cos(t + 3\pi/2)) \\ &\quad - e^{-3\pi s/2} L(\cos t + \sin t), \end{aligned}$$

$$(s^2 + 9)Y(s) = \frac{1}{s^2 + 1} - e^{-3\pi s/2} \frac{s + 1}{s^2 + 1}, \quad \text{so} \quad Y(s) = \frac{1}{(s^2 + 1)(s^2 + 9)} - e^{-3\pi s/2} \frac{s + 1}{(s^2 + 1)(s^2 + 9)}.$$

$$\frac{1}{(s^2 + 1)(s^2 + 9)} = \frac{1}{8} \left( \frac{1}{s^2 + 1} - \frac{1}{s^2 + 9} \right) \leftrightarrow \frac{1}{8} \left( \sin t - \frac{1}{3} \sin 3t \right) \quad \text{and}$$

$$\frac{s}{(s^2 + 1)(s^2 + 9)} = \frac{1}{8} \left( \frac{s}{s^2 + 1} - \frac{s}{s^2 + 9} \right) \leftrightarrow \frac{1}{8} (\cos t - \cos 3t).$$

$$\begin{aligned} \frac{s + 1}{(s^2 + 1)(s^2 + 9)} &= \frac{s + 1}{(s^2 + 1)(s^2 + 9)} = \frac{s + 1}{8} \left( \frac{s + 1}{s^2 + 1} - \frac{s + 1}{s^2 + 9} \right) \\ &\leftrightarrow \frac{1}{8} \left( \cos t + \sin t - \cos 3t - \frac{1}{3} \sin 3t \right), \quad \text{so} \end{aligned}$$

$$\begin{aligned} e^{-3\pi s/2} \frac{s + 1}{(s^2 + 1)(s^2 + 9)} &\leftrightarrow \frac{u(t - 3\pi/2)}{8} (\cos(t - 3\pi/2) + \sin(t - 3\pi/2) \\ &\quad - \cos 3(t - 3\pi/2) - \frac{1}{3} \sin 3(t - \pi/2)) \\ &= \frac{u(t - 3\pi/2)}{8} \left( \sin t - \cos t + \sin 3t - \frac{1}{3} \cos 3t \right). \end{aligned}$$

Therefore,  $y = \frac{1}{8} (\cos t - \cos 3t) - \frac{1}{8} u(t - 3\pi/2) \left( \sin t - \cos t + \sin 3t - \frac{1}{3} \cos 3t \right)$ .

**8.5.10.**  $y'' + y = t - 2u(t - \pi)t$ ,  $y(0) = 0$ ,  $y'(0) = 0$ . Since

$$L(u(t - \pi)t) = e^{-\pi s} L(t + \pi) = e^{-\pi s} \left( \frac{1}{s^2} + \frac{\pi}{s} \right),$$

$$(s^2 + 1)Y(s) = \frac{1}{s^2} - 2e^{-\pi s} \left( \frac{1}{s^2} + \frac{\pi}{s} \right);$$

$$\begin{aligned} Y(s) &= \frac{1}{s^2(s^2 + 1)} - 2e^{-\pi s} \left( \frac{1}{s^2(s^2 + 1)} + \frac{\pi}{s(s^2 + 1)} \right) \\ &= \left( \frac{1}{s^2} - \frac{1}{s^2 + 1} \right) - 2e^{-\pi s} \left( \frac{1}{s^2} - \frac{1}{s^2 + 1} \right) - 2\pi e^{-\pi s} \left( \frac{1}{s} - \frac{s}{s^2 + 1} \right). \end{aligned}$$

Since

$$\begin{aligned} \frac{1}{s^2} - \frac{1}{s^2 + 1} &\leftrightarrow t - \sin t \Rightarrow e^{-\pi s} \left( \frac{1}{s^2} - \frac{1}{s^2 + 1} \right) \\ &\leftrightarrow u(t - \pi)(t - \pi - \sin(t - \pi)) = u(t - \pi)(t - \pi + \sin t) \end{aligned}$$

and

$$\begin{aligned}\frac{1}{s} - \frac{s}{s^2 + 1} &\leftrightarrow 1 - \cos t \Rightarrow e^{-\pi s} \left( \frac{1}{s} - \frac{s}{s^2 + 1} \right) \\ &\leftrightarrow u(t - \pi)(1 - \cos(t - \pi)) = u(t - \pi)(1 + \cos t),\end{aligned}$$

$$y = t - \sin t - 2u(t - \pi)(t + \sin t + \pi \cos t).$$

**8.5.12.**  $y'' + y = t - 3u(t - 2\pi)t$ ,  $y(0) = 1$ ,  $y'(0) = 2$ ;

$$L(u(t - 2\pi)t) = e^{-2\pi s}L(t + 2\pi) = e^{-2\pi s} \left( \frac{1}{s^2} + \frac{2\pi}{s} \right);$$

$$(s^2 + 1)Y(s) = \frac{1 - 3e^{-2\pi s}}{s^2} - \frac{6\pi e^{-2\pi s}}{s} + 2 + s;$$

$$Y(s) = \frac{1 - 3e^{-2\pi s}}{s^2(s^2 + 1)} - \frac{6\pi e^{-2\pi s}}{s(s^2 + 1)} + \frac{2 + s}{s^2 + 1};$$

$$\frac{1}{s^2(s^2 + 1)} = \frac{1}{s^2} - \frac{1}{s^2 + 1} \leftrightarrow t - \sin t;$$

$$\frac{e^{-2\pi s}}{s^2(s^2 + 1)} \leftrightarrow u(t - 2\pi)((t - 2\pi - \sin(t - 2\pi)) = u(t - 2\pi)(t - 2\pi - \sin t);$$

$$\frac{1}{s(s^2 + 1)} = \frac{1}{s} - \frac{s}{s^2 + 1} \leftrightarrow 1 - \cos t;$$

$$\frac{e^{-2\pi s}}{s(s^2 + 1)} \leftrightarrow u(t - 2\pi)(1 - \cos(t - 2\pi)) = u(t - 2\pi)(1 - \cos t);$$

$$\frac{2 + s}{s^2 + 1} \leftrightarrow 2 \sin t + \cos t;$$

$$y = t + \sin t + \cos t - u(t - 2\pi)(3t - 3 \sin t - 6\pi \cos t).$$

**8.5.14.**  $y'' - 4y' + 3y = -1 + 2u(t - 1)$ ,  $y(0) = 0$ ,  $y'(0) = 0$ ;

$$(s^2 - 4s + 3)Y(s) = \frac{-1 + 2e^{-s}}{s}; Y(s) = \frac{-1 + 2e^{-s}}{s(s - 1)(s - 3)};$$

$$\frac{1}{s(s - 1)(s - 3)} = \frac{1}{3s} + \frac{1}{6} \frac{1}{s - 3} - \frac{1}{2} \frac{1}{s - 1} \leftrightarrow \frac{1}{3} + \frac{1}{6}e^{3t} - \frac{1}{2}e^t;$$

$$\frac{e^{-s}}{s(s - 1)(s - 3)} \leftrightarrow u(t - 1) \left( \frac{1}{3} + \frac{1}{6}e^{3(t-1)} - \frac{1}{2}e^{t-1} \right);$$

$$y = -\frac{1}{3} - \frac{1}{6}e^{3t} + \frac{1}{2}e^t + u(t - 1) \left( \frac{2}{3} + \frac{1}{3}e^{3(t-1)} - e^{t-1} \right).$$

**8.5.16.**  $y'' + 2y' + y = 4e^t - 4u(t - 1)e^t$ ,  $y(0) = 0$ ,  $y'(0) = 0$ . Since

$$L(4u(t - 1)e^t) = e^{-s}L(4e^{(t+1)}) = \frac{4e^{-s+1}}{s - 1},$$

$$(s^2 + 2s + 1)Y(s) = \frac{4}{s - 1} - \frac{4e^{-s+1}}{s - 1}, \quad \text{so}$$

$$Y(s) = \frac{4}{(s - 1)(s + 1)^2} - \frac{4e^{-s+1}}{(s - 1)(s + 1)^2}.$$

$$\frac{1}{(s-1)(s+1)^2} = \frac{A}{s-1} + \frac{B}{s+1} + \frac{C}{(s+1)^2},$$

where

$$A(s+1)^2 + B(s-1)(s+1) + C(s-1) = 4.$$

$$\begin{aligned} A &= 1 && (\text{set } s = 1); \\ C &= -2 && (\text{set } s = -1); \\ A + B &= 0 && (\text{equate coefficients of } s^2). \end{aligned}$$

Solving this system yields  $A = 1$ ,  $B = -1$ ,  $C = -2$ . Therefore,

$$\frac{1}{(s-1)(s+1)^2} = \frac{1}{s-1} - \frac{1}{s+1} - \frac{2}{(s+1)^2} \quad \text{and}$$

$$\begin{aligned} y &= e^t - e^{-t} - 2te^{-t} - eu(t-1)(e^{t-1} - e^{-(t-1)} - 2(t-1)e^{-(t-1)}) \\ &= e^t - e^{-t} - 2te^{-t} - u(t-1)(e^t - e^{-(t-2)} - 2(t-1)e^{-(t-2)}). \end{aligned}$$

**8.5.18.**  $y'' - 4y' + 4y = e^{2t} - 2u(t-2)e^{2t}$ ,  $y(0) = 0$ ,  $y'(0) = -1$ . Since

$$L(u(t-2)e^{2t}) = e^{-2s}L(e^{2t+4}) = \frac{e^{-2s+4}}{s-2},$$

$$(s^2 - 4s + 4)Y(s) = \frac{1}{s-2} - \frac{2e^{-2s+4}}{s-2} - 1, \quad \text{so}$$

$$Y(s) = \frac{1}{(s-2)^3} - \frac{2e^{-2s+4}}{(s-2)^3} - \frac{1}{(s-2)^2}.$$

$$\frac{1}{(s-2)^3} \leftrightarrow \frac{t^2 e^{2t}}{2} \Rightarrow \frac{e^{-2s+4}}{(s-2)^3} \leftrightarrow \frac{e^4}{2} u(t-2)e^{2(t-2)}(t-2)^2 = u(t-2)\frac{(t-2)^2 e^{2t}}{2};$$

therefore  $y = \frac{t^2 e^{2t}}{2} - te^{2t} - u(t-2)(t-2)^2 e^{2t}$ .

**8.5.20.**  $y'' + 2y' + 2y = 1 + u(t-2\pi)(t-1) - u(t-3\pi)(t+1)$ ,  $y(0) = 2$ ,  $y'(0) = -1$ ;

$$L(u(t-2\pi)(t-1)) = e^{-2\pi s}L((t+2\pi-1)) = e^{-2\pi s}\left(\frac{1}{s^2} + \frac{2\pi-1}{s}\right);$$

$$L(u(t-3\pi)(t+1)) = e^{-3\pi s}L((t+3\pi+1)) = e^{-3\pi s}\left(\frac{1}{s^2} + \frac{3\pi+1}{s}\right);$$

$$(s^2 + 2s + 2)Y(s) = \frac{1}{s} + e^{-2\pi s}\left(\frac{1}{s^2} + \frac{2\pi-1}{s}\right) - e^{-3\pi s}\left(\frac{1}{s^2} + \frac{3\pi+1}{s}\right) + (-1 + 2s) + 4.$$

Let  $G(s) = \frac{1}{s(s^2 + 2s + 2)}$ ,  $H(s) = \frac{1}{s(s^2 + 2s + 2)}$ ; then

$$Y(s) = Y_1(s) + e^{-2\pi s}Y_2(s) - e^{-3\pi s}Y_3(s), \tag{A}$$

where

$$Y_1(s) = G(s) + \frac{2s+3}{s^2+2s+2}, \quad (\text{B})$$

$$Y_2(s) = H(s) + (2\pi-1)G(s), \quad (\text{C})$$

$$Y_3(s) = H(s) + (3\pi+1)G(s). \quad (\text{D})$$

Let  $y_i(t) = L^{-1}(Y_i(s))$ , ( $i = 1, 2, 3$ ). From (A),

$$y(t) = y_1(t) + u(t-2\pi)y_2(t-2\pi) - u(t-3\pi)y_3(t-3\pi). \quad (\text{E})$$

Find  $L^{-1}(G(s))$ :

$$G(s) = \frac{A}{s} + \frac{B(s+1)+C}{(s+1)^2+1}$$

where  $A((s+1)^2+1) + (B(s+1)+C)s = 1$ . Setting  $s = 0$  yields  $A = 1/2$ ; setting  $s = -1$  yields  $A - C = 1$ , so  $C = -1/2$ ; since  $A + B = 0$  (coefficient of  $s^2$ ),  $B = -1/2$ . Therefore,

$$G(s) = \frac{1}{2} \left( \frac{1}{s} - \frac{(s+1)+1}{(s+1)^2+1} \right) \leftrightarrow \frac{1}{2} - \frac{1}{2}e^{-t}(\cos t + \sin t). \quad (\text{F})$$

Find  $L^{-1}(H(s))$ :

$$H(s) = \frac{A}{s} + \frac{B}{s+2} + \frac{C(s+1)+D}{(s+1)^2+1}$$

where  $(As+B)((s+1)^2+1) + (C(s+1)+D)s^2 = 1$ .

$$\begin{aligned} 2B &= 1 & (\text{set } s=0); \\ -A+B+D &= 1 & (\text{set } s=-1); \\ 5A+5B+2C+D &= 1 & (\text{set } s=1); \\ A+B &= 0 & (\text{equate coefficients of } s^3). \end{aligned}$$

Solving this system yields  $A = -1/2$ ,  $B = 1/2$ ,  $C = 1/2$ ,  $D = 0$ ; therefore

$$H(s) = -\frac{1}{2} \left( \frac{1}{s} - \frac{1}{s^2} - \frac{s+1}{(s+1)^2+1} \right) \leftrightarrow -\frac{1}{2}(1-t-e^{-t}\cos t). \quad (\text{G})$$

Since

$$\frac{2s+3}{s^2+2s+2} = \frac{2(s+1)+1}{(s+1)^2+1} \leftrightarrow e^{-t}(2\cos t + \sin t),$$

(B) and (F)) imply that

$$y_1(t) = \frac{1}{2}e^{-t}(3\cos t + \sin t) + \frac{1}{2}. \quad (\text{H})$$

From (C), (F), and (G),

$$y_2(t) = \pi - 1 + \frac{t}{2} + (\pi-1)e^{-t}\cos t - \frac{2\pi-1}{2}e^{-t}\sin t,$$

so

$$y_2(t-2\pi) = - \left( e^{-(t-2\pi)} \left( (\pi-1)\cos t + \frac{2\pi-1}{2}\sin t \right) + 1 - \frac{t}{2} \right). \quad (\text{I})$$

From (D), (F)), and (G),

$$y_3(t) = \frac{1}{2} (-e^{-t}(3\pi \cos t + (3\pi + 1) \sin t + t + 3\pi)),$$

so

$$y_3(t - 3\pi) = \frac{1}{2} (e^{-(t-3\pi)}(3\pi \cos t + (3\pi + 1) \sin t + t)). \quad (\text{J})$$

Now (E), (20), (I), and (J)

$$\begin{aligned} y &= \frac{1}{2} e^{-t}(3 \cos t + \sin t) + \frac{1}{2} \text{ imply that} \\ &-u(t-2\pi) \left( e^{-(t-2\pi)} \left( (\pi-1) \cos t + \frac{2\pi-1}{2} \sin t \right) + 1 - \frac{t}{2} \right) \\ &-\frac{1}{2} u(t-3\pi) (e^{-(t-3\pi)}(3\pi \cos t + (3\pi + 1) \sin t + t)). \end{aligned}$$

**8.5.22.** (a)  $f(t) = \sum_{n=0}^{\infty} u(t-n\pi)$ ;  $F(s) = \frac{1}{s} \sum_{n=0}^{\infty} e^{-n\pi s}$ ;  $Y(s) = \frac{1}{s(s^2+1)} \sum_{n=0}^{\infty} e^{-n\pi s}$ ;  $\frac{1}{s} - \frac{s}{s^2+1} \leftrightarrow 1 - \cos t$ ;  $\frac{e^{-n\pi s}}{s(s^2+1)} \leftrightarrow u(t-n\pi)(1 - \cos(t-n\pi)) = u(t-n\pi)(1 - (-1)^n \cos t)$ ;

$y(t) = \sum_{n=0}^{\infty} u(t-n\pi)(1 - (-1)^n \cos t)$ . If  $m\pi \leq t < (m+1)\pi$ ,  $y(t) = \sum_{n=0}^m (1 - (-1)^n \cos t)$ . Therefore,

$$y(t) = \begin{cases} 2m+1 - \cos t, & 2m\pi \leq t < (2m+1)\pi \quad (m=0,1,\dots) \\ 2m, & (2m-1)\pi \leq t < 2m\pi \quad (m=1,2,\dots) \end{cases}.$$

(b)  $f(t) = \sum_{n=0}^{\infty} u(t-2n\pi)$ ;  $F(s) = \sum_{n=0}^{\infty} e^{-2n\pi s} L(t+2n\pi s) = \sum_{n=0}^{\infty} e^{-2n\pi s} \left( \frac{1}{s^2} + \frac{2n\pi}{s} \right)$ ;  $Y(s) = \sum_{n=0}^{\infty} e^{-2n\pi s} Y_n(s)$ , where  $Y_n(s) = \frac{1}{s^2(s^2+1)} + \frac{2n\pi}{s(s^2+1)} = \frac{1}{s^2} - \frac{1}{s^2+1} + \frac{2n\pi}{s} - \frac{2n\pi}{s^2+1} \leftrightarrow y_n(t) = t - \sin t + 2n\pi - 2n\pi \cos t$ . Since  $\cos(t-2n\pi) = \cos t$  and  $\sin(t-2n\pi) = \sin t$ ,  $e^{-2n\pi s} Y_n(s) \leftrightarrow u(t-2n\pi)y_n(t) = u(t-2n\pi)(t - \sin t - 2n\pi \cos t)$ ; therefore  $y(t) = \sum_{n=0}^{\infty} u(t-2n\pi)(t - \sin t - 2n\pi \cos t)$ .

If  $2m\pi \leq t < 2(m+1)\pi$ , then

$$y(t) = \sum_{n=0}^m (t - \sin t - 2n\pi \cos t) = (m+1)(t - \sin t - m\pi \cos t).$$

(c)  $f(t) = 1 + 2 \sum_{n=1}^{\infty} (-1)^n u(t-n\pi)$ ;  $F(s) = \frac{1}{s} \left( 1 + 2 \sum_{n=1}^{\infty} (-1)^n e^{-n\pi s} \right)$ ;  $Y(s) = \frac{1}{s(s^2+1)} \left( 1 + 2 \sum_{n=1}^{\infty} (-1)^n e^{-n\pi s} \right)$ ;

$\frac{1}{s(s^2+1)} = \frac{1}{s} - \frac{s}{s^2+1} \leftrightarrow 1 - \cos t$ ;  $\frac{e^{-n\pi s}}{s(s^2+1)} \leftrightarrow u(t-n\pi)(1 - \cos(t-n\pi)) = u(t-n\pi)(1 - (-1)^n \cos t)$ ;

$y(t) = 1 - \cos t + 2 \sum_{n=1}^{\infty} (-1)^n u(t-n\pi)(1 - (-1)^n \cos t)$ . If  $m\pi \leq t < (m+1)\pi$ ,

$$y(t) = 1 - \cos t + 2 \sum_{n=1}^m (-1)^n (1 - (-1)^n \cos t) = (-1)^m - (2m+1) \cos t.$$

(d)  $f(t) = \sum_{n=0}^{\infty} u(t-n)$ ;  $F(s) = \frac{1}{s} \sum_{n=0}^{\infty} e^{-ns}$ ;  $Y(s) = \frac{1}{s(s^2-1)} \sum_{n=0}^{\infty} e^{-ns}$ ;  $\frac{1}{s(s^2-1)} = \frac{1}{2} \frac{1}{s-1} +$

$$\frac{1}{2} \frac{1}{s+1} - \frac{1}{s} \leftrightarrow \frac{1}{2}(e^t + e^{-t} - 2); \frac{e^{-ns}}{s(s^2-1)} \leftrightarrow \frac{u(t-n)}{2}(e^t + e^{-t} - 2); y(t) = \frac{1}{2} \sum_{n=0}^{\infty} u(t-n)(e^{t-n} + e^{-(t-n)} - 2).$$

If  $m \leq t < (m+1)$ ,

$$\begin{aligned} y(t) &= \frac{1}{2} \sum_{n=0}^m (e^{t-n} + e^{-(t-n)} - 2) = \frac{1}{2}(e^{t-m} + e^{-t}) \sum_{n=0}^m e^n - m - 1 \\ &= \frac{1 - e^{m+1}}{2(1 - e)}(e^{t-m} + e^{-t}) - m - 1. \end{aligned}$$

$$\begin{aligned} \text{(e) } f(t) &= (\sin t + 2 \cos t) \sum_{n=0}^{\infty} u(t-2n\pi); F(s) = \frac{1+2s}{s^2+1} \sum_{n=0}^{\infty} e^{-2n\pi s}; Y(s) = \frac{1+2s}{(s^2+1)(s^2+2s+2)} \sum_{n=0}^{\infty} e^{-2n\pi s}; \\ &\frac{1+2s}{(s^2+1)(s^2+2s+2)} = \frac{As+B}{s^2+1} + \frac{C(s+1)+D}{(s+1)^2+1} \end{aligned}$$

where

$$\begin{aligned} (As+B)((s+1)^2+1) + (C(s+1)+D)(s^2+1) &= 1+2s. \\ 2B+C+D &= 1 \quad (\text{set } s=0); \\ -A+B+2D &= -1 \quad (\text{set } s=-1); \\ 5A+5B+4C+2D &= 3 \quad (\text{set } s=1); \\ A+C &= 0 \quad (\text{equate coefficients of } s^3). \end{aligned}$$

Solving this system yields  $A=0$ ,  $B=1$ ,  $C=0$ ,  $D=-1$ . Therefore,

$$\begin{aligned} \frac{1+2s}{(s^2+1)(s^2+2s+2)} &= \frac{1}{s^2+1} - \frac{1}{(s+1)^2+1} \\ &\leftrightarrow (1 - e^{-t}) \sin t. \end{aligned}$$

Since  $\sin(t-2n\pi) = \sin t$ ,

$$e^{-2n\pi s} \frac{1+2s}{(s^2+1)(s^2+2s+2)} \leftrightarrow u(t-2n\pi)(1 - e^{-(t-2n\pi)}) \sin t,$$

so

$$y(t) = \sin t \sum_{n=0}^{\infty} u(t-2n\pi)(1 - e^{-(t-2n\pi)}).$$

If  $2m\pi \leq t < 2(m+1)\pi$ ,

$$y(t) = \sin t \sum_{n=0}^m (1 - e^{-(t-2n\pi)}) = \left( m+1 - \left( \frac{1 - e^{2(m+1)\pi}}{1 - e^{2\pi}} \right) e^{-t} \right) \sin t.$$

$$\text{(f) } f(t) = \sum_{n=0}^{\infty} u(t-n); F(s) = \frac{1}{s} \sum_{n=0}^{\infty} e^{-ns}; Y(s) = \frac{1}{s(s-1)(s-2)};$$

$$\frac{1}{s(s-1)(s-2)} = \frac{1}{2s} - \frac{1}{s-1} + \frac{1}{2} \frac{1}{s-2} \leftrightarrow \frac{1}{2}(1 - 2e^t + e^{2t});$$

$$\frac{e^{-ns}}{s(s-1)(s-2)} \leftrightarrow \frac{1}{2}u(t-n)(1-2e^{t-n}+e^{2(t-n)});$$

$$y(t) = \frac{1}{2} \sum_{n=0}^{\infty} u(t-n)(1-2e^{t-n}+e^{2(t-n)}).$$

If  $m \leq t < m+1$ ,

$$\begin{aligned} y(t) &= \sum_{n=0}^m (1-2e^{t-n}+e^{2(t-n)}) = \frac{m+1}{2} - e^{t-m} \sum_{n=0}^m e^n + \frac{1}{2}e^{2(t-m)} \sum_{n=0}^m e^{2n} \\ &= \frac{m+1}{2} - e^{t-m} \frac{1-e^{m+1}}{1-e} + \frac{1}{2}e^{2(t-m)} \frac{1-e^{2m+2}}{1-e^2}. \end{aligned}$$

**8.5.24.** (a) The assumptions imply that  $y''(t) = \frac{f(t) - by'(t) - cy(t)}{a}$  on  $(\alpha, t_0)$  and  $(t_0, \beta)$ ,  $y''(t_0+) = \frac{f(t_0+) - by'(t_0) - cy(t_0)}{a}$ , and  $y''(t_0-) = \frac{f(t_0-) - by'(t_0) - cy(t_0)}{a}$ . This implies the conclusion.

(b) Since  $y''$  has a jump discontinuity at  $t_0$ , applying Exercise 8.4.23(c) to  $y'$  shows that  $y'$  is not differentiable at  $t_0$ . Therefore,  $y$  cannot satisfy (A) on  $(\alpha, \beta)$  if  $f$  has a jump discontinuity at some  $t_0$  in  $(\alpha, \beta)$ .

**8.5.26.** If  $0 \leq t < t_0$ , then  $y(t) = z_0(t)$ . Therefore,  $y(0) = z_0(0) = k_0$  and  $y'(0) = z'(0) = k_1$ , and

$$ay'' + by' + cy = az''_0 + bz'_0 + cz_0 = f_0(t) = f(t), \quad 0 < t < t_0.$$

Now suppose that  $1 \leq m \leq n$ . For convenience, define  $t_{n+1} = \infty$ . If  $t_m \leq t < t_{m+1}$ , then  $y(t) = \sum_{k=0}^m z_m(t)$ , so

$$ay'' + by' + cy = \sum_{k=0}^m (az''_k + bz'_k + cz_k) = f_0 + \sum_{k=1}^m (f_k - f_{k-1}) = f_m = f, \quad t_m < t < t_{m+1}.$$

Thus,  $y$  satisfies  $ay'' + by' + cy = f$  on any open interval that does not contain any of the points  $t_1, t_2, \dots, t_n$ .

Since  $z(t_m) = z'(t_m)$  for  $m = 1, 2, \dots$ ,  $y$  and  $y'$  are continuous on  $[0, \infty)$ . Since  $y''(t) = -(by'(t) + cy(t))/a$  if  $t \neq t_m$  ( $m = 1, 2, \dots$ ),  $y''$  has limits from the left at  $t_1, \dots, t_n$ .

## 8.6 Convolution

**8.6.2.** (a)  $\sin at \leftrightarrow \frac{a}{s^2 + a^2}$  and  $\cos bt \leftrightarrow \frac{s}{s^2 + b^2}$ , so  $H(s) = \frac{as}{(s^2 + a^2)(s^2 + b^2)}$ .

(b)  $e^t \leftrightarrow \frac{1}{s-1}$  and  $\sin at \leftrightarrow \frac{a}{s^2 + a^2}$ , so  $H(s) = \frac{a}{(s-1)(s^2 + a^2)}$ .

(c)  $\sinh at \leftrightarrow \frac{a}{s^2 - a^2}$  and  $\cosh at \leftrightarrow \frac{1}{s^2 - a^2}$ , so  $H(s) = \frac{as}{(s^2 - a^2)^2}$ .

(d)  $t \sin \omega t \leftrightarrow \frac{2\omega s}{(s^2 + \omega^2)^2}$  and  $t \cos \omega t \leftrightarrow \frac{s^2 - \omega^2}{(s^2 + \omega^2)^2}$ , so  $H(s) = \frac{2\omega s(s^2 - \omega^2)}{(s^2 + \omega^2)^4}$ .

(e)  $e^t \int_0^t \sin \omega \tau \cos \omega(t-\tau) d\tau = \int_0^t (e^\tau \sin \omega \tau) (e^{(t-\tau)} \cos \omega(t-\tau)) d\tau$ ;  $e^t \sin \omega t \leftrightarrow \frac{\omega}{(s-1)^2 + \omega^2}$

and  $e^t \cos \omega t \leftrightarrow \frac{s-1}{(s-1)^2 + \omega^2}$ , so  $H(s) = \frac{(s-1)\omega}{((s-1)^2 + \omega^2)^2}$ .



(f)  $e^t \int_0^t \tau^2(t-\tau)e^\tau d\tau = \int_0^t \tau^2 e^{2\tau}(t-\tau)e^{(t-\tau)} d\tau$ ;  $t^2 e^{2t} \leftrightarrow \frac{2}{(s-2)^3}$  and  $te^t \leftrightarrow \frac{1}{(s-1)^2}$ , so

$$H(s) = \frac{2}{(s-2)^3(s-1)^2}.$$

(g)  $e^{-t} \int_0^t e^{-\tau} \tau \cos \omega(t-\tau) d\tau = \int_0^t \tau e^{-2\tau} e^{-(t-\tau)} \cos \omega(t-\tau) d\tau$ ;  $te^{-2t} \leftrightarrow \frac{1}{(s+2)^2}$  and  $e^{-t} \cos \omega t \leftrightarrow \frac{s+1}{(s+1)^2 + \omega^2}$ , so  $H(s) = \frac{s+1}{(s+2)^2[(s+1)^2 + \omega^2]}$ .

(h)  $e^t \int_0^t e^{2\tau} \sinh(t-\tau) d\tau = \int_0^t e^{3\tau} (e^{(t-\tau)} \sinh(t-\tau)) d\tau$ ;  $e^{3t} \leftrightarrow \frac{1}{s-3}$  and  $e^t \sinh t \leftrightarrow \frac{1}{(s-1)^2 - 1}$ , so  $H(s) = \frac{1}{(s-3)((s-1)^2 - 1)}$ .

(i)  $te^{2t} \leftrightarrow \frac{1}{(s-2)^2}$  and  $\sin 2t \leftrightarrow \frac{2}{s^2 + 4}$ , so  $H(s) = \frac{2}{(s-2)^2(s^2 + 4)}$ .

(j)  $t^3 \leftrightarrow \frac{6}{s^4}$  and  $e^t \leftrightarrow \frac{1}{s-1}$ , so  $H(s) = \frac{6}{s^4(s-1)}$ .

(k)  $t^6 \leftrightarrow \frac{6!}{s^7}$  and  $e^{-t} \sin 3t \leftrightarrow \frac{3}{(s+1)^2 + 9}$ , so  $H(s) = \frac{3 \cdot 6!}{s^7[(s+1)^2 + 9]}$ .

(l)  $t^2 \leftrightarrow \frac{2}{s^3}$  and  $t^3 \leftrightarrow \frac{6}{s^4}$ , so  $H(s) = \frac{12}{s^7}$ .

(m)  $t^7 \leftrightarrow \frac{7!}{s^8}$  and  $e^{-t} \sin 2t \leftrightarrow \frac{2}{(s+1)^2 + 4}$ , so  $H(s) = \frac{2 \cdot 7!}{s^8[(s+1)^2 + 4]}$ .

(n)  $t^4 \leftrightarrow \frac{24}{s^5}$  and  $\sin 2t \leftrightarrow \frac{2}{s^2 + 4}$ , so  $H(s) = \frac{48}{s^5(s^2 + 4)}$ .

**8.6.4.** (a)  $Y(s) = \frac{1}{s^2} - \frac{Y(s)}{s^2 + 1}$ ;  $Y(s) \left(1 + \frac{1}{s^2}\right) = \frac{1}{s^2}$ ;  $Y(s) \frac{s^2 + 1}{s^2} = \frac{1}{s^2}$ ;  $Y(s) = \frac{1}{s^2 + 1}$ , so  $y = \sin t$ .

(b)  $Y(s) = \frac{1}{s^2 + 1} - \frac{2sY(s)}{s^2 + 1}$ ;  $Y(s) \left(1 + \frac{2s}{s^2 + 1}\right) = \frac{1}{s^2 + 1}$ ;  $Y(s) \frac{(s+1)^2}{s^2 + 1} = \frac{1}{s^2 + 1}$ ;  $Y(s) = \frac{1}{(s+1)^2}$ , so  $y = te^{-t}$ .

(c)  $Y(s) = \frac{1}{s} + \frac{2sY(s)}{s^2 + 1}$ ;  $Y(s) \left(1 - \frac{2s}{s^2 + 1}\right) = \frac{1}{s}$ ;  $Y(s) \frac{(s-1)^2}{s^2 + 1} = \frac{1}{s}$ ;  $Y(s) = \frac{(s^2 + 1)}{s(s-1)^2} = \frac{A}{s} + \frac{B}{s^2} + \frac{C}{(s-1)^2}$ , where  $A(s-1)^2 + Bs(s-1) + Cs = s^2 + 1$ . Setting  $s = 0$  and  $s = 1$  shows that  $A = 1$  and  $C = 2$ ; equating coefficients of  $s^2$  yields  $A + B = 1$ , so  $B = 0$ . Therefore,  $Y(s) = \frac{1}{s} + \frac{1}{(s-1)^2}$ , so  $y = 1 + 2te^t$ .

(d)  $Y(s) = \frac{1}{s^2} + \frac{Y(s)}{s+1}$ ;  $Y(s) \left(1 - \frac{1}{s+1}\right) = \frac{1}{s^2}$ ;  $Y(s) \left(\frac{s}{s+1}\right) = \frac{1}{s^2}$ ;  $Y(s) = \frac{s+1}{s^3} = \frac{1}{s^2} + \frac{1}{s^3}$ , so  $y = t + \frac{t^2}{2}$ .

(e)  $sY(s) - 4 = \frac{1}{s^2} + \frac{sY(s)}{s^2 + 1}$ ;  $Y(s) \left(s - \frac{s}{s^2 + 1}\right) = 4 + \frac{1}{s^2}$ ;  $Y(s) \frac{s^3}{s^2 + 1} = \frac{4s^2 + 1}{s^2}$ ;  $Y(s) = \frac{(4s^2 + 1)(s^2 + 1)}{s^5} = \frac{4s^4 + 5s^2 + 1}{s^5} = \frac{4}{s} + \frac{5}{s^3} + \frac{1}{s^5}$ , so  $y = 4 + \frac{5}{2}t^2 + \frac{1}{24}t^4$ .

(f)  $Y(s) = \frac{s-1}{s^2+1} + \frac{Y(s)}{s^2+1}$ ;  $Y(s)\left(1 - \frac{1}{s^2+1}\right) = \frac{s-1}{s^2+1}$ ;  $Y(s)\frac{s^2}{s^2+1} = \frac{s-1}{s^2+1}$ ;  $Y(s) = \frac{s-1}{s^2} = \frac{1}{s} - \frac{1}{s^2}$ , so  $y = 1 - t$ .

**8.6.6.** Substituting  $x = t - \tau$  yields  $\int_0^t f(t-\tau)g(\tau) d\tau = -\int_t^0 f(x)g(t-x)(-dx) = \int_0^t f(x)g(t-x) dx = \int_0^t f(\tau)g(t-\tau) d\tau$ .

**8.6.8.**  $p(s)Y(s) = F(s) + a(k_1 + k_0s) + bk_0$ , so (A)  $Y(s) = \frac{F(s)}{p(s)} + \frac{k_0(as+b) + k_1a}{p(s)}$ . Since  $p(s) = a(s-r_1)(s-r_2)$  and therefore  $b = -a(r_1+r_2)$ , (A) can be rewritten as

$$Y(s) = \frac{F(s)}{a(s-r_1)(s-r_2)} + \frac{k_0(s-r_1-r_2)}{(s-r_1)(s-r_2)} + \frac{k_1}{(s-r_1)(s-r_2)}.$$

$$\frac{1}{(s-r_1)(s-r_2)} = \frac{1}{r_2-r_1} \left( \frac{1}{s-r_2} - \frac{1}{s-r_1} \right) \leftrightarrow \frac{e^{r_2t} - e^{r_1t}}{r_2-r_1},$$

so the convolution theorem implies that

$$\frac{F(s)}{a(s-r_1)(s-r_2)} \leftrightarrow \frac{1}{a} \int_0^t \frac{e^{r_2\tau} - e^{r_1\tau}}{r_2-r_1} f(t-\tau) d\tau.$$

$$\frac{s-r_1-r_2}{(s-r_1)(s-r_2)} = \frac{r_2}{r_2-r_1} \frac{1}{s-r_1} - \frac{r_1}{r_2-r_1} \frac{1}{s-r_2} \leftrightarrow \frac{r_2e^{r_1t} - r_1e^{r_2t}}{r_2-r_1}.$$

Therefore,

$$y(t) = k_0 \frac{r_2e^{r_1t} - r_1e^{r_2t}}{r_2-r_1} + k_1 \frac{e^{r_2t} - e^{r_1t}}{r_2-r_1} + \frac{1}{a} \int_0^t \frac{e^{r_2\tau} - e^{r_1\tau}}{r_2-r_1} f(t-\tau) d\tau.$$

**8.6.10.**  $p(s)Y(s) = F(s) + a(k_1 + k_0s) + bk_0$ , so (A)  $Y(s) = \frac{F(s)}{p(s)} + \frac{k_0(as+b) + k_1a}{p(s)}$ . Since  $p(s) = a(s-\lambda)^2 + \omega^2$  and therefore  $b = -2a\lambda$ , (A) can be rewritten as

$$Y(s) = \frac{F(s)}{a[(s-\lambda)^2 + \omega^2]} + \frac{k_0(s-2\lambda)}{(s-\lambda)^2 + \omega^2} + \frac{k_1}{(s-\lambda)^2 + \omega^2}.$$

$\frac{1}{(s-\lambda)^2 + \omega^2} \leftrightarrow \frac{1}{\omega} e^{\lambda t} \sin \omega t$ , so the convolution theorem implies that

$$\frac{F(s)}{a[(s-\lambda)^2 + \omega^2]} \leftrightarrow \frac{1}{a\omega} \int_0^t e^{\lambda\tau} f(t-\tau) \sin \omega\tau d\tau.$$

$$\frac{s-2\lambda}{(s-\lambda)^2 + \omega^2} = \frac{(s-\lambda) - \lambda}{(s-\lambda)^2 + \omega^2} \leftrightarrow e^{\lambda t} \left( \cos \omega t - \frac{\lambda}{\omega} \sin \omega t \right).$$

Therefore,

$$y(t) = e^{\lambda t} \left[ k_0 \left( \cos \omega t - \frac{\lambda}{\omega} \sin \omega t \right) + \frac{k_1}{\omega} \sin \omega t \right] + \frac{1}{a\omega} \int_0^t e^{\lambda\tau} f(t-\tau) \sin \omega\tau d\tau.$$

8.6.12. (a)

$$\begin{aligned}
ay'' + by' + cy &= f_0(t) + u(t - t_1)(f_1(t) - f_0(t)), \quad y(0) = 0, \quad y'(0) = 0; \\
p(s)Y(s) &= F_0(s) + L(u(t - t_1)(f_1(t) - f_0(t))) = F_0(s) + e^{-st_1}L(g); \\
Y(s) &= \frac{F_0(s) + e^{-st_1}G(s)}{p(s)}. \tag{B}
\end{aligned}$$

(b) Since  $F_0(s) \leftrightarrow f_0(t)$ ,  $G(s) \leftrightarrow g(t)$ , and  $\frac{1}{p(s)} \leftrightarrow w(t)$ , the convolution theorem implies that

$$\frac{F_0(s)}{p(s)} \leftrightarrow \int_0^t w(t - \tau)f_0(\tau) d\tau \quad \text{and} \quad \frac{G(s)}{p(s)} \leftrightarrow \int_0^t w(t - \tau)g(\tau) d\tau.$$

Now Theorem 8.4.2 implies that  $\frac{e^{-st_1}G(s)}{p(s)} \leftrightarrow u(t - t_1) \int_0^t w(t - t_1 - \tau)g(\tau) d\tau$ , and (B) implies that

$$y(t) = \int_0^t w(t - \tau)f_0(\tau) d\tau + u(t - t_1) \int_0^{t-t_1} w(t - t_1 - \tau)g(\tau) d\tau.$$

(c) Let  $z_0(t) = \int_0^t w(t - \tau)f_0(\tau) d\tau$  and  $z_1(t) = \int_0^t w(t - \tau)g(\tau) d\tau$ . Then  $y(t) = z_0(t) + u(t - t_1)z_1(t - t_1)$ . Using Leibniz's rule as in the solution of Exercise 8.6.11(b) shows that

$$z'_0(t) = \int_0^t w'(t - \tau)f_0(\tau) d\tau, \quad z'_1(t) = \int_0^t w'(t - \tau)g(\tau) d\tau, \quad t > 0,$$

$$z''_0(t) = \frac{f_0(t)}{a} + \int_0^t w''(t - \tau)f_0(\tau) d\tau, \quad z''_1(t) = \frac{g(t)}{a} + \int_0^t w''(t - \tau)g(\tau) d\tau, \quad t > 0,$$

if  $t > 0$ , and that

$$az''_0 + bz'_0 + cz_0 = f_0(t) \quad \text{and} \quad az''_1 + bz'_1 + cz_1 = f_1(t + t_1) - f_0(t + t_1), \quad t > 0.$$

This implies the stated conclusion for  $y'$  and  $y''$  on  $(0, t)$  and  $(t, \infty)$ , and that  $ay'' + by' + cy = f(t)$  on these intervals.

(d) Since the functions  $z_0(t)$  and  $h(t) = u(t - t_1)z_1(t - t_1)$  are both continuous on  $[0, \infty)$  and  $h(t) = 0$  if  $0 \leq t \leq t_1$ ,  $y$  is continuous on  $[0, \infty)$ . From (c),  $y'$  is continuous on  $[0, t_1)$  and  $(t_1, \infty)$ , so we need only show that  $y'$  is continuous at  $t_1$ . For this it suffices to show that  $h'(t_1) = 0$ . Since  $h(t_1) = 0$  if  $t \leq t_1$ , (B)  $\lim_{t \rightarrow t_1^-} \frac{h(t) - h(t_1)}{t - t_1} = 0$ . If  $t > t_1$ , then  $h(t) = \int_0^{t-t_1} w(t - t_1 - \tau)g(\tau) d\tau$ . Since  $h(t_1) = 0$ ,

$$\left| \frac{h(t) - h(t_1)}{t - t_1} \right| \leq \int_0^{t-t_1} |w(t - t_1 - \tau)g(\tau)| d\tau. \tag{B}$$

Since  $g$  is continuous from the right at 0, we can choose constants  $T > 0$  and  $M > 0$  so that  $|g(\tau)| < M$  if  $0 \leq \tau \leq T$ . Then (B) implies that

$$\left| \frac{h(t) - h(t_1)}{t - t_1} \right| \leq M \int_0^{t-t_1} |w(t - t_1 - \tau)| d\tau, \quad t_1 < t < t_1 + T. \tag{C}$$

Now suppose  $\epsilon > 0$ . Since  $w(0) = 0$ , we can choose  $T_1$  such that  $0 < T_1 < T$  and  $|w(x)| < \epsilon/M$  if  $0 \leq x < T_1$ . If  $t_1 < t < t_1 + T_1$  and  $0 \leq \tau \leq t - t_1$ , then  $0 \leq t - t_1 - \tau < T_1$ , so (C) implies that

$$\left| \frac{h(t) - h(t_1)}{t - t_1} \right| < \epsilon, \quad t_1 < t < t_1 + T.$$

Therefore,  $\lim_{t \rightarrow t_1+} \frac{h(t) - h(t_1)}{t - t_1} = 0$ . This and (B) imply that  $h'(t_1) = 0$ .

## 8.7 Constant Coefficient Equations with Impulses

$$\begin{aligned} \text{8.7.2. } (s^2 + s - 2)\hat{Y}(s) &= -\frac{10}{s+1} + (-9+7s) + 7; \hat{Y}(s) = \frac{-10 + (s+1)(7s-2)}{(s-1)(s+2)(s+1)} = \frac{2}{s+2} + \frac{5}{s+1}; \\ \hat{y} &= 2e^{-2t} + 5e^{-t}; \frac{1}{p(s)} = \frac{1}{(s+2)(s-1)} = \frac{1}{3} \left( \frac{1}{s-1} - \frac{1}{s+2} \right); w = L^{-1} \left( \frac{1}{p(s)} \right) = \frac{e^t - e^{-2t}}{3}; \\ y &= 2e^{-2t} + 5e^{-t} + \frac{5}{3}u(t-1)(e^{(t-1)} - e^{-2(t-1)}). \end{aligned}$$

$$\begin{aligned} \text{8.7.4. } (s^2 + 1)\hat{Y}(s) &= \frac{3}{s^2+9} - 1 + s; \\ \hat{Y}(s) &= \frac{3}{(s^2+1)(s^2+9)} + \frac{s-1}{s^2+1} = \frac{3}{8} \left( \frac{1}{s^2+1} - \frac{1}{s^2+9} \right) + \frac{s-1}{s^2+1} = \frac{1}{8} \left( \frac{8s-5}{s^2+1} - \frac{3}{s^2+9} \right); \\ \hat{y} &= \frac{1}{8}(8\cos t - 5\sin t - \sin 3t); \frac{1}{p(s)} = \frac{1}{s^2+1}; w = L^{-1} \left( \frac{1}{p(s)} \right) = \sin t; y = \frac{1}{8}(8\cos t - 5\sin t - \sin 3t) - 2u(t-\pi/2)\cos t. \end{aligned}$$

$$\begin{aligned} \text{8.7.6. } (s^2 - 1)\hat{Y}(s) &= \frac{8}{s} + 1 - s; \hat{Y}(s) = \frac{8 + s(1-s)}{s(s-1)(s+1)} = \frac{4}{s-1} + \frac{3}{s+1} - \frac{8}{s}; \hat{y} = 4e^t + 3e^{-t} - 8; \\ \frac{1}{p(s)} &= \frac{1}{(s-1)(s+1)} = \frac{1}{2} \frac{1}{s-1} - \frac{1}{s+1}; w = L^{-1} \left( \frac{1}{p(s)} \right) = \frac{e^t + e^{-t}}{2} = \sinh t; y = 4e^t + 3e^{-t} - 8 + 2u(t-2)\sinh(t-2); \end{aligned}$$

$$\begin{aligned} \text{8.7.8. } (s^2 + 4)\hat{Y}(s) &= \frac{8}{s-2} + 8s; \text{ (A) } \hat{Y}(s) = \frac{8}{(s-2)(s^2+4)} + \frac{8s}{s^2+4}; \frac{8}{(s-2)(s^2+4)} = \frac{A}{s-2} + \frac{Bs+C}{s^2+4} \text{ where } A(s^2+4) + (Bs+C)(s-2) = 8. \text{ Setting } s=2 \text{ yields } A=1; \text{ setting } s=0 \\ \text{yields } 4A-2C &= 8, \text{ so } C=-2; A+B=0 \text{ (coefficient of } x^2), \text{ so } B=-A=-1; \text{ therefore } \frac{8}{(s-2)(s^2+4)} = \frac{1}{s-2} - \frac{s+2}{s^2+4}, \text{ so (A) implies that } \hat{y} = e^{2t} + 7\cos 2t - \sin 2t; \frac{1}{p(s)} = \frac{1}{s^2+4}; \\ w &= L^{-1} \left( \frac{1}{p(s)} \right) = \frac{1}{2}\sin 2t. \text{ Since } \sin(2t-\pi) = -\sin 2t, y = e^{2t} + 7\cos 2t - \sin 2t - \frac{1}{2}u(t-\pi/2)\sin 2t. \end{aligned}$$

$$\begin{aligned} \text{8.7.10. } (s^2 + 2s + 1)\hat{Y}(s) &= \frac{1}{s-1} + (2-s) - 2; \hat{Y}(s) = \frac{1-s(s-1)}{(s-1)(s+1)^2} = \frac{A}{s-1} + \frac{B}{s+1} + \frac{C}{(s+1)^2} \\ \text{where } A(s+1)^2 + (B(s+1) + C)(s-1) &= 1-s(s-1). \text{ Setting } s=1 \text{ yields } A=1/4; \text{ setting } s=-1 \\ \text{yields } C=1/2; \text{ since } A+B &= -1 \text{ (coefficient of } s^2), B=-1-A=-5/4. \text{ Therefore, } \hat{Y}(s) = \frac{1}{4} \frac{1}{s-1} - \frac{5}{4} \frac{1}{s+1} + \frac{1}{2} \frac{1}{(s+1)^2}; \hat{y} = \frac{1}{4}e^t + \frac{1}{4}e^{-t}(2t-5); \frac{1}{p(s)} = \frac{1}{(s+1)^2}; w = L^{-1} \left( \frac{1}{p(s)} \right) = te^{-t}; \\ y &= \frac{1}{4}e^t + \frac{1}{4}e^{-t}(2t-5) + 2u(t-2)(t-2)e^{-(t-2)}. \end{aligned}$$

$$\mathbf{8.7.12.} \quad (s^2 + 2s + 2)\hat{Y}(s) = (2 - s) - 2; \quad Y(s) = \frac{-(s+1)+1}{(s+1)^2+1}; \quad \hat{y} = e^{-t}(\sin t - \cos t); \quad \frac{1}{p(s)} =$$

$$\frac{1}{(s+1)^2+1}; \quad w = L^{-1}\left(\frac{1}{p(s)}\right) = e^{-t}\sin t. \quad \text{Since } \sin(t-\pi) = -\sin t \text{ and } \sin(t-2\pi) = \sin t,$$

$$y = e^{-t}(\sin t - \cos t) - e^{-(t-\pi)}u(t-\pi)\sin t - 3u(t-2\pi)e^{-(t-2\pi)}\sin t.$$

$$\mathbf{8.7.14.} \quad (2s^2-3s-2)\hat{Y}(s) = \frac{1}{s} + 2(2-s)+3; \quad \hat{Y}(s) = \frac{1+s(7-2s)}{2s(s+1/2)(s-2)} = \frac{7}{10}\frac{1}{s-2} - \frac{6}{5}\frac{1}{s+1/2} - \frac{1}{2s};$$

$$\hat{y} = \frac{7}{10}e^{2t} - \frac{6}{5}e^{-t/2} - \frac{1}{2}; \quad \frac{1}{p(s)} = \frac{1}{2(s+1/2)(s-2)} = \frac{1}{5}\left(\frac{1}{s-2} - \frac{1}{s+1/2}\right); \quad w = L^{-1}\left(\frac{1}{p(s)}\right) =$$

$$\frac{1}{5}(e^{2t} - e^{-t/2}); \quad y = \frac{7}{10}e^{2t} - \frac{6}{5}e^{-t/2} - \frac{1}{2} + \frac{1}{5}u(t-2)(e^{2(t-2)} - e^{-(t-2)/2});$$

$$\mathbf{8.7.16.} \quad (s^2 + 1)\hat{Y}(s) = \frac{s}{s^2+4} - 1; \quad \hat{Y}(s) = \frac{s}{(s^2+1)(s^2+4)} - \frac{1}{s^2+1} = \frac{1}{3}\left(\frac{s}{s^2+1} - \frac{s}{s^2+4}\right) -$$

$$\frac{1}{s^2+1}; \quad \hat{y} = \frac{1}{3}(\cos t - \cos 2t - 3\sin t); \quad \frac{1}{p(s)} = \frac{1}{s^2+1}; \quad w = L^{-1}\left(\frac{1}{p(s)}\right) = \sin t. \quad \text{Since } \sin(t-\pi/2) =$$

$$-\cos t \text{ and } \sin(t-\pi) = -\sin t,$$

$$y = \frac{1}{3}(\cos t - \cos 2t - 3\sin t) - 2u(t-\pi/2)\cos t + 3u(t-\pi)\sin t.$$

$$\mathbf{8.7.18.} \quad (s^2 + 2s + 1)\hat{Y}(s) = \frac{1}{s-1} - 1; \quad (\text{A}) \quad \hat{Y}(s) = \frac{1}{(s-1)(s+1)^2} - \frac{1}{(s+1)^2};$$

$$\frac{1}{(s-1)(s+1)^2} = \frac{A}{s-1} + \frac{B}{s+1} + \frac{C}{(s+1)^2}$$

where  $A(s+1)^2 + (B(s+1) + C)(s-1) = 1$ . Setting  $s = 1$  yields  $A = 1/4$ ; setting  $s = -1$  yields  $C = -1/2$ ; since  $A + B = 0$  (coefficient of  $s^2$ ),  $B = -A = -1/4$ . Therefore,

$$\frac{1}{(s-1)(s+1)^2} = \frac{1}{4}\frac{1}{s-1} - \frac{1}{4}\frac{1}{s+1} - \frac{1}{2}\frac{1}{(s+1)^2}.$$

This and (A) imply that

$$\hat{Y}(s) = \frac{1}{4}\frac{1}{s-1} - \frac{1}{4}\frac{1}{s+1} - \frac{3}{2}\frac{1}{(s+1)^2};$$

$$\hat{y} = \frac{1}{4}(e^t - e^{-t}(1+6t)); \quad \frac{1}{p(s)} = \frac{1}{(s+1)^2}; \quad w = L^{-1}\left(\frac{1}{p(s)}\right) = te^{-t};$$

$$y = \frac{1}{4}(e^t - e^{-t}(1+6t)) - u(t-1)(t-1)e^{-(t-1)} + 2u(t-2)(t-2)e^{-(t-2)}.$$

$$\mathbf{8.7.20.} \quad y'' + 4y = 1 - 2u(t-\pi/2) + \delta(t-\pi) - 3\delta(t-3\pi/2), \quad y(0) = 1, \quad y'(0) = -1. \quad (s^2+4)\hat{Y}(s) =$$

$$\frac{1-2e^{-\pi s/2}}{s} + s-1; \quad \hat{Y}(s) = \frac{1-2e^{-\pi s/2}}{s(s^2+4)} + \frac{s-1}{s^2+4}. \quad \text{Since } \frac{1}{s(s^2+4)} = \frac{1}{4}\left(\frac{1}{s} - \frac{s}{s^2+4}\right), \quad \hat{Y}(s) =$$

$$\frac{1}{4s} + \frac{3}{4}\frac{s}{s^2+4} - \frac{1}{s^2+4} - 2e^{-\pi s/2}\left(\frac{1}{s} - \frac{s}{s^2+4}\right). \quad \hat{y} = \frac{3}{4}\cos 2t - \frac{1}{2}\sin 2t + \frac{1}{4} + \frac{1}{4}u(t-\pi/2)(1+\cos 2t).$$

$$w = L^{-1}\left(\frac{1}{p(s)}\right) = \frac{1}{2}\sin 2t. \quad \text{Since } \sin 2(t-\pi) = \sin 2t \text{ and } \sin 2(t-3\pi/2) = -\sin 2t,$$

$$y = \frac{3}{4}\cos 2t - \frac{1}{2}\sin 2t + \frac{1}{4} + \frac{1}{4}u(t-\pi/2)(1+\cos 2t) + \frac{1}{2}u(t-\pi)\sin 2t + \frac{3}{2}u(t-3\pi/2)\sin 2t.$$

$$\begin{aligned} \mathbf{8.7.26.} \quad w(t) &= e^{-t} \sin t; f_h(t) = \frac{u(t-t_0) - u(t-t_0-h)}{h}; (s^2+2s+2)Y_h(s) = \frac{1}{h} \frac{e^{-st_0} - e^{-s(t_0+h)}}{s}; \\ Y_h(s) &= \frac{1}{h} \frac{e^{-st_0} - e^{-s(t_0+h)}}{s(s^2+2s+2)}; \frac{1}{s(s^2+2s+2)} = -\frac{(s+1)+1}{2((s+1)^2+1)} + \frac{1}{2s} \leftrightarrow \frac{1}{2}(1 - e^{-t}(\cos t + \sin t)); \end{aligned}$$

$$y_h(t) = \begin{cases} 0, & 0 \leq t < t_0, \\ \frac{1}{2h} [1 - e^{-(t-t_0)}(\cos(t-t_0) + \sin(t-t_0))], & t_0 \leq t < t_0 + h, \\ \frac{e^{-(t-t_0)}}{2h} [e^h(\cos(t-t_0-h) + \sin(t-t_0-h)) - \cos(t-t_0) - \sin(t-t_0)], & t \geq t_0 + h. \end{cases}$$

$$\begin{aligned} \mathbf{8.7.28.} \quad w(t) &= e^{-t} - e^{-2t}; f_h(t) = \frac{u(t-t_0) - u(t-t_0-h)}{h}; (s^2+3s+2)Y_h(s) = \frac{1}{h} \frac{e^{-st_0} - e^{-s(t_0+h)}}{s}; \\ Y_h(s) &= \frac{1}{h} \frac{e^{-st_0} - e^{-s(t_0+h)}}{s(s+1)(s+2)}; \frac{1}{s(s+1)(s+2)} = \frac{1}{2(s+2)} - \frac{1}{s+1} + \frac{1}{2s} \leftrightarrow \frac{e^{-2t}}{2} - e^{-t} + \frac{1}{2} = \frac{(e^{-t} - 1)^2}{2}; \end{aligned}$$

$$y_h(t) = \begin{cases} 0, & 0 \leq t < t_0, \\ \frac{(e^{-(t-t_0)} - 1)^2}{2h}, & t_0 \leq t < t_0 + h, \\ \frac{(e^{-(t-t_0)} - 1)^2 - (e^{-(t-t_0-h)} - 1)^2}{2h}, & t \geq t_0 + h. \end{cases}$$

$$\begin{aligned} \mathbf{8.7.30.} \quad (\text{a}) \quad (s^2 - 1)\hat{Y}(s) &= 1, \text{ so } \hat{y} = w = L^{-1}\left(\frac{1}{s^2 - 1}\right) = \frac{1}{2}(e^t - e^{-t}); y = \hat{y} + \sum_{k=0}^{\infty} u(t-k)w(t-k) \\ &= \frac{1}{2} \sum_{k=0}^{\infty} u(t-k)(e^{t-k} - e^{-t-k}). \text{ If } m \leq t < m+1, \text{ then } y = \frac{1}{2} \sum_{k=0}^m (e^{t-k} - e^{-t-k}) = \\ &= \frac{1}{2}(e^{t-m} - e^{-t}) \sum_{k=0}^m e^k \frac{e^{m+1} - 1}{2(e-1)}(e^{t-m} - e^{-t}). \end{aligned}$$

$$\begin{aligned} (\text{b}) \quad (s^2 + 1)\hat{Y}(s) &= 1, \text{ so } \hat{y} = w = L^{-1}\left(\frac{1}{s^2 + 1}\right) = \sin t; y = \hat{y} + \sum_{k=0}^{\infty} u(t-2k\pi)w(t-2k\pi) = \\ &= \sin t \sum_{k=0}^{\infty} u(t-2k\pi). \text{ If } 2m\pi \leq t < 2(m+1)\pi, \text{ then } y = (m+1) \sin t. \end{aligned}$$

$$\begin{aligned} (\text{c}) \quad (s^2 - 3s + 2)\hat{Y}(s) &= 1, \text{ so } \hat{y} = w = L^{-1}\left(\frac{1}{(s-1)(s-2)}\right) = (e^{2t} - e^t); y = \hat{y} + \sum_{k=0}^{\infty} u(t-k)w(t-k) \\ &= \sum_{k=0}^{\infty} u(t-k)(e^{2(t-k)} - e^{t-k}). \text{ If } m \leq t < m+1, \text{ then } y = \sum_{k=0}^m (e^{2(t-k)} - e^{t-k}) = \\ &= e^{2(t-m)} \sum_{k=0}^m e^{2k} - e^{t-m} \sum_{k=0}^m e^k = e^{2(t-m)} \frac{e^{2m+2} - 1}{e^2 - 1} - e^{(t-m)} \frac{e^{m+1} - 1}{e - 1}. \end{aligned}$$

$$\begin{aligned} (\text{d}) \quad w &= L^{-1}\left(\frac{1}{s^2 + 1}\right) = \sin t; y = \sum_{k=1}^{\infty} u(t-k\pi)w(t-k\pi) = \sin t \sum_{k=1}^{\infty} (-1)^k u(t-k\pi), \text{ so } y = \\ &= \begin{cases} 0, & 2m\pi \leq t < (2m+1)\pi, \\ -\sin t, & (2m+1)\pi \leq t < (2m+2)\pi, \end{cases} \quad (m = 0, 1, \dots). \end{aligned}$$

# Chapter 9

## Linear Higher Order Equations

### 9.1 Introduction to Linear Higher Order Equations

**9.1.2.** From Example 9.1.1,  $y = c_1x^2 + c_2x^3 + \frac{c_3}{x}$ ,  $y' = 2c_1x + 3c_2x^2 - \frac{c_3}{x^2}$ , and  $y'' = 2c_1 + 6c_2x + \frac{2c_3}{x^3}$ , where

$$\begin{aligned} c_1 - c_2 - c_3 &= 4 \\ -2c_1 + 3c_2 - c_3 &= -14 \\ 2c_1 - 6c_2 - 2c_3 &= 20, \end{aligned}$$

so  $c_1 = 2$ ,  $c_2 = -3$ ,  $c_3 = 1$ , and  $y = 2x^2 - 3x^3 + \frac{1}{x}$ .

**9.1.4.** The general solution of  $y^{(n)} = 0$  can be written as  $y(x) = \sum_{m=0}^{n-1} c_m(x-x_0)^m$ . Since  $y^{(j)}(x) =$

$$\sum_{m=j}^{n-1} m(m-1)\cdots(m-j+1)c_m(x-x_0)^{m-j}, \quad y^{(j)}(x_0) = j!c_j. \quad \text{Therefore, } y_i = \frac{(x-x_0)^{i-1}}{(i-1)!}, \quad 1 \leq i \leq n.$$

**9.1.6.** We omit the verification that the given functions are solutions of the given equations.

(a) The equation is normal on  $(-\infty, \infty)$ .  $W(x) = \begin{vmatrix} e^x & e^{-x} & xe^{-x} \\ e^x & -e^{-x} & e^{-x}(1-x) \\ e^x & e^{-x} & e^{-x}(x-2) \end{vmatrix}; W(0) = \begin{vmatrix} 1 & 1 & 0 \\ 1 & -1 & 1 \\ 1 & 1 & -2 \end{vmatrix} =$

4. Apply Theorem 9.1.4.

(b) The equation is normal on  $(-\infty, \infty)$ .

$$W(x) = \begin{vmatrix} e^x & e^x \cos 2x & e^x \sin 2x \\ e^x & e^x(\cos 2x - 2 \sin 2x) & e^x(2 \cos 2x + \sin 2x) \\ e^x & -e^x(3 \cos 2x + 4 \sin 2x) & e^x(4 \cos 2x - 3 \sin 2x) \end{vmatrix};$$

$$W(0) = \begin{vmatrix} 1 & 1 & 0 \\ 1 & 1 & 2 \\ 1 & -3 & 4 \end{vmatrix} = 8. \quad \text{Apply Theorem 9.1.4.}$$

(c) The equation is normal on  $(-\infty, 0)$  and  $(0, \infty)$ .

$$W(x) = \begin{vmatrix} e^x & e^{-x} & x \\ e^x & -e^{-x} & 1 \\ e^x & e^{-x} & 0 \end{vmatrix} = 2x. \quad \text{Apply Theorem 9.1.4.}$$

(d) The equation is normal on  $(-\infty, 0)$  and  $(0, \infty)$ .

$$W(x) = \begin{vmatrix} e^x/x & e^{-x}/x & 1 \\ e^x(1/x - 1/x^2) & -e^{-x}(x^2 + 2x + 2)/x^2 & 0 \\ e^x(1/x - 2/x^2 + 2/x^3) & e^{-x}(x^2 + 2x + 2)/x^3 & 0 \end{vmatrix} = 2/x^2. \quad \text{Apply Theorem 9.1.4.}$$

(e) The equation is normal on  $(-\infty, \infty)$ .  $W(x) = \begin{vmatrix} x & x^2 & e^x \\ 1 & 2x & e^x \\ 0 & 2 & e^x \end{vmatrix}$ ;  $W(0) = \begin{vmatrix} 0 & 0 & 1 \\ 1 & 0 & 1 \\ 0 & 2 & 1 \end{vmatrix} = 2$ ;

Apply Theorem 9.1.4.

(f) The equation is normal on  $(-\infty, 1/2)$  and  $(1/2, \infty)$ .

$$W(x) = \begin{vmatrix} x & e^x & e^{-x} & e^{2x} \\ 1 & e^x & -e^{-x} & 2e^{2x} \\ 0 & e^x & e^{-x} & 4e^{2x} \\ 0 & e^x & -e^{-x} & 8e^{2x} \end{vmatrix} = e^{2x}(12x - 6). \text{ Apply Theorem 9.1.4.}$$

(g) The equation is normal on  $(-\infty, 0)$  and  $(0, \infty)$ .

$$W(x) = \begin{vmatrix} 1 & x^2 & e^{2x} & e^{-2x} \\ 0 & 2x & 2e^{2x} & -2e^{-2x} \\ 0 & 2 & 4e^{2x} & 4e^{-2x} \\ 0 & 0 & 8e^{2x} & -8e^{-2x} \end{vmatrix} = -128x. \text{ Apply Theorem 9.1.4.}$$

**9.1.8.** From Abel's formula, (A)  $W(x) = W(\pi/2) \exp\left(-\int_{\pi/4}^x \tan t \, dt\right)$ ;  $\int_{\pi/4}^x \tan t \, dt = -\ln \cos t \Big|_{\pi/4}^x = -\ln(\sqrt{2} \cos x)$ ; therefore (A) implies that  $W(x) = \sqrt{2}K \cos x$ .

**9.1.10.** (a)  $W(x) = \begin{vmatrix} 1 & e^x & e^{-x} \\ 0 & e^x & -e^{-x} \\ 0 & e^x & e^{-x} \end{vmatrix} = (e^x)(e^{-x}) \begin{vmatrix} 1 & 1 & 1 \\ 0 & 1 & -1 \\ 0 & 1 & 1 \end{vmatrix} = 2.$

$$\begin{aligned} \text{(b) } W(x) &= \begin{vmatrix} e^x & e^x \sin x & e^x \cos x \\ e^x & e^x(\cos x + \sin x) & e^x(\cos x - \sin x) \\ e^x & 2e^x \cos x & -2e^x \sin x \end{vmatrix} = \\ &= e^{3x} \begin{vmatrix} 1 & \sin x & \cos x \\ 1 & \cos x + \sin x & \cos x - \sin x \\ 1 & 2 \cos x & -2 \sin x \end{vmatrix} \\ &= e^{3x} \begin{vmatrix} 1 & \sin x & \cos x \\ 0 & \cos x & -\sin x \\ 1 & 2 \cos x - \sin x & -2 \sin x - \cos x \end{vmatrix} \\ &= e^{3x} \begin{vmatrix} 1 & \sin x & \cos x \\ 0 & \cos x & -\sin x \\ 0 & -\sin x & -\cos x \end{vmatrix} = -e^{3x} \end{aligned}$$

(c)  $W(x) = \begin{vmatrix} 2 & x+1 & x^2+2 \\ 0 & 1 & 2x \\ 0 & 0 & 2 \end{vmatrix} = 4.$



$$\begin{aligned}
\text{(d) } W(x) &= \begin{vmatrix} x & x \ln |x| & 1/x \\ 1 & \ln |x| + 1 & -1/x^2 \\ 0 & 1/x & 2/x^3 \end{vmatrix} = \begin{vmatrix} 1 & \ln |x| & 1/x^2 \\ 1 & \ln |x| + 1 & -1/x^2 \\ 0 & 1 & 2/x^2 \end{vmatrix} \\
&= \frac{1}{x^2} \begin{vmatrix} 1 & \ln |x| & 1 \\ 1 & \ln |x| + 1 & -1 \\ 0 & 1 & 2 \end{vmatrix} = \frac{1}{x^2} \begin{vmatrix} 1 & \ln |x| & 1 \\ 0 & 1 & -2 \\ 0 & 1 & 2 \end{vmatrix} \\
&= \frac{1}{x^2} \begin{vmatrix} 1 & \ln |x| & 1 \\ 0 & 1 & -2 \\ 0 & 0 & 4 \end{vmatrix} = 4/x^2.
\end{aligned}$$

$$\text{(e) } W(x) = \begin{vmatrix} 1 & x & x^2/2 & x^3/3 & \cdots & x^n/n! \\ 0 & 1 & x & x^2/2 & \cdots & x^{n-1}/(n-1)! \\ 0 & 0 & 1 & x & \cdots & x^{n-2}/(n-2)! \\ 0 & 0 & 0 & 1 & \cdots & x^{n-3}/(n-3)! \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & 1 \end{vmatrix} = 1.$$

$$\text{(f) } W(x) = \begin{vmatrix} e^x & e^{-x} & x \\ e^x & -e^{-x} & 1 \\ e^x & e^{-x} & 0 \end{vmatrix} = \begin{vmatrix} 1 & 1 & x \\ 1 & -1 & 1 \\ 1 & 1 & 0 \end{vmatrix} = \begin{vmatrix} 1 & 1 & x \\ 0 & -2 & 1-x \\ 0 & 0 & -x \end{vmatrix} = 2x.$$

$$\begin{aligned}
\text{(g) } W(x) &= \begin{vmatrix} e^x/x & e^{-x}/x & 1 \\ e^x/x - e^x/x^2 & -e^{-x}/x - e^{-x}/x^2 & 0 \\ e^x/x - 2e^x/x^2 + 2e^x/x^3 & e^{-x}/x + 2e^{-x}/x^2 + 2e^{-x}/x^3 & 0 \end{vmatrix} \\
&= \begin{vmatrix} 1/x & 1/x & 1 \\ 1/x - 1/x^2 & -1/x - 1/x^2 & 0 \\ 1/x - 2/x^2 + 2/x^3 & 1/x + 2/x^2 + 2/x^3 & 0 \end{vmatrix} \\
&= \begin{vmatrix} 1/x - 1/x^2 & -1/x - 1/x^2 \\ 1/x - 2/x^2 + 2/x^3 & 1/x + 2/x^2 + 2/x^3 \end{vmatrix} \\
&= \begin{vmatrix} 1/x - 1/x^2 & -2/x \\ 1/x - 2/x^2 + 2/x^3 & 4/x^2 \end{vmatrix} = 2/x^2.
\end{aligned}$$

$$\begin{aligned}
\text{(h) } W(x) &= \begin{vmatrix} x & x^2 & e^x \\ 1 & 2x & e^x \\ 0 & 2 & e^x \end{vmatrix} = e^x \begin{vmatrix} x & x^2 & 1 \\ 1 & 2x & 1 \\ 0 & 2 & 1 \end{vmatrix} = e^x \left( x \begin{vmatrix} 2x & 1 \\ 2 & 1 \end{vmatrix} - \begin{vmatrix} x^2 & 1 \\ 2 & 1 \end{vmatrix} \right) \\
&= e^x(x^2 - 2x + 2).
\end{aligned}$$

$$\begin{aligned}
\text{(i)} \quad W(x) &= \begin{vmatrix} x & x^3 & 1/x & 1/x^2 \\ 1 & 3x^2 & -1/x^2 & -2/x^3 \\ 0 & 6x & 2/x^3 & 6/x^4 \\ 0 & 6 & -6/x^4 & -24/x^5 \end{vmatrix} = \begin{vmatrix} 0 & -2x^3 & 2/x & 3/x^2 \\ 1 & 3x^2 & -1/x^2 & -2/x^3 \\ 0 & 6x & 2/x^3 & 6/x^4 \\ 0 & 6 & -6/x^4 & -24/x^5 \end{vmatrix} \\
&= - \begin{vmatrix} -2x^3 & 2/x & 3/x^2 \\ 6x & 2/x^3 & 6/x^4 \\ 6 & -6/x^4 & -24/x^5 \end{vmatrix} = -x^4 \begin{vmatrix} -2 & 2/x^4 & 3/x^5 \\ 6 & 2/x^4 & 6/x^5 \\ 6 & -6/x^4 & -24/x^5 \end{vmatrix} \\
&= -x^4 \begin{vmatrix} -2 & 2/x^4 & 3/x^5 \\ 0 & 8/x^4 & 15/x^5 \\ 0 & 0 & -15/x^5 \end{vmatrix} = -240/x^5.
\end{aligned}$$

$$\begin{aligned}
\text{(j)} \quad W(x) &= \begin{vmatrix} e^x & e^{-x} & x & e^{2x} \\ e^x & -e^{-x} & 1 & 2e^{2x} \\ e^x & e^{-x} & 0 & 4e^{2x} \\ e^x & -e^{-x} & 0 & 8e^{2x} \end{vmatrix} = e^{2x} \begin{vmatrix} 1 & 1 & x & 1 \\ 1 & -1 & 1 & 2 \\ 1 & 1 & 0 & 4 \\ 1 & -1 & 0 & 8 \end{vmatrix} \\
&= e^{2x} \begin{vmatrix} 1-x & 1+x & 0 & 1-2x \\ 1 & -1 & 1 & 2 \\ 1 & 1 & 0 & 4 \\ 1 & -1 & 0 & 8 \end{vmatrix} = -e^{2x} \begin{vmatrix} 1-x & 1+x & 1-2x \\ 1 & 1 & 4 \\ 1 & -1 & 8 \end{vmatrix} \\
&= -e^{2x} \begin{vmatrix} 2 & 1+x & 1-2x \\ 2 & 1 & 4 \\ 0 & -1 & 8 \end{vmatrix} = -e^{2x} \begin{vmatrix} 2 & 1+x & 1-2x \\ 0 & -x & 3+2x \\ 0 & -1 & 8 \end{vmatrix} = 6e^{2x}(2x-1).
\end{aligned}$$

$$\begin{aligned}
\text{(k)} \quad W(x) &= \begin{vmatrix} e^{2x} & e^{-2x} & 1 & x^2 \\ 2e^{2x} & -2e^{-2x} & 0 & 2x \\ 4e^{2x} & 4e^{-2x} & 0 & 2 \\ 8e^{2x} & -8e^{-2x} & 0 & 0 \end{vmatrix} = \begin{vmatrix} 1 & 1 & 1 & x^2 \\ 2 & -2 & 0 & 2x \\ 4 & 4 & 0 & 2 \\ 8 & -8 & 0 & 0 \end{vmatrix} \\
&= \begin{vmatrix} 2 & -2 & 2x \\ 4 & 4 & 2 \\ 8 & -8 & 0 \end{vmatrix} = \begin{vmatrix} 0 & -2 & 2x \\ 8 & 4 & 2 \\ 0 & -8 & 0 \end{vmatrix} = -128x.
\end{aligned}$$

**9.1.12.** Let  $y$  be an arbitrary solution of  $Ly = 0$  on  $(a, b)$ . Since  $\{z_1, \dots, z_n\}$  is a fundamental set of solutions of  $Ly = 0$  on  $(a, b)$ , there are constants  $c_1, c_2, \dots, c_n$  such that  $y = \sum_{i=1}^n c_i y_i$ . Therefore,

$y = \sum_{i=1}^n c_i \sum_{j=1}^n a_{ij} y_j = \sum_{j=1}^n C_j y_j$ , with  $C_j = \sum_{i=1}^n a_{ij} c_i$ . Hence  $\{y_1, \dots, y_n\}$  is a fundamental set of solutions of  $Ly = 0$  on  $(a, b)$ .

**9.1.14.** Let  $y$  be a given solution of  $Ly = 0$  and  $z = \sum_{j=1}^n y^{(j-1)}(x_0) y_j$ . Then  $z^{(r)}(x_0) = y^{(r)}(x_0)$ ,  $r = 0, \dots, n-1$ . Since the solution of every initial value problem is unique (Theorem 9.1.1),  $z = y$ .

**9.1.16.** If  $\{y_1, y_2, \dots, y_n\}$  is linearly dependent on  $(a, b)$  there are constants  $c_1, \dots, c_n$ , not all zeros, such that  $c_1 y_1 + c_2 y_2 + \dots + c_n y_n = 0$ . Let  $k$  be the smallest integer such that  $c_k \neq 0$ . If  $k = 1$ , then

$y_1 = \frac{1}{c_1}(c_2y_2 + \cdots + c_ny_n)$ ; if  $1 < k < n$ , then  $y_k = 0 \cdot y_1 + \cdots + 0 \cdot y_{k-1} + \frac{1}{c_k}(c_{k+1}y_{k+1} + \cdots + c_ny_n)$ ; if  $k = n$ , then  $y_n = 0$ , so  $y_n = 0 \cdot y_1 + 0 \cdot y_2 + \cdots + 0 \cdot y_n$ .

**9.1.18.** Since  $F = \sum \pm f_{1i_1} f_{2i_2}, \dots, f_{ni_n}$ ,

$$\begin{aligned} F' &= \sum \pm f'_{1i_1} f_{2i_2}, \dots, f_{ni_n} + \sum \pm f_{1i_1} f'_{2i_2}, \dots, f_{ni_n} + \cdots + \sum \pm f_{1i_1} f_{2i_2}, \dots, f'_{ni_n} \\ &= F_1 + F_2 + \cdots + F_n. \end{aligned}$$

**9.1.20.** Since  $y_j^{(n)} = -\sum_{k=1}^n (P_k/P_0)y_j^{(n-k)}$ , Exercise 9.1.19 implies that

$$W' = - \begin{vmatrix} y_1 & y_2 & \cdots & y_n \\ y'_1 & y'_2 & \cdots & y'_n \\ \vdots & \vdots & \ddots & \vdots \\ y_1^{(n-2)} & y_2^{(n-2)} & \cdots & y_n^{(n-2)} \\ \sum_{k=1}^n (P_k/P_0)y_1^{(n-k)} & \sum_{k=1}^n (P_k/P_0)y_2^{(n-k)} & \cdots & \sum_{k=1}^n (P_k/P_0)y_n^{(n-k)} \end{vmatrix},$$

so Exercise 9.1.17 implies that

$$W' = - \sum_{k=1}^n \frac{P_k}{P_0} \begin{vmatrix} y_1 & y_2 & \cdots & y_n \\ y'_1 & y'_2 & \cdots & y'_n \\ \vdots & \vdots & \ddots & \vdots \\ y_1^{(n-2)} & y_2^{(n-2)} & \cdots & y_n^{(n-2)} \\ y_1^{(n-k)} & y_2^{(n-k)} & \cdots & y_n^{(n-k)} \end{vmatrix}.$$

However, the determinants on the right each have two identical rows if  $k = 2, \dots, n$ . Therefore,  $W' = -\frac{P_1 W}{P_0}$ . Separating variables yields  $\frac{W'}{W} = -\frac{P_1}{P_0}$ ; hence  $\ln \frac{W(x)}{W(x_0)} = -\int_{x_0}^x \frac{P_1(t)}{P_0(t)} dt$ , which implies

Abel's formula.

**9.1.22.** See the proof of Theorem 5.3.3.

**9.1.24.** (a)

$$P_0(x) = - \begin{vmatrix} x & x^2 - 1 & x^2 + 1 \\ 1 & 2x & 2x \\ 0 & 2 & 2 \end{vmatrix} = - \begin{vmatrix} x & x^2 - 1 & x^2 + 1 \\ 1 & 0 & 0 \\ 0 & 2 & 2 \end{vmatrix} = \begin{vmatrix} x^2 - 1 & x^2 + 1 \\ 2 & 2 \end{vmatrix} = -4;$$

$$P_1(x) = \begin{vmatrix} x & x^2 - 1 & x^2 + 1 \\ 1 & 2x & 2x \\ 0 & 0 & 0 \end{vmatrix} = 0; P_2(x) = - \begin{vmatrix} x & x^2 - 1 & x^2 + 1 \\ 0 & 2 & 2 \\ 0 & 0 & 0 \end{vmatrix} = 0; P_3(x) = \begin{vmatrix} 1 & 2x & 2x \\ 0 & 2 & 2 \\ 0 & 0 & 0 \end{vmatrix} =$$

0. Therefore,  $-4y''' = 0$ , which is equivalent to  $y''' = 0$ .

(b)

$$P_0 = - \begin{vmatrix} e^x & e^{-x} & x \\ e^x & -e^{-x} & 1 \\ e^x & e^{-x} & 0 \end{vmatrix} = - \begin{vmatrix} 1 & 1 & x \\ 1 & -1 & 1 \\ 1 & 1 & 0 \end{vmatrix} = - \begin{vmatrix} 0 & 0 & x \\ 0 & -2 & 1 \\ 1 & 1 & 0 \end{vmatrix} = -2x;$$

$$\begin{aligned}
P_1 &= \begin{vmatrix} e^x & e^{-x} & x \\ e^x & -e^{-x} & 1 \\ e^x & -e^{-x} & 0 \end{vmatrix} = \begin{vmatrix} 1 & 1 & x \\ 1 & -1 & 1 \\ 1 & -1 & 0 \end{vmatrix} = \begin{vmatrix} 0 & 2 & x \\ 0 & 0 & 1 \\ 1 & -1 & 0 \end{vmatrix} = 2; \\
P_2 &= - \begin{vmatrix} e^x & e^{-x} & x \\ e^x & e^{-x} & 0 \\ e^x & -e^{-x} & 0 \end{vmatrix} = - \begin{vmatrix} 1 & 1 & x \\ 1 & 1 & 0 \\ 1 & -1 & 0 \end{vmatrix} = - \begin{vmatrix} 0 & 0 & x \\ 1 & 1 & 0 \\ 0 & -2 & 0 \end{vmatrix} = 2x; \\
P_3 &= \begin{vmatrix} e^x & -e^{-x} & 1 \\ e^x & e^{-x} & 0 \\ e^x & -e^{-x} & 0 \end{vmatrix} = \begin{vmatrix} 1 & -1 & 1 \\ 1 & 1 & 0 \\ 1 & -1 & 0 \end{vmatrix} = \begin{vmatrix} 0 & 0 & 1 \\ 1 & 2 & 0 \\ 1 & -1 & 0 \end{vmatrix} = -2.
\end{aligned}$$

Therefore,  $-2xy''' + 2y'' + 2xy' - 2y = 0$ , which is equivalent to  $xy''' - y'' - xy' + y = 0$ .

(c)

$$\begin{aligned}
P_0(x) &= - \begin{vmatrix} e^x & xe^{-x} & 1 \\ e^x & e^{-x}(1-x) & 0 \\ e^x & e^{-x}(x-2) & 0 \end{vmatrix} = - \begin{vmatrix} 1 & x & 1 \\ 1 & 1-x & 0 \\ 1 & x-2 & 0 \end{vmatrix} = - \begin{vmatrix} 1 & 1-x \\ 1 & x-2 \end{vmatrix} = 3-2x; \\
P_1(x) &= \begin{vmatrix} e^x & xe^{-x} & 1 \\ e^x & e^{-x}(1-x) & 0 \\ e^x & e^{-x}(3-x) & 0 \end{vmatrix} = \begin{vmatrix} 1 & x & 1 \\ 1 & 1-x & 0 \\ 1 & 3-x & 0 \end{vmatrix} = \begin{vmatrix} 1 & 1-x \\ 1 & 3-x \end{vmatrix} = 2; \\
P_2(x) &= - \begin{vmatrix} e^x & xe^{-x} & 1 \\ e^x & e^{-x}(x-2) & 0 \\ e^x & e^{-x}(3-x) & 0 \end{vmatrix} = - \begin{vmatrix} 1 & x & 1 \\ 1 & x-2 & 0 \\ 1 & 3-x & 0 \end{vmatrix} = - \begin{vmatrix} 1 & x-2 \\ 1 & 3-x \end{vmatrix} = 2x-5; \\
P_3(x) &= \begin{vmatrix} e^x & e^{-x}(1-x) & 0 \\ e^x & e^{-x}(x-2) & 0 \\ e^x & e^{-x}(3-x) & 0 \end{vmatrix} = 0.
\end{aligned}$$

Therefore,  $(3-2x)y''' + 2y'' + (2x-5)y' = 0$ .

(d)

$$\begin{aligned}
P_0(x) &= - \begin{vmatrix} x & x^2 & e^x \\ 1 & 2x & e^x \\ 0 & 2 & e^x \end{vmatrix} = -e^x \begin{vmatrix} x & x^2 & 1 \\ 1 & 2x & 1 \\ 0 & 2 & 1 \end{vmatrix} = -e^x \begin{vmatrix} x & x^2-2 & 0 \\ 1 & 2x-2 & 0 \\ 0 & 2 & 1 \end{vmatrix} \\
&= -e^x \begin{vmatrix} x & x^2-2 \\ 1 & 2x-2 \end{vmatrix} = -e^x(x^2-2x+2); \\
P_1(x) &= \begin{vmatrix} x & x^2 & e^x \\ 1 & 2x & e^x \\ 0 & 0 & e^x \end{vmatrix} = e^x \begin{vmatrix} x & x^2 \\ 1 & 2x \end{vmatrix} = x^2e^x; \\
P_2(x) &= - \begin{vmatrix} x & x^2 & e^x \\ 0 & 2 & e^x \\ 0 & 0 & e^x \end{vmatrix} = -e^x \begin{vmatrix} x & x^2 \\ 0 & 2 \end{vmatrix} = -2xe^x; \\
P_3(x) &= \begin{vmatrix} 1 & 2x & e^x \\ 0 & 2 & e^x \\ 0 & 0 & e^x \end{vmatrix} = e^x \begin{vmatrix} 1 & 2x \\ 0 & 2 \end{vmatrix} = 2e^x.
\end{aligned}$$

Therefore,  $-e^x(x^2 - 2x + 2)y''' + x^2e^xy'' - 2xe^xy' + 2e^xy = 0$ ; which is equivalent to  $(x^2 - 2x + 2)y''' - x^2y'' + 2xy' - 2y = 0$ .

(e)

$$\begin{aligned} P_0(x) &= - \begin{vmatrix} x & x^2 & 1/x \\ 1 & 2x & -1/x^2 \\ 0 & 2 & 2/x^3 \end{vmatrix} = -x \begin{vmatrix} 1 & x & 1/x^2 \\ 1 & 2x & -1/x^2 \\ 0 & 2 & 2/x^3 \end{vmatrix} = -x \begin{vmatrix} 1 & x & 1/x^2 \\ 0 & x & -2/x^2 \\ 0 & 2 & 2/x^3 \end{vmatrix} \\ &= -x \begin{vmatrix} x & -2/x^2 \\ 2 & 2/x^3 \end{vmatrix} = -\frac{6}{x}; \end{aligned}$$

$$P_1(x) = \begin{vmatrix} x & x^2 & 1/x \\ 1 & 2x & -1/x^2 \\ 0 & 0 & -6/x^4 \end{vmatrix} = -\frac{6}{x^4} \begin{vmatrix} x & x^2 \\ 1 & 2x \end{vmatrix} = -\frac{6}{x^2};$$

$$P_2(x) = - \begin{vmatrix} x & x^2 & 1/x \\ 0 & 2 & 2/x^3 \\ 0 & 0 & -6/x^4 \end{vmatrix} = \frac{6}{x^4} \begin{vmatrix} x & x^2 \\ 0 & 2 \end{vmatrix} = \frac{12}{x^3};$$

$$P_3(x) = \begin{vmatrix} 1 & 2x & -1/x^2 \\ 0 & 2 & 2/x^3 \\ 0 & 0 & -6/x^4 \end{vmatrix} = -\frac{6}{x^4} \begin{vmatrix} 1 & 2x \\ 0 & 2 \end{vmatrix} = -\frac{12}{x^4}.$$

Therefore,  $-\frac{6}{x}y''' - \frac{6}{x^2}y'' + \frac{12}{x^3}y' - \frac{12}{x^4}y = 0$ , which is equivalent to  $x^3y''' + x^2y'' - 2xy' + 2y = 0$ .

(f)

$$\begin{aligned} P_0(x) &= - \begin{vmatrix} x+1 & e^x & e^{3x} \\ 1 & e^x & 3e^{3x} \\ 0 & e^x & 9e^{3x} \end{vmatrix} = -e^{4x} \begin{vmatrix} x+1 & 1 & 1 \\ 1 & 1 & 3 \\ 0 & 1 & 9 \end{vmatrix} = -e^{4x} \begin{vmatrix} x+1 & 0 & -8 \\ 1 & 0 & -6 \\ 0 & 1 & 9 \end{vmatrix} \\ &= e^{4x} \begin{vmatrix} x+1 & -8 \\ 1 & -6 \end{vmatrix} = 2e^{4x}(1-3x); \end{aligned}$$

$$\begin{aligned} P_1(x) &= \begin{vmatrix} x+1 & e^x & e^{3x} \\ 1 & e^x & 3e^{3x} \\ 0 & e^x & 27e^{3x} \end{vmatrix} = e^{4x} \begin{vmatrix} x+1 & 1 & 1 \\ 1 & 1 & 3 \\ 0 & 1 & 27 \end{vmatrix} = e^{4x} \begin{vmatrix} x+1 & 0 & -26 \\ 1 & 0 & -24 \\ 0 & 1 & 27 \end{vmatrix} \\ &= -e^{4x} \begin{vmatrix} x+1 & -26 \\ 1 & -24 \end{vmatrix} = 2e^{4x}(12x-1); \end{aligned}$$

$$\begin{aligned} P_2(x) &= - \begin{vmatrix} x+1 & e^x & e^{3x} \\ 0 & e^x & 9e^{3x} \\ 0 & e^x & 27e^{3x} \end{vmatrix} = -e^{4x} \begin{vmatrix} x+1 & 1 & 1 \\ 0 & 1 & 9 \\ 0 & 1 & 27 \end{vmatrix} \\ &= -e^{4x} \begin{vmatrix} x+1 & 1 & 1 \\ 0 & 1 & 9 \\ 0 & 0 & 18 \end{vmatrix} = -18e^{4x}(x+1); \end{aligned}$$

$$P_3(x) = \begin{vmatrix} 1 & e^x & 3e^{3x} \\ 0 & e^x & 9e^{3x} \\ 0 & e^x & 27e^{3x} \end{vmatrix} = e^{4x} \begin{vmatrix} 1 & 1 & 3 \\ 0 & 1 & 9 \\ 0 & 1 & 27 \end{vmatrix} = e^{4x} \begin{vmatrix} 1 & 1 & 3 \\ 0 & 1 & 9 \\ 0 & 0 & 18 \end{vmatrix} = 18e^{4x}.$$

Therefore,

$$2e^{4x}(1-3x)y''' + 2e^{4x}(12x-1)y'' - 18e^{4x}(x+1)y' + 18e^{4x}y = 0,$$

which is equivalent to

$$(3x-1)y''' - (12x-1)y'' + 9(x+1)y' - 9y = 0.$$

(g)

$$\begin{aligned} P_0(x) &= \begin{vmatrix} x & x^3 & 1/x & 1/x^2 \\ 1 & 3x^2 & -1/x^2 & -2/x^3 \\ 0 & 6x & 2/x^3 & 6/x^4 \\ 0 & 6 & -6/x^4 & -24/x^5 \end{vmatrix} = x \begin{vmatrix} 1 & x^2 & 1/x^2 & 1/x^3 \\ 1 & 3x^2 & -1/x^2 & -2/x^3 \\ 0 & 6x & 2/x^3 & 6/x^4 \\ 0 & 6 & -6/x^4 & -24/x^5 \end{vmatrix} \\ &= x \begin{vmatrix} 1 & x^2 & 1/x^2 & 1/x^3 \\ 0 & 2x^2 & -2/x^2 & -3/x^3 \\ 0 & 6x & 2/x^3 & 6/x^4 \\ 0 & 6 & -6/x^4 & -24/x^5 \end{vmatrix} = x \begin{vmatrix} 2x^2 & -2/x^2 & -3/x^3 \\ 6x & 2/x^3 & 6/x^4 \\ 6 & -6/x^4 & -24/x^5 \end{vmatrix} \\ &= x^2 \begin{vmatrix} 2x & -2/x^3 & -3/x^4 \\ 6x & 2/x^3 & 6/x^4 \\ 6 & -6/x^4 & -24/x^5 \end{vmatrix} = x^2 \begin{vmatrix} 2x & -2/x^3 & -3/x^4 \\ 0 & 8/x^3 & 15/x^4 \\ 0 & 0 & -15/x^5 \end{vmatrix} = -\frac{240}{x^5} \end{aligned}$$

$$\begin{aligned} P_1(x) &= - \begin{vmatrix} x & x^3 & 1/x & 1/x^2 \\ 1 & 3x^2 & -1/x^2 & -2/x^3 \\ 0 & 6x & 2/x^3 & 6/x^4 \\ 0 & 0 & 24/x^5 & 120/x^6 \end{vmatrix} = -x \begin{vmatrix} 1 & x^2 & 1/x^2 & 1/x^3 \\ 1 & 3x^2 & -1/x^2 & -2/x^3 \\ 0 & 6x & 2/x^3 & 6/x^4 \\ 0 & 0 & 24/x^5 & 120/x^6 \end{vmatrix} \\ &= -x \begin{vmatrix} 1 & x^2 & 1/x^2 & 1/x^3 \\ 0 & 2x^2 & -2/x^2 & -3/x^3 \\ 0 & 6x & 2/x^3 & 6/x^4 \\ 0 & 0 & 24/x^5 & 120/x^6 \end{vmatrix} = -x^2 \begin{vmatrix} 2x & -2/x^3 & -3/x^4 \\ 6x & 2/x^3 & 6/x^4 \\ 0 & 24/x^5 & 120/x^6 \end{vmatrix} \\ &= -x^2 \begin{vmatrix} 2x & -2/x^3 & -3/x^4 \\ 0 & 8/x^3 & 15/x^4 \\ 0 & 24/x^5 & 120/x^6 \end{vmatrix} = -2x^3 \begin{vmatrix} 8/x^3 & 15/x^4 \\ 24/x^5 & 120/x^6 \end{vmatrix} = -\frac{1200}{x^6}; \end{aligned}$$

$$\begin{aligned}
P_2(x) &= \begin{vmatrix} x & x^3 & 1/x & 1/x^2 \\ 1 & 3x^2 & -1/x^2 & -2/x^3 \\ 0 & 6 & -6/x^4 & -24/x^5 \\ 0 & 0 & 24/x^5 & 120/x^6 \end{vmatrix} = x \begin{vmatrix} 1 & x^2 & 1/x^2 & 1/x^3 \\ 1 & 3x^2 & -1/x^2 & -2/x^3 \\ 0 & 6 & -6/x^4 & -24/x^5 \\ 0 & 0 & 24/x^5 & 120/x^6 \end{vmatrix} \\
&= x \begin{vmatrix} 1 & x^2 & 1/x^2 & 1/x^3 \\ 0 & 2x^2 & -2/x^2 & -3/x^3 \\ 0 & 6 & -6/x^4 & -24/x^5 \\ 0 & 0 & 24/x^5 & 120/x^6 \end{vmatrix} = x^3 \begin{vmatrix} 2 & -2/x^4 & -3/x^5 \\ 6 & -6/x^4 & -24/x^5 \\ 0 & 24/x^5 & 120/x^6 \end{vmatrix} \\
&= x^3 \begin{vmatrix} 2 & -2/x^4 & -3/x^5 \\ 0 & 0 & -15/x^5 \\ 0 & 24/x^5 & 120/x^6 \end{vmatrix} = 2x^3 \begin{vmatrix} 0 & -15/x^5 \\ 24/x^5 & 120/x^6 \end{vmatrix} = \frac{720}{x^7};
\end{aligned}$$

$$\begin{aligned}
P_3(x) &= - \begin{vmatrix} x & x^3 & 1/x & 1/x^2 \\ 0 & 6x & 2/x^3 & 6/x^4 \\ 0 & 6 & -6/x^4 & -24/x^5 \\ 0 & 0 & 24/x^5 & 120/x^6 \end{vmatrix} = -x^2 \begin{vmatrix} 6 & 2/x^4 & 6/x^5 \\ 6 & -6/x^4 & -24/x^5 \\ 0 & 24/x^5 & 120/x^6 \end{vmatrix} \\
&= -x^2 \begin{vmatrix} 6 & 2/x^4 & 6/x^5 \\ 0 & -8/x^4 & -30/x^5 \\ 0 & 24/x^5 & 120/x^6 \end{vmatrix} = -6x^2 \begin{vmatrix} -8/x^4 & -30/x^5 \\ 24/x^5 & 120/x^6 \end{vmatrix} = \frac{1440}{x^6};
\end{aligned}$$

$$\begin{aligned}
P_4(x) &= \begin{vmatrix} 1 & 3x^2 & -1/x^2 & -2/x^3 \\ 0 & 6x & 2/x^3 & 6/x^4 \\ 0 & 6 & -6/x^4 & -24/x^5 \\ 0 & 0 & 24/x^5 & 120/x^6 \end{vmatrix} = x \begin{vmatrix} 6 & 2/x^4 & 6/x^5 \\ 6 & -6/x^4 & -24/x^5 \\ 0 & 0 & 24/x^5 & 120/x^6 \end{vmatrix} \\
&= x \begin{vmatrix} 6 & 2/x^4 & 6/x^5 \\ 0 & -8/x^4 & -30/x^5 \\ 0 & 0 & 24/x^5 & 120/x^6 \end{vmatrix} = 6x \begin{vmatrix} -8/x^4 & -30/x^5 \\ 24/x^5 & 120/x^6 \end{vmatrix} = -\frac{1440}{x^9}.
\end{aligned}$$

Therefore,

$$-\frac{240}{x^5}y^{(4)} - \frac{1200}{x^6}y''' + \frac{720}{x^7}y'' + \frac{1440}{x^8}y' - \frac{1440}{x^9}y = 0,$$

which is equivalent to  $x^4y^{(4)} + 5x^3y''' - 3x^2y'' - 6xy' + 6y = 0$ .

(h)

$$\begin{aligned}
P_0(x) &= \begin{vmatrix} x & x \ln|x| & 1/x & x^2 \\ 1 & \ln|x| + 1 & -1/x^2 & 2x \\ 0 & 1/x & 2/x^3 & 2 \\ 0 & -1/x^2 & -6/x^4 & 0 \end{vmatrix} = x \begin{vmatrix} 1 & \ln|x| & 1/x^2 & x \\ 1 & \ln|x| + 1 & -1/x^2 & 2x \\ 0 & 1/x & 2/x^3 & 2 \\ 0 & -1/x^2 & -6/x^4 & 0 \end{vmatrix} \\
&= x \begin{vmatrix} 1 & \ln|x| & 1/x^2 & x \\ 0 & 1 & -2/x^2 & x \\ 0 & 1/x & 2/x^3 & 2 \\ 0 & -1/x^2 & -6/x^4 & 0 \end{vmatrix} = x \begin{vmatrix} 1 & -2/x^2 & x \\ 1/x & 2/x^3 & 2 \\ -1/x^2 & -6/x^4 & 0 \end{vmatrix} \\
&= x \begin{vmatrix} 1 & -2/x^2 & x \\ 1/x & 2/x^3 & 2 \\ -1/x^2 & -6/x^4 & 0 \end{vmatrix} = x \begin{vmatrix} 1 & -2/x^2 & x \\ 0 & 4/x^3 & 1 \\ 0 & -8/x^4 & 1/x \end{vmatrix} = x \begin{vmatrix} 4/x^3 & 1 \\ -8/x^4 & 1/x \end{vmatrix} = \frac{12}{x^3};
\end{aligned}$$

$$\begin{aligned}
P_1(x) &= - \begin{vmatrix} x & x \ln |x| & 1/x & x^2 \\ 1 & \ln |x| + 1 & -1/x^2 & 2x \\ 0 & 1/x & 2/x^3 & 2 \\ 0 & 2/x^3 & 24/x^5 & 0 \end{vmatrix} = -x \begin{vmatrix} 1 & \ln |x| & 1/x^2 & x \\ 1 & \ln |x| + 1 & -1/x^2 & 2x \\ 0 & 1/x & 2/x^3 & 2 \\ 0 & 2/x^3 & 24/x^5 & 0 \end{vmatrix} \\
&= -x \begin{vmatrix} 1 & \ln |x| & 1/x^2 & x \\ 0 & 1 & -2/x^2 & x \\ 0 & 1/x & 2/x^3 & 2 \\ 0 & 2/x^3 & 24/x^5 & 0 \end{vmatrix} = -x \begin{vmatrix} 1 & -2/x^2 & x \\ 1/x & 2/x^3 & 2 \\ 2/x^3 & 24/x^5 & 0 \end{vmatrix} \\
&= -x \begin{vmatrix} 1 & -2/x^2 & x \\ 0 & 4/x^3 & 1 \\ 0 & 28/x^5 & -2/x^2 \end{vmatrix} = -x \begin{vmatrix} 4/x^3 & 1 \\ 28/x^5 & -2/x^2 \end{vmatrix} = \frac{36}{x^4};
\end{aligned}$$

$$\begin{aligned}
P_2(x) &= \begin{vmatrix} x & x \ln |x| & 1/x & x^2 \\ 1 & \ln |x| + 1 & -1/x^2 & 2x \\ 0 & -1/x^2 & -6/x^4 & 0 \\ 0 & 2/x^3 & 24/x^5 & 0 \end{vmatrix} = x \begin{vmatrix} 1 & \ln |x| & 1/x^2 & x \\ 1 & \ln |x| + 1 & -1/x^2 & 2x \\ 0 & -1/x^2 & -6/x^4 & 0 \\ 0 & 2/x^3 & 24/x^5 & 0 \end{vmatrix} \\
&= x \begin{vmatrix} 1 & \ln |x| & 1/x^2 & x \\ 0 & 1 & -2/x^2 & x \\ 0 & -1/x^2 & -6/x^4 & 0 \\ 0 & 2/x^3 & 24/x^5 & 0 \end{vmatrix} = x \begin{vmatrix} 1 & -2/x^2 & x \\ -1/x^2 & -6/x^4 & 0 \\ 2/x^3 & 24/x^5 & 0 \end{vmatrix} \\
&= x \begin{vmatrix} 1 & -2/x^2 & x \\ 0 & -8/x^4 & 1/x \\ 0 & 28/x^5 & -2/x^2 \end{vmatrix} = x \begin{vmatrix} -8/x^4 & 1/x \\ 28/x^5 & -2/x^2 \end{vmatrix} = -\frac{12}{x^5};
\end{aligned}$$

$$\begin{aligned}
P_3(x) &= - \begin{vmatrix} x & x \ln |x| & 1/x & x^2 \\ 0 & 1/x & 2/x^3 & 2 \\ 0 & -1/x^2 & -6/x^4 & 0 \\ 0 & 2/x^3 & 24/x^5 & 0 \end{vmatrix} = -x \begin{vmatrix} 1/x & 2/x^3 & 2 \\ -1/x^2 & -6/x^4 & 0 \\ 2/x^3 & 24/x^5 & 0 \end{vmatrix} \\
&= -x \begin{vmatrix} 1/x & 2/x^3 & 2 \\ 0 & -4/x^4 & 2/x \\ 0 & 20/x^5 & -4/x^2 \end{vmatrix} = - \begin{vmatrix} -4/x^4 & 2/x \\ 20/x^5 & -4/x^2 \end{vmatrix} = \frac{24}{x^6};
\end{aligned}$$

$$\begin{aligned}
P_4(x) &= \begin{vmatrix} 1 & \ln |x| + 1 & -1/x^2 & 2x \\ 0 & 1/x & 2/x^3 & 2 \\ 0 & -1/x^2 & -6/x^4 & 0 \\ 0 & 2/x^3 & 24/x^5 & 0 \end{vmatrix} = \begin{vmatrix} 1/x & 2/x^3 & 2 \\ -1/x^2 & -6/x^4 & 0 \\ 2/x^3 & 24/x^5 & 0 \end{vmatrix} \\
&= 2 \begin{vmatrix} -1/x^2 & -6/x^4 \\ 2/x^3 & 24/x^5 \end{vmatrix} = -\frac{24}{x^7}.
\end{aligned}$$

Therefore,

$$\frac{12}{x^3}y^{(4)} + \frac{36}{x^4}y''' - \frac{12}{x^5}y'' + \frac{24}{x^6}y' - \frac{24}{x^7}y = 0,$$

which is equivalent to  $x^4y^{(4)} + 3x^2y''' - x^2y'' + 2xy' - 2y = 0$ .



(i)

$$\begin{aligned}
P_0(x) &= \begin{vmatrix} e^x & e^{-x} & x & e^{2x} \\ e^x & -e^{-x} & 1 & 2e^{2x} \\ e^x & e^{-x} & 0 & 4e^{2x} \\ e^x & -e^{-x} & 0 & 8e^{2x} \end{vmatrix} = e^{2x} \begin{vmatrix} 1 & 1 & x & 1 \\ 1 & -1 & 1 & 2 \\ 1 & 1 & 0 & 4 \\ 1 & -1 & 0 & 8 \end{vmatrix} \\
&= e^{2x} \begin{vmatrix} 2 & 1 & x & 1 \\ 0 & -1 & 1 & 2 \\ 2 & 1 & 0 & 4 \\ 0 & -1 & 0 & 8 \end{vmatrix} = e^{2x} \begin{vmatrix} 2 & 1 & x & 1 \\ 0 & -1 & 1 & 2 \\ 0 & 0 & -x & 3 \\ 0 & -1 & 0 & 8 \end{vmatrix} \\
&= 2e^{2x} \begin{vmatrix} -1 & 1 & 2 \\ 0 & -x & 3 \\ -1 & 0 & 8 \end{vmatrix} = 2e^{2x} \begin{vmatrix} -1 & 1 & 2 \\ 0 & -x & 3 \\ 0 & -1 & 6 \end{vmatrix} \\
&= -2e^{2x} \begin{vmatrix} -x & 3 \\ -1 & 6 \end{vmatrix} = 6e^{2x}(2x-1);
\end{aligned}$$

$$\begin{aligned}
P_1(x) &= - \begin{vmatrix} e^x & e^{-x} & x & e^{2x} \\ e^x & -e^{-x} & 1 & 2e^{2x} \\ e^x & e^{-x} & 0 & 4e^{2x} \\ e^x & e^{-x} & 0 & 16e^{2x} \end{vmatrix} = -e^{2x} \begin{vmatrix} 1 & 1 & x & 1 \\ 1 & -1 & 1 & 2 \\ 1 & 1 & 0 & 4 \\ 1 & 1 & 0 & 16 \end{vmatrix} \\
&= -e^{2x} \begin{vmatrix} 0 & 1 & x & 1 \\ 2 & -1 & 1 & 2 \\ 0 & 1 & 0 & 4 \\ 0 & 1 & 0 & 16 \end{vmatrix} = 2e^{2x} \begin{vmatrix} 1 & x & 1 \\ 1 & 0 & 4 \\ 1 & 0 & 16 \end{vmatrix} \\
&= 2e^{2x} \begin{vmatrix} 1 & x & 1 \\ 0 & -x & 3 \\ 0 & -x & 15 \end{vmatrix} = 2e^{2x} \begin{vmatrix} -x & 3 \\ -x & 15 \end{vmatrix} = -24xe^{2x};
\end{aligned}$$

$$\begin{aligned}
P_2(x) &= \begin{vmatrix} e^x & e^{-x} & x & e^{2x} \\ e^x & -e^{-x} & 1 & 2e^{2x} \\ e^x & -e^{-x} & 0 & 8e^{2x} \\ e^x & e^{-x} & 0 & 16e^{2x} \end{vmatrix} = e^{2x} \begin{vmatrix} 1 & 1 & x & 1 \\ 1 & -1 & 1 & 2 \\ 1 & -1 & 0 & 8 \\ 1 & 1 & 0 & 16 \end{vmatrix} \\
&= e^{2x} \begin{vmatrix} 0 & 1 & x & 1 \\ 2 & -1 & 1 & 2 \\ 2 & -1 & 0 & 8 \\ 0 & 1 & 0 & 16 \end{vmatrix} = e^{2x} \begin{vmatrix} 0 & 1 & x & 1 \\ 2 & -1 & 1 & 2 \\ 0 & 0 & -1 & 6 \\ 0 & 1 & 0 & 16 \end{vmatrix} = -2e^{2x} \begin{vmatrix} 1 & x & 1 \\ 0 & -1 & 6 \\ 1 & 0 & 16 \end{vmatrix} \\
&= -2e^{2x} \begin{vmatrix} 1 & x & 1 \\ 0 & -1 & 6 \\ 0 & -x & 15 \end{vmatrix} = -2e^{2x} \begin{vmatrix} -1 & 6 \\ -x & 15 \end{vmatrix} = 6e^{2x}(5-2x);
\end{aligned}$$

$$\begin{aligned}
P_3(x) &= - \begin{vmatrix} e^x & e^{-x} & x & e^{2x} \\ e^x & e^{-x} & 0 & 4e^{2x} \\ e^x & -e^{-x} & 0 & 8e^{2x} \\ e^x & e^{-x} & 0 & 16e^{2x} \end{vmatrix} = -e^{2x} \begin{vmatrix} 1 & 1 & x & 1 \\ 1 & 1 & 0 & 4 \\ 1 & -1 & 0 & 8 \\ 1 & 1 & 0 & 16 \end{vmatrix} \\
&= -e^{2x} \begin{vmatrix} 0 & 1 & x & 1 \\ 0 & 1 & 0 & 4 \\ 2 & -1 & 0 & 8 \\ 0 & 1 & 0 & 16 \end{vmatrix} = -2e^{2x} \begin{vmatrix} 1 & x & 1 \\ 0 & -x & 3 \\ 0 & -x & 15 \end{vmatrix} = -2e^{2x} \begin{vmatrix} -x & 3 \\ -x & 15 \end{vmatrix} = 24xe^{2x};
\end{aligned}$$

$$\begin{aligned}
P_4(x) &= \begin{vmatrix} e^x & -e^{-x} & 1 & 2e^{2x} \\ e^x & e^{-x} & 0 & 4e^{2x} \\ e^x & -e^{-x} & 0 & 8e^{2x} \\ e^x & e^{-x} & 0 & 16e^{2x} \end{vmatrix} = e^{2x} \begin{vmatrix} 1 & -1 & 1 & 2 \\ 1 & 1 & 0 & 4 \\ 1 & -1 & 0 & 8 \\ 1 & 1 & 0 & 16 \end{vmatrix} \\
&= e^{2x} \begin{vmatrix} 0 & -1 & 1 & 2 \\ 2 & 1 & 0 & 4 \\ 0 & -1 & 0 & 8 \\ 2 & 1 & 0 & 16 \end{vmatrix} = e^{2x} \begin{vmatrix} 0 & -1 & 1 & 2 \\ 2 & 1 & 0 & 4 \\ 0 & -1 & 0 & 8 \\ 0 & 0 & 0 & 12 \end{vmatrix} \\
&= -2e^{2x} \begin{vmatrix} -1 & 1 & 2 \\ -1 & 0 & 8 \\ 0 & 0 & 12 \end{vmatrix} = -2e^{2x} \begin{vmatrix} -1 & 1 & 2 \\ 0 & -1 & 6 \\ 0 & 0 & 12 \end{vmatrix} = -24e^{2x}.
\end{aligned}$$

Therefore,

$$6e^{2x}(2x-1)y^{(4)} - 24xe^{2x}y''' + 6e^{2x}(5-2x)y'' + 24xe^{2x}y' - 24e^{2x}y = 0,$$

which is equivalent to  $(2x-1)y^{(4)} - 4xy''' + (5-2x)y'' + 4xy' - 4y = 0$ .

**9.1.24.** (j)

$$\begin{aligned}
P_0(x) &= \begin{vmatrix} e^{2x} & e^{-2x} & 1 & x^2 \\ 2e^{2x} & -2e^{-2x} & 0 & 2x \\ 4e^{2x} & 4e^{-2x} & 0 & 2 \\ 8e^{2x} & -8e^{-2x} & 0 & 0 \end{vmatrix} = \begin{vmatrix} 1 & 1 & 1 & x^2 \\ 2 & -2 & 0 & 2x \\ 4 & 4 & 0 & 2 \\ 8 & -8 & 0 & 0 \end{vmatrix} \\
&= \begin{vmatrix} 2 & -2 & 2x \\ 4 & 4 & 2 \\ 8 & -8 & 0 \end{vmatrix} = \begin{vmatrix} 2 & -2 & 2x \\ 0 & 8 & 2-4x \\ 0 & 0 & -8x \end{vmatrix} = -128x
\end{aligned}$$

$$\begin{aligned}
P_1(x) &= - \begin{vmatrix} e^{2x} & e^{-2x} & 1 & x^2 \\ 2e^{2x} & -2e^{-2x} & 0 & 2x \\ 4e^{2x} & 4e^{-2x} & 0 & 2 \\ 16e^{2x} & 16e^{-2x} & 0 & 0 \end{vmatrix} = - \begin{vmatrix} 1 & 1 & 1 & x^2 \\ 2 & -2 & 0 & 2x \\ 4 & 4 & 0 & 2 \\ 16 & 16 & 0 & 0 \end{vmatrix} \\
&= - \begin{vmatrix} 2 & -2 & 2x \\ 4 & 4 & 2 \\ 16 & 16 & 0 \end{vmatrix} = - \begin{vmatrix} 2 & -2 & 2x \\ 0 & 8 & 2-4x \\ 0 & 0 & -8 \end{vmatrix} = 128;
\end{aligned}$$

$$\begin{aligned}
P_2(x) &= \begin{vmatrix} e^{2x} & e^{-2x} & 1 & x^2 \\ 2e^{2x} & -2e^{-2x} & 0 & 2x \\ 8e^{2x} & -8e^{-2x} & 0 & 0 \\ 16e^{2x} & 16e^{-2x} & 0 & 0 \end{vmatrix} = \begin{vmatrix} 1 & 1 & 1 & x^2 \\ 2 & -2 & 0 & 2x \\ 8 & -8 & 0 & 0 \\ 16 & 16 & 0 & 0 \end{vmatrix} \\
&= \begin{vmatrix} 2 & -2 & 2x \\ 8 & -8 & 0 \\ 16 & 16 & 0 \end{vmatrix} = 2x \begin{vmatrix} 8 & -8 \\ 16 & 16 \end{vmatrix} = 512x; \\
P_3(x) &= - \begin{vmatrix} e^{2x} & e^{-2x} & 1 & x^2 \\ 4e^{2x} & 4e^{-2x} & 0 & 2 \\ 8e^{2x} & -8e^{-2x} & 0 & 0 \\ 16e^{2x} & 16e^{-2x} & 0 & 0 \end{vmatrix} = - \begin{vmatrix} 1 & 1 & 1 & x^2 \\ 4 & 4 & 0 & 2 \\ 8 & -8 & 0 & 0 \\ 16 & 16 & 0 & 0 \end{vmatrix} \\
&= - \begin{vmatrix} 4 & 4 & 2 \\ 8 & -8 & 0 \\ 16 & 16 & 0 \end{vmatrix} = -2 \begin{vmatrix} 8 & -8 \\ 16 & 16 \end{vmatrix} = -512; \\
P_4(x) &= \begin{vmatrix} 2e^{2x} & -2e^{-2x} & 0 & 2x \\ 4e^{2x} & 4e^{-2x} & 0 & 2 \\ 8e^{2x} & -8e^{-2x} & 0 & 0 \\ 16e^{2x} & 16e^{-2x} & 0 & 0 \end{vmatrix} = 0.
\end{aligned}$$

Therefore,  $-128xy^{(4)} + 128y''' + 512xy'' - 512y = 0$ , which is equivalent to  $xy^{(4)} - y''' - 4xy'' + 4y' = 0$ .

## 9.2 Higher Order Constant Coefficient Homogeneous Equations

**9.2.2.**  $p(r) = r^4 + 8r^2 - 9 = (r-1)(r+1)(r^2+9)$ ;  $y = c_1e^x + c_2e^{-x} + c_3\cos 3x + c_4\sin 3x$ .

**9.2.4.**  $p(r) = 2r^3 + 3r^2 - 2r - 3 = (r-1)(r+1)(2r+3)$ ;  $y = c_1e^x + c_2e^{-x} + c_3e^{-3x/2}$ .

**9.2.6.**  $p(r) = 4r^3 - 8r^2 + 5r - 1 = (r-1)(2r-1)^2$ ;  $y = c_1e^x + e^{x/2}(c_2 + c_3x)$ .

**9.2.8.**  $p(r) = r^4 + r^2 = r^2(r^2+1)$ ;  $y = c_1 + c_2x + c_3\cos x + c_4\sin x$ .

**9.2.10.**  $p(r) = r^4 + 12r^2 + 36 = (r^2+6)^2$ ;  $y = (c_1 + c_2x)\cos\sqrt{6}x + (c_3 + c_4x)\sin\sqrt{6}x$ .

**9.2.12.**  $p(r) = 6r^4 + 5r^3 + 7r^2 + 5r + 1 = (2r+1)(3r+1)(r^2+1)$ ;  $y = c_1e^{-x/2} + c_2e^{-x/3} + c_3\cos x + c_4\sin x$ .

**9.2.14.**  $p(r) = r^4 - 4r^3 + 7r^2 - 6r + 2 = (r-1)^2(r^2-2r+2)$ ;  $y = e^x(c_1 + c_2x + c_3\cos x + c_4\sin x)$ .

**9.2.16.**  $p(r) = r^3 + 3r^2 - r - 3 = (r-1)(r+1)(r+3)$ ;

$$\begin{aligned}
y &= c_1e^x + c_2e^{-x} + c_3e^{-3x} & c_1 + c_2 + c_3 &= 0 \\
y' &= c_1e^x - c_2e^{-x} - 3c_3e^{-3x} & c_1 - c_2 - 3c_3 &= 14 \\
y'' &= c_1e^x + c_2e^{-x} + 9c_3e^{-3x} & c_1 + c_2 + 9c_3 &= -40
\end{aligned}$$

$c_1 = 2, c_2 = 3, c_3 = -5$ ;  $y = 2e^x + 3e^{-x} - 5e^{-3x}$ .

**9.2.18.**  $p(r) = r^3 - 2r - 4 = (r-2)(r^2+2r+2)$ ;

$$\begin{aligned}
y &= e^{-x}(c_1\cos x + c_2\sin x) + c_3e^{2x} & c_1 + c_3 &= 6 \\
y' &= -e^{-x}((c_1 - c_2)\cos x + (c_1 + c_2)\sin x) + 2c_3e^{2x} & -c_1 + c_2 + 2c_3 &= 3 \\
y'' &= e^{-x}(2c_1\sin x - 2c_2\cos x) + 4c_3e^{2x} & -2c_2 + 4c_3 &= 22
\end{aligned}$$

$$c_1 = 2, c_2 = -3, c_3 = 4; y = 2e^{-x} \cos x - 3e^{-x} \sin x + 4e^{2x}.$$

$$\mathbf{9.2.20.} \quad p(r) = r^3 - 6r^2 + 12r - 8 = (r - 2)^3;$$

$$\begin{aligned} y &= e^{2x}(c_1 + c_2x + c_3x^2) & c_1 &= 1 \\ y' &= e^{2x}(2c_1 + c_2 + (2c_2 + 2c_3)x + 2c_3x^2) & 2c_1 + c_2 &= -1 \\ y'' &= 2e^{2x}(2c_1 + 2c_2 + c_3 + 2(c_2 + 2c_3)x + 2c_3x^2) & 4c_1 + 4c_2 + 2c_3 &= -4 \end{aligned}$$

$$c_1 = 1, c_2 = -3, c_3 = 2; y = e^{2x}(1 - 3x + 2x^2).$$

$$\mathbf{9.2.22.} \quad p(r) = 8r^3 - 4r^2 - 2r + 1 = (2r + 1)(2r - 1)^2;$$

$$\begin{aligned} y &= e^{x/2}(c_1 + c_2x) + c_3e^{-x/2} & c_1 + c_3 &= 4 \\ y' &= \frac{1}{2}e^{x/2}(c_1 + 2c_2 + c_2x) - \frac{1}{2}c_3e^{-x/2} & \frac{1}{2}c_1 + c_2 - \frac{1}{2}c_3 &= -3 \\ y'' &= \frac{1}{4}e^{x/2}(c_1 + 4c_2 + c_2x) + \frac{1}{4}c_3e^{-x/2} & \frac{1}{4}c_1 + c_2 + \frac{1}{4}c_3 &= -1 \end{aligned}$$

$$c_1 = 1, c_2 = -2, c_3 = 3; y = e^{x/2}(1 - 2x) + 3e^{-x/2}.$$

$$\mathbf{9.2.24.} \quad p(r) = r^4 - 6r^3 + 7r^2 + 6r - 8 = (r - 1)(r - 2)(r - 4)(r + 1);$$

$$\begin{aligned} y &= c_1e^x + c_2e^{2x} + c_3e^{4x} + c_4e^{-x} & c_1 + c_2 + c_3 + c_4 &= -2 \\ y' &= c_1e^x + 2c_2e^{2x} + 4c_3e^{4x} - c_4e^{-x} & c_1 + 2c_2 + 4c_3 - c_4 &= -8 \\ y'' &= c_1e^x + 4c_2e^{2x} + 16c_3e^{4x} + c_4e^{-x} & c_1 + 4c_2 + 16c_3 + c_4 &= -14 \\ y''' &= c_1e^x + 8c_2e^{2x} + 64c_3e^{4x} - c_4e^{-x} & c_1 + 8c_2 + 64c_3 - c_4 &= -62 \end{aligned}$$

$$c_1 = -4, c_2 = 1, c_3 = -1, c_4 = 2; y = -4e^x + e^{2x} - e^{4x} + 2e^{-x}.$$

$$\mathbf{9.2.26.} \quad p(r) = r^4 + 2r^3 - 2r^2 - 8r - 8 = (r - 2)(r + 2)(r^2 + 2r + 2);$$

$$\begin{aligned} y &= c_1e^{2x} + c_2e^{-2x} + e^{-x}(c_3 \cos x + c_4 \sin x) \\ y' &= 2c_1e^{2x} - 2c_2e^{-2x} - e^{-x}((c_3 - c_4) \cos x + (c_3 + c_4) \sin x) \\ y'' &= 4c_1e^{2x} + 4c_2e^{-2x} + e^{-x}(2c_3 \sin x - 2c_4 \cos x) \\ y''' &= 8c_1e^{2x} - 8c_2e^{-2x} + e^{-x}((2c_3 + 2c_4) \cos x + 2(c_4 - c_3) \sin x) \end{aligned}$$

$$\begin{aligned} c_1 + c_2 + c_3 &= 5 \\ 2c_1 - 2c_2 - c_3 + c_4 &= -2 \\ 4c_1 + 4c_2 - 2c_4 &= 6 \\ 8c_1 - 8c_2 + 2c_3 + 2c_4 &= 8 \end{aligned}$$

$$c_1 = 1, c_2 = 1, c_3 = 3, c_4 = 1; y = e^{2x} + e^{-2x} + e^{-x}(3 \cos x + \sin x).$$

$$\mathbf{9.2.28.} \quad (\text{a}) \quad W(x) = \begin{vmatrix} e^x & xe^x & e^{2x} \\ e^x & e^x(x+1) & 2e^{2x} \\ e^x & e^x(x+2) & 4e^{2x} \end{vmatrix}; \quad W(0) = \begin{vmatrix} 1 & 0 & 1 \\ 1 & 1 & 2 \\ 1 & 2 & 4 \end{vmatrix} = \begin{vmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 2 & 3 \end{vmatrix} = \begin{vmatrix} 1 & 1 \\ 2 & 3 \end{vmatrix} = 1.$$

$$(\text{b}) \quad W(x) = \begin{vmatrix} \cos 2x & \sin 2x & e^{3x} \\ -2 \sin 2x & 2 \cos 2x & 3e^{3x} \\ -4 \cos 2x & -4 \sin 2x & 9e^{3x} \end{vmatrix}; \quad W(0) = \begin{vmatrix} 1 & 0 & 1 \\ 0 & 2 & 3 \\ -4 & 0 & 9 \end{vmatrix} = \begin{vmatrix} 1 & 0 & 0 \\ 0 & 2 & 3 \\ -4 & 0 & 13 \end{vmatrix} = \begin{vmatrix} 2 & 3 \\ 0 & 13 \end{vmatrix} =$$

26.

$$(\text{c}) \quad W(x) = \begin{vmatrix} e^{-x} \cos x & e^{-x} \sin x & e^x \\ -e^{-x}(\cos x + \sin x) & e^{-x}(\cos x - \sin x) & e^x \\ 2e^{-x} \sin x & -2e^{-x} \cos x & e^x \end{vmatrix}; \quad W(0) = \begin{vmatrix} 1 & 0 & 1 \\ -1 & 1 & 1 \\ 0 & -2 & 1 \end{vmatrix} = \begin{vmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 0 & -2 & 1 \end{vmatrix} =$$

$$\begin{vmatrix} 1 & 2 \\ -2 & 1 \end{vmatrix} = 5.$$

$$(d) W(x) = \begin{vmatrix} 1 & x & x^2 & e^x \\ 0 & 1 & 2x & e^x \\ 0 & 0 & 2 & e^x \\ 0 & 0 & 0 & e^x \end{vmatrix}; W(0) = \begin{vmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 1 \end{vmatrix} = 1.$$

$$(e) W(x) = \begin{vmatrix} e^x & e^{-x} & \cos x & \sin x \\ e^x & -e^{-x} & -\sin x & \cos x \\ e^x & e^{-x} & -\cos x & -\sin x \\ e^x & -e^{-x} & \sin x & -\cos x \end{vmatrix};$$

$$\begin{aligned} W(0) &= \begin{vmatrix} 1 & 1 & 1 & 0 \\ 1 & -1 & 0 & 1 \\ 1 & 1 & -1 & 0 \\ 1 & -1 & 0 & -1 \end{vmatrix} = \begin{vmatrix} 1 & 1 & 1 & 0 \\ 1 & -1 & 0 & 1 \\ 1 & 1 & -1 & 0 \\ 0 & 0 & 0 & -2 \end{vmatrix} = -2 \begin{vmatrix} 1 & 1 & 1 \\ 1 & -1 & 0 \\ 1 & 1 & -1 \end{vmatrix} \\ &= -2 \begin{vmatrix} 1 & 1 & 1 \\ 1 & -1 & 0 \\ 0 & 0 & -2 \end{vmatrix} = 4 \begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix} = -8. \end{aligned}$$

$$(f) W(x) = \begin{vmatrix} \cos x & \sin x & e^x \cos x & e^x \sin x \\ -\sin x & \cos x & e^x(\cos x - \sin x) & e^x(\cos x + \sin x) \\ -\cos x & -\sin x & -2e^x \sin x & 2e^x \cos x \\ \sin x & -\cos x & -e^x(2 \cos x + 2 \sin x) & e^x(2 \cos x - 2 \sin x) \end{vmatrix};$$

$$\begin{aligned} W(0) &= \begin{vmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ -1 & 0 & 0 & 2 \\ 0 & -1 & -2 & 2 \end{vmatrix} = \begin{vmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & -1 & -2 & 2 \end{vmatrix} = \begin{vmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ -1 & -2 & 2 \end{vmatrix} \\ &= \begin{vmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & -1 & 3 \end{vmatrix} = \begin{vmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 5 \end{vmatrix} = 5. \end{aligned}$$

**9.2.40.** (a) Since  $y = Q_1(D)P_1(D)y + Q_2(D)P_2(D)y$  and  $P_1(D)y = P_2(D)y = 0$ , it follows that  $y = 0$ .

(b) Suppose that (A)  $a_1u_1 + \cdots + a_ru_r + b_1v_1 + \cdots + b_sv_s = 0$ , where  $a_1, \dots, a_r$  and  $b_1, \dots, b_s$  are constants. Denote  $u = a_1u_1 + \cdots + a_ru_r$  and  $v = b_1v_1 + \cdots + b_sv_s$ . Then (B)  $P_1(D)u = 0$  and (C)  $P_2(D)v = 0$ . Since  $u + v = 0$ ,  $P_2(D)(u + v) = 0$ . Therefore,  $0 = P_2(D)(u + v) = P_2(D)u + P_2(D)v$ . Now (C) implies that  $P_2(D)u = 0$ . This, (B), and (a) imply that  $u = a_1u_1 + \cdots + a_ru_r = 0$ , so  $a_1 = \cdots = a_r = 0$ , since  $u_1, \dots, u_r$  are linearly independent. Now (A) reduces to  $b_1v_1 + \cdots + b_sv_s = 0$ , so  $b_1 = \cdots = b_s = 0$ , since  $v_1, \dots, v_s$  are linearly independent. Therefore,  $u_1, \dots, u_r, v_1, \dots, v_r$  are linearly independent.

(c) It suffices to show that  $\{y_1, y_2, \dots, y_n\}$  is linearly independent. Suppose that  $c_1y_1 + \cdots + c_ny_n = 0$ . We may assume that  $y_1, \dots, y_r$  are linearly independent solutions of  $p_1(D)y = 0$  and  $y_{r+1}, \dots, y_n$  are solutions of  $P_2(D) = p_2(D) \cdots p_k(D)y = 0$ . Since  $p_1(r)$  and  $P_2(r)$  have no common factors, (b) implies that (A)  $c_1y_1 + \cdots + c_ry_r = 0$  and (B)  $c_{r+1}y_{r+1} + \cdots + c_ny_n = 0$ . Now (A) implies that  $c_1 = \cdots = c_r = 0$ , since  $y_1, \dots, y_r$  are linearly independent. If  $k = 2$ , then  $y_{r+1}, \dots, y_n$  are linearly independent, so  $c_{r+1} = \cdots = c_n = 0$ , and the proof is complete. If  $k > 2$  repeat this argument, starting from (B), with  $p_1$  replaced by  $p_2$ , and  $P_2$  replaced by  $P_3 = p_3 \cdots p_n$ .

**9.2.42.** (a)

$$\begin{aligned} (\cos A + i \sin A)(\cos B + i \sin B) &= (\cos A \cos B - \sin A \sin B) \\ &\quad + (\cos A \sin B + \sin A \cos B) \\ &= \cos(A + B) + i \sin(A + B). \end{aligned}$$

(b) Obvious for  $n = 0$ . If  $n = -1$  write

$$\begin{aligned} \frac{1}{\cos \theta + i \sin \theta} &= \frac{1}{\cos \theta + i \sin \theta} \frac{\cos \theta - i \sin \theta}{\cos \theta - i \sin \theta} \\ &= \frac{\cos \theta - i \sin \theta}{\cos^2 \theta + \sin^2 \theta} = \cos \theta - i \sin \theta = \cos(-\theta) + i \sin(-\theta). \end{aligned}$$

(d) If  $n$  is a negative integer, then (B)  $(\cos \theta + i \sin \theta)^n = \frac{1}{(\cos \theta + i \sin \theta)^{|n|}}$ . From the hint, (C)

$\frac{1}{(\cos \theta + i \sin \theta)^{|n|}} = (\cos \theta - i \sin \theta)^{|n|} = (\cos(-\theta) + i \sin(-\theta))^{|n|}$ . Replacing  $\theta$  by  $-\theta$  and  $n$  by  $|n|$  in (A) shows that (D)  $(\cos(-\theta) + i \sin(-\theta))^{|n|} = \cos(-|n|\theta) + i \sin(-|n|\theta)$ . Since  $|n| = -n$ , (E)  $\cos(-|n|\theta) + i \sin(-|n|\theta) = \cos n\theta + i \sin n\theta$ . Now (B), (C), (D), and (E) imply (A).

(e) From (A),  $z_k^n = \cos 2k\pi + i \sin 2k\pi = 1$  and  $\zeta_k^n = \cos(2k+1)\pi + i \sin(2k+1)\pi = \cos(2k+1)\pi = \cos \pi = -1$ .

(f) From (e),  $\rho^{1/n} z_0, \dots, \rho^{1/n} z_{n-1}$  are all zeros of  $z^n - \rho$ . Since they are distinct numbers,  $z^n - \rho$  has the stated factorization.

From (e),  $\rho^{1/n} \zeta_0, \dots, \rho^{1/n} \zeta_{n-1}$  are all zeros of  $z^n + \rho$ . Since they are distinct numbers,  $z^n + \rho$  has the stated factorization.

**9.2.43.** (a)  $p(r) = r^3 - 1 = (r - z_0)(r - z_1)(r - z_2)$  where  $z_k = \cos \frac{2k\pi}{3} + i \sin \frac{2k\pi}{3}$ ,  $k = 0, 1, 2$ .

Hence,  $z_0 = 1$ ,  $z_1 = -\frac{1}{2} + i\frac{\sqrt{3}}{2}$ , and  $z_2 = -\frac{1}{2} - i\frac{\sqrt{3}}{2}$ . Therefore,  $p(r) = (r - 1) \left( \left( r + \frac{1}{2} \right)^2 + \frac{3}{4} \right)$ , so  $\left\{ e^x, e^{-x/2} \cos \left( \frac{\sqrt{3}}{2} x \right), e^{-x/2} \sin \left( \frac{\sqrt{3}}{2} x \right) \right\}$  is a fundamental set of solutions.

(b)  $p(r) = r^3 + 1 = (r - \zeta_0)(r - \zeta_1)(r - \zeta_2)$  where  $\zeta_k = \cos \frac{(2k+1)\pi}{3} + i \sin \frac{(2k+1)\pi}{3}$ ,  $k = 0, 1, 2$ .

Hence,  $\zeta_0 = \frac{1}{2} + i\frac{\sqrt{3}}{2}$ ,  $\zeta_1 = -1$ ,  $\zeta_2 = \frac{1}{2} - i\frac{\sqrt{3}}{2}$ . Therefore,  $p(r) = (r + 1) \left( \left( r - \frac{1}{2} \right)^2 + \frac{3}{4} \right)$ , so  $\left\{ e^{-x}, e^{x/2} \cos \left( \frac{\sqrt{3}}{2} x \right), e^{x/2} \sin \left( \frac{\sqrt{3}}{2} x \right) \right\}$  is a fundamental set of solutions.

(c)  $p(r) = r^4 + 64 = (r - 2\sqrt{2}\zeta_0)(r - 2\sqrt{2}\zeta_1)(r - 2\sqrt{2}\zeta_2)(r - 2\sqrt{2}\zeta_3)$ , where  $\zeta_k = \cos \frac{(2k+1)\pi}{4} + i \sin \frac{(2k+1)\pi}{4}$ ,  $k = 0, 1, 2, 3$ . Therefore,  $\zeta_0 = \frac{1+i}{\sqrt{2}}$ ,  $\zeta_1 = \frac{-1+i}{\sqrt{2}}$ ,  $\zeta_2 = \frac{-1-i}{\sqrt{2}}$ , and  $\zeta_3 = \frac{1-i}{\sqrt{2}}$ , so  $p(r) = ((r-2)^2+4)((r+2)^2+4)$  and  $\{e^{2x} \cos 2x, e^{2x} \sin 2x, e^{-2x} \cos 2x, e^{-2x} \sin 2x\}$  is a fundamental set of solutions.

(d)  $p(r) = r^6 - 1 = (r - z_0)(r - z_1)(r - z_2)(r - z_3)(r - z_4)(r - z_5)$  where  $z_k = \cos \frac{2k\pi}{6} + i \sin \frac{2k\pi}{6}$ ,  $k = 0, 1, 2, 3, 4, 5$ . Therefore,  $z_0 = 1$ ,  $z_1 = \frac{1}{2} + i\frac{\sqrt{3}}{2}$ ,  $z_2 = -\frac{1}{2} + i\frac{\sqrt{3}}{2}$ ,  $z_3 = -1$ ,  $z_4 = -\frac{1}{2} - i\frac{\sqrt{3}}{2}$ , and

$z_5 = \frac{1}{2} - i\frac{\sqrt{3}}{2}$ , so  $p(r) = (r-1)(r+1) \left( \left( r - \frac{1}{2} \right)^2 + \frac{3}{4} \right) \left( \left( r + \frac{1}{2} \right)^2 + \frac{3}{4} \right)$  and  $\left\{ e^x, e^{-x}, e^{x/2} \cos \left( \frac{\sqrt{3}}{2} x \right), e^{x/2} \sin \left( \frac{\sqrt{3}}{2} x \right), e^{-x/2} \cos \left( \frac{\sqrt{3}}{2} x \right), e^{-x/2} \sin \left( \frac{\sqrt{3}}{2} x \right) \right\}$  is a fundamental set of solutions.

(e)  $p(r) = r^6 + 64 = (r-2\zeta_0)(r-2\zeta_1)(r-2\zeta_2)(r-2\zeta_3)(r-2\zeta_4)(r-2\zeta_5)$  where  $\zeta_k = \cos \frac{(2k+1)\pi}{6} + i \sin \frac{(2k+1)\pi}{6}$ ,  $k = 0, 1, 2, 3, 4, 5$ . Therefore,  $\zeta_0 = \frac{\sqrt{3}}{2} + \frac{i}{2}$ ,  $\zeta_1 = i$ ,  $\zeta_2 = -\frac{\sqrt{3}}{2} + \frac{i}{2}$ ,  $\zeta_3 = -\frac{\sqrt{3}}{2} - \frac{i}{2}$ ,  $\zeta_4 = -i$ , and  $\zeta_5 = \frac{\sqrt{3}}{2} - \frac{i}{2}$ , so  $p(r) = (r^2 + 4)((r - \sqrt{3})^2 + 1)((r + \sqrt{3})^2 + 1)$  and  $\{\cos 2x, \sin 2x, e^{-\sqrt{3}x} \cos x, e^{-\sqrt{3}x} \sin x, e^{\sqrt{3}x} \cos x, e^{\sqrt{3}x} \sin x\}$  is a fundamental set of solutions.

(f)  $p(r) = (r-1)^6 - 1 = (r-1-z_0)(r-1-z_1)(r-1-z_2)(r-1-z_3)(r-1-z_4)(r-1-z_5)$  where  $z_k = \cos \frac{2k\pi}{6} + i \sin \frac{2k\pi}{6}$ ,  $k = 0, 1, 2, 3, 4, 5$ . Therefore,  $z_0 = 1$ ,  $z_1 = \frac{1}{2} + i\frac{\sqrt{3}}{2}$ ,  $z_2 = -\frac{1}{2} + i\frac{\sqrt{3}}{2}$ ,  $z_3 = -1$ ,  $z_4 = -\frac{1}{2} - i\frac{\sqrt{3}}{2}$ , and  $z_5 = \frac{1}{2} - i\frac{\sqrt{3}}{2}$ , so  $p(r) = r(r-2) \left( \left( r - \frac{3}{2} \right)^2 + \frac{3}{4} \right) \left( \left( r - \frac{1}{2} \right)^2 + \frac{3}{4} \right)$  and  $\left\{ 1, e^{2x}, e^{3x/2} \cos \left( \frac{\sqrt{3}}{2} x \right), e^{3x/2} \sin \left( \frac{\sqrt{3}}{2} x \right), e^{x/2} \cos \left( \frac{\sqrt{3}}{2} x \right), e^{x/2} \sin \left( \frac{\sqrt{3}}{2} x \right) \right\}$  is a fundamental set of solutions.

(g)  $p(r) = r^5 + r^4 + r^3 + r^2 + r + 1 = \frac{r^6 - 1}{r - 1}$ . Therefore, from the solution of (d)  $p(r) = (r+1) \left( \left( r - \frac{1}{2} \right)^2 + \frac{3}{4} \right) \left( \left( r + \frac{1}{2} \right)^2 + \frac{3}{4} \right)$  and  $\left\{ e^{-x}, e^{x/2} \cos \left( \frac{\sqrt{3}}{2} x \right), e^{x/2} \sin \left( \frac{\sqrt{3}}{2} x \right), e^{-x/2} \cos \left( \frac{\sqrt{3}}{2} x \right), e^{-x/2} \sin \left( \frac{\sqrt{3}}{2} x \right) \right\}$  is a fundamental set of solutions.

### 9.3 Undetermined Coefficients for Higher Order Equations

**9.3.2.** If  $y = u^{-3x}$ , then  $y''' - 2y'' - 5y' + 6y = e^{-3x}[(u''' - 11u'' + 34u' - 24u) - 2(u'' - 6u' + 9u) - 5(u' - 3u) + 6u] = e^{-3x}(u''' - 11u'' + 34u' - 24u)$ . Let  $u_p = A + Bx + Cx^2$ , where  $(-24A + 34B - 22C) + (-24B + 68C)x - 24Cx^2 = 32 - 23x + 6x^2$ . Then  $C = -1/4$ ,  $B = 1/4$ ,  $A = -3/4$  and  $y_p = -\frac{e^{-3x}}{4}(3 - x + x^2)$ .

**9.3.4.** If  $y = ue^{-2x}$ , then  $y''' + 3y'' - y' - 3y = e^{-2x}[(u''' - 6u'' + 12u' - 8u) + 3(u'' - 4u' + 4u) - (u' - 2u) - 3u] = e^{-2x}(u''' - 3u'' - u' + 3u)$ . Let  $u_p = A + Bx + Cx^2$ , where  $(3A - B - 6C) + (3B - 2C)x + 3Cx^2 = 2 - 17x + 3x^2$ . Then  $C = 1$ ,  $B = -5$ ,  $A = 1$ , and  $y_p = e^{-2x}(1 - 5x + x^2)$ .

**9.3.6.** If  $y = ue^x$ , then  $y''' + y'' - 2y = e^x[(u''' + 3u'' + 3u' + u) + (u'' + 2u' + u) - 2u] = e^x(u''' + 4u'' + 5u')$ . Let  $u_p = x(A + Bx + Cx^2)$ , where  $(5A + 8B + 6C) + (10B + 24C)x + 15Cx^2 = 14 + 34x + 15x^2$ . Then  $C = 1$ ,  $B = 1$ ,  $A = 0$ , and  $y_p = x^2e^x(1 + x)$ .

**9.3.8.** If  $y = ue^x$ , then  $y''' - y'' - y' + y = e^x[(u''' + 3u'' + 3u' + u) - (u'' + 2u' + u) - (u' + u) + u] = e^x(u''' + 2u'')$ . Let  $u_p = x^2(A + Bx)$  where  $(4A + 6B) + 12Bx = 7 + 6x$ . Then  $B = 1/2$ ,  $A = 1$ , and

$$y_p = \frac{x^2 e^x}{2}(2+x).$$

**9.3.10.** If  $y = ue^{3x}$ , then  $y''' - 5y'' + 3y' + 9y = e^{3x}[(u''' + 9u'' + 27u' + 27u) - 5(u'' + 6u' + 9u) + 3(u' + 3u) + 9u] = e^{3x}(u''' + 4u'')$ . Let  $u_p = x^2(A + Bx + Cx^2)$ , where  $(8A + 6B) + (24B + 24C)x + 48Cx^2 = 22 - 48x^2$ . Then  $C = -1$ ,  $B = 1$ ,  $A = 2$ , and  $y_p = x^2 e^{3x}(2 + x - x^2)$ .

**9.3.12.** If  $y = ue^{x/2}$ , then  $8y''' - 12y'' + 6y' - y = e^{x/2}[8(u''' + 3u''/2 + 3u'/4 + u/8) - 12(u'' + u' + u/4) + 6(u' + u/2) - u] = 8e^{x/2}u'''$ , so  $u''' = \frac{1+4x}{8}$ . Integrating three times and taking the

constants of integration to be zero yields  $u_p = \frac{x^3}{48}(1+x)$ . Therefore,  $y_p = \frac{x^3 e^{x/2}}{48}(1+x)$ .

**9.3.14.** If  $y = ue^{2x}$ , then  $y^{(4)} + 3y''' + y'' - 3y' - 2y = e^{2x}[(u^{(4)} + 8u''' + 24u'' + 32u' + 16u) + 3(u''' + 6u'' + 12u' + 8u) + (u'' + 4u' + 4u) - 3(u' + 2u) - 2u] = e^{2x}(u^{(4)} + 11u''' + 43u'' + 69u' + 36u)$ . Let  $u_p = A + Bx$  where  $(36A + 69B) + 36Bx = -33 - 36x$ . Then  $B = -1$ ,  $A = 1$ , and  $y_p = e^{2x}(1 - x)$ .

**9.3.16.** If  $y = ue^x$ , then  $4y^{(4)} - 11y''' - 9y'' - 2y = e^x[4(u^{(4)} + 4u''' + 6u'' + 4u' + u) - 11(u''' + 2u'' + u) - 9(u' + u) - 2u] = e^x(4u^{(4)} + 16u''' + 13u'' - 15u' - 18u)$ . Let  $u_p = A + Bx$  where  $-(18A + 15B) - 18Bx = -1 + 6x$ . Then  $B = -1/3$ ,  $A = 1/3$ , and  $y_p = \frac{e^x}{3}(1 - x)$ .

**9.3.18.** If  $y = ue^x$ , then  $y^{(4)} - 4y''' + 6y'' - 4y' + 2y = e^x[(u^{(4)} + 4u''' + 6u'' + 4u' + u) - 4(u''' + 3u'' + 3u' + u) + 6(u'' + 2u' + u) - 4(u' + u) + 2u] = e^x(u^{(4)} + u)$ . Let  $u_p = A + Bx + Cx^2 + Dx^3 + Ex^4$  where  $(A + 24E) + Bx + Cx^2 + Dx^3 + Ex^4 = 24 + x + x^4$ . Then  $E = 1$ ,  $D = 0$ ,  $C = 0$ ,  $B = 1$ ,  $A = 0$ , and  $y_p = xe^x(1 + x^3)$ .

**9.3.20.** If  $y = ue^{2x}$ , then  $y^{(4)} + y''' - 2y'' - 6y' - 4y = e^{2x}[(u^{(4)} + 8u''' + 24u'' + 32u' + 16u) + (u''' + 6u'' + 12u' + 8u) - 2(u'' + 4u' + 4u) - 6(u' + 2u) - 4u] = e^{2x}(u^{(4)} + 9u''' + 28u'' + 30u')$ . Let  $u_p = x(A + Bx + Cx^2)$  where  $(30A + 56B + 54C) + (60B + 168C)x + 90Cx^2 = -(4 + 28x + 15x^2)$ .

Then  $C = -1/6$ ,  $B = 0$ ,  $A = 1/6$ , and  $y_p = \frac{xe^{2x}}{6}(1 - x^2)$ .

**9.3.22.** If  $y = ue^x$ , then  $y^{(4)} - 5y'' + 4y = e^x[(u^{(4)} + 4u''' + 6u'' + 4u' + u) - 5(u'' + 2u' + u) + 4u] = e^x(u^{(4)} + 4u''' + u'' - 6u')$ . Let  $u_p = x(A + Bx + Cx^2)$  where  $(-6A + 2B + 24C) + (-12B + 6C)x - 18Cx^2 = 3 + x - 3x^2$ , so  $C = 1/6$ ,  $B = 0$ ,  $A = 1/6$ . Then  $y_p = \frac{xe^x}{6}(1 + x^2)$ .

**9.3.24.** If  $y = ue^{2x}$ , then  $y^{(4)} - 3y''' + 4y' = e^{2x}[(u^{(4)} + 8u''' + 24u'' + 32u' + 16u) - 3(u''' + 6u'' + 12u' + 8u) + 4(u' + 2u)] = e^{2x}(u^{(4)} + 5u''' + 6u'')$ . Let  $u_p = x^2(A + Bx + Cx^2)$  where  $(12A + 30B + 24C) + (36B + 120C)x + 72Cx^2 = 15 + 26x + 12x^2$ . Then  $C = 1/6$ ,  $B = 1/6$ ,  $A = 1/2$ , and  $y_p = \frac{x^2 e^{2x}}{6}(3 + x + x^2)$ .

**9.3.26.** If  $y = ue^x$ , then  $2y^{(4)} - 5y''' + 3y'' + y' - y = e^x[2(u^{(4)} + 4u''' + 6u'' + 4u' + u) - 5(u''' + 3u'' + 3u' + u) + 3(u'' + 2u' + u) + (u' + u) - u] = e^x(2u^{(4)} + 3u''')$ . Let  $u_p = x^3(A + Bx)$ , where  $(18A + 48B) + 72Bx = 11 + 12x$ . Then  $B = 1/6$ ,  $A = 1/6$ , and  $y_p = \frac{x^3 e^x}{6}(1 + x)$ .

**9.3.28.** If  $y = ue^{2x}$ , then  $y^{(4)} - 7y''' + 18y'' - 20y' + 8y = e^{2x}[(u^{(4)} + 8u''' + 24u'' + 32u' + 16u) - 7(u''' + 6u'' + 12u' + 8u) + 18(u'' + 4u' + 4u) - 20(u' + 2u) + 8u] = e^{2x}(u^{(4)} + u''')$ . Let  $u_p = x^3(A + Bx + Cx^2)$  where  $(6A + 24B) + (24B + 120C)x + 60Cx^2 = 3 - 8x - 5x^2$ . Then so  $C = -1/12$ ,  $B = 1/12$ ,  $A = 1/6$ , and  $y_p = \frac{x^3 e^{2x}}{12}(2 + x - x^2)$ .

**9.3.30.** If  $y = ue^{-x}$ , then  $y''' + y'' - 4y' - 4y = e^{-x}[(u''' - 3u'' + 3u' - u) + (u'' - 2u' + u) - 4(u' - u) - 4u] =$



$e^{-x}(u''' - 2u'' - 3u')$ . Let  $u_p = (A_0 + A_1x) \cos 2x + (B_0 + B_1x) \sin 2x$ , where

$$\begin{aligned} 8A_1 - 14B_1 &= -22 \\ 14A_1 + 8B_1 &= -6 \\ 8A_0 - 14B_0 - 15A_1 - 8B_1 &= 1 \\ 14A_0 + 8B_0 + 8A_1 - 15B_1 &= -1. \end{aligned}$$

Then  $A_1 = -1$ ,  $B_1 = 1$ ,  $A_0 = 1$ ,  $B_0 = 1$ , and  $y_p = e^{-x}[(1-x) \cos 2x + (1+x) \sin 2x]$ .

**9.3.32.** If  $y = ue^x$ , then  $y''' - 2y'' + y' - 2y = e^x[(u''' + 3u'' + 3u' + u) - 2(u'' + 2u' + u) + (u' + u) - 2u] = e^x(u''' + u'' - 2u)$ . Let  $u_p = (A_0 + A_1x + A_2x^2) \cos 2x + (B_0 + B_1x + B_2x^2) \sin 2x$  where

$$\begin{aligned} -6A_2 - 8B_2 &= -4 \\ 8A_2 - 6B_2 &= -3 \\ -6A_1 - 8B_1 - 24A_2 + 8B_2 &= 5 \\ 8A_1 - 6B_1 - 8A_2 - 24B_2 &= -5 \\ -6A_0 - 8B_0 - 12A_1 + 4B_1 + 2A_2 + 12B_2 &= -9 \\ 8A_0 - 6B_0 - 4A_1 - 12B_1 - 12A_2 + 2B_2 &= 6. \end{aligned}$$

Then  $A_2 = 0$ ,  $B_2 = 1/2$ ;  $A_1 = 1/2$ ,  $B_1 = -1/2$ ;  $A_0 = 1/1$ ,  $B_0 = 1/2$ ; and  $y_p = \frac{e^x}{2}[(1+x) \cos 2x + (1-x+x^2) \sin 2x]$ .

**9.3.34.** If  $y = ue^x$ , then  $y''' - y'' + 2y = e^x[(u''' + 3u'' + 3u' + u) - (u'' + 2u' + u) + 2u] = e^x(u''' + 2u'' + u' + 2u)$ . Since  $\cos x$  and  $\sin x$  satisfy  $u''' + 2u'' + u' + 2u = 0$ , let  $u_p = x[(A_0 + A_1x) \cos x + (B_0 + B_1x) \sin x]$  where

$$\begin{aligned} -4A_1 + 8B_1 &= 4 \\ -8A_1 - 4B_1 &= -12 \\ -2A_0 + 4B_0 + 4A_1 + 6B_1 &= 20 \\ -4A_0 - 2B_0 - 6A_1 + 4B_1 &= -12. \end{aligned}$$

Then  $A_1 = 1$ ,  $B_1 = 1$ ;  $A_0 = 1$ ,  $B_0 = 3$ ; and  $y_p = xe^x[(1+x) \cos x + (3+x) \sin x]$ .

**9.3.36.** If  $y = ue^{3x}$ , then  $y''' - 3y'' + 2y' - 4y = e^{3x}[(u''' + 9u'' + 27u' + 27u) - 6(u'' + 6u' + 9u) + 18(u' + 3u)] = e^{3x}(u''' + 3u'' + 9u' + 27u)$ . Since  $\cos 3x$  and  $\sin 3x$  satisfy  $u''' + 3u'' + 9u' + 27u = 0$ , let  $u_p = x[(A_0 + A_1x) \cos 3x + (B_0 + B_1x) \sin 3x]$  where

$$\begin{aligned} -36A_1 + 36B_1 &= 3 \\ -36A_1 - 36B_1 &= 3 \\ -18A_0 + 18B_0 + 6A_1 + 18B_1 &= -2 \\ -18A_0 - 18B_0 - 18A_1 + 6B_1 &= 3. \end{aligned}$$

Then  $A_1 = -1/12$ ,  $B_1 = 0$ ;  $A_0 = 1$ ,  $B_0 = -1/12$ ; and  $y_p = -\frac{xe^{3x}}{12}(x \cos 3x + \sin 3x)$ .

**9.3.38.** If  $y = ue^x$ , then  $y^{(4)} - 3y''' + 2y'' + 2y' - 4y = e^x[(u^{(4)} + 4u''' + 6u'' + 4u' + u) - 3(u''' + 3u'' + 3u' + u) + 2(u'' + 2u' + u) + 4(u' + u) + u] = e^x(u^{(4)} + u''' - u'' + u' - 2u)$ . Let  $u_p = A \cos 2x + B \sin 2x$  where  $18A - 6B = 2$  and  $6A + 18B = -1$ . Then  $A = 1/12$ ,  $B = -1/12$ , and  $y_p = \frac{e^x}{12}(\cos 2x - \sin 2x)$ .

**9.3.40.** If  $y = ue^{-x}$ , then  $y^{(4)} + 6y''' + 13y'' + 12y' + 4y = e^{-x}[(u^{(4)} - 4u''' + 6u'' - 4u' + u) + 6(u''' - 3u'' + 3u' - u) + 13(u'' - 2u' + u) + 12u' - u + 4u] = e^{-x}(u^{(4)} + 2u''' + u'')$ . Let  $u_p =$

$(A_0 + A_1x) \cos x + (B_0 + B_1x) \sin x$  where

$$\begin{aligned} -2B_1 &= -1 \\ 2A_1 &= -1 \\ -2B_0 - 6A_1 - 2B_1 &= 4 \\ 2A_0 + 2A_1 - 6B_1 &= -5. \end{aligned}$$

Then  $A_1 = -1/2$ ,  $B_1 = 1/2$ ,  $A_0 = -1/2$ ,  $B_0 = -1$ , and  $y_p = -\frac{e^{-x}}{2}[(1+x) \cos x + (2-x) \sin x]$ .

**9.3.42.** If  $y = ue^x$ , then  $y^{(4)} - 5y''' + 13y'' - 19y' + 10y = e^x[(u^{(4)} + 4u''' + 6u'' + 4u' + u) - 5(u''' + 3u'' + 3u' + u) + 13(u'' + 2u' + u) - 19(u' + u) + 10u] = e^x(u^{(4)} - u''' + 4u'' - 4u')$ . Since  $\cos 2x$  and  $\sin 2x$  satisfy  $u^{(4)} - u''' + 4u'' - 4u' = 0$ , let  $u_p = x(A \cos 2x + B \sin 2x)$  where  $8A - 16B = 1$  and  $16A + 8B = 1$ . Then  $A = 3/40$ ,  $B_0 = -1/40$ , and  $y_p = \frac{xe^x}{40}(3 \cos 2x - \sin 2x)$ .

**9.3.44.** If  $y = ue^x$ , then  $y^{(4)} - 5y''' + 13y'' - 19y' + 10y = e^x[(u^{(4)} + 4u''' + 6u'' + 4u' + u) - 5(u''' + 3u'' + 3u' + u) + 13(u'' + 2u' + u) - 19(u' + u) + 10u] = e^x(u^{(4)} - u''' + 4u'' - 4u')$ . Since  $\cos 2x$  and  $\sin 2x$  satisfy  $u^{(4)} - u''' + 4u'' - 4u' = 0$ , let  $u_p = x[(A_0 + A_1x) \cos 2x + (B_0 + B_1x) \sin 2x]$  where

$$\begin{aligned} 16A_1 - 32B_1 &= 8 \\ 32A_1 + 16B_1 &= -4 \\ 8A_0 - 16B_0 - 40A_1 - 12B_1 &= 7 \\ 16A_0 + 8B_0 + 12A_1 - 40B_1 &= 8. \end{aligned}$$

Then  $A_1 = 0$ ,  $B_1 = -1/4$ ;  $A_0 = 0$ ,  $B_0 = -1/4$ , and  $y_p = -\frac{xe^x}{4}(1+x) \sin 2x$ .

**9.3.46.** If  $y = ue^{2x}$ , then  $y^{(4)} - 8y''' + 32y'' - 64y' + 64y + 4y = e^{2x}[(u^{(4)} + 8u''' + 24u'' + 32u' + 16u) - 8(u''' + 6u'' + 12u' + 8u) + 32(u'' + 4u' + 4u) - 64(u' + 2u) + 64u] = e^{2x}(u^{(4)} + 8u'' + 16u)$ . Since  $\cos 2x$ ,  $\sin 2x$ ,  $x \cos 2x$ , and  $x \sin 2x$  satisfy  $u^{(4)} + 8u'' + 16u = 0$ , let  $u_p = x^2(A \cos 2x + B \sin 2x)$

where  $-32A = 1$  and  $-32B = -1$ . Then  $A = -1/32$ ,  $B = 1/32$ , and  $y_p = -\frac{x^2 e^{2x}}{32}(\cos 2x - \sin 2x)$ .

**9.3.48.** Find particular solutions of (a)  $y''' - 4y'' + 5y' - 2y = -4e^x$ , (b)  $y''' - 4y'' + 5y' - 2y = e^{2x}$ , and (c)  $y''' - 4y'' + 5y' - 2y = -2 \cos x + 4 \sin x$ .

(a) If  $y = ue^x$ , then  $y''' - 4y'' + 5y' - 2y = e^x[(u''' + 3u'' + 3u' + u) - 4(u'' + 2u' + u) + 5(u' + u) - 2u] = e^x(u''' - u'')$ . Let  $u_{1p} = Ax^2$  where  $-2A = -4$ . Then  $A = 2$ , and  $y_{1p} = 2x^2 e^x$ .

(b) If  $y = ue^{2x}$ , then  $y''' - 4y'' + 5y' - 2y = e^{2x}[(u''' + 6u'' + 12u' + 8u) - 4(u'' + 4u' + 4u) + 5(u' + 2u) - 2u] = e^{2x}(u''' + 2u'' + u')$ . Let  $u_{2p} = x$ . Then  $y_{2p} = xe^{2x}$ .

(c) If  $y_{3p} = A \cos x + B \sin x$ , then  $y_{3p}''' - 4y_{3p}'' + 5y_{3p}' - 2y_{3p} = (2A + 4B) \cos x + (-4A + 2B) \sin x = -2 \cos x + 4 \sin x$  if  $A = -1$  and  $B = 0$ , so  $y_{3p} = -\cos x$ .

From the principle of superposition,  $y_p = 2x^2 e^x + xe^{2x} - \cos x$ .

**9.3.50.** Find particular solutions of (a)  $y''' - y' = -2(1+x)$ , (b)  $y''' - y' = 4e^x$ , (c)  $y''' - y' = -6e^{-x}$ , and (d)  $y''' - y' = 96e^{3x}$

(a) Let  $y_{1p} = x(A + Bx)$ . Then  $y_{1p}''' - y_{1p}' = -A - 2Bx = -2(1+x)$  if  $A = 2$  and  $B = 1$ ; therefore  $y_{1p} = 2x + 2x^2$ .

(b) If  $y = ue^x$ , then  $y''' - y' = e^x[(u''' + 3u'' + 3u' + u) - (u' + u)] = e^x(u''' + 3u'' + 2u')$ . Let  $u_{2p} = 4x$ . Then  $y_{2p} = 4xe^x$ .

(c) If  $y = ue^{-x}$ , then  $y''' - y' = e^{-x}[(u''' - 3u'' + 3u' - u) - (u' - u)] = e^{-x}(u''' - 3u'' + 2u')$ . Let  $u_{2p} = -3x$ . Then  $y_{2p} = -6xe^{-x}$ .

(d) Since  $e^{3x}$  does not satisfy the complementary equation, let  $y_{4p} = Ae^{3x}$ . Then  $y_{4p}''' - y_{4p}' = 24Ae^{3x}$ . Let  $A = 4$ ; then  $y_{4p} = 4e^{4x}$ .

From the principle of superposition,  $y_p = 2x + x^2 + 2xe^x - 3xe^{-x} + 4e^{3x}$ .

**9.3.52.** Find particular solutions of (a)  $y''' + 3y'' + 3y' + y = 12e^{-x}$  and (b)  $y''' + 3y'' + 3y' + y = 9\cos 2x - 13\sin 2x$ .

(a) If  $y = ue^{-2x}$ , then  $y''' + 3y'' + 3y' + y = e^{-2x}[(u''' - 3u'' + 3u' - u) + 3(u'' - 2u' + u) + 3(u' - u) + u] = e^{-x}u'''$ . Let  $u_{1p}''' = 12$ . Integrating three times and taking the constants of integration to be zero yields  $u_{1p} = 2x^3$ . Therefore,  $y_{1p} = 2x^3$ .

(b) Let  $y_{2p} = A\cos 2x + B\sin 2x$  where  $-11A - 2B = 9$  and  $2A - 11B = -13$ . Then  $A = -1$ ,  $B = 1$ , and  $y_{2p} = -\cos 2x + \sin 2x$ .

From the principle of superposition,  $y_p = 2x^3e^{-2x} - \cos 2x + \sin 2x$ .

**9.3.54.** Find particular solutions of (a)  $y^{(4)} - 5y'' + 4y = -12e^x$ , (b)  $y^{(4)} - 5y'' + 4y = 6e^{-x}$ , and (c)  $y^{(4)} - 5y'' + 4y = 10\cos x$ .

(a) If  $y = ue^x$ , then  $y^{(4)} - 5y'' + 4y = e^x[(u^{(4)} + 4u''' + 6u'' + 4u' + u) - 5(u'' + 2u' + u) + 4u] = e^x(u^{(4)} + 4u''' + u'' - 6u')$ . Let  $u_{1p} = 2x$ . Then  $y_{1p} = 2xe^x$ .

(b) If  $y = ue^{-x}$ , then  $y^{(4)} - 5y'' + 4y = e^{-x}[(u^{(4)} - 4u''' + 6u'' - 4u' + u) - 5(u'' - 2u' + u) + 4u] = e^{-x}(u^{(4)} - 4u''' + u'' + 6u')$ . Let  $u_{2p} = x$ . Then  $y_{2p} = xe^{-x}$ .

(c) Let  $y_{3p} = A\cos x + B\sin x$  where  $10A = 10$  and  $10B = 0$ . Then  $A = 1$ ,  $B = 0$ , and  $y_{3p} = \cos x$ .

From the principle of superposition,  $y_p = 2xe^x + xe^{-x} + \cos x$ .

**9.3.56.** Find particular solutions of (a)  $y^{(4)} + 2y''' - 3y'' - 4y' + 4y = 2e^x(1+x)$  and (b)  $y^{(4)} + 2y''' - 3y'' - 4y' + 4y = e^{-2x}$ .

(a) If  $y = ue^x$ , then  $y^{(4)} + 2y''' - 3y'' - 4y' + 4y = e^x[(u^{(4)} + 4u''' + 6u'' + 4u' + u) + 2(u''' + 3u'' + 3u' + u) - 3(u'' + 2u' + u) - 4(u' + u) + 4u] = e^x(u^{(4)} + 6u''' + 9u'')$ . Let  $u_{1p} = x^2(A + Bx)$  where

$(18A + 36B) + 54Bx = 2 + 2x$ . Then  $B = 1/27$ ,  $A = 1/27$ , and  $y_{1p} = \frac{x^2}{27}(1+x)e^x$ .

(b) If  $y = ue^{-2x}$ , then  $y^{(4)} + 2y''' - 3y'' - 4y' + 4y = e^{-2x}[(u^{(4)} - 8u''' + 24u'' - 32u' + 16u) + 2(u''' - 6u'' + 12u' - 8u) - 3(u'' - 4u' + 4u) - 4(u' - 2u) + 4u] = e^{-2x}(u^{(4)} - 6u''' + 9u'')$ . Let  $u_{2p} = Ax^2$

where  $18A = 1$ . Then  $A = 1/18$  and  $y_p = \frac{x^2}{18}e^{-2x}$ .

From the principle of superposition,  $y_p = \frac{x^2}{54}[(2+2x)e^x + 3e^{-2x}]$ .

**9.3.58.** Find particular solutions of (a)  $y^{(4)} + 5y''' + 9y'' + 7y' + 2y = e^{-x}(30 + 24x)$  and (b)  $y^{(4)} + 5y''' + 9y'' + 7y' + 2y = -e^{-2x}$ .

(a) If  $y = ue^{-x}$ , then  $y^{(4)} + 5y''' + 9y'' + 7y' + 2y = e^{-x}[(u^{(4)} - 4u''' + 6u'' - 4u' + u) + 5(u''' - 3u'' + 3u' - u) + 9(u'' - 2u' + u) + 7(u' - u) + 2u] = e^{-x}(u^{(4)} + u''')$ . Let  $u_{1p} = x^3(A + Bx)$  where  $(6A + 24B) + 24Bx = 30 + 24x$ . The  $B = 1$ ,  $A = 1$ , and  $y_{1p} = x^3(1+x)e^{-x}$ .

(b) If  $y = ue^{-2x}$ , then  $y^{(4)} + 5y''' + 9y'' + 7y' + 2y = e^{-2x}[(u^{(4)} - 8u''' + 24u'' - 32u' + 16u) + 5(u''' - 6u'' + 12u' - 8u) + 9(u'' - 4u' + 4u) + 7(u' - 2u) + 2u] = e^{-2x}(u^{(4)} - 3u''' + 3u'' - u')$ . Let  $u_{2p} = x$ . Then  $y_{2p} = xe^{-2x}$ .

From the principle of superposition,  $y_p = x^3(1+x)e^{-x} + xe^{-2x}$ .

**9.3.60.** If  $y = ue^{2x}$ , then  $y''' - y'' - y' + y = e^{2x}[(u''' + 6u'' + 12u' + 8u) - (u'' + 4u' + 4u) - (u' + 2u) + u] = e^{2x}(u''' + 5u'' + 7u' + 3u)$ . Let  $u_p = A + Bx$ , where  $(3A + 7B) + 3x = 10 + 3x$ . Then  $B = 1$ ,  $A = 1$  and  $y_p = e^{2x}(1+x)$ . Since  $p(r) = (r+1)(r-1)^2$ ,  $y = e^{2x}(1+x) + c_1e^{-x} + e^x(c_2 + c_3x)$

**9.3.62.** If  $y = ue^{2x}$ , then  $y''' - 6y'' + 11y' - 6y = e^{2x}[(u''' + 6u'' + 12u' + 8u) - 6(u'' + 4u' + 4u) + 11(u' + 2u) - 6u] = e^{2x}(u''' - u')$ . Let  $u_p = x(A + Bx + Cx^2)$  where  $(-A + 6C) - 2Bx - 3Cx^2 =$

$5 - 4x - 3x^2$ . Then  $C = 1$ ,  $B = 2$ ,  $A = 1$ , and  $y_p = xe^{2x}(1+x)^2$ . Since  $p(r) = (r-1)(r-2)(r-3)$ ,  $y = xe^{2x}(1+x)^2 + c_1e^x + c_2e^{2x} + c_3e^{3x}$ .

**9.3.64.** If  $y = ue^x$ , then  $y''' - 3y'' + 3y' - y = e^x[(u''' + 3u'' + 3u' + u) - 3(u'' + 2u' + u) + 3(u' + u) - u] = e^xu'''$ . Let  $u''' = 1 + x$ . Integrating three times and taking the constants of integration to be zero yields  $u = \frac{x^3}{24}(4+x)$ . Therefore,  $y_p = \frac{x^3e^x}{24}(4+x)$ . Since  $p(r) = (r-1)^3$ ,

$$y = \frac{x^3e^x}{24}(4+x) + e^x(c_1 + c_2x + c_3x^2).$$

**9.3.66.** If  $y = ue^{-2x}$ , then  $y''' + 2y'' - y' - 2y = e^{-2x}[(u''' - 6u'' + 12u' - 8u) + 2(u'' - 4u' + 4u) - (u' - 2u) - 2u] = e^{-2x}(u''' - 4u'' + 3u')$ . Let  $u_p = (A_0 + A_1x)\cos x + (B_0 + B_1x)\sin x$  where

$$\begin{aligned} 4A_1 + 2B_1 &= -2 \\ -2A_1 + 4B_1 &= -9 \\ 4A_0 + 2B_0 - 8B_1 &= 23 \\ -2A_0 + 4B_0 + 8A_1 &= 8. \end{aligned}$$

Then  $A_1 = 1/2$ ,  $B_1 = -2$ ;  $A_0 = 1$ ,  $B_0 = 3/2$ , and  $y_p = e^{-2x}\left[\left(1 + \frac{x}{2}\right)\cos x + \left(\frac{3}{2} - 2x\right)\sin x\right]$ .

Since  $p(r) = (r-1)(r+1)(r+2)$ ,  $y = e^{-2x}\left[\left(1 + \frac{x}{2}\right)\cos x + \left(\frac{3}{2} - 2x\right)\sin x\right] + c_1e^x + c_2e^{-x} + c_3e^{-2x}$ .

**9.3.68.** If  $y = ue^x$ , then  $y^{(4)} - 4y''' + 14y'' - 20y' + 25y = e^x[(u^{(4)} + 4u''' + 6u'' + 4u' + u) - 4(u''' + 3u'' + 3u' + u) + 14(u'' + 2u' + u) - 20(u' + u) + 25u] = e^x(u^{(4)} + 8u'' + 16u)$ . Since  $\cos 2x$ ,  $\sin 2x$ ,  $x\cos 2x$ , and  $x\sin 2x$  satisfy  $u^{(4)} + 8u'' + 16u = 0$ , let  $u_p = x^2[(A_0 + A_1x)\cos 2x + (B_0 + B_1x)\sin 2x]$  where

$$\begin{aligned} -96A_1 &= 6 \\ -96B_1 &= 0 \\ -32A_0 + 48B_1 &= 2 \\ -48A_1 - 32B_0 &= 3. \end{aligned}$$

Then  $A_1 = -1/16$ ,  $B_1 = 0$ ;  $A_0 = -1/16$ ,  $B_0 = 0$ , and  $y_l = -\frac{x^2e^x}{16}(1+x)\cos 2x$ . Since  $p(r) =$

$$[(r-1)^2 + 1]^2, y = -\frac{x^2e^x}{16}(1+x)\cos 2x + e^x[(c_1 + c_2x)\cos 2x + (c_3 + c_4x)\sin 2x].$$

**9.3.70.** If  $y = ue^{-x}$ , then  $y''' - y'' - y' + y = e^{-x}[(u''' - 3u'' + 3u' - u) - (u'' - 2u' + u) - (u' - u) + u] = e^{-x}(u''' - 4u'' + 4u')$ . Let  $u_p = x(A + Bx)$ , where  $(4A - 8B) + 8Bx = -4 + 8x$ . Then  $B = 1$ ,  $A = 1$ , and  $y_p = x(1+x)e^{-x}$ . Since  $p(r) = (r+1)(r-1)^2$  the general solution is  $y = x(1+x)e^{-x} + c_1e^{-x} + c_2e^x + c_3xe^x$ . Therefore,

$$\begin{bmatrix} y \\ y' \\ y'' \end{bmatrix} = \begin{bmatrix} x(1+x)e^{-x} \\ -e^{-x}(x^2 - x - 1) \\ e^{-x}(x^2 - 3x) \end{bmatrix} + \begin{bmatrix} e^{-x} & e^x & xe^x \\ -e^{-x} & e^x & e^x(x+1) \\ e^{-x} & e^x & e^x(x+2) \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}.$$

Setting  $x = 0$  and imposing the initial conditions yields

$$\begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 1 & 1 & 0 \\ -1 & 1 & 1 \\ 1 & 1 & 2 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix},$$

so  $c_1 = 1$ ,  $c_2 = 1$ ,  $c_3 = -1$ , and  $y = e^{-x}(1+x+x^2) + (1-x)e^x$ .

**9.3.72.** If  $y = ue^{-x}$ , then  $y''' - 2y'' - 5y' + 6y = e^{-x}[(u^{(4)} - 4u''' + 6u'' - 4u' + u) + 2(u''' - 3u'' + 3u' - u) + 2(u'' - 2u' + u) + 2(u' - u) + u] = e^{-x}(u^{(4)} - 2u''' + 2u'')$ . Let  $u_p = x^2(A + Bx)$ , where  $(4A - 12B) + 12Bx = 20 - 12x$ . Then  $B = -1$ ,  $A = 2$ , and  $y_p = x^2(2 - x)e^{-x}$ . Since  $p(r) = (r+1)^2(r^2+1)$ , the general solution is  $y = x^2(2 - x)e^{-x} + e^{-x}(c_1 + c_2x) + c_3 \cos x + c_4 \sin x$ . Therefore,

$$\begin{bmatrix} y \\ y' \\ y'' \\ y''' \end{bmatrix} = \begin{bmatrix} x^2(2-x)e^{-x} \\ x(x^2-5x+4)e^{-x} \\ -(x^3-8x^2+14x-4)e^{-x} \\ (x^3-11x^2+30x-18)e^{-x} \end{bmatrix} + \begin{bmatrix} e^{-x} & xe^{-x} & \cos x & \sin x \\ -e^{-x} & (1-x)e^{-x} & -\sin x & \cos x \\ e^{-x} & (x-2)e^{-x} & -\cos x & -\sin x \\ -e^{-x} & (3-x)e^{-x} & \sin x & -\cos x \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{bmatrix}$$

Setting  $x = 0$  and imposing the initial conditions yields

$$\begin{bmatrix} 3 \\ -4 \\ 7 \\ -22 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 4 \\ -18 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 1 & 0 \\ -1 & 1 & 0 & 1 \\ 1 & -2 & -1 & 0 \\ -1 & 3 & 0 & -1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{bmatrix},$$

so  $c_1 = 2$ ,  $c_2 = -1$ ,  $c_3 = 1$ ,  $c_4 = -1$ , and  $y = (2 - x)(x^2 + 1)e^{-x} + \cos x - \sin x$ .

**9.3.74.** If  $y = ue^x$ , then  $y^{(4)} - 3y''' + 5y'' - 2y' = e^x[(u^{(4)} + 4u''' + 6u'' + 4u' + u) - 3(u''' + 3u'' + 3u' + u) + 4(u'' + 2u' + u) - 2(u' + u)] = e^x(u^{(4)} + u''' + u'' + u')$ . Since  $\cos x$  and  $\sin x$  satisfy  $u^{(4)} + u''' + u'' + u' = 0$ , let  $u_p = x(A \cos x + B \sin x)$  where  $-2A - 2B = -2$  and  $2A - 2B = 2$ . Then  $A = 1$ ,  $B = 0$ , and  $y_p = e^x \cos x$ . Since  $p(r) = r(r-1)[(r-1)^2 + 1]$  the general solution is  $y = e^x \cos x + c_1 + e^x(c_2 + c_3 \cos x + c_4 \sin x)$ . Therefore,

$$\begin{bmatrix} y \\ y' \\ y'' \\ y''' \end{bmatrix} = \begin{bmatrix} xe^x \cos x \\ e^x((x+1) \cos x - x \sin x) \\ e^x(2 \cos x - 2(x+1) \sin x) \\ -e^x(2x \cos x + 2(x+3) \sin x) \end{bmatrix} + \begin{bmatrix} 1 & e^x & e^x \cos x & e^x \sin x \\ 0 & e^x & e^x(\cos x - \sin x) & e^x(\cos x + \sin x) \\ 0 & e^x & -2e^x \sin x & 2e^x \cos x \\ 0 & e^x & -e^x(2 \cos x + 2 \sin x) & e^x(2 \cos x - 2 \sin x) \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{bmatrix}$$

Setting  $x = 0$  and imposing the initial conditions yields

$$\begin{bmatrix} 2 \\ 0 \\ -1 \\ -5 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 2 \\ 0 \end{bmatrix} + \begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 1 & -2 & 2 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{bmatrix},$$

so  $c_1 = 2$ ,  $c_2 = -1$ ,  $c_3 = 1$ ,  $c_4 = -1$ , and  $2 + e^x[(1 + x) \cos x - \sin x - 1]$ .

## 9.4 Variation of Parameters for Higher Order Equations

$$\begin{aligned} \mathbf{9.4.2.} \quad W &= \begin{vmatrix} e^{-x^2} & xe^{-x^2} & x^2e^{-x^2} \\ -2xe^{-x^2} & e^{-x^2}(1-2x^2) & 2xe^{-x^2}(1-x^2) \\ e^{-x^2}(4x^2-2) & 2xe^{-x^2}(2x^2-3) & 2e^{-x^2}(2x^4-5x^2+1) \end{vmatrix} = 2e^{-3x^2}; \\ W_1 &= \begin{vmatrix} xe^{-x^2} & x^2e^{-x^2} \\ e^{-x^2}(1-2x^2) & 2xe^{-x^2}(1-x^2) \end{vmatrix} = x^2e^{-2x^2}; \quad W_2 = \begin{vmatrix} e^{-x^2} & x^2e^{-x^2} \\ -2xe^{-x^2} & 2xe^{-x^2}(1-x^2) \end{vmatrix} = \\ 2xe^{-2x^2}; \quad W_3 &= \begin{vmatrix} e^{-x^2} & xe^{-x^2} \\ -2xe^{-x^2} & e^{-x^2}(1-2x^2) \end{vmatrix} = e^{-2x^2}; \quad u'_1 = \frac{FW_1}{P_0W} = \frac{1}{2}x^{5/2}; \quad u'_2 = -\frac{FW_2}{P_0W} = -x^{3/2}; \end{aligned}$$

$$u'_3 = \frac{FW_2}{P_0W} = \sqrt{x}/2; u_1 = x^{7/2}/7; u_2 = -\frac{2}{5}x^{5/2}; u_3 = x^{3/2}/3; y_p = u_1y_1 + u_2y_2 + u_3y_3 = \frac{8}{105}e^{-x^2}x^{7/2}.$$

$$\begin{aligned} \mathbf{9.4.4.} \quad W &= \begin{vmatrix} 1 & \frac{e^x}{x} & \frac{e^{-x}}{x} \\ 0 & \frac{e^x(x-1)}{x^2} & -\frac{e^{-x}(x+1)}{x^2} \\ 0 & \frac{e^x(x^2-2x+2)}{x^3} & \frac{e^{-x}(x^2+2x+2)}{x^3} \end{vmatrix} = 2/x^2; W_1 = \begin{vmatrix} \frac{e^x}{x} & \frac{e^{-x}}{x} \\ \frac{e^x(x-1)}{x^2} & -\frac{e^{-x}(x+1)}{x^2} \end{vmatrix} = \\ -\frac{2}{x^2}; W_2 &= \begin{vmatrix} 1 & \frac{e^{-x}}{x} \\ 0 & -\frac{e^{-x}(x+1)}{x^2} \end{vmatrix} = -\frac{e^{-x}(x+1)}{x^2}; W_3 = \begin{vmatrix} 1 & \frac{e^x}{x} \\ 0 & \frac{e^x(x-1)}{x^2} \end{vmatrix} = \frac{e^x(x-1)}{x^2}; u'_1 = \\ \frac{FW_1}{P_0W} &= -2; u'_2 = -\frac{FW_2}{P_0W} = e^{-x}(x+1); u'_3 = \frac{FW_3}{P_0W} = e^x(x-1); u_1 = -2x; u_2 = -e^{-x}(x+2); \\ u_3 &= e^x(x-2); y_p = u_1y_1 + u_2y_2 + u_3y_3 = -2(x^2+2)/x. \end{aligned}$$

$$\begin{aligned} \mathbf{9.4.6.} \quad W &= \begin{vmatrix} e^x & e^{-x} & \frac{1}{x} \\ e^x & -e^{-x} & -1/x^2 \\ e^x & e^{-x} & \frac{2}{x^3} \end{vmatrix} = \frac{2(x^2-2)}{x^3}; W_1 = \begin{vmatrix} e^{-x} & \frac{1}{x} \\ -e^{-x} & -\frac{1}{x^2} \end{vmatrix} = \frac{e^{-x}(x-1)}{x^2}; W_2 = \\ \begin{vmatrix} e^x & \frac{1}{x} \\ e^x & -\frac{1}{x^2} \end{vmatrix} &= -\frac{e^x(x+1)}{x^2}; W_3 = \begin{vmatrix} e^x & e^{-x} \\ e^x & -e^{-x} \end{vmatrix} = -2; u'_1 = \frac{FW_1}{P_0W} = e^{-x}(x-1); u'_2 = -\frac{FW_2}{P_0W} = \\ e^x(x+1); u'_3 &= \frac{FW_3}{P_0W} = -2x^2; y_p = u_1y_1 + u_2y_2 + u_3y_3 = -2\frac{x^2}{3}. \end{aligned}$$

$$\begin{aligned} \mathbf{9.4.8.} \quad W &= \begin{vmatrix} \sqrt{x} & \frac{1}{\sqrt{x}} & x^2 \\ \frac{1}{2\sqrt{x}} & -\frac{1}{2x^{3/2}} & 2x \\ -\frac{1}{4x^{3/2}} & \frac{1}{4x^{5/2}} & 2 \end{vmatrix} = -\frac{15}{4x}; W_1 = \begin{vmatrix} \frac{1}{\sqrt{x}} & x^2 \\ -\frac{1}{2x^{3/2}} & 2x \end{vmatrix} = \frac{5\sqrt{x}}{2}; W_2 = \begin{vmatrix} \sqrt{x} & x^2 \\ \frac{1}{2\sqrt{x}} & 2x \end{vmatrix} = \\ \frac{3x^{3/2}}{2}; W_3 &= \begin{vmatrix} \sqrt{x} & \frac{1}{\sqrt{x}} \\ \frac{1}{2\sqrt{x}} & -\frac{1}{2x^{3/2}} \end{vmatrix} = -\frac{1}{x}; u'_1 = \frac{FW_1}{P_0W} = -5\sqrt{x}; u'_2 = -\frac{FW_2}{P_0W} = 3x^{3/2}; u'_3 = \frac{FW_3}{P_0W} = \\ \frac{2}{x}; u_1 &= -\frac{10}{3}x^{3/2}; u_2 = \frac{6}{5}x^{5/2}; u_3 = 2\ln|x|; y_p = u_1y_1 + u_2y_2 + u_3y_3 = 2x^2\ln|x| - \frac{32}{15}x^2. \text{ Since } \\ -\frac{32}{15}x^2 &\text{ satisfies the complementary equation we take } y_p = \ln|x|. \end{aligned}$$

$$\mathbf{9.4.10.} \quad W = \begin{vmatrix} x & 1/x & \frac{e^x}{x} \\ 1 & -\frac{1}{x^2} & \frac{e^x(x-1)}{x^2} \\ 0 & \frac{2}{x^3} & \frac{e^x(x^2-2x+2)}{x^3} \end{vmatrix} = \frac{2e^x(1-x)}{x^3}; W_1 = \begin{vmatrix} \frac{1}{x} & \frac{e^x}{x} \\ -\frac{1}{x^2} & \frac{e^x(x-1)}{x^2} \end{vmatrix} = \frac{e^x}{x^2};$$

$$W_2 = \begin{vmatrix} x & \frac{e^x}{x} \\ 1 & \frac{e^x(x-1)}{x^2} \end{vmatrix} = \frac{e^x(x-2)}{x}; W_3 = \begin{vmatrix} x & \frac{1}{x} \\ 1 & -\frac{1}{x^2} \end{vmatrix} = -\frac{2}{x}; u'_1 = \frac{FW_1}{P_0W} = 1; u'_2 = -\frac{FW_2}{P_0W} = x(2-x); u'_3 = \frac{FW_3}{P_0W} = -2xe^{-x}; u_1 = x; u_2 = \frac{x^2(3-x)}{3}; u_3 = 2e^{-x}(x+1); y_p = u_1y_1 + u_2y_2 + u_3y_3 = \frac{2x^3 + 3x^2 + 6x + 6}{3x}. \text{ Since } x + \frac{2}{x} \text{ satisfies the complementary equation we take } y_p = \frac{2x^2 + 6}{3}.$$

$$\mathbf{9.4.12.} \quad W = \begin{vmatrix} x & e^x & e^{-x} \\ 1 & e^x & -e^{-x} \\ 0 & e^x & e^{-x} \end{vmatrix} = 2x; W_1 = \begin{vmatrix} e^x & e^{-x} \\ e^x & -e^{-x} \end{vmatrix} = -2; W_2 = \begin{vmatrix} x & e^{-x} \\ 1 & -e^{-x} \end{vmatrix} = -e^{-x}(x+1); W_3 = \begin{vmatrix} x & e^x \\ 1 & e^x \end{vmatrix} = e^x(x-1); u'_1 = \frac{FW_1}{P_0W} = -1; u'_2 = -\frac{FW_2}{P_0W} = e^{-x}(x+1)/2; u'_3 = \frac{FW_3}{P_0W} = e^x(x-1)/2; u_1 = -x; u_2 = -e^{-x}(x+2)/2; u_3 = e^x(x-2)/2; y_p = u_1y_1 + u_2y_2 + u_3y_3 = -x^2 - 2.$$

$$\mathbf{9.4.14.} \quad W = \begin{vmatrix} \sqrt{x} & 1/\sqrt{x} & x^{3/2} & \frac{1}{x^{3/2}} \\ \frac{1}{2\sqrt{x}} & \frac{1}{2x^{3/2}} & \frac{3\sqrt{x}}{2} & -\frac{2x^{5/2}}{15} \\ -\frac{4x^{3/2}}{3} & \frac{4x^{5/2}}{15} & \frac{4\sqrt{x}}{3} & \frac{4x^{7/2}}{105} \\ \frac{8x^{5/2}}{8x^{5/2}} & -\frac{8x^{7/2}}{8x^{7/2}} & -\frac{8x^{3/2}}{8x^{3/2}} & -\frac{8x^{9/2}}{8x^{9/2}} \end{vmatrix} = \frac{12}{x^6}; W_1 = \begin{vmatrix} \frac{1}{\sqrt{x}} & x^{3/2} & \frac{1}{x^{3/2}} \\ -\frac{1}{2x^{3/2}} & \frac{3\sqrt{x}}{2} & -\frac{3}{2x^{5/2}} \\ \frac{4x^{3/2}}{3} & \frac{4x^{5/2}}{15} & \frac{4x^{7/2}}{105} \end{vmatrix} = \frac{6}{x^{5/2}}; W_2 = \begin{vmatrix} \sqrt{x} & x^{3/2} & \frac{1}{x^{3/2}} \\ \frac{1}{2\sqrt{x}} & \frac{3\sqrt{x}}{2} & -\frac{3}{2x^{5/2}} \\ -\frac{1}{4x^{3/2}} & \frac{3}{4\sqrt{x}} & \frac{15}{4x^{7/2}} \end{vmatrix} = -\frac{2}{x^{9/2}}; W_3 = \begin{vmatrix} \sqrt{x} & \frac{1}{\sqrt{x}} & \frac{1}{x^{3/2}} \\ \frac{1}{2\sqrt{x}} & \frac{1}{2x^{3/2}} & -\frac{3}{2x^{5/2}} \\ -\frac{1}{4x^{3/2}} & \frac{3}{4x^{5/2}} & \frac{15}{4x^{7/2}} \end{vmatrix} = -\frac{2}{x^{3/2}}; u'_1 = -\frac{FW_1}{P_0W} = -3x; u'_2 = \frac{FW_2}{P_0W} = 3x^2; u'_3 = -\frac{FW_3}{P_0W} = 1; u'_4 = \frac{FW_4}{P_0W} = -x^3; u_1 = -\frac{3x^2}{2}; u_2 = x^3; u_3 = x; u_4 = -\frac{x^4}{4}; y_p = u_1y_1 + u_2y_2 + u_3y_3 = \frac{x^{5/2}}{4}.$$

$$\mathbf{9.4.16.} \quad W = \begin{vmatrix} x & x^2 & x^3 & x^4 \\ 1 & 2x & 3x^2 & 4x^3 \\ 0 & 2 & 6x & 12x^2 \\ 0 & 0 & 6 & 24x \end{vmatrix} = 12x^4; W_1 = \begin{vmatrix} x^2 & x^3 & x^4 \\ 2x & 3x^2 & 4x^3 \\ 2 & 6x & 12x^2 \end{vmatrix} = 2x^6; W_2 = \begin{vmatrix} x & x^3 & x^4 \\ 1 & 3x^2 & 4x^3 \\ 0 & 6x & 12x^2 \end{vmatrix} = 6x^5; W_3 = \begin{vmatrix} x & x^2 & x^3 \\ 1 & 2x & 3x^2 \\ 0 & 2 & 6x \end{vmatrix} = 6x^4; W_4 = \begin{vmatrix} x & x^2 & x^3 \\ 1 & 2x & 3x^2 \\ 0 & 2 & 6x \end{vmatrix} = 2x^3; u'_1 = -\frac{FW_1}{P_0W} = -\frac{x^2}{6}; u'_2 = \frac{FW_2}{P_0W} = \frac{x}{2}; u'_3 = -\frac{FW_3}{P_0W} = -\frac{1}{2}; u'_4 = \frac{FW_4}{P_0W} = \frac{1}{6x}; u_1 = -\frac{x^3}{18}; u_2 = \frac{x^2}{4}; u_3 = -\frac{x}{2}; u_4 = \frac{\ln|x|}{6};$$

$y_p = u_1 y_1 + u_2 y_2 + u_3 y_3 = \frac{x^4 \ln |x|}{6} - \frac{11x^4}{36}$ . Since  $-\frac{11x^4}{36}$  satisfies the complementary equation we take  $y_p = \frac{x^4 \ln |x|}{6}$ .

$$\begin{aligned} \mathbf{9.4.18.} \quad W &= \begin{vmatrix} x & x^2 & 1/x & 1/x^2 \\ 1 & 2x & -1/x^2 & -2/x^3 \\ 0 & 2 & 2/x^3 & 6/x^4 \\ 0 & 0 & -6/x^4 & -24/x^5 \end{vmatrix} = -72/x^6; \quad W_1 = \begin{vmatrix} x^2 & 1/x & 1/x^2 \\ 2x & -1/x^2 & -2/x^3 \\ 2 & 2/x^3 & 6/x^4 \end{vmatrix} = -12/x^4; \\ W_2 &= \begin{vmatrix} x & 1/x & 1/x^2 \\ 1 & -1/x^2 & -2/x^3 \\ 0 & 2/x^3 & 6/x^4 \end{vmatrix} = -6/x^5; \quad W_3 = \begin{vmatrix} x & x^2 & 1/x^2 \\ 1 & 2x & -2/x^3 \\ 0 & 2 & 6/x^4 \end{vmatrix} = 12/x^2; \quad W_4 = \begin{vmatrix} x & x^2 & 1/x \\ 1 & 2x & -1/x^2 \\ 0 & 2 & 2/x^3 \end{vmatrix} = \\ 6/x; \quad u'_1 &= -\frac{FW_1}{P_0 W} = -2; \quad u'_2 = \frac{FW_2}{P_0 W} = 1/x; \quad u'_3 = -\frac{FW_3}{P_0 W} = 2x^2; \quad u'_4 = \frac{FW_4}{P_0 W} = -x^3; \quad u_1 = -2x; \\ u_2 &= \ln |x|; \quad u_3 = 2x^3/3; \quad u_4 = -x^4/4; \quad y_p = u_1 y_1 + u_2 y_2 + u_3 y_3 = x^2 \ln |x| - 19x^2/12. \text{ Since } -19x^2/12 \\ &\text{satisfies the complementary equation we take } y_p = x^2 \ln |x|. \end{aligned}$$

$$\mathbf{9.4.20.} \quad W = \begin{vmatrix} e^x & e^{2x} & \frac{e^x}{x} & \frac{e^{2x}}{x} \\ e^x & 2e^{2x} & \frac{e^x(x-1)}{x^2} & \frac{e^{2x}(2x-1)}{x^2} \\ e^x & 4e^{2x} & \frac{e^x(x^2-2x+2)}{x^3} & \frac{2e^{2x}(2x^2-2x+1)}{x^3} \\ e^x & 8e^{2x} & \frac{e^x(x^3-3x^2+6x-6)}{x^4} & \frac{2e^{2x}(4x^3-6x^2+6x-3)}{x^4} \end{vmatrix} = -\frac{e^{6x}}{x^4};$$

$$W_1 = \begin{vmatrix} e^{2x} & \frac{e^x}{x} & \frac{e^{2x}}{x} \\ 2e^{2x} & \frac{e^x(x-1)}{x^2} & \frac{e^{2x}(2x-1)}{x^2} \\ 4e^{2x} & \frac{e^x(x^2-2x+2)}{x^3} & \frac{2e^{2x}(2x^2-2x+1)}{x^3} \end{vmatrix} = \frac{e^{5x}}{x^3};$$

$$W_2 = \begin{vmatrix} e^x & \frac{e^x}{x} & \frac{e^{2x}}{x^3} \\ e^x & \frac{e^x(x-1)}{x^2} & \frac{e^{2x}(2x-1)}{x^3} \\ e^x & \frac{e^x(x^2-2x+2)}{x^3} & \frac{2e^{2x}(2x^2-2x+1)}{x^3} \end{vmatrix} = -\frac{e^{4x}}{x^3};$$

$$W_3 = \begin{vmatrix} e^x & e^{2x} & \frac{e^{2x}}{x^3} \\ e^x & 2e^{2x} & \frac{e^{2x}(2x-1)}{x^3} \\ e^x & 4e^{2x} & \frac{2e^{2x}(2x^2-2x+1)}{x^3} \end{vmatrix} = \frac{e^{5x}(2-x)}{x^3};$$

$$W_4 = \begin{vmatrix} e^x & e^{2x} & \frac{e^x}{x^3} \\ e^x & 2e^{2x} & \frac{e^x(x-1)}{x^3} \\ e^x & 4e^{2x} & \frac{e^x(x^2-2x+2)}{x^3} \end{vmatrix} = \frac{e^{4x}(x+2)}{x^3};$$

$$u'_1 = -\frac{FW_1}{P_0 W} = 3; \quad u'_2 = \frac{FW_2}{P_0 W} = 3e^{-x}; \quad u'_3 = -\frac{FW_3}{P_0 W} = 3(2-x); \quad u'_4 = \frac{FW_4}{P_0 W} = -3e^{-x}(x+2); \quad u_1 =$$



$$3x; u_2 = -3e^{-x}; u_3 = 3x(4-x)/2; u_4 = 3e^{-x}(x+3); y_p = u_1y_1 + u_2y_2 + u_3y_3 = \frac{3e^x(x^2 + 4x + 6)}{2x}.$$

Since  $\frac{3e^x(2x+3)}{x}$  is a solution of the complementary equation we take  $y_p = \frac{3xe^x}{2}$ .

$$\begin{aligned} \mathbf{9.4.22.} \quad W &= \begin{vmatrix} x & x^3 & x \ln x \\ 1 & 3x^2 & 1 + \ln x \\ 0 & 6x & 1/x \end{vmatrix} = -4x^2; \quad W_1 = \begin{vmatrix} x^3 & x \ln x \\ 3x^2 & 1 + \ln x \end{vmatrix} = x^3 - 2x^3 \ln x; \quad W_2 = \\ &\begin{vmatrix} x & x \ln x \\ 1 & 1 + \ln x \end{vmatrix} = x; \quad W_3 = \begin{vmatrix} x & x^3 \\ 1 & 3x^2 \end{vmatrix} = 2x^3; \quad u'_1 = \frac{FW_1}{P_0W} = 2 \ln x/x - \frac{1}{x}; \quad u'_2 = -\frac{FW_2}{P_0W} = \frac{1}{x^3}; \\ u'_3 &= \frac{FW_3}{P_0W} = -\frac{2}{x}; \quad u_1 = (\ln x)^2 - \ln x; \quad u_2 = -\frac{1}{2x^2}; \quad u_3 = -2 \ln x; \quad y_p = u_1y_1 + u_2y_2 + u_3y_3 = \\ &-x(\ln x)^2 - x \ln x - \frac{x}{2}. \quad \text{Since } -x \ln x - \frac{x}{2} \text{ satisfies the complementary equation we take } y_p = -x(\ln x)^2 \end{aligned}$$

The general solution is  $y = -x(\ln x)^2 + c_1x + c_2x^3 + c_3x \ln x$ , so

$$\begin{bmatrix} y \\ y' \\ y'' \end{bmatrix} = \begin{bmatrix} -x(\ln x)^2 \\ -(\ln x)^2 - 2 \ln x \\ -\frac{2 \ln x}{x} - \frac{2}{x} \end{bmatrix} + \begin{bmatrix} x & x^3 & x \ln x \\ 1 & 3x^2 & 1 + \ln x \\ 0 & 6x & \frac{1}{x} \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}$$

Setting  $x = 1$  and imposing the initial conditions yields

$$\begin{bmatrix} 4 \\ 4 \\ -2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ -2 \end{bmatrix} + \begin{bmatrix} 1 & 1 & 0 \\ 1 & 3 & 1 \\ 0 & 6 & 1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}.$$

Solving this system yields  $c_1 = 3, c_2 = 1, c_3 = -2$ . Therefore,  $y = -x(\ln x)^2 + 3x + x^3 - 2x \ln x$ .

$$\begin{aligned} \mathbf{9.4.24.} \quad W &= \begin{vmatrix} e^x & e^{2x} & xe^{-x} \\ e^x & 2e^{2x} & e^{-x}(1-x) \\ e^x & 4e^{2x} & e^{-x}(x-2) \end{vmatrix} = e^{2x}(6x-5); \quad W_1 = \begin{vmatrix} e^{2x} & xe^{-x} \\ 2e^{2x} & e^{-x}(1-x) \end{vmatrix} = e^x(1-3x); \\ W_2 &= \begin{vmatrix} e^x & xe^{-x} \\ e^x & e^{-x}(1-x) \end{vmatrix} = 1-2x; \quad W_3 = \begin{vmatrix} e^x & e^{2x} \\ e^x & 2e^{2x} \end{vmatrix} = e^{3x}; \quad u'_1 = \frac{FW_1}{P_0W} = 1-3x; \quad u'_2 = \\ &-\frac{FW_2}{P_0W} = e^{-x}(2x-1); \quad u'_3 = \frac{FW_3}{P_0W} = e^{2x}; \quad u_1 = \frac{x(2-3x)}{2}; \quad u_2 = -e^{-x}(2x+1); \quad u_3 = \frac{e^{2x}}{2}; \\ y_p &= u_1y_1 + u_2y_2 + u_3y_3 = -e^x(3x^2 + x + 2)/2. \quad \text{Since } -\frac{e^x}{2} \text{ is a solution of the complementary} \\ &\text{equation we take } y_p = -\frac{e^x(3x+1)x}{2}. \end{aligned}$$

The general solution is  $y = -\frac{e^x(3x+1)x}{2} + c_1e^x + c_2e^{2x} + c_3xe^{-x}$ , so

$$\begin{bmatrix} y \\ y' \\ y'' \end{bmatrix} = \begin{bmatrix} -\frac{e^x(3x+1)x}{2} \\ -\frac{e^x(3x^2+7x+1)}{2} \\ -\frac{e^x(3x^2+13x+8)}{2} \end{bmatrix} + \begin{bmatrix} e^x & e^{2x} & xe^{-x} \\ e^x & 2e^{2x} & e^{-x}(1-x) \\ e^x & 4e^{2x} & e^{-x}(x-2) \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}.$$

Setting  $x = 0$  and imposing the initial conditions yields

$$\begin{bmatrix} -4 \\ -\frac{3}{2} \\ -19 \end{bmatrix} = \begin{bmatrix} 0 \\ -\frac{1}{2} \\ -4 \end{bmatrix} + \begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 1 & 4 & -2 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}.$$

Solving this system yields  $c_1 = -3$ ,  $c_2 = -1$ ,  $c_3 = 4$ . Therefore,  $y = -\frac{e^x(3x+1)x}{2} - 3e^x - e^{2x} + 4xe^{-x}$ .

**9.4.26.**  $W = \begin{vmatrix} x & x^2 & e^x \\ 1 & 2x & e^x \\ 0 & 2 & e^x \end{vmatrix} = e^x(x^2 - 2x + 2)$ ;  $W_1 = \begin{vmatrix} x^2 & e^x \\ 2x & e^x \end{vmatrix} = e^x(x^2 - 2x)$ ;  $W_2 = \begin{vmatrix} x & e^x \\ 1 & e^x \end{vmatrix} = e^x(x - 1)$ ;  $W_3 = \begin{vmatrix} x & x^2 \\ 1 & 2x \end{vmatrix} = x^2$ ;  $u'_1 = \frac{FW_1}{P_0W} = x(x - 2)$ ;  $u'_2 = -\frac{FW_2}{P_0W} = 1 - x$ ;  $u'_3 = \frac{FW_2}{P_0W} = x^2e^{-x}$ ;  $y_p = u_1y_1 + u_2y_2 + u_3y_3 = -\frac{x^4 + 6x^2 + 12x + 12}{6}$ . Since  $-\frac{6x^2 + 12x}{6}$  is a solution of the complementary equations we take  $y_p = -\frac{x^4 + 12}{6}$ .

The general solution is  $y = -\frac{x^4 + 12}{6} + c_1x + c_2x^2 + c_3e^x$ , so

$$\begin{bmatrix} y \\ y' \\ y'' \end{bmatrix} = -\begin{bmatrix} (x^4 + 12)/6 \\ 2x^3/3 \\ 2x^2 \end{bmatrix} + \begin{bmatrix} x & x^2 & e^x \\ 1 & 2x & e^x \\ 0 & 2 & e^x \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}.$$

Setting  $x = 0$  and imposing the initial conditions yields

$$\begin{bmatrix} 0 \\ 5 \\ 0 \end{bmatrix} = -\begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 1 \\ 0 & 2 & 1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}.$$

Solving this system yields  $c_1 = 3$ ,  $c_2 = -1$ ,  $c_3 = 2$ . Therefore,  $y = -\frac{x^4 + 12}{6} + 3x - x^2 + 2e^x$ .

**9.4.28.**  $W = \begin{vmatrix} x+1 & e^x & e^{3x} \\ 1 & e^x & 3e^{3x} \\ 0 & e^x & 9e^{3x} \end{vmatrix} = 2e^{4x}(3x-1)$ ;  $W_1 = \begin{vmatrix} e^x & e^{3x} \\ e^x & 3e^{3x} \end{vmatrix} = 2e^{4x}$ ;  $W_2 = \begin{vmatrix} x+1 & e^{3x} \\ 1 & 3e^{3x} \end{vmatrix} = e^{3x}(3x+2)$ ;  $W_3 = \begin{vmatrix} x+1 & e^x \\ 1 & e^x \end{vmatrix} = xe^x$ ;  $u'_1 = \frac{FW_1}{P_0W} = 2e^x$ ;  $u'_2 = -\frac{FW_2}{P_0W} = -3x - 2$ ;  $u'_3 = \frac{FW_2}{P_0W} = xe^{-2x}$ ;  $u_1 = 2e^x$ ;  $u_2 = -\frac{x(3x+4)}{2}$ ;  $u_3 = -\frac{e^{-2x}(2x+1)}{4}$ ;  $y_p = u_1y_1 + u_2y_2 + u_3y_3 = -\frac{e^x(6x^2 + 2x - 7)}{4}$ . Since  $\frac{7e^x}{4}$  is a solution of the complementary equation we take  $y_p = -\frac{xe^x(3x+1)}{2}$ .

The general solution is  $y = -\frac{xe^x(3x+1)}{2} + c_1(x+1) + c_2e^x + c_3e^{2x}$ , so

$$\begin{bmatrix} y \\ y' \\ y'' \end{bmatrix} = \begin{bmatrix} -xe^x(3x+1)/2 \\ -e^x(3x^2 + 7x + 1)/2 \\ -e^x(3x^2 + 13x + 8)/2 \end{bmatrix} + \begin{bmatrix} x+1 & e^x & e^{3x} \\ 1 & e^x & 3e^{3x} \\ 0 & e^x & 9e^{3x} \end{bmatrix}.$$

Setting  $x = 0$  and imposing the initial conditions yields

$$\begin{bmatrix} \frac{3}{4} \\ \frac{5}{4} \\ \frac{1}{4} \end{bmatrix} = \begin{bmatrix} 0 \\ -\frac{1}{2} \\ -4 \end{bmatrix} + \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 3 \\ 0 & 1 & 9 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}.$$

Solving this system yields  $c_1 = \frac{1}{2}$ ,  $c_2 = -\frac{1}{4}$ ,  $c_3 = \frac{1}{2}$ . Therefore,  $y = -\frac{xe^x(3x+1)}{2} + \frac{x+1}{2} - \frac{e^x}{4} + \frac{e^{2x}}{2}$ .

$$\begin{aligned} \mathbf{9.4.30.} \quad W &= \begin{vmatrix} x & x^2 & \frac{1}{x} & x \ln x \\ 1 & 2x & -\frac{1}{x^2} & \ln x + 1 \\ 0 & 2 & \frac{2}{x^3} & \frac{1}{x} \\ 0 & 0 & -\frac{6}{x^2} & -\frac{1}{x^2} \end{vmatrix} = -\frac{12}{x^3}; \quad W_1 = \begin{vmatrix} x^2 & \frac{1}{x} & x \ln x \\ 2x & -\frac{1}{x^2} & \ln x + 1 \\ 2 & \frac{2}{x^3} & \frac{1}{x} \end{vmatrix} = 6\frac{\ln x}{x} - \frac{3}{x}; \\ W_2 &= \begin{vmatrix} x & \frac{1}{x} & x \ln x \\ 1 & -\frac{1}{x^2} & \ln x + 1 \\ 0 & \frac{2}{x^3} & \frac{1}{x} \end{vmatrix} = -\frac{4}{x^2}; \quad W_3 = \begin{vmatrix} x & x^2 & x \ln x \\ 1 & 2x & \ln x + 1 \\ 0 & 2 & \frac{1}{x} \end{vmatrix} = -x; \quad W_4 = \begin{vmatrix} x & x^2 & \frac{1}{x} \\ 1 & 2x & -\frac{1}{x^2} \\ 0 & 2 & \frac{2}{x^3} \end{vmatrix} = \frac{6}{x}; \\ u'_1 &= -\frac{FW_1}{P_0W} = 9\ln x/2 - \frac{9}{4}; \quad u'_2 = \frac{FW_2}{P_0W} = \frac{3}{x}; \quad u'_3 = -\frac{FW_3}{P_0W} = -\frac{3x^2}{4}; \quad u'_4 = \frac{FW_4}{P_0W} = -\frac{9}{2}; \\ u_1 &= \frac{9x \ln x}{2} - \frac{27x}{4}; \quad u_2 = 3\ln x; \quad u_3 = -\frac{x^3}{4}; \quad u_4 = -\frac{9x}{2}; \quad y_p = u_1y_1 + u_2y_2 + u_3y_3 = 3x^2 \ln x - 7x^2. \end{aligned}$$

Since  $-7x^2$  satisfies the complementary equation we take  $y_p = 3x^2 \ln x$ .

The general solution is  $y = 3x^2 \ln x + c_1x + c_2x^2 + \frac{c_3}{x} + c_4x \ln x$ , so

$$\begin{bmatrix} y \\ y' \\ y'' \\ y''' \end{bmatrix} = \begin{bmatrix} 3x^2 \ln x \\ 6x \ln x + 3x \\ 6 \ln x + 9 \\ \frac{6}{x} \end{bmatrix} + \begin{vmatrix} x & x^2 & \frac{1}{x} & x \ln x \\ 1 & 2x & -\frac{1}{x^2} & \ln x + 1 \\ 0 & 2 & \frac{2}{x^3} & \frac{1}{x} \\ 0 & 0 & -\frac{6}{x^2} & -\frac{1}{x^2} \end{vmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{bmatrix}.$$

Setting  $x = 1$  and imposing the initial conditions yields

$$\begin{bmatrix} -7 \\ -11 \\ -5 \\ 6 \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \\ 9 \\ 6 \end{bmatrix} + \begin{vmatrix} 1 & 1 & 1 & 0 \\ 1 & 2 & -1 & 1 \\ 0 & 2 & 2 & 1 \\ 0 & 0 & -6 & -1 \end{vmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{bmatrix}.$$

Solving this system yields  $c_1 = 0$ ,  $c_2 = -7$ ,  $c_3 = 0$ ,  $c_4 = 0$ . Therefore,  $y = 3x^2 \ln x - 7x^2$ .

$$\mathbf{9.4.32.} \quad W = \begin{vmatrix} x & \sqrt{x} & 1/x & 1/\sqrt{x} \\ 1 & 1/2\sqrt{x} & -1/x^2 & -1/2x^{3/2} \\ 0 & -1/4x^{3/2} & 2/x^3 & 3/4x^{5/2} \\ 0 & 3/8x^{5/2} & -6/x^4 & -15/8x^{7/2} \end{vmatrix} = -9/8x^6; \quad W_1 = \begin{vmatrix} \sqrt{x} & 1/x & 1/\sqrt{x} \\ 1/2\sqrt{x} & -1/x^2 & -1/2x^{3/2} \\ -1/4x^{3/2} & 2/x^3 & 3/4x^{5/2} \end{vmatrix} =$$

$$\begin{aligned}
3/4x^4; W_2 &= \begin{vmatrix} x & 1/x & 1/\sqrt{x} \\ 1 & -1/x^2 & -1/2x^{3/2} \\ 0 & 2/x^3 & 3/4x^{5/2} \end{vmatrix} = 3/2x^{7/2}; W_3 = \begin{vmatrix} x & \sqrt{x} & 1/\sqrt{x} \\ 1 & 1/2\sqrt{x} & -1/2x^{3/2} \\ 0 & -1/4x^{3/2} & 3/4x^{5/2} \end{vmatrix} = -3/4x^2; \\
W_4 &= \begin{vmatrix} x & \sqrt{x} & 1/x \\ 1 & 1/2\sqrt{x} & -1/x^2 \\ 0 & -1/4x^{3/2} & 2/x^3 \end{vmatrix} = -3/2x^{5/2}; u'_1 = -\frac{FW_1}{P_0W} = 1/x; u'_2 = \frac{FW_2}{P_0W} = -2/\sqrt{x}; \\
u'_3 &= -\frac{FW_3}{P_0W} = -x; u'_4 = \frac{FW_4}{P_0W} = 2\sqrt{x}; u_1 = \ln x; u_2 = -4\sqrt{x}; u_3 = -x^2/2; u_4 = 4x^{3/2}/3; \\
y_p &= u_1y_1 + u_2y_2 + u_3y_3 = x \ln x - 19x/6. \text{ since } -19x/6 \text{ satisfies the complementary equation we} \\
&\text{take } y_p = x \ln x.
\end{aligned}$$

The general solution is  $y = x \ln x + c_1x + c_2\sqrt{x} + c_3/x + c_4/\sqrt{x}$ , so

$$\begin{bmatrix} y \\ y' \\ y'' \\ y''' \end{bmatrix} = \begin{bmatrix} x \ln x \\ \ln x + 1 \\ 1/x \\ -1/x^2 \end{bmatrix} + \begin{bmatrix} x & \sqrt{x} & 1/x & 1/\sqrt{x} \\ 1 & 1/2\sqrt{x} & -1/x^2 & -1/2x^{3/2} \\ 0 & -1/4x^{3/2} & 2/x^3 & 3/4x^{5/2} \\ 0 & 3/8x^{5/2} & -6/x^4 & -15/8x^{7/2} \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{bmatrix}.$$

Setting  $x = 1$  and imposing the initial conditions yields

$$\begin{bmatrix} 2 \\ 0 \\ 4 \\ \frac{37}{4} \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 1 \\ -1 \end{bmatrix} + \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1/2 & -1 & -1/2 \\ 0 & -1/4 & 2 & 3/4 \\ 0 & 3/8 & -6 & -15/8 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{bmatrix}.$$

Solving this system yields  $c_1 = 1, c_2 = -1, c_3 = 1, c_4 = 1$ . Therefore,  $y = x \ln x + x - \sqrt{x} + \frac{1}{x} + \frac{1}{\sqrt{x}}$ .

**9.4.34.** (a) Since  $u'_j = (-1)^{n-j} \frac{FW_j}{P_0W}$  ( $1 \leq j \leq n$ ), the argument used in the derivation of the method of variation of parameters implies that  $y_p$  is a solution of (A).

(b) Follows immediately from (a), since  $u_j(x_0) = 0, j = 1, 2, \dots, n$ .

(c) Expand the determinant in cofactors of its  $n$ th row.

(d) Just differentiate the determinant  $n - 1$  times.

(e) If  $0 \leq j \leq n - 2$ , then  $\left. \frac{\partial^j G(x, t)}{\partial x^j} \right|_{x=t}$  has two identical rows, and is therefore zero, while

$$\left. \frac{\partial^{n-1} G(x, t)}{\partial x^j} \right|_{x=t} = W(t)$$

(f) Since  $y_p(x) = \int_{x_0}^x G(x, t) F(t) dt$ ,  $y'_p(x) = G(x, x) F(x) + \int_{x_0}^x \frac{\partial G(x, t)}{\partial x} F(t) dt$ . But  $G(x, x) = 0$

from (e), so  $y'_p(x) = \int_{x_0}^x \frac{\partial G(x, t)}{\partial x} F(t) dt$ . Repeating this argument for  $j = 1, \dots, n$  and invoking (e) each time yields the conclusion.

9.4.36.

$$\begin{aligned}
\begin{vmatrix} y_1(t) & y_1(t) & y_2(t) \\ y_1'(t) & y_1'(t) & y_2'(t) \\ y_1(x) & y_1(x) & y_2(x) \end{vmatrix} &= \begin{vmatrix} t & t^2 & 1/t \\ 1 & 2t & -1/t^2 \\ x & x^2 & 1/x \end{vmatrix} \\
&= x \begin{vmatrix} t^2 & 1/t \\ 2t & -1/t^2 \end{vmatrix} - x^2 \begin{vmatrix} t & 1/t \\ 1 & -1/t^2 \end{vmatrix} + \frac{1}{x} \begin{vmatrix} t & t^2 \\ 1 & 2t \end{vmatrix} \\
&= -3x + 2\frac{x^2}{t} + \frac{t^2}{x} = \frac{(x-t)^2(2x+t)}{xt}.
\end{aligned}$$

Since  $P_0(t) = t^3$  and  $W(t) = \begin{vmatrix} t & t^2 & 1/t \\ 1 & 2t & -1/t^2 \\ x & x^2 & 1/x \end{vmatrix} = \frac{6}{t}$ ,  $G(x, t) = \frac{(x-t)^2(2x+t)}{6xt^3}$ , so  $y_p = \int_{x_0}^x \frac{(x-t)^2(2x+t)}{6xt^3} F(t) dt$ .

9.4.38.

$$\begin{aligned}
\begin{vmatrix} y_1(t) & y_1(t) & y_2(t) \\ y_1'(t) & y_1'(t) & y_2'(t) \\ y_1(x) & y_1(x) & y_2(x) \end{vmatrix} &= \begin{vmatrix} t & 1/t & e^t/t \\ 1 & -1/t^2 & e^t(1/t - 1/t^2) \\ x & 1/x & e^x/x \end{vmatrix} \\
&= x \begin{vmatrix} 1/t & e^t/t \\ -1/t^2 & e^t(1/t - 1/t^2) \end{vmatrix} - \frac{1}{x} \begin{vmatrix} t & e^t/t \\ 1 & e^t(1/t - 1/t^2) \end{vmatrix} + \frac{e^x}{x} \begin{vmatrix} t & 1/t \\ 1 & -1/t^2 \end{vmatrix} \\
&= \frac{xe^t}{t^2} - \frac{e^t(t-2)}{xt} - \frac{2e^x}{xt} = \frac{x^2e^t - e^tt(t-2) - 2te^x}{xt^2}
\end{aligned}$$

Since  $P_0(t) = t(1-t)$  and  $W(t) = \begin{vmatrix} t & 1/t & e^t/t \\ 1 & -1/t^2 & e^t(1/t - 1/t^2) \\ 0 & 2/t^3 & e^t(1/t - 2/t^2 + 2/t^3) \end{vmatrix} = \frac{2e^t(1-t)}{t^3}$ ,  $G(x, t) = \frac{x^2 - t(t-2) - 2te^{(x-t)}}{2x(t-1)^2}$ , so  $y_p = \int_{x_0}^x \frac{x^2 - t(t-2) - 2te^{(x-t)}}{2x(t-1)^2} F(t) dt$ .

9.4.40.

$$\begin{aligned}
\begin{vmatrix} y_1(t) & y_1(t) & y_2(t) & y_3(t) \\ y_1'(t) & y_1'(t) & y_2'(t) & y_3'(t) \\ y_1''(t) & y_1''(t) & y_2''(t) & y_3''(t) \\ y_1(x) & y_1(x) & y_2(x) & y_3(x) \end{vmatrix} &= \begin{vmatrix} 1 & t & t^2 & 1/t \\ 0 & 1 & 2t & -1/t^2 \\ 0 & 0 & 2 & 2/t^3 \\ 1 & x & x^2 & 1/x \end{vmatrix} \\
&= - \begin{vmatrix} t & t^2 & 1/t \\ 1 & 2t & -1/t^2 \\ 0 & 2 & 2/t^3 \end{vmatrix} + x \begin{vmatrix} 1 & t^2 & 1/t \\ 0 & 2t & -1/t^2 \\ 0 & 2 & 2/t^3 \end{vmatrix} \\
&\quad - x^2 \begin{vmatrix} 1 & t & 1/t \\ 0 & 1 & -1/t^2 \\ 0 & 0 & 2/t^3 \end{vmatrix} + \frac{1}{x} \begin{vmatrix} 1 & t & t^2 \\ 0 & 1 & 2t \\ 0 & 0 & 2 \end{vmatrix} \\
&= -\frac{6}{t} + \frac{6x}{t^2} - \frac{2x^2}{t^3} + \frac{2}{x} = \frac{2(t-x)^3}{xt^3}.
\end{aligned}$$

Since  $P_0(t) = t$  and  $W(t) = \begin{vmatrix} 1 & t & t^2 & 1/t \\ 0 & 1 & 2t & -1/t^2 \\ 0 & 0 & 2 & 2/t^3 \\ 0 & 0 & 0 & -6/t^4 \end{vmatrix} = -\frac{12}{t^4}$ ,  $G(x, t) = \frac{(x-t)^3}{6x}$ , so  $y_p = \int_{x_0}^x \frac{(x-t)^3}{6x} F(t) dt$ .

## 9.4.42.

$$\begin{aligned}
\begin{vmatrix} y_1(t) & y_1(t) & y_2(t) & y_3(t) \\ y_1'(t) & y_1'(t) & y_2'(t) & y_3'(t) \\ y_1''(t) & y_1''(t) & y_2''(t) & y_3''(t) \\ y_1(x) & y_1(x) & y_2(x) & y_3(x) \end{vmatrix} &= \begin{vmatrix} 1 & t^2 & e^{2t} & e^{-2t} \\ 0 & 2t & 2e^{2t} & -2e^{-2t} \\ 0 & 2 & 4e^{2t} & 4e^{-2t} \\ 1 & x^2 & e^{2x} & e^{-2x} \end{vmatrix} \\
&= - \begin{vmatrix} t^2 & e^{2t} & e^{-2t} \\ 2t & 2e^{2t} & -2e^{-2t} \\ 2 & 4e^{2t} & 4e^{-2t} \end{vmatrix} + x^2 \begin{vmatrix} 1 & e^{2t} & e^{-2t} \\ 0 & 2e^{2t} & -2e^{-2t} \\ 0 & 4e^{2t} & 4e^{-2t} \end{vmatrix} \\
&\quad - e^{2x} \begin{vmatrix} 1 & t^2 & e^{-2t} \\ 0 & 2t & -2e^{-2t} \\ 0 & 2 & 4e^{-2t} \end{vmatrix} + e^{-2x} \begin{vmatrix} 1 & t^2 & e^{2t} \\ 0 & 2t & 2e^{2t} \\ 0 & 2 & 4e^{2t} \end{vmatrix} \\
&= -(16t^2 - 8) + 16x^2 - e^{2(x-t)t}(8t + 4) + e^{-2(x-t)}(8t - 4).
\end{aligned}$$

Since  $P_0(t) = t$  and  $W(t) = \begin{vmatrix} 1 & t^2 & e^{2t} & e^{-2t} \\ 0 & 2t & 2e^{2t} & -2e^{-2t} \\ 0 & 2 & 4e^{2t} & 4e^{-2t} \\ 0 & 0 & 8e^{2t} & -8e^{-2t} \end{vmatrix} = -128t$ ,

$$\begin{aligned}
G(x, t) &= \frac{e^{2(x-t)}(1 + 2t) + e^{-2(x-t)}(1 - 2t) - 4x^2 + 4t^2 - 2}{32t^2}, \text{ so} \\
y_p &= \int_{x_0}^x \frac{e^{2(x-t)}(1 + 2t) + e^{-2(x-t)}(1 - 2t) - 4x^2 + 4t^2 - 2}{32t^2} F(t) dt.
\end{aligned}$$

## Chapter 10

# Linear Systems of Differential Equations

### 10.1 Introduction to Systems of Differential Equations

**10.1.2.**  $Q'_1 = (\text{rate in})_1 - (\text{rate out})_1$  and  $Q'_2 = (\text{rate in})_2 - (\text{rate out})_2$ .

The volumes of the solutions in  $T_1$  and  $T_2$  are  $V_1(t) = 100 + 2t$  and  $V_2(t) = 100 + 3t$ , respectively.  $T_1$  receives salt from the external source at the rate of  $(2 \text{ lb/gal}) \times (6 \text{ gal/min}) = 12 \text{ lb/min}$ , and from  $T_2$  at the rate of  $(\text{lb/gal in } T_2) \times (1 \text{ gal/min}) = \frac{1}{100 + 3t}Q_2 \text{ lb/min}$ . Therefore, (A)

$(\text{rate in})_1 = 12 + \frac{1}{100 + 3t}Q_2$ . Solution leaves  $T_1$  at 5 gal/min, since 3 gal/min are drained and 2 gal/min are pumped to  $T_2$ ; hence (B)  $(\text{rate out})_1 = (\text{lb/gal in } T_1) \times (5 \text{ gal/min}) = \frac{1}{100 + 2t}Q_1 \times 5 = \frac{5}{100 + 2t}Q_1$ . Now (A) and (B) imply that (C)  $Q'_1 = 12 - \frac{5}{100 + 2t}Q_1 + \frac{1}{100 + 3t}Q_2$ .

$T_2$  receives salt from the external source at the rate of  $(1 \text{ lb/gal}) \times (5 \text{ gal/min}) = 5 \text{ lb/min}$ , and from  $T_1$  at the rate of  $(\text{lb/gal in } T_1) \times (2 \text{ gal/min}) = \frac{1}{100 + 2t}Q_1 \times 2 = \frac{1}{50 + t}Q_1 \text{ lb/min}$ .

Therefore, (D)  $(\text{rate in})_2 = 5 + \frac{1}{50 + t}Q_1$ . Solution leaves  $T_2$  at 4 gal/min, since 3 gal/min are drained and 1 gal/min is pumped to  $T_1$ ; hence (E)  $(\text{rate out})_2 = (\text{lb/gal in } T_2) \times (4 \text{ gal/min}) = \frac{1}{100 + 3t}Q_2 \times 4 = \frac{4}{100 + 3t}Q_2$ . Now (D) and (E) imply that (F)  $Q'_2 = 5 + \frac{1}{50 + t}Q_1 - \frac{4}{100 + 3t}Q_2$ . Now (C) and (F) form the desired system.

**10.1.4.**  $m\mathbf{X}'' = -\alpha\mathbf{X}' - mgR^2 \frac{\mathbf{X}}{\|\mathbf{X}\|^3}$ ; see Example 10.1.3.

**10.1.8.**

$$\begin{aligned}I_{1i} &= g_1(t_i, y_{1i}, y_{2i}), \\J_{1i} &= g_2(t_i, y_{1i}, y_{2i}), \\I_{2i} &= g_1(t_i + h, y_{1i} + hI_{1i}, y_{2i} + hJ_{1i}), \\J_{2i} &= g_2(t_i + h, y_{1i} + hI_{1i}, y_{2i} + hJ_{1i}), \\y_{1,i+1} &= y_{1i} + \frac{h}{2}(I_{1i} + I_{2i}), \\y_{2,i+1} &= y_{2i} + \frac{h}{2}(J_{1i} + J_{2i}).\end{aligned}$$



## 10.2 Linear Systems of Differential Equations

**10.2.6.** Let  $y_i = y^{(i-1)}$ ,  $i = 1, 2, \dots, n$ ; then  $y'_i = y_{i+1}$ ,  $i = 1, 2, \dots, n-1$  and  $P_0(t)y'_n + P_1(t)y_n + \dots + P_n(t)y_1 = F(t)$ , so

$$A = -\frac{1}{P_0} \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ P_n & P_{n-1} & P_{n-2} & \cdots & P_1 \end{bmatrix} \quad \text{and} \quad \mathbf{f} = \frac{1}{P_0} \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ F \end{bmatrix}.$$

If  $P_0, P_1, \dots, P_n$  and  $F$  are continuous and  $P_0$  has no zeros on  $(a, b)$ , then  $P_1/P_0, \dots, P_n/P_0$  and  $F/P_0$  are continuous on  $(a, b)$ .

**10.2.7.** (a)  $(c_1P + c_2Q)'_{ij} = (c_1p_{ij} + c_2q_{ij})' = c_1p'_{ij} + c_2q'_{ij} = (c_1P' + c_2Q')_{ij}$ ; hence  $(c_1P + c_2Q)' = c_1P' + c_2Q'$ .

(b) Let  $P$  be  $k \times r$  and  $Q$  be  $r \times s$ ; then  $PQ$  is  $k \times s$  and  $(PQ)_{ij} = \sum_{l=1}^r p_{il}q_{lj}$ . Therefore,

$$(PQ)'_{ij} = \sum_{l=1}^r p'_{il}q_{lj} + \sum_{l=1}^r p_{il}q'_{lj} = (P'Q)_{ij} + (PQ')_{ij}. \quad \text{Therefore, } (PQ)' = P'Q + PQ'.$$

**10.2.10.** (a) From Exercise 10.2.7(b) with  $P = Q = X$ ,  $(X^2)' = (XX)' = X'X + XX'$ .

(b) By starting from Exercise 10.2.7(b) and using induction it can be shown if  $P_1, P_2, \dots, P_n$  are square matrices of the same order, then  $(P_1P_2 \cdots P_n)' = P'_1P_2 \cdots P_n + P_1P'_2 \cdots P_n + \dots + P_1P_2 \cdots P'_n$ . Taking  $P_1 = P_2 = \dots = P_n = X$  yields (A)  $(Y^n)' = Y'Y^{n-1} + Y Y'Y^{n-2} + Y^2 Y'Y^{n-3} + \dots + Y^{n-1} Y' = \sum_{r=0}^{n-1} Y^r Y' Y^{n-r-1}$ .

(c) If  $Y$  is a scalar function, then (A) reduces to the familiar result  $(Y^n)' = nY^{n-1}Y'$ .

**10.2.12.** From Exercise 10.2.6, the initial value problem (A)  $P_0(x)y^{(n)} + P_1(x)y^{(n-1)} + \dots + P_n(x)y = F(x)$ ,  $y(x_0) = k_0$ ,  $y'(x_0) = k_1, \dots$ ,  $y^{(n-1)}(x_0) = k_{n-1}$  is equivalent to the initial value problem (B)  $\mathbf{y}' = A(t)\mathbf{y} + \mathbf{f}(t)$ , with

$$A = -\frac{1}{P_0} \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ P_n & P_{n-1} & P_{n-2} & \cdots & P_1 \end{bmatrix}, \quad \mathbf{f} = \frac{1}{P_0} \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ F \end{bmatrix}, \quad \text{and} \quad \mathbf{k} = \begin{bmatrix} k_0 \\ k_1 \\ \vdots \\ k_{n-1} \end{bmatrix}.$$

Since Theorem 10.2.1 implies that (B) has a unique solution on  $(a, b)$ , it follows that (A) does also.

## 10.3 Basic Theory of Homogeneous Linear System

**10.3.2.** (a) The system equivalent of (A) is (B)  $\mathbf{y}' = -\frac{1}{P_0(x)} \begin{bmatrix} 0 & 1 \\ P_2(x) & P_1(x) \end{bmatrix} \mathbf{y}$ , where  $\mathbf{y} = \begin{bmatrix} y \\ y' \end{bmatrix}$ .

Let  $\mathbf{y}_1 = \begin{bmatrix} y_1 \\ y'_1 \end{bmatrix}$  and  $\mathbf{y}_2 = \begin{bmatrix} y_2 \\ y'_2 \end{bmatrix}$ . Then the Wronskian of  $\{\mathbf{y}_1, \mathbf{y}_2\}$  as defined in this section is  $\begin{bmatrix} y_1 & y_2 \\ y'_1 & y'_2 \end{bmatrix} = W$ .

(b) The trace of the matrix in (B) is  $-P_1(x)/P_0(x)$ , so Eqn. 10.3.6 implies that  $W(x) = W(x_0) \exp \left\{ - \int_{x_0}^x \frac{P_1(s)}{P_0(s)} ds \right\}$ .

**10.3.4.** (a) See the solution of Exercise 9.1.18.

$$(c) \begin{vmatrix} y'_{11} & y'_{12} \\ y_{21} & y_{22} \end{vmatrix} = \begin{vmatrix} a_{11}y_{11} + a_{12}y_{21} & a_{11}y_{12} + a_{12}y_{22} \\ y_{21} & y_{22} \end{vmatrix} = a_{11} \begin{vmatrix} y_{11} & y_{12} \\ y_{21} & y_{22} \end{vmatrix} + a_{12} \begin{vmatrix} y_{21} & y_{22} \\ y_{21} & y_{22} \end{vmatrix} = a_{11}W + a_{12}0 = a_{11}W. \text{ Similarly, } \begin{bmatrix} y_{11} & y_{12} \\ y'_{21} & y'_{22} \end{bmatrix} = a_{22}W.$$

**10.3.6.** (a) From the equivalence of Theorem 10.3.3(b) and (e),  $Y(t_0)$  is invertible.

(b) From the equivalence of Theorem 10.3.3(a) and (b), the solution of the initial value problem is  $\mathbf{y} = Y(t)\mathbf{c}$ , where  $\mathbf{c}$  is a constant vector. To satisfy  $\mathbf{y}(t_0) = \mathbf{k}$ , we must have  $Y(t_0)\mathbf{c} = \mathbf{k}$ , so  $\mathbf{c} = Y^{-1}(t_0)\mathbf{k}$  and  $\mathbf{y} = Y^{-1}(t_0)Y(t)\mathbf{k}$ .

**10.3.8.** (b)  $\mathbf{y} = \begin{bmatrix} e^{-4t} \\ e^{-4t} \end{bmatrix} + c_2 \begin{bmatrix} -2e^{3t} \\ 5e^{3t} \end{bmatrix}$  where  $\begin{matrix} c_1 - 2c_2 = 10 \\ c_1 + 5c_2 = -4 \end{matrix}$ , so  $c_1 = 6$ ,  $c_2 = -2$ , and

$$\mathbf{y} = \begin{bmatrix} 6e^{-4t} + 4e^{3t} \\ 6e^{-4t} - 10e^{3t} \end{bmatrix}.$$

(c)  $Y(t) = \begin{bmatrix} e^{-4t} & -2e^{3t} \\ e^{-4t} & 5e^{3t} \end{bmatrix}$ ;  $Y(0) = \begin{bmatrix} 1 & -2 \\ 1 & 5 \end{bmatrix}$ ;  $Y^{-1}(0) = \frac{1}{7} \begin{bmatrix} 5 & 2 \\ -1 & 1 \end{bmatrix}$ ;  $\mathbf{y} = Y(t)Y^{-1}(0)\mathbf{k} = \frac{1}{7} \begin{bmatrix} 5e^{-4t} + 2e^{3t} & 2e^{-4t} - 2e^{3t} \\ 5e^{-4t} - 5e^{3t} & 2e^{-4t} + 5e^{3t} \end{bmatrix} \mathbf{k}$ .

**10.3.10.** (b)  $\mathbf{y}_1 = c_1 \begin{bmatrix} e^{3t} \\ e^{3t} \end{bmatrix} + c_2 \begin{bmatrix} e^t \\ -e^t \end{bmatrix}$ , where  $\begin{matrix} c_1 + c_2 = 2 \\ c_1 - c_2 = 8 \end{matrix}$ , so  $c_1 = 5$ ,  $c_2 = -3$ , and

$$\mathbf{y} = \begin{bmatrix} 5e^{3t} - 3e^t \\ 5e^{3t} + 3e^t \end{bmatrix}.$$

(c)  $Y(t) = \begin{bmatrix} e^{3t} & e^t \\ 3e^{3t} & -e^t \end{bmatrix}$ ;  $Y(0) = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$ ;  
 $Y^{-1}(0) = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$ ;  $\mathbf{y} = Y(t)Y^{-1}(0)\mathbf{k} = \frac{1}{2} \begin{bmatrix} e^{3t} + e^t & e^{3t} - e^t \\ e^{3t} - e^t & e^{3t} + e^t \end{bmatrix} \mathbf{k}$ .

**10.3.12.** (b)  $\mathbf{y} = c_1 \begin{bmatrix} -e^{-2t} \\ 0 \\ e^{-2t} \end{bmatrix} + c_2 \begin{bmatrix} -e^{-2t} \\ e^{-2t} \\ 0 \end{bmatrix} + c_3 \begin{bmatrix} e^{4t} \\ e^{4t} \\ e^{4t} \end{bmatrix}$ , where  $\begin{matrix} -c_1 - c_2 + c_3 = 0 \\ c_2 + c_3 = -9 \\ c_1 + c_3 = 12 \end{matrix}$ , so

$$c_1 = 11, c_2 = -10, c_3 = 1, \text{ and } \mathbf{y} = \frac{1}{3} \begin{bmatrix} -e^{-2t} + e^{4t} \\ -10e^{-2t} + e^{4t} \\ 11e^{-2t} + e^{4t} \end{bmatrix}.$$

(c)  $Y(t) = \begin{bmatrix} -e^{-2t} & -e^{-2t} & e^{4t} \\ 0 & e^{-2t} & e^{4t} \\ e^{-2t} & 0 & e^{4t} \end{bmatrix}$ ;  $Y(0) = \begin{bmatrix} -1 & -1 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$ ;  $Y^{-1}(0) = \frac{1}{3} \begin{bmatrix} -1 & -1 & 2 \\ -1 & 2 & -1 \\ 1 & 1 & 1 \end{bmatrix}$ ;

$$\mathbf{y} = Y(t)Y^{-1}(0)\mathbf{k} = \frac{1}{3} \begin{bmatrix} 2e^{-2t} + e^{4t} & -e^{-2t} + e^{4t} & -e^{-2t} + e^{4t} \\ -e^{-2t} + e^{4t} & 2e^{-2t} + e^{4t} & -e^{-2t} + e^{4t} \\ -e^{-2t} + e^{4t} & -e^{-2t} + e^{4t} & 2e^{-2t} + e^{4t} \end{bmatrix} \mathbf{k}.$$

**10.3.14.** If  $Y$  and  $Z$  are both fundamental matrices for  $\mathbf{y}' = A(t)\mathbf{y}$ , then  $Z = CY$ , where  $C$  is a constant invertible matrix. Therefore,  $ZY^{-1} = C$  and  $YZ^{-1} = C^{-1}$ .

**10.3.16.** (a) The Wronskian of  $\{\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_n\}$  equals one when  $t = t_0$ . Apply Theorem 10.3.3.

(b) Let  $Y$  be the matrix with columns  $\{\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_n\}$ . From (a),  $Y$  is a fundamental matrix for

$\mathbf{y}' = A(t)\mathbf{y}$  on  $(a, b)$ . From Exercise 10.3.15(b), so is  $Z = YC$  if  $C$  is any invertible constant matrix.

**10.3.18.** (a)  $\Gamma_1'(t) = Z'(t)Z(s) = AZ(t)Z(s) = A\Gamma(t)$  and  $\Gamma_1(0) = Z(s)$ , since  $Z(0) = I$ .  $\Gamma_2'(t) = Z'(t+s) = AZ(t+s) = A\Gamma_2(t)$  (since  $A$  is constant) and  $\Gamma_2(0) = Z(s)$ . Applying Theorem 10.2.1 to the columns of  $\Gamma_1$  and  $\Gamma_2$  shows that  $\Gamma_1 = \Gamma_2$ .

(b) With  $s = -t$ , (a) implies that  $Z(t)Z(-t) = Z(0) = I$ ; therefore  $(Z(t))^{-1} = Z(-t)$ .

(c)  $e^{0 \cdot A} = I$  is analogous to  $e^{0 \cdot a} = e^0 = 1$  when  $a$  is a scalar, while  $e^{(t+s)A} = e^{tA}e^{sA}$  is analogous to  $e^{(t+s)a} = e^{ta}e^{sa}$  when  $a$  is a scalar.

## 10.4 Constant Coefficient Homogeneous Systems I

**10.4.2.**  $\frac{1}{4} \begin{vmatrix} -5-4\lambda & 3 \\ 3 & -5-4\lambda \end{vmatrix} = (\lambda+1/2)(\lambda+2)$ . Eigenvectors associated with  $\lambda_1 = -1/2$  satisfy

$\begin{bmatrix} -3 & 3 \\ -4 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ , so  $x_1 = x_2$ . Taking  $x_2 = 1$  yields  $\mathbf{y}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{-t/2}$ . Eigenvectors

associated with  $\lambda_2 = -2$  satisfy  $\begin{bmatrix} \frac{3}{4} & \frac{3}{4} \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ , so  $x_1 = -x_2$ . Taking  $x_2 = 1$  yields

$\mathbf{y}_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix} e^{-2t}$ . Hence  $\mathbf{y} = c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{-t/2} + c_2 \begin{bmatrix} -1 \\ 1 \end{bmatrix} e^{-2t}$ .

**10.4.4.**  $\begin{vmatrix} -1-\lambda & -4 \\ -1 & -1-\lambda \end{vmatrix} = (\lambda-1)(\lambda+3)$ . Eigenvectors associated with  $\lambda_1 = -3$  satisfy

$\begin{bmatrix} 2 & -4 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ , so  $x_1 = 2x_2$ . Taking  $x_2 = 1$  yields  $\mathbf{y}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix} e^{-3t}$ . Eigenvectors

associated with  $\lambda_2 = 1$  satisfy  $\begin{bmatrix} -2 & -4 \\ -1 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ , so  $x_1 = -2x_2$ . Taking  $x_2 = 1$  yields

$\mathbf{y}_2 = \begin{bmatrix} -2 \\ 1 \end{bmatrix} e^t$ . Hence  $\mathbf{y} = c_1 \begin{bmatrix} 2 \\ 1 \end{bmatrix} e^{-3t} + c_2 \begin{bmatrix} -2 \\ 1 \end{bmatrix} e^t$ .

**10.4.6.**  $\begin{vmatrix} 4-\lambda & -3 \\ 2 & -1-\lambda \end{vmatrix} = (\lambda-2)(\lambda-1)$ . Eigenvectors associated with  $\lambda_1 = 2$  satisfy  $\begin{bmatrix} 2 & -3 \\ 2 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ ,

so  $x_1 = \frac{3}{2}x_2$ . Taking  $x_2 = 2$  yields  $\mathbf{y}_1 = \begin{bmatrix} 3 \\ 2 \end{bmatrix} e^{2t}$ . Eigenvectors associated with  $\lambda_2 = 1$

satisfy  $\begin{bmatrix} 3 & -3 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ , so  $x_1 = x_2$ . Taking  $x_2 = 1$  yields  $\mathbf{y}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^t$ . Hence

$\mathbf{y} = c_1 \begin{bmatrix} 3 \\ 2 \end{bmatrix} e^{2t} + c_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^t$ .

**10.4.8.**  $\begin{vmatrix} 1-\lambda & -1 & -2 \\ 1 & -2-\lambda & -3 \\ -4 & 1 & -1-\lambda \end{vmatrix} = -(\lambda+3)(\lambda+1)(\lambda-2)$ . The eigenvectors associated with

with  $\lambda_1 = -3$  satisfy the system with augmented matrix  $\begin{bmatrix} 4 & -1 & -2 & : & 0 \\ 1 & 1 & -3 & : & 0 \\ -4 & 1 & 2 & : & 0 \end{bmatrix}$ , which is row

equivalent to  $\begin{bmatrix} 1 & 0 & -1 & : & 0 \\ 0 & 1 & -2 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$ . Hence  $x_1 = x_3$  and  $x_2 = 2x_3$ . Taking  $x_3 = 1$  yields  $\mathbf{y}_1 =$

$\begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} e^{-3t}$ . The eigenvectors associated with  $\lambda_2 = -1$  satisfy the system with augmented

matrix  $\begin{bmatrix} 2 & -1 & -2 & : & 0 \\ 1 & -1 & -3 & : & 0 \\ -4 & 1 & 0 & : & 0 \end{bmatrix}$ , which is row equivalent to  $\begin{bmatrix} 1 & 0 & 1 & : & 0 \\ 0 & 1 & 4 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$ . Hence  $x_1 = -x_3$

and  $x_2 = -4x_3$ . Taking  $x_3 = 1$  yields  $\mathbf{y}_2 = \begin{bmatrix} -1 \\ -4 \\ 1 \end{bmatrix} e^{-t}$ . The eigenvectors associated with

$\lambda_3 = 2$  satisfy the system with augmented matrix  $\begin{bmatrix} -1 & -1 & -2 & : & 0 \\ 1 & -4 & -3 & : & 0 \\ -4 & 1 & -3 & : & 0 \end{bmatrix}$ , which is row equivalent

to  $\begin{bmatrix} 1 & 0 & 1 & : & 0 \\ 0 & 1 & 1 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$ . Hence  $x_1 = -x_3$  and  $x_2 = -x_3$ . Taking  $x_3 = 1$  yields  $\mathbf{y}_3 = \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} e^{2t}$ .

Hence  $\mathbf{y} = c_1 \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} e^{-3t} + c_2 \begin{bmatrix} -1 \\ -4 \\ 1 \end{bmatrix} e^{-t} + c_3 \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} e^{2t}$ .

**10.4.10.**  $\begin{vmatrix} 3-\lambda & 5 & 8 \\ 1 & -1-\lambda & -2 \\ -1 & -1 & -1-\lambda \end{vmatrix} = -(\lambda-1)(\lambda+2)(\lambda-2)$ . The eigenvectors associated with

with  $\lambda_1 = 1$  satisfy the system with augmented matrix  $\begin{bmatrix} 2 & 5 & 8 & : & 0 \\ 1 & -2 & -2 & : & 0 \\ -1 & -1 & -2 & : & 0 \end{bmatrix}$ , which is row

equivalent to  $\begin{bmatrix} 1 & 0 & \frac{2}{3} & : & 0 \\ 0 & 1 & \frac{4}{3} & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$ . Hence  $x_1 = -\frac{2}{3}x_3$  and  $x_2 = -\frac{4}{3}x_3$ . Taking  $x_3 = 3$  yields

$\mathbf{y}_1 = \begin{bmatrix} -2 \\ -4 \\ 3 \end{bmatrix} e^t$ . The eigenvectors associated with  $\lambda_2 = -2$  satisfy the system with augmented

matrix  $\begin{bmatrix} 5 & 5 & 8 & : & 0 \\ 1 & 1 & -2 & : & 0 \\ -1 & -1 & 1 & : & 0 \end{bmatrix}$ , which is row equivalent to  $\begin{bmatrix} 1 & 1 & 0 & : & 0 \\ 0 & 0 & 1 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$ . Hence  $x_1 = -x_2$

and  $x_3 = 0$ . Taking  $x_2 = 1$  yields  $\mathbf{y}_2 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} e^{-2t}$ . The eigenvectors associated with  $\lambda_3 = 2$

satisfy the system with augmented matrix  $\begin{bmatrix} 1 & 5 & 8 & : & 0 \\ 1 & -3 & -2 & : & 0 \\ -1 & -1 & -3 & : & 0 \end{bmatrix}$ , which is row equivalent to

$\begin{bmatrix} 1 & 0 & \frac{7}{4} & : & 0 \\ 0 & 1 & \frac{5}{4} & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$ . Hence  $x_1 = -\frac{7}{4}x_3$  and  $x_2 = -\frac{5}{4}x_3$ . Taking  $x_3 = 4$  yields  $\mathbf{y}_3 = \begin{bmatrix} -7 \\ -5 \\ 4 \end{bmatrix} e^{2t}$ .

Hence  $\mathbf{y} = c_1 \begin{bmatrix} -2 \\ -4 \\ 3 \end{bmatrix} e^t + c_2 \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} e^{-2t} + c_3 \begin{bmatrix} -7 \\ -5 \\ 4 \end{bmatrix} e^{2t}$ .

**10.4.12.**  $\begin{vmatrix} 4-\lambda & -1 & -4 \\ 4 & -3-\lambda & -2 \\ 1 & -1 & -1-\lambda \end{vmatrix} = -(\lambda-3)(\lambda+2)(\lambda+1)$ . The eigenvectors associated

with  $\lambda_1 = 3$  satisfy the system with augmented matrix  $\begin{bmatrix} 1 & -1 & -4 & : & 0 \\ 4 & -6 & -2 & : & 0 \\ 1 & -1 & -4 & : & 0 \end{bmatrix}$ , which is row

equivalent to  $\begin{bmatrix} 1 & 0 & -11 & : & 0 \\ 0 & 1 & -7 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$ . Hence  $x_1 = 11x_3$  and  $x_2 = 7x_3$ . Taking  $x_3 = 1$  yields

$\mathbf{y}_1 = \begin{bmatrix} 11 \\ 7 \\ 1 \end{bmatrix} e^{3t}$ . The eigenvectors associated with  $\lambda_2 = -2$  satisfy the system with augmented ma-

trix  $\begin{bmatrix} 6 & -1 & -4 & : & 0 \\ 4 & -1 & -2 & : & 0 \\ 1 & -1 & 1 & : & 0 \end{bmatrix}$ , which is row equivalent to  $\begin{bmatrix} 1 & 0 & -1 & : & 0 \\ 0 & 1 & -2 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$ . Hence  $x_1 = x_3$  and

$x_2 = 2x_3$ . Taking  $x_3 = 1$  yields  $\mathbf{y}_2 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} e^{-2t}$ . The eigenvectors associated with  $\lambda_3 = -1$

satisfy the system with augmented matrix  $\begin{bmatrix} 5 & -1 & -4 & : & 0 \\ 4 & -2 & -2 & : & 0 \\ 1 & -1 & 0 & : & 0 \end{bmatrix}$ , which is row equivalent to

$\begin{bmatrix} 1 & 0 & -1 & : & 0 \\ 0 & 1 & -1 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$ . Hence  $x_1 = x_3$  and  $x_2 = x_3$ . Taking  $x_3 = 1$  yields  $\mathbf{y}_3 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} e^{-t}$ .

Hence  $\mathbf{y} = c_1 \begin{bmatrix} 11 \\ 7 \\ 1 \end{bmatrix} e^{3t} + c_2 \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} e^{-2t} + c_3 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} e^{-t}$ .

**10.4.14.**  $\begin{vmatrix} 3-\lambda & 2 & -2 \\ -2 & 7-\lambda & -2 \\ -10 & 10 & -5-\lambda \end{vmatrix} = -(\lambda+5)(\lambda-5)^2$ . The eigenvectors associated with  $\lambda_1 =$

$-5$  satisfy the system with augmented matrix  $\begin{bmatrix} 8 & 2 & -2 & : & 0 \\ -2 & 12 & -2 & : & 0 \\ -10 & 10 & 0 & : & 0 \end{bmatrix}$ , which is row equivalent

to  $\begin{bmatrix} 1 & 0 & -\frac{1}{5} & : & 0 \\ 0 & 1 & -\frac{1}{5} & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$ . Hence  $x_1 = \frac{1}{5}x_3$  and  $x_2 = \frac{1}{5}x_3$ . Taking  $x_3 = 5$  yields  $\mathbf{y}_1 =$

$\begin{bmatrix} 1 \\ 1 \\ 5 \end{bmatrix} e^{-5t}$ . The eigenvectors associated with  $\lambda_2 = 5$  satisfy the system with augmented matrix

$\begin{bmatrix} -2 & 2 & -2 & : & 0 \\ -2 & 2 & -2 & : & 0 \\ -10 & 10 & -10 & : & 0 \end{bmatrix}$ , which is row equivalent to  $\begin{bmatrix} 1 & -1 & 1 & : & 0 \\ 0 & 0 & 0 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$ . Hence  $x_1 = x_2 - x_3$ .

Taking  $x_2 = 0$  and  $x_3 = 1$  yields  $\mathbf{y}_2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} e^{5t}$ . Taking  $x_2 = 1$  and  $x_3 = 0$  yields  $\mathbf{y}_3 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} e^{5t}$ .

Hence  $\mathbf{y} = c_1 \begin{bmatrix} 1 \\ 1 \\ 5 \end{bmatrix} e^{-5t} + c_2 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} e^{5t} + c_3 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} e^{5t}$ .

**10.4.16.**  $\begin{vmatrix} -7-\lambda & 4 \\ -6 & 7-\lambda \end{vmatrix} = (\lambda-5)(\lambda+5)$ . Eigenvectors associated with  $\lambda_1 = 5$  satisfy  $\begin{bmatrix} -12 & 4 \\ -6 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ , so  $x_1 = \frac{x_2}{3}$ . Taking  $x_2 = 3$  yields  $\mathbf{y}_1 = \begin{bmatrix} 1 \\ 3 \end{bmatrix} e^{5t}$ . Eigenvectors associated with  $\lambda_2 = 5$  satisfy  $\begin{bmatrix} -2 & 4 \\ -6 & 12 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ , so  $x_1 = 2x_2$ . Taking  $x_2 = 1$  yields  $\mathbf{y}_2 = \begin{bmatrix} 2 \\ 1 \end{bmatrix} e^{-5t}$ . The general solution is  $\mathbf{y} = c_1 \begin{bmatrix} 1 \\ 3 \end{bmatrix} e^{5t} + c_2 \begin{bmatrix} 2 \\ 1 \end{bmatrix} e^{-5t}$ . Now  $\mathbf{y}(0) = \begin{bmatrix} 2 \\ -4 \end{bmatrix} \Rightarrow c_1 \begin{bmatrix} 1 \\ 3 \end{bmatrix} + c_2 \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ -4 \end{bmatrix}$ , so  $c_1 = -2$  and  $c_2 = 2$ . Therefore,  $\mathbf{y} = -\begin{bmatrix} 2 \\ 6 \end{bmatrix} e^{5t} + \begin{bmatrix} 4 \\ 2 \end{bmatrix} e^{-5t}$ .

**10.4.18.**  $\begin{vmatrix} 21-\lambda & -12 \\ 24 & -15-\lambda \end{vmatrix} = (\lambda-9)(\lambda+3)$ . Eigenvectors associated with  $\lambda_1 = 9$  satisfy  $\begin{bmatrix} 12 & -12 \\ 24 & -24 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ , so  $x_1 = x_2$ . Taking  $x_2 = 1$  yields  $\mathbf{y}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{9t}$ . Eigenvectors associated with  $\lambda_2 = -3$   $\begin{bmatrix} 24 & -12 \\ 24 & -12 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ , so  $x_1 = \frac{1}{2}x_2$ . Taking  $x_2 = 2$  yields  $\mathbf{y}_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix} e^{-3t}$ . The general solution is  $\mathbf{y} = c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{9t} + c_2 \begin{bmatrix} 1 \\ 2 \end{bmatrix} e^{-3t}$ . Now  $\mathbf{y}(0) = \begin{bmatrix} 5 \\ 3 \end{bmatrix} \Rightarrow c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 5 \\ 3 \end{bmatrix}$ , so  $c_1 = 7$  and  $c_2 = -2$ . Therefore,  $\mathbf{y} = \begin{bmatrix} 7 \\ 7 \end{bmatrix} e^{9t} - \begin{bmatrix} 2 \\ 4 \end{bmatrix} e^{-3t}$ .

**10.4.20.**  $\begin{vmatrix} \frac{1}{6}-\lambda & \frac{1}{3} & 0 \\ \frac{2}{3} & -\frac{1}{6}-\lambda & 0 \\ 0 & 0 & \frac{1}{2}-\lambda \end{vmatrix} = -(\lambda+1/2)(\lambda-1/2)^2$ . The eigenvectors associated with

$\lambda_1 = -1/2$  satisfy the system with augmented matrix  $\begin{bmatrix} \frac{2}{3} & \frac{1}{3} & 0 & : & 0 \\ \frac{2}{3} & \frac{1}{3} & 0 & : & 0 \\ 0 & 0 & 1 & : & 0 \end{bmatrix}$ , which is row equivalent

to  $\begin{bmatrix} 1 & \frac{1}{2} & 0 & : & 0 \\ 0 & 0 & 1 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$ . Hence  $x_1 = -\frac{x_2}{2}$  and  $x_3 = 0$ . Taking  $x_2 = 2$  yields  $\mathbf{y}_1 = \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix} e^{-t/2}$ .

The eigenvectors associated with  $\lambda_2 = \lambda_3 = 1/2$  satisfy the system with augmented matrix

$\begin{bmatrix} -\frac{1}{3} & \frac{1}{3} & 0 & : & 0 \\ \frac{2}{3} & -\frac{2}{3} & 0 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$ , which is row equivalent to  $\begin{bmatrix} 1 & -1 & 0 & : & 0 \\ 0 & 0 & 0 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$ . Hence  $x_1 = x_2$  and  $x_3$

is arbitrary. Taking  $x_2 = 1$  and  $x_3 = 0$  yields  $\mathbf{y}_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} e^{t/2}$ . Taking  $x_2 = 0$  and  $x_3 = 1$

yields  $\mathbf{y}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} e^{t/2}$ . The general solution is  $\mathbf{y} = c_1 \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix} e^{-t/2} + c_2 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} e^{t/2} + c_3 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} e^{t/2}$ . Now

$\mathbf{y}(0) = \begin{bmatrix} 4 \\ 7 \\ 1 \end{bmatrix} \Rightarrow c_1 \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + c_3 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} e^{t/2} = \begin{bmatrix} 4 \\ 7 \\ 1 \end{bmatrix}$ , so  $c_1 = 1$ ,  $c_2 = 5$ , and  $c_3 = 1$ . Hence

$$\mathbf{y} = \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix} e^{-t/2} + \begin{bmatrix} 5 \\ 5 \\ 0 \end{bmatrix} e^{t/2} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} e^{t/2}.$$

**10.4.22.**  $\begin{vmatrix} 6-\lambda & -3 & -8 \\ 2 & 1-\lambda & -2 \\ 3 & -3 & -5-\lambda \end{vmatrix} = -(\lambda-1)(\lambda+2)(\lambda-3)$ . The eigenvectors associated with

$\lambda_1 = 1$  satisfy the system with augmented matrix  $\begin{bmatrix} 5 & -3 & -8 & : & 0 \\ 2 & 0 & -2 & : & 0 \\ 3 & -3 & -6 & : & 0 \end{bmatrix}$ , which is row equivalent

to  $\begin{bmatrix} 1 & 0 & -1 & : & 0 \\ 0 & 1 & 1 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$ . Hence  $x_1 = x_3$  and  $x_2 = -x_3$ . Taking  $x_3 = 1$  yields  $\mathbf{y}_1 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} e^t$ . The

eigenvectors associated with  $\lambda_2 = -2$  satisfy the system with augmented matrix  $\begin{bmatrix} 8 & -3 & -8 & : & 0 \\ 2 & 3 & -2 & : & 0 \\ 3 & -3 & -3 & : & 0 \end{bmatrix}$ ,

which is row equivalent to  $\begin{bmatrix} 1 & 0 & -1 & : & 0 \\ 0 & 1 & 0 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$ . Hence  $x_1 = x_3$  and  $x_2 = 0$ . Taking  $x_3 = 1$  yields

$\mathbf{y}_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} e^{-2t}$ . The eigenvectors associated with  $\lambda_3 = 3$  satisfy the system with augmented

matrix  $\begin{bmatrix} 3 & -3 & -8 & : & 0 \\ 2 & -2 & -2 & : & 0 \\ 3 & -3 & -8 & : & 0 \end{bmatrix}$ , which is row equivalent to  $\begin{bmatrix} 1 & -1 & 0 & : & 0 \\ 0 & 0 & 1 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$ . Hence  $x_1 = x_2$

and  $x_3 = 0$ . Taking  $x_2 = 1$  yields  $\mathbf{y}_3 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} e^{3t}$ . The general solution is  $\mathbf{y} = c_1 \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} e^t +$

$c_2 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} e^{-2t} + c_3 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} e^{3t}$ . Now  $\mathbf{y}(0) = \begin{bmatrix} 0 \\ -1 \\ -1 \end{bmatrix} \Rightarrow c_1 \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + c_3 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \\ -1 \end{bmatrix}$ , so

$c_1 = 2$ ,  $c_2 = -3$ , and  $c_3 = 1$ . Therefore,  $\mathbf{y} = \begin{bmatrix} 2 \\ -2 \\ 2 \end{bmatrix} e^t - \begin{bmatrix} 3 \\ 0 \\ 3 \end{bmatrix} e^{-2t} + \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} e^{3t}$ .

**10.4.24.**  $\begin{vmatrix} 3-\lambda & 0 & 1 \\ 11 & -2-\lambda & 7 \\ 1 & 0 & 3-\lambda \end{vmatrix} = -(\lambda-2)(\lambda+2)(\lambda-4)$ . The eigenvectors associated with

with  $\lambda_1 = 2$  satisfy the system with augmented matrix  $\begin{bmatrix} 1 & 0 & 1 & : & 0 \\ 11 & -4 & 7 & : & 0 \\ 1 & 0 & 1 & : & 0 \end{bmatrix}$ , which is row

equivalent to  $\begin{bmatrix} 1 & 0 & 1 & : & 0 \\ 0 & 1 & 1 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$ . Hence  $x_1 = -x_3$  and  $x_2 = -x_3$ . Taking  $x_3 = 1$  yields

$\mathbf{y}_1 = \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} e^{2t}$ . The eigenvectors associated with  $\lambda_2 = -2$  satisfy the system with augmented

matrix  $\begin{bmatrix} 5 & 0 & 1 & : & 0 \\ 11 & 0 & 7 & : & 0 \\ 1 & 0 & 5 & : & 0 \end{bmatrix}$ , which is row equivalent to  $\begin{bmatrix} 1 & 0 & 0 & : & 0 \\ 0 & 0 & 1 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$ . Hence  $x_1 = x_3 = 0$

and  $x_2$  is arbitrary. Taking  $x_3 = 1$  yields  $\mathbf{y}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} e^{-2t}$ . The eigenvectors associated with

with  $\lambda_3 = 4$  satisfy the system with augmented matrix  $\begin{bmatrix} -1 & 0 & 1 & : & 0 \\ 11 & -6 & 7 & : & 0 \\ 1 & 0 & -1 & : & 0 \end{bmatrix}$ , which is row

equivalent to  $\begin{bmatrix} 1 & 0 & -1 & : & 0 \\ 0 & 1 & -3 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$ . Hence  $x_1 = x_3$  and  $x_2 = 3x_3$ . Taking  $x_3 = 1$  yields

$\mathbf{y}_3 = \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix} e^{4t}$ . The general solution is  $\mathbf{y} = c_1 \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} e^{2t} + c_2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} e^{-2t} + c_3 \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix} e^{4t}$ . Now

$\mathbf{y}(0) = \begin{bmatrix} 2 \\ 7 \\ 6 \end{bmatrix} \Rightarrow c_1 \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + c_3 \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 7 \\ 6 \end{bmatrix}$ , so  $c_1 = 2$ ,  $c_2 = -3$ , and  $c_3 = 4$ . Hence

$$\mathbf{y} = \begin{bmatrix} -2 \\ -2 \\ 2 \end{bmatrix} e^{2t} - \begin{bmatrix} 0 \\ 3 \\ 0 \end{bmatrix} e^{-2t} + \begin{bmatrix} 4 \\ 12 \\ 4 \end{bmatrix} e^{4t}$$

**10.4.26.**  $\begin{vmatrix} 3-\lambda & -1 & 0 \\ 4 & -2-\lambda & 0 \\ 4 & -4 & 2-\lambda \end{vmatrix} = -(\lambda+1)(\lambda-2)^2$ . The eigenvectors associated with  $\lambda_1 =$

$-1$  satisfy the system with augmented matrix  $\begin{bmatrix} 4 & -1 & 0 & : & 0 \\ 4 & -1 & 0 & : & 0 \\ 4 & -4 & 3 & : & 0 \end{bmatrix}$ , which is row equivalent to

$\begin{bmatrix} 1 & 0 & -\frac{1}{4} & : & 0 \\ 0 & 1 & -1 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$ . Hence  $x_1 = x_2/4$  and  $x_2 = x_3$ . Taking  $x_3 = 4$  yields  $\mathbf{y}_1 = \begin{bmatrix} 1 \\ 4 \\ 4 \end{bmatrix} e^{-t}$ .

The eigenvectors associated with  $\lambda_2 = \lambda_3 = 2$  satisfy the system with augmented matrix

$\begin{bmatrix} 1 & -1 & 0 & : & 0 \\ 4 & -4 & 0 & : & 0 \\ 4 & -4 & 0 & : & 0 \end{bmatrix}$ , which is row equivalent to  $\begin{bmatrix} 1 & -1 & 0 & : & 0 \\ 0 & 0 & 0 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$ . Hence  $x_1 = x_2$  and  $x_3$

is arbitrary. Taking  $x_2 = 1$  and  $x_3 = 0$  yields  $\mathbf{y}_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} e^{2t}$ . Taking  $x_2 = 0$  and  $x_3 = 1$

yields  $\mathbf{y}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} e^{2t}$ . The general solution is  $\mathbf{y} = c_1 \begin{bmatrix} 1 \\ 4 \\ 4 \end{bmatrix} e^{-t} + c_2 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} e^{2t} + c_3 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} e^{2t}$ . Now



$$\mathbf{y}(0) = \begin{bmatrix} 7 \\ 10 \\ 2 \end{bmatrix} \Rightarrow c_1 \begin{bmatrix} 1 \\ 4 \\ 4 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + c_3 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 7 \\ 10 \\ 2 \end{bmatrix}, \text{ so } c_1 = 1, c_2 = 6, \text{ and } c_3 = -2. \text{ Hence}$$

$$\mathbf{y} = \begin{bmatrix} 1 \\ 4 \\ 4 \end{bmatrix} e^{-t} + \begin{bmatrix} 6 \\ 6 \\ -2 \end{bmatrix} e^{2t}.$$

**10.4.28. (a)** If  $\mathbf{y}(t_0) = 0$ , then  $\mathbf{y}$  is the solution of the initial value problem  $\mathbf{y}' = A\mathbf{y}$ ,  $\mathbf{y}(t_0) = 0$ . Since  $\mathbf{y} \equiv 0$  is a solution of this problem, Theorem 10.2.1 implies the conclusion.

**(b)** It is given that  $\mathbf{y}'_1(t) = A\mathbf{y}_1(t)$  for all  $t$ . Replacing  $t$  by  $t-\tau$  shows that  $\mathbf{y}'_1(t-\tau) = A\mathbf{y}_1(t-\tau) = A\mathbf{y}_2(t)$  for all  $t$ . Since  $\mathbf{y}'_2(t) = \mathbf{y}'_1(t-\tau)$  by the chain rule, this implies that  $\mathbf{y}'_2(t) = A\mathbf{y}_2(t)$  for all  $t$ .

**(c)** If  $\mathbf{z}(t) = \mathbf{y}_1(t-\tau)$ , then  $\mathbf{z}(t_2) = \mathbf{y}_1(t_1) = \mathbf{y}_2(t_2)$ ; therefore  $\mathbf{z}$  and  $\mathbf{y}_2$  are both solutions of the initial value problem  $\mathbf{y}' = A\mathbf{y}$ ,  $\mathbf{y}(t_2) = \mathbf{k}$ , where  $\mathbf{k} = \mathbf{y}_2(t_2)$ .

**10.4.42.** The characteristic polynomial of  $A$  is  $p(\lambda) = \lambda^2 - (a+b)\lambda + ab - \alpha\beta$ , so the eigenvalues of  $A$  are  $\lambda_1 = \frac{a+b-\gamma}{2}$  and  $\lambda_2 = \frac{a+b+\gamma}{2}$ , where  $\gamma = \sqrt{(a-b)^2 + 4\alpha\beta}$ ;  $\mathbf{x}_1 = \begin{bmatrix} b-a+\gamma \\ 2\beta \end{bmatrix}$  and

$\mathbf{x}_2 = \begin{bmatrix} b-a-\gamma \\ 2\beta \end{bmatrix}$  are associated eigenvectors. Since  $\gamma > |b-a|$ , if  $L_1$  and  $L_2$  are lines through the origin parallel to  $\mathbf{x}_1$  and  $\mathbf{x}_2$ , then  $L_1$  is in the first and third quadrants and  $L_2$  is in the second and fourth quadrants. The slope of  $L_1$  is  $\rho = \frac{2\beta}{b-a+\gamma} > 0$ . If  $Q_0 = \rho P_0$  there are three possibilities:

(i) if  $\alpha\beta = ab$ , then  $\lambda_1 = 0$  and  $P(t) = P_0$ ,  $Q(t) = Q_0$  for all  $t > 0$ ; (ii) if  $\alpha\beta < ab$ , then  $\lambda_1 > 0$  and  $\lim_{t \rightarrow \infty} P(t) = \lim_{t \rightarrow \infty} Q(t) = \infty$  (monotonically); (iii) if  $\alpha\beta > ab$ , then  $\lambda_1 < 0$  and  $\lim_{t \rightarrow \infty} P(t) = \lim_{t \rightarrow \infty} Q(t) = 0$  (monotonically). Now suppose  $Q_0 \neq \rho P_0$ , so that the trajectory cannot intersect  $L_1$ , and assume for the moment that (A) makes sense for all  $t > 0$ ; that is, even if one or the other of  $P$  and  $Q$  is negative. Since  $\lambda_2 > 0$  it follows that either  $\lim_{t \rightarrow \infty} P(t) = \infty$  or  $\lim_{t \rightarrow \infty} Q(t) = \infty$  (or both), and the trajectory is asymptotically parallel to  $L_2$ . Therefore, the trajectory must cross into the third quadrant (so  $P(T) = 0$  and  $Q(T) > 0$  for some finite  $T$ ) if  $Q_0 > \rho P_0$ , or into the fourth quadrant (so  $Q(T) = 0$  and  $P(T) > 0$  for some finite  $T$ ) if  $Q_0 < \rho P_0$ .

## 10.5 Constant Coefficient Homogeneous Systems II

**10.5.2.**  $\begin{vmatrix} -\lambda & -1 \\ 1 & -2-\lambda \end{vmatrix} = (\lambda+1)^2$ . Hence  $\lambda_1 = -1$ . Eigenvectors satisfy  $\begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ , so

$x_1 = x_2$ . Taking  $x_2 = 1$  yields  $\mathbf{y}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{-t}$ . For a second solution we need a vector  $\mathbf{u}$  such that

$\begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ . Let  $u_1 = 1$  and  $u_2 = 0$ . Then  $\mathbf{y}_2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} e^{-t} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} t e^{-t}$ . The general solution

is  $\mathbf{y} = c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{-t} + c_2 \left( \begin{bmatrix} 1 \\ 0 \end{bmatrix} e^{-t} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} t e^{-t} \right)$ .

**10.5.4.**  $\mathbf{y}' = \begin{vmatrix} 3-\lambda & 1 \\ -1 & 1-\lambda \end{vmatrix} = (\lambda-2)^2$ . Hence  $\lambda_1 = 2$ . Eigenvectors satisfy  $\begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ ,

so  $x_1 = -x_2$ . Taking  $x_2 = 1$  yields  $\mathbf{y}_1 = \begin{bmatrix} -1 \\ 1 \end{bmatrix} e^{2t}$ . For a second solution we need a vector  $\mathbf{u}$  such

that  $\begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$ . Let  $u_1 = -1$  and  $u_2 = 0$ . Then  $\mathbf{y}_2 = \begin{bmatrix} -1 \\ 0 \end{bmatrix} e^{2t} + \begin{bmatrix} -1 \\ 1 \end{bmatrix} t e^{2t}$ . The

general solution is  $\mathbf{y} = c_1 \begin{bmatrix} -1 \\ 1 \end{bmatrix} e^{2t} + c_2 \left( \begin{bmatrix} -1 \\ 0 \end{bmatrix} e^{2t} + \begin{bmatrix} -1 \\ 1 \end{bmatrix} t e^{2t} \right)$ .

**10.5.6.**  $\begin{vmatrix} -10-\lambda & 9 \\ -4 & 2-\lambda \end{vmatrix} = (\lambda+4)^2$ . Hence  $\lambda_1 = -4$ . Eigenvectors satisfy  $\begin{bmatrix} -6 & 9 \\ -4 & 6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ ,

so  $x_1 = \frac{3}{2}x_2$ . Taking  $x_2 = 2$  yields  $\mathbf{y}_1 = \begin{bmatrix} 3 \\ 2 \end{bmatrix} e^{-4t}$ . For a second solution we need a vector  $\mathbf{u}$  such

that  $\begin{bmatrix} -6 & 9 \\ -4 & 6 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$ . Let  $u_1 = -\frac{1}{2}$  and  $u_2 = 0$ . Then  $\mathbf{y}_2 = \begin{bmatrix} -1 \\ 0 \end{bmatrix} \frac{e^{-4t}}{2} + \begin{bmatrix} 3 \\ 2 \end{bmatrix} t e^{-4t}$ . The general

solution is  $\mathbf{y} = c_1 \begin{bmatrix} 3 \\ 2 \end{bmatrix} e^{-4t} + c_2 \left( \begin{bmatrix} -1 \\ 0 \end{bmatrix} \frac{e^{-4t}}{2} + \begin{bmatrix} 3 \\ 2 \end{bmatrix} t e^{-4t} \right)$ .

**10.5.8.**  $\begin{bmatrix} -\lambda & 2 & 1 \\ -4 & 6-\lambda & 1 \\ 0 & 4 & 2-\lambda \end{bmatrix} = -\lambda(\lambda-4)^2$ . Hence  $\lambda_1 = 0$  and  $\lambda_2 = \lambda_3 = 4$ . The eigenvectors

associated with  $\lambda_1 = 0$  satisfy the system with augmented matrix  $\begin{bmatrix} 0 & 2 & 1 & : & 0 \\ -4 & 6 & 1 & : & 0 \\ 0 & 4 & 2 & : & 0 \end{bmatrix}$ , which

is row equivalent to  $\begin{bmatrix} 1 & 0 & \frac{1}{2} & : & 0 \\ 0 & 1 & \frac{1}{2} & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$ . Hence  $x_1 = -\frac{1}{2}x_3$  and  $x_2 = -\frac{1}{2}x_3$ . Taking  $x_3 = 2$

yields  $\mathbf{y}_1 = \begin{bmatrix} -1 \\ -1 \\ 2 \end{bmatrix}$ . The eigenvectors associated with  $\lambda_2 = 4$  satisfy the system with augmented

matrix  $\begin{bmatrix} -4 & 2 & 1 & : & 0 \\ -4 & 2 & 1 & : & 0 \\ 0 & 4 & -2 & : & 0 \end{bmatrix}$ , which is row equivalent to  $\begin{bmatrix} 1 & 0 & -\frac{1}{2} & : & 0 \\ 0 & 1 & -\frac{1}{2} & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$ . Hence  $x_1 = \frac{1}{2}x_3$

and  $x_2 = \frac{1}{2}x_3$ . Taking  $x_3 = 2$  yields  $\mathbf{y}_2 = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} e^{4t}$ . For a third solution we need a vector  $\mathbf{u}$

such that  $\begin{bmatrix} -4 & 2 & 1 \\ -4 & 2 & 1 \\ 0 & 4 & -2 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$ . The augmented matrix of this system is row equivalent to

$\begin{bmatrix} 1 & 0 & -\frac{1}{2} & : & 0 \\ 0 & 1 & -\frac{1}{2} & : & \frac{1}{2} \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$ . Let  $u_3 = 0$ ,  $u_1 = 0$ , and  $u_2 = \frac{1}{2}$ . Then  $\mathbf{y}_3 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \frac{e^{4t}}{2} + \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} t e^{4t}$ . The general

solution is

$$\mathbf{y} = c_1 \begin{bmatrix} -1 \\ -1 \\ 2 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} e^{4t} + c_3 \left( \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \frac{e^{4t}}{2} + \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} t e^{4t} \right).$$

**10.5.10.**  $\begin{bmatrix} -1-\lambda & 1 & -1 \\ -2 & -\lambda & 2 \\ -1 & 3 & -1-\lambda \end{bmatrix} = -(\lambda-2)(\lambda+2)^2$ . Hence  $\lambda_1 = 2$  and  $\lambda_2 = \lambda_3 = -2$ . The eigen-

vectors associated with  $\lambda_1 = 2$  satisfy the system with augmented matrix  $\begin{bmatrix} -3 & 1 & -1 & : & 0 \\ -2 & -2 & 2 & : & 0 \\ -1 & 3 & -3 & : & 0 \end{bmatrix}$ ,

which is row equivalent to  $\begin{bmatrix} 1 & 0 & 0 & : & 0 \\ 0 & 1 & -1 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$ . Hence  $x_1 = 0$  and  $x_2 = x_3$ . Taking  $x_3 = 1$  yields

$\mathbf{y}_1 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} e^{2t}$ . The eigenvectors associated with  $\lambda_2 = -2$  satisfy the system with augmented matrix  $\begin{bmatrix} 1 & 1 & -1 & : & 0 \\ -2 & 2 & 2 & : & 0 \\ -1 & 3 & 1 & : & 0 \end{bmatrix}$ , which is row equivalent to  $\begin{bmatrix} 1 & 0 & -1 & : & 0 \\ 0 & 1 & 0 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$ . Hence  $x_1 = x_3$

and  $x_2 = 0$ . Taking  $x_3 = 1$  yields  $\mathbf{y}_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} e^{-2t}$ . For a third solution we need a vector  $\mathbf{u}$

such that  $\begin{bmatrix} 1 & 1 & -1 \\ -2 & 2 & 2 \\ -1 & 3 & 1 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$ . The augmented matrix of this system is row equivalent to

$\begin{bmatrix} 1 & 0 & -1 & : & \frac{1}{2} \\ 0 & 1 & 0 & : & \frac{1}{2} \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$ . Let  $u_3 = 0$ ,  $u_1 = \frac{1}{2}$ , and  $u_2 = \frac{1}{2}$ . Then  $\mathbf{y}_3 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \frac{e^{-2t}}{2} + \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} te^{-2t}$ . The general solution is

$$\mathbf{y} = c_1 \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} e^{2t} + c_2 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} e^{-2t} + c_3 \left( \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \frac{e^{-2t}}{2} + \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} te^{-2t} \right).$$

**10.5.12.**  $\begin{bmatrix} 6-\lambda & -5 & 3 \\ 2 & -1-\lambda & 3 \\ 2 & 1 & 1-\lambda \end{bmatrix} = -(\lambda+2)(\lambda-4)^2$ . Hence  $\lambda_1 = -2$  and  $\lambda_2 = \lambda_3 = 4$ . The eigen-

vectors associated with  $\lambda_1 = -2$  satisfy the system with augmented matrix  $\begin{bmatrix} 8 & -5 & 3 & : & 0 \\ 2 & 1 & 3 & : & 0 \\ 2 & 1 & 3 & : & 0 \end{bmatrix}$ ,

which is row equivalent to  $\begin{bmatrix} 1 & 0 & 1 & : & 0 \\ 0 & 1 & 1 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$ . Hence  $x_1 = -x_3$  and  $x_2 = -x_3$ . Taking  $x_3 = 1$

yields  $\mathbf{y}_1 = \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} e^{-2t}$ . The eigenvectors associated with  $\lambda_2 = 4$  satisfy the system with augmented matrix

$\begin{bmatrix} 2 & -5 & 3 & : & 0 \\ 2 & -5 & 3 & : & 0 \\ 2 & 1 & -3 & : & 0 \end{bmatrix}$ , which is row equivalent to  $\begin{bmatrix} 1 & 0 & -1 & : & 0 \\ 0 & 1 & -1 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$ . Hence

$x_1 = x_3$  and  $x_2 = x_3$ . Taking  $x_3 = 1$  yields  $\mathbf{y}_2 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} e^{4t}$ . For a third solution we need a vector

$\mathbf{u}$  such that  $\begin{bmatrix} 2 & -5 & 3 \\ 2 & -5 & 3 \\ 2 & 1 & -3 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ . The augmented matrix of this system is row equivalent

to  $\begin{bmatrix} 1 & 0 & -1 & : & \frac{1}{2} \\ 0 & 1 & -1 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$ . Let  $u_3 = 0$ ,  $u_1 = \frac{1}{2}$ , and  $u_2 = 0$ . Then  $\mathbf{y}_3 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \frac{e^{4t}}{2} + \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} te^{4t}$ . The

general solution is  $\mathbf{y} = c_1 \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} e^{-2t} + c_2 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} e^{4t} + c_3 \left( \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \frac{e^{4t}}{2} + \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} te^{4t} \right)$ .

**10.5.14.**  $\begin{vmatrix} 15 - \lambda & -9 \\ 16 & -9 - \lambda \end{vmatrix} = (\lambda - 3)^2$ . Hence  $\lambda_1 = 3$ . Eigenvectors satisfy  $\begin{bmatrix} 12 & -9 \\ 16 & -12 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ ,

so  $x_1 = \frac{3}{4}x_2$ . Taking  $x_2 = 4$  yields  $\mathbf{y}_1 = \begin{bmatrix} 3 \\ 4 \\ 4 \end{bmatrix} e^{3t}$ . For a second solution we need a vector  $\mathbf{u}$  such

that  $\begin{bmatrix} 12 & -9 \\ 16 & -12 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$ . Let  $u_1 = \frac{1}{4}$  and  $u_2 = 0$ . Then  $\mathbf{y}_2 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \frac{e^{3t}}{4} + \begin{bmatrix} 3 \\ 4 \\ 4 \end{bmatrix} te^{3t}$ . The general

solution is  $\mathbf{y} = c_1 \begin{bmatrix} 3 \\ 4 \\ 4 \end{bmatrix} e^{3t} + c_2 \left( \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \frac{e^{3t}}{4} + \begin{bmatrix} 3 \\ 4 \\ 4 \end{bmatrix} te^{3t} \right)$ . Now  $\mathbf{y}(0) = \begin{bmatrix} 5 \\ 8 \\ 8 \end{bmatrix} \Rightarrow c_1 \begin{bmatrix} 3 \\ 4 \\ 4 \end{bmatrix} + c_2 \begin{bmatrix} \frac{1}{4} \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 5 \\ 8 \\ 8 \end{bmatrix}$ , so

$c_1 = 2$  and  $c_2 = -4$ . Therefore,  $\mathbf{y} = \begin{bmatrix} 5 \\ 8 \\ 8 \end{bmatrix} e^{3t} - \begin{bmatrix} 12 \\ 16 \\ 16 \end{bmatrix} te^{3t}$ .

**10.5.16.**  $\begin{vmatrix} -7 - \lambda & 24 \\ -6 & 17 - \lambda \end{vmatrix} = (\lambda - 5)^2$ . Hence  $\lambda_1 = 5$ . Eigenvectors satisfy  $\begin{bmatrix} -12 & 24 \\ -6 & 12 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ ,

so  $x_1 = 2x_2$ . Taking  $x_2 = 1$  yields  $\mathbf{y}_1 = \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix} e^{5t}$ . For a second solution we need a vector  $\mathbf{u}$  such

that  $\begin{bmatrix} -12 & 24 \\ -6 & 12 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ . Let  $u_1 = \frac{1}{6}$  and  $u_2 = 0$ . Then  $\mathbf{y}_2 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \frac{e^{5t}}{6} + \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix} te^{5t}$ . The general

solution is  $\mathbf{y} = c_1 \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix} e^{5t} + c_2 \left( \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \frac{e^{5t}}{6} + \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix} te^{5t} \right)$ . Now  $\mathbf{y}(0) = \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix} \Rightarrow c_1 \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} \frac{1}{6} \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$ , so

$c_1 = 1$  and  $c_2 = 6$ . Therefore,  $\mathbf{y} = \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix} e^{5t} - \begin{bmatrix} 12 \\ 6 \\ 6 \end{bmatrix} te^{5t}$ .

**10.5.18.**  $\begin{bmatrix} -1 - \lambda & 1 & 0 \\ 1 & -1 - \lambda & -2 \\ -1 & -1 & -1 - \lambda \end{bmatrix} = -(\lambda - 1)(\lambda + 2)^2$ . Hence  $\lambda_1 = 1$  and  $\lambda_2 = \lambda_3 = -2$ . The

eigenvectors associated with  $\lambda_1 = 1$  satisfy the system with augmented matrix  $\begin{bmatrix} -2 & 1 & 0 & : & 0 \\ 1 & -2 & -2 & : & 0 \\ -1 & -1 & -2 & : & 0 \end{bmatrix}$ ,

which is row equivalent to  $\begin{bmatrix} 1 & 0 & \frac{2}{3} & : & 0 \\ 0 & 1 & \frac{4}{3} & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$ . Hence  $x_1 = -\frac{2}{3}x_3$  and  $x_2 = -\frac{4}{3}x_3$ . Taking

$x_3 = 3$  yields The eigenvectors associated with  $\lambda_2 = -2$  satisfy the system with augmented matrix

$\begin{bmatrix} 1 & 1 & 0 & : & 0 \\ 1 & 1 & -2 & : & 0 \\ -1 & -1 & 1 & : & 0 \end{bmatrix}$ , which is row equivalent to  $\begin{bmatrix} 1 & 1 & 0 & : & 0 \\ 0 & 0 & 1 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$ . Hence  $x_1 = -x_2$  and

$x_3 = 0$ . Taking  $x_2 = 1$  yields  $\mathbf{y}_2 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} e^{-2t}$ . For a third solution we need a vector  $\mathbf{u}$  such

that  $\begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & -2 \\ -1 & -1 & 1 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$ . The augmented matrix of this system is row equivalent to

$\begin{bmatrix} 1 & 1 & 0 & \vdots & -1 \\ 0 & 0 & 1 & \vdots & -1 \\ 0 & 0 & 0 & \vdots & 0 \end{bmatrix}$ . Let  $u_2 = 0$ ,  $u_1 = -1$ , and  $u_3 = -1$ . Then  $\mathbf{y}_3 = \begin{bmatrix} -1 \\ 0 \\ -1 \end{bmatrix} e^{-2t} + \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} t e^{-2t}$ .

The general solution is  $\mathbf{y} = c_1 \begin{bmatrix} -2 \\ -4 \\ 3 \end{bmatrix} e^t + c_2 \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} e^{-2t} + c_3 \left( \begin{bmatrix} -1 \\ 0 \\ -1 \end{bmatrix} e^{-2t} + \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} t e^{-2t} \right)$ . Now  $\mathbf{y}(0) =$

$$\begin{bmatrix} 6 \\ 5 \\ -7 \end{bmatrix} \Rightarrow c_1 \begin{bmatrix} -2 \\ -4 \\ 3 \end{bmatrix} + c_2 \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + c_3 \begin{bmatrix} -1 \\ 0 \\ -1 \end{bmatrix} = \begin{bmatrix} 6 \\ 5 \\ -7 \end{bmatrix}, \text{ so } c_1 = -2, c_2 = -3, \text{ and } c_3 = 1. \text{ Therefore,}$$

$$\mathbf{y} = \begin{bmatrix} 4 \\ 8 \\ -6 \end{bmatrix} e^t + \begin{bmatrix} 2 \\ -3 \\ -1 \end{bmatrix} e^{-2t} + \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} t e^{-2t}.$$

**10.5.20.**  $\begin{bmatrix} -7-\lambda & -4 & 4 \\ -1 & 0-\lambda & 1 \\ -9 & -5 & 6-\lambda \end{bmatrix} = -(\lambda+3)(\lambda-1)^2$ . Hence  $\lambda_1 = -3$  and  $\lambda_2 = \lambda_3 = 1$ . The eigen-

vectors associated with  $\lambda_1 = -3$  satisfy the system with augmented matrix  $\begin{bmatrix} -4 & -4 & 4 & \vdots & 0 \\ -1 & 3 & 1 & \vdots & 0 \\ -9 & -5 & 9 & \vdots & 0 \end{bmatrix}$ ,

which is row equivalent to  $\begin{bmatrix} 1 & 0 & -1 & \vdots & 0 \\ 0 & 1 & 0 & \vdots & 0 \\ 0 & 0 & 0 & \vdots & 0 \end{bmatrix}$ . Hence  $x_1 = x_3$  and  $x_2 = 0$ . Taking  $x_3 = 1$  yields

$\mathbf{y}_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} e^{-3t}$ . The eigenvectors associated with  $\lambda_2 = 1$  satisfy the system with augmented matrix

$\begin{bmatrix} -8 & -4 & 4 & \vdots & 0 \\ -1 & -1 & 1 & \vdots & 0 \\ -9 & -5 & -5 & \vdots & 0 \end{bmatrix}$ , which is row equivalent to  $\begin{bmatrix} 1 & 0 & 0 & \vdots & 0 \\ 0 & 1 & -1 & \vdots & 0 \\ 0 & 0 & 0 & \vdots & 0 \end{bmatrix}$ . Hence  $x_1 = 0$

and  $x_2 = x_3$ . Taking  $x_3 = 1$  yields  $\mathbf{y}_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} e^t$ . For a third solution we need a vector  $\mathbf{u}$  such

that  $\begin{bmatrix} -8 & -4 & 4 \\ -1 & -1 & 1 \\ -9 & -5 & 5 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$ . The augmented matrix of this system is row equivalent to

$\begin{bmatrix} 1 & 0 & 0 & \vdots & 1 \\ 0 & 1 & -1 & \vdots & -2 \\ 0 & 0 & 0 & \vdots & 0 \end{bmatrix}$ . Let  $u_3 = 0$ ,  $u_1 = 1$ , and  $u_2 = -2$ . Then  $\mathbf{y}_3 = \begin{bmatrix} 1 \\ -2 \\ 0 \end{bmatrix} e^t + \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} t e^t$ . The

general solution is  $\mathbf{y} = c_1 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} e^{-3t} + c_2 \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} e^t + c_3 \left( \begin{bmatrix} 1 \\ -2 \\ 0 \end{bmatrix} e^t + \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} t e^t \right)$ . Now  $\mathbf{y}(0) = \begin{bmatrix} -6 \\ 9 \\ -1 \end{bmatrix} \Rightarrow$

$c_1 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} + c_3 \begin{bmatrix} 1 \\ -2 \\ 0 \end{bmatrix} = \begin{bmatrix} -6 \\ 9 \\ -1 \end{bmatrix}$ , so  $c_1 = -2$ ,  $c_2 = 1$ , and  $c_3 = -4$ . Therefore,  $\mathbf{y} =$

$$- \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix} e^{-3t} + \begin{bmatrix} -4 \\ 9 \\ 1 \end{bmatrix} e^t - \begin{bmatrix} 0 \\ 4 \\ 4 \end{bmatrix} t e^t.$$

**10.5.22.**  $\begin{bmatrix} 4-\lambda & -8 & -4 \\ -3 & -1-\lambda & -3 \\ 1 & -1 & 9-\lambda \end{bmatrix} = -(\lambda+4)(\lambda-8)^2$ . Hence  $\lambda_1 = -4$  and  $\lambda_2 = \lambda_3 = 8$ . The eigen-

vectors associated with  $\lambda_1 = -4$  satisfy the system with augmented matrix  $\begin{bmatrix} 8 & -8 & -4 & : & 0 \\ -3 & 3 & -3 & : & 0 \\ 1 & -1 & 13 & : & 0 \end{bmatrix}$ ,

which is row equivalent to  $\begin{bmatrix} 1 & -1 & 0 & : & 0 \\ 0 & 0 & 1 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$ . Hence  $x_1 = x_2$  and  $x_3 = 0$ . Taking  $x_2 = 1$  yields

$\mathbf{y}_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} e^t$ . The eigenvectors associated with  $\lambda_2 = 8$  satisfy the system with augmented matrix

$\begin{bmatrix} -4 & -8 & -4 & : & 0 \\ -3 & -9 & -3 & : & 0 \\ 1 & -1 & 1 & : & 0 \end{bmatrix}$ , which is row equivalent to  $\begin{bmatrix} 1 & 0 & 1 & : & 0 \\ 0 & 1 & 0 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$ . Hence  $x_1 = -x_3$  and

$x_2 = 0$ . Taking  $x_3 = 1$  yields  $\mathbf{y}_2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} e^{8t}$ . For a third solution we need a vector  $\mathbf{u}$  such

that  $\begin{bmatrix} -4 & -8 & -4 \\ -3 & -9 & -3 \\ 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$ . The augmented matrix of this system is row equivalent to

$\begin{bmatrix} 1 & 0 & 1 & : & \frac{3}{4} \\ 0 & 1 & 0 & : & -\frac{1}{4} \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$ . Let  $u_3 = 0$ ,  $u_1 = \frac{3}{4}$ , and  $u_2 = -\frac{1}{4}$ . Then  $\mathbf{y}_3 = \begin{bmatrix} 3 \\ -1 \\ 0 \end{bmatrix} \frac{e^{8t}}{4} + \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} t e^{8t}$ .

The general solution is  $c_1 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} e^t + c_2 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} e^{8t} + c_3 \left( \begin{bmatrix} 3 \\ -1 \\ 0 \end{bmatrix} \frac{e^{8t}}{4} + \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} t e^{8t} \right)$ . Now  $\mathbf{y}(0) = \begin{bmatrix} -4 \\ 1 \\ -3 \end{bmatrix} \Rightarrow$

$c_1 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} + c_3 \begin{bmatrix} \frac{3}{4} \\ -\frac{1}{4} \\ 0 \end{bmatrix} = \begin{bmatrix} -4 \\ 1 \\ -3 \end{bmatrix}$ , so  $c_1 = -1$ ,  $c_2 = -3$ , and  $c_3 = -8$ . Therefore,  $\mathbf{y} =$

$-\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} e^{-4t} + \begin{bmatrix} -3 \\ 2 \\ -3 \end{bmatrix} e^{8t} + \begin{bmatrix} 8 \\ 0 \\ -8 \end{bmatrix} t e^{8t}$ .

**10.5.24.**  $\begin{bmatrix} 5-\lambda & -1 & 1 \\ -1 & 9-\lambda & -3 \\ -2 & 2 & 4-\lambda \end{bmatrix} = -(\lambda-6)^3$ . Hence  $\lambda_1 = 6$ . The eigenvectors satisfy the system

with augmented matrix  $\begin{bmatrix} 1 & -1 & 1 & : & 0 \\ -1 & 3 & -3 & : & 0 \\ -2 & 2 & -2 & : & 0 \end{bmatrix}$ , which is row equivalent to  $\begin{bmatrix} 1 & 0 & 0 & : & 0 \\ 0 & 1 & -1 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$ .

Hence  $x_1 = 0$  and  $x_2 = x_3$ . Taking  $x_3 = 1$  yields  $\mathbf{y}_1 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} e^{6t}$ . For a second solution we

need a vector  $\mathbf{u}$  such that  $\begin{bmatrix} 1 & -1 & 1 \\ -1 & 3 & -3 \\ -2 & 2 & -2 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$ . The augmented matrix of this system

is row equivalent to  $\begin{bmatrix} 1 & 0 & 0 & : & -\frac{1}{4} \\ 0 & 1 & -1 & : & \frac{1}{4} \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$ . Let  $u_3 = 0$ ,  $u_1 = -\frac{1}{4}$ , and  $u_2 = \frac{1}{4}$ . Then  $\mathbf{y}_2 =$

$\begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \frac{e^{6t}}{4} + \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} te^{6t}$ . For a third solution we need a vector  $\mathbf{v}$  such that  $\begin{bmatrix} 1 & -1 & 1 \\ -1 & 3 & -3 \\ -2 & 2 & -2 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} -\frac{1}{4} \\ \frac{1}{4} \\ 0 \end{bmatrix}$ . The augmented matrix of this system is row equivalent to  $\begin{bmatrix} 1 & 0 & 0 & : & \frac{1}{8} \\ 0 & 1 & -1 & : & \frac{1}{8} \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$ . Let

$v_3 = 0$ ,  $v_1 = \frac{1}{8}$ , and  $v_2 = \frac{1}{8}$ . Then  $\mathbf{y}_3 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \frac{e^{6t}}{8} + \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \frac{te^{6t}}{4} + \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \frac{t^2e^{6t}}{2}$ . The general solution is

$$\mathbf{y} = c_1 \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} e^{6t} + c_2 \left( \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \frac{e^{6t}}{4} + \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} te^{6t} \right) + c_3 \left( \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \frac{e^{6t}}{8} + \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \frac{te^{6t}}{4} + \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \frac{t^2e^{6t}}{2} \right).$$

**10.5.26.**  $\begin{bmatrix} -6-\lambda & -4 & -4 \\ 2 & -1-\lambda & 1 \\ 2 & 3 & 1-\lambda \end{bmatrix} = -(\lambda+2)^3$ . Hence  $\lambda_1 = -2$ . The eigenvectors satisfy the sys-

tem with augmented matrix  $\begin{bmatrix} -4 & -4 & -4 & : & 0 \\ 2 & 1 & 1 & : & 0 \\ 2 & 3 & 3 & : & 0 \end{bmatrix}$ , which is row equivalent to  $\begin{bmatrix} 1 & 0 & 0 & : & 0 \\ 0 & 1 & 1 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$ .

Hence  $x_1 = 0$  and  $x_2 = -x_3$ . Taking  $x_3 = 1$  yields  $\mathbf{y}_1 = \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} e^{-2t}$ . For a second solution we need

a vector  $\mathbf{u}$  such that  $\begin{bmatrix} -4 & -4 & -4 \\ 2 & 1 & 1 \\ 2 & 3 & 3 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}$ . The augmented matrix of this system is row

equivalent to  $\begin{bmatrix} 1 & 0 & 0 & : & -1 \\ 0 & 1 & 1 & : & 1 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$ . Let  $u_3 = 0$ ,  $u_1 = -1$ , and  $u_2 = 1$ . Then  $\mathbf{y}_2 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} e^{-2t} +$

$\begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} te^{-2t}$ . For a third solution we need a vector  $\mathbf{v}$  such that  $\begin{bmatrix} -4 & -4 & -4 \\ 2 & 1 & 1 \\ 2 & 3 & 3 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$ .

The augmented matrix of this system is row equivalent to  $\begin{bmatrix} 1 & 0 & 0 & : & \frac{3}{4} \\ 0 & 1 & 1 & : & -\frac{1}{2} \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$ . Let  $v_3 = 0$ ,

$v_1 = \frac{3}{4}$ , and  $v_2 = -\frac{1}{2}$ . Then  $\mathbf{y}_3 = \begin{bmatrix} 3 \\ -2 \\ 0 \end{bmatrix} \frac{e^{-2t}}{4} + \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} te^{-2t} + \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} \frac{t^2e^{-2t}}{2}$ . The general solution

$$\text{is } \mathbf{y} = c_1 \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} e^{-2t} + c_2 \left( \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} e^{-2t} + \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} t e^{-2t} \right) \\ + c_3 \left( \begin{bmatrix} 3 \\ -2 \\ 0 \end{bmatrix} \frac{e^{-2t}}{4} + \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} t e^{-2t} + \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} \frac{t^2 e^{-2t}}{2} \right)$$

**10.5.28.**  $\begin{bmatrix} -2-\lambda & -12 & 10 \\ 2 & -24-\lambda & 11 \\ 2 & -24 & 8-\lambda \end{bmatrix} = -(\lambda+6)^3$ . Hence  $\lambda_1 = -6$ . The eigenvectors satisfy the

system with augmented matrix  $\begin{bmatrix} 4 & -12 & 10 & : & 0 \\ 2 & -18 & 11 & : & 0 \\ 2 & -24 & 14 & : & 0 \end{bmatrix}$ , which is row equivalent to  $\begin{bmatrix} 1 & 0 & 1 & : & 0 \\ 0 & 1 & -\frac{1}{2} & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$ .

Hence  $x_1 = -x_3$  and  $x_2 = \frac{x_3}{2}$ . Taking  $x_3 = 2$  yields  $\mathbf{y}_1 = \begin{bmatrix} -2 \\ 1 \\ 2 \end{bmatrix} e^{-6t}$ . For a second solution we

need a vector  $\mathbf{u}$  such that  $\begin{bmatrix} 4 & -12 & 10 \\ 2 & -18 & 11 \\ 2 & -24 & 14 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \begin{bmatrix} -2 \\ 1 \\ 2 \end{bmatrix}$ . The augmented matrix of this system

is row equivalent to  $\begin{bmatrix} 1 & 0 & 1 & : & -1 \\ 0 & 1 & -\frac{1}{2} & : & -\frac{1}{6} \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$ . Let  $u_3 = 0$ ,  $u_1 = -1$ , and  $u_2 = -\frac{1}{6}$ . Then  $\mathbf{y}_2 =$

$-\begin{bmatrix} 6 \\ 1 \\ 0 \end{bmatrix} \frac{e^{-6t}}{6} + \begin{bmatrix} -2 \\ 1 \\ 2 \end{bmatrix} t e^{-6t}$ . For a third solution we need a vector  $\mathbf{v}$  such that  $\begin{bmatrix} 4 & -12 & 10 \\ 2 & -18 & 11 \\ 2 & -24 & 14 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} =$

$\begin{bmatrix} -1 \\ -\frac{1}{6} \\ 0 \end{bmatrix}$ . The augmented matrix of this system is row equivalent to  $\begin{bmatrix} 1 & 0 & 1 & : & -\frac{1}{3} \\ 0 & 1 & -\frac{1}{2} & : & -\frac{1}{36} \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$ . Let

$v_3 = 0$ ,  $v_1 = -\frac{1}{3}$ , and  $v_2 = -\frac{1}{36}$ . Then  $\mathbf{y}_3 = -\begin{bmatrix} 12 \\ 1 \\ 0 \end{bmatrix} \frac{e^{-6t}}{36} - \begin{bmatrix} 6 \\ 1 \\ 0 \end{bmatrix} \frac{t e^{-6t}}{6} + \begin{bmatrix} -2 \\ 1 \\ 2 \end{bmatrix} \frac{t^2 e^{-6t}}{2}$ . The general

solution is  $\mathbf{y} = c_1 \begin{bmatrix} -2 \\ 1 \\ 2 \end{bmatrix} e^{-6t} + c_2 \left( -\begin{bmatrix} 6 \\ 1 \\ 0 \end{bmatrix} \frac{e^{-6t}}{6} + \begin{bmatrix} -2 \\ 1 \\ 2 \end{bmatrix} t e^{-6t} \right) \\ + c_3 \left( -\begin{bmatrix} 12 \\ 1 \\ 0 \end{bmatrix} \frac{e^{-6t}}{36} - \begin{bmatrix} 6 \\ 1 \\ 0 \end{bmatrix} \frac{t e^{-6t}}{6} + \begin{bmatrix} -2 \\ 1 \\ 2 \end{bmatrix} \frac{t^2 e^{-6t}}{2} \right).$

**10.5.30.**  $\begin{bmatrix} -4-\lambda & 0 & -1 \\ -1 & -3-\lambda & -1 \\ 1 & 0 & -2-\lambda \end{bmatrix} = -(\lambda+3)^3$ . Hence  $\lambda_1 = 3$ . The eigenvectors satisfy the sys-

tem with augmented matrix  $\begin{bmatrix} -1 & 0 & -1 & : & 0 \\ -1 & 0 & -1 & : & 0 \\ 1 & 0 & 1 & : & 0 \end{bmatrix}$ , which is row equivalent to  $\begin{bmatrix} 1 & 0 & 1 & : & 0 \\ 0 & 0 & 0 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$ .



Hence  $x_1 = -x_3$  and  $x_2$  is arbitrary. Taking  $x_2 = 0$  and  $x_3 = 1$  yields  $\mathbf{y}_1 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} e^{-3t}$ . Taking

$x_2 = 1$  and  $x_3 = 0$  yields  $\mathbf{y}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} e^{-3t}$ . For a third solution we need constants  $\alpha$  and  $\beta$  and a vector

$\mathbf{u}$  such that  $\begin{bmatrix} -1 & 0 & -1 \\ -1 & 0 & -1 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \alpha \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} + \beta \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$ . The augmented matrix of this system is row

equivalent to  $\begin{bmatrix} 1 & 0 & 1 & : & \alpha \\ 0 & 0 & 0 & : & \alpha + \beta \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$ ; hence the system has a solution if  $\alpha = -\beta = 1$ , which yields

the eigenvector  $\mathbf{x}_3 = \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}$ . Taking  $u_1 = 1$  and  $u_2 = u_3 = 0$  yields the solution  $\mathbf{y}_3 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} e^{-3t} +$

$\begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} te^{-3t}$ . The general solution is  $\mathbf{y} = c_1 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} e^{-3t} + c_2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} e^{-3t} + c_3 \left( \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} e^{-3t} + \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} te^{-3t} \right)$

**10.5.32.**  $\begin{bmatrix} -3-\lambda & -1 & 0 \\ 1 & -1-\lambda & 0 \\ -1 & -1 & -2-\lambda \end{bmatrix} = -(\lambda+2)^3$ . Hence  $\lambda_1 = -2$ . The eigenvectors satisfy the

system with augmented matrix  $\begin{bmatrix} -1 & -1 & 0 & : & 0 \\ 1 & 1 & 0 & : & 0 \\ -1 & -1 & 0 & : & 0 \end{bmatrix}$ , which is row equivalent to  $\begin{bmatrix} 1 & 1 & 0 & : & 0 \\ 0 & 0 & 0 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$ .

Hence  $x_1 = -x_2$  and  $x_3$  is arbitrary. Taking  $x_2 = 1$  and  $x_3 = 0$  yields  $\mathbf{y}_1 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} e^{-2t}$ . Taking

$x_2 = 0$  and  $x_3 = 1$  yields  $\mathbf{y}_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} e^{-2t}$ . For a third solution we need constants  $\alpha$  and  $\beta$  and a vector

$\mathbf{u}$  such that  $\begin{bmatrix} -1 & -1 & 0 \\ 1 & 1 & 0 \\ -1 & -1 & 0 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \alpha \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$ . The augmented matrix of this system is row

equivalent to  $\begin{bmatrix} 1 & 1 & 0 & : & \alpha \\ 0 & 0 & 0 & : & \alpha + \beta \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$ ; hence the system has a solution if  $\alpha = -\beta = 1$ , which yields

the eigenvector  $\mathbf{x}_3 = \begin{bmatrix} -1 \\ 1 \\ -1 \end{bmatrix}$ . Taking  $u_1 = 1$  and  $u_2 = u_3 = 0$  yields the solution  $\mathbf{y}_3 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} e^{-2t} +$

$\begin{bmatrix} -1 \\ 1 \\ -1 \end{bmatrix} te^{-2t}$ . The general solution is  $\mathbf{y} = c_1 \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} e^{-2t} + c_2 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} e^{-2t} + c_3 \left( \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} e^{-2t} + \begin{bmatrix} -1 \\ 1 \\ -1 \end{bmatrix} te^{-2t} \right)$ .

**10.5.34.**

$$\begin{aligned}
\mathbf{y}'_3 - A\mathbf{y}_3 &= (\lambda_1 I - A)\mathbf{v}e^{\lambda_1 t} + (\lambda_1 I - A)\mathbf{u}te^{\lambda_1 t} + \mathbf{u}e^{\lambda_1 t} \\
&\quad + (\lambda_1 I - A)\mathbf{x}\frac{t^2 e^{\lambda_1 t}}{2} + \mathbf{x}te^{\lambda_1 t} \\
&= -\mathbf{u}e^{\lambda_1 t} - \mathbf{x}te^{\lambda_1 t} + \mathbf{u}e^{\lambda_1 t} + 0 + \mathbf{x}te^{\lambda_1 t} = 0.
\end{aligned}$$

Now suppose that  $c_1\mathbf{y}_1 + c_2\mathbf{y}_2 + c_3\mathbf{y}_3 = 0$ . Then

$$c_1\mathbf{x} + c_2(\mathbf{u} + t\mathbf{x}) + c_3\left(\mathbf{v} + t\mathbf{u} + \frac{t^2}{2}\mathbf{x}\right) = 0. \quad (\text{A})$$

Differentiating this twice yields  $c_3\mathbf{x} = 0$ , so  $c_3 = 0$  since  $\mathbf{x} \neq 0$ . Therefore, (A) reduces to (B)  $c_1\mathbf{x} + c_2(\mathbf{u} + t\mathbf{x}) = 0$ . Differentiating this yields  $c_2\mathbf{x} = 0$ , so  $c_2 = 0$  since  $\mathbf{x} \neq 0$ . Therefore, (B) reduces to  $c_1\mathbf{x} = 0$ , so  $c_1 = 0$  since  $\mathbf{x} \neq 0$ . Therefore,  $\mathbf{y}_1$ ,  $\mathbf{y}_2$ , and  $\mathbf{y}_3$  are linearly independent.

**10.6 Constant Coefficient Homogeneous Systems III**

**10.6.2.**  $\begin{vmatrix} -11-\lambda & 4 \\ -26 & 9-\lambda \end{vmatrix} = (\lambda+1)^2 + 4$ . The augmented matrix of  $(A - (-1+2i)I)\mathbf{x} = 0$  is  $\begin{bmatrix} -10-2i & 4 & \vdots & 0 \\ -26 & 10-2i & \vdots & 0 \end{bmatrix}$ , which is row equivalent to  $\begin{bmatrix} 1 & \frac{-5+i}{13} & \vdots & 0 \\ 0 & 0 & \vdots & 0 \end{bmatrix}$ . Therefore,  $x_1 = (5-i)x_2/13$ . Taking  $x_2 = 13$  yields the eigenvector  $\mathbf{x} = \begin{bmatrix} 5-i \\ 13 \end{bmatrix}$ . Taking real and imaginary parts of  $e^{-t}(\cos 2t + i \sin 2t)$   $\begin{bmatrix} 5-i \\ 13 \end{bmatrix}$  yields

$$\mathbf{y} = c_1 e^{-t} \begin{bmatrix} 5 \cos 2t + \sin 2t \\ 13 \cos 2t \end{bmatrix} + c_2 e^{-t} \begin{bmatrix} 5 \sin 2t - \cos 2t \\ 13 \sin 2t \end{bmatrix}.$$

**10.6.4.**  $\begin{vmatrix} 5-\lambda & -6 \\ 3 & -1-\lambda \end{vmatrix} = (\lambda-2)^2 + 9$ . Hence,  $\lambda = 2+3i$  is an eigenvalue of  $A$ . The associated eigenvectors satisfy  $(A - (2+3i)I)\mathbf{x} = 0$ . The augmented matrix of this system is  $\begin{bmatrix} 3-3i & -6 & \vdots & 0 \\ 3 & -3-3i & \vdots & 0 \end{bmatrix}$ , which is row equivalent to  $\begin{bmatrix} 1 & -1-i & \vdots & 0 \\ 0 & 0 & \vdots & 0 \end{bmatrix}$ . Therefore,  $x_1 = (1+i)x_2$ . Taking  $x_2 = 1$  yields  $x_1 = 1+i$ , so  $\mathbf{x} = \begin{bmatrix} 1+i \\ 1 \end{bmatrix}$  is an eigenvector. Taking real and imaginary parts of  $e^{2t}(\cos 3t + i \sin 3t)$   $\begin{bmatrix} 1+i \\ 1 \end{bmatrix}$  yields  $\mathbf{y} = c_1 e^{2t} \begin{bmatrix} \cos 3t - \sin 3t \\ \cos 3t \end{bmatrix} + c_2 e^{2t} \begin{bmatrix} \sin 3t + \cos 3t \\ \sin 3t \end{bmatrix}$ .

**10.6.6.**  $\begin{vmatrix} -3-\lambda & 3 & 1 \\ 1 & -5-\lambda & -3 \\ -3 & 7 & 3-\lambda \end{vmatrix} = -(\lambda+1)((\lambda+2)^2 + 4)$ . The augmented matrix of  $(A + I)\mathbf{x} = 0$  is  $\begin{bmatrix} -2 & 3 & 1 & \vdots & 0 \\ 1 & -4 & -3 & \vdots & 0 \\ -3 & 7 & 4 & \vdots & 0 \end{bmatrix}$ , which is row equivalent to  $\begin{bmatrix} 1 & 0 & 1 & \vdots & 0 \\ 0 & 1 & 1 & \vdots & 0 \\ 0 & 0 & 0 & \vdots & 0 \end{bmatrix}$ . Therefore,  $x_1 =$

$x_2 = -x_3$ . Taking  $x_3 = 1$  yields  $\mathbf{y}_1 = \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} e^{-t}$ . The augmented matrix of  $(A - (-2 + 2i)I)\mathbf{x} = 0$

is  $\begin{bmatrix} -1-2i & 3 & 1 & : & 0 \\ 1 & -3-2i & -3 & : & 0 \\ -3 & 7 & 5-2i & : & 0 \end{bmatrix}$ , which is row equivalent to  $\begin{bmatrix} 1 & 0 & -\frac{1+i}{2} & : & 0 \\ 0 & 1 & \frac{1-i}{2} & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$ . Therefore,  $x_1 =$

$\frac{(1+i)}{2}x_3$  and  $x_2 = -\frac{(1-i)}{2}x_3$ . Taking  $x_3 = 2$  yields the eigenvector  $\mathbf{x}_2 = \begin{bmatrix} 1+i \\ -1+i \\ 2 \end{bmatrix}$ . The

real and imaginary parts of  $e^{-2t}(\cos 2t + i \sin 2t) \begin{bmatrix} 1+i \\ -1+i \\ 2 \end{bmatrix}$  are  $\mathbf{y}_2 = e^{-2t} \begin{bmatrix} \cos 2t - \sin 2t \\ -\cos 2t - \sin 2t \\ 2 \cos 2t \end{bmatrix}$  and

$\mathbf{y}_3 = e^{-2t} \begin{bmatrix} \sin 2t + \cos 2t \\ -\sin 2t + \cos 2t \\ 2 \sin 2t \end{bmatrix}$ . Therefore,

$$\mathbf{y} = c_1 \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} e^{-t} + c_2 e^{-2t} \begin{bmatrix} \cos 2t - \sin 2t \\ -\cos 2t - \sin 2t \\ 2 \cos 2t \end{bmatrix} + c_3 e^{-2t} \begin{bmatrix} \sin 2t + \cos 2t \\ -\sin 2t + \cos 2t \\ 2 \sin 2t \end{bmatrix}.$$

**10.6.8.**  $\begin{vmatrix} -3-\lambda & 1 & -3 \\ 4 & -1-\lambda & 2 \\ 4 & -2 & 3-\lambda \end{vmatrix} = -(\lambda-1)((\lambda+1)^2+4)$ . The augmented matrix of  $(A -$

$I)\mathbf{x} = 0$  is  $\begin{bmatrix} -4 & 1 & -3 & : & 0 \\ 4 & -2 & 2 & : & 0 \\ 4 & -2 & 2 & : & 0 \end{bmatrix}$ , which is row equivalent to  $\begin{bmatrix} 1 & 0 & 1 & : & 0 \\ 0 & 1 & 1 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$ . Therefore,  $x_1 =$

$x_2 = -x_3$ . Taking  $x_3 = 1$  yields  $\mathbf{y}_1 = \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} e^t$ . The augmented matrix of  $(A - (-1 + 2i)I)\mathbf{x} = 0$  is

$\begin{bmatrix} -2-2i & 1 & -3 & : & 0 \\ 4 & -2i & 2 & : & 0 \\ 4 & -2 & 4-2i & : & 0 \end{bmatrix}$ , which is row equivalent to  $\begin{bmatrix} 1 & 0 & \frac{1-i}{2} & : & 0 \\ 0 & 1 & -1 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$ . Therefore,  $x_1 =$

$-\frac{1-i}{2}x_3$  and  $x_2 = x_3$ . Taking  $x_3 = 2$  yields the eigenvector  $\mathbf{x}_2 = \begin{bmatrix} -1+i \\ 2 \\ 2 \end{bmatrix}$ . The real and

imaginary parts of  $e^{-t}(\cos 2t + i \sin 2t) \begin{bmatrix} -1+i \\ 2 \\ 2 \end{bmatrix}$  are  $\mathbf{y}_2 = e^{-t} \begin{bmatrix} -\sin 2t - \cos 2t \\ 2 \cos 2t \\ 2 \cos 2t \end{bmatrix}$  and  $\mathbf{y}_3 =$

$e^{-t} \begin{bmatrix} \cos 2t - \sin 2t \\ 2 \sin 2t \\ 2 \sin 2t \end{bmatrix}$ . Therefore,

$$\mathbf{y} = c_1 \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} e^{-t} + c_2 e^{-t} \begin{bmatrix} -\sin 2t - \cos 2t \\ 2 \cos 2t \\ 2 \cos 2t \end{bmatrix} + c_3 e^{-t} \begin{bmatrix} \cos 2t - \sin 2t \\ 2 \sin 2t \\ 2 \sin 2t \end{bmatrix}.$$

**10.6.10.**  $\frac{1}{3} \begin{vmatrix} 7-3\lambda & -5 \\ 2 & 5-3\lambda \end{vmatrix} = (\lambda-2)^2 + 1$ . The augmented matrix of  $(A - (2+i)I)\mathbf{x} = 0$  is  $\frac{1}{3} \begin{bmatrix} 1-3i & -5 & \vdots & 0 \\ 2 & -1-3i & \vdots & 0 \end{bmatrix}$ , which is row equivalent to  $\begin{bmatrix} 1 & -\frac{1+3i}{2} & \vdots & 0 \\ 0 & 0 & \vdots & 0 \end{bmatrix}$ . Therefore,  $x_1 = \frac{1+3i}{2}x_2$ . Taking  $x_2 = 2$  yields the eigenvector  $\mathbf{x} = \begin{bmatrix} 1+3i \\ 2 \end{bmatrix}$ . Taking real and imaginary parts of  $e^{2t}(\cos t + i \sin t) \begin{bmatrix} 1+3i \\ 2 \end{bmatrix}$  yields

$$\mathbf{y} = c_1 e^{2t} \begin{bmatrix} \cos t - 3 \sin t \\ 2 \cos t \end{bmatrix} + c_2 e^{2t} \begin{bmatrix} \sin t + 3 \cos t \\ 2 \sin t \end{bmatrix}.$$

**10.6.12.**  $\begin{vmatrix} 34-\lambda & 52 \\ -20 & -30-\lambda \end{vmatrix} = (\lambda-2)^2 + 16$ . The augmented matrix of  $(A - (2+4i)I)\mathbf{x} = 0$  is  $\begin{bmatrix} 32-4i & 52 & \vdots & 0 \\ -20 & -32-4i & \vdots & 0 \end{bmatrix}$ , which is row equivalent to  $\begin{bmatrix} 1 & \frac{8+i}{5} & \vdots & 0 \\ 0 & 0 & \vdots & 0 \end{bmatrix}$ . Therefore,  $x_1 = -\frac{(8+i)}{5}x_2$ . Taking  $x_2 = 5$  yields the eigenvector  $\mathbf{x} = \begin{bmatrix} -8-i \\ 5 \end{bmatrix}$ . Taking real and imaginary parts of  $e^{2t}(\cos 4t + i \sin 4t) \begin{bmatrix} -8-i \\ 5 \end{bmatrix}$  yields  $\mathbf{y} = c_1 e^{2t} \begin{bmatrix} \sin 4t - 8 \cos 4t \\ 5 \cos 4t \end{bmatrix} + c_2 e^{2t} \begin{bmatrix} -\cos 4t - 8 \sin 4t \\ 5 \sin 4t \end{bmatrix}$ .

**10.6.14.**  $\begin{vmatrix} 3-\lambda & -4 & -2 \\ -5 & 7-\lambda & -8 \\ -10 & 13 & -8-\lambda \end{vmatrix} = -(\lambda+2)((\lambda-2)^2 + 9)$ . The augmented matrix of  $(A + 2I)\mathbf{x} = 0$  is  $\begin{bmatrix} 5 & -4 & -2 & \vdots & 0 \\ -5 & 9 & -8 & \vdots & 0 \\ -10 & 13 & -6 & \vdots & 0 \end{bmatrix}$ , which is row equivalent to  $\begin{bmatrix} 1 & 0 & -2 & \vdots & 0 \\ 0 & 1 & -2 & \vdots & 0 \\ 0 & 0 & 0 & \vdots & 0 \end{bmatrix}$ . Therefore,  $x_1 = x_2 = 2x_3$ . Taking  $x_3 = 1$  yields  $\mathbf{y}_1 = \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix} e^{-2t}$ . The augmented matrix of  $(A - (2+3i)I)\mathbf{x} = 0$  is

$\begin{bmatrix} 1-3i & -4 & -2 & \vdots & 0 \\ -5 & 5-3i & -8 & \vdots & 0 \\ -10 & 13 & -10-3i & \vdots & 0 \end{bmatrix}$ , which is row equivalent to  $\begin{bmatrix} 1 & 0 & 1-i & \vdots & 0 \\ 0 & 1 & -i & \vdots & 0 \\ 0 & 0 & 0 & \vdots & 0 \end{bmatrix}$ . Therefore,  $x_1 =$

$-(1-i)x_3$  and  $x_2 = ix_3$ . Taking  $x_3 = 1$  yields the eigenvector  $\mathbf{x}_2 = \begin{bmatrix} -1+i \\ i \\ 1 \end{bmatrix}$ . The real

and imaginary parts of  $e^{2t}(\cos 3t + i \sin 3t) \begin{bmatrix} -1+i \\ i \\ 1 \end{bmatrix}$  are  $\mathbf{y}_2 = e^{2t} \begin{bmatrix} -\cos 3t - \sin 3t \\ -\sin 3t \\ \cos 3t \end{bmatrix}$  and

$\mathbf{y}_3 = c_3 e^{2t} \begin{bmatrix} -\sin 3t + \cos 3t \\ \cos 3t \\ \sin 3t \end{bmatrix}$ . Therefore,

$$\mathbf{y} = c_1 \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix} e^{-2t} + c_2 e^{2t} \begin{bmatrix} -\cos 3t - \sin 3t \\ -\sin 3t \\ \cos 3t \end{bmatrix} + c_3 e^{2t} \begin{bmatrix} -\sin 3t + \cos 3t \\ \cos 3t \\ \sin 3t \end{bmatrix}.$$

**10.6.16.**  $\begin{vmatrix} 1-\lambda & 2 & -2 \\ 0 & 2-\lambda & -1 \\ 1 & 0 & -\lambda \end{vmatrix} = -(\lambda-2)((\lambda-1)^2+1)$ . The augmented matrix of  $(A-I)\mathbf{x} = 0$

is  $\begin{bmatrix} 0 & 2 & -2 & : & 0 \\ 0 & 1 & -1 & : & 0 \\ 1 & 0 & -1 & : & 0 \end{bmatrix}$ , which is row equivalent to  $\begin{bmatrix} 1 & 0 & -1 & : & 0 \\ 0 & 1 & -1 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$ . Therefore,  $x_1 = x_2 = 1$ .

Taking  $x_3 = 1$  yields  $\mathbf{y}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} e^t$ . The augmented matrix of  $(A - (1+i)I)\mathbf{x} = 0$  is  $\begin{bmatrix} -i & 2 & -2 & : & 0 \\ 0 & 1-i & -1 & : & 0 \\ 1 & 0 & -1-i & : & 0 \end{bmatrix}$ ,

which is row equivalent to  $\begin{bmatrix} 1 & 0 & -1-i & : & 0 \\ 0 & 1 & -\frac{1+i}{2} & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$ . Therefore,  $x_1 = (1+i)x_3$  and  $x_2 = \frac{(1+i)}{2}x_3$ .

Taking  $x_3 = 2$  yields the eigenvector  $\mathbf{x}_2 = \begin{bmatrix} 2+2i \\ 1+i \\ 2 \end{bmatrix}$ . The real and imaginary parts of  $e^{4t}(\cos t +$

$i \sin t) \begin{bmatrix} 2+2i \\ 1+i \\ 2 \end{bmatrix}$  are  $\mathbf{y}_2 = e^t \begin{bmatrix} 2 \cos t - 2 \sin t \\ \cos t - \sin t \\ 2 \cos t \end{bmatrix}$  and  $\mathbf{y}_3 = c_3 e^t \begin{bmatrix} 2 \sin t + 2 \cos t \\ \cos t + \sin t \\ 2 \sin t \end{bmatrix}$ . Therefore,

$$\mathbf{y} = c_1 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} e^t + c_2 e^t \begin{bmatrix} 2 \cos t - 2 \sin t \\ \cos t - \sin t \\ 2 \cos t \end{bmatrix} + c_3 e^t \begin{bmatrix} 2 \sin t + 2 \cos t \\ \cos t + \sin t \\ 2 \sin t \end{bmatrix}.$$

**10.6.18.**  $\begin{vmatrix} 7-\lambda & 15 \\ -3 & 1-\lambda \end{vmatrix} = (\lambda-4)^2 + 36$ . The augmented matrix of  $(A - (4+6i)I)\mathbf{x} = 0$  is  $\begin{bmatrix} 3-6i & 15 & : & 0 \\ -3 & -3-6i & : & 0 \end{bmatrix}$ , which is row equivalent to  $\begin{bmatrix} 1 & 1+2i & : & 0 \\ 0 & 0 & : & 0 \end{bmatrix}$ . Therefore,  $x_1 =$

$-(1+2i)x_2$ . Taking  $x_2 = 1$  yields the eigenvector  $\mathbf{x} = \begin{bmatrix} -1-2i \\ 1 \end{bmatrix}$ . Taking real and imaginary parts

of  $e^{4t}(\cos 6t + i \sin 6t) \begin{bmatrix} -1-2i \\ 1 \end{bmatrix}$  yields  $\mathbf{y} = c_1 e^{4t} \begin{bmatrix} 2 \sin 6t - \cos 6t \\ \cos 6t \end{bmatrix} + c_2 e^{4t} \begin{bmatrix} -2 \cos 6t - \sin 6t \\ \sin 6t \end{bmatrix}$ .

Now  $\mathbf{y}(0) = \begin{bmatrix} 5 \\ 1 \end{bmatrix} \Rightarrow \begin{bmatrix} -1 & -2 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 5 \\ 1 \end{bmatrix}$ , so  $c_1 = 1$ ,  $c_2 = -3$ , and  $\mathbf{y} = e^{4t} \begin{bmatrix} 5 \cos 6t + 5 \sin 6t \\ \cos 6t - 3 \sin 6t \end{bmatrix}$ .

**10.6.20.**  $\frac{1}{6} \begin{vmatrix} 4-6\lambda & -2 \\ 5 & 2-6\lambda \end{vmatrix} = \left(\lambda - \frac{1}{2}\right)^2 + \frac{1}{4}$ . The augmented matrix of  $\left(A - \frac{1+i}{2}I\right)\mathbf{x} = 0$

is  $\frac{1}{6} \begin{bmatrix} 1-3i & -2 & : & 0 \\ 5 & -1-3i & : & 0 \end{bmatrix}$ , which is row equivalent to  $\begin{bmatrix} 1 & -\frac{1+3i}{5} & : & 0 \\ 0 & 0 & : & 0 \end{bmatrix}$ . Therefore,  $x_1 =$

$\frac{1+3i}{5}x_2$ . Taking  $x_2 = 5$  yields the eigenvector  $\mathbf{x} = \begin{bmatrix} 1+3i \\ 5 \end{bmatrix}$ . Taking real and imaginary parts of

$e^{t/2}(\cos t/2 + i \sin t/2) \begin{bmatrix} 1+3i \\ 5 \end{bmatrix}$  yields  $\mathbf{y} = c_1 e^{t/2} \begin{bmatrix} \cos t/2 - 3 \sin t/2 \\ 5 \cos t/2 \end{bmatrix} + c_2 e^{t/2} \begin{bmatrix} \sin t/2 + 3 \cos t/2 \\ 5 \sin t/2 \end{bmatrix}$ .

Now  $\mathbf{y}(0) = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 3 \\ 5 & 0 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ , so  $c_1 = -\frac{1}{5}$ ,  $c_2 = \frac{2}{5}$ , and  $\mathbf{y} = e^{t/2} \begin{bmatrix} \cos(t/2) + \sin(t/2) \\ -\cos(t/2) + 2 \sin(t/2) \end{bmatrix}$ .

**10.6.22.**  $\begin{vmatrix} 4-\lambda & 4 & 0 \\ 8 & 10-\lambda & -20 \\ 2 & 3 & -2-\lambda \end{vmatrix} = -(\lambda-8)((\lambda-2)^2+4)$ . The augmented matrix of  $(A -$

$8I)\mathbf{x} = 0$  is  $\begin{bmatrix} 0 & 4 & 0 & : & 0 \\ 8 & 6 & -20 & : & 0 \\ 2 & 3 & -6 & : & 0 \end{bmatrix}$ , which is row equivalent to  $\begin{bmatrix} 1 & 0 & -2 & : & 0 \\ 0 & 1 & -2 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$ . Therefore,  $x_1 =$

$x_2 = 2x_3$ . Taking  $x_3 = 2$  yields  $\mathbf{y}_1 = \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix} e^{8t}$ . The augmented matrix of  $(A - (2 + 2i)I)\mathbf{x} = 0$  is

$\begin{bmatrix} 2-2i & 4 & 0 & : & 0 \\ 8 & 8-2i & -20 & : & 0 \\ 2 & 3 & -4-2i & : & 0 \end{bmatrix}$ , which is row equivalent to  $\begin{bmatrix} 1 & 0 & -2+2i & : & 0 \\ 0 & 1 & -2i & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$ . Therefore,  $x_1 =$

$(2-2i)x_3$  and  $x_2 = 2ix_3$ . Taking  $x_3 = 1$  yields the eigenvector  $\mathbf{x}_2 = \begin{bmatrix} 2-2i \\ 2i \\ 1 \end{bmatrix}$ . The real and

imaginary parts of  $e^{2t}(\cos 2t + i \sin 2t) \begin{bmatrix} 2-2i \\ 2i \\ 1 \end{bmatrix}$  are  $\mathbf{y}_2 = e^{2t} \begin{bmatrix} 2 \cos 2t + 2 \sin 2t \\ -2 \sin 2t \\ 2 \cos 2t \end{bmatrix}$  and  $\mathbf{y}_3 =$

$e^{2t} \begin{bmatrix} 2 \sin 2t + 2 \cos 2t \\ 2 \cos 2t \\ \sin 2t \end{bmatrix}$ , so the general solution is  $\mathbf{y} = c_1 \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix} e^{8t} + c_2 e^{2t} \begin{bmatrix} 2 \cos 2t + 2 \sin 2t \\ -2 \sin 2t \\ 2 \cos 2t \end{bmatrix} +$

$c_3 e^{2t} \begin{bmatrix} 2 \sin 2t + 2 \cos 2t \\ 2 \cos 2t \\ \sin 2t \end{bmatrix}$ . Now  $\mathbf{y}(0) = \begin{bmatrix} 8 \\ 6 \\ 5 \end{bmatrix} \Rightarrow \begin{bmatrix} 2 & 2 & -2 \\ 2 & 0 & 2 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} 8 \\ 6 \\ 5 \end{bmatrix}$ , so  $c_1 = 2$ ,  $c_2 = 3$ ,

$c_3 = 1$ , and  $\mathbf{y} = \begin{bmatrix} 4 \\ 4 \\ 2 \end{bmatrix} e^{8t} + e^{2t} \begin{bmatrix} 4 \cos 2t + 8 \sin 2t \\ -6 \sin 2t + 2 \cos 2t \\ 3 \cos 2t + \sin 2t \end{bmatrix}$ .

**10.6.24.**  $\begin{vmatrix} 4-\lambda & -4 & 4 \\ -10 & 3-\lambda & 15 \\ 2 & -3 & 1-\lambda \end{vmatrix} = -(\lambda-8)(\lambda^2+16)$ . The augmented matrix of  $(A-8I)\mathbf{x} = 0$

is  $\begin{bmatrix} -4 & -4 & 4 & : & 0 \\ -10 & -5 & 15 & : & 0 \\ 2 & -3 & -7 & : & 0 \end{bmatrix}$ , which is row equivalent to  $\begin{bmatrix} 1 & 0 & -2 & : & 0 \\ 0 & 1 & 1 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$ . Therefore,  $x_1 = 2x_3$

and  $x_2 = -x_3$ . Taking  $x_3 = 1$  yields  $\mathbf{y}_1 = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix} e^{8t}$ . The augmented matrix of  $(A-4iI)\mathbf{x} = 0$  is

$\begin{bmatrix} 4-4i & -4 & 4 & : & 0 \\ -10 & 3-4i & 15 & : & 0 \\ 2 & -3 & 1-4i & : & 0 \end{bmatrix}$ , which is row equivalent to  $\begin{bmatrix} 1 & 0 & -1+i & : & 0 \\ 0 & 1 & -1+2i & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$ . Therefore,  $x_1 =$

$(1-i)x_3$  and  $x_2 = (1-2i)x_3$ . Taking  $x_3 = 1$  yields the eigenvector  $\mathbf{x}_2 = \begin{bmatrix} 1-i \\ 1-2i \\ 1 \end{bmatrix}$ . The

real and imaginary parts of  $(\cos 4t + i \sin 4t) \begin{bmatrix} 1-i \\ 1-2i \\ 1 \end{bmatrix}$  are  $\mathbf{y}_2 = \begin{bmatrix} \cos 4t + \sin 4t \\ \cos 4t + 2 \sin 4t \\ \cos 4t \end{bmatrix}$  and  $\mathbf{y}_3 =$

$\begin{bmatrix} \sin 4t - \cos 4t \\ \sin 4t - 2 \cos 4t \\ \sin 4t \end{bmatrix}$ , so the general solution is  $\mathbf{y} = c_1 \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix} e^{8t} + c_2 \begin{bmatrix} \cos 4t + \sin 4t \\ \cos 4t + 2 \sin 4t \\ \cos 4t \end{bmatrix} + c_3 \begin{bmatrix} \sin 4t - \cos 4t \\ \sin 4t - 2 \cos 4t \\ \sin 4t \end{bmatrix}$ .

Now  $\mathbf{y}(0) = \begin{bmatrix} 16 \\ 14 \\ 6 \end{bmatrix} \Rightarrow \begin{bmatrix} 2 & 1 & -1 \\ -1 & 1 & -2 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} 16 \\ 14 \\ 6 \end{bmatrix}$ , so  $c_1 = 3$ ,  $c_2 = 3$ ,  $c_3 = -7$ , and  $\mathbf{y} = \begin{bmatrix} 6 \\ -3 \\ 3 \end{bmatrix} e^{8t} + \begin{bmatrix} 10 \cos 4t - 4 \sin 4t \\ 17 \cos 4t - \sin 4t \\ 3 \cos 4t - 7 \sin 4t \end{bmatrix}$

**10.6.28. (a)** From the quadratic formula the roots are

$$\begin{aligned} k_1 &= \frac{\|\mathbf{u}\|^2 - \|\mathbf{v}\|^2 + \sqrt{(\|\mathbf{u}\|^2 - \|\mathbf{v}\|^2)^2 + 4(\mathbf{u}, \mathbf{v})^2}}{2(\mathbf{u}, \mathbf{v})} \\ k_2 &= \frac{\|\mathbf{u}\|^2 - \|\mathbf{v}\|^2 - \sqrt{(\|\mathbf{u}\|^2 - \|\mathbf{v}\|^2)^2 + 4(\mathbf{u}, \mathbf{v})^2}}{2(\mathbf{u}, \mathbf{v})}. \end{aligned}$$

Clearly  $k_1 > 0$  and  $k_2 < 0$ . Moreover,

$$k_1 k_2 = \frac{(\|\mathbf{u}\|^2 - \|\mathbf{v}\|^2)^2 - [(\|\mathbf{u}\|^2 - \|\mathbf{v}\|^2)^2 + 4(\mathbf{u}, \mathbf{v})^2]}{4(\mathbf{u}, \mathbf{v})^2} = -1.$$

**(b)** Since  $k_2 = -1/k_1$ ,

$$\begin{aligned} \mathbf{u}_1^{(2)} &= \mathbf{u} - k_2 \mathbf{v} = \mathbf{u} + \frac{1}{k_1} \mathbf{v} = \frac{1}{k_1} (\mathbf{v} + k_1 \mathbf{u}) = \frac{1}{k_1} \mathbf{v}_1^{(1)} \\ \mathbf{v}_1^{(2)} &= \mathbf{v} + k_2 \mathbf{u} = \mathbf{v} - \frac{1}{k_1} \mathbf{u} = -\frac{1}{k_1} (\mathbf{u} - k_1 \mathbf{v}) = -\frac{1}{k_1} \mathbf{u}_1^{(1)}. \end{aligned}$$

**10.6.30.**  $\begin{vmatrix} -15 - \lambda & 10 \\ -25 & 15 - \lambda \end{vmatrix} = \lambda^2 + 25$ . The augmented matrix of  $(A - 5iI)\mathbf{x} = 0$  is  $\begin{bmatrix} -15 - 5i & 10 & : & 0 \\ -25 & 15 - 5i & : & 0 \end{bmatrix}$ ,

which is row equivalent to  $\begin{bmatrix} 1 & \frac{-3+i}{5} & : & 0 \\ 0 & 0 & : & 0 \end{bmatrix}$ . Therefore,  $x_1 = \frac{(3-i)}{5} x_2$ . Taking  $x_2 = 5$  yields the eigenvector  $\mathbf{x} = \begin{bmatrix} 3-i \\ 5 \end{bmatrix}$ , so  $\mathbf{u} = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$  and  $\mathbf{v} = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$ . The quadratic equation is  $-3k^2 - 33k + 3 = 0$ ,

with positive root  $k \approx .0902$ . Routine calculations yield  $\mathbf{U} \approx \begin{bmatrix} .5257 \\ .8507 \end{bmatrix}$ ,  $\mathbf{V} \approx \begin{bmatrix} -.8507 \\ .5257 \end{bmatrix}$ .

**10.6.32.**  $\begin{vmatrix} -3 - \lambda & -15 \\ 3 & 3 - \lambda \end{vmatrix} = \lambda^2 + 36$ . The augmented matrix of  $(A - 6iI)\mathbf{x} = 0$  is  $\begin{bmatrix} -3 - 6i & -15 & : & 0 \\ 3 & 3 - 6i & : & 0 \end{bmatrix}$ ,

which is row equivalent to  $\begin{bmatrix} 1 & 1-2i & : & 0 \\ 0 & 0 & : & 0 \end{bmatrix}$ . Therefore,  $x_1 = -(1-2i)x_2$ . Taking  $x_2 = 1$  yields the eigenvector  $\mathbf{x} = \begin{bmatrix} -1+2i \\ 1 \end{bmatrix}$ , so  $\mathbf{u} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$  and  $\mathbf{v} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$ . The quadratic equation is

$-2k^2 + 2k + 2 = 0$ , with positive root  $k \approx 1.6180$ . Routine calculations yield  $\mathbf{U} \approx \begin{bmatrix} -.9732 \\ .2298 \end{bmatrix}$ ,  $\mathbf{V} \approx \begin{bmatrix} .2298 \\ .9732 \end{bmatrix}$ .

**10.6.34.**  $\begin{vmatrix} 5 - \lambda & -12 \\ 6 & -7 - \lambda \end{vmatrix} = (\lambda + 1)^2 + 36$ . The augmented matrix of  $(A - (-1 + 6i)I)\mathbf{x} = 0$  is  $\begin{bmatrix} 6 - 6i & -12 & : & 0 \\ 6 & -6 - 6i & : & 0 \end{bmatrix}$ , which is row equivalent to  $\begin{bmatrix} 1 & -(1+i) & : & 0 \\ 0 & 0 & : & 0 \end{bmatrix}$ . Therefore,  $x_1 = (1+i)x_2$ . Taking  $x_2 = 1$  yields the eigenvector  $\mathbf{x} = \begin{bmatrix} 1+i \\ 1 \end{bmatrix}$ , so  $\mathbf{u} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$  and  $\mathbf{v} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ . The

quadratic equation is  $k^2 - k - 1 = 0$ , with positive root  $k \approx 1.6180$ . Routine calculations yield

$$\mathbf{U} \approx \begin{bmatrix} -.5257 \\ .8507 \end{bmatrix}, \mathbf{V} \approx \begin{bmatrix} .8507 \\ .5257 \end{bmatrix}.$$

**10.6.36.**  $\begin{vmatrix} -4-\lambda & 9 \\ -5 & 2-\lambda \end{vmatrix} = (\lambda+1)^2 + 36$ . The augmented matrix of  $(A - (-1+6i)I)\mathbf{x} = 0$  is  $\begin{bmatrix} -3-6i & 9 & \vdots & 0 \\ -5 & 3-6i & \vdots & 0 \end{bmatrix}$ , which is row equivalent to  $\begin{bmatrix} 1 & -\frac{3-6i}{5} & \vdots & 0 \\ 0 & 0 & \vdots & 0 \end{bmatrix}$ . Therefore,  $x_1 = \frac{3-6i}{5}x_2$ . Taking  $x_2 = 5$  yields the eigenvector  $\mathbf{x} = \begin{bmatrix} 3-6i \\ 5 \end{bmatrix}$ , so  $\mathbf{u} = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$  and  $\mathbf{v} = \begin{bmatrix} -6 \\ 0 \end{bmatrix}$ . The

quadratic equation is  $-18k^2 + 2k + 18 = 0$ , with positive root  $k \approx 1.0571$ . Routine calculations yield  $\mathbf{U} \approx \begin{bmatrix} .8817 \\ .4719 \end{bmatrix}, \mathbf{V} \approx \begin{bmatrix} -.4719 \\ .8817 \end{bmatrix}$ .

**10.6.38.**  $\begin{vmatrix} -1-\lambda & -5 \\ 20 & -1-\lambda \end{vmatrix} = (\lambda+1)^2 + 100$ . The augmented matrix of  $(A - (-1+10i)I)\mathbf{x} = 0$  is  $\begin{bmatrix} -10i & -5 & \vdots & 0 \\ 20 & -10i & \vdots & 0 \end{bmatrix}$ , which is row equivalent to  $\begin{bmatrix} 1 & -\frac{i}{2} & \vdots & 0 \\ 0 & 0 & \vdots & 0 \end{bmatrix}$ . Therefore,  $x_1 = \frac{i}{2}x_2$ . Taking  $x_2 = 2$  yields the eigenvector  $\mathbf{x} = \begin{bmatrix} i \\ 2 \end{bmatrix}$ , so  $\mathbf{u} = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$  and  $\mathbf{v} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ . Since  $(\mathbf{u}, \mathbf{v}) = 0$  we just normalize  $\mathbf{u}$  and  $\mathbf{v}$  to obtain  $\mathbf{U} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \mathbf{V} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ .

**10.6.40.**  $\begin{vmatrix} -7-\lambda & 6 \\ -12 & 5-\lambda \end{vmatrix} = (\lambda+1)^2 + 36$ . The augmented matrix of  $(A - (-1+6i)I)\mathbf{x} = 0$  is  $\begin{bmatrix} -6-6i & 6 & \vdots & 0 \\ -12 & 6-6i & \vdots & 0 \end{bmatrix}$ , which is row equivalent to  $\begin{bmatrix} 1 & -\frac{1-i}{2} & \vdots & 0 \\ 0 & 0 & \vdots & 0 \end{bmatrix}$ . Therefore,  $x_1 = \frac{1-i}{2}x_2$ . Taking  $x_2 = 2$  yields the eigenvector  $\mathbf{x} = \begin{bmatrix} 1-i \\ 2 \end{bmatrix}$ , so  $\mathbf{u} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$  and  $\mathbf{v} = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$ . The quadratic equation is  $-k^2 - 4k + 1 = 0$ , with positive root  $k \approx .2361$ . Routine calculations yield  $\mathbf{U} \approx \begin{bmatrix} .5257 \\ .8507 \end{bmatrix}, \mathbf{V} \approx \begin{bmatrix} -.8507 \\ .5257 \end{bmatrix}$ .

## 10.7 Variation of Parameters for Nonhomogeneous Linear Systems

$$\mathbf{10.7.2.} \quad Y = \begin{bmatrix} -3e^{-t} & -e^{-2t} \\ e^{-t} & 2e^{-2t} \end{bmatrix}; \mathbf{u}' = Y^{-1}\mathbf{f} = \frac{1}{5} \begin{bmatrix} -2e^t & -e^t \\ e^{2t} & 3e^{2t} \end{bmatrix} \begin{bmatrix} 50e^{3t} \\ 10e^{-3t} \end{bmatrix} = \begin{bmatrix} -20e^{4t} - 2e^{-2t} \\ 10e^{5t} + 6e^{-t} \end{bmatrix};$$

$$\mathbf{u} = \begin{bmatrix} e^{-2t} - 5e^{4t} \\ 2e^{5t} - 6e^{-t} \end{bmatrix}; \mathbf{y}_p = Y\mathbf{u} = \begin{bmatrix} 13e^{3t} + 3e^{-3t} \\ -e^{3t} - 11e^{-3t} \end{bmatrix}.$$

$$\mathbf{10.7.4.} \quad Y = \begin{bmatrix} -e^{2t} & -e^{-t} \\ 2e^{2t} & e^{-t} \end{bmatrix}; \mathbf{u}' = Y^{-1}\mathbf{f} = \begin{bmatrix} e^{-2t} & e^{-2t} \\ -2e^t & -e^t \end{bmatrix} \begin{bmatrix} 2 \\ -2e^t \end{bmatrix} = \begin{bmatrix} 2e^{-2t} - 2e^{-t} \\ 2e^{2t} - 4e^t \end{bmatrix}; \mathbf{u} = \begin{bmatrix} 2e^{-t} - e^{-2t} \\ e^{2t} - 4e^t \end{bmatrix}; \mathbf{y}_p = Y\mathbf{u} = \begin{bmatrix} 5 - 3e^t \\ 5e^t - 6 \end{bmatrix}.$$

$$\mathbf{10.7.6.} \quad Y = \begin{bmatrix} \sin t & -\cos t \\ \cos t & \sin t \end{bmatrix}; \mathbf{u}' = Y^{-1}\mathbf{f} = \begin{bmatrix} \sin t & \cos t \\ -\cos t & \sin t \end{bmatrix} \begin{bmatrix} 1 \\ t \end{bmatrix} = \begin{bmatrix} t \cos t + \sin t \\ t \sin t - \cos t \end{bmatrix}; \mathbf{u} =$$



$$\begin{bmatrix} t \sin t \\ -t \cos t \end{bmatrix}; \mathbf{y}_p = Y\mathbf{u} = \begin{bmatrix} t \\ 0 \end{bmatrix}.$$

$$\begin{aligned} 10.7.8. \quad Y &= \begin{bmatrix} e^{3t} & e^{2t} & e^{-t} \\ -e^{3t} & 0 & -3e^{-t} \\ e^{3t} & e^{2t} & 7e^{-t} \end{bmatrix}; \mathbf{u}' = Y^{-1}\mathbf{f} = \frac{1}{6} \begin{bmatrix} 3e^{-3t} & -6e^{-3t} & -3e^{-3t} \\ 4e^{-2t} & 6e^{-2t} & 2e^{-2t} \\ -e^t & 0 & e^t \end{bmatrix} \begin{bmatrix} 1 \\ e^t \\ e^t \end{bmatrix} = \\ \frac{1}{6} \begin{bmatrix} 3e^{-3t} - 9e^{-2t} \\ 8e^{-t} + 4e^{-2t} \\ e^{2t} - e^t \end{bmatrix}; \mathbf{u} &= \frac{1}{12} \begin{bmatrix} 9e^{-2t} - 2e^{-3t} \\ -16e^{-t} - 4e^{-2t} \\ e^{2t} - 2e^t \end{bmatrix}; \mathbf{y}_p = Y\mathbf{u} = -\frac{1}{6} \begin{bmatrix} 3e^t + 4 \\ 6e^t - 4 \\ 10 \end{bmatrix}. \end{aligned}$$

$$\begin{aligned} 10.7.10. \quad Y &= \begin{bmatrix} -e^t & e^{-t} & te^{-t} \\ e^t & -e^{-t} & 3e^{-t} - te^{-t} \\ e^t & e^{-t} & te^{-t} \end{bmatrix}; \mathbf{u}' = Y^{-1}\mathbf{f} = \frac{1}{18} \begin{bmatrix} -9e^{-t} & 0 & 9e^{-t} \\ 3e^t(3-2t) & -6te^t & 9e^t \\ 6e^t & 6e^t & 0 \end{bmatrix} \begin{bmatrix} e^t \\ e^t \\ e^t \end{bmatrix} = \\ \frac{1}{3} \begin{bmatrix} 0 \\ e^{2t}(3-2t) \\ 2e^{2t} \end{bmatrix}; \mathbf{u} &= \frac{1}{3} \begin{bmatrix} 0 \\ e^{2t}(2-t) \\ e^{2t} \end{bmatrix}; \mathbf{y}_p = Y\mathbf{u} = \frac{1}{3} \begin{bmatrix} 2e^t \\ e^t \\ 2e^t \end{bmatrix}. \end{aligned}$$

$$\begin{aligned} 10.7.12. \quad \mathbf{u}' &= Y^{-1}\mathbf{f} = \frac{1}{2t} \begin{bmatrix} e^{-t} & e^{-t} \\ e^t & -e^t \end{bmatrix} \begin{bmatrix} t \\ t^2 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} e^{-t}(t+1) \\ e^t(1-t) \end{bmatrix}; \mathbf{u} = \frac{1}{2} \begin{bmatrix} -e^{-t}(t+2) \\ e^t(2-t) \end{bmatrix}; \mathbf{y}_p = \\ Y\mathbf{u} &= \frac{t}{2} \begin{bmatrix} e^t & e^{-t} \\ e^t & -e^{-t} \end{bmatrix} \begin{bmatrix} -e^{-t}(t+2) \\ e^t(2-t) \end{bmatrix} = - \begin{bmatrix} t^2 \\ 2t \end{bmatrix}. \end{aligned}$$

$$\begin{aligned} 10.7.14. \quad \mathbf{u}' &= Y^{-1}\mathbf{f} = \frac{1}{3} \begin{bmatrix} 2 & -e^{-t} \\ -e^t & 2 \end{bmatrix} \begin{bmatrix} e^{2t} \\ e^{-2t} \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 2e^{2t} - e^{-3t} \\ 2e^{-2t} - e^{3t} \end{bmatrix}; \mathbf{u} = \frac{1}{9} \begin{bmatrix} 3e^{2t} + e^{-3t} \\ -e^{3t} - 3e^{-2t} \end{bmatrix}; \\ \mathbf{y}_p &= Y\mathbf{u} = \frac{1}{9} \begin{bmatrix} 2 & e^{-t} \\ e^t & 2 \end{bmatrix} \begin{bmatrix} 3e^{2t} + e^{-3t} \\ -e^{3t} - 3e^{-2t} \end{bmatrix} = \frac{1}{9} \begin{bmatrix} 5e^{2t} - e^{-3t} \\ e^{3t} - 5e^{-2t} \end{bmatrix}. \end{aligned}$$

$$\begin{aligned} 10.7.16. \quad \mathbf{u}' &= Y^{-1}\mathbf{f} = \frac{1}{t^2-1} \begin{bmatrix} t & -e^{-t} \\ -e^t & t \end{bmatrix} \begin{bmatrix} t^2-1 \\ t^2-1 \end{bmatrix} = \begin{bmatrix} t-e^{-t} \\ t-e^t \end{bmatrix}; \mathbf{u} = \frac{1}{2} \begin{bmatrix} 2e^{-t} + t^2 \\ t^2 - 2e^t \end{bmatrix}; \\ \mathbf{y}_p &= Y\mathbf{u} = \frac{1}{2} \begin{bmatrix} t & e^{-t} \\ e^t & t \end{bmatrix} \begin{bmatrix} 2e^{-t} + t^2 \\ t^2 - 2e^t \end{bmatrix} = \frac{1}{2} \begin{bmatrix} te^{-t}(t+2) + t^3 - 2 \\ te^t(t-2) + t^3 + 2 \end{bmatrix}. \end{aligned}$$

$$\begin{aligned} 10.7.18. \quad \mathbf{u}' &= Y^{-1}\mathbf{f} = \frac{1}{3} \begin{bmatrix} e^{-5t} & e^{-4t} & e^{-3t} \\ 2e^{-2t} & -e^{-t} & -1 \\ -e^{-2t} & 2e^{-t} & -1 \end{bmatrix} \begin{bmatrix} e^t \\ 0 \\ 0 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} e^{-4t} \\ 2e^{-t} \\ -e^{-t} \end{bmatrix}; \mathbf{u} = \frac{1}{12} \begin{bmatrix} -e^{-4t} \\ -8e^{-t} \\ 4e^{-t} \end{bmatrix}; \\ \mathbf{y}_p &= Y\mathbf{u} = \frac{1}{12} \begin{bmatrix} e^{5t} & e^{2t} & 0 \\ e^{4t} & 0 & e^t \\ e^{3t} & -1 & -1 \end{bmatrix} \begin{bmatrix} -e^{-4t} \\ -8e^{-t} \\ 4e^{-t} \end{bmatrix} = \frac{1}{4} \begin{bmatrix} -3e^t \\ 1 \\ e^{-t} \end{bmatrix}. \end{aligned}$$

$$\begin{aligned} 10.7.20. \quad \mathbf{u}' &= Y^{-1}\mathbf{f} = \frac{1}{2t} \begin{bmatrix} -e^{-2t} & te^{-t} & te^{-t} + e^{-2t} \\ 1 & -te^t & te^t - 1 \\ 2 & 0 & -2 \end{bmatrix} \begin{bmatrix} e^t \\ 0 \\ e^t \end{bmatrix} = \begin{bmatrix} 1 \\ e^{2t} \\ 0 \end{bmatrix}; \mathbf{u} = \frac{1}{4} \begin{bmatrix} 2t \\ e^{2t} \\ 0 \end{bmatrix}; \end{aligned}$$

$$\mathbf{y}_p = Y\mathbf{u} = Y = \frac{1}{4t} \begin{bmatrix} e^t & e^{-t} & t \\ e^t & -e^{-t} & e^{-t} \\ e^t & e^{-t} & 0 \end{bmatrix} \begin{bmatrix} 2t \\ e^{2t} \\ 0 \end{bmatrix} = \frac{e^t}{4t} \begin{bmatrix} 2t+1 \\ 2t-1 \\ 2t+1 \end{bmatrix}.$$

10.7.22. (c) If  $\mathbf{y}_p = Y\mathbf{u}$ , then  $\mathbf{y}_p' = Y'\mathbf{u} + Y\mathbf{u}' = AY\mathbf{u} + Y\mathbf{u}'$ , so (E)  $\mathbf{y}_p' = A\mathbf{y}_p + Y\mathbf{u}'$ . However, from the derivation of the method of variation of parameters in Section 9.4,  $Y\mathbf{u}' = \mathbf{f}$  as defined in the solution of (a). This and (E) imply the conclusion.

(d) Since  $Y\mathbf{u}' = \mathbf{f}$  with  $\mathbf{f}$  as defined in the solution of (a),  $u_1, u_2, \dots, u_n$  satisfy the conditions required in the derivation of the method of variation of parameters in Section 9.4; hence,  $\mathbf{y}_p = c_1\mathbf{y}_1 + c_2\mathbf{y}_2 + \dots + c_n\mathbf{y}_n$  is a particular solution of (A).

## Chapter 11

# Boundary Value Problems and Fourier Expansions

### 11.1 Eigenvalue Problems for $y'' + \lambda y = 0$

**11.1.2.** From Theorem 11.1.2 with  $L = \pi$ ,  $\lambda_n = n^2$ ,  $y_n = \sin nx$ ,  $n = 1, 2, 3, \dots$

**11.1.4.** From Theorem 11.1.4 with  $L = \pi$ ,  $\lambda_n = \frac{(2n-1)^2}{4}$ ,  $y_n = \sin \frac{(2n-1)x}{2}$ ,  $n = 1, 2, 3, \dots$ ,

**11.1.6.** From Theorem 11.1.6 with  $L = \pi$ ,  $\lambda_0 = 0$ ,  $y_0 = 1$ ,  $\lambda_n = n^2$ ,  $y_{1n} = \cos nx$ ,  $y_{2n} = \sin nx$ ,  $n = 1, 2, 3, \dots$

**11.1.8.** From Theorem 11.1.5 with  $L = 1$ ,  $\lambda_n = \frac{(2n-1)^2\pi^2}{4}$ ,  $y_n = \cos \frac{(2n-1)\pi x}{2}$ ,  $n = 1, 2, 3, \dots$

**11.1.10.** From Theorem 11.1.6 with  $L = 1$ ,  $\lambda_0 = 0$ ,  $y_0 = 1$ ,  $\lambda_n = n^2\pi^2$ ,  $y_{1n} = \cos n\pi x$ ,  $y_{2n} = \sin n\pi x$ ,  $n = 1, 2, 3, \dots$

**11.1.12.** From Theorem 11.1.6 with  $L = 2$ ,  $\lambda_0 = 0$ ,  $y_0 = 1$ ,  $\lambda_n = \frac{n^2\pi^2}{4}$ ,  $y_{1n} = \cos \frac{n\pi x}{2}$ ,  $y_{2n} = \sin \frac{n\pi x}{2}$ ,  $n = 1, 2, 3, \dots$

**11.1.14.** From Theorem 11.1.5 with  $L = 3$ ,  $\lambda_n = \frac{(2n-1)^2\pi^2}{36}$ ,  $y_n = \cos \frac{(2n-1)\pi x}{6}$ ,  $n = 1, 2, 3, \dots$

**11.1.16.** From Theorem 11.1.3 with  $L = 5$ ,  $\lambda_n = \frac{n^2\pi^2}{25}$ ,  $y_n = \cos \frac{n\pi x}{5}$ ,  $n = 1, 2, 3, \dots$

**11.1.18.** From Theorem 11.1.1, any eigenvalues of Problem 11.1.4 must be positive. If  $\lambda > 0$ , then every solution of  $y'' + \lambda y = 0$  is of the form  $y = c_1 \cos \sqrt{\lambda}x + c_2 \sin \sqrt{\lambda}x$  where  $c_1$  and  $c_2$  are constants. Therefore,  $y' = \sqrt{\lambda}(-c_1 \sin \sqrt{\lambda}x + c_2 \cos \sqrt{\lambda}x)$ . Since  $y'(0) = 0$ ,  $c_2 = 0$ . Therefore,  $y = c_1 \cos \sqrt{\lambda}x$ . Since  $y(L) = 0$ ,  $c_1 \cos \sqrt{\lambda}L = 0$ . To make  $c_1 \cos \sqrt{\lambda}L = 0$  with  $c_1 \neq 0$  we must choose  $\sqrt{\lambda} = \frac{(2n-1)\pi}{2L}$ , where  $n$  is a positive integer. Therefore,  $\lambda_n = \frac{(2n-1)^2\pi^2}{4L^2}$  is an eigenvalue and  $y_n = \cos \frac{(2n-1)\pi x}{2L}$  is an associated eigenfunction.

**11.1.20.** If  $r$  is a positive integer, then  $\int_{-L}^L \cos \frac{r\pi x}{L} dx = \frac{L}{r\pi} \sin \frac{r\pi x}{L} \Big|_{-L}^L = 0$ , so  $y_0 = 1$  is orthogonal

to all the other eigenfunctions. If  $m$  and  $n$  are distinct positive integers, then  $\int_0^L \cos \frac{m\pi x}{L} \cos \frac{n\pi x}{L} dx = \frac{1}{2} \int_{-L}^L \cos \frac{m\pi x}{L} \cos \frac{n\pi x}{L} dx = 0$ , from Example 11.1.4.

**11.1.22.** Let  $m$  and  $n$  be distinct positive integers. From the identity  $\cos A \cos B = \frac{1}{2}[\cos(A - B) + \cos(A + B)]$  with  $A = (2m - 1)\pi x/2L$  and  $B = (2n - 1)\pi x/2L$ ,

$$\int_0^L \cos \frac{(2m-1)\pi x}{2L} \cos \frac{(2n-1)\pi x}{2L} dx = \frac{1}{2} \int_0^L \left[ \cos \frac{(m-n)\pi x}{L} + \cos \frac{(m+n-1)\pi x}{L} \right] dx = 0.$$

**11.1.24.** If  $y = c_1 + c_2 x$ , then  $y'(0) = 0$  implies that  $c_2 = 0$ , so  $y = c_1$ . Now  $\int_0^L y(x) dx =$

$c_1 \int_0^L dx = c_1 L = 0$  only if  $c_1 = 0$ . Therefore, zero is not an eigenvalue.

If  $y = c_1 \cosh kx + c_2 \sinh kx$ , then  $y'(0) = 0$  implies that  $c_2 = 0$ , so  $y = c_1 \cosh kx$ . Now  $\int_0^L y(x) dx = c_1 \int_0^L \cosh kx dx = c_1 \frac{\sinh kL}{k} = 0$  with  $k > 0$  only if  $c_1 = 0$ . Therefore, there are no negative eigenvalues.

If  $y = c_1 \cos kx + c_2 \sin kx$ , then  $y'(0) = 0$  implies that  $c_2 = 0$ , so  $y = c_1 \cos kx$ . Now  $\int_0^L y(x) dx = c_1 \int_0^L \cos kx dx = c_2 \frac{\sin kL}{k} = 0$  if  $k = \frac{n\pi}{L}$ , where  $n$  is a positive integer. Therefore,  $\lambda_n = \frac{n^2\pi^2}{L^2}$  and  $y_n = \cos \frac{n\pi x}{L}$ ,  $n = 1, 2, 3, \dots$

**11.1.26.** If  $y = c_1 + c_2(x - L)$ , then  $y'(L) = 0$  implies that  $c_2 = 0$ , so  $y = c_1$ . Now  $\int_0^L y(x) dx =$

$c_1 \int_0^L dx = c_1 L = 0$  only if  $c_1 = 0$ . Therefore, zero is not an eigenvalue.

If  $y = c_1 \cosh k(x - L) + c_2 \sinh k(x - L)$ , then  $y'(L) = 0$  implies that  $c_2 = 0$ , so  $y = c_1 \cosh k(x - L)$ . Now  $\int_0^L y(x) dx = c_1 \int_0^L \cosh k(x - L) dx = c_1 \frac{\sinh kL}{k} = 0$  with  $k > 0$  only if  $c_1 = 0$ . Therefore, there are no negative eigenvalues.

If  $y = c_1 \cos k(x - L) + c_2 \sin k(x - L)$ , then  $y'(L) = 0$  implies that  $c_2 = 0$ , so  $y = c_1 \cos k(x - L)$ . Now  $\int_0^L y(x) dx = c_1 \int_0^L \cos k(x - L) dx = c_2 \frac{\sin kL}{k} = 0$  if  $k = \frac{n\pi}{L}$ , where  $n$  is a positive integer. Therefore,  $\lambda_n = \frac{n^2\pi^2}{L^2}$  and  $y_n = \cos \frac{n\pi(x - L)}{L}$ , or, equivalently,  $y_n = \cos \frac{n\pi x}{L}$ ,  $n = 1, 2, 3, \dots$

## 11.2 Fourier Expansions I

### 11.2.2.

$$\begin{aligned}
 a_0 &= \frac{1}{2} \int_{-1}^1 (2-x) dx = \int_0^1 2 dx = 2; \\
 a_n &= \int_{-1}^1 (2-x) \cos n\pi x dx = 4 \int_0^1 \cos n\pi x dx = \frac{4}{n\pi} \sin n\pi x \Big|_0^1 = 0; \\
 b_n &= \int_{-1}^1 (2-x) \sin n\pi x dx = -2 \int_0^1 x \sin n\pi x dx \\
 &= \frac{2}{n\pi} \left[ x \cos n\pi x \Big|_0^1 - \int_0^1 \cos n\pi x dx \right] \\
 &= \frac{2}{n\pi} \left[ \cos n\pi - \frac{1}{n\pi} \sin n\pi x \Big|_0^1 \right] = (-1)^n \frac{2}{n\pi};
 \end{aligned}$$

$F(x) = 2 + \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^n}{n} \sin n\pi x$ . From Theorem 11.2.4,

$$F(x) = \begin{cases} 2, & x = -1, \\ 2-x, & -1 < x < 1, \\ 2, & x = 1. \end{cases}$$

**11.2.4.** Since  $f$  is even,  $b_n = 0$  for  $n \geq 1$ ;  $a_0 = \int_0^1 (1-3x^2) dx = (x-x^3) \Big|_0^1 = 0$ ; if  $n \geq 1$ , then

$$\begin{aligned}
 a_n &= 2 \int_0^1 (1-3x^2) \cos n\pi x dx = \frac{2}{n\pi} \left[ (1-3x^2) \sin n\pi x \Big|_0^1 + 6 \int_0^1 x \sin n\pi x dx \right] \\
 &= -\frac{12}{n^2\pi^2} \left[ x \cos n\pi x \Big|_0^1 - \int_0^1 \cos n\pi x dx \right] \\
 &= -\frac{12}{n^2\pi^2} \left[ \cos n\pi - \frac{1}{n\pi} \sin n\pi x \Big|_0^1 \right] = (-1)^{n+1} \frac{12}{n^2\pi^2};
 \end{aligned}$$

$F(x) = -\frac{12}{\pi^2} \sum_{n=1}^{\infty} (-1)^n \frac{\cos n\pi x}{n^2}$ . From Theorem 11.2.4,  $F(x) = 1-3x^2$ ,  $-1 \leq x \leq 1$ .

**11.2.6.** Since  $f$  is odd,  $a_n = 0$  if  $n \geq 0$ ;

$$\begin{aligned}
 b_1 &= \frac{2}{\pi} \int_0^{\pi} x \cos x \sin x dx = \frac{1}{\pi} \int_0^{\pi} x \sin 2x dx \\
 &= -\frac{1}{2\pi} \left[ x \cos 2x \Big|_0^{\pi} - \int_0^{\pi} \cos 2x dx \right] = -\frac{1}{2\pi} \left[ \pi - \frac{\sin 2x}{2} \Big|_0^{\pi} \right] = -\frac{1}{2}.
 \end{aligned}$$

if  $n \geq 2$ , then

$$\begin{aligned} b_n &= \frac{2}{\pi} \int_0^\pi x \cos x \sin nx \, dx = \frac{1}{\pi} \int_0^\pi x [\sin(n+1)x + \sin(n-1)x] \, dx \\ &= -\frac{1}{\pi} \left[ x \left[ \frac{\cos(n+1)x}{n+1} + \frac{\cos(n-1)x}{n-1} \right] \right]_0^\pi - \int_0^\pi \left[ \frac{\cos(n+1)x}{n+1} + \frac{\cos(n-1)x}{n-1} \right] dx \\ &= (-1)^n \left[ \frac{1}{n+1} + \frac{1}{n-1} \right] + \frac{1}{\pi} \left[ \frac{\sin(n+1)x}{(n+1)^2} + \frac{\sin(n-1)x}{(n-1)^2} \right] \bigg|_0^\pi = (-1)^n \frac{2n}{n^2-1}; \end{aligned}$$

$F(x) = -\frac{1}{2} \sin x + 2 \sum_{n=2}^{\infty} (-1)^n \frac{n}{n^2-1} \sin nx$ . From Theorem 11.2.4,  $F(x) = x \cos x$ ,  $-\pi \leq x \leq \pi$ .

**11.2.8.** Since  $f$  is even,  $b_n = 0$  if  $n \geq 1$ ;  $a_0 = \frac{1}{\pi} \int_0^\pi x \sin x \, dx = -\frac{1}{\pi} \left[ x \cos x \right]_0^\pi - \int_0^\pi \cos x \, dx = 1 + \frac{\sin x}{\pi} \bigg|_0^\pi = 1$ ;  $a_1 = \frac{2}{\pi} \int_0^\pi x \sin x \cos x \, dx = \frac{1}{\pi} \int_0^\pi x \sin 2x \, dx = -\frac{1}{2\pi} \left[ x \cos 2x \right]_0^\pi - \int_0^\pi \cos 2x \, dx = -\frac{1}{2} + \frac{\sin 2x}{4\pi} \bigg|_0^\pi = -\frac{1}{2}$ ; if  $n \geq 2$ , then

$$\begin{aligned} a_n &= \frac{2}{\pi} \int_0^\pi x \sin x \cos nx \, dx = \frac{1}{\pi} \int_0^\pi x [\sin(n+1)x - \sin(n-1)x] \, dx \\ &= \frac{1}{\pi} \left[ x \left[ \frac{\cos(n-1)x}{n-1} - \frac{\cos(n+1)x}{n+1} \right] \right]_0^\pi - \int_0^\pi \left[ \frac{\cos(n-1)x}{n-1} - \frac{\cos(n+1)x}{n+1} \right] dx \\ &= (-1)^{n+1} \left[ \frac{1}{n-1} - \frac{1}{n+1} \right] - \frac{1}{\pi} \left[ \frac{\sin(n-1)x}{(n-1)^2} - \frac{\sin(n+1)x}{(n+1)^2} \right] \bigg|_0^\pi = (-1)^{n+1} \frac{2}{n^2-1}; \end{aligned}$$

$F(x) = 1 - \frac{1}{2} \cos x - 2 \sum_{n=2}^{\infty} \frac{(-1)^n}{n^2-1} \cos nx$ . From Theorem 11.2.4,  $F(x) = x \sin x$ ,  $-\pi \leq x \leq \pi$ .

**11.2.10.** Since  $f$  is even,  $b_n = 0$  if  $n \geq 0$ ;  $a_0 = \int_0^{1/2} \cos \pi x \, dx = \frac{\sin \pi x}{\pi} \bigg|_0^{1/2} = \frac{1}{\pi}$ ;  $a_1 = 2 \int_0^{1/2} \cos^2 \pi x \, dx = \int_0^{1/2} (1 + \cos 2\pi x) \, dx = \frac{1}{2} + \frac{\sin 2\pi x}{2\pi} \bigg|_0^{1/2} = \frac{1}{2}$ ; if  $n \geq 2$ , then

$$\begin{aligned} a_n &= 2 \int_0^{1/2} \cos \pi x \cos n\pi x \, dx = \int_0^{1/2} [\cos(n+1)x + \cos(n-1)x] \, dx \\ &= \frac{1}{\pi} \left[ \frac{\sin(n+1)\pi x}{n+1} + \frac{\sin(n-1)\pi x}{n-1} \right] \bigg|_0^{1/2} = \frac{1}{\pi} \left[ \frac{1}{n+1} - \frac{1}{n-1} \right] \cos \frac{n\pi}{2} \\ &= -\frac{2}{(n^2-1)\pi} \cos \frac{n\pi}{2} = \begin{cases} (-1)^{m+1} \frac{2}{(4m^2-1)\pi} & \text{if } n = 2m, \\ 0 & \text{if } n = 2m+1; \end{cases} \end{aligned}$$

$F(x) = \frac{1}{\pi} + \frac{1}{2} \cos \pi x - \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^n}{4n^2 - 1} \cos 2n\pi x$ . From Theorem 11.2.4,  $F(x) = f(x)$ ,  $-1 \leq x \leq 1$ .

**11.2.12.** Since  $f$  is odd,  $a_n = 0$  if  $n \geq 0$ ;  $b_1 = 2 \int_0^{1/2} \sin^2 2\pi x \, dx = \int_0^{1/2} (1 - \cos 4\pi x) \, dx =$

$\frac{1}{2} - \frac{\sin 4\pi x}{4\pi} \Big|_0^{1/2} = \frac{1}{2}$ ; if  $n \geq 2$ , then

$$\begin{aligned} b_n &= 2 \int_0^{1/2} \sin \pi x \sin n\pi x \, dx = \int_0^{1/2} [\cos(n-1)\pi x - \cos(n+1)\pi x] \, dx \\ &= \frac{1}{\pi} \left[ \frac{\sin(n-1)\pi x}{n-1} - \frac{\sin(n+1)\pi x}{n+1} \right] \Big|_0^{1/2} \\ &= -\frac{2n}{(n^2-1)\pi} \cos \frac{n\pi}{2} = \begin{cases} (-1)^{m+1} \frac{4m}{4m^2-1} & \text{if } n = 2m, \\ 0 & \text{if } n = 2m+1; \end{cases} \end{aligned}$$

$F(x) = \frac{1}{2} \sin \pi x - \frac{4}{\pi} \sum_{n=1}^{\infty} (-1)^n \frac{n}{4n^2 - 1} \sin 2n\pi x$ . From Theorem 11.2.4,

$$F(x) = \begin{cases} 0, & -1 \leq x < \frac{1}{2}, \\ -\frac{1}{2}, & x = -\frac{1}{2}, \\ \sin \pi x, & -\frac{1}{2} < x < \frac{1}{2}, \\ \frac{1}{2}, & x = \frac{1}{2}, \\ 0, & \frac{1}{2} < x \leq 1. \end{cases}$$

**11.2.14.** Since  $f$  is even,  $b_n = 0$  if  $n \geq 1$ ;

$$a_0 = \int_0^{1/2} x \sin \pi x \, dx = -\frac{1}{\pi} \left[ x \cos \pi x \Big|_0^{1/2} - \int_0^{1/2} \cos \pi x \, dx \right] = \frac{\sin \pi x}{\pi^2} \Big|_0^{1/2} = \frac{1}{\pi^2};$$

$$\begin{aligned} a_1 &= 2 \int_0^{1/2} x \sin \pi x \cos \pi x \, dx = \int_0^{1/2} x \sin 2\pi x \, dx \\ &= -\frac{1}{2\pi} \left[ x \cos 2\pi x \Big|_0^{1/2} - \int_0^{1/2} \cos 2\pi x \, dx \right] = \frac{1}{4\pi} + \frac{\sin 2\pi x}{4\pi^2} \Big|_0^{1/2} = \frac{1}{4\pi}; \end{aligned}$$

if  $n \geq 2$ , then

$$\begin{aligned}
 a_n &= 2 \int_0^{1/2} x \sin \pi x \cos n\pi x \, dx = \int_0^{1/2} x [\sin(n+1)\pi x - \sin(n-1)\pi x] \, dx \\
 &= \frac{1}{\pi} \left[ x \left[ \frac{\cos(n-1)\pi x}{n-1} - \frac{\cos(n+1)\pi x}{n+1} \right] \right]_0^{1/2} - \int_0^{1/2} \left[ \frac{\cos(n-1)\pi x}{n-1} - \frac{\cos(n+1)\pi x}{n+1} \right] dx \\
 &= \frac{1}{2\pi} \left[ \frac{\cos(n-1)\pi/2}{n-1} - \frac{\cos(n+1)\pi/2}{n+1} \right] - \frac{1}{\pi^2} \left[ \frac{\sin(n-1)\pi/2}{(n-1)^2} - \frac{\sin(n+1)\pi/2}{(n+1)^2} \right] \\
 &= \frac{1}{\pi} \frac{n}{n^2-1} \sin \frac{n\pi}{2} + \frac{2}{\pi^2} \frac{n^2+1}{(n^2-1)^2} \cos \frac{n\pi}{2} = \begin{cases} (-1)^m \frac{2}{\pi^2} \frac{4m^2+1}{(4m^2-1)^2} & \text{if } n = 2m, \\ (-1)^m \frac{1}{4\pi} \frac{2m+1}{m(m+1)} & \text{if } n = 2m+1; \end{cases} \\
 F(x) &= \frac{1}{\pi^2} + \frac{1}{4\pi} \cos \pi x + \frac{2}{\pi^2} \sum_{n=1}^{\infty} (-1)^n \frac{4n^2+1}{(4n^2-1)^2} \cos 2n\pi x + \frac{1}{4\pi} \sum_{n=1}^{\infty} (-1)^n \frac{2n+1}{n(n+1)} \cos(2n+1)\pi x.
 \end{aligned}$$

From Theorem 11.2.4,

$$F(x) = \begin{cases} 0, & -1 \leq x < \frac{1}{2}, \\ \frac{1}{4}, & x = -\frac{1}{2}, \\ x \sin \pi x, & -\frac{1}{2} < x < \frac{1}{2}, \\ \frac{1}{4}, & x = \frac{1}{2}, \\ 0, & \frac{1}{2} < x \leq 1. \end{cases}$$

**11.2.16.** Note that  $\int_{-1}^0 x^2 g(x) \, dx = \int_0^1 x^2 g(-x) \, dx$ ; therefore,  $a_0 = \frac{1}{2} \left[ \int_{-1}^0 x^2 \, dx + \int_0^1 (1-x^2) \, dx \right] = \frac{1}{2} \int_0^1 dx = \frac{1}{2}$ , and if  $n \geq 1$ , then

$$a_n = \int_{-1}^0 x^2 \cos n\pi x \, dx + \int_0^1 (1-x^2) \cos n\pi x \, dx = \int_0^1 \cos n\pi x \, dx = \left. \frac{\sin n\pi x}{n\pi} \right|_0^1 = 0$$

and

$$\begin{aligned}
 b_n &= \int_{-1}^0 x^2 \sin n\pi x \, dx + \int_0^1 (1-x^2) \sin n\pi x \, dx = \int_0^1 (1-2x^2) \sin n\pi x \, dx \\
 &= -\frac{1}{n\pi} \left[ (1-2x^2) \cos n\pi x \right]_0^1 + 4 \int_0^1 x \cos n\pi x \, dx \\
 &= \frac{1 + \cos n\pi}{n\pi} - \frac{4}{n^2\pi^2} \left[ x \sin n\pi x \right]_0^1 - \int_0^1 \sin n\pi x \, dx \\
 &= \frac{1 + \cos n\pi}{n\pi} - \frac{4 \cos n\pi x}{n^3\pi^3} \Big|_0^1 = \frac{1 + \cos n\pi}{n\pi} + \frac{4(1 - \cos n\pi)}{n^3\pi^3} \\
 &= \begin{cases} \frac{1}{8\pi} & \text{if } n = 2m, \\ \frac{1}{(2m+1)^3\pi^3} & \text{if } n = 2m+1; \end{cases}
 \end{aligned}$$

$$F(x) = \frac{1}{2} + \frac{1}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} \sin 2n\pi x + \frac{8}{\pi^3} \sum_{n=0}^{\infty} \frac{1}{(2n+1)^3} \sin(2n+1)\pi x.$$

From Theorem 11.2.4,

$$F(x) = \begin{cases} \frac{1}{2}, & x = -1, \\ x^2, & -1 < x < 0, \\ \frac{1}{2}, & x = 0, \\ 1 - x^2, & 0 < x < 1, \\ \frac{1}{2}, & x = 1. \end{cases}$$

**11.2.18.**  $a_0 = \frac{1}{6} \left[ \int_{-3}^{-2} 2 dx + \int_{-2}^2 3 dx + \int_2^3 1 dx \right] = \frac{5}{2}$ . If  $n \geq 1$ , then

$$\begin{aligned} a_n &= \frac{1}{3} \int_{-3}^3 f(x) \cos \frac{n\pi x}{3} dx \\ &= \frac{1}{3} \left[ \int_{-3}^{-2} 2 \cos \frac{n\pi x}{3} dx + \int_{-2}^2 3 \cos \frac{n\pi x}{3} dx + \int_2^3 \cos \frac{n\pi x}{3} dx \right] = \frac{3}{n\pi} \sin \frac{2n\pi}{3}, \\ b_n &= \frac{1}{3} \left[ \int_{-3}^{-2} 2 \sin \frac{n\pi x}{3} dx + \int_{-2}^2 3 \sin \frac{n\pi x}{3} dx + \int_2^3 \sin \frac{n\pi x}{3} dx \right] = \frac{1}{n\pi} \left( \cos n\pi - \cos \frac{2n\pi}{3} \right); \end{aligned}$$

$$F(x) = \frac{5}{2} + \frac{3}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} \sin \frac{2n\pi}{3} \cos \frac{n\pi x}{3} + \frac{1}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} \left( \cos n\pi - \cos \frac{2n\pi}{3} \right) \sin \frac{n\pi x}{3}.$$

**11.2.20.** (a)  $a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^x dx = \frac{e^x}{2\pi} \Big|_{-\pi}^{\pi} = \frac{\sinh \pi}{\pi}$ . If  $n \geq 1$  then (A)  $a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} e^x \cos nx dx$  and

(B)  $b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} e^x \sin nx dx$ . Integrating (B) by parts yields

$$b_n = \frac{1}{\pi} \left[ e^x \sin nx \Big|_{-\pi}^{\pi} - n \int_{-\pi}^{\pi} e^x \cos nx dx \right] = -na_n. \quad (\text{C})$$

Integrating (A) by parts yields

$$a_n = \frac{1}{\pi} \left[ e^x \cos nx \Big|_{-\pi}^{\pi} + n \int_{-\pi}^{\pi} e^x \sin nx dx \right] = (-1)^n \frac{2 \sinh \pi}{\pi} + nb_n = (-1)^n \frac{2 \sinh \pi}{\pi} - n^2 a_n,$$

from (C). Therefore,  $a_n = \frac{2 \sinh \pi}{\pi} \frac{(-1)^n}{n^2 + 1}$ . Now (C) implies that  $b_n = \frac{2 \sinh \pi}{\pi} \frac{(-1)^{n+1} n}{n^2 + 1}$ . Therefore,

$$F(x) = \frac{\sinh \pi}{\pi} \left( 1 + 2 \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2 + 1} \cos nx - 2 \sum_{n=1}^{\infty} \frac{(-1)^n n}{n^2 + 1} \sin nx \right).$$

(b) From Theorem 11.2.4,  $F(\pi) = \cosh \pi$ , so  $\frac{\sinh \pi}{\pi} \left( 1 + 2 \sum_{n=1}^{\infty} \frac{1}{n^2 + 1} \right) = \cosh \pi$ , which implies the stated result.



**11.2.24.** Since  $f$  is even,  $b_n = 0$ ,  $n \geq 1$ ,  $a_0 = \frac{1}{\pi} \int_0^\pi \cos kx \, dx = \frac{\sin kx}{k\pi} \Big|_0^\pi = \frac{\sin k\pi}{k\pi}$ ; if  $n \geq 1$  then

$$\begin{aligned} a_n &= \frac{2}{\pi} \int_0^\pi \cos kx \cos nx \, dx = \frac{1}{\pi} \int_0^\pi [\cos(n-k)x + \cos(n+k)x] \, dx \\ &= \frac{1}{\pi} \left[ \frac{\sin(n-k)x}{n-k} + \frac{\sin(n+k)x}{n+k} \right] \Big|_0^\pi \\ &= \frac{\cos n\pi \sin kx}{\pi} \left[ \frac{1}{n+k} - \frac{1}{n-k} \right] = (-1)^{n+1} \frac{2k \sin k\pi}{(n^2 - k^2)\pi}; \end{aligned}$$

$$F(x) = \frac{\sin k\pi}{\pi} \left[ \frac{1}{k} - 2k \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2 - k^2} \cos nx \right].$$

**11.2.26.** Since  $f$  is continuous on  $[-L, L]$  and  $f(-L) = f(L)$ , Theorem 11.2.4 implies that

$$f(x) = a_0 + \sum_{n=1}^{\infty} \left( a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right), \quad -L \leq x \leq L,$$

if  $a_0 = \frac{1}{2L} \int_{-L}^L f(x) \, dx$ , and, for  $n \geq 1$ ,

$$\begin{aligned} a_n &= \frac{1}{L} \int_{-L}^L f(x) \cos \frac{n\pi x}{L} \, dx = \frac{1}{n\pi} \left[ f(x) \sin \frac{n\pi x}{L} \Big|_{-L}^L - \int_{-L}^L f'(x) \sin \frac{n\pi x}{L} \, dx \right] \\ &= \frac{L}{n^2\pi^2} \left[ f'(x) \cos \frac{n\pi x}{L} \Big|_{-L}^L - \int_{-L}^L f''(x) \cos \frac{n\pi x}{L} \, dx \right] = -\frac{L}{n^2\pi^2} \int_{-L}^L f''(x) \cos \frac{n\pi x}{L} \, dx \end{aligned}$$

(since  $f'(-L) = f'(L)$ ), and

$$\begin{aligned} b_n &= \frac{1}{L} \int_{-L}^L f(x) \sin \frac{n\pi x}{L} \, dx = -\frac{1}{n\pi} \left[ f(x) \cos \frac{n\pi x}{L} \Big|_{-L}^L - \int_{-L}^L f'(x) \cos \frac{n\pi x}{L} \, dx \right] \\ &= \frac{1}{n\pi} \int_{-L}^L f'(x) \cos \frac{n\pi x}{L} \, dx \quad (\text{since } f(-L) = f(L)) \\ &= \frac{L}{n^2\pi^2} \left[ f'(x) \sin \frac{n\pi x}{L} \Big|_{-L}^L - \int_{-L}^L f''(x) \sin \frac{n\pi x}{L} \, dx \right] = -\frac{L}{n^2\pi^2} \int_{-L}^L f''(x) \sin \frac{n\pi x}{L} \, dx. \end{aligned}$$

If  $f'''$  is integrable on  $[-L, L]$ , then

$$\begin{aligned} a_n &= -\frac{L}{n^2\pi^2} \int_{-L}^L f''(x) \cos \frac{n\pi x}{L} \, dx = -\frac{L^2}{n^3\pi^3} \left[ f'''(x) \sin \frac{n\pi x}{L} \Big|_{-L}^L - \int_{-L}^L f'''(x) \sin \frac{n\pi x}{L} \, dx \right] \\ &= \frac{L^2}{n^3\pi^3} \int_{-L}^L f'''(x) \sin \frac{n\pi x}{L} \, dx. \end{aligned}$$

**11.2.28.** The Fourier series is  $a_0 + \sum_{n=1}^{\infty} \left( a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right)$  where

$$a_0 = \frac{1}{2L} \int_{-L}^L f(x) dx = \frac{1}{2L} \left[ \int_{-L}^0 f(x) dx + \int_0^L f(x) dx \right]. \quad (\text{A})$$

Since  $\int_{-L}^0 f(x) dx = - \int_{-L}^0 f(x+L) dx = - \int_0^L f(x) dx$ , (A) implies that  $a_0 = 0$ . If  $n \geq 1$ , then

$$a_n = \frac{1}{L} \int_{-L}^L f(x) \cos \frac{n\pi x}{L} dx = \frac{1}{L} \left[ \int_{-L}^0 f(x) \cos \frac{n\pi x}{L} dx + \int_0^L f(x) \cos \frac{n\pi x}{L} dx \right]. \quad (\text{B})$$

Since

$$\begin{aligned} \int_{-L}^0 f(x) \cos \frac{n\pi x}{L} dx &= - \int_{-L}^0 f(x+L) \cos \frac{n\pi x}{L} dx = - \int_0^L f(x) \cos \frac{n\pi(x+L)}{L} dx \\ &= (-1)^{n+1} \int_0^L f(x) \cos \frac{n\pi x}{L} dx, \end{aligned}$$

(B) implies that  $a_{2n-1} = A_n$  and  $a_{2n} = 0$ ,  $n \geq 1$ . A similar argument shows that  $b_{2n-1} = B_n$  and  $b_{2n} = 0$ ,  $n \geq 1$ .

**11.2.30.(b)** Let  $\phi_0 = 1$  and  $\phi_{2m} = \cos \frac{m\pi x}{L}$ ,  $\phi_{2m-1} = \sin \frac{m\pi x}{L}$ ,  $m \geq 1$ . Then  $c_0 = a_0$  and  $c_{2m} = a_m$ ,  $c_{2m-1} = b_m$ ,  $m \geq 1$ . Since  $\int_{-L}^L \phi_0^2(x) dx = 2L$  and  $\int_{-L}^L \phi_{2m-1}^2(x) dx = \frac{1}{2} \int_{-L}^L \left( 1 - \cos \frac{2mx}{L} \right) dx = L$ ,  $\int_{-L}^L \phi_{2m}^2(x) dx = \frac{1}{2} \int_{-L}^L \left( 1 + \cos \frac{2mx}{L} \right) dx = L$ ,  $m \geq 1$ , Exercise 12.2.29(d) implies the conclusion.

## 11.3 Fourier Expansions II

**11.3.2.**  $a_0 = \int_0^1 (1-x) dx = -\frac{(1-x)^2}{2} \Big|_0^1 = \frac{1}{2}$ ; if  $n \geq 1$ ,

$$\begin{aligned} a_n &= 2 \int_0^1 (1-x) \cos n\pi x dx = \frac{2}{n\pi} \left[ (1-x) \sin n\pi x \Big|_0^1 + \int_0^1 \sin n\pi x dx \right] \\ &= -\frac{2}{n^2\pi^2} \cos n\pi x \Big|_0^1 = \frac{2}{n^2\pi^2} [1 - (-1)^n] = \begin{cases} \frac{4}{(2m-1)^2\pi^2} & \text{if } n = 2m-1, \\ 0 & \text{if } n = 2m; \end{cases} \end{aligned}$$

$$C(x) = \frac{1}{2} + \frac{4}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} \cos(2n-1)\pi x.$$

$$11.3.4. \quad a_0 = \frac{1}{\pi} \int_0^\pi \sin kx \, dx = -\frac{\cos kx}{k} \Big|_0^\pi = \frac{1 - \cos k\pi}{k\pi}; \text{ if } n \geq 1, \text{ then}$$

$$\begin{aligned} a_n &= \frac{2}{\pi} \int_0^\pi \sin kx \cos nx \, dx = \frac{1}{\pi} \int_0^\pi [\sin(n+k)x - \sin(n-k)x] \, dx \\ &= \frac{1}{\pi} \left[ \frac{\cos(n-k)x}{n-k} - \frac{\cos(n+k)x}{n+k} \right] \Big|_0^\pi = \frac{1}{\pi} \left[ \frac{\cos(n-k)\pi - 1}{n-k} - \frac{\cos(n+k)\pi - 1}{n+k} \right] \\ &= -\frac{2k[1 - (-1)^n \cos k\pi]}{(n^2 - k^2)\pi}; \end{aligned}$$

$$C(x) = \frac{1 - \cos k\pi}{k\pi} - \frac{2k}{\pi} \sum_{n=1}^{\infty} \frac{[1 - (-1)^n \cos k\pi]}{n^2 - k^2} \cos nx.$$

$$11.3.6. \quad a_0 = \frac{1}{L} \int_0^L (x^2 - L^2) \, dx = \frac{1}{L} \left( \frac{x^3}{3} - L^2 x \right) \Big|_0^L = -\frac{2L^2}{3}; \text{ if } n \geq 1,$$

$$\begin{aligned} a_n &= \frac{2}{L} \int_0^L (x^2 - L^2) \cos \frac{n\pi x}{L} \, dx = \frac{2}{n\pi} \left[ (x^2 - L^2) \sin \frac{n\pi x}{L} \Big|_0^L - 2 \int_0^L x \sin \frac{n\pi x}{L} \, dx \right] \\ &= \frac{4L}{n^2 \pi^2} \left[ x \cos \frac{n\pi x}{L} \Big|_0^L - \int_0^L \cos \frac{n\pi x}{L} \, dx \right] = (-1)^n \frac{4L^2}{n^2 \pi^2} - \frac{4L^2}{n^3 \pi^3} \sin \frac{n\pi x}{L} \Big|_0^L = (-1)^n \frac{4L^2}{n^2 \pi^2}; \end{aligned}$$

$$C(x) = -\frac{2L^2}{3} + \frac{4L^2}{\pi^2} \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} \cos \frac{n\pi x}{L}.$$

$$11.3.8. \quad a_0 = \int_0^\pi e^x \, dx = e^x \Big|_0^\pi = \frac{e^\pi - 1}{\pi}; \text{ if } n \geq 1, \text{ then}$$

$$\begin{aligned} a_n &= \frac{2}{\pi} \int_0^\pi e^x \cos nx \, dx = \frac{2}{\pi} \left[ e^x \cos nx \Big|_0^\pi - n \int_0^\pi e^x \sin nx \, dx \right] \\ &= \frac{2}{\pi} \left[ (-1)^n e^\pi - 1 - n e^x \sin nx \Big|_0^\pi - n^2 \int_0^\pi e^x \cos nx \, dx \right] = \frac{2}{\pi} [(-1)^n e^\pi - 1] - n^2 a_n; \end{aligned}$$

$$(1 + n^2)a_n = \frac{2}{\pi} [(-1)^n e^\pi - 1]; \quad a_n = \frac{2}{(n^2 + 1)\pi} [(-1)^n e^\pi - 1];$$

$$C(x) = \frac{e^\pi - 1}{\pi} + \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{[(-1)^n e^\pi - 1]}{(n^2 + 1)} \cos nx.$$

$$11.3.10. \quad a_0 = \frac{1}{L} \int_0^L (x^2 - 2Lx) \, dx = \frac{1}{L} \left( \frac{x^3}{3} - Lx^2 \right) \Big|_0^L = -\frac{2L^2}{3}; \text{ if } n \geq 1,$$

$$\begin{aligned} a_n &= \frac{2}{L} \int_0^L (x^2 - 2Lx) \cos \frac{n\pi x}{L} \, dx = \frac{2}{n\pi} \left[ (x^2 - 2Lx) \sin \frac{n\pi x}{L} \Big|_0^L - 2 \int_0^L (x - L) \sin \frac{n\pi x}{L} \, dx \right] \\ &= \frac{4L}{n^2 \pi^2} \left[ (x - L) \cos \frac{n\pi x}{L} \Big|_0^L - \int_0^L \cos \frac{n\pi x}{L} \, dx \right] = \frac{4L^2}{n^2 \pi^2} - \frac{4L^3}{n^3 \pi^3} \sin \frac{n\pi x}{L} \Big|_0^L = \frac{4L^2}{n^2 \pi^2}; \end{aligned}$$

$$C(x) = -\frac{2L^2}{3} + \frac{4L^2}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^2} \cos \frac{n\pi x}{L}.$$

$$\begin{aligned} \mathbf{11.3.12.} \quad b_n &= 2 \int_0^1 (1-x) \sin n\pi x \, dx = -\frac{2}{n\pi} \left[ (1-x) \cos n\pi x \right]_0^1 + \int_0^1 \cos n\pi x \, dx = \frac{2}{n\pi} + \\ &\quad \frac{2}{n^2\pi^2} \sin n\pi x \Big|_0^1 = \frac{2}{n\pi}; \quad S(x) = \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} \sin n\pi x. \end{aligned}$$

$$\mathbf{11.3.14.} \quad b_n = \frac{2}{L} \int_0^{L/2} \sin \frac{n\pi x}{L} \, dx = -\frac{2}{n\pi} \cos \frac{n\pi x}{L} \Big|_0^{L/2} = \frac{2}{n\pi} \left[ 1 - \cos \frac{n\pi}{2} \right];$$

$$S(x) = \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} \left[ 1 - \cos \frac{n\pi}{2} \right] \sin \frac{n\pi x}{L}.$$

**11.3.16.**

$$\begin{aligned} b_1 &= \frac{2}{\pi} \int_0^{\pi} x \sin^2 x \, dx = \frac{1}{\pi} \int_0^{\pi} x(1 - \cos 2x) \, dx = \frac{x^2}{2\pi} \Big|_0^{\pi} - \frac{1}{\pi} \int_0^{\pi} x \cos 2x \, dx \\ &= \frac{\pi}{2} - \frac{1}{2\pi} \left[ x \sin 2x \Big|_0^{\pi} - \int_0^{\pi} \cos 2x \, dx \right] = \frac{\pi}{2} + \frac{\sin 2x}{4\pi} \Big|_0^{\pi} = \frac{\pi}{2}; \end{aligned}$$

if  $n \geq 2$ , then

$$\begin{aligned} b_n &= \frac{2}{\pi} \int_0^{\pi} x \sin x \sin nx \, dx = \frac{1}{\pi} \int_0^{\pi} x [\cos(n-1)x - \cos(n+1)x] \, dx \\ &= \frac{1}{\pi} \left[ x \left[ \frac{\sin(n-1)x}{n-1} - \frac{\sin(n+1)x}{n+1} \right] \Big|_0^{\pi} - \int_0^{\pi} \left[ \frac{\sin(n-1)x}{n-1} - \frac{\sin(n+1)x}{n+1} \right] \, dx \right] \\ &= \frac{1}{\pi} \left[ \frac{\cos(n-1)x}{(n-1)^2} - \frac{\cos(n+1)x}{(n+1)^2} \right] \Big|_0^{\pi} = \frac{1}{\pi} \left[ \frac{1}{(n-1)^2} - \frac{1}{(n+1)^2} \right] [(-1)^{n+1} - 1] \\ &= \frac{4n}{(n^2-1)^2\pi} [(-1)^{n+1} - 1] = \begin{cases} 0 & \text{if } n = 2m-1, \\ -\frac{16m}{(4m^2-1)\pi} & \text{if } n = 2m; \end{cases} \end{aligned}$$

$$S(x) = \frac{\pi}{2} \sin x - \frac{16}{\pi} \sum_{n=1}^{\infty} \frac{n}{(4n^2-1)^2} \sin 2nx.$$

$$\mathbf{11.3.18.} \quad c_n = \frac{2}{L} \int_0^L \cos \frac{(2n-1)\pi x}{2L} \, dx = \frac{4}{(2n-1)\pi} \sin \frac{(2n-1)\pi x}{2L} \Big|_0^L = (-1)^{n+1} \frac{4}{(2n-1)\pi};$$

$$C_M(x) = -\frac{4}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^n}{2n-1} \cos \frac{(2n-1)\pi x}{2L}.$$

## 11.3.20.

$$\begin{aligned}
d_n &= 2 \int_0^1 x \cos \frac{(2n-1)\pi x}{2} dx \\
&= \frac{4}{(2n-1)\pi} \left[ x \sin \frac{(2n-1)\pi x}{2} \Big|_0^1 - \int_0^1 \sin \frac{(2n-1)\pi x}{2} dx \right] \\
&= \frac{4}{(2n-1)\pi} \left[ (-1)^{n+1} + \frac{2}{(2n-1)\pi} \cos \frac{(2n-1)\pi x}{2} \Big|_0^1 \right] \\
&= -\frac{4}{(2n-1)\pi} \left[ (-1)^n + \frac{2}{(2n-1)\pi} \right]; \\
C_M(x) &= -\frac{4}{\pi} \sum_{n=1}^{\infty} \left[ (-1)^n + \frac{2}{(2n-1)\pi} \right] \cos \frac{(2n-1)\pi x}{2}.
\end{aligned}$$

## 11.3.22.

$$\begin{aligned}
c_n &= \frac{2}{\pi} \int_0^{\pi} \cos x \cos \frac{(2n-1)x}{2} dx = \frac{1}{\pi} \int_0^{\pi} \left[ \frac{\cos(2n+1)x}{2} + \frac{\cos(2n-3)x}{2} \right] dx \\
&= \frac{2}{\pi} \left[ \frac{\sin(2n+1)x/2}{2n+1} + \frac{\sin(2n-3)x/2}{2n-3} \right] \Big|_0^{\pi} \\
&= (-1)^n \frac{2}{\pi} \left[ \frac{1}{2n+1} + \frac{1}{2n-3} \right] = (-1)^n \frac{4(2n-1)}{(2n-3)(2n+1)\pi}; \\
C_M(x) &= \frac{4}{\pi} \sum_{n=1}^{\infty} (-1)^n \frac{2n-1}{(2n-3)(2n+1)} \cos \frac{(2n-1)x}{2}.
\end{aligned}$$

## 11.3.24.

$$\begin{aligned}
c_n &= \frac{2}{L} \int_0^L (Lx - x^2) \cos \frac{(2n-1)\pi x}{2L} dx \\
&= \frac{4}{(2n-1)\pi} \left[ (Lx - x^2) \sin \frac{(2n-1)\pi x}{2L} \Big|_0^L - \int_0^L (L - 2x) \sin \frac{(2n-1)\pi x}{2L} dx \right] \\
&= \frac{8L}{(2n-1)^2\pi^2} \left[ (L - 2x) \cos \frac{(2n-1)\pi x}{2L} \Big|_0^L + 2 \int_0^L \cos \frac{(2n-1)\pi x}{2L} dx \right] \\
&= -\frac{8L^2}{(2n-1)^2\pi^2} + \frac{32L^2}{(2n-1)^3\pi^3} \sin \frac{(2n-1)\pi x}{2L} \Big|_0^L = \frac{32L^2}{(2n-1)^3\pi^3} \sin \frac{(2n-1)\pi x}{2} \\
&= -\frac{8L^2}{(2n-1)^2\pi^2} + (-1)^{n-1} \frac{32L^2}{(2n-1)^3\pi^3}; \\
C_M(x) &= -\frac{8L^2}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} \left[ 1 + \frac{4(-1)^n}{(2n-1)\pi} \right] \cos \frac{(2n-1)\pi x}{2L}.
\end{aligned}$$

**11.3.26.**

$$\begin{aligned}
d_n &= \frac{2}{L} \int_0^L x^2 \sin \frac{(2n-1)\pi x}{2L} dx \\
&= -\frac{4}{(2n-1)\pi} \left[ x^2 \cos \frac{(2n-1)\pi x}{2L} \Big|_0^L - 2 \int_0^L x \cos \frac{(2n-1)\pi x}{2L} dx \right] \\
&= \frac{16L}{(2n-1)^2\pi^2} \left[ x \sin \frac{(2n-1)\pi x}{2L} \Big|_0^L - \int_0^L \sin \frac{(2n-1)\pi x}{2L} dx \right] \\
&= (-1)^{n+1} \frac{16L^2}{(2n-1)^2\pi^2} + \frac{32L^2}{(2n-1)^3\pi^3} \cos \frac{(2n-1)\pi x}{2L} \Big|_0^L \\
&= (-1)^{n+1} \frac{16L^2}{(2n-1)^2\pi^2} - \frac{32L^2}{(2n-1)^3\pi^3}; \\
S_M(x) &= -\frac{16L^2}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} \left[ (-1)^n + \frac{2}{(2n-1)\pi} \right] \sin \frac{(2n-1)\pi x}{2L}.
\end{aligned}$$

**11.3.28.**

$$\begin{aligned}
d_n &= \frac{2}{\pi} \int_0^{\pi} \cos x \sin \frac{(2n-1)x}{2} dx = \frac{1}{\pi} \int_0^{\pi} \left[ \frac{\sin(2n+1)x}{2} + \frac{\sin(2n-3)x}{2} \right] dx \\
&= -\frac{2}{\pi} \left[ \frac{\cos(2n+1)x/2}{2n-1} + \frac{\cos(2n-3)x/2}{2n-3} \right] \Big|_0^{\pi} \\
&= \frac{2}{\pi} \left[ \frac{1}{2n+1} + \frac{1}{2n-3} \right] = \frac{4(2n-1)}{(2n-3)(2n+1)\pi}; \\
S_M(x) &= \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{2n-1}{(2n-3)(2n+1)} \sin \frac{(2n-1)x}{2}.
\end{aligned}$$

**11.3.30.**

$$\begin{aligned}
d_n &= \frac{2}{L} \int_0^L (Lx - x^2) \sin \frac{(2n-1)\pi x}{2L} dx \\
&= -\frac{4}{(2n-1)\pi} \left[ (Lx - x^2) \cos \frac{(2n-1)\pi x}{2L} \Big|_0^L - \int_0^L (L - 2x) \cos \frac{(2n-1)\pi x}{2L} dx \right] \\
&= \frac{8L}{(2n-1)^2\pi^2} \left[ (L - 2x) \sin \frac{(2n-1)\pi x}{2L} \Big|_0^L + 2 \int_0^L \sin \frac{(2n-1)\pi x}{2L} dx \right] \\
&= (-1)^n \frac{8L^2}{(2n-1)^2\pi^2} - \frac{32L^2}{(2n-1)^3\pi^3} \cos \frac{(2n-1)\pi x}{2L} \Big|_0^L \\
&= (-1)^n \frac{8L^2}{(2n-1)^2\pi^2} + \frac{32L^2}{(2n-1)^3\pi^3}; \\
S_M(x) &= \frac{8L^2}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} \left[ (-1)^n + \frac{4}{(2n-1)\pi} \right] \sin \frac{(2n-1)\pi x}{2L}.
\end{aligned}$$

**11.3.32.**  $a_0 = \frac{1}{L} \int_0^L (3x^4 - 4Lx^3) dx = \frac{1}{L} \left( \frac{3x^5}{5} - Lx^4 \right) \Big|_0^L = -\frac{2L^4}{5}$ . Since  $f'(0) = f'(L) = 0$  and  $f'''(x) = 24(3x - L)$ ,

$$\begin{aligned} a_n &= \frac{48L^2}{n^3\pi^3} \int_0^L (3x - L) \sin \frac{n\pi x}{L} dx = -\frac{48L^3}{n^4\pi^4} \left[ (3x - L) \cos \frac{n\pi x}{L} \Big|_0^L - 3 \int_0^L \cos \frac{n\pi x}{L} dx \right] \\ &= -\frac{48L^3}{n^4\pi^4} [(-1)^n 2L + L] + \frac{144L^4}{n^5\pi^5} \sin \frac{n\pi x}{L} \Big|_0^L = -\frac{48L^4}{n^4\pi^4} [1 + (-1)^n 2], \quad n \geq 1; \end{aligned}$$

$$C(x) = -\frac{2L^4}{5} - \frac{48L^4}{\pi^4} \sum_{n=1}^{\infty} \frac{1 + (-1)^n 2}{n^4} \cos \frac{n\pi x}{L}.$$

**11.3.34.**  $a_0 = \frac{1}{L} \int_0^L (x^4 - 2Lx^3 + L^2x^2) dx = \frac{1}{L} \left( \frac{x^5}{5} - \frac{Lx^4}{2} + \frac{L^2x^3}{3} \right) \Big|_0^L = \frac{L^4}{30}$ . Since  $f'(0) = f'(L) = 0$  and  $f'''(x) = 12(2x - L)$ ,

$$\begin{aligned} a_n &= \frac{24L^2}{n^3\pi^3} \int_0^L (2x - L) \sin \frac{n\pi x}{L} dx = -\frac{24L^3}{n^4\pi^4} \left[ (2x - L) \cos \frac{n\pi x}{L} \Big|_0^L - 2 \int_0^L \cos \frac{n\pi x}{L} dx \right] \\ &= -\frac{24L^3}{n^4\pi^4} [(-1)^n L + L] + \frac{48L^4}{n^5\pi^5} \sin \frac{n\pi x}{L} \Big|_0^L = -\frac{24L^4}{n^4\pi^4} [1 + (-1)^n] \\ &= \begin{cases} 0 & \text{if } n = 2m - 1, \\ -\frac{3L^4}{m^4\pi^4} & \text{if } n = 2m, \end{cases} \quad n \geq 1. \end{aligned}$$

$$C(x) = \frac{L^4}{30} - \frac{3L^4}{\pi^4} \sum_{n=1}^{\infty} \frac{1}{n^4} \cos \frac{2n\pi x}{L}.$$

**11.3.36.** Since  $f(0) = f(L) = 0$  and  $f''(x) = -2$ ,

$$\begin{aligned} b_n &= \frac{4L}{n^2\pi^2} \int_0^L \sin \frac{n\pi x}{L} dx = -\frac{4L^2}{n^3\pi^3} \cos \frac{n\pi x}{L} \Big|_0^L = -\frac{4L^2}{n^3\pi^2} (\cos n\pi - 1) \\ &= \begin{cases} \frac{8L^2}{(2m-1)^3\pi^3}, & \text{if } n = 2m - 1, \\ 0, & \text{if } n = 2m; \end{cases} \end{aligned}$$

$$S(x) = \frac{8L^2}{\pi^3} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^3} \sin \frac{(2n-1)\pi x}{L}.$$

**11.3.38.** Since  $f(0) = f(L) = 0$  and  $f''(x) = -6x$ ,

$$b_n = \frac{12L}{n^2\pi^2} \int_0^L x \sin \frac{n\pi x}{L} dx = -\frac{12L^2}{n^3\pi^3} \left[ x \cos \frac{n\pi x}{L} \Big|_0^L - \int_0^L \cos \frac{n\pi x}{L} dx \right] = (-1)^{n+1} \frac{12L^3}{n^3\pi^3};$$

$$S(x) = -\frac{12L^3}{\pi^3} \sum_{n=1}^{\infty} \frac{(-1)^n}{n^3} \sin \frac{n\pi x}{L}.$$

**11.3.40.** Since  $f(0) = f(L) = f''(0) = f''(L) = 0$  and  $f^{(4)} = 360x$ ,

$$\begin{aligned} b_n &= \frac{720L^3}{n^4\pi^4} \int_0^L x \sin \frac{n\pi x}{L} dx = -\frac{720L^4}{n^5\pi^5} \left[ x \cos \frac{n\pi x}{L} \Big|_0^L - \int_0^L \cos \frac{n\pi x}{L} dx \right] \\ &= (-1)^{n+1} \frac{720L^5}{n^5\pi^5} + \frac{720L^5}{n^6\pi^6} \sin \frac{n\pi x}{L} \Big|_0^L = (-1)^{n+1} \frac{720L^5}{n^5\pi^5}; \\ S(x) &= -\frac{720L^5}{\pi^5} \sum_{n=1}^{\infty} \frac{(-1)^n}{n^5} \sin \frac{n\pi x}{L}. \end{aligned}$$

**11.3.42.** (a) Since  $f$  is continuous on  $[0, L]$  and  $f(L) = 0$ , Theorem 11.3.3 implies that

$$f(x) = \sum_{n=1}^{\infty} c_n \cos \frac{(2n-1)\pi x}{2L}, \quad -L \leq x \leq L, \text{ with}$$

$$\begin{aligned} c_n &= \frac{2}{L} \int_0^L f(x) \cos \frac{(2n-1)\pi x}{2L} dx \\ &= \frac{4}{(2n-1)\pi} \left[ f(x) \sin \frac{(2n-1)\pi x}{2L} \Big|_0^L - \int_0^L f'(x) \sin \frac{(2n-1)\pi x}{2L} dx \right] \\ &= -\frac{4}{(2n-1)\pi} \int_0^L f'(x) \sin \frac{(2n-1)\pi x}{2L} dx \quad (\text{since } f(L) = 0) \\ &= \frac{8L^2}{(2n-1)^2\pi^2} \left[ f'(x) \cos \frac{(2n-1)\pi x}{2L} \Big|_0^L - \int_0^L f''(x) \cos \frac{(2n-1)\pi x}{2L} dx \right] \\ &= -\frac{8L}{(2n-1)^2\pi^2} \int_0^L f''(x) \cos \frac{(2n-1)\pi x}{2L} dx \quad (\text{since } f'(0) = 0). \end{aligned}$$

(b) Continuing the integration by parts yields

$$\begin{aligned} c_n &= -\frac{16L^2}{(2n-1)^3\pi^3} \left[ f''(x) \sin \frac{(2n-1)\pi x}{2L} \Big|_0^L - \int_0^L f'''(x) \sin \frac{(2n-1)\pi x}{2L} dx \right] \\ &= \frac{16L^2}{(2n-1)^3\pi^3} \int_0^L f'''(x) \sin \frac{(2n-1)\pi x}{2L} dx. \end{aligned}$$

**11.3.44.** Since  $f'(0) = f(L) = 0$  and  $f''(x) = -2$ ,

$$\begin{aligned} c_n &= \frac{16L}{(2n-1)^2\pi^2} \int_0^L \cos \frac{(2n-1)\pi x}{2L} dx \\ &= \frac{32L^2}{(2n-1)^3\pi^3} \sin \frac{(2n-1)\pi x}{2L} \Big|_0^L = (-1)^{n+1} \frac{32L^2}{(2n-1)^3\pi^3}; \\ C_M(x) &= -\frac{32L^2}{\pi^3} \sum_{n=1}^{\infty} \frac{(-1)^n}{(2n-1)^3} \cos \frac{(2n-1)\pi x}{2L}. \end{aligned}$$



**11.3.46.** Since  $f'(0) = f(L) = 0$  and  $f''(x) = 6(2x + L)$ ,

$$\begin{aligned}
 c_n &= -\frac{48L}{(2n-1)^2\pi^2} \int_0^L (2x+L) \cos \frac{(2n-1)\pi x}{2L} dx \\
 &= -\frac{96L^2}{(2n-1)^3\pi^3} \left[ (2x+L) \sin \frac{(2n-1)\pi x}{2L} \Big|_0^L - 2 \int_0^L \sin \frac{(2n-1)\pi x}{2L} dx \right] \\
 &= -\frac{96L^2}{(2n-1)^3\pi^3} \left[ (-1)^{n+1} 3L - \frac{4L}{(2n-1)\pi} \cos \frac{(2n-1)\pi x}{2L} \Big|_0^L \right] \\
 &= \frac{96L^3}{(2n-1)^3\pi^3} \left[ (-1)^n 3 + \frac{4}{(2n-1)\pi} \right]; \\
 C_M(x) &= \frac{96L^3}{\pi^3} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^3} \left[ (-1)^n 3 + \frac{4}{(2n-1)\pi} \right] \cos \frac{(2n-1)\pi x}{2L}.
 \end{aligned}$$

**11.3.48.** Since  $f'(0) = f(L) = f''(L) = 0$  and  $f'''(x) = 12(2x - L)$ ,

$$\begin{aligned}
 c_n &= \frac{192L^2}{(2n-1)^3\pi^3} \int_0^L (2x-L) \sin \frac{(2n-1)\pi x}{2L} dx \\
 &= -\frac{384L^3}{(2n-1)^4\pi^4} \left[ (2x-L) \cos \frac{(2n-1)\pi x}{2L} \Big|_0^L - 2 \int_0^L \cos \frac{(2n-1)\pi x}{2L} dx \right] \\
 &= -\frac{384L^3}{(2n-1)^4\pi^4} \left[ L - \frac{4L}{(2n-1)\pi} \sin \frac{(2n-1)\pi x}{2L} \Big|_0^L \right] \\
 &= -\frac{384L^4}{(2n-1)^4\pi^4} \left[ 1 + \frac{(-1)^n 4}{(2n-1)\pi} \right]; \\
 C_M(x) &= -\frac{384L^4}{\pi^4} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^4} \left[ 1 + \frac{(-1)^n 4}{(2n-1)\pi} \right] \cos \frac{(2n-1)\pi x}{2L}.
 \end{aligned}$$

**11.3.50.** (a) Since  $f$  is continuous on  $[0, L]$  and  $f(0) = 0$ , Theorem 11.3.4 implies that

$$f(x) = \sum_{n=1}^{\infty} d_n \sin \frac{(2n-1)\pi x}{2L}, \quad -L \leq x \leq L, \text{ with}$$

$$\begin{aligned}
 d_n &= \frac{2}{L} \int_0^L f(x) \sin \frac{(2n-1)\pi x}{2L} dx \\
 &= -\frac{4}{(2n-1)\pi} \left[ f(x) \cos \frac{(2n-1)\pi x}{2L} \Big|_0^L - \int_0^L f'(x) \cos \frac{(2n-1)\pi x}{2L} dx \right] \\
 &= \frac{4}{(2n-1)\pi} \int_0^L f'(x) \cos \frac{(2n-1)\pi x}{2L} dx \quad (\text{since } f(0) = 0) \\
 &= \frac{8L}{(2n-1)^2\pi^2} \left[ f'(x) \sin \frac{(2n-1)\pi x}{2L} \Big|_0^L - \int_0^L f''(x) \sin \frac{(2n-1)\pi x}{2L} dx \right] \\
 &= -\frac{8L}{(2n-1)^2\pi^2} \int_0^L f''(x) \sin \frac{(2n-1)\pi x}{2L} dx \quad \text{since } f'(L) = 0.
 \end{aligned}$$

(b) Continuing the integration by parts yields

$$\begin{aligned} d_n &= \frac{16L^2}{(2n-1)^3\pi^3} \left[ f''(x) \cos \frac{(2n-1)\pi x}{2L} \Big|_0^L - \int_0^L f'''(x) \cos \frac{(2n-1)\pi x}{2L} dx \right] \\ &= -\frac{16L^2}{(2n-1)^3\pi^3} \int_0^L f'''(x) \cos \frac{(2n-1)\pi x}{2L} dx. \end{aligned}$$

**11.3.52.** Since  $f(0) = f'(L) = 0$ , and  $f''(x) = 6(L-2x)$

$$\begin{aligned} d_n &= -\frac{48L}{(2n-1)^2\pi^2} \int_0^L (L-2x) \sin \frac{(2n-1)\pi x}{2L} dx \\ &= \frac{96L^2}{(2n-1)^3\pi^3} \left[ (L-2x) \cos \frac{(2n-1)\pi x}{2L} \Big|_0^L + 2 \int_0^L \cos \frac{(2n-1)\pi x}{2L} dx \right] \\ &= \frac{96L^2}{(2n-1)^3\pi^3} \left[ -L + \frac{4L}{(2n-1)\pi} \sin \frac{(2n-1)\pi x}{2L} \Big|_0^L \right] \\ &= -\frac{96L^3}{(2n-1)^3\pi^3} \left[ 1 + (-1)^n \frac{4}{(2n-1)\pi} \right]; \\ S_M(x) &= -\frac{96L^3}{\pi^3} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^3} \left[ 1 + (-1)^n \frac{4}{(2n-1)\pi} \right] \sin \frac{(2n-1)\pi x}{2L}. \end{aligned}$$

**11.3.54.** Since  $f(0) = f'(L) = f''(0) = 0$  and  $f'''(x) = 6$ ,

$$\begin{aligned} d_n &= -\frac{96L^2}{(2n-1)^3\pi^3} \int_0^L \cos \frac{(2n-1)\pi x}{2L} dx \\ &= -\frac{192L^3}{(2n-1)^4\pi^4} \sin \frac{(2n-1)\pi x}{2L} \Big|_0^L = (-1)^n \frac{192L^3}{(2n-1)^4\pi^4}; \\ S_M(x) &= \frac{192L^3}{\pi^4} \sum_{n=1}^{\infty} \frac{(-1)^n}{(2n-1)^4} \sin \frac{(2n-1)\pi x}{2L}. \end{aligned}$$

**11.3.56.** Since  $f(0) = f'(L) = f''(0) = 0$  and  $f'''(x) = 12(2x-L)$ ,

$$\begin{aligned} d_n &= -\frac{192L^2}{(2n-1)^3\pi^3} \int_0^L (2x-L) \cos \frac{(2n-1)\pi x}{2L} dx \\ &= -\frac{384L^3}{(2n-1)^4\pi^4} \left[ (2x-L) \sin \frac{(2n-1)\pi x}{2L} \Big|_0^L - 2 \int_0^L \sin \frac{(2n-1)\pi x}{2L} dx \right] \\ &= -\frac{384L^3}{(2n-1)^4\pi^4} \left[ (-1)^{n+1}L + \frac{4L}{(2n-1)\pi} \cos \frac{(2n-1)\pi x}{2L} \Big|_0^L \right] \\ &= -\frac{384L^3}{(2n-1)^4\pi^4} \left[ (-1)^{n+1}L - \frac{4L}{(2n-1)\pi} \right] \\ &= \frac{384L^4}{(2n-1)^4\pi^4} \left[ (-1)^n + \frac{4}{(2n-1)\pi} \right]; \end{aligned}$$

$$S_M(x) = \frac{384L^4}{\pi^4} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^4} \left[ (-1)^n + \frac{4}{(2n-1)\pi} \right] \sin \frac{(2n-1)\pi x}{2L}.$$

**11.3.58.** The Fourier sine series of  $f_4$  on  $[0, 2L]$  is  $\sum_{n=1}^{\infty} B_n \sin \frac{n\pi x}{2L}$ , where

$$B_n = \frac{1}{L} \int_0^{2L} f_4(x) \sin \frac{n\pi x}{2L} dx = \frac{1}{L} \left[ \int_0^L f(x) \sin \frac{n\pi x}{2L} dx + \int_L^{2L} f(2L-x) \sin \frac{n\pi x}{2L} dx \right].$$

Replacing  $x$  by  $2L-x$  yields  $\int_L^{2L} f(2L-x) \sin \frac{n\pi x}{2L} dx = \int_0^L f(x) \sin \frac{n\pi(2L-x)}{2L} dx$ . Since  $\sin \frac{n\pi(2L-x)}{2L} = (-1)^{n+1} \sin \frac{n\pi x}{2L}$ ,

$$\int_L^{2L} f(2L-x) \sin \frac{n\pi x}{2L} dx = (-1)^{n+1} \int_0^L f(x) \sin \frac{n\pi x}{2L} dx,$$

so

$$B_n = \frac{1 + (-1)^{n+1}}{L} \int_0^L f(x) \sin \frac{n\pi x}{2L} dx = \begin{cases} \frac{2}{L} \int_0^L f(x) \sin \frac{(2m-1)\pi x}{2L} dx & \text{if } n = 2m-1, \\ 0 & \text{if } n = 2m. \end{cases}$$

Therefore, the Fourier sine series of  $f_4$  on  $[0, 2L]$  is  $\sum_{n=1}^{\infty} d_n \sin \frac{(2n-1)\pi x}{2L}$  with

$$d_n = \frac{2}{L} \int_0^L f(x) \sin \frac{(2n-1)\pi x}{2L} dx.$$

**11.3.60.** The Fourier cosine series of  $f_4$  on  $[0, 2L]$  is  $A_0 + \sum_{n=1}^{\infty} A_n \cos \frac{n\pi x}{2L}$ , where

$$A_0 = \frac{1}{2L} \int_0^{2L} f_4(x) dx = \frac{1}{2L} \left[ \int_0^L f(x) dx + \int_L^{2L} f(2L-x) dx \right] = \frac{1}{L} \int_0^L f(x) dx$$

and

$$A_n = \frac{1}{L} \int_0^{2L} f_4(x) \cos \frac{n\pi x}{2L} dx = \frac{1}{L} \left[ \int_0^L f(x) \cos \frac{n\pi x}{2L} dx + \int_L^{2L} f(2L-x) \cos \frac{n\pi x}{2L} dx \right].$$

Replacing  $x$  by  $2L-x$  yields

$$\int_L^{2L} f(2L-x) \cos \frac{n\pi x}{2L} dx = - \int_L^0 f(x) \cos \frac{n\pi(2L-x)}{2L} dx = \int_0^L f(x) \cos \frac{n\pi(2L-x)}{2L} dx.$$

Since  $\cos \frac{n\pi(2L-x)}{2L} = \cos n\pi \cos \frac{n\pi x}{2L} = (-1)^n \cos \frac{n\pi x}{2L}$ ,

$$A_n = \frac{1 + (-1)^n}{L} \int_0^L f(x) \cos \frac{n\pi x}{2L} dx = \begin{cases} 0 & \text{if } n = 2m-1 \\ \frac{2}{L} \int_0^L f(x) \cos \frac{m\pi x}{L} dx & \text{if } n = 2m. \end{cases}$$

Therefore, the Fourier cosine series of  $f_4$  on  $[0, 2L]$  is  $A_0 + \sum_{n=0}^{\infty} A_{2n} \cos \frac{n\pi x}{L} = a_0 + \sum_{n=0}^{\infty} a_n \cos \frac{n\pi x}{L}$ .

## Chapter 12

# Fourier Solutions of Partial Differential

### 12.1 The Heat Equation

**12.1.2.**  $X(x)T(t)$  satisfies  $u_t = a^2 u_{xx}$  if  $X'' + \lambda X = 0$  and (A)  $T' = -a^2 \lambda T$  for the same value of  $\lambda$ . The product also satisfies the boundary conditions  $u(0, t) = u_x(L, t) = 0$ ,  $t > 0$ , if and only if  $X(0) = X'(L) = 0$ . Since we are interested in nontrivial solutions,  $X$  must be a nontrivial solution of (B)  $X'' + \lambda X = 0$ ,  $X(0) = 0$ ,  $X'(L) = 0$ . From Theorem 11.1.4,  $\lambda_n = (2n-1)^2 \pi^2 / 4L^2$  is an eigenvalue of (B) with associated eigenfunction  $X_n = \sin \frac{(2n-1)\pi x}{2L}$ ,  $n = 1, 2, 3, \dots$ . Substituting  $\lambda = (2n-1)^2 \pi^2 / 4L^2$  into (A) yields  $T' = -((2n-1)^2 \pi^2 a^2 / 4L^2)T$ , which has the solution  $T_n = e^{-(2n-1)^2 \pi^2 a^2 t / 4L^2}$ .

We have now shown that the functions  $u_n(x, t) = e^{-(2n-1)^2 \pi^2 a^2 t / 4L^2} \sin \frac{(2n-1)\pi x}{2L}$ ,  $n = 1, 2, 3, \dots$  satisfy  $u_t = a^2 u_{xx}$  and the boundary conditions  $u(0, t) = u_x(L, t) = 0$ ,  $t > 0$ . Any finite sum  $\sum_{n=1}^m d_n e^{-(2n-1)^2 \pi^2 a^2 t / 4L^2} \sin \frac{(2n-1)\pi x}{2L}$  also has these properties. Therefore, it is plausible to expect

that that this is also true of the infinite series (C)  $u(x, t) = \sum_{n=1}^{\infty} d_n e^{-(2n-1)^2 \pi^2 a^2 t / 4L^2} \sin \frac{(2n-1)\pi x}{2L}$

under suitable conditions on the coefficients  $\{d_n\}$ . Since  $u(x, 0) = \sum_{n=1}^{\infty} d_n \sin \frac{(2n-1)\pi x}{2L}$ , if  $\{d_n\}$  are the mixed Fourier sine coefficients of  $f$  on  $[0, L]$ , then  $u(x, 0) = f(x)$  at all points  $x$  in  $[0, L]$  where the mixed Fourier sine series converges to  $f(x)$ . In this case (C) is a formal solution of the initial-boundary value problem of Definition 12.1.3.

**12.1.8.** Since  $f(0) = f(1) = 0$  and  $f''(x) = -2$ , Theorem 11.3.5(b) implies that

$$\begin{aligned}\alpha_n &= \frac{4}{n^2\pi^2} \int_0^1 \sin n\pi x \, dx = -\frac{4}{n^3\pi^3} \cos n\pi x \Big|_0^1 = -\frac{4}{n^3\pi^2} (\cos n\pi - 1) \\ &= \begin{cases} \frac{8}{(2m-1)^3\pi^3}, & \text{if } n = 2m-1, \\ 0, & \text{if } n = 2m; \end{cases}\end{aligned}$$

$$S(x) = \frac{8}{\pi^3} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^3} \sin \frac{(2n-1)\pi x}{L}. \text{ From Definition 12.1.1,}$$

$$u(x, t) = \frac{8}{\pi^3} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^3} e^{-(2n-1)^2\pi^2 t} \sin(2n-1)\pi x.$$

**12.1.10.**

$$\begin{aligned}
\alpha_1 &= \frac{2}{\pi} \int_0^\pi x \sin^2 x \, dx = \frac{1}{\pi} \int_0^\pi x(1 - \cos 2x) \, dx = \frac{x^2}{2\pi} \Big|_0^\pi - \frac{1}{\pi} \int_0^\pi x \cos 2x \, dx \\
&= \frac{\pi}{2} - \frac{1}{2\pi} \left[ x \sin 2x \Big|_0^\pi - \int_0^\pi \cos 2x \, dx \right] = \frac{\pi}{2} + \frac{\sin 2x}{4\pi} \Big|_0^\pi = \frac{\pi}{2};
\end{aligned}$$

if  $n \geq 2$ , then

$$\begin{aligned}
\alpha_n &= \frac{2}{\pi} \int_0^\pi x \sin x \sin nx \, dx = \frac{1}{\pi} \int_0^\pi x [\cos(n-1)x - \cos(n+1)x] \, dx \\
&= \frac{1}{\pi} \left[ x \left[ \frac{\sin(n-1)x}{n-1} - \frac{\sin(n+1)x}{n+1} \right] \Big|_0^\pi - \int_0^\pi \left[ \frac{\sin(n-1)x}{n-1} - \frac{\sin(n+1)x}{n+1} \right] dx \right] \\
&= \frac{1}{\pi} \left[ \frac{\cos(n-1)x}{(n-1)^2} - \frac{\cos(n+1)x}{(n+1)^2} \right] \Big|_0^\pi = \frac{1}{\pi} \left[ \frac{1}{(n-1)^2} - \frac{1}{(n+1)^2} \right] [(-1)^{n+1} - 1] \\
&= \frac{4n}{(n^2-1)^2\pi} [(-1)^{n+1} - 1] = \begin{cases} 0 & \text{if } n = 2m-1, \\ -\frac{16m}{(4m^2-1)\pi} & \text{if } n = 2m; \end{cases}
\end{aligned}$$

$$S(x) = \frac{\pi}{2} \sin x - \frac{16}{\pi} \sum_{n=1}^{\infty} \frac{n}{(4n^2-1)^2} \sin 2nx. \text{ From Definition 12.1.1,}$$

$$u(x, t) = \frac{\pi}{2} e^{-3t} \sin x - \frac{16}{\pi} \sum_{n=1}^{\infty} \frac{n}{(4n^2-1)^2} e^{-12n^2t} \sin 2nx.$$

**12.1.12.** Since  $f(0) = f(L) = 0$  and  $f''(x) = -6x$ , Theorem 11.3.5(b) implies that

$$\begin{aligned}
\alpha_n &= \frac{36}{n^2\pi^2} \int_0^3 x \sin \frac{n\pi x}{3} \, dx = -\frac{108}{n^3\pi^3} \left[ x \cos \frac{n\pi x}{3} \Big|_0^3 - \int_0^3 \cos \frac{n\pi x}{3} \, dx \right] \\
&= (-1)^{n+1} \frac{108}{n^3\pi^3} + \frac{108}{n^4\pi^4} \sin \frac{n\pi x}{3} \Big|_0^3 = (-1)^{n+1} \frac{324}{n^3\pi^3};
\end{aligned}$$

$$S(x) = -\frac{324}{\pi^3} \sum_{n=1}^{\infty} \frac{(-1)^n}{n^3} \sin \frac{n\pi x}{3}. \text{ From Definition 12.1.1, } u(x, t) = -\frac{324}{\pi^3} \sum_{n=1}^{\infty} \frac{(-1)^n}{n^3} e^{-4n^2\pi^2t/9} \sin \frac{n\pi x}{3}.$$

**12.1.14.** Since  $f(0) = f(1) = f''(0) = f''(L) = 0$  and  $f^{(4)} = 360x$ , Theorem 11.3.5(b) and Exercise 35(b) of Section 11.3 imply that

$$\begin{aligned}
\alpha_n &= \frac{720}{n^4\pi^4} \int_0^1 x \sin n\pi x \, dx = -\frac{720}{n^5\pi^5} \left[ x \cos n\pi x \Big|_0^1 - \int_0^1 \cos n\pi x \, dx \right] \\
&= (-1)^{n+1} \frac{720}{n^5\pi^5} + \frac{720}{n^6\pi^6} \sin \frac{n\pi x}{L} \Big|_0^1 = (-1)^{n+1} \frac{720}{n^5\pi^5};
\end{aligned}$$

$$S(x) = -\frac{720}{\pi^5} \sum_{n=1}^{\infty} \frac{(-1)^n}{n^5} \sin n\pi x. \text{ From Definition 12.1.1, } u(x, t) = -\frac{720}{\pi^5} \sum_{n=1}^{\infty} \frac{(-1)^n}{n^5} e^{-7n^2\pi^2t} \sin n\pi x.$$

**12.1.16.** Since  $f(0) = f(1) = f''(0) = f''(L) = 0$  and  $f^{(4)} = 120(3x - 1)$ , Theorem 11.3.5(b) and Exercise 35(b) of Section 11.3 imply that

$$\begin{aligned}\alpha_n &= \frac{240}{n^4\pi^4} \int_0^1 (3x - 1) \sin n\pi x \, dx = -\frac{240}{n^5\pi^5} \left[ (3x - 1) \cos n\pi x \right]_0^1 - 3 \int_0^1 \cos n\pi x \, dx \\ &= -\frac{240}{n^5\pi^5} [(-1)^n 2 + 1] + \frac{720}{n^6\pi^6} \sin n\pi x \Big|_0^1 = -\frac{240}{n^5\pi^5} [1 + (-1)^n 2];\end{aligned}$$

$$S(x) = -\frac{240}{\pi^5} \sum_{n=1}^{\infty} \frac{1 + (-1)^n 2}{n^5} \sin n\pi x. \text{ From Definition 12.1.1,}$$

$$u(x, t) = -\frac{240}{\pi^5} \sum_{n=1}^{\infty} \frac{1 + (-1)^n 2}{n^5} e^{-2n^2\pi^2 t} \sin n\pi x.$$

**12.1.18.**  $\alpha_0 = \frac{1}{2} \int_0^2 (x^2 - 4x) \, dx = \frac{1}{L} \left( \frac{x^3}{3} - 2x^2 \right) \Big|_0^2 = -\frac{8}{3};$  if  $n \geq 1$ ,

$$\begin{aligned}\alpha_n &= \int_0^2 (x^2 - 4x) \cos \frac{n\pi x}{2} \, dx = \frac{2}{n\pi} \left[ (x^2 - 4x) \sin \frac{n\pi x}{2} \right]_0^2 - 2 \int_0^2 (x - 2) \sin \frac{n\pi x}{2} \, dx \\ &= \frac{8}{n^2\pi^2} \left[ (x - 2) \cos \frac{n\pi x}{2} \right]_0^2 - \int_0^2 \cos \frac{n\pi x}{2} \, dx = \frac{16}{n^2\pi^2} - \frac{32}{n^3\pi^3} \sin \frac{n\pi x}{2} \Big|_0^2 = \frac{16}{n^2\pi^2};\end{aligned}$$

$$C(x) = -\frac{8}{3} + \frac{16}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^2} \cos \frac{n\pi x}{2}. \text{ From Definition 12.1.3, } u(x, t) = -\frac{8}{3} + \frac{16}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^2} e^{-n^2\pi^2 t} \cos \frac{n\pi x}{2}.$$

**12.1.20.** From Example 11.3.5,  $C(x) = 4 - \frac{384}{\pi^4} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^4} \cos \frac{(2n-1)\pi x}{2}$ . From Definition 12.1.3,  $u(x, t) = 4 - \frac{384}{\pi^4} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^4} e^{-3(2n-1)^2\pi^2 t/4} \cos \frac{(2n-1)\pi x}{2}$ .

**12.1.22.**  $\alpha_0 = \frac{1}{L} \int_0^1 (3x^4 - 4Lx^3) \, dx = \frac{1}{L} \left( \frac{3x^5}{5} - x^4 \right) \Big|_0^1 = -\frac{2}{5}$ . Since  $f'(0) = f'(1) = 0$  and  $f'''(x) = 24(3x - 1)$ , Theorem 11.3.5(a) implies that

$$\begin{aligned}\alpha_n &= \frac{48}{n^3\pi^3} \int_0^1 (3x - 1) \sin n\pi x \, dx = -\frac{48}{n^4\pi^4} \left[ (3x - 1) \cos n\pi x \right]_0^1 - 3 \int_0^1 \cos n\pi x \, dx \\ &= -\frac{48}{n^4\pi^4} [(-1)^n 2 + 1] + \frac{144}{n^5\pi^5} \sin n\pi x \Big|_0^1 = -\frac{48}{n^4\pi^4} [1 + (-1)^n 2], \quad n \geq 1;\end{aligned}$$

$$C(x) = -\frac{2}{5} - \frac{48}{\pi^4} \sum_{n=1}^{\infty} \frac{1 + (-1)^n 2}{n^4} \cos n\pi x. \text{ From Definition 12.1.3,}$$

$$u(x, t) = -\frac{2}{5} - \frac{48}{\pi^4} \sum_{n=1}^{\infty} \frac{1 + (-1)^n 2}{n^4} e^{-3n^2\pi^2 t} \cos n\pi x.$$



**12.1.24.**  $\alpha_0 = \frac{1}{\pi} \int_0^\pi (x^4 - 2\pi x^3 + \pi^2 x^2) dx = \frac{1}{\pi} \left( \frac{x^5}{5} - \frac{\pi x^4}{2} + \frac{\pi^2 x^3}{3} \right) \Big|_0^\pi = \frac{\pi^4}{30}$ . Since  $f'(0) = f'(\pi) = 0$  and  $f'''(x) = 12(2x - \pi)$ , Theorem 11.3.5(a) implies that

$$\begin{aligned} \alpha_n &= \frac{24}{n^3 \pi} \int_0^\pi (2x - \pi) \sin nx \, dx = -\frac{24}{n^4 \pi} \left[ (2x - \pi) \cos nx \Big|_0^\pi - 2 \int_0^\pi \cos nx \, dx \right] \\ &= -\frac{24}{n^4 \pi} [(-1)^n \pi + \pi] + \frac{48}{n^5 \pi} \sin nx \Big|_0^\pi = -\frac{24}{n^4} [1 + (-1)^n] \\ &= \begin{cases} 0 & \text{if } n = 2m - 1, \\ -\frac{3}{m^4} & \text{if } n = 2m, \end{cases} \quad n \geq 1; \end{aligned}$$

$C(x) = \frac{\pi^4}{30} - 3 \sum_{n=1}^{\infty} \frac{1}{n^4} \cos 2nx$ . From Definition 12.1.3,  $u(x, t) = \frac{\pi^4}{30} - 3 \sum_{n=1}^{\infty} \frac{1}{n^4} e^{-4n^2 t} \cos 2nx$ .

**12.1.26.**

$$\begin{aligned} \alpha_n &= \frac{2}{\pi} \int_0^\pi (\pi x - x^2) \sin \frac{(2n-1)x}{2} \, dx \\ &= -\frac{4}{(2n-1)\pi} \left[ (\pi x - x^2) \cos \frac{(2n-1)x}{2} \Big|_0^\pi - \int_0^\pi (\pi - 2x) \cos \frac{(2n-1)x}{2} \, dx \right] \\ &= \frac{8}{(2n-1)^2 \pi} \left[ (\pi - 2x) \sin \frac{(2n-1)x}{2} \Big|_0^\pi + 2 \int_0^\pi \sin \frac{(2n-1)x}{2} \, dx \right] \\ &= (-1)^n \frac{8}{(2n-1)^2} - \frac{32}{(2n-1)^3 \pi} \cos \frac{(2n-1)x}{2} \Big|_0^\pi \\ &= (-1)^n \frac{8}{(2n-1)^2} + \frac{32}{(2n-1)^3 \pi}; \end{aligned}$$

$S_M(x) = 8 \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} \left[ (-1)^n + \frac{4}{(2n-1)\pi} \right] \sin \frac{(2n-1)x}{2}$ . From Definition 12.1.4,

$$u(x, t) = 8 \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} \left[ (-1)^n + \frac{4}{(2n-1)\pi} \right] e^{-3(2n-1)^2 t/4} \sin \frac{(2n-1)x}{2}.$$

**12.1.28.** Since  $f(0) = f'(1) = 0$ , and  $f''(x) = 6(1 - 2x)$ , Theorem 11.3.5(d) implies that

$$\begin{aligned} \alpha_n &= -\frac{48}{(2n-1)^2 \pi^2} \int_0^1 (1 - 2x) \sin \frac{(2n-1)\pi x}{2} \, dx \\ &= \frac{96}{(2n-1)^3 \pi^3} \left[ (1 - 2x) \cos \frac{(2n-1)\pi x}{2} \Big|_0^1 + 2 \int_0^1 \cos \frac{(2n-1)\pi x}{2} \, dx \right] \\ &= \frac{96}{(2n-1)^3 \pi^3} \left[ -1 + \frac{4}{(2n-1)\pi} \sin \frac{(2n-1)\pi x}{2} \Big|_0^1 \right] \\ &= -\frac{96}{(2n-1)^3 \pi^3} \left[ 1 + (-1)^n \frac{4}{(2n-1)\pi} \right]; \end{aligned}$$

$$S_M(x) = -\frac{96}{\pi^3} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^3} \left[ 1 + (-1)^n \frac{4}{(2n-1)\pi} \right] \sin \frac{(2n-1)\pi x}{2}. \text{ From Definition 12.1.4,}$$

$$u(x, t) = -\frac{96}{\pi^3} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^3} \left[ 1 + (-1)^n \frac{4}{(2n-1)\pi} \right] e^{-(2n-1)^2 \pi^2 t} \sin \frac{(2n-1)\pi x}{2}.$$

**12.1.30.** Since  $f(0) = f'(L) = f''(0) = 0$  and  $f'''(x) = 6$ , Theorem 11.3.5(d) and Exercise 11.3.50(b) imply that

$$\begin{aligned} \alpha_n &= -\frac{96}{(2n-1)^3 \pi^3} \int_0^1 \cos \frac{(2n-1)\pi x}{2} dx \\ &= -\frac{192}{(2n-1)^4 \pi^4} \sin \frac{(2n-1)\pi x}{2} \Big|_0^1 = (-1)^n \frac{192}{(2n-1)^4 \pi^4}; \end{aligned}$$

$$S_M(x) = \frac{192}{\pi^4} \sum_{n=1}^{\infty} \frac{(-1)^n}{(2n-1)^4} \sin \frac{(2n-1)\pi x}{2}. \text{ From Definition 12.1.4,}$$

$$u(x, t) = \frac{192}{\pi^4} \sum_{n=1}^{\infty} \frac{(-1)^n}{(2n-1)^4} e^{-(2n-1)^2 \pi^2 t} \sin \frac{(2n-1)\pi x}{2}.$$

**12.1.32.** Since  $f(0) = f'(1) = f''(0) = 0$  and  $f'''(x) = 12(2x-1)$ , Theorem 11.3.5(d) and Exercise 11.3.50(b) imply that

$$\begin{aligned} \alpha_n &= -\frac{192}{(2n-1)^3 \pi^3} \int_0^1 (2x-1) \cos \frac{(2n-1)\pi x}{2} dx \\ &= -\frac{384}{(2n-1)^4 \pi^4} \left[ (2x-1) \sin \frac{(2n-1)\pi x}{2} \Big|_0^1 - 2 \int_0^1 \sin \frac{(2n-1)\pi x}{2} dx \right] \\ &= -\frac{384}{(2n-1)^4 \pi^4} \left[ (-1)^{n+1} 1 + \frac{4}{(2n-1)\pi} \cos \frac{(2n-1)\pi x}{2} \Big|_0^1 \right] \\ &= -\frac{384}{(2n-1)^4 \pi^4} \left[ (-1)^{n+1} 1 - \frac{4}{(2n-1)\pi} \right] = \frac{384}{(2n-1)^4 \pi^4} \left[ (-1)^n + \frac{4}{(2n-1)\pi} \right]; \end{aligned}$$

$$S_M(x) = \frac{384}{\pi^4} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^4} \left[ (-1)^n + \frac{4}{(2n-1)\pi} \right] \sin \frac{(2n-1)\pi x}{2}. \text{ From Definition 12.1.4,}$$

$$u(x, t) = \frac{384}{\pi^4} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^4} \left[ (-1)^n + \frac{4}{(2n-1)\pi} \right] e^{-(2n-1)^2 \pi^2 t} \sin \frac{(2n-1)\pi x}{2}.$$

**12.1.36.** From Example 11.3.3,  $C_M(x) = \frac{8}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} \cos \frac{(2n-1)\pi x}{2}$ . From Definition 12.1.5,

$$u(x, t) = \frac{8}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} e^{-3(2n-1)^2 \pi^2 t/4} \cos \frac{(2n-1)\pi x}{2}.$$

**12.1.38.** Since  $f'(0) = f(\pi) = 0$  and  $f''(x) = -2$ , Theorem 11.3.5(c) implies that

$$\alpha_n = \frac{16}{(2n-1)^2\pi} \int_0^\pi \cos \frac{(2n-1)x}{2} dx = \frac{32}{(2n-1)^3\pi} \sin \frac{(2n-1)x}{2} \Big|_0^\pi = (-1)^{n+1} \frac{32}{(2n-1)^3\pi};$$

$$C_M(x) = -\frac{32}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^n}{(2n-1)^3} \cos \frac{(2n-1)x}{2}. \text{ From Definition 12.1.5,}$$

$$u(x, t) = -\frac{32}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^n}{(2n-1)^3} e^{-7(2n-1)^2 t/4} \cos \frac{(2n-1)x}{2}.$$

**12.1.40.** Since  $f'(0) = f(1) = 0$  and  $f''(x) = 6(2x+1)$ , Theorem 11.3.5(c) implies that

$$\begin{aligned} \alpha_n &= -\frac{48L}{(2n-1)^2\pi^2} \int_0^1 (2x+1) \cos \frac{(2n-1)\pi x}{2} dx \\ &= -\frac{96}{(2n-1)^3\pi^3} \left[ (2x+1) \sin \frac{(2n-1)\pi x}{2} \Big|_0^1 - 2 \int_0^1 \sin \frac{(2n-1)\pi x}{2} dx \right] \\ &= -\frac{96}{(2n-1)^3\pi^3} \left[ (-1)^{n+1} 3 - \frac{4}{(2n-1)\pi} \cos \frac{(2n-1)\pi x}{2} \Big|_0^1 \right] \\ &= \frac{96}{(2n-1)^3\pi^3} \left[ (-1)^n 3 + \frac{4}{(2n-1)\pi} \right]; \end{aligned}$$

$$C_M(x) = \frac{96}{\pi^3} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^3} \left[ (-1)^n 3 + \frac{4}{(2n-1)\pi} \right] \cos \frac{(2n-1)\pi x}{2}. \text{ From Definition 12.1.5,}$$

$$u(x, t) = \frac{96}{\pi^3} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^3} \left[ (-1)^n 3 + \frac{4}{(2n-1)\pi} \right] e^{-(2n-1)^2 \pi^2 t/4} \cos \frac{(2n-1)\pi x}{2}.$$

**12.1.42.** Theorem 11.3.5(c) and Exercise 11.3.42(b) imply that Since  $f'(0) = f(1) = f''(1) = 0$  and  $f'''(x) = 12(2x-1)$ ,

$$\begin{aligned} \alpha_n &= \frac{192}{(2n-1)^3\pi^3} \int_0^1 (2x-1) \sin \frac{(2n-1)\pi x}{2} dx \\ &= -\frac{384}{(2n-1)^4\pi^4} \left[ (2x-1) \cos \frac{(2n-1)\pi x}{2} \Big|_0^1 - 2 \int_0^1 \cos \frac{(2n-1)\pi x}{2} dx \right] \\ &= -\frac{384}{(2n-1)^4\pi^4} \left[ 1 - \frac{4}{(2n-1)\pi} \sin \frac{(2n-1)\pi x}{2} \Big|_0^1 \right] = -\frac{384}{(2n-1)^4\pi^4} \left[ 1 + \frac{(-1)^n 4}{(2n-1)\pi} \right]; \end{aligned}$$

$$C_M(x) = -\frac{384}{\pi^4} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^4} \left[ 1 + \frac{(-1)^n 4}{(2n-1)\pi} \right] \cos \frac{(2n-1)\pi x}{2}. \text{ From Definition 12.1.5,}$$

$$u(x, t) = -\frac{384}{\pi^4} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^4} \left[ 1 + \frac{(-1)^n 4}{(2n-1)\pi} \right] e^{-(2n-1)^2 \pi^2 t/4} \cos \frac{(2n-1)\pi x}{2}.$$

$$12.1.44. \alpha_n = \frac{2}{L} \int_0^{L/2} \sin \frac{n\pi x}{L} dx = -\frac{2}{n\pi} \cos \frac{n\pi x}{L} \Big|_0^{L/2} = \frac{2}{n\pi} \left[ 1 - \cos \frac{n\pi}{2} \right];$$

$$S(x) = \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} \left[ 1 - \cos \frac{n\pi}{2} \right] \sin \frac{n\pi x}{L}. \text{ From Definition 12.1.1,}$$

$$u(x, t) = \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} \left[ 1 - \cos \frac{n\pi}{2} \right] e^{-n^2 \pi^2 t^2 / L^2} \sin \frac{n\pi x}{L}.$$

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$$\begin{aligned} \alpha_n &= \frac{2}{L} \int_0^{L/2} \sin \frac{(2n-1)\pi x}{2L} dx = -\frac{4}{(2n-1)\pi} \cos \frac{(2n-1)\pi x}{2L} \Big|_0^{L/2} \\ &= \frac{4}{(2n-1)\pi} \left[ 1 - \cos \frac{(2n-1)\pi}{4} \right]; \end{aligned}$$

$$S_M(x) = \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{1}{2n-1} \left[ 1 - \cos \frac{(2n-1)\pi}{4} \right] \sin \frac{(2n-1)\pi x}{2L}. \text{ From Definition 12.1.4,}$$

$$u(x, t) = \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{1}{2n-1} \left[ 1 - \cos \frac{(2n-1)\pi}{4} \right] e^{-(2n-1)^2 \pi^2 a^2 t / 4L^2} \sin \frac{(2n-1)\pi x}{2L}.$$

12.1.48. Let  $u(x, t) = v(x, t) + q(x)$ ; then  $u_t = v_t$  and  $u_{xx} = v_{xx} + q''$ , so

$$\begin{aligned} v_t &= 9v_{xx} + 9q'' - 54x, \quad 0 < x < 4, \quad t > 0, \\ v(0, t) &= 1 - q(0), \quad v(4, t) = 61 - q(4), \quad t > 0, \\ v(x, 0) &= 2 - x + x^3 - q(x), \quad 0 \leq x \leq 4. \end{aligned} \tag{A}$$

We want  $q''(x) = 6x$ ,  $q(0) = 1$ ,  $q(4) = 61$ ;  $q(x) = x^3 + a_1 + a_2x$ ;  $q(0) = 1 \Rightarrow a_1 = 1$ ;  $q(x) = x^3 + 1 + a_2x$ ;  $q(4) = 61 \Rightarrow a_2 = -1$ ;  $q(x) = x^3 + 1 - x$ . Now (A) reduces to

$$\begin{aligned} v_t &= 9v_{xx}, \quad 0 < x < 4, \quad t > 0, \\ v(0, t) &= 0, \quad v(4, t) = 0, \quad t > 0, \\ v(x, 0) &= 1, \quad 0 \leq x \leq 4, \end{aligned}$$

which we solve by separation of variables.

$$\begin{aligned} \alpha_n &= \frac{1}{2} \int_0^L \sin \frac{n\pi x}{4} dx = -\frac{2}{n\pi} \cos \frac{n\pi x}{4} \Big|_0^4 \\ &= \frac{2}{n\pi} [1 - (-1)^n] = \begin{cases} \frac{4}{(2m-1)\pi} & \text{if } n = 2m-1, \\ 0 & \text{if } n = 2m; \end{cases} \end{aligned}$$

$$S(x) = \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{1}{(2n-1)} \sin \frac{(2n-1)\pi x}{4}. \text{ From Definition 12.1.1,}$$

$$v(x, t) = \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{1}{(2n-1)} e^{-9\pi^2 n^2 t / 16} \sin \frac{(2n-1)\pi x}{4}.$$

Therefore,

$$u(x, t) = 1 - x + x^3 + \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{e^{-9\pi^2 n^2 t/16}}{(2n-1)} \sin \frac{(2n-1)\pi x}{4}.$$

**12.1.50.** Let  $u(x, t) = v(x, t) + q(x)$ ; then  $u_t = v_t$  and  $u_{xx} = v_{xx} + q''$ , so

$$\begin{aligned} v_t &= 3u_{xx} + 3q'' - 18x, & 0 < x < 1, & \quad t > 0, \\ v_x(0, t) &= -1 - q'(0), & v(1, t) &= -1 - q(1), & \quad t > 0, \\ v(x, 0) &= x^3 - 2x - q(x), & 0 \leq x \leq 1. \end{aligned} \quad (\text{A})$$

We want  $q''(x) = 6x$ ,  $q'(0) = -1$ ,  $q(1) = -1$ ;  $q'(x) = 3x^2 + a_2$ ;  $q'(0) = -1 \Rightarrow a_2 = -1$ ;  $q'(x) = 3x^2 - 1$ ;  $q(x) = x^3 - x + a_1x$ ;  $q(1) = -1 \Rightarrow a_1 = -1$ ;  $q(x) = x^3 - x - 1$ . Now (A) reduces to

$$\begin{aligned} v_t &= 3u_{xx}, & 0 < x < 1, & \quad t > 0, \\ v_x(0, t) &= 0, & v(1, t) &= 0, & \quad t > 0, \\ v(x, 0) &= 1 - x, & 0 \leq x \leq 1. \end{aligned}$$

From Example 11.3.3,  $C_M(x) = \frac{8}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} \cos \frac{(2n-1)\pi x}{2}$ . From Definition 12.1.5,  $v(x, t) = \frac{8}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} e^{-3(2n-1)^2 \pi^2 t/4} \cos \frac{(2n-1)\pi x}{2}$ . Therefore,

$$u(x, t) = -1 - x + x^3 + \frac{8}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} e^{-3(2n-1)^2 \pi^2 t/4} \cos \frac{(2n-1)\pi x}{2}.$$

**12.1.52.** Let  $u(x, t) = v(x, t) + q(x)$ ; then  $u_t = v_t$  and  $u_{xx} = v_{xx} + q''$ , so

$$\begin{aligned} v_t &= v_{xx} + q'' + \pi^2 \sin \pi x, & 0 < x < 1, & \quad t > 0, \\ v(0, t) &= -q(0), & v_x(1, t) &= -\pi - q'(1), & \quad t > 0, \\ v(x, 0) &= 2 \sin \pi x - q(x), & 0 \leq x \leq 1. \end{aligned} \quad (\text{A})$$

We want  $q''(x) = -\pi^2 \sin \pi x$ ,  $q(0) = 0$ ,  $q'(1) = -\pi$ ;  $q'(x) = \pi \cos \pi x + a_2$ ;  $q'(1) = -\pi \Rightarrow a_2 = 0$ ;  $q'(x) = \pi \cos \pi x$ ;  $q(x) = \sin \pi x + a_1$ ;  $q(0) = 0 \Rightarrow a_1 = 0$ ;  $q(x) = \sin \pi x$ . Now (A) reduces to

$$\begin{aligned} v_t &= v_{xx}, & 0 < x < 1, & \quad t > 0, \\ v(0, t) &= 0, & v_x(1, t) &= 0, & \quad t > 0, \\ v(x, 0) &= \sin \pi x, & 0 \leq x \leq 1. \end{aligned}$$

$$\begin{aligned} \alpha_n &= 2 \int_0^1 \sin \pi x \sin \frac{(2n-1)\pi x}{2} dx = \int_0^1 \left[ \frac{\cos(2n-3)\pi x}{2} - \frac{\cos(2n+1)\pi x}{2} \right] dx \\ &= \frac{2}{\pi} \left[ \frac{\sin(2n-3)\pi x/2}{(2n-3)} - \frac{\sin(2n+1)\pi x/2}{(2n+1)} \right] \Big|_0^1 \\ &= (-1)^n \frac{2}{\pi} \left[ \frac{1}{2n-3} - \frac{1}{2n+1} \right] = (-1)^n \frac{8}{\pi (2n+1)(2n-3)}; \end{aligned}$$

$S_M(x) = \frac{8}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^n}{(2n+1)(2n-3)} \sin \frac{(2n-1)\pi x}{2}$ . From Definition 12.1.4,

$$v(x, t) = \frac{8}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^n}{(2n+1)(2n-3)} e^{-(2n-1)^2 \pi^2 t/4} \sin \frac{(2n-1)\pi x}{2}.$$

Therefore,  $u(x, t) = \sin \pi x + \frac{8}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^n}{(2n+1)(2n-3)} e^{-(2n-1)^2 \pi^2 t/4} \sin \frac{(2n-1)\pi x}{2}$ .

**12.1.54.** (a) Since  $f$  is piecewise smooth of  $[0, L]$ , there is a constant  $K$  such that  $|f(x)| \leq K$ ,  $0 \leq x \leq L$ . Therefore,  $|\alpha_n| = \frac{2}{L} \left| \int_0^L f(x) \sin \frac{n\pi x}{L} dx \right| \leq \frac{2}{L} \int_0^L |f(x)| dx = 2K$ . Hence,  $|\alpha_n e^{-n^2 \pi^2 a^2 t/L^2}| \leq 2K e^{-n^2 \pi^2 a^2 t/L^2}$ , so  $u(x, t)$  converges for all  $x$  if  $t > 0$ , by the comparison test.

(b) Let  $t$  be a fixed positive number. Apply Theorem 12.1.2 with  $z = x$  and  $w_n(x) = \alpha_n e^{-n^2 \pi^2 t/L^2} \sin \frac{n\pi x}{L}$ .

Then  $w'_n(x) = \frac{\pi}{L} n \alpha_n e^{-n^2 \pi^2 t/L^2} \cos \frac{n\pi x}{L}$ , so  $|w'_n(x)| \leq \frac{2K\pi}{L} n e^{-n^2 \pi^2 t/L^2}$ ,  $-\infty < x < \infty$ . Since  $\sum_{n=1}^{\infty} n e^{-n^2 \pi^2 a^2 t/L^2}$  converges if  $t > 0$ , Theorem 12.1.1 (with  $z_1 = x_1$  and  $z_2 = x_2$  arbitrary) implies the conclusion.

(c) Since  $\sum_{n=1}^{\infty} n^2 e^{-n^2 \pi^2 a^2 t/L^2}$  also converges if  $t > 0$ , an argument like that in (b) with  $w_n(x) = n \alpha_n e^{-n^2 \pi^2 t/L^2} \cos \frac{n\pi x}{L}$  yields the conclusion.

(d) Let  $x$  be arbitrary, but fixed. Apply Theorem 12.1.2 with  $z = t$  and  $w_n(t) = \alpha_n e^{-n^2 \pi^2 a^2 t/L^2} \sin \frac{n\pi x}{L}$ . Then  $w'_n(t) = -\frac{\pi^2 a^2}{L^2} n^2 \alpha_n e^{-n^2 \pi^2 a^2 t/L^2} \sin \frac{n\pi x}{L}$ , so  $|w'_n(t)| \leq \frac{2K\pi^2 a^2}{L} n^2 e^{-n^2 \pi^2 a^2 t_0/L^2}$  if  $t > t_0$ . Since  $\sum_{n=1}^{\infty} n^2 e^{-n^2 \pi^2 a^2 t_0/L^2}$  converges, Theorem 12.1.1 (with  $z_1 = t_0 > 0$  and  $z_2 = t_1$  arbitrary) implies the conclusion for  $t \geq t_0$ . However, since  $t_0$  is an arbitrary positive number, this holds for  $t > 0$ .

## 12.2 The Wave Equation

**12.2.1.**  $\beta_n = 2 \left[ \int_0^{1/2} x \sin n\pi x + \int_{1/2}^1 (1-x) \sin n\pi x dx \right];$

$$\begin{aligned} \int_0^{1/2} x \sin n\pi x dx &= -\frac{1}{n\pi} \left[ x \cos n\pi x \right]_0^{1/2} - \int_0^{1/2} \cos n\pi x dx \\ &= -\frac{1}{2n\pi} \cos \frac{n\pi}{2} + \frac{1}{n^2 \pi^2} \sin n\pi x \Big|_0^{1/2} = -\frac{1}{2n\pi} \cos \frac{n\pi}{2} + \frac{1}{n^2 \pi^2} \sin \frac{n\pi}{2}; \end{aligned}$$

$$\begin{aligned} \int_0^{1/2} (1-x) \sin n\pi x dx &= -\frac{1}{n\pi} \left[ (1-x) \cos n\pi x \right]_{1/2}^1 + \int_0^{1/2} \cos n\pi x dx \\ &= \frac{1}{2n\pi} \cos \frac{n\pi}{2} - \frac{1}{n^2 \pi^2} \sin n\pi x \Big|_{1/2}^1 = \frac{1}{2n\pi} \cos \frac{n\pi}{2} + \frac{1}{n^2 \pi^2} \sin \frac{n\pi}{2}; \end{aligned}$$

$$b_n = \frac{4}{n^2\pi^2} \sin \frac{n\pi}{2} = \begin{cases} (-1)^{m+1} \frac{4}{(2m-1)^2\pi^2} & \text{if } n = 2m-1 \\ 0 & \text{if } n = 2m; \end{cases}$$

$$S_g(x) = \frac{4}{\pi^2} \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{(2n-1)^2} \sin(2n-1)\pi x. \text{ From Definition 12.1.1,}$$

$$u(x, t) = \frac{4}{3\pi^3} \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{(2n-1)^3} \sin 3(2n-1)\pi t \sin(2n-1)\pi x.$$

**12.2.2.** Since  $f(0) = f(1) = 0$  and  $f''(x) = -2$ , Theorem 11.3.5(b) implies that

$$\begin{aligned} \alpha_n &= \frac{4}{n^2\pi^2} \int_0^1 \sin n\pi x \, dx = -\frac{4}{n^3\pi^3} \cos n\pi x \Big|_0^1 = -\frac{4}{n^3\pi^2} (\cos n\pi - 1) \\ &= \begin{cases} \frac{8}{(2m-1)^3\pi^3}, & \text{if } n = 2m-1, \\ 0, & \text{if } n = 2m; \end{cases} \end{aligned}$$

$$S_f(x) = \frac{8}{\pi^3} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^3} \sin(2n-1)\pi x. \text{ From Definition 12.1.1,}$$

$$u(x, t) = \frac{8}{\pi^3} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^3} \cos 3(2n-1)\pi t \sin(2n-1)\pi x.$$

**12.2.4.** Since  $g(0) = g(1) = 0$  and  $g''(x) = -2$ , Theorem 11.3.5(b) implies that

$$\begin{aligned} \beta_n &= \frac{4}{n^2\pi^2} \int_0^1 \sin n\pi x \, dx = -\frac{4}{n^3\pi^3} \cos n\pi x \Big|_0^1 = -\frac{4}{n^3\pi^2} (\cos n\pi - 1) \\ &= \begin{cases} \frac{8}{(2m-1)^3\pi^3}, & \text{if } n = 2m-1, \\ 0, & \text{if } n = 2m; \end{cases} \end{aligned}$$

$$S_g(x) = \frac{8}{\pi^3} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^3} \sin(2n-1)\pi x. \text{ From Definition 12.1.1,}$$

$$u(x, t) = \frac{8}{3\pi^4} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^4} \sin 3(2n-1)\pi t \sin(2n-1)\pi x.$$

**12.2.6.** From Example 11.2.6,  $S_f(x) = \frac{324}{\pi^3} \sum_{n=1}^{\infty} \frac{(-1)^n}{n^3} \sin \frac{n\pi x}{3}$ . From Definition 12.1.1,

$$u(x, t) = \frac{324}{\pi^3} \sum_{n=1}^{\infty} \frac{(-1)^n}{n^3} \cos \frac{8\pi nt}{3} \sin \frac{n\pi x}{3}.$$

**12.2.8.** From Example 11.2.6  $S_g(x) = \frac{324}{\pi^3} \sum_{n=1}^{\infty} \frac{(-1)^n}{n^3} \sin \frac{n\pi x}{3}$ . From Definition 12.1.1,

$$u(x, t) = \frac{81}{2\pi^4} \sum_{n=1}^{\infty} \frac{(-1)^n}{n^4} \sin \frac{8\pi nt}{3} \sin \frac{n\pi x}{3}.$$

**12.2.10.**

$$\begin{aligned}
\alpha_1 &= \frac{2}{\pi} \int_0^\pi x \sin^2 x \, dx = \frac{1}{\pi} \int_0^\pi x(1 - \cos 2x) \, dx = \frac{x^2}{2\pi} \Big|_0^\pi - \frac{1}{\pi} \int_0^\pi x \cos 2x \, dx \\
&= \frac{\pi}{2} - \frac{1}{2\pi} \left[ x \sin 2x \Big|_0^\pi - \int_0^\pi \cos 2x \, dx \right] = \frac{\pi}{2} + \frac{\sin 2x}{4\pi} \Big|_0^\pi = \frac{\pi}{2};
\end{aligned}$$

if  $n \geq 2$ , then

$$\begin{aligned}
\alpha_n &= \frac{2}{\pi} \int_0^\pi x \sin x \sin nx \, dx = \frac{1}{\pi} \int_0^\pi x [\cos(n-1)x - \cos(n+1)x] \, dx \\
&= \frac{1}{\pi} \left[ x \left[ \frac{\sin(n-1)x}{n-1} - \frac{\sin(n+1)x}{n+1} \right] \Big|_0^\pi - \int_0^\pi \left[ \frac{\sin(n-1)x}{n-1} - \frac{\sin(n+1)x}{n+1} \right] \, dx \right] \\
&= \frac{1}{\pi} \left[ \frac{\cos(n-1)x}{(n-1)^2} - \frac{\cos(n+1)x}{(n+1)^2} \right] \Big|_0^\pi = \frac{1}{\pi} \left[ \frac{1}{(n-1)^2} - \frac{1}{(n+1)^2} \right] [(-1)^{n+1} - 1] \\
&= \frac{4n}{(n^2-1)^2\pi} [(-1)^{n+1} - 1] = \begin{cases} 0 & \text{if } n = 2m-1, \\ -\frac{16m}{(4m^2-1)\pi} & \text{if } n = 2m; \end{cases}
\end{aligned}$$

$$S_f(x) = \frac{\pi}{2} \sin x - \frac{16}{\pi} \sum_{n=1}^{\infty} \frac{n}{(4n^2-1)^2} \sin 2nx. \text{ From Definition 12.1.1,}$$

$$u(x, t) = \frac{\pi}{2} \cos \sqrt{5} t \sin x - \frac{16}{\pi} \sum_{n=1}^{\infty} \frac{n}{(4n^2-1)^2} \cos 2n\sqrt{5} t \sin 2nx.$$

**12.2.12.**

$$\begin{aligned}
\beta_1 &= \frac{2}{\pi} \int_0^\pi x \sin^2 x \, dx = \frac{1}{\pi} \int_0^\pi x(1 - \cos 2x) \, dx = \frac{x^2}{2\pi} \Big|_0^\pi - \frac{1}{\pi} \int_0^\pi x \cos 2x \, dx \\
&= \frac{\pi}{2} - \frac{1}{2\pi} \left[ x \sin 2x \Big|_0^\pi - \int_0^\pi \cos 2x \, dx \right] = \frac{\pi}{2} + \frac{\sin 2x}{4\pi} \Big|_0^\pi = \frac{\pi}{2};
\end{aligned}$$

if  $n \geq 2$  then

$$\begin{aligned}
\beta_n &= \frac{2}{\pi} \int_0^\pi x \sin x \sin nx \, dx = \frac{1}{\pi} \int_0^\pi x [\cos(n-1)x - \cos(n+1)x] \, dx \\
&= \frac{1}{\pi} \left[ x \left[ \frac{\sin(n-1)x}{n-1} - \frac{\sin(n+1)x}{n+1} \right] \Big|_0^\pi - \int_0^\pi \left[ \frac{\sin(n-1)x}{n-1} - \frac{\sin(n+1)x}{n+1} \right] \, dx \right] \\
&= \frac{1}{\pi} \left[ \frac{\cos(n-1)x}{(n-1)^2} - \frac{\cos(n+1)x}{(n+1)^2} \right] \Big|_0^\pi = \frac{1}{\pi} \left[ \frac{1}{(n-1)^2} - \frac{1}{(n+1)^2} \right] [(-1)^{n+1} - 1] \\
&= \frac{4n}{(n^2-1)^2\pi} [(-1)^{n+1} - 1] = \begin{cases} 0 & \text{if } n = 2m-1, \\ -\frac{16m}{(4m^2-1)\pi} & \text{if } n = 2m; \end{cases}
\end{aligned}$$



$S_g(x) = \frac{\pi}{2} \sin x - \frac{16}{\pi} \sum_{n=1}^{\infty} \frac{n}{(4n^2-1)^2} \sin 2nx$ . From Definition 12.1.1,

$$u(x, t) = \frac{\pi}{2\sqrt{5}} \sin \sqrt{5} t \sin x - \frac{8}{\pi\sqrt{5}} \sum_{n=1}^{\infty} \frac{1}{(4n^2-1)^2} \sin 2n\sqrt{5} t \sin 2nx.$$

**12.2.14.** Since  $f(0) = f(1) = f''(0) = f''(L) = 0$  and  $f^{(4)} = 360x$ , Theorem 11.3.5(b) and Exercise 35(b) of Section 11.3 imply that

$$\begin{aligned} \alpha_n &= \frac{720}{n^4\pi^4} \int_0^1 x \sin n\pi x \, dx = -\frac{720}{n^5\pi^5} \left[ x \cos n\pi x \Big|_0^1 - \int_0^1 \cos n\pi x \, dx \right] \\ &= (-1)^{n+1} \frac{720}{n^5\pi^5} + \frac{720}{n^6\pi^6} \sin \frac{n\pi x}{L} \Big|_0^1 = (-1)^{n+1} \frac{720}{n^5\pi^5}; \end{aligned}$$

$S_f(x) = -\frac{720}{\pi^5} \sum_{n=1}^{\infty} \frac{(-1)^n}{n^5} \sin n\pi x$ . From Definition 12.1.1,  $u(x, t) = -\frac{720}{\pi^5} \sum_{n=1}^{\infty} \frac{(-1)^n}{n^5} \cos 3n\pi t \sin n\pi x$ .

**12.2.16.** (a)  $t$  must be in some interval of the form  $[mL/a, (m+1)L/a]$ . If  $\frac{mL}{a} \leq t \leq \left(m + \frac{1}{2}\right) \frac{L}{a}$ , then (i) holds with  $0 \leq \tau \leq L/2a$ . If  $\left(m + \frac{1}{2}\right) \frac{L}{a} \leq t \leq \frac{(m+1)L}{a}$ , then (ii) holds with  $0 \leq \tau \leq L/2a$ .

(b) Suppose that (i) holds. Since

$$\cos \frac{(2n-1)\pi a}{L} \left( \tau + \frac{mL}{a} \right) = \cos \frac{(2n-1)\pi a\tau}{L} \cos(2n-1)m\pi = (-1)^m \cos \frac{(2n-1)\pi a\tau}{L},$$

(A) implies that  $u(x, t) = (-1)^m u(x, \tau)$ .

Suppose that (ii) holds. Since

$$\begin{aligned} \cos \frac{(2n-1)\pi a}{L} \left( -\tau + \frac{(m+1)L}{a} \right) &= \cos \frac{(2n-1)\pi a\tau}{L} \cos(2n-1)(m+1)\pi \\ &= (-1)^{m+1} \cos \frac{(2n-1)\pi a\tau}{L}, \end{aligned}$$

(B) implies that  $u(x, t) = (-1)^{m+1} u(x, \tau)$ .

**12.2.18.** Since  $f'(0) = f(2) = 0$  and  $f''(x) = -2$ , Theorem 11.3.5(c) implies that

$$\begin{aligned} \alpha_n &= \frac{32}{(2n-1)^2\pi^2} \int_0^2 \cos \frac{(2n-1)\pi x}{4} \, dx \\ &= \frac{128}{(2n-1)^3\pi^3} \sin \frac{(2n-1)\pi x}{4} \Big|_0^4 = (-1)^{n+1} \frac{128}{(2n-1)^3\pi^3}; \end{aligned}$$

$C_{Mf}(x) = -\frac{128}{\pi^3} \sum_{n=1}^{\infty} \frac{(-1)^n}{(2n-1)^3} \cos \frac{(2n-1)\pi x}{4}$ . From Exercise 12.2.17,

$$u(x, t) = -\frac{128}{\pi^3} \sum_{n=1}^{\infty} \frac{(-1)^n}{(2n-1)^3} \cos \frac{3(2n-1)\pi t}{4} \cos \frac{(2n-1)\pi x}{4}.$$

**12.2.20.** Since  $g'(0) = g(2) = 0$  and  $g''(x) = -2$ , Theorem 11.3.5(c) implies that

$$\begin{aligned}\beta_n &= \frac{32}{(2n-1)^2\pi^2} \int_0^2 \cos \frac{(2n-1)\pi x}{4} dx \\ &= \frac{128}{(2n-1)^3\pi^3} \sin \frac{(2n-1)\pi x}{4} \Big|_0^4 = (-1)^{n+1} \frac{128}{(2n-1)^3\pi^3};\end{aligned}$$

$$C_{Mf}(x) = -\frac{128}{\pi^3} \sum_{n=1}^{\infty} \frac{(-1)^n}{(2n-1)^3} \cos \frac{(2n-1)\pi x}{4}.$$
 From Exercise 12.2.17,

$$u(x, t) = -\frac{512}{3\pi^4} \sum_{n=1}^{\infty} \frac{(-1)^n}{(2n-1)^4} \sin \frac{3(2n-1)\pi t}{4} \cos \frac{(2n-1)\pi x}{4}.$$

**12.2.22.** Since  $f'(0) = f(1) = 0$  and  $f''(x) = 6(2x+1)$ , Theorem 11.3.5(c) implies that

$$\begin{aligned}\alpha_n &= -\frac{48L}{(2n-1)^2\pi^2} \int_0^1 (2x+1) \cos \frac{(2n-1)\pi x}{2} dx \\ &= -\frac{96}{(2n-1)^3\pi^3} \left[ (2x+1) \sin \frac{(2n-1)\pi x}{2} \Big|_0^1 - 2 \int_0^1 \sin \frac{(2n-1)\pi x}{2} dx \right] \\ &= -\frac{96}{(2n-1)^3\pi^3} \left[ (-1)^{n+1} 3 - \frac{4}{(2n-1)\pi} \cos \frac{(2n-1)\pi x}{2} \Big|_0^1 \right] \\ &= \frac{96}{(2n-1)^3\pi^3} \left[ (-1)^n 3 + \frac{4}{(2n-1)\pi} \right]; \\ C_{Mf}(x) &= \frac{96}{\pi^3} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^3} \left[ (-1)^n 3 + \frac{4}{(2n-1)\pi} \right] \cos \frac{(2n-1)\pi x}{2}.\end{aligned}$$

From Exercise 12.2.17,

$$u(x, t) = \frac{96}{\pi^3} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^3} \left[ (-1)^n 3 + \frac{4}{(2n-1)\pi} \right] \cos \frac{(2n-1)\sqrt{5}\pi t}{2} \cos \frac{(2n-1)\pi x}{2}.$$

**12.2.24.** Since  $g'(0) = g(1) = 0$  and  $g''(x) = 6(2x+1)$ , Theorem 11.3.5(c) implies that

$$\begin{aligned}\beta_n &= -\frac{48L}{(2n-1)^2\pi^2} \int_0^1 (2x+1) \cos \frac{(2n-1)\pi x}{2} dx \\ &= -\frac{96}{(2n-1)^3\pi^3} \left[ (2x+1) \sin \frac{(2n-1)\pi x}{2} \Big|_0^1 - 2 \int_0^1 \sin \frac{(2n-1)\pi x}{2} dx \right] \\ &= -\frac{96}{(2n-1)^3\pi^3} \left[ (-1)^{n+1} 3 - \frac{4}{(2n-1)\pi} \cos \frac{(2n-1)\pi x}{2} \Big|_0^1 \right] \\ &= \frac{96}{(2n-1)^3\pi^3} \left[ (-1)^n 3 + \frac{4}{(2n-1)\pi} \right];\end{aligned}$$

$$C_{Mg}(x) = \frac{96}{\pi^3} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^3} \left[ (-1)^n 3 + \frac{4}{(2n-1)\pi} \right] \cos \frac{(2n-1)\pi x}{2}. \text{ From Exercise 12.2.17,}$$

$$u(x, t) = \frac{192}{\pi^4 \sqrt{5}} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^4} \left[ (-1)^n 3 + \frac{4}{(2n-1)\pi} \right] \sin \frac{(2n-1)\sqrt{5}\pi t}{2} \cos \frac{(2n-1)\pi x}{2}.$$

**12.2.26.** Since  $f'(0) = f(1) = f''(1) = 0$  and  $f'''(x) = 24(x-1)$ , Theorem 11.3.5(c) and Exercise 42(b) of Section 11.3 imply that

$$\begin{aligned} \alpha_n &= \frac{384}{(2n-1)^3 \pi^3} \int_0^1 (x-1) \sin \frac{(2n-1)\pi x}{2} dx \\ &= -\frac{768}{(2n-1)^4 \pi^4} \left[ (x-1) \cos \frac{(2n-1)\pi x}{2} \Big|_0^1 - \int_0^1 \cos \frac{(2n-1)\pi x}{2} dx \right] \\ &= -\frac{768}{(2n-1)^4 \pi^4} \left[ 1 - \frac{2}{(2n-1)\pi} \sin \frac{(2n-1)\pi x}{2} \Big|_0^1 \right] \\ &= -\frac{768}{(2n-1)^4 \pi^4} \left[ 1 + \frac{(-1)^n 2}{(2n-1)\pi} \right]; \end{aligned}$$

$$C_{Mf}(x) = -\frac{384}{\pi^4} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^4} \left[ 1 + \frac{(-1)^n 4}{(2n-1)\pi} \right] \cos \frac{(2n-1)\pi x}{2}. \text{ From Exercise 12.2.17,}$$

$$u(x, t) = -\frac{384}{\pi^4} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^4} \left[ 1 + \frac{(-1)^n 4}{(2n-1)\pi} \right] \cos \frac{3(2n-1)\pi t}{2} \cos \frac{(2n-1)\pi x}{2}.$$

**12.2.28.** Since  $g'(0) = g(1) = g''(1) = 0$  and  $g'''(x) = 24(x-1)$ , Theorem 11.3.5(c) and Exercise 11.2.42(b) imply that

$$\begin{aligned} \beta_n &= \frac{384}{(2n-1)^3 \pi^3} \int_0^1 (x-1) \sin \frac{(2n-1)\pi x}{2} dx \\ &= -\frac{768}{(2n-1)^4 \pi^4} \left[ (x-1) \cos \frac{(2n-1)\pi x}{2} \Big|_0^1 - \int_0^1 \cos \frac{(2n-1)\pi x}{2} dx \right] \\ &= -\frac{768}{(2n-1)^4 \pi^4} \left[ 1 - \frac{2}{(2n-1)\pi} \sin \frac{(2n-1)\pi x}{2} \Big|_0^1 \right] \\ &= -\frac{768}{(2n-1)^4 \pi^4} \left[ 1 + \frac{(-1)^n 2}{(2n-1)\pi} \right]; \end{aligned}$$

$$C_{Mg}(x) = -\frac{384}{\pi^4} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^4} \left[ 1 + \frac{(-1)^n 4}{(2n-1)\pi} \right] \cos \frac{(2n-1)\pi x}{2}.$$

From Exercise 12.2.17,

$$u(x, t) = -\frac{768}{3\pi^5} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^5} \left[ 1 + \frac{(-1)^n 4}{(2n-1)\pi} \right] \sin \frac{3(2n-1)\pi t}{2} \cos \frac{(2n-1)\pi x}{2}.$$

**12.2.30.** Since  $f'(0) = f(1) = f''(1) = 0$  and  $f'''(x) = 24(x-1)$ , Theorem 11.3.5(c) and Exercise 42(b) of Section 11.3 imply that

$$\begin{aligned}\alpha_n &= \frac{384}{(2n-1)^3\pi^3} \int_0^1 (x-1) \sin \frac{(2n-1)\pi x}{2} dx \\ &= -\frac{768}{(2n-1)^4\pi^4} \left[ (x-1) \cos \frac{(2n-1)\pi x}{2} \Big|_0^1 - \int_0^1 \cos \frac{(2n-1)\pi x}{2} dx \right] \\ &= -\frac{768}{(2n-1)^4\pi^4} \left[ 1 - \frac{2}{(2n-1)\pi} \sin \frac{(2n-1)\pi x}{2} \Big|_0^1 \right] \\ &= -\frac{768}{(2n-1)^4\pi^4} \left[ 1 + \frac{(-1)^n 2}{(2n-1)\pi} \right];\end{aligned}$$

$$C_{Mf}(x) = -\frac{768}{\pi^4} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^4} \left[ 1 + \frac{(-1)^n 2}{(2n-1)\pi} \right] \cos \frac{(2n-1)\pi x}{2}. \text{ From Exercise 12.2.17,}$$

$$u(x, t) = -\frac{768}{\pi^4} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^4} \left[ 1 + \frac{(-1)^n 2}{(2n-1)\pi} \right] \cos \frac{(2n-1)\pi t}{2} \cos \frac{(2n-1)\pi x}{2}.$$

**12.2.32.** Setting  $A = (2n-1)\pi x/2L$  and  $B = (2n-1)\pi at/2L$  in the identities

$\cos A \cos B = \frac{1}{2}[\cos(A+B) + \cos(A-B)]$  and  $\cos A \sin B = \frac{1}{2}[\sin(A+B) - \sin(A-B)]$  yields

$$\cos \frac{(2n-1)\pi at}{2L} \cos \frac{(2n-1)\pi x}{2L} = \frac{1}{2} \left[ \cos \frac{(2n-1)\pi(x+at)}{2L} + \cos \frac{(2n-1)\pi(x-at)}{2L} \right] \quad (\text{A})$$

and

$$\begin{aligned}\sin \frac{(2n-1)\pi at}{2L} \cos \frac{(2n-1)\pi x}{2L} &= \frac{1}{2} \left[ \sin \frac{(2n-1)\pi(x+at)}{2L} - \sin \frac{(2n-1)\pi(x-at)}{2L} \right] \\ &= \frac{(2n-1)\pi}{4L} \int_{x-at}^{x+at} \cos \frac{(2n-1)\pi \tau}{2L} d\tau.\end{aligned} \quad (\text{B})$$

Since  $C_{Mf}(x) = \sum_{n=1}^{\infty} \alpha_n \cos \frac{(2n-1)\pi x}{2L}$ , (A) implies that

$$\sum_{n=1}^{\infty} \alpha_n \cos \frac{(2n-1)\pi at}{2L} \cos \frac{(2n-1)\pi x}{2L} = \frac{1}{2} [C_{Mf}(x+at) + C_{Mf}(x-at)]. \quad (\text{C})$$

Since it can be shown that a mixed Fourier cosine series can be integrated term by term between any two limits, (B) implies that

$$\begin{aligned}\sum_{n=1}^{\infty} \frac{2L\beta_n}{(2n-1)\pi a} \sin \frac{(2n-1)\pi at}{2L} \cos \frac{(2n-1)\pi x}{2L} &= \frac{1}{2a} \sum_{n=1}^{\infty} \beta_n \int_{x-at}^{x+at} \cos \frac{(2n-1)\pi \tau}{2L} d\tau \\ &= \frac{1}{2a} \int_{x-at}^{x+at} \left( \sum_{n=1}^{\infty} \beta_n \cos \frac{(2n-1)\pi \tau}{2L} \right) d\tau \\ &= \frac{1}{2a} \int_{x-at}^{x+at} C_{Mg}(\tau) d\tau.\end{aligned}$$

This and (C) imply that

$$u(x, t) = \frac{1}{2}[C_{Mf}(x + at) + C_{Mf}(x - at)] + \frac{1}{2a} \int_{x-at}^{x+at} C_{Mg}(\tau) d\tau.$$

**12.2.34.** We begin by looking for functions of the form  $v(x, t) = X(x)T(t)$  that are not identically zero and satisfy  $v_{tt} = a^2 v_{xx}$ ,  $v(0, t) = 0$ ,  $v_x(L, t) = 0$  for all  $(x, t)$ . As shown in the text,  $X$  and  $T$  must satisfy  $X'' + \lambda X = 0$  and (B)  $T'' + a^2 \lambda T = 0$  for the same value of  $\lambda$ . Since  $v(0, t) = X(0)T(t)$  and  $v_x(L, t) = X'(L)T(t)$  and we don't want  $T$  to be identically zero,  $X(0) = 0$  and  $X'(L) = 0$ . Therefore,  $\lambda$  must be an eigenvalue of (C)  $X'' + \lambda X = 0$ ,  $X(0) = 0$ ,  $X'(L) = 0$ , and  $X$  must be a  $\lambda$ -eigenfunction. From Theorem 11.1.4, the eigenvalues of (C) are  $\lambda_n = \frac{(2n-1)^2}{\pi^2/4L^2}$ , integer), with associated eigenfunctions  $X_n = \sin \frac{(2n-1)\pi x}{2L}$ ,  $n = 1, 2, 3, \dots$ . Substituting  $\lambda = \frac{(2n-1)^2\pi^2}{4L^2}$  into (B) yields  $T'' + ((2n-1)^2\pi^2 a^2/4L^2)T = 0$ , which has the general solution

$$T_n = \alpha_n \cos \frac{(2n-1)\pi at}{2L} + \frac{2\beta_n L}{(2n-1)\pi a} \sin \frac{(2n-1)\pi at}{2L},$$

where  $\alpha_n$  and  $\beta_n$  are constants. Now let

$$v_n(x, t) = X_n(x)T_n(t) = \left( \alpha_n \cos \frac{(2n-1)\pi at}{2L} + \frac{2\beta_n L}{(2n-1)\pi a} \sin \frac{(2n-1)\pi at}{2L} \right) \sin \frac{(2n-1)\pi x}{2L}.$$

Then

$$\frac{\partial v_n}{\partial t}(x, t) = \left( -\frac{(2n-1)\pi a}{2L} \alpha_n \sin \frac{(2n-1)\pi at}{2L} + \beta_n \cos \frac{(2n-1)\pi at}{2L} \right) \sin \frac{(2n-1)\pi x}{2L},$$

so

$$v_n(x, 0) = \alpha_n \sin \frac{(2n-1)\pi x}{2L} \quad \text{and} \quad \frac{\partial v_n}{\partial t}(x, 0) = \beta_n \sin \frac{(2n-1)\pi x}{2L}.$$

Therefore,  $v_n$  satisfies (A) with  $f(x) = \alpha_n \sin \frac{(2n-1)\pi x}{2L}$  and  $g(x) = \beta_n \sin \frac{(2n-1)\pi x}{2L}$ . More generally, if  $\alpha_1, \alpha_2, \dots, \alpha_m$  and  $\beta_1, \beta_2, \dots, \beta_m$  are constants and

$$u_m(x, t) = \sum_{n=1}^m \left( \alpha_n \cos \frac{(2n-1)\pi at}{2L} + \frac{2\beta_n L}{(2n-1)\pi a} \sin \frac{(2n-1)\pi at}{2L} \right) \sin \frac{(2n-1)\pi x}{2L},$$

then  $u_m$  satisfies (A) with

$$f(x) = \sum_{n=1}^m \alpha_n \sin \frac{(2n-1)\pi x}{2L} \quad \text{and} \quad g(x) = \sum_{n=1}^m \beta_n \sin \frac{(2n-1)\pi x}{2L}.$$

This motivates the definition.

**12.2.36.** Since  $f(0) = f'(1) = 0$ , and  $f''(x) = 6(1 - 2x)$ , Theorem 11.3.5(d) implies that

$$\begin{aligned}
 \alpha_n &= -\frac{48}{(2n-1)^2\pi^2} \int_0^1 (1-2x) \sin \frac{(2n-1)\pi x}{2} dx \\
 &= \frac{96}{(2n-1)^3\pi^3} \left[ (1-2x) \cos \frac{(2n-1)\pi x}{2} \Big|_0^1 + 2 \int_0^1 \cos \frac{(2n-1)\pi x}{2} dx \right] \\
 &= \frac{96}{(2n-1)^3\pi^3} \left[ -1 + \frac{4}{(2n-1)\pi} \sin \frac{(2n-1)\pi x}{2} \Big|_0^1 \right] \\
 &= -\frac{96}{(2n-1)^3\pi^3} \left[ 1 + (-1)^n \frac{4}{(2n-1)\pi} \right]; \\
 S_{Mf}(x) &= -\frac{96}{\pi^3} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^3} \left[ 1 + (-1)^n \frac{4}{(2n-1)\pi} \right] \sin \frac{(2n-1)\pi x}{2}. \text{ From Exercise 12.2.34,} \\
 u(x, t) &= -\frac{96}{\pi^3} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^3} \left[ 1 + (-1)^n \frac{4}{(2n-1)\pi} \right] \cos \frac{3(2n-1)\pi t}{2} \sin \frac{(2n-1)\pi x}{2}.
 \end{aligned}$$

**12.2.38.** Since  $g(0) = g'(1) = 0$ , and  $g''(x) = 6(1 - 2x)$ , Theorem 11.3.5(d) implies that

$$\begin{aligned}
 \beta_n &= -\frac{48}{(2n-1)^2\pi^2} \int_0^1 (1-2x) \sin \frac{(2n-1)\pi x}{2} dx \\
 &= \frac{96}{(2n-1)^3\pi^3} \left[ (1-2x) \cos \frac{(2n-1)\pi x}{2} \Big|_0^1 + 2 \int_0^1 \cos \frac{(2n-1)\pi x}{2} dx \right] \\
 &= \frac{96}{(2n-1)^3\pi^3} \left[ -1 + \frac{4}{(2n-1)\pi} \sin \frac{(2n-1)\pi x}{2} \Big|_0^1 \right] \\
 &= -\frac{96}{(2n-1)^3\pi^3} \left[ 1 + (-1)^n \frac{4}{(2n-1)\pi} \right]; \\
 S_{Mg}(x) &= -\frac{96}{\pi^3} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^3} \left[ 1 + (-1)^n \frac{4}{(2n-1)\pi} \right] \sin \frac{(2n-1)\pi x}{2}. \text{ From Exercise 12.2.34,} \\
 u(x, t) &= -\frac{64}{\pi^4} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^4} \left[ 1 + (-1)^n \frac{4}{(2n-1)\pi} \right] \sin \frac{3(2n-1)\pi t}{2} \sin \frac{(2n-1)\pi x}{2}.
 \end{aligned}$$

**12.2.40.** Since  $f(0) = f'(\pi) = f''(0) = 0$  and  $f'''(x) = 6$ , Theorem 11.3.5(d) and Exercise 11.3.50(b) imply that

$$\begin{aligned}
 \alpha_n &= -\frac{96}{(2n-1)^3\pi} \int_0^\pi \cos \frac{(2n-1)x}{2} dx = -\frac{192}{(2n-1)^4\pi} \sin \frac{(2n-1)x}{2} \Big|_0^\pi = (-1)^n \frac{192}{(2n-1)^4\pi}; \\
 S_{Mf}(x) &= \frac{192}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^n}{(2n-1)^4} \sin \frac{(2n-1)x}{2}. \text{ From Exercise 12.2.34,} \\
 u(x, t) &= \frac{192}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^n}{(2n-1)^4} \cos \frac{(2n-1)\sqrt{3}t}{2} \sin \frac{(2n-1)x}{2}.
 \end{aligned}$$

**12.2.42.** Since  $g(0) = g'(\pi) = g''(0) = 0$  and  $g'''(x) = 6$ , Theorem 11.3.5(d) and Exercise 50(b) imply that

$$\beta_n = -\frac{96}{(2n-1)^3\pi} \int_0^\pi \cos \frac{(2n-1)x}{2} dx = -\frac{192}{(2n-1)^4\pi} \sin \frac{(2n-1)x}{2} \Big|_0^\pi = (-1)^n \frac{192}{(2n-1)^4\pi};$$

$$S_{Mg}(x) = \frac{192}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^n}{(2n-1)^4} \sin \frac{(2n-1)x}{2}. \text{ From Exercise 12.2.34,}$$

$$u(x, t) = \frac{384}{\sqrt{3}\pi} \sum_{n=1}^{\infty} \frac{(-1)^n}{(2n-1)^5} \sin \frac{(2n-1)\sqrt{3}t}{2} \sin \frac{(2n-1)x}{2}.$$

**12.2.44.** Since  $f(0) = f'(1) = f''(0) = 0$  and  $f'''(x) = 12(2x-1)$ , Theorem 11.3.5(d) and Exercise 11.3.50(b) imply that

$$\begin{aligned} \alpha_n &= -\frac{192}{(2n-1)^3\pi^3} \int_0^1 (2x-1) \cos \frac{(2n-1)\pi x}{2} dx \\ &= -\frac{384}{(2n-1)^4\pi^4} \left[ (2x-1) \sin \frac{(2n-1)\pi x}{2} \Big|_0^1 - 2 \int_0^1 \sin \frac{(2n-1)\pi x}{2} dx \right] \\ &= -\frac{384}{(2n-1)^4\pi^4} \left[ (-1)^{n+1} + \frac{4}{(2n-1)\pi} \cos \frac{(2n-1)\pi x}{2} \Big|_0^1 \right] \\ &= -\frac{384}{(2n-1)^4\pi^4} \left[ (-1)^{n+1} - \frac{4}{(2n-1)\pi} \right] = \frac{384}{(2n-1)^4\pi^4} \left[ (-1)^n + \frac{4}{(2n-1)\pi} \right]; \end{aligned}$$

$$S_{Mf}(x) = \frac{384}{\pi^4} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^4} \left[ (-1)^n + \frac{4}{(2n-1)\pi} \right] \sin \frac{(2n-1)\pi x}{2}. \text{ From Exercise 12.2.34,}$$

$$u(x, t) = \frac{384}{\pi^4} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^4} \left[ (-1)^n + \frac{4}{(2n-1)\pi} \right] \cos(2n-1)\pi t \sin \frac{(2n-1)\pi x}{2}.$$

**12.2.46.** Since  $g(0) = g'(1) = g''(0) = 0$  and  $g'''(x) = 12(2x-1)$ , Theorem 11.3.5(d) and Exercise 11.3.50(b) imply that

$$\begin{aligned} \beta_n &= -\frac{192}{(2n-1)^3\pi^3} \int_0^1 (2x-1) \cos \frac{(2n-1)\pi x}{2} dx \\ &= -\frac{384}{(2n-1)^4\pi^4} \left[ (2x-1) \sin \frac{(2n-1)\pi x}{2} \Big|_0^1 - 2 \int_0^1 \sin \frac{(2n-1)\pi x}{2} dx \right] \\ &= -\frac{384}{(2n-1)^4\pi^4} \left[ (-1)^{n+1} + \frac{4}{(2n-1)\pi} \cos \frac{(2n-1)\pi x}{2} \Big|_0^1 \right] \\ &= -\frac{384}{(2n-1)^4\pi^4} \left[ (-1)^{n+1} - \frac{4}{(2n-1)\pi} \right] = \frac{384}{(2n-1)^4\pi^4} \left[ (-1)^n + \frac{4}{(2n-1)\pi} \right]; \end{aligned}$$

$$S_{Mg}(x) = \frac{384}{\pi^4} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^4} \left[ (-1)^n + \frac{4}{(2n-1)\pi} \right] \sin \frac{(2n-1)\pi x}{2}. \text{ From Exercise 12.2.34,}$$

$$u(x, t) = \frac{384}{\pi^5} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^5} \left[ (-1)^n + \frac{4}{(2n-1)\pi} \right] \sin(2n-1)\pi t \sin \frac{(2n-1)\pi x}{2}.$$

**12.2.48.** Since  $f$  is continuous on  $[0, L]$  and  $f'(L) = 0$ , Theorem 11.3.4 implies that  $S_{Mf}(x) = f(x)$ ,  $0 \leq x \leq L$ . From Exercise 11.3.58,  $S_{Mf}$  is the odd periodic extension (with period  $2L$ ) of the function  $r(x) = \begin{cases} f(x), & 0 \leq x \leq L, \\ f(2L-x), & L < x \leq 2L, \end{cases}$  which is continuous on  $[0, 2L]$ . Since  $r(0) = r(2L) = f(0) = 0$ ,  $S_{Mf}$  is continuous on  $(-\infty, \infty)$ . Moreover,  $r'(x) = \begin{cases} f'(x), & 0 < x < L, \\ -f'(2L-x), & L < x < 2L, \end{cases}$   $r'_+(0) = f'_+(0)$ ,  $r'_-(2L) = -f'_+(0)$ , and, since  $f'_-(L) = 0$ ,  $r'(L) = 0$ . Hence,  $r$  is differentiable on  $[0, 2L]$ . Since  $r(0) = r(2L) = f(0) = 0$ , Theorem 12.2.3(a) with  $h = r$ ,  $p = S_{Mf}$ , and  $L$  replaced by  $2L$  implies that  $S_{Mf}$  is differentiable on  $(-\infty, \infty)$ . Similarly,  $S_{Mg}$  is differentiable on  $(-\infty, \infty)$ .

Now we note that  $r''(x) = \begin{cases} f''(x), & 0 < x < L, \\ f''(2L-x), & L < x < 2L, \end{cases}$   $r''(L) = f''(L)$ , and  $r''_+(0) = r''_-(2L) = f''_+(0) = 0$ . Since  $S'_{Mf}$  is the even periodic extension of  $r'$ , Theorem 12.2.3(b) with  $h = r'$ ,  $q = S'_{Mf}$ , and  $L$  replaced by  $2L$  implies that  $S'_{Mf}$  is differentiable on  $(-\infty, \infty)$ . Now follow the argument used to complete the proof of Theorem 12.2.4.

**12.2.50.** From Example 11.3.5,  $C_f(x) = 4 - \frac{768}{\pi^4} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^4} \cos \frac{(2n-1)\pi x}{2}$ . From Exercise 12.2.49,

$$u(x, t) = 4 - \frac{768}{\pi^4} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^4} \cos \frac{\sqrt{5}(2n-1)\pi t}{2} \cos \frac{(2n-1)\pi x}{2}.$$

**12.2.52.**  $\alpha_0 = \int_0^\pi (3x^4 - 4Lx^3) dx = \left( \frac{3x^5}{5} - \pi x^4 \right) \Big|_0^\pi = -\frac{2\pi^4}{5}$ . Since  $f'(0) = f'(\pi) = 0$  and  $f'''(x) = 24(3x - \pi)$ , Theorem 11.3.5(a) implies that

$$\begin{aligned} \alpha_n &= \frac{48}{n^3\pi} \int_0^\pi (3x - \pi) \sin nx \, dx = -\frac{48}{n^4\pi} \left[ (3x - \pi) \cos nx \Big|_0^\pi - 3 \int_0^\pi \cos nx \, dx \right] \\ &= -\frac{48}{n^4\pi} [(-1)^n 2\pi + \pi] + \frac{144}{n^5\pi} \sin nx \Big|_0^\pi = -\frac{48}{n^4} [1 + (-1)^n 2], \quad n \geq 1; \end{aligned}$$

$C_f(x) = -\frac{2\pi^4}{5} - 48 \sum_{n=1}^{\infty} \frac{1 + (-1)^n 2}{n^4} \cos nx$ . From Exercise 12.2.49,

$$u(x, t) = -\frac{2\pi^4}{5} - 48 \sum_{n=1}^{\infty} \frac{1 + (-1)^n 2}{n^4} \cos 2nt \cos nx.$$

**12.2.54.**  $\beta_0 = \int_0^\pi (3x^4 - 4Lx^3) dx = \frac{1}{\pi} \left( \frac{3x^5}{5} - \pi x^4 \right) \Big|_0^\pi = -\frac{2\pi^4}{5}$ . Since  $g'(0) = g'(\pi) = 0$  and



$g'''(x) = 24(3x - \pi)$ , Theorem 11.3.5(a) implies that

$$\begin{aligned}\beta_n &= \frac{48}{n^3\pi} \int_0^\pi (3x - \pi) \sin nx \, dx = -\frac{48}{n^4\pi} \left[ (3x - \pi) \cos nx \Big|_0^\pi - 3 \int_0^\pi \cos nx \, dx \right] \\ &= -\frac{48}{n^4\pi} [(-1)^n 2\pi + \pi] + \frac{144}{n^5\pi} \sin nx \Big|_0^\pi = -\frac{48}{n^4} [1 + (-1)^n 2], \quad n \geq 1;\end{aligned}$$

$$C_g(x) = -\frac{2\pi^4}{5} - 48 \sum_{n=1}^{\infty} \frac{1 + (-1)^n 2}{n^4} \cos nx. \text{ From Exercise 12.2.49,}$$

$$u(x, t) = -\frac{2\pi^4 t}{5} - 24 \sum_{n=1}^{\infty} \frac{1 + (-1)^n 2}{n^5} \sin 2nt \cos nx.$$

**12.2.56.**  $\alpha_0 = \frac{1}{\pi} \int_0^\pi (x^4 - 2\pi x^3 + \pi^2 x^2) \, dx = \frac{1}{\pi} \left( \frac{x^5}{5} - \frac{\pi x^4}{2} + \frac{\pi^2 x^3}{3} \right) \Big|_0^\pi = \frac{\pi^4}{30}$ . Since  $f'(0) = f'(\pi) = 0$  and  $f'''(x) = 12(2x - \pi)$ , Theorem 11.3.5(a) implies that

$$\begin{aligned}\alpha_n &= \frac{24}{n^3\pi} \int_0^\pi (2x - \pi) \sin nx \, dx = -\frac{24}{n^4\pi} \left[ (2x - \pi) \cos nx \Big|_0^\pi - 2 \int_0^\pi \cos nx \, dx \right] \\ &= -\frac{24}{n^4\pi} [(-1)^n \pi + \pi] + \frac{48}{n^5\pi} \sin nx \Big|_0^\pi = -\frac{24}{n^4} [1 + (-1)^n] \\ &= \begin{cases} 0 & \text{if } n = 2m - 1, \\ -\frac{3}{m^4} & \text{if } n = 2m, \end{cases} \quad n \geq 1;\end{aligned}$$

$$C_f(x) = \frac{\pi^4}{30} - 3 \sum_{n=1}^{\infty} \frac{1}{n^4} \cos 2nx. \text{ From Exercise 12.2.49, } u(x, t) = \frac{\pi^4}{30} - 3 \sum_{n=1}^{\infty} \frac{1}{n^4} \cos 8nt \cos 2nx.$$

**12.2.58.**  $\beta_0 = \frac{1}{\pi} \int_0^\pi (x^4 - 2\pi x^3 + \pi^2 x^2) \, dx = \frac{1}{\pi} \left( \frac{x^5}{5} - \frac{\pi x^4}{2} + \frac{\pi^2 x^3}{3} \right) \Big|_0^\pi = \frac{\pi^4}{30}$ . Since  $g'(0) = g'(\pi) = 0$  and  $g'''(x) = 12(2x - \pi)$ , Theorem 11.3.5(a) implies that

$$\begin{aligned}\beta_n &= \frac{24}{n^3\pi} \int_0^\pi (2x - \pi) \sin nx \, dx = -\frac{24}{n^4\pi} \left[ (2x - \pi) \cos nx \Big|_0^\pi - 2 \int_0^\pi \cos nx \, dx \right] \\ &= -\frac{24}{n^4\pi} [(-1)^n \pi + \pi] + \frac{48}{n^5\pi} \sin nx \Big|_0^\pi = -\frac{24}{n^4} [1 + (-1)^n] \\ &= \begin{cases} 0 & \text{if } n = 2m - 1, \\ -\frac{3}{m^4} & \text{if } n = 2m, \end{cases} \quad n \geq 1;\end{aligned}$$

$$C_g(x) = \frac{\pi^4}{30} - 3 \sum_{n=1}^{\infty} \frac{1}{n^4} \cos 2nx. \text{ From Exercise 12.2.49, } u(x, t) = \frac{\pi^4 t}{30} - \frac{3}{8} \sum_{n=1}^{\infty} \frac{1}{n^5} \sin 8nt \cos 2nx.$$

**12.2.60.** Setting  $A = n\pi x/L$  and  $B = n\pi at/L$  in the identities  $\cos A \cos B = \frac{1}{2}[\cos(A + B) +$

$\cos(A - B)$ ] and  $\cos A \sin B = \frac{1}{2}[\sin(A + B) - \sin(A - B)]$  yields

$$\cos \frac{n\pi at}{L} \cos \frac{n\pi x}{L} = \frac{1}{2} \left[ \cos \frac{n\pi(x + at)}{L} + \cos \frac{n\pi(x - at)}{L} \right] \quad (\text{A})$$

and

$$\begin{aligned} \sin \frac{n\pi at}{L} \cos \frac{n\pi x}{L} &= \frac{1}{2} \left[ \sin \frac{n\pi(x + at)}{L} - \sin \frac{n\pi(x - at)}{L} \right] \\ &= \frac{n\pi}{2L} \int_{x-at}^{x+at} \cos \frac{n\pi\tau}{L} d\tau. \end{aligned} \quad (\text{B})$$

Since  $C_f(x) = \alpha_0 + \sum_{n=1}^{\infty} \alpha_n \sin \frac{n\pi x}{L}$ , (A) implies that

$$\alpha_0 + \sum_{n=1}^{\infty} \alpha_n \cos \frac{n\pi at}{L} \cos \frac{n\pi x}{L} = \frac{1}{2} [C_f(x + at) + C_f(x - at)]. \quad (\text{C})$$

Since it can be shown that a Fourier sine series can be integrated term by term between any two limits, (B) implies that

$$\begin{aligned} \beta_0 t + \sum_{n=1}^{\infty} \frac{\beta_n L}{n\pi a} \sin \frac{n\pi at}{L} \cos \frac{n\pi x}{L} &= \beta_0 t + \frac{1}{2a} \sum_{n=1}^{\infty} \beta_n \int_{x-at}^{x+at} \cos \frac{n\pi\tau}{L} d\tau \\ &= \frac{1}{2a} \int_{x-at}^{x+at} \left( \beta_0 + \sum_{n=1}^{\infty} \beta_n \cos \frac{n\pi\tau}{L} \right) d\tau \\ &= \frac{1}{2a} \int_{x-at}^{x+at} C_g(\tau) d\tau. \end{aligned}$$

This and (C) imply that

$$u(x, t) = \frac{1}{2} [C_f(x + at) + C_f(x - at)] + \frac{1}{2a} \int_{x-at}^{x+at} C_g(\tau) d\tau.$$

**12.2.62.(a).** Since  $|p_n(x)| \leq 1$  and  $|q_n(t)| \leq 1$  for all  $t$ ,  $|k_n p_n(x) q_n(t)| \leq |k_n|$  for all  $(x, t)$ , and the comparison test implies the conclusion.

(b) If  $t$  is fixed but arbitrary, then  $|k_n p'_n(x) q_n(t)| \leq |\lambda| n |k_n|$ , so Theorem 12.1.2 with  $z = x$  and  $w_n(x) = k_n p_n(x) q_n(t)$  justifies term by term differentiation with respect to  $x$  on  $(-\infty, \infty)$ . If  $x$  is fixed but arbitrary, then  $|k_n p_n(x) q'_n(t)| \leq |\mu| n |k_n|$ , so Theorem 12.1.2 with  $z = t$  and  $w_n(t) = k_n p_n(x) q_n(t)$  justifies term by term differentiation with respect to  $t$  on  $(-\infty, \infty)$ .

(c) The argument is similar to argument use in (b).

(d) Apply (b) and (c) to the series  $\sum_{n=1}^{\infty} \alpha_n \cos \frac{n\pi at}{L} \sin \frac{n\pi x}{L}$  and  $\sum_{n=1}^{\infty} \frac{\beta_n L}{n\pi a} \sin \frac{n\pi at}{L} \sin \frac{n\pi x}{L}$ , recalling that the individual terms in the series satisfy  $u_{tt} = a^2 u_{xx}$  for all  $(x, t)$ .

$$(d) \quad u(x, t) = \frac{f(x + ct) + f(x - ct)}{2} + \frac{1}{2a} \int_{x-at}^{x+at} g(u) du.$$

**12.2.64.**

$$\begin{aligned} u(x, t) &= \frac{(x + at) + (x - at)}{2} + \frac{1}{2a} \int_{x-at}^{x+at} 4au du = x + 2 \int_{x-at}^{x+at} u du = x + u^2 \Big|_{x-at}^{x+at} \\ &= x + (x + at)^2 - (x - at)^2 = x(1 + 4at). \end{aligned}$$

**12.2.66.**

$$\begin{aligned}
u(x, t) &= \frac{\sin(x + at) + \sin(x - at)}{2} + \frac{1}{2a} \int_{x-at}^{x+at} a \cos u \, du \\
&= \frac{\sin(x + at) + \sin(x - at)}{2} + \frac{\sin(x + at) - \sin(x - at)}{2} = \sin(x + at).
\end{aligned}$$

**12.2.68.**

$$\begin{aligned}
u(x, t) &= \frac{(x + at) \sin(x + at) + (x - at) \sin(x - at)}{2} + \frac{1}{2a} \int_{x-at}^{x+at} \sin u \, du \\
&= \frac{x[\sin(x + at) + \sin(x - at)]}{2} + \frac{at[\sin(x + at) - \sin(x - at)]}{2} \\
&\quad + \frac{\cos(x - at) - \cos(x + at)}{2a} \\
&= x \sin x \cos at + at \cos x \sin at + \frac{\sin x \sin at}{a}.
\end{aligned}$$

## 12.3 Laplace's Equation in Rectangular Coordinates

**12.3.2.** Since  $f(0) = f(1) = 0$  and  $f''(x) = 2 - 6x$ , Theorem 11.3.5(b) implies that

$$\begin{aligned}
\alpha_n &= -\frac{4}{n^2 \pi^2} \int_0^1 (4 - 6x) \sin \frac{n\pi x}{2} \, dx = \frac{8}{n^3 \pi^3} \left[ (4 - 6x) \cos \frac{n\pi x}{2} \Big|_0^2 + 6 \int_0^2 \cos \frac{n\pi x}{2} \, dx \right] \\
&= -\frac{32}{n^3 \pi^3} (1 + (-1)^n 2) + \frac{96}{n^4 \pi^4} \sin \frac{n\pi x}{2} \Big|_0^2 = -\frac{32}{n^3 \pi^3} [1 + (-1)^n 2];
\end{aligned}$$

$$S(x) = -\frac{32}{\pi^3} \sum_{n=1}^{\infty} \frac{[1 + (-1)^n 2]}{n^3} \sin \frac{n\pi x}{2}. \text{ From Example 12.3.1,}$$

$$u(x, y) = -\frac{32}{\pi^3} \sum_{n=1}^{\infty} \frac{[1 + (-1)^n 2] \sinh n\pi(3 - y)/2}{n^3 \sinh 3n\pi/2} \sin \frac{n\pi x}{2}.$$

**12.3.4.**

$$\begin{aligned}
\alpha_1 &= \frac{2}{\pi} \int_0^{\pi} x \sin^2 x \, dx = \frac{1}{\pi} \int_0^{\pi} x(1 - \cos 2x) \, dx = \frac{x^2}{2\pi} \Big|_0^{\pi} - \frac{1}{\pi} \int_0^{\pi} x \cos 2x \, dx \\
&= \frac{\pi}{2} - \frac{1}{2\pi} \left[ x \sin 2x \Big|_0^{\pi} - \int_0^{\pi} \cos 2x \, dx \right] = \frac{\pi}{2} + \frac{\sin 2x}{4\pi} \Big|_0^{\pi} = \frac{\pi}{2};
\end{aligned}$$

if  $n \geq 2$ , then

$$\begin{aligned}
 \alpha_n &= \frac{2}{\pi} \int_0^\pi x \sin x \sin nx \, dx = \frac{1}{\pi} \int_0^\pi x [\cos(n-1)x - \cos(n+1)x] \, dx \\
 &= \frac{1}{\pi} \left[ x \left[ \frac{\sin(n-1)x}{n-1} - \frac{\sin(n+1)x}{n+1} \right] \right]_0^\pi - \int_0^\pi \left[ \frac{\sin(n-1)x}{n-1} - \frac{\sin(n+1)x}{n+1} \right] dx \\
 &= \frac{1}{\pi} \left[ \frac{\cos(n-1)x}{(n-1)^2} - \frac{\cos(n+1)x}{(n+1)^2} \right]_0^\pi = \frac{1}{\pi} \left[ \frac{1}{(n-1)^2} - \frac{1}{(n+1)^2} \right] [(-1)^{n+1} - 1] \\
 &= \frac{4n}{(n^2-1)^2\pi} [(-1)^{n+1} - 1] = \begin{cases} 0 & \text{if } n = 2m-1, \\ -\frac{16m}{(4m^2-1)\pi} & \text{if } n = 2m; \end{cases}
 \end{aligned}$$

$$S(x) = \frac{\pi}{2} \sin x - \frac{16}{\pi} \sum_{n=1}^{\infty} \frac{n}{(4n^2-1)^2} \sin 2nx. \text{ From Example 12.3.1,}$$

$$u(x, y) = \frac{\pi \sinh(1-y)}{2 \sinh 1} \sin x - \frac{16}{\pi} \sum_{n=1}^{\infty} \frac{n \sinh 2n(1-y)}{(4n^2-1)^2 \sinh 2n} \sin 2nx.$$

$$\mathbf{12.3.6.} \quad \alpha_0 = \int_0^1 (1-x) \, dx = -\frac{(1-x)^2}{2} \Big|_0^1 = \frac{1}{2}; \text{ if } n \geq 1,$$

$$\begin{aligned}
 \alpha_n &= 2 \int_0^1 (1-x) \cos n\pi x \, dx = \frac{2}{n\pi} \left[ (1-x) \sin n\pi x \right]_0^1 + \int_0^1 \sin n\pi x \, dx \\
 &= -\frac{2}{n^2\pi^2} \cos n\pi x \Big|_0^1 = \frac{2}{n^2\pi^2} [1 - (-1)^n] = \begin{cases} \frac{4}{(2m-1)^2\pi^2} & \text{if } n = 2m-1, \\ 0 & \text{if } n = 2m; \end{cases}
 \end{aligned}$$

$$C(x) = \frac{1}{2} + \frac{4}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} \cos(2n-1)\pi x. \text{ From Example 12.3.3,}$$

$$u(x, y) = \frac{y}{2} + \frac{4}{\pi^3} \sum_{n=1}^{\infty} \frac{\sinh(2n-1)\pi y}{(2n-1)^3 \cosh 2(2n-1)\pi} \cos(2n-1)\pi x.$$

$$\mathbf{12.3.8.} \quad \alpha_0 = \int_0^1 (x-1)^2 \, dx = \frac{(x-1)^3}{3} \Big|_0^1 = \frac{1}{3}; \text{ if } n \geq 1, \text{ then}$$

$$\begin{aligned}
 \alpha_n &= 2 \int_0^1 (x-1)^2 \cos n\pi x \, dx = \frac{2}{n\pi} \left[ (x-1)^2 \sin n\pi x \right]_0^1 - 2 \int_0^1 (x-1) \sin n\pi x \, dx \\
 &= \frac{4}{n^2\pi^2} \left[ (x-1) \cos n\pi x \right]_0^1 - \int_0^1 \cos n\pi x \, dx = \frac{4}{n^2\pi^2} - \frac{4}{n^3\pi^3} \sin n\pi x \Big|_0^1 = \frac{4}{n^2\pi^2};
 \end{aligned}$$

$$C(x) = \frac{1}{3} + \frac{4}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^2} \cos n\pi x. \text{ From Example 12.3.3, } u(x, y) = \frac{y}{3} + \frac{4}{\pi^3} \sum_{n=1}^{\infty} \frac{\sinh n\pi y}{n^3 \cosh n\pi} \cos n\pi x.$$

**12.3.10.** Since  $g(0) = g'(1) = 0$ , and  $g''(y) = 6(1 - 2y)$ , Theorem 11.3.5(d) implies that

$$\begin{aligned}
 \alpha_n &= -\frac{48}{(2n-1)^2\pi^2} \int_0^1 (1-2y) \sin \frac{(2n-1)\pi y}{2} dy \\
 &= \frac{96}{(2n-1)^3\pi^3} \left[ (1-2y) \cos \frac{(2n-1)\pi y}{2} \Big|_0^1 + 2 \int_0^1 \cos \frac{(2n-1)\pi y}{2} dy \right] \\
 &= \frac{96}{(2n-1)^3\pi^3} \left[ -1 + \frac{4}{(2n-1)\pi} \sin \frac{(2n-1)\pi y}{2} \Big|_0^1 \right] \\
 &= -\frac{96}{(2n-1)^3\pi^3} \left[ 1 + (-1)^n \frac{4}{(2n-1)\pi} \right]; \\
 S_M(y) &= -\frac{96}{\pi^3} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^3} \left[ 1 + (-1)^n \frac{4}{(2n-1)\pi} \right] \sin \frac{(2n-1)\pi y}{2}. \text{ From Example 12.3.5,} \\
 u(x, y) &= -\frac{96}{\pi^3} \sum_{n=1}^{\infty} \left[ 1 + (-1)^n \frac{4}{(2n-1)\pi} \right] \frac{\cosh(2n-1)\pi(x-2)/2}{(2n-1)^3 \cosh 2(2n-1)\pi/2} \sin \frac{(2n-1)\pi y}{2}.
 \end{aligned}$$

**12.3.12.** From Example 11.3.8.3,

$$S_M(y) = \frac{96}{\pi^3} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^3} \left[ 3 + (-1)^n \frac{4}{(2n-1)\pi} \right] \sin \frac{(2n-1)\pi y}{2}.$$

From Example 12.3.5,

$$u(x, y) = \frac{96}{\pi^3} \sum_{n=1}^{\infty} \left[ 3 + (-1)^n \frac{4}{(2n-1)\pi} \right] \frac{\cosh(2n-1)\pi(x-3)/2}{(2n-1)^3 \cosh 3(2n-1)\pi/2} \sin \frac{(2n-1)\pi y}{2}.$$

**12.3.14.**

$$\begin{aligned}
 c_n &= \frac{2}{3} \int_0^3 (3y - y^2) \cos \frac{(2n-1)\pi y}{6} dy \\
 &= \frac{4}{(2n-1)\pi} \left[ (3y - y^2) \sin \frac{(2n-1)\pi y}{6} \Big|_0^3 - \int_0^3 (3-2y) \sin \frac{(2n-1)\pi y}{6} dy \right] \\
 &= \frac{24}{(2n-1)^2\pi^2} \left[ (3-2y) \cos \frac{(2n-1)\pi y}{6} \Big|_0^3 + 2 \int_0^3 \cos \frac{(2n-1)\pi y}{6} dy \right] \\
 &= -\frac{72}{(2n-1)^2\pi^2} + \frac{288}{(2n-1)^3\pi^3} \sin \frac{(2n-1)\pi y}{6} \Big|_0^3 = \frac{288}{(2n-1)^3\pi^3} \sin \frac{(2n-1)\pi y}{2} \\
 &= -\frac{72}{(2n-1)^2\pi^2} + (-1)^{n-1} \frac{288}{(2n-1)^3\pi^3}; \\
 C_M(y) &= -\frac{72}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} \left[ 1 + \frac{4(-1)^n}{(2n-1)\pi} \right] \cos \frac{(2n-1)\pi y}{6}. \text{ From Example 12.3.7,} \\
 u(x, y) &= -\frac{432}{\pi^3} \sum_{n=1}^{\infty} \left[ 1 + \frac{4(-1)^n}{(2n-1)\pi} \right] \frac{\cosh(2n-1)\pi x/6}{(2n-1)^3 \sinh(2n-1)\pi/3} \cos \frac{(2n-1)\pi y}{6}.
 \end{aligned}$$

**12.3.16.** Since  $g'(0) = g(1) = 0$  and  $g''(y) = -6y$ , Theorem 11.3.5(c) implies that

$$\begin{aligned}\alpha_n &= \frac{48}{(2n-1)^2\pi^2} \int_0^1 y \cos \frac{(2n-1)\pi y}{2} dy \\ &= \frac{96}{(2n-1)^3\pi^3} \left[ y \sin \frac{(2n-1)\pi y}{2} \Big|_0^1 - \int_0^1 \sin \frac{(2n-1)\pi y}{2} dy \right] \\ &= \frac{96}{(2n-1)^3\pi^3} \left[ (-1)^{n+1} + \frac{2}{(2n-1)\pi} \cos \frac{(2n-1)\pi y}{2} \Big|_0^1 \right] \\ &= -\frac{96}{(2n-1)^3\pi^3} \left[ (-1)^n + \frac{2}{(2n-1)\pi} \right] dy;\end{aligned}$$

$$C_M(y) = -\frac{96}{\pi^3} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^3} \left[ (-1)^n + \frac{2}{(2n-1)\pi} \right] \cos \frac{(2n-1)\pi y}{2}. \text{ From Example 12.3.7,}$$

$$u(x, y) = -\frac{192}{\pi^4} \sum_{n=1}^{\infty} \frac{\cosh(2n-1)\pi x/2}{(2n-1)^4 \sinh(2n-1)\pi/2} \left[ (-1)^n + \frac{2}{(2n-1)\pi} \right] \cos \frac{(2n-1)\pi y}{2}.$$

**12.3.18.** The boundary conditions require products  $v(x, y) = X(x)Y(y)$  such that (A)  $X'' + \lambda X = 0$ ,  $X'(0) = 0$ ,  $X'(a) = 0$ , and (B)  $Y'' - \lambda Y = 0$ ,  $Y(0) = 1$ ,  $Y(b) = 0$ . From Theorem 11.1.3, the eigenvalues of (A) are  $\lambda = 0$ , with associated eigenfunction  $X_0 = 1$ , and  $\lambda_n = \frac{n^2\pi^2}{a^2}$ , with associated eigenfunctions  $Y_n = \cos \frac{n\pi x}{a}$ ,  $n = 1, 2, 3, \dots$ . Substituting  $\lambda = 0$  into

(B) yields  $Y_0'' = 0$ ,  $Y_0(0) = 1$ ,  $Y_0(b) = 0$ , so  $Y_0(y) = 1 - \frac{y}{b}$ . Substituting  $\lambda = \frac{n^2\pi^2}{a^2}$  into

(B) yields  $Y_n'' - (n^2\pi^2/a^2)Y_n = 0$ ,  $Y_n(0) = 1$ ,  $Y_n(b) = 0$ , so  $Y_n = \frac{\sinh n\pi(b-y)/a}{\sinh n\pi b/a}$ . Then

$v_n(x, y) = X_n(x)Y_n(y) = \frac{\sinh n\pi(b-y)/a}{\sinh n\pi b/a} \cos \frac{n\pi x}{a}$ , so  $v_n(x, 0) = \cos \frac{n\pi x}{a}$ . Therefore,  $v_n$  is so-

lution of the given problem with  $f(x) = \cos \frac{n\pi x}{a}$ . More generally, if  $\alpha_0, \dots, \alpha_m$  are arbitrary

constants, then  $u_m(x, y) = \alpha_0 \left(1 - \frac{y}{b}\right) + \sum_{n=1}^m \alpha_n \frac{\sinh n\pi(b-y)/a}{\sinh n\pi b/a} \cos \frac{n\pi x}{a}$  is a solution of the given

problem with  $f(x) = \alpha_0 + \sum_{n=1}^m \alpha_n \cos \frac{n\pi x}{a}$ . Therefore, if  $f$  is an arbitrary piecewise smooth func-

tion on  $[0, a]$  we define the formal solution of the given problem to be  $u(x, y) = \alpha_0 \left(1 - \frac{y}{b}\right) +$

$\sum_{n=1}^{\infty} \alpha_n \frac{\sinh n\pi(b-y)/a}{\sinh n\pi b/a} \cos \frac{n\pi x}{a}$ , where  $C(x) = \alpha_0 + \sum_{n=1}^{\infty} \alpha_n \cos \frac{n\pi x}{a}$  is the Fourier cosine series of

$f$  on  $[0, a]$ ; that is,  $\alpha_0 = \frac{1}{a} \int_0^a f(x) dx$  and  $\alpha_n = \frac{2}{a} \int_0^a f(x) \cos \frac{n\pi x}{a} dx$ ,  $n \geq 1$ .

Now consider the special case.  $\alpha_0 = \frac{1}{2} \int_0^2 (x^4 - 4x^3 + 4x^2) dx = \frac{1}{2} \left( \frac{x^5}{5} - x^4 + \frac{4x^3}{3} \right) \Big|_0^2 = \frac{6}{5}$ .

Since  $f'(0) = f'(2) = 0$  and  $f'''(x) = 12(2x - 2)$ , Theorem 11.3.5(a) implies that

$$\begin{aligned}\alpha_n &= \frac{96}{n^3\pi^3} \int_0^2 (2x-2) \sin \frac{n\pi x}{2} dx = -\frac{192}{n^4\pi^4} \left[ (2x-2) \cos \frac{n\pi x}{2} \Big|_0^2 - 2 \int_0^2 \cos \frac{n\pi x}{2} dx \right] \\ &= -\frac{192}{n^4\pi^4} [(-1)^n 2 + 2] + \frac{768}{n^5\pi^5} \sin \frac{n\pi x}{2} \Big|_0^2 = -\frac{384}{n^4\pi^4} [1 + (-1)^n] \\ &= \begin{cases} 0 & \text{if } n = 2m-1, \\ -\frac{48}{m^4\pi^4} & \text{if } n = 2m, \end{cases} \quad n \geq 1.\end{aligned}$$

$$C(x) = \frac{8}{15} - \frac{48}{\pi^4} \sum_{n=1}^{\infty} \frac{1}{n^4} \cos n\pi x; \quad u(x, y) = \frac{8(1-y)}{15} - \frac{48}{\pi^4} \sum_{n=1}^{\infty} \frac{1}{n^4} \frac{\sinh n\pi(1-y)}{\sinh n\pi} \cos n\pi x.$$

**12.3.20.** The boundary conditions require products  $v(x, y) = X(x)Y(y)$  such that (A)  $X'' + \lambda X = 0$ ,  $X(0) = 0$ ,  $X'(a) = 0$ , and (B)  $Y'' - \lambda Y = 0$ ,  $Y(0) = 1$ ,  $Y(b) = 0$ . From Theorem 11.1.4, the eigenvalues of (A) are  $\lambda_n = \frac{(2n-1)^2\pi^2}{4a^2}$ , with associated eigenfunctions  $Y_n = \sin \frac{(2n-1)\pi x}{2a}$ ,  $n = 1, 2, 3, \dots$ . Substituting  $\lambda = \frac{(2n-1)^2\pi^2}{4a^2}$  into (B) yields  $Y_n'' - ((2n-1)^2\pi^2/4a^2)Y_n = 0$ ,  $Y_n(0) = 1$ ,  $Y_n(b) = 0$ , so  $Y_n = \frac{\sinh(2n-1)\pi(b-y)/2a}{\sinh(2n-1)\pi b/2a}$ . Then  $v_n(x, y) = X_n(x)Y_n(y) = \frac{\sinh(2n-1)\pi(b-y)/2a}{\sinh(2n-1)\pi b/2a} \sin \frac{(2n-1)\pi x}{2a}$ , so  $v_n(x, 0) = \sin \frac{(2n-1)\pi x}{2a}$ . Therefore,  $v_n$  is solution of the given problem with  $f(x) = \sin \frac{(2n-1)\pi x}{2a}$ . More generally, if  $\alpha_1, \dots, \alpha_m$  are arbitrary constants, then  $u_m(x, y) = \sum_{n=1}^m \alpha_n \frac{\sinh(2n-1)\pi(b-y)/2a}{\sinh(2n-1)\pi b/2a} \cos \frac{(2n-1)\pi x}{2a}$  is a solution of the given problem with  $f(x) = \sum_{n=1}^m \alpha_n \sin \frac{(2n-1)\pi x}{2a}$ . Therefore, if  $f$  is an arbitrary piecewise smooth function on  $[0, a]$  we define the formal solution of the given problem to be  $u(x, y) = \sum_{n=1}^{\infty} \alpha_n \frac{\sinh(2n-1)\pi(b-y)/2a}{\sinh(2n-1)\pi b/2a} \sin \frac{(2n-1)\pi x}{2a}$ , where  $S_m(x) = \sum_{n=1}^m \alpha_n \sin \frac{(2n-1)\pi x}{2a}$  is the mixed Fourier sine series of  $f$  on  $[0, a]$ ; that is,  $\alpha_n = \frac{2}{a} \int_0^a f(x) \sin \frac{(2n-1)\pi x}{2a} dx$ .

Now consider the special case. Since  $f(0) = f'(L) = 0$  and  $f''(x) = -2$ , Theorem 11.3.5(d) implies that

$$\begin{aligned}\alpha_n &= \frac{48}{(2n-1)^2\pi^2} \int_0^3 \sin \frac{(2n-1)\pi x}{6} dx = -\frac{288}{(2n-1)^3\pi^3} \cos \frac{(2n-1)\pi x}{6} \Big|_0^3 = \frac{288}{(2n-1)^3\pi^3}; \\ S_M(x) &= \frac{288}{\pi^3} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^3} \sin \frac{(2n-1)\pi x}{6}; \\ u(x, y) &= \frac{288}{\pi^3} \sum_{n=1}^{\infty} \frac{\sinh(2n-1)\pi(2-y)/6}{(2n-1)^3 \sinh(2n-1)\pi/3} \sin \frac{(2n-1)\pi x}{6}.\end{aligned}$$

**12.3.22.** The boundary conditions require products  $v(x, y) = X(x)Y(y)$  such that (A)  $X'' + \lambda X = 0$ ,  $X'(0) = 0$ ,  $X'(a) = 0$ , and (B)  $Y'' - \lambda Y = 0$ ,  $Y'(0) = 0$ ,  $Y(b) = 1$ . From Theorem 11.1.3, the eigenvalues of (A) are  $\lambda = 0$ , with associated eigenfunction  $X_0 = 1$ , and  $\lambda_n = \frac{n^2\pi^2}{a^2}$ , with associated eigenfunctions  $Y_n = \cos \frac{n\pi x}{a}$ ,  $n = 1, 2, 3, \dots$ . Substituting  $\lambda = 0$  into (B) yields  $Y_0'' = 0$ ,  $Y_0'(0) = 0$ ,  $Y_0(b) = 1$ , so  $Y_0 = 1$ . Substituting  $\lambda = \frac{n^2\pi^2}{a^2}$  into (B) yields  $Y_n'' - (n^2\pi^2/a^2)Y_n = 0$ ,  $Y_n'(0) = 0$ ,  $Y_n(b) = 1$ , so  $Y_n = \frac{\cosh n\pi y/a}{\cosh n\pi b/a}$ . Then  $v_n(x, y) = X_n(x)Y_n(y) = \frac{\sinh n\pi y/a}{\cosh n\pi b/a} \cos \frac{n\pi x}{a}$ , so  $v_n(x, b) = \cos \frac{n\pi x}{a}$ . Therefore,  $v_n$  is solution of the given problem with  $f(x) = \cos \frac{n\pi x}{a}$ . More generally, if  $\alpha_0, \dots, \alpha_m$  are arbitrary constants, then  $u_m(x, y) = \alpha_0 + \sum_{n=1}^m \alpha_n \frac{\cosh n\pi y/a}{\cosh n\pi b/a} \cos \frac{n\pi x}{a}$  is a solution of the given problem with  $f(x) = \alpha_0 + \sum_{n=1}^m \alpha_n \cos \frac{n\pi x}{a}$ . Therefore, if  $f$  is an arbitrary piecewise smooth function on  $[0, a]$  we define the formal solution of the given problem to be  $u(x, y) = \alpha_0 + \sum_{n=1}^{\infty} \alpha_n \frac{\cosh n\pi y/a}{\cosh n\pi b/a} \cos \frac{n\pi x}{a}$ , where  $C(x) = \alpha_0 + \sum_{n=1}^{\infty} \alpha_n \cos \frac{n\pi x}{a}$  is the Fourier cosine series of  $f$  on  $[0, a]$ ; that is,  $\alpha_0 = \frac{1}{a} \int_0^a f(x) dx$  and  $\alpha_n = \frac{2}{a} \int_0^a f(x) \cos \frac{n\pi x}{a} dx$ ,  $n \geq 1$ .

Now consider the special case.

$$\alpha_0 = \frac{1}{\pi} \int_0^{\pi} (x^4 - 2\pi x^3 + \pi^2 x^2) dx = \frac{1}{\pi} \left( \frac{x^5}{5} - \frac{\pi x^4}{2} + \frac{\pi^2 x^3}{3} \right) \Big|_0^{\pi} = \frac{\pi^4}{30}.$$

Since  $f'(0) = f'(\pi) = 0$  and  $f'''(x) = 12(2x - \pi)$ , Theorem 11.3.5(a) implies that

$$\begin{aligned} \alpha_n &= \frac{24}{n^3\pi} \int_0^{\pi} (2x - \pi) \sin nx dx = -\frac{24}{n^4\pi} \left[ (2x - \pi) \cos nx \Big|_0^{\pi} - 2 \int_0^{\pi} \cos nx dx \right] \\ &= -\frac{24}{n^4\pi} [(-1)^n\pi + \pi] + \frac{48}{n^5\pi} \sin nx \Big|_0^{\pi} = -\frac{24}{n^4} [1 + (-1)^n] \\ &= \begin{cases} 0 & \text{if } n = 2m - 1, \\ -\frac{3}{m^4} & \text{if } n = 2m, \end{cases} \quad n \geq 1; \end{aligned}$$

$$C(x) = \frac{\pi^4}{30} - 3 \sum_{n=1}^{\infty} \frac{1}{n^4} \cos 2nx; \quad u(x, y) = \frac{\pi^4}{30} - 3 \sum_{n=1}^{\infty} \frac{1}{n^4} \frac{\cosh 2ny}{\cosh 2n} \cos 2nx$$

**12.3.24.** The boundary conditions require products  $v(x, y) = X(x)Y(y)$  such that (A)  $X'' - \lambda X = 0$ ,  $X'(0) = 0$ ,  $X(a) = 1$ , and (B)  $Y'' + \lambda Y = 0$ ,  $Y(0) = 0$ ,  $Y(b) = 0$ . From Theorem 11.1.2, the eigenvalues of (B) are  $\lambda_n = \frac{n^2\pi^2}{b^2}$ , with associated eigenfunctions  $Y_n = \sin \frac{n\pi y}{b}$ ,  $n = 1, 2, 3, \dots$ . Substituting  $\lambda = \frac{n^2\pi^2}{b^2}$  into (A) yields  $X_n'' - (n^2\pi^2/b^2)X_n = 0$ ,  $X_n'(0) = 0$ ,  $X_n(a) = 1$ , so  $X_n = \frac{\cosh n\pi x/b}{\cosh n\pi a/b}$ . Then  $v_n(x, y) = X_n(x)Y_n(y) = \frac{\cosh n\pi x/b}{\cosh n\pi a/b} \sin \frac{n\pi y}{b}$ , so  $v_n(a, y) = \sin \frac{n\pi y}{b}$ .



Therefore,  $v_n$  is solution of the given problem with  $g(y) = \sin \frac{n\pi y}{b}$ . More generally, if  $\alpha_1, \dots, \alpha_m$  are arbitrary constants, then  $u_m(x, y) = \sum_{n=1}^m \alpha_n \frac{\cosh n\pi x/b}{\cosh n\pi a/b} \sin \frac{n\pi y}{b}$  is a solution of the given problem with  $g(y) = \sum_{n=1}^m \alpha_n \sin \frac{n\pi y}{b}$ . Therefore, if  $g$  is an arbitrary piecewise smooth function on  $[0, b]$  we define the formal solution of the given problem to be  $u(x, y) = \sum_{n=1}^{\infty} \alpha_n \frac{\cosh n\pi x/b}{\cosh n\pi a/b} \sin \frac{n\pi y}{b}$ , where  $S(y) = \sum_{n=1}^{\infty} \alpha_n \sin \frac{n\pi y}{b}$  is the Fourier sine series of  $g$  on  $[0, b]$ ; that is,  $\alpha_n = \frac{2}{b} \int_0^b g(y) \sin \frac{n\pi y}{b} dy$ .

Now consider the special case. Since  $g(0) = g(1) = g''(0) = g''(L) = 0$  and  $f^{(4)}(y) = 24$ , Theorem 11.3.5(b) and Exercise 35(b) of Section 11.3 imply that

$$\begin{aligned} \alpha_n &= \frac{48}{n^4 \pi^4} \int_0^1 \sin n\pi y dy = -\frac{48}{n^5 \pi^5} \cos n\pi y \Big|_0^1 \\ &= -\frac{48}{n^5 \pi^5} [(-1)^n - 1] = \begin{cases} \frac{96}{(2m-1)^5 \pi^5} & \text{if } n = 2m-1 \\ 0 & \text{if } n = 2m; \end{cases} \end{aligned}$$

$$S(y) = \frac{96}{\pi^5} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^5} \sin(2n-1)\pi y; \quad u(x, y) = \frac{96}{\pi^5} \sum_{n=1}^{\infty} \frac{\cosh(2n-1)\pi x}{(2n-1)^5 \cosh(2n-1)\pi} \sin(2n-1)\pi y.$$

**12.3.26.** The boundary conditions require products  $v(x, y) = X(x)Y(y)$  such that (A)  $X'' - \lambda X = 0$ ,  $X'(0) = 0$ ,  $X'(a) = 1$ , and (B)  $Y'' + \lambda Y = 0$ ,  $Y(0) = 0$ ,  $Y(b) = 0$ . From Theorem 11.1.2, the eigenvalues of (B) are  $\lambda_n = \frac{n^2 \pi^2}{b^2}$ , with associated eigenfunctions  $Y_n = \sin \frac{n\pi y}{b}$ ,  $n = 1, 2, 3, \dots$

Substituting  $\lambda = \frac{n^2 \pi^2}{b^2}$  into (A) yields  $X_n'' - (n^2 \pi^2/b^2)X_n = 0$ ,  $X_n'(0) = 0$ ,  $X_n'(a) = 1$ , so  $X_n = \frac{b}{n\pi} \frac{\cosh n\pi x/b}{\sinh n\pi a/b}$ . Then  $v_n(x, y) = X_n(x)Y_n(y) = \frac{b}{n\pi} \frac{\cosh n\pi x/b}{\sinh n\pi a/b} \sin \frac{n\pi y}{b}$ , so  $\frac{\partial v_n}{\partial x}(a, y) = \sin \frac{n\pi y}{b}$ .

Therefore,  $v_n$  is solution of the given problem with  $g(y) = \sin \frac{n\pi y}{b}$ . More generally, if  $\alpha_1, \dots, \alpha_m$  are arbitrary constants, then  $u_m(x, y) = \frac{b}{\pi} \sum_{n=1}^m \alpha_n \frac{\cosh n\pi x/b}{\sinh n\pi a/b} \sin \frac{n\pi y}{b}$  is a solution of the given problem with  $g(y) = \sum_{n=1}^m \alpha_n \sin \frac{n\pi y}{b}$ . Therefore, if  $g$  is an arbitrary piecewise smooth function on  $[0, b]$  we

define the formal solution of the given problem to be  $u(x, y) = \frac{b}{\pi} \sum_{n=1}^{\infty} \alpha_n \frac{\cosh n\pi x/b}{\sinh n\pi a/b} \sin \frac{n\pi y}{b}$ , where  $S(y) = \sum_{n=1}^{\infty} \alpha_n \sin \frac{n\pi y}{b}$  is the Fourier sine series of  $g$  on  $[0, b]$ ; that is,  $\alpha_n = \frac{2}{b} \int_0^b g(y) \sin \frac{n\pi y}{b} dy$ .

Now consider the special case.  $\alpha_n = \frac{1}{2} \left[ \int_0^2 y \sin \frac{n\pi y}{4} dy + \int_2^4 (4-y) \sin \frac{n\pi y}{4} dy \right];$

$$\begin{aligned} \int_0^2 y \sin \frac{n\pi y}{4} dy &= -\frac{4}{n\pi} \left[ y \cos \frac{n\pi y}{4} \Big|_0^2 - \int_0^2 \cos \frac{n\pi y}{4} dy \right] \\ &= -\frac{2}{n\pi} \cos \frac{n\pi}{2} + \frac{4}{n^2\pi^2} \sin \frac{n\pi}{2} \Big|_0^2 = -\frac{2}{n\pi} \cos \frac{n\pi}{2} + \frac{4}{n^2\pi^2} \sin \frac{n\pi}{2}; \end{aligned}$$

$$\begin{aligned} \int_0^2 (4-y) \sin \frac{n\pi y}{4} dy &= -\frac{2}{n\pi} \left[ (4-y) \cos \frac{n\pi y}{4} \Big|_2^4 + \int_0^2 \cos \frac{n\pi y}{4} dy \right] \\ &= \frac{2}{n\pi} \cos \frac{n\pi}{2} - \frac{4}{n^2\pi^2} \sin \frac{n\pi}{2} \Big|_2^4 = \frac{2}{n\pi} \cos \frac{n\pi}{2} + \frac{4}{n^2\pi^2} \sin \frac{n\pi}{2}; \end{aligned}$$

$$\alpha_n = \frac{16}{n^2\pi^2} \sin \frac{n\pi}{2} = \begin{cases} (-1)^{m+1} \frac{16}{(2m-1)^2\pi^2} & \text{if } n = 2m-1 \\ 0 & \text{if } n = 2m; \end{cases}$$

$$S(y) = \frac{16}{\pi^2} \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{(2n-1)^2} \sin \frac{(2n-1)\pi y}{4};$$

$$u(x, y) = \frac{64}{\pi^3} \sum_{n=1}^{\infty} (-1)^{n+1} \frac{\cosh(2n-1)\pi x/4}{(2n-1)^3 \sinh(2n-1)\pi/4} \sin \frac{(2n-1)\pi y}{4}.$$

**12.3.28.** The boundary conditions require products  $v(x, y) = X(x)Y(y)$  such that (A)  $X'' - \lambda X = 0$ ,  $X'(0) = 1$ ,  $X(a) = 0$ , and (B)  $Y'' + \lambda Y = 0$ ,  $Y'(0) = 0$ ,  $Y'(b) = 0$ . From Theorem 11.1.3, the eigenvalues of (B) are  $\lambda_0 = 0$ , with associated eigenfunction  $Y_0 = 1$ , and  $\lambda_n = \frac{n^2\pi^2}{b^2}$ , with associated eigenfunctions  $Y_n = \cos \frac{n\pi y}{b}$ ,  $n = 1, 2, 3, \dots$ . Substituting  $\lambda_0 = 0$  into (A) yields  $X_0'' = 0$ ,  $X_0'(0) = 1$ ,  $X_0(a) = 0$ , so  $X_0 = x - a$ . Substituting  $\lambda = \frac{n^2\pi^2}{b^2}$  into (A) yields  $X_n'' - (n^2\pi^2/b^2)X_n = 0$ ,  $X_n'(0) = 1$ ,  $X_n(a) = 0$ , so  $X_n = \frac{b}{n\pi} \frac{\sinh n\pi(x-a)/b}{\cosh n\pi a/b}$ . Then  $v_n(x, y) = X_n(x)Y_n(y) = \frac{b}{n\pi} \frac{\sinh n\pi(x-a)/b}{\cosh n\pi a/b} \cos \frac{n\pi y}{b}$ , so  $\frac{\partial v_n}{\partial x}(0, y) = \cos \frac{n\pi y}{b}$ . Therefore,  $v_n$  is solution of the given problem with  $g(y) = \cos \frac{n\pi y}{b}$ . More generally, if  $\alpha_0, \dots, \alpha_m$  are arbitrary constants, then  $u_m(x, y) = \alpha_0(x-a) + \frac{b}{\pi} \sum_{n=1}^m \alpha_n \frac{\sinh n\pi(x-a)/b}{n \cosh n\pi a/b} \cos \frac{n\pi y}{b}$  is a solution of the given problem with  $g(y) = \sum_{n=1}^m \alpha_n \cos \frac{n\pi y}{b}$ . Therefore, if  $g$  is an arbitrary piecewise smooth function on  $[0, b]$  we define the formal solution of the given problem to be  $u(x, y) = \alpha_0(x-a) + \frac{b}{\pi} \sum_{n=1}^{\infty} \alpha_n \frac{\sinh n\pi(x-a)/b}{n \cosh n\pi a/b} \cos \frac{n\pi y}{b}$  where  $C(y) = \alpha_0 + \sum_{n=1}^{\infty} \alpha_n \cos \frac{n\pi y}{b}$  is the Fourier cosine series of  $g$  on  $[0, b]$ ; that is,  $\alpha_0 = \frac{1}{b} \int_0^b g(y) dy$ ,  $\alpha_n = \frac{2}{b} \int_0^b g(y) \cos \frac{n\pi y}{b} dy$ ,  $n \geq 1$ .

Now consider the special case. From Example 11.3.1,

$$C(y) = \frac{\pi}{2} - \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} \cos(2n-1)y;$$

$$u(x, y) = \frac{\pi(x-2)}{2} - \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{\sinh(2n-1)(x-2)}{(2n-1)^3 \cosh 2(2n-1)} \cos(2n-1)y.$$

**12.3.30.** The boundary conditions require products  $v(x, y) = X(x)Y(y)$  such that (A)  $X'' + \lambda X = 0$ ,  $X'(0) = 0$ ,  $X(a) = 0$ , and (B)  $Y'' - \lambda Y = 0$ ,  $Y(0) = 1$ , and  $Y$  is bounded. From Theorem 11.1.5, the eigenvalues of (A) are  $\lambda_n = \frac{(2n-1)^2\pi^2}{4a^2}$ , with associated eigenfunctions  $Y_n = \cos \frac{(2n-1)\pi x}{2a}$ ,  $n = 1, 2, 3, \dots$ . Substituting  $\lambda = \frac{(2n-1)^2\pi^2}{4a^2}$  into (B) yields  $Y_n'' - ((2n-1)^2\pi^2/4a^2)Y_n = 0$ ,  $Y_n(0) = 1$ , so  $Y_n = e^{-(2n-1)\pi y/2a}$ . Then  $v_n(x, y) = X_n(x)Y_n(y) = e^{-(2n-1)\pi y/2a} \cos \frac{(2n-1)\pi x}{2a}$ , so  $v_n(x, 0) = \cos \frac{(2n-1)\pi x}{2a}$ . Therefore,  $v_n$  is solution of the given problem with  $f(x) = \cos \frac{(2n-1)\pi x}{2a}$ . More generally, if  $\alpha_1, \dots, \alpha_m$  are arbitrary constants, then  $u_m(x, y) = \sum_{n=1}^m \alpha_n e^{-(2n-1)\pi y/2a} \cos \frac{(2n-1)\pi x}{2a}$  is a solution of the given problem with  $f(x) = \sum_{n=1}^m \alpha_n \cos \frac{(2n-1)\pi x}{2a}$ . Therefore, if  $f$  is an arbitrary piecewise smooth function on  $[0, a]$  we define the formal solution of the given problem to be  $u(x, y) = \sum_{n=1}^{\infty} \alpha_n e^{-(2n-1)\pi y/2a} \cos \frac{(2n-1)\pi x}{2a}$ , where  $C_m(x) = \sum_{n=1}^m \alpha_n \cos \frac{(2n-1)\pi x}{2a}$  is the mixed Fourier cosine series of  $f$  on  $[0, a]$ ; that is,  $\alpha_n = \frac{2}{a} \int_0^a f(x) \cos \frac{(2n-1)\pi x}{2a} dx$ .

Now consider the special case. Since  $f'(0) = f(L) = 0$  and  $f''(x) = -2$ , Theorem 11.3.5(d) implies that

$$\begin{aligned} \alpha_n &= \frac{16L}{(2n-1)^2\pi^2} \int_0^3 \cos \frac{(2n-1)\pi x}{6} dx \\ &= \frac{288}{(2n-1)^3\pi^3} \sin \frac{(2n-1)\pi x}{6} \Big|_0^3 = (-1)^{n+1} \frac{288}{(2n-1)^3\pi^3}; \\ C_M(x) &= -\frac{288}{\pi^3} \sum_{n=1}^{\infty} \frac{(-1)^n}{(2n-1)^3} \cos \frac{(2n-1)\pi x}{6}; \\ u(x, y) &= -\frac{288}{\pi^3} \sum_{n=1}^{\infty} \frac{(-1)^n}{(2n-1)^3} e^{-(2n-1)\pi y/6} \cos \frac{(2n-1)\pi x}{6}. \end{aligned}$$

**12.3.32.** The boundary conditions require products  $v(x, y) = X(x)Y(y)$  such that (A)  $X'' + \lambda X = 0$ ,  $X(0) = 0$ ,  $X(a) = 0$ , and (B)  $Y'' - \lambda Y = 0$ ,  $Y'(0) = 1$ , and  $Y$  is bounded. From Theorem 11.1.2, the eigenvalues of (A) are  $\lambda_n = \frac{n^2\pi^2}{a^2}$ , with associated eigenfunctions  $Y_n = \sin \frac{n\pi x}{a}$ ,  $n = 1, 2, 3, \dots$ . Substituting  $\lambda = \frac{n^2\pi^2}{a^2}$  into (B) yields  $Y_n'' - (n^2\pi^2/a^2)Y_n = 0$ ,  $Y_n'(0) = 1$ , so  $Y_n = -\frac{a}{n\pi} e^{-n\pi y/a}$ .

Then  $v_n(x, y) = X_n(x)Y_n(y) = -\frac{a}{n\pi}e^{-n\pi y/a}\sin\frac{n\pi x}{a}$ , so  $\frac{\partial v_n}{\partial y}(x, 0) = \sin\frac{n\pi x}{a}$ . Therefore,  $v_n$  is solution of the given problem with  $f(x) = \sin\frac{n\pi x}{a}$ . More generally, if  $\alpha_1, \dots, \alpha_m$  are arbitrary constants, then  $u_m(x, y) = -\frac{a}{\pi}\sum_{n=1}^m\frac{\alpha_n}{n}e^{-n\pi y/a}\sin\frac{n\pi x}{a}$  is a solution of the given problem with  $f(x) = \sum_{n=1}^m\alpha_n\sin\frac{n\pi x}{a}$ . Therefore, if  $f$  is an arbitrary piecewise smooth function on  $[0, a]$  we

define the formal solution of the given problem to be  $u(x, y) = -\frac{a}{\pi}\sum_{n=1}^{\infty}\frac{\alpha_n}{n}e^{-n\pi y/a}\sin\frac{n\pi x}{a}$ , where  $C(x) = \sum_{n=1}^{\infty}\alpha_n\sin\frac{n\pi x}{a}$  is the Fourier sine series of  $f$  on  $[0, a]$ ; that is,  $\alpha_n = \frac{2}{a}\int_0^a f(x)\sin\frac{n\pi x}{a}dx$ .

Now consider the special case. Since  $f(0) = f(\pi) = 0$  and  $f''(x) = 2\pi - 6x$ , Theorem 11.3.5(b) implies that

$$\begin{aligned}\alpha_n &= -\frac{2}{n^2\pi}\int_0^\pi (2\pi - 6x)\sin nx\,dx = \frac{2}{n^3\pi}\left[(2\pi - 6x)\cos nx\right]_0^\pi + 6\int_0^\pi \cos nx\,dx \\ &= -\frac{4}{n^3}[1 + (-1)^n 2] + \frac{12}{n^4\pi}\sin nx\Big|_0^\pi = -\frac{4}{n^3}[1 + (-1)^n 2];\end{aligned}$$

$$S(x) = -4\sum_{n=1}^{\infty}\frac{[1 + (-1)^n 2]}{n^3}\sin nx; \quad u(x) = 4\sum_{n=1}^{\infty}\frac{[1 + (-1)^n 2]}{n^4}e^{-ny}\sin nx;$$

**12.3.34.** The boundary conditions require products  $v(x, y) = X(x)Y(y)$  such that (A)  $X'' + \lambda X = 0$ ,  $X(0) = 0$ ,  $X'(a) = 0$ , and (B)  $Y'' - \lambda Y = 0$ ,  $Y'(0) = 1$ , and  $Y$  is bounded. From Theo-

rem 11.1.4, the eigenvalues of (A) are  $\lambda_n = \frac{(2n-1)^2\pi^2}{4a^2}$ , with associated eigenfunctions  $Y_n = \sin\frac{(2n-1)\pi x}{2a}$ ,  $n = 1, 2, 3, \dots$ . Substituting  $\lambda = \frac{(2n-1)^2\pi^2}{4a^2}$  into (B) yields  $Y_n'' - ((2n-1)^2\pi^2/4a^2)Y_n = 0$ ,  $Y_n'(0) = 1$ , so  $Y_n = -\frac{2a}{(2n-1)\pi}e^{-(2n-1)\pi y/2a}$ . Then  $v_n(x, y) = X_n(x)Y_n(y) =$

$-\frac{2a}{(2n-1)\pi}e^{-(2n-1)\pi y/2a}\sin\frac{(2n-1)\pi x}{2a}$ , so  $\frac{\partial v_n}{\partial y}(x, 0) = \sin\frac{(2n-1)\pi x}{2a}$ . Therefore,  $v_n$  is solution of the given problem with  $f(x) = \sin\frac{(2n-1)\pi x}{2a}$ . More generally, if  $\alpha_1, \dots, \alpha_m$  are arbitrary constants,

then  $u_m(x, y) = -\frac{2a}{\pi}\sum_{n=1}^m\frac{\alpha_n}{2n-1}e^{-(2n-1)\pi y/2a}\sin\frac{(2n-1)\pi x}{2a}$  is a solution of the given problem with  $f(x) = \sum_{n=1}^m\alpha_n\sin\frac{(2n-1)\pi x}{2a}$ . Therefore, if  $f$  is an arbitrary piecewise smooth function on  $[0, a]$  we

define the formal solution of the given problem to be  $u(x, y) = -\frac{2a}{\pi}\sum_{n=1}^{\infty}\frac{\alpha_n}{2n-1}e^{-(2n-1)\pi y/2a}\sin\frac{(2n-1)\pi x}{2a}$ , where

$S_m(x) = \sum_{n=1}^{\infty}\alpha_n\sin\frac{(2n-1)\pi x}{2a}$  is the mixed Fourier sine series of  $f$  on  $[0, a]$ ; that is,  $\alpha_n = \frac{2}{a}\int_0^a f(x)\sin\frac{(2n-1)\pi x}{2a}dx$ .

Now consider the special case.

$$\begin{aligned}
 \alpha_n &= \frac{2}{5} \int_0^5 (5x - x^2) \sin \frac{(2n-1)\pi x}{10} dx \\
 &= -\frac{4}{(2n-1)\pi} \left[ (5x - x^2) \cos \frac{(2n-1)\pi x}{10} \Big|_0^5 - \int_0^5 (5-2x) \cos \frac{(2n-1)\pi x}{10} dx \right] \\
 &= \frac{40}{(2n-1)^2 \pi^2} \left[ (5-2x) \sin \frac{(2n-1)\pi x}{10} \Big|_0^5 + 2 \int_0^5 \sin \frac{(2n-1)\pi x}{10} dx \right] \\
 &= (-1)^n \frac{200}{(2n-1)^2 \pi^2} - \frac{800}{(2n-1)^3 \pi^3} \cos \frac{(2n-1)\pi x}{10} \Big|_0^5 \\
 &= (-1)^n \frac{200}{(2n-1)^2 \pi^2} + \frac{800}{(2n-1)^3 \pi^3}; \\
 S_M(x) &= \frac{200}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} \left[ (-1)^n + \frac{4}{(2n-1)\pi} \right] \sin \frac{(2n-1)\pi x}{10}; \\
 u(x, y) &= -\frac{2000}{\pi^3} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^3} \left[ (-1)^n + \frac{4}{(2n-1)\pi} \right] e^{-(2n-1)\pi y/10} \sin \frac{(2n-1)\pi x}{10}.
 \end{aligned}$$

**12.3.36.** Solving BVP(1, 1, 1, 1)( $f_0, 0, 0, 0$ ) requires products  $X(x)Y(y)$  such that

$$X'' + \lambda X = 0, \quad X'(0) = 0, \quad X'(a) = 0; \quad Y'' - \lambda Y = 0, \quad Y'(b) = 0, \quad Y'(0) = 1.$$

Hence,  $X_n = \cos \frac{n\pi x}{a}$ ,  $Y_n = -\frac{a \cosh n\pi(y-b)/a}{n\pi \sinh n\pi b/a}$ , and  $c_1 - \frac{a}{\pi} \sum_{n=1}^{\infty} A_n \frac{\cosh n\pi(y-b)/a}{n\pi \sinh n\pi b/a} \cos \frac{n\pi x}{a}$  is a formal solution of BVP(1, 1, 1, 1)( $f_0, 0, 0, 0$ ) if  $c_1$  is any constant and  $\sum_{n=1}^{\infty} A_n \cos \frac{n\pi x}{a}$  is the Fourier cosine expansion of  $f_0$  on  $[0, a]$ , which is possible if and only if  $\int_0^a f_0(x) dx = 0$ .

Similarly,  $c_2 + \frac{a}{\pi} \sum_{n=1}^{\infty} B_n \frac{\cosh n\pi y/a}{n\pi \sinh n\pi b/a} \cos \frac{n\pi x}{a}$  is a formal solution of BVP(1, 1, 1, 1)(0,  $f_1, 0, 0$ ) if  $c_2$  is any constant and  $\sum_{n=1}^{\infty} B_n \cos \frac{n\pi x}{a}$  is the Fourier cosine expansion of  $f_1$  on  $[0, a]$ , which is possible if and only if  $\int_0^a f_1(x) dx = 0$ .

Interchanging  $x$  and  $y$  and  $a$  and  $b$  shows that  $c_3 - \frac{b}{\pi} \sum_{n=1}^{\infty} C_n \frac{\cosh n\pi(x-a)/b}{n\pi \sinh n\pi a/b} \cos \frac{n\pi y}{b}$  is a formal solution of BVP(1, 1, 1, 1)(0, 0,  $g_0, 0$ ) if  $c_3$  is any constant and  $\sum_{n=1}^{\infty} C_n \cos \frac{n\pi y}{b}$  is the Fourier cosine expansion of  $g_0$  on  $[0, b]$ , which is possible if and only if  $\int_0^b g_0(x) dx = 0$ , and  $c_4 + \frac{b}{\pi} \sum_{n=1}^{\infty} D_n \frac{\cosh n\pi x/b}{n\pi \sinh n\pi a/b} \cos \frac{n\pi y}{b}$  is a formal solution of BVP(1, 1, 1, 1)(0, 0, 0,  $g_1$ ) if  $c_4$  is any con-

stant and  $\sum_{n=1}^{\infty} D_n \cos \frac{n\pi y}{b}$  is the Fourier cosine expansion of  $g_1$  on  $[0, b]$ , which is possible if and only if  $\int_0^b g_1(x) dx = 0$ .

Adding the four solutions yields

$$u(x, y) = C + \frac{a}{\pi} \sum_{n=1}^{\infty} \frac{B_n \cosh n\pi y/a - A_n \cosh n\pi(y-b)/a}{n \sinh n\pi b/a} \cos \frac{n\pi x}{a} \\ + \frac{b}{\pi} \sum_{n=1}^{\infty} \frac{D_n \cosh n\pi x/b - C_n \cosh n\pi(x-a)/b}{n \sinh n\pi a/b} \cos \frac{n\pi y}{b},$$

where  $C$  is an arbitrary constant.

## 12.4 Laplace's Equation in Polar Coordinates

**12.4.2.**  $v(r, \theta) = R(r)\Theta(\theta)$  where (A)  $r^2 R'' + rR' - \lambda R = 0$  and  $\Theta'' + \lambda\Theta = 0$ ,  $\Theta(0) = 0$ ,  $\Theta(\gamma) = 0$ . From Theorem 11.1.2,  $\lambda_n = \frac{n^2\pi^2}{\gamma^2}$ ,  $\Theta_n = \sin \frac{n\pi\theta}{\gamma}$ ,  $n = 1, 2, 3, \dots$ . Substituting  $\lambda = \frac{n^2\pi^2}{\gamma^2}$  into

(A) yields the Euler equation  $r^2 R'' + rR' - \frac{n^2\pi^2}{\gamma^2} R = 0$  for  $R_n$ . The indicial polynomial is

$\left(s - \frac{n\pi}{\gamma}\right)\left(s + \frac{n\pi}{\gamma}\right)$ , so  $R_n = c_1 r^{n\pi/\gamma} + c_2 r^{-n\pi/\gamma}$ , by Theorem 7.4.3. We want  $R_n(\rho) = 1$  and

$R_n(\rho_0) = 0$ , so  $R_n(r) = \frac{\rho_0^{-n\pi/\gamma} r^{n\pi/\gamma} - \rho_0^{n\pi/\gamma} r^{-n\pi/\gamma}}{\rho_0^{-n\pi/\gamma} \rho^{n\pi/\gamma} - \rho_0^{n\pi/\gamma} \rho^{-n\pi/\gamma}};$

$$v_n(r, \theta) = \frac{\rho_0^{-n\pi/\gamma} r^{n\pi/\gamma} - \rho_0^{n\pi/\gamma} r^{-n\pi/\gamma}}{\rho_0^{-n\pi/\gamma} \rho^{n\pi/\gamma} - \rho_0^{n\pi/\gamma} \rho^{-n\pi/\gamma}} \sin \frac{n\pi\theta}{\gamma};$$

$u(r, \theta) = \sum_{n=1}^{\infty} \alpha_n \frac{\rho_0^{-n\pi/\gamma} r^{n\pi/\gamma} - \rho_0^{n\pi/\gamma} r^{-n\pi/\gamma}}{\rho_0^{-n\pi/\gamma} \rho^{n\pi/\gamma} - \rho_0^{n\pi/\gamma} \rho^{-n\pi/\gamma}} \sin \frac{n\pi\theta}{\gamma}$ , where  $S(\theta) = \sum_{n=1}^{\infty} \gamma_n \sin \frac{n\pi\theta}{\gamma}$  is the Fourier sine

series if  $f$  on  $[0, \gamma]$ ; that is,  $\alpha_n = \frac{1}{\gamma} \int_0^\gamma f(\theta) \sin \frac{n\pi\theta}{\gamma} d\theta$ ,  $n = 1, 2, 3, \dots$ .

**12.4.4.**  $v(r, \theta) = R(r)\Theta(\theta)$  where (A)  $r^2 R'' + rR' - \lambda R = 0$  and  $\Theta'' + \lambda\Theta = 0$ ,  $\Theta'(0) = 0$ ,  $\Theta(\gamma) = 0$ .

From Theorem 11.1.5,  $\lambda_n = \frac{(2n-1)^2\pi^2}{4\gamma^2}$ ,  $\Theta_n = \cos \frac{(2n-1)\pi\theta}{2\gamma}$ ,  $n = 1, 2, 3, \dots$ . Substituting  $\lambda =$

$\frac{(2n-1)^2\pi^2}{4\gamma^2}$  into (A) yields the Euler equation  $r^2 R'' + rR' - \frac{(2n-1)^2\pi^2}{4\gamma^2} R = 0$  for  $R_n$ . The indicial

polynomial is  $\left(s - \frac{(2n-1)\pi}{2\gamma}\right)\left(s + \frac{(2n-1)\pi}{2\gamma}\right)$ , so  $R_n = c_1 r^{(2n-1)\pi/2\gamma} + c_2 r^{-(2n-1)\pi/2\gamma}$ , by Theo-

rem 7.4.3. We want  $R_n$  to be bounded as  $r \rightarrow 0+$  and  $R_n(\rho) = 1$ , so we take  $R_n(r) = \frac{r^{(2n-1)\pi/2\gamma}}{\rho^{(2n-1)\pi/2\gamma}};$

$v_n(r, \theta) = \frac{r^{(2n-1)\pi/2\gamma}}{\rho^{(2n-1)\pi/2\gamma}} \cos \frac{(2n-1)\pi\theta}{2\gamma};$   $u(r, \theta) = \sum_{n=1}^{\infty} \alpha_n \frac{r^{(2n-1)\pi/2\gamma}}{\rho^{(2n-1)\pi/2\gamma}} \cos \frac{(2n-1)\pi\theta}{2\gamma}$ , where  $C_M(\theta) =$

$\sum_{n=1}^{\infty} \alpha_n \cos \frac{(2n-1)\pi\theta}{2\gamma}$  is the mixed Fourier cosine series of  $f$  on  $[0, \gamma]$ ; that is,  $\alpha_n = \frac{2}{\gamma} \int_0^{\gamma} f(\theta) \cos \frac{(2n-1)\pi\theta}{2\gamma} d\theta$ ,  $n = 1, 2, 3, \dots$ .

**12.4.6.**  $v(r, \theta) = R(r)\Theta(\theta)$  where (A)  $r^2 R'' + rR' - \lambda R = 0$  and  $\Theta'' + \lambda\Theta = 0$ ,  $\Theta'(0) = 0$ ,  $\Theta'(\gamma) = 0$ .

From Theorem 11.1.3,  $\lambda_0 = 0$ ,  $\Theta_0 = 1$ ;  $\lambda_n = \frac{n^2\pi^2}{\gamma^2}$ ,  $\Theta_n = \cos \frac{n\pi\theta}{\gamma}$ ,  $n = 1, 2, 3, \dots$ . Substituting  $\lambda = 0$  into (A) yields the equation  $r^2 R_0'' + rR_0' = 0$  for  $R_0$ ;  $R_0 = c_1 + c_2 \ln r$ . Since we want  $R_0$  to be bounded as  $r \rightarrow 0+$  and  $R_0(\rho) = 1$ ,  $R_0(r) = 1$ ; therefore  $v_0(r, \theta) = 1$ .

Substituting  $\lambda = \frac{n^2\pi^2}{\gamma^2}$  into (A) yields the Euler equation  $r^2 R_n'' + rR_n' - \frac{n^2\pi^2}{\gamma^2} R_n = 0$  for  $R_n$ . The indicial polynomial is  $\left(s - \frac{n\pi}{\gamma}\right) \left(s + \frac{n\pi}{\gamma}\right)$ , so  $R_n = c_1 r^{n\pi/\gamma} + c_2 r^{-n\pi/\gamma}$ , by Theorem 7.4.3. Since we want  $R_n$  to be bounded as  $r \rightarrow 0+$  and  $R_n(\rho) = 1$ ,  $R_n(r) = \frac{r^{n\pi/\gamma}}{\rho^{n\pi/\gamma}}$ ;  $v_n(r, \theta) = \frac{r^{n\pi/\gamma}}{\rho^{n\pi/\gamma}} \cos \frac{n\pi\theta}{\gamma}$ ,  $n = 1, 2, 3, \dots$ ;  $u(r, \theta) = \alpha_0 + \sum_{n=1}^{\infty} \alpha_n \frac{r^{n\pi/\gamma}}{\rho^{n\pi/\gamma}} \cos \frac{n\pi\theta}{\gamma}$ , where  $F(\theta) = \alpha_0 + \sum_{n=1}^{\infty} \alpha_n \cos \frac{n\pi\theta}{\gamma}$  is the Fourier cosine series of  $f$  on  $[0, \gamma]$ ; that is,  $\alpha_0 = \frac{1}{\gamma} \int_0^{\gamma} f(\theta) d\theta$  and  $\alpha_n = \frac{2}{\gamma} \int_0^{\gamma} f(\theta) \cos \frac{n\pi\theta}{\gamma} d\theta$ ,  $n = 1, 2, 3, \dots$ .

## Chapter 13

# Boundary Value Problems for Second Order Ordinary Differential Equations

### 13.1 Boundary Value Problems

**13.1.2.** By inspection,  $y_p = -x$ ;  $y = -x + c_1 e^x + c_2 e^{-x}$ ;  $y(0) = -2 \implies c_1 + c_2 = -2$ ;  
 $y(1) = 1 \implies -1 + c_1 e + c_2/e = 1$ ;  $\begin{bmatrix} 1 & 1 \\ e & 1/e \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} -2 \\ 2 \end{bmatrix}$ ;  $c_1 = \frac{2}{e-1}$ ;  $c_2 = \frac{2e}{1-e}$ ;

$$y = -x + \frac{2(e^x - e^{-(x-1)})}{e-1}$$

**13.1.4.** By inspection,  $y_p = -x$ ;  $y = -x + c_1 e^x + c_2 e^{-x}$ ;  $y' = -1 + c_1 e^x - c_2 e^{-x}$ ;  $y(0) + y'(0) = 3 \implies -1 + 2c_1 = 3$ ;  $c_1 = 2$ ;  $y(1) - y'(1) = 2 \implies \frac{2c_2}{e} = 2$ ;  $c_2 = e$ ;  $y = -x + 2e^x + e^{-(x-1)}$

**13.1.6.**  $y_p = Ax^2 e^x$ ;  $y'_p = A(x^2 + 2xe^x)$ ;  $y''_p = A(x^2 e^x + 4xe^x + 2e^x)$ ;  $y''_p - 2y'_p + y_p = 2Ae^x = e^x$  if  $A = 1$ ;  $y_p = x^2 e^x$ ;  $y = (x^2 + c_1 + c_2 x)e^x$ ;  $y' = (x^2 + 2x + c_1 + c_2 + c_2 x)e^x$ ;  $B_1(y) = 3$  and  $B_2(y) = 6e \implies -c_1 - 2c_2 = 3$ ,  $2c_1 + 3c_2 = 24$ ;  $c_1 = 13$ ;  $c_2 = -8$ ;  $y = (x^2 - 8x + 13)e^x$ .

**13.1.8.**  $B_1(y) = y(0)$ ;  $B_2(y) = y(1) - y'(1)$ . Let  $y_1 = x$ ,  $y_2 = 1$ ;  $B_1(y_1) = B_2(y_1) = 0$ . By variation of parameters, if  $y_p = u_1 x + u_2$  where  $u'_1 x + u'_2 = 0$  and  $u'_1 = F$ , then  $y''_p = F(x)$ . Let  $u'_1 = F$ ,  $u'_2 = -xF$ ;  $u_1 = -\int_x^1 F(t) dt$ ,  $u_2 = -\int_0^x tF(t) dt$ ;  $y_p = -x \int_x^1 F(t) dt - \int_0^x tF(t) dt$ ;  $y'_p = -\int_x^1 F(t) dt$ ;

$y = y_p + c_1 x + c_2$ . Since  $B_1(y_p) = 0$ ,  $B_1(x) = 0$  and  $B_1(1) = 1$ ,  $B_1(y) = 0 \implies c_2 = 0$ ; hence,  $y = y_p + c_1 x$ . Since  $B_2(y_p) = -\int_0^1 tF(t) dt$  and  $B_2(x) = 0$ ,  $B_2(y) = 0 \implies \int_0^1 tF(t) dt = 0$ . There

is no solution if this conditions does not hold. If it does hold, then the solutions are  $y = y_p + c_1 x$ , with  $c_1$  arbitrary.

**13.1.10.** (a) The condition is  $b-a \neq (k+1/2)\pi$  ( $k = \text{integer}$ ). Let  $y_1 = \sin(x-a)$  and  $y_2 = \cos(x-b)$ . Then  $y_1(a) = y'_2(b) = 0$  and  $\{y_1, y_2\}$  is linearly independent if  $b-a \neq (k+1/2)\pi$  ( $k = \text{integer}$ ),



since

$$\begin{vmatrix} \sin(x-a) & \cos(x-b) \\ \cos(x-a) & -\sin(x-b) \end{vmatrix} = -\cos(b-a) \neq 0.$$

Now Theorem 13.1.2 implies that (A) has a unique solution for any continuous  $F$  and constants  $k_1$  and  $k_2$ . If  $y = u_1 \sin(x - a) + u_2 \cos(x - b)$  where

$$\begin{aligned} u'_1 \sin(x - a) + u'_2 \cos(x - b) &= 0 \\ u'_1 \cos(x - a) - u'_2 \sin(x - b) &= F, \end{aligned}$$

then  $y'' + y = F$ .

$$\begin{aligned} u'_1 &= F(x) \frac{\cos(x - b)}{\cos(b - a)}, \quad u'_2 = -F(x) \frac{\sin(x - a)}{\cos(b - a)}, \\ u_1 &= -\frac{1}{\cos(b - a)} \int_x^b F(t) \cos(t - b) dt, \quad u_2 = -\frac{1}{\cos(b - a)} \int_a^x F(t) \sin(t - b) dt; \\ y &= -\frac{\sin(x - a)}{\cos(b - a)} \int_x^b F(t) \cos(t - b) dt - \frac{\cos(x - b)}{\cos(b - a)} \int_a^x F(t) \sin(t - a) dt. \end{aligned}$$

(b) If  $b - a = (k + 1/2)\pi$  ( $k = \text{integer}$ ), then  $y_1 = \sin(x - a)$  satisfies both boundary conditions  $y(a) = 0$  and  $y'(b) = 0$ . Let  $y_2 = \cos(x - a)$ . If  $y_p = u_1 \sin(x - a) + u_2 \cos(x - a)$  where

$$\begin{aligned} u'_1 \sin(x - a) + u'_2 \cos(x - a) &= 0, \\ u'_1 \cos(x - a) - u'_2 \sin(x - a) &= F, \end{aligned}$$

then  $y''_p + y_p = F$ ;  $u'_1 = F \cos(x - a)$ ;  $u'_2 = -F \sin(x - a)$ ;

$$\begin{aligned} u_1 &= -\int_x^b F(t) \cos(t - a) dt, \quad u_2 = -\int_a^x F(t) \sin(t - a) dt; \\ y_p &= -\sin(x - a) \int_x^b F(t) \cos(t - a) dt - \cos(x - a) \int_a^x F(t) \sin(t - a) dt; \\ y'_p &= -\cos(x - a) \int_x^b F(t) \cos(t - a) dt + \sin(x - a) \int_a^x F(t) \sin(t - a) dt. \end{aligned}$$

The general solution of  $y'' + y = F$  is  $y = y_p + c_1 \sin(x - a) + c_2 \cos(x - a)$ . Since  $b - a = (k + 1/2)\pi$ ,  $y'(b) = 0 \implies y'_p(b) = (-1)^k \int_a^b F(t) \sin(t - a) dt = 0$ ; therefore,  $F$  must satisfy  $\int_a^b F(t) \sin(t - a) dt = 0$ . In this case, the solutions of the boundary value problem are  $y = y_p + c_1 \sin(x - a)$ , with  $c_1$  arbitrary.

**13.1.12.** Let  $y_1 = \sinh(x - a)$  and  $y_2 = \sinh(x - b)$ . Then  $y_1(a) = 0$ ,  $y_2(b) = 0$ , and

$$W(x) = \begin{vmatrix} \sinh(x - a) & \sinh(x - b) \\ \cosh(x - a) & \cosh(x - b) \end{vmatrix} = \sinh(b - a) \neq 0.$$

(Since  $W$  is constant (Theorem 5.1.4), evaluate it by setting  $x = b$ .) From Theorem 13.1.2, (A) has a unique solution for any continuous  $F$  and constants  $k_1$  and  $k_2$ . If  $y = u_1 \sinh(x - a) + u_2 \sinh(x - b)$  where

$$\begin{aligned} u'_1 \sinh(x - a) + u'_2 \sinh(x - b) &= 0 \\ u'_1 \cosh(x - a) + u'_2 \cosh(x - b) &= F, \end{aligned}$$

then  $y'' - y = F$ .

$$\begin{aligned} u'_1 &= -F(x) \frac{\sinh(x-b)}{\sinh(b-a)}, & u'_2 &= F(x) \frac{\sinh(x-a)}{\sinh(b-a)}, \\ u_1 &= \frac{1}{\sinh(b-a)} \int_x^b F(t) \sinh(t-b) dt, & u_2 &= \frac{1}{\sinh(b-a)} \int_a^x F(t) \sinh(t-a) dt; \\ y &= \frac{\sinh(x-a)}{\sinh(b-a)} \int_x^b F(t) \sinh(t-b) dt + \frac{\sinh(x-b)}{\sinh(b-a)} \int_a^x F(t) \sinh(t-a) dt. \end{aligned}$$

**13.1.14.** Let  $y_1 = \cosh(x-a)$  and  $y_2 = \cosh(x-b)$ . Then  $y'_1(a) = y'_2(b) = 0$  and

$$W(x) = \begin{vmatrix} \cosh(x-a) & \cosh(x-b) \\ \sinh(x-a) & \sinh(x-b) \end{vmatrix} = -\sinh(b-a) \neq 0.$$

(Since  $W$  is constant (Theorem 5.1.40, evaluate it by setting  $x = b$ .) If  $y = u_1 \cosh(x-a) + u_2 \cosh(x-b)$  where

$$\begin{aligned} u'_1 \cosh(x-a) + u'_2 \cosh(x-b) &= 0 \\ u'_1 \sinh(x-a) + u'_2 \sinh(x-b) &= F, \end{aligned}$$

then  $y'' - y = F$ .

$$\begin{aligned} u'_1 &= F(x) \frac{\cosh(x-b)}{\sinh(b-a)}, & u'_2 &= -F(x) \frac{\cosh(x-a)}{\sinh(b-a)}, \\ u_1 &= -\frac{1}{\sinh(b-a)} \int_x^b F(t) \cosh(t-b) dt, & u_2 &= -\frac{1}{\sinh(b-a)} \int_a^x F(t) \cosh(t-a) dt; \\ y &= -\frac{\cosh(x-a)}{\sinh(b-a)} \int_x^b F(t) \cosh(t-b) dt - \frac{\cosh(x-b)}{\sinh(b-a)} \int_a^x F(t) \cosh(t-a) dt. \end{aligned}$$

**13.1.16.** Let  $y_1 = \sin \omega x$ ,  $y_2 = \sin \omega(x-\pi)$ ; then  $y_1(0) = 0$ ,  $y_2(\pi) = 0$ ,

$$W(x) = \begin{vmatrix} \sin \omega x & \sin \omega(x-\pi) \\ \omega \cos \omega x & \omega \cos \omega(x-\pi) \end{vmatrix} = \omega \sin \omega \pi \neq 0$$

if and only if  $\omega$  is not an integer. If this is so, then  $y = u_1 \sin \omega x + u_2 \sin \omega(x-\pi)$  if

$$\begin{aligned} u'_1 \sin \omega x + u'_2 \sin \omega(x-\pi) &= 0 \\ \omega(u'_1 \cos \omega x + u'_2 \cos \omega(x-\pi)) &= F; \\ u'_1 &= -F \frac{\sin \omega(x-\pi)}{\omega \sin \omega \pi}, & u'_2 &= F \frac{\sin \omega x}{\omega \sin \omega \pi}; \\ u_1 &= \frac{1}{\omega \sin \omega \pi} \int_x^\pi F(t) \sin \omega(t-\pi) dt; & u_2 &= \frac{1}{\omega \sin \omega \pi} \int_0^x F(t) \sin \omega t dt; \\ y &= \frac{1}{\omega \sin \omega \pi} \left( \sin \omega x \int_x^\pi F(t) \sin \omega(t-\pi) dt + \sin \omega(x-\pi) \int_0^x F(t) \sin \omega t dt \right). \end{aligned}$$

If  $\omega = n$  (positive integer), then  $y_1 = \sin nx$  is a nontrivial solution of  $y'' + y = 0$ ,  $y(0) = 0$ ,  $y(\pi) = 0$ . Let  $y_2 = \cos nx$ ; then  $W(x) = \begin{vmatrix} \sin nx & \cos nx \\ n \cos nx & -n \sin nx \end{vmatrix} = -n$ , and  $y_p = u_1 \sin nx + u_2 \cos nx$  satisfies  $y_p'' + n^2 y_p = 0$  if

$$\begin{aligned} u_1' \sin nx + u_2' \cos nx &= 0 \\ nu_1' \cos nx - nu_2' \sin nx &= F; \end{aligned}$$

$$u_1' = \frac{1}{n} F \cos nx, \quad u_2' = -\frac{1}{n} F \sin nx; \quad u_1 = -\frac{1}{n} \int_x^\pi F(t) \cos nt \, dt; \quad u_2 = -\frac{1}{n} \int_0^x F(t) \sin nt \, dt;$$

$$y_p = -\frac{1}{n} \left( \sin nx \int_x^\pi F(t) \cos nt \, dt + \cos nx \int_0^x F(t) \sin nt \, dt \right);$$

$y = y_p + c_1 \sin nx + c_2 \cos nx$ . Since  $y_p = 0$ ,  $y(0) = 0$ , so  $c_2 = 0$ ;  $y = y_p + c_1 \sin nx$ . Since  $y(\pi) = 0$ ,  $\int_0^\pi F(t) \sin nt \, dt = 0$  is necessary for existence of a solution. If this hold, then the solutions are  $y = y_p + c_1 \sin nx$ , with  $c_1$  arbitrary.

**13.1.18.** Let  $y_1 = \cos \omega x$ ;  $y_2 = \sin \omega(x - \pi)$ ; then  $y_1'(0) = y_2(\pi) = 0$ , and

$$W(x) = \begin{vmatrix} \cos \omega x & \sin \omega(x - \pi) \\ -\omega \sin \omega x & \omega \cos \omega(x - \pi) \end{vmatrix} = \omega \cos \omega \pi \neq 0$$

if and only if  $\omega \neq n + 1/2$  ( $n = \text{integer}$ ). If this is so, then  $y = u_1 \cos \omega x + u_2 \sin \omega(x - \pi)$  satisfies  $y'' + \omega^2 y = F(x)$  if

$$\begin{aligned} u_1' \cos \omega x + u_2' \sin \omega(x - \pi) &= 0 \\ \omega(-u_1' \sin \omega x + u_2' \cos \omega(x - \pi)) &= F; \end{aligned}$$

then

$$\begin{aligned} u_1' &= -\frac{F \sin \omega(x - \pi)}{\omega \cos \omega \pi}, \quad u_2' = \frac{F \cos \omega x}{\omega \cos \omega \pi}, \\ u_1 &= \frac{1}{\omega \cos \omega \pi} \int_x^\pi F(t) \sin \omega(t - \pi) \, dt, \quad u_2 = \frac{1}{\omega \cos \omega \pi} \int_0^x F(t) \cos \omega t \, dt, \\ y &= \frac{1}{\omega \cos \omega \pi} \left( \sin \omega x \int_x^\pi F(t) \sin \omega(t - \pi) \, dt + \sin \omega(x - \pi) \int_0^x F(t) \cos \omega t \, dt \right). \end{aligned}$$

If  $\omega = n + 1/2$  ( $n = \text{integer}$ ), then  $y_1 = \cos(n + 1/2)x$  is a nontrivial solution  $y'' + y = 0$ ,  $y'(0) = y(\pi) = 0$ . Let  $y_2 = \sin(n + 1/2)x$ ; then

$$W(x) = \begin{vmatrix} \cos(n + 1/2)x & \sin(n + 1/2)x \\ -(n + 1/2) \sin(n + 1/2)x & (n + 1/2) \cos(n + 1/2)x \end{vmatrix} = n + 1/2,$$

so  $y_p = u_1 \cos(n + 1/2)x + u_2 \sin(n + 1/2)x$  satisfies  $y_p'' + (n + 1/2)^2 y_p = F$  if

$$\begin{aligned} u_1' \cos(n + 1/2)x + u_2' \sin(n + 1/2)x &= 0 \\ -(n + 1/2)u_1' \sin(n + 1/2)x + (n + 1/2)u_2' \cos(n + 1/2)x &= F. \end{aligned}$$

$$\begin{aligned}
u'_1 &= -\frac{F \sin(n+1/2)x}{n+1/2}; & u'_2 &= \frac{F \cos(n+1/2)x}{n+1/2}; \\
u_1 &= \int_x^\pi \frac{F(t) \sin(n+1/2)t}{n+1/2} dt; & u_2 &= \int_0^x \frac{F(t) \cos(n+1/2)t}{n+1/2} dt; \\
y_p &= \frac{1}{n+1/2} \left( \cos(n+1/2)x \int_x^\pi F(t) \sin(n+1/2)t dt + \sin(n+1/2)x \int_0^x F(t) \cos(n+1/2)t dt \right); \\
y'_p &= -\sin(n+1/2)x \int_x^\pi F(t) \sin(n+1/2)t dt + \cos(n+1/2)x \int_0^x F(t) \cos(n+1/2)t dt; \\
y &= y_p + c_1 \cos(n+1/2)x + c_2 \sin(n+1/2)x; \\
y' &= y'_p + (n+1/2)(-c_1 \sin(n+1/2)x + c_2 \cos(n+1/2)x).
\end{aligned}$$

Since  $y'_p(0) = 0$ ,  $y'(0) = 0 \implies c_2 = 0$ ;  $y = y_p + c_1 \cos(n+1/2)x$ ;

$y' = y'_p + (n+1/2)c_1 \cos(n+1/2)x$ ;  $y(\pi) = 0 \implies y_p(\pi) = 0$ . Hence,  $\int_0^\pi F(y) \cos(n+1/2)t dt = 0$  is a necessary condition for existence of a solution. If this holds, then the solutions are  $y = y_p + c_1 \cos(n+1/2)x$  with  $c_1$  arbitrary.

**13.1.20.** Suppose  $y = c_1 z_1 + c_2 z_2$  is a nontrivial solution of the homogeneous boundary value problem. Then  $B_1(y) = c_1 B_1(z_1) + c_2 B_1(z_2) = 0$ . From Theorem 13.1.1 we may assume without loss of generality that  $B_1(z_2) \neq 0$ . Then  $c_2 = -\frac{B_1(z_1)}{B_1(z_2)} c_1$ . Therefore,  $y$  is constant multiple of

$y_0 = B_1(z_2)z_1 - B_1(z_1)z_2 \neq 0$ . To that check  $y$  satisfies the boundary conditions, note that  $B_1(y_0) = B_1(z_2)B_1(z_1) - B_1(z_1)B_1(z_2) = 0$   $B_2(y_0) = B_1(z_2)B_2(z_1) - B_1(z_1)B_2(z_2) = 0$ , by Theorem 13.1.2.

**13.1.22.**  $y_1 = a_1 + a_2 x$ ;  $y_1(0) - 2y'_1(0) = a_1 - 2a_2 = 0$  if  $a_1 = 2$ ,  $a_2 = 1$ ;  $y_1 = 2 + x$ .  $y_2 = b_1 + b_2 x$ ;  $y_2(1) = 2y'_1(1) = b_1 + 3b_2 = 0$  if  $b_1 = 3$ ,  $b_2 = -1$ ;  $y_2 = 3 - x$ .

$$\begin{aligned}
W(x) &= \begin{vmatrix} 2+x & 3-x \\ 1 & -1 \end{vmatrix} = -5; & G(x, t) &= \begin{cases} -\frac{(2+t)(3-x)}{5}, & 0 \leq t \leq x, \\ -\frac{(2+x)(3-t)}{5}, & x \leq t \leq 1. \end{cases} \\
y &= -\frac{1}{5} \left[ (2+x) \int_x^1 (3-t)F(t) dt + (3-x) \int_0^x (2+t)F(t) dt \right]. \quad (B)
\end{aligned}$$

(a) With  $F(x) = 1$ , (B) becomes

$$\begin{aligned}
y &= -\frac{1}{5} \left[ (2+x) \int_x^1 (3-t) dt + (3-x) \int_0^x (2+t) dt \right] \\
&= -\frac{1}{5} \left[ (2+x) \left( \frac{x^2 - 6x + 5}{2} \right) + (3-x) \left( \frac{x^2 + 4x}{2} \right) \right] \\
&= \frac{x^2 - x - 2}{2}.
\end{aligned}$$

(b) With  $F(x) = x$ , (B) becomes

$$\begin{aligned} y &= -\frac{1}{5} \left[ (2+x) \int_x^1 (3t-t^2) dt + (3-x) \int_0^x (2t+t^2) dt \right] \\ &= -\frac{1}{5} \left[ (2+x) \left( \frac{2x^3-9x^2+7}{6} \right) + (3-x) \left( \frac{x^3+3x^2}{3} \right) \right] \\ &= \frac{5x^3-7x-14}{30}. \end{aligned}$$

(b) With  $F(x) = x^2$ , (B) becomes

$$\begin{aligned} y &= -\frac{1}{5} \left[ (2+x) \int_x^1 (3t^2-t^3) dt + (3-x) \int_0^x (2t^2+t^3) dt \right] \\ &= -\frac{1}{5} \left[ (2+x) \left( \frac{x^4-4x^2+3}{4} \right) + (3-x) \left( \frac{3x^4+8x^3}{12} \right) \right] \\ &= \frac{5x^4-9x-18}{60}. \end{aligned}$$

**13.1.24.**  $y_1 = x^2 - x$ ,  $y_2 = x^2 - 2x$ ; then  $y_1(1) = 0$ ,  $y_2(2) = 0$ ;  $W(x) = \begin{vmatrix} x^2 - x & x^2 - 2x \\ 2x - 1 & 2x - 2 \end{vmatrix} = x^2$ .

Since  $P_0(x) = x^2$ ,  $G(x, t) = \begin{cases} \frac{(t-1)x(x-2)}{t^3}, & 1 \leq t \leq x, \\ \frac{x(x-1)(t-2)}{t^3}, & x \leq t \leq 2. \end{cases}$

$$y = x(x-1) \int_x^2 \frac{t-2}{t^3} F(t) dt + x(x-2) \int_1^x F(t) dt. \quad (\text{B})$$

(a) With  $F(x) = 2x^3$ , (B) becomes

$$\begin{aligned} y &= 2x(x-1) \int_x^2 (t-2) dt + 2x(x-2) \int_1^x (t-1) dt \\ &= -x(x-1)(x-2)^2 + x(x-2)(x-1)^2 = x(x-1)(x-2). \end{aligned}$$

(b) With  $F(x) = 6x^4$ , (B) becomes

$$\begin{aligned} y &= 6x(x-1) \int_x^2 (t-2)t dt + 6x(x-2) \int_1^x (t-1)t dt \\ &= -2x(x-1)(x+1)(x-2)^2 + x(x-2)(x-1)^2(2x+1) = x(x-1)(x-2)(x+3). \end{aligned}$$

**13.1.26.**  $y_1 = a_1 + a_2x$ ;  $y_1' = a_2$ ;  $B_1(y_1) = \alpha a_1 + \beta a_2 = 0$  if  $a_1 = \beta$ ,  $a_2 = -\alpha$ ;  $y_1 = \beta - \alpha x$ .  
 $y_2 = b_1 + b_2x$ ;  $y_2' = b_2$ ;  $B_2(y_2) = \rho b_1 + (\rho + \delta)b_2 = 0$  if  $b_1 = \rho + \delta$ ,  $b_2 = -\rho$ ;  $y_2 = \rho + \delta - \rho x$ ;  
 $W(x) = \begin{bmatrix} \beta - \alpha x & \rho + \delta - \rho x \\ -\alpha & -\rho \end{bmatrix} = \alpha(\rho + \delta) - \beta\rho$ . From Theorem 13.1.2, (A) has a unique solution

if and only if  $\alpha(\rho + \delta) - \beta\rho \neq 0$ . Then

$$G(x, t) = \begin{cases} \frac{(\beta - \alpha t)(\rho + \delta - \rho x)}{\alpha(\rho + \delta) - \beta\rho}, & 0 \leq t \leq x, \\ \frac{(\beta - \alpha x)(\rho + \delta - \rho t)}{\alpha(\rho + \delta) - \beta\rho}, & x \leq t \leq 1. \end{cases}$$

**13.1.28.**  $y_1 = a_1 \cos x + a_2 \sin x$ ;  $y'_1 = -a_1 \sin x + a_2 \cos x$ ;  $B_1(y_1) = \alpha a_1 + \beta a_2 = 0$  if  $a_1 = \beta$ ,  $a_2 = -\alpha$ .  $y_1 = \beta \cos x - \alpha \sin x$ .  $y_2 = b_1 \cos x + b_2 \sin x$ ;  $y'_2 = -b_1 \sin x + b_2 \cos x$ ;  $B_2(y_2) = \rho b_2 - \delta b_1 = 0$  if  $b_1 = \rho$ ,  $b_2 = \delta$ ;  $y_2 = \rho \cos x + \delta \sin x$ ;  $W(x) = \begin{bmatrix} \beta \cos x - \alpha \sin x & \rho \cos x + \delta \sin x \\ -\beta \sin x - \alpha \cos x & -\rho \sin x + \delta \cos x \end{bmatrix}$  Since  $W$  is constant, we can evaluate it with  $x = 0$ :  $W = \begin{bmatrix} \beta & \rho \\ -\alpha & \delta \end{bmatrix} = \alpha\rho + \beta\delta$ . From Theorem 13.1.2, (A) has a unique solution if and only if  $\alpha\rho + \beta\delta \neq 0$ . Then

$$G(x, t) = \begin{cases} \frac{(\beta \cos t - \alpha \sin t)(\rho \cos x + \delta \sin x)}{\alpha\rho + \beta\delta}, & 0 \leq t \leq x, \\ \frac{(\beta \cos x - \alpha \sin x)(\rho \cos t + \delta \sin t)}{\alpha\rho + \beta\delta}, & x \leq t \leq \pi. \end{cases}$$

**13.1.30.**  $y_1 = e^x(a_1 \cos x + a_2 \sin x)$ ;  $y'_1 = e^x[a_1(\cos x - \sin x) + a_2(\sin x + \cos x)]$ ;

$B_1(y_1) = (\alpha + \beta)a_1 + \beta a_2 = 0$  if  $a_1 = \beta$ ,  $a_2 = -(\alpha + \beta)$ .

$y_1 = e^x(\beta \cos x - (\alpha + \beta) \sin x)$ .  $y_2 = e^x(b_1 \cos x + b_2 \sin x)$ ;

$y'_2 = e^x[(b_1(\cos x - \sin x)) + b_2(\sin x + \cos x)]$ ;

$B_2(y_2) = -e^{\pi/2}[(\rho + \delta)b_2 - \delta b_1] = 0$  if  $b_1 = \delta$ ,  $b_2 = (\rho + \delta)$ ;

$y_2 = e^x[(\rho + \delta) \cos x + \delta \sin x]$ ;

To evaluate  $W(x)$ , we write  $y_1 = e^x v_1$  and  $y_2 = e^x v_2$ , where

$v_1 = \beta \cos x - (\alpha + \beta) \sin x$  and  $v_2 = (\rho + \delta) \cos x + \delta \sin x$ .

Then  $y'_1 = y_1 + e^x v'_1$  and  $y_2 = y_2 + e^x v'_2$ ,

$$\begin{aligned} W(x) &= \begin{vmatrix} y_1 & y_2 \\ y_1 + e^x v'_1 & y_2 + e^x v'_2 \end{vmatrix} = \begin{vmatrix} y_1 & y_2 \\ e^x v'_1 & x v'_2 \end{vmatrix} = e^{2x} \begin{vmatrix} v_1 & v_2 \\ v'_1 & x v'_2 \end{vmatrix} \\ &= \begin{vmatrix} \beta \cos x - (\alpha + \beta) \sin x & (\rho + \delta) \cos x + \delta \sin x \\ (-\beta \sin x - (\alpha + \beta) \cos x & -(\rho + \delta) \sin x + \delta \cos x \end{vmatrix}. \end{aligned}$$

Since  $v''_i + v_i = 0$ ,  $i = 1, 2$ , Theorem 5.1.4 implies that  $W(x) = K e^{2x}$ , where  $K$  is a constant that can be determined by setting  $x = 0$  in the determinant:

$$W(x)e^{2x} \begin{vmatrix} \beta & \rho + \delta \\ -\alpha - \beta & \delta \end{vmatrix} = [\beta\delta + (\alpha + \beta)(\rho + \delta)].$$

From Theorem 13.1.2, the boundary value problem has a unique solution if and only if  $\beta\delta + (\alpha + \beta)(\rho + \delta) \neq 0$ . In this case the Green's function is

$$G(x, t) = \begin{cases} e^{x-t} \frac{[\beta \cos t - (\alpha + \beta) \sin t][\rho + \delta) \cos x + \delta \sin x]}{\beta\delta + (\alpha + \beta)(\rho + \delta)}, & a \leq t \leq x \\ e^{x-t} \frac{[\beta \cos x - (\alpha + \beta) \sin x][\rho + \delta) \cos t + \delta \sin t]}{\beta\delta + (\alpha + \beta)(\rho + \delta)}, & x \leq t \leq \pi/2. \end{cases}$$

**13.1.32.** Let  $y_p = \int_a^b G(x, t)F(t) dt$ . From Theorem 13.1.3,  $Ly_p = F$ ,  $B_1(y_p) = 0$ , and  $B_2(y_p) = 0$ . The solution of  $Ly = F$ ,  $B_1(y) = k_1$ , and  $B_2(y) = k_2$  is of the form  $y = y_p + c_1y_1 + c_2y_2$ . Since  $B_1(y_p) = 0$  and  $B_1(y_1) = 0$ ,  $B_1(y) = k_1 \implies k_1 = c_2B_1(y_2) \implies c_2 = \frac{k_1}{B_1(y_2)}$ . Since  $B_2(y_p) = 0$  and  $B_2(y_2) = 0$ ,  $B_2(y) = k_2 \implies k_2 = c_1B_2(y_1) \implies c_1 = \frac{k_2}{B_2(y_1)}$ .

## 13.2 Sturm-Liouville Problems

**13.2.2.**  $y'' + \frac{1}{x}y' + \left(1 - \frac{\nu^2}{x^2}\right)y = 0$ ;  $\frac{p'}{p} = \frac{1}{x}$ ;  $\ln|p| = \ln|x|$ ;  $p = x$ ;  $xy'' + y' = \left(x - \frac{\nu^2}{x}\right)y = 0$ ;  
 $(xy')' + \left(x - \frac{\nu^2}{x}\right)y = 0$ .

**13.2.4.**  $y'' + \frac{b}{x}y' + \frac{c}{x^2}y = 0$ ;  $\frac{p'}{p} = \frac{b}{x}$ ;  $\ln|p| = b \ln|x|$ ;  $p = x^b$ ;  
 $x^b y'' + bx^{b-1}y' + cx^{b-2}y = 0$ ;  $(x^b y')' + cx^{b-2}y = 0$ .

**13.2.6.**  $xy'' + (1-x)y' + \alpha y = 0$ ;  $y'' + \left(\frac{1}{x} - 1\right)y' + \frac{\alpha}{x}y = 0$ ;  $\frac{p'}{p} = \frac{1}{x} - 1$ ;  $\ln|p| = \ln|x| - x$ ;  
 $p = xe^{-x}$ ;  $xe^{-x}y'' + (1-x)y' + \alpha e^{-x}y = 0$ ;  $(xe^{-x}y')' + \alpha e^{-x}y = 0$ .

**13.2.8.** If  $\lambda$  is an eigenvalue of (A) and  $y$  is a  $\lambda$ -eigenfunction, multiplying the differential equation in (B) by  $y$  yields  $(xy')y + \frac{\lambda}{x}y^2 = 0$ ;

$$\lambda \int_1^2 \frac{y^2(x)}{x^2} dx = - \int_1^2 (xy'(x))' y(x) dx = -xy'(x)y(x) \Big|_1^2 + \int_1^2 x(y'(x))^2 dx;$$

$$y(1) = y(2) = 0 \implies xy'(x)y(x) \Big|_1^2 = 0; \quad \lambda \int_1^2 \frac{y^2(x)}{x} dx = \int_1^2 x(y'(x))^2 dx.$$

Therefore  $\lambda \geq 0$ . We must still show that  $\lambda = 0$  is not an eigenvalue. To this end, suppose that  $(xy')' = 0$ ; then  $xy' = c_1$ ;  $y' = \frac{c_1}{x}$ ;  $y = c_1 \ln|x| + c_2$ ;  $y(1) = 0 \implies c_2 = 0$ ;  $y = c_1 \ln|x|$ ;  
 $y(2) = 0 \implies c_1 = 0$ ;  $y \equiv 0$ ; therefore  $\lambda = 0$  is not an eigenvalue.

**13.2.10.** Characteristic equation:  $r^2 + 2r + 1 + \lambda = 0$ ;  $r = -1 \pm \sqrt{-\lambda}$ .

$\lambda = 0$ :  $y = e^{-x}(c_1 + c_2x)$ ;  $y' = -e^{-x}(c_1 - c_2 + c_2x)$ ;  $y'(0) = 0 \implies c_1 = c_2$ ;  $y' = -c_2xe^{-x}$ ;  
 $y'(1) = 0 \implies -c_2/e = 0 \implies c_2 = 0$ ;  $\lambda = 0$  is not an eigenvalue.

$\lambda = -k^2$ ,  $k > 0$ :  $r = -1 \pm k$ ;  $y = e^{-x}(c_1 \cosh kx + c_2 \sinh kx)$ ;

$$y' = -c_1e^{-x}(-\cosh kx + k \sinh kx) + c_2e^{-x}(-\sinh kx + k \cosh kx).$$

The boundary conditions require that

$$c_1 + c_2k = 0 \quad \text{and} \quad (-\cosh k + k \sinh k)c_1 + (-\sinh k + k \cosh k)c_2 = 0.$$

This system has a nontrivial solution if and only if  $(1 - k^2) \sinh k = 0$ . Let  $k = 1$  and  $c_1 = c_2 = 1$ ; then  $\lambda = -1$  is the only negative eigenvalue, with associated eigenfunction  $y = 1$ .

$\lambda = k^2$ ,  $k > 0$ :  $r = -1 \pm ik$ ;  $y = e^{-x}(c_1 \cos kx + c_2 \sin kx)$ ;

$y' = c_1e^{-x}(-\cos kx - k \sin kx) + c_2e^{-x}(-\sin kx + k \cos kx)$ . The boundary conditions require that  
 $-c_1 + c_2k = 0 \quad \text{and} \quad (-\cos k - k \sin k)c_1 + (-\sin k + k \cos k)c_2 = 0$ .

This system has a nontrivial solution if and only if  $(1 + k^2) \sin k = 0$ . Let  $k = n\pi$  ( $k$  a positive



integer) and  $c_1 = n\pi$ ,  $c_2 = 1$ ; then  $\lambda_n = n^2\pi^2$  is an eigenvalue, with associated eigenfunction  $y_n = e^{-x}(n\pi \cos n\pi x + \sin n\pi x)$ .

**13.2.12.** Characteristic equation:  $r^2 + \lambda = 0$ .

$\lambda = 0$ :  $y = c_1 + c_2x$ .  $y(0) = 0 \implies c_1 = 0$ , so  $y = c_2x$ . Now  $y(1) - 2y'(1) = 0 \implies c_2 = 0$ . Therefore  $\lambda = 0$  is not an eigenvalue.

$\lambda = -k^2$ ,  $k > 0$ :  $y = c_1 \cosh kx + c_2 \sinh kx$ ;  $y' = k(c_1 \sinh kx + c_2 \cosh kx)$ .  $y'(0) \implies c_2 = 0$ , so  $y = c_1 \cosh kx$ . Now  $y(1) - 2y'(1) = 0 \implies c_1(\cosh k - 2k \sinh k) = 0$ , which is possible with  $c_1 \neq 0$  if and only if  $\tanh k = \frac{1}{2k}$ . Graphing both sides of this equation on the same axes show that it has one positive solution  $k_0$ ;  $y_0 = \cosh k_0x$  is a  $-k_0^2$ -eigenfunction.

$\lambda = k^2$ ,  $k > 0$ :  $y = c_1 \cos kx + c_2 \sin kx$ ;  $y' = k(-c_1 \sin kx + c_2 \cos kx)$ .  $y'(0) \implies c_2 = 0$ , so  $y = c_1 \cos kx$ . Now  $y(1) - 2y'(1) = 0 \implies c_1(\cos k + 2k \sin k) = 0$ , which is possible with  $c_1 \neq 0$  if and only if  $\tan k = -\frac{1}{2k}$ . Graphing both sides of this equation on the same axes shows that it has a solution  $k_n$  in  $((2n-1)\pi/2, n\pi)$ ,  $n = 1, 2, 3, \dots$ ;  $y_n = \cos k_nx$  is a  $k_n^2$ -eigenfunction.

**13.2.14.** Characteristic equation:  $r^2 + \lambda = 0$ .

$\lambda = 0$ :  $y = c_1 + c_2x$ . The boundary conditions require that  $c_1 + 2c_2 = 0$  and  $c_1 + \pi c_2 = 0$ , which imply that  $c_1 = c_2 = 0$ , so  $\lambda = 0$  is not an eigenvalue.

$\lambda = -k^2$ ,  $k > 0$ :  $y = c_1 \cosh kx + c_2 \sinh kx$ ;  $y' = k(c_1 \sinh kx + c_2 \cosh kx)$ . The boundary conditions require that

$$c_1 + 2kc_2 = 0 \quad \text{and} \quad c_1 \cosh k\pi + c_2 \sinh 2k\pi = 0.$$

This system has a nontrivial solution if and only if  $\tanh k\pi = 2k$ . Graphing both sides of this equation on the same axes shows that it has a solution  $k_0$  in  $(0, \pi)$ ;  $y_0 = 2k_0 \cosh k_0x - \sinh k_0x$  is a  $-k_0^2$ -eigenfunction.

$\lambda = k^2$ ,  $k > 0$ :  $y = c_1 \cos kx + c_2 \sin kx$ ;  $y' = k(-c_1 \sin kx + c_2 \cos kx)$ . The boundary conditions require that

$$c_1 + 2kc_2 = 0 \quad \text{and} \quad c_1 \cos k\pi + c_2 \sin k\pi = 0.$$

This system has a nontrivial solution if and only if  $\tan k\pi = 2k$ . Graphing both sides of this equation on the same axes shows that it has a solution  $k_n$  in  $(n, n+1/2)$ ,  $n = 1, 2, 3, \dots$ ;  $y_n = 2k_n \cos k_nx - \sin k_nx$  is a  $k_n^2$ -eigenfunction.

**13.2.16.** Characteristic equation:  $r^2 + \lambda = 0$ .

$\lambda = 0$ :  $y = c_1 + c_2x$ . The boundary conditions require that  $c_1 + c_2 = 0$  and  $c_1 + 4c_2 = 0$ , so  $c_1 = c_2 = 0$ . Therefore  $\lambda = 0$  is not an eigenvalue.

$\lambda = -k^2$ ,  $k > 0$ :  $y = c_1 \cosh kx + c_2 \sinh kx$ ;  $y' = k(c_1 \sinh kx + c_2 \cosh kx)$ . The boundary conditions require that

$$c_1 + kc_2 = 0 \quad \text{and} \quad (\cosh 2k + 2k \sinh 2k)c_1 + (\sinh 2k + 2k \cosh 2k)c_2 = 0.$$

This system has a nontrivial solution if and only if  $\tanh 2k = -\frac{k}{1-2k^2}$ . Graphing both sides of this equation on the same axes shows that it has a solution  $k_0$  in  $(1/\sqrt{2}, 1)$ ;  $y_0 = k_0 \cosh k_0x - \sinh k_0x$  is a  $-k_0^2$ -eigenfunction.

$\lambda = k^2$ ,  $k > 0$ :  $y = c_1 \cos kx + c_2 \sin kx$ ;  $y' = k(-c_1 \sin kx + c_2 \cos kx)$ . The boundary conditions require that

$$c_1 + kc_2 = 0 \quad \text{and} \quad (\cos 2k - 2k \sin 2k)c_1 + (\sin 2k + 2k \cos 2k)c_2 = 0.$$

This system has a nontrivial solution if and only if  $\tan 2k = -\frac{k}{1+2k^2}$ . Graphing both sides of this equation on the same axes shows that it has a solution  $k_n$  in  $((2n-1)\pi/4, n\pi/2)$ ,  $n = 1, 2, 3, \dots$ ;  $y_n = k_n \cos k_nx - \sin k_nx$  is a  $k_n^2$ -eigenfunction.

**13.2.18.** Characteristic equation:  $r^2 + \lambda = 0$ .

$\lambda = 0$ :  $y = c_1 + c_2x$ . The boundary conditions require that  $3c_1 + 2c_2 = 0$  and  $3c_1 + 4c_2 = 0$ , so  $c_1 = c_2 = 0$ . Therefore  $\lambda = 0$  is not an eigenvalue.

$\lambda = -k^2$ ,  $k > 0$ :  $y = c_1 \cosh kx + c_2 \sinh kx$ ;  $y' = k(c_1 \sinh kx + c_2 \cosh kx)$ . The boundary conditions require that

$$3c_1 + kc_2 = 0 \quad \text{and} \quad (3 \cosh 2k - 2k \sinh 2k)c_1 + (3 \sinh 3k - 2k \cosh 2k)c_2 = 0.$$

This system has a nontrivial solution if and only if  $\tanh 2k = \frac{9k}{9 + 2k^2}$ . Graphing both sides of this equation on the same axes shows that it has solutions  $y_1$  in  $(1, 2)$  and  $y_2$  in  $(5/2, 7/2)$ ;  $y_n = k_n \cosh k_n x - 3 \sinh k_n x$  is a  $-k_n^2$ -eigenfunction,  $k = 1, 2$ .

$\lambda = k^2$ ,  $k > 0$ :  $y = c_1 \cos kx + c_2 \sin kx$ ;  $y' = k(-c_1 \sin kx + c_2 \cos kx)$ . The boundary conditions require that

$$3c_1 + kc_2 = 0 \quad \text{and} \quad (3 \cos 2k + 2k \sin 2k)c_1 + (3 \sin 2k - 2k \cos 2k)c_2 = 0.$$

This system has a nontrivial solution if and only if  $\tan 2k = -\frac{9k}{9 - 2k^2}$ . Graphing both sides of this equation on the same axes shows that it has solutions  $k_0$  in  $(3/\sqrt{2}, \pi)$  and  $k_n$  in  $((2n+3)\pi/4, (n+2)\pi/3)$ ,  $n = 1, 2, 3, \dots$ ;  $y_n = k_n \cos k_n x - 3 \sin k_n x$  is a  $k_n^2$ -eigenfunction.

**13.2.20.** Characteristic equation:  $r^2 + \lambda = 0$ .

$\lambda = 0$ :  $y = c_1 + c_2x$ . The boundary conditions require that  $5c_1 + 2c_2 = 0$  and  $5c_1 + 3c_2 = 0$ , so  $c_1 = c_2 = 0$ . Therefore  $\lambda = 0$  is not an eigenvalue.

$\lambda = -k^2$ ,  $k > 0$ :  $y = c_1 \cosh kx + c_2 \sinh kx$ ;  $y' = k(c_1 \sinh kx + c_2 \cosh kx)$ . The boundary conditions require that

$$5c_1 + 2kc_2 = 0 \quad \text{and} \quad (5 \cosh k - 2k \sinh k)c_1 + (5 \sinh k - 2k \cosh k)c_2 = 0.$$

This system has a nontrivial solution if and only if  $\tanh k = \frac{20k}{25 + 4k^2}$ . Graphing both sides of this equation on the same axes shows that it has solutions  $k_1$  in  $(1, 2)$  and  $k_2$  in  $(5/2, 7/2)$ ;  $y_n = 2k_n \cosh k_n x - \sinh k_n x$  is  $-k_n$ -eigenfunction,  $n = 1, 2$ .

$\lambda = k^2$ ,  $k > 0$ :  $y = c_1 \cos kx + c_2 \sin kx$ ;  $y' = k(-c_1 \sin kx + c_2 \cos kx)$ . The boundary conditions require that

$$5c_1 + 2kc_2 = 0 \quad \text{and} \quad (5 \cos k + 2k \sin k)c_1 + (5 \sin k - 2k \cos k)c_2 = 0.$$

This system has a nontrivial solution if and only if  $\tan k = -\frac{20k}{25 - 4k^2}$ . Graphing both sides of this equation on the same axes shows that it has a solution  $k_n$  in

$((2n+1)\pi/2, (n+1)\pi)$ ,  $n = 1, 2, 3, \dots$ ;  $y_n = 2k_n \cos k_n x - 3 \sin k_n x$  is a  $k_n^2$ -eigenfunction.

**13.2.22.**  $\lambda = 0$ :  $x^2 y'' - 2xy' + 2y = 0$  is an Euler equation with indicial equation  $r(r-1) - 2r + 2 = (r-1)(r-2) = 0$ .  $y = x(c_1 + c_2x)$ ;  $y(1) = y(2) = 0 \implies c_1 + c_2 = c_1 + 2c_2 = 0 \implies c_1 = c_2 = 0$ , so  $\lambda = 0$  is not an eigenvalue.

$\lambda = -k^2$ ;  $k > 0$ :  $y = x(c_1 \cosh k(x-1) + c_2 \sinh k(x-1))$ ;  $y(1) = 0 \implies c_1 = 0$ ;  $y = c_2 x \sinh k(x-1)$ ;  $y(2) = 0 \implies 2c_2 \sinh k = 0 \implies c_2 = 0$ ;  $\lambda$  is not an eigenvalue.

$\lambda = k^2$ ;  $k > 0$ :  $y = x(c_1 \cos k(x-1) + c_2 \sin k(x-1))$ ;  $y(1) = 0 \implies c_1 = 0$ ;  $y = c_2 x \sin k(x-1)$ ;  $y(2) = 0$  with  $c_2 \neq 0$  if  $k = n\pi$  ( $n$  a positive integer);  $\lambda_n = n^2\pi^2$ ;  $y_n = x \sin n\pi(x-1)$  is a  $k_n^2$ -eigenfunction.

**13.2.24.**  $\lambda = 0$ :  $x^2 y'' - 2xy' + 2y = 0$  is an Euler equation with indicial equation  $r(r-1) - 2r + 2 = (r-1)(r-2) = 0$ .  $y = x(c_1 + c_2x)$ ;  $y' = c_1 + 2c_2x$ ;  $y(1) = y'(2) = 0 \implies c_1 + c_2 = c_1 + 4c_2 = 0 \implies c_1 = c_2 = 0$ , so  $\lambda = 0$  is not an eigenvalue.

$\lambda = -k^2$ ;  $k > 0$ :  $y = x(c_1 \cosh k(x-1) + c_2 \sinh k(x-1))$ ;  $y(1) = 0 \implies c_1 = 0$ ;  $y = c_2 x \sinh k(x-1)$ ;  $y'(2) = 0 \implies c_2(\sinh k + k \cosh k) = 0 \implies c_2 = 0$ ;  $\lambda$  is

not an eigenvalue.

$\lambda = k^2$ ;  $k > 0$ :  $y = x(c_1 \cos k(x-1) + c_2 \sin k(x-1))$ ;  $y(1) = 0 \implies c_1 = 0$ ;  $y = c_2 x \sin k(x-1)$ ;  $y' = c_2(\sin k(x-1) + kx \cos k(x-1))$ ;  $y'(2) = 0$  with  $c_2 \neq 0$  if and only if  $\sin k + 2k \cos k = 0$  or, equivalently,  $\tan k = -2k$ . Graphing both sides of this equation on the same axes shows that it has a solution  $k_n$  in  $((2n-1)\pi/2, n\pi)$ ,  $n = 1, 2, 3, \dots$ ;  $y_n = x \sin k_n(x-1)$  is a  $k_n^2$ -eigenfunction.

**13.2.26.**  $\lambda = 0$ :  $y = c_1 + c_2 x$ . The boundary conditions require that  $c_1 + \alpha c_2 = 0$  and  $c_1 + (\pi + \alpha)c_2 = 0$ , so  $c_1 = c_2 = 0$ . Therefore  $\lambda = 0$  is not an eigenvalue of (A).

$\lambda = -k^2$ ,  $k > 0$ :  $y = c_1 \cosh kx + c_2 \sinh kx$ ;  $y' = k(c_1 \sinh kx + c_2 \cosh kx)$ . The boundary conditions require that

$$\begin{aligned} c_1 + \alpha k c_2 &= 0 \\ (\cosh k\pi + \alpha k \sinh k\pi)c_1 + (\sinh k\pi + \alpha k \cosh k\pi)c_2 &= 0. \end{aligned} \quad (\text{D})$$

This system has a nontrivial solution if and only if  $(1 - k^2 \alpha^2) \sinh k\pi = 0$ , which holds with  $k > 0$  if and only if  $k^2 = \pm 1/\alpha$ . Therefore  $\lambda = -1/\alpha^2$  is the only negative eigenvalue. We can choose  $k = \pm 1/\alpha$ . Either way, the first equation in (D) implies that  $e^{-x/\alpha}$  is an associated eigenfunction.

$\lambda = k^2$ ,  $k > 0$ :  $y = c_1 \cos kx + c_2 \sin kx$ ;  $y' = k(-c_1 \sin kx + c_2 \cos kx)$ . The boundary conditions require that

$$\begin{aligned} c_1 + \alpha k c_2 &= 0 \\ (\cos k\pi - \alpha k \sin k\pi)c_1 + (\sin k\pi + \alpha k \cos k\pi)c_2 &= 0. \end{aligned} \quad (\text{E})$$

This system has a nontrivial solution if and only if  $(1 + k^2 \alpha^2) \sin k\pi = 0$ . Choosing  $k = n$  produces eigenvalues  $\lambda_n = n^2 \pi^2$ . Setting  $k = n$  in the first equation in (E) yields  $c_1 + \alpha n c_2 = 0$ , so  $y_n = n\alpha \cos nx - \sin nx$ .

**13.2.28.**  $y = c_1 + c_2 x$ . The boundary conditions require that

$$\alpha c_1 + \beta c_2 = 0 \quad \text{and} \quad \rho c_1 + (\rho L + \delta)c_2 = 0.$$

This system has a nontrivial solution if and only if  $\alpha(\rho L + \delta) - \beta\rho = 0$ .

**13.2.30.** (a)  $y = c_1 \cos kx + c_2 \sin kx$ ;  $y' = k(-c_1 \sin kx + c_2 \cos kx)$ . The boundary conditions require that

$$\alpha c_1 + \beta k c_2 = 0 \quad \text{and} \quad (\rho \cos kL - \delta k \sin kL)c_1 + (\rho \sin kL + \delta k \cos kL)c_2 = 0.$$

This system has a nontrivial solution if and only if its determinant is zero. This implies the conclusion.

(b) If  $\alpha\delta - \beta\rho = 0$ , (A) reduces to

$$(\alpha\rho + k^2\beta\delta) \sin kL = 0. \quad (\text{B})$$

From the solution of Exercise 13.2.29(b),  $\alpha\rho + k^2\beta\delta > 0$  for all  $k > 0$ . Therefore the positive zeros of (B) are  $k_n = n\pi/L$ ,  $n = 1, 2, 3, \dots$ , so the positive eigenvalues (SL) are  $\lambda_n = n^2 \pi^2 / L^2$ ,  $n = 1, 2, 3, \dots$

**13.2.32.** Suppose  $\lambda$  is an eigenvalue and  $y$  is an associated eigenfunction. From the solution of Exercise 13.2.31,

$$\lambda \int_a^b r(x)y^2(x) dx = p(a)y(a)y'(a) - p(b)y(b)y'(b) + \int_a^b p(x)(y'(x))^2 dx. \quad (\text{A})$$

If  $\alpha\beta = 0$  then either  $y(a) = 0$  or  $y'(a) = 0$ , so  $y(a)y'(a) = 0$ . If  $\alpha\beta < 0$  then  $y(a) = -\frac{\beta}{\alpha}y'(a)$ , so

$$y(a)y'(a) = -\frac{\beta}{\alpha}(y'(a))^2. \quad (\text{B})$$

Moreover,  $y'(a) \neq 0$  because if  $y'(a) = 0$  then  $y(a) = 0$ , from (B), and  $y \equiv 0$ , a contradiction. Since  $-\frac{\beta}{\alpha} > 0$  if  $\alpha\beta < 0$ , we conclude that if  $\alpha\beta \leq 0$ , then

$$p(a)y(a)y'(a) \geq 0, \quad (\text{C})$$

with equality if and only if  $\rho\delta = 0$ . A similar argument shows that if  $\rho\delta \geq 0$ , then

$$p(b)y(b)y'(b) \geq 0, \quad (\text{D})$$

with equality if and only if  $\alpha\beta = 0$ . Since  $(\alpha\beta)^2 + (\rho\delta)^2 > 0$ , the inequality must hold in at least one of (C) and (D). Now (A) implies that  $\lambda > 0$ .